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Chapter

1

Rational Numbers

INTRODUCTION

As learnt in class VII, the sum, the difference and the product of two integers is an integer. What happens if we divide an integer by another integer?

Consider $-3 \div 7 = -\frac{3}{7}$ which is not an integer.

$2 \div -4 = -\frac{1}{2}$ which is not an integer.

Thus, we need to extend our number system to include such numbers. This new system of numbers is called rational numbers. In this chapter, we will study the different properties of rational numbers. These properties are similar to properties of integers which we have learnt in the previous class.

Rational Numbers: A number which is in the form of $\frac{m}{n}$ where $n \neq 0$ and both m and n are integers are called rational numbers.

Rational numbers are denoted by Q .

For example :

$\frac{5}{8}, \frac{7}{3}, \frac{-6}{5}, \frac{1}{2}$ are all rational numbers.

0.12 is rational because $0.12 = \frac{12}{100}$.

Hence, all integers and decimal numbers are also rational numbers.

Natural Numbers: The counting numbers 1, 2, 3 ... are called natural numbers.

Whole Numbers: If we include '0' with natural numbers then the set of numbers 0, 1, 2, 3, are called whole numbers. It is denoted by (W).

Integers: All natural numbers, 0 and negative of natural numbers are called integers. They are denoted by I or Z . Thus -3, -2, -1, 0, 1, 2, 3 etc are all integers.

PROPERTIES OF RATIONAL NUMBERS

Closure

(a) **Addition :** For any two rational numbers, say, $\frac{a}{b}$ and $\frac{c}{d}$; $\left(\frac{a}{b} + \frac{c}{d}\right)$ is also a rational number.

Example : $\left(+\frac{5}{3}\right) + \left(+\frac{1}{6}\right) = \left(+\frac{11}{6}\right)$ which is rational.

(b) **Subtraction :** For any two rational numbers, say, $\frac{a}{b}$ and $\frac{c}{d}$; $\left(\frac{a}{b} - \frac{c}{d}\right)$ is also a rational number.

Example : $\frac{1}{2} - \frac{3}{2} = \frac{-2}{2} = -1$ which is rational.

(c) **Multiplication :** For any two rational numbers, say, $\frac{a}{b}$ and $\frac{c}{d}$; $\left(\frac{a}{b} \times \frac{c}{d}\right)$ is also a rational number.

Example : $\frac{1}{2} \times \left(\frac{-1}{10}\right) = \frac{-1}{20}$ which is a rational.

(d) **Division :** For any two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$; $\left[\frac{a}{b} \div \frac{c}{d}\right]$ is also a rational number.

Example : $\frac{2}{3} \div \left(\frac{-5}{9}\right) = \frac{2}{3} \times \left(\frac{-9}{5}\right) = \frac{-6}{5}$ Which is a rational.

Commutativity

(a) **Addition :** For any two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$; $\frac{a}{b} + \frac{c}{d} = \frac{c}{d} + \frac{a}{b}$

Example : $\left(+\frac{4}{5}\right) + \left(+\frac{7}{2}\right) = \left(+\frac{7}{2}\right) + \left(+\frac{4}{5}\right)$

(b) **Subtraction** : Subtraction of rational numbers is not commutative.

Example : $\frac{1}{7} - \frac{1}{5} \neq \frac{1}{5} - \frac{1}{7}$

(c) **Multiplication** : For any two rational numbers $\frac{a}{b}$ and $\frac{c}{d}$; $\frac{a}{b} \times \frac{c}{d} = \frac{c}{d} \times \frac{a}{b}$.

Example : $\left(\frac{-5}{7}\right) \times \left(\frac{-4}{7}\right) = \left(\frac{-4}{7}\right) \times \left(\frac{-5}{7}\right)$

(d) **Division** : Division of rational numbers is not commutative.

Example : $\frac{1}{12} \div \frac{5}{4} \neq \frac{5}{4} \div \frac{1}{12}$

Associativity

(a) **Addition** : For any three rational numbers $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$; $\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f}\right) = \left(\frac{a}{b} + \frac{c}{d}\right) + \frac{e}{f}$

Example : $\left(\frac{1}{8} + \frac{3}{7}\right) + \left(\frac{-1}{7}\right) = \frac{1}{8} + \left\{\frac{3}{7} + \left(\frac{-1}{7}\right)\right\}$

(b) **Subtraction** : Subtraction is not associative.

Example : $\left(\frac{4}{7} - \frac{4}{3}\right) - \frac{4}{5} \neq \frac{4}{7} - \left(\frac{4}{3} - \frac{4}{5}\right)$

(c) **Multiplication** : For any three rational numbers $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$; $\frac{a}{b} \times \left(\frac{c}{d} \times \frac{e}{f}\right) = \left(\frac{a}{b} \times \frac{c}{d}\right) \times \frac{e}{f}$

Example : $\left(\frac{1}{8} \times \frac{3}{7}\right) \times \left(\frac{-1}{7}\right) = \frac{1}{8} \times \left\{\frac{3}{7} \times \left(\frac{-1}{7}\right)\right\}$

(d) **Division** : For any three rational numbers $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$, division is not associative, i.e., $\left(\frac{a}{b} \div \frac{c}{d}\right) \div \frac{e}{f} \neq \frac{a}{b} \div \left(\frac{c}{d} \div \frac{e}{f}\right)$

Example : $\left(\frac{1}{2} \div \frac{3}{4}\right) \div \frac{7}{8} \neq \frac{1}{2} \div \left(\frac{3}{4} \div \frac{7}{8}\right)$

ILLUSTRATION : 1

Show that $\left(\frac{1}{5} \div \frac{3}{4}\right) \div \frac{1}{2} \neq \frac{1}{5} \div \left(\frac{3}{4} \div \frac{1}{2}\right)$

SOLUTION :

L.H.S. $\left(\frac{1}{5} \times \frac{4}{3}\right) \div \frac{1}{2} = \frac{4}{15} \div \frac{1}{2}$

$= \frac{4}{15} \times 2 = \frac{8}{15}$

R.H.S. $\frac{1}{5} \div \left(\frac{3}{4} \times 2\right) = \frac{1}{5} \div \frac{3}{2}$

$= \frac{1}{5} \times \frac{2}{3} = \frac{2}{15}$

So, L.H.S. $\neq$ R. H. S

THE ROLE OF 0

Additive Identity

For every rational number $\frac{a}{b}$, $\frac{a}{b} + 0 = 0 + \frac{a}{b} = \frac{a}{b}$, then 0 is called additive identity.

Example : $\frac{1}{7} + 0 = 0 + \frac{1}{7} = \frac{1}{7}$

Additive Inverse

For every rational number $\frac{a}{b}$, we have $\frac{a}{b} + \left(\frac{-a}{b}\right) = 0$, then $\frac{a}{b}$ and $\frac{-a}{b}$ are the additive inverse of each other.

Zero Property of Multiplication

For every rational number $\frac{a}{b}$, we have $\frac{a}{b} \times 0 = 0 \times \frac{a}{b} = 0$

Example : $\frac{1}{9} \times 0 = 0$

Division By 0

Division by zero is not defined i.e., $a \div 0 = \text{not defined}$

Example : $\frac{1}{3} \div 0 = \text{not defined}$

But for every rational number $\frac{a}{b}$, $0 \div \frac{a}{b} = 0$

Example : $0 \div \frac{3}{4} = 0$

THE ROLE OF 1

Multiplicative Identity

For every rational number $\frac{a}{b}$, $\frac{a}{b} \times 1 = 1 \times \frac{a}{b} = \frac{a}{b}$

1 is called the multiplicative identity.

Division By 1

For every rational number $\frac{a}{b}$, $\frac{a}{b} \div 1 = \frac{a}{b}$

Example : $\frac{1}{2} \div 1 = \frac{1}{2}$

NEGATIVE OF A NUMBER

For any rational number $\frac{a}{b}$, there is a rational number, $-\left(\frac{a}{b}\right)$ such that $\frac{a}{b} + \left(-\frac{a}{b}\right) = 0$.

We say $-\left(\frac{a}{b}\right)$ is the negative of $\frac{a}{b}$ and $\frac{a}{b}$ is called the negative of $-\left(\frac{a}{b}\right)$.

Example: $\frac{5}{7} + \left(\frac{-5}{7}\right) = \frac{5+(-5)}{7} = \frac{0}{7} = 0$

MULTIPLICATIVE INVERSE

For any rational number $\frac{a}{b}$, if $\frac{a}{b} \times \frac{b}{a} = 1$, then $\frac{a}{b}$ and $\frac{b}{a}$ are the multiplicative inverse of each other and $\frac{b}{a}$ is called the reciprocal of $\frac{a}{b}$.

Therefore, the reciprocal of $\frac{-5}{8}$ is $\frac{8}{-5}$ or $\frac{-8}{5}$.

DISTRIBUTION OF MULTIPLICATION OVER ADDITION

For any three rational numbers, $\frac{a}{b}, \frac{c}{d}$ and $\frac{e}{f}$, $\frac{a}{b} \times \left(\frac{c}{d} + \frac{e}{f}\right) = \left(\frac{a}{b} \times \frac{c}{d}\right) + \left(\frac{a}{b} \times \frac{e}{f}\right)$

ILLUSTRATION : 2

Show that $\frac{2}{3} \left(\frac{5}{4} + \frac{7}{3}\right) = \left(\frac{2}{3} \times \frac{5}{4}\right) + \left(\frac{2}{3} \times \frac{7}{3}\right)$

SOLUTION :

$$\text{L.H.S.} \quad \frac{2}{3} \left(\frac{5}{4} + \frac{7}{3}\right) = \frac{2}{3} \left(\frac{15+28}{12}\right) = \frac{2}{3} \left(\frac{43}{12}\right) = \frac{43}{18}$$

$$\begin{aligned} \text{R.H.S.} \quad \left(\frac{2}{3} \times \frac{5}{4}\right) + \left(\frac{2}{3} \times \frac{7}{3}\right) &= \frac{5}{6} + \frac{14}{9} \\ &= \frac{15+28}{18} = \frac{43}{18} \end{aligned}$$

So, L.H.S = R.H.S.

ILLUSTRATION : 3

Simplify $\left(\frac{3}{11} \times \frac{5}{6}\right) - \left(\frac{9}{12} \times \frac{4}{3}\right) + \left(\frac{5}{13} \times \frac{6}{5}\right)$

SOLUTION :

$$\frac{15}{66} - \frac{36}{36} + \frac{30}{65} = \frac{5}{22} - 1 + \frac{6}{13} = \frac{-89}{286}$$

REPRESENTATION OF RATIONAL NUMBERS ON NUMBER LINE

Representation of Rational Numbers in the Form of $\frac{p}{q}$ on Number Line:

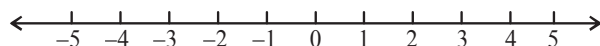
Like whole numbers, integers, rational numbers, in the form of $\frac{p}{q}$ where p and q are integers can be represented on number line.

ILLUSTRATION : 4

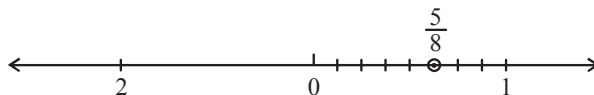
Represent $\frac{5}{8}$, $-\frac{3}{5}$ and $\frac{8}{3}$ on number line.

SOLUTION :

Draw a number line as shown below:



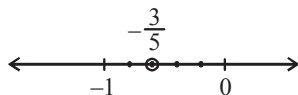
Now in $\frac{5}{8}$, Numerator (5) < Denominator (8) and number is positive. $\frac{5}{8}$ represent a number between 0 and 1. Mark 7 (= 8 - 1) points at equal distance which divide the distances between 0 and 1 in 8 (denominator) equal parts, as shown below (Magnified).



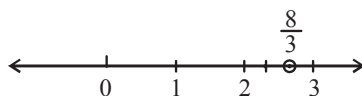
Out of these 7 points the 5th (numerator) point represent the rational number $\frac{5}{8}$.

In $-\frac{3}{5}$, which is a -ve rational number, absolute value of the numerator (3) < denominator (5). So it represent a number between 0 and -1.

To represent it on number line, mark 4 (= 5 - 1) points between 0 and -1 which divide the distance between 0 and 1 in 5 (denominator) equal parts. Out of these 4 points. 3rd (absolute value of the numerator) point from right between 0 and -1 represents the rational number $-\frac{3}{5}$.



$\frac{8}{3}$ can be written as $2 + \frac{2}{3}$. Clearly this number lies between 2 and 3 eg. on the number line. Mark 2 (= 3 - 1) points between 2 and 3, which divides the distance between 2 and 3 in 3 (denominator) equal parts. Of these two points, the 2nd point from left between 2 and 3 represent $8 \left(= 2 + \frac{2}{3} \right)$



RATIONAL NUMBERS BETWEEN TWO RATIONAL NUMBERS

Between two rational numbers, there are infinitely many rational numbers.

Method - 1 : Let a and b be two rational numbers $q_1 = \frac{a+b}{2} \Rightarrow a < q_1 < b$

q_1 is the rational number between a and b .

$$q_2 = \frac{a+q_1}{2} \Rightarrow a < q_2 < q_1 < b$$

q_2 is the rational number between a and q_1 .

$$q_3 = \frac{q_1+b}{2} \Rightarrow a < q_2 < q_1 < q_3 < b$$

q_3 is the rational number between q_1 and q_2 .

In this manner, we can find infinite rational numbers between two given distinct rational numbers.

Method – 2 : $\frac{a}{b}$ and $\frac{c}{d}$ be two rational numbers.

Step – 1 : Make denominators equal in both rational numbers.

Step – 2 : If we have to find n rational numbers between $\frac{ad}{bd}$ and $\frac{cb}{bd}$, then multiply numerators and denominators by such a number so that the difference between the numerators is at least n .

ILLUSTRATION : 5

Insert 5 rational numbers between $\frac{1}{5}$ and $\frac{2}{5}$.

SOLUTION :

Since the denominator of $\frac{1}{5}$ and $\frac{2}{5}$ is same and we do not have five integers between 1 and 2, therefore change their denominators such that, there exist at least five integers between numerator part.

Thus,

$$\frac{1}{5} = \frac{1 \times 7}{5 \times 7} = \frac{7}{35}$$

$$\frac{2}{5} = \frac{2 \times 7}{5 \times 7} = \frac{14}{35}$$

Now we can write

$$\frac{7}{35} < \frac{8}{35} < \frac{9}{35} < \frac{10}{35} < \frac{11}{35} < \frac{12}{35} < \frac{14}{35}$$

$$\therefore \frac{8}{35}, \frac{9}{35}, \frac{10}{35}, \frac{11}{35}, \frac{12}{35} \text{ lie between } \frac{7}{35} \text{ and } \frac{14}{35}.$$

MISCELLANEOUS

SOLVED EXAMPLES

1. The sum of two rational numbers is $\frac{-3}{5}$. If one of the numbers is $\frac{-9}{20}$, find the other.

Sol. It is given that sum of the numbers $= \frac{-3}{5}$ and one of the numbers $= \frac{-9}{20}$

Suppose the other rational number is x . Since the sum is $\frac{-3}{5}$

$$\therefore x + \left(\frac{-9}{20}\right) = \frac{-3}{5} \Rightarrow x = \frac{-3}{5} - \left(\frac{-9}{20}\right) \Rightarrow x = \frac{-3}{5} + \frac{9}{20} \quad \left[-\left(\frac{-9}{20}\right) = \frac{9}{20} \right]$$

$$\Rightarrow x = \frac{(-3) \times 4 + 9 \times 1}{20} \Rightarrow x = \frac{(-3) \times 4 + 9 \times 1}{20} = \frac{-12 + 9}{20} = \frac{-3}{20}$$

2. Simplify : $\left(\frac{-7}{18} \times \frac{15}{-7}\right) - \left(1 \times \frac{1}{4}\right) + \left(\frac{1}{2} \times \frac{1}{4}\right)$

$$\begin{aligned} \text{Sol. } \left(\frac{-7}{18} \times \frac{15}{-7}\right) - \left(1 \times \frac{1}{4}\right) + \left(\frac{1}{2} \times \frac{1}{4}\right) &= \left(\frac{-7}{18} \times \frac{15}{-7}\right) - \left(\frac{1}{1} \times \frac{1}{4}\right) + \left(\frac{1}{2} \times \frac{1}{4}\right) \\ &= \frac{5}{6} - \frac{1}{4} + \frac{1}{8} = \frac{5}{6} + \frac{-1}{4} + \frac{1}{8} = \frac{5 \times 4 + (-1) \times 6 + 1 \times 3}{24} = \frac{20 + (-6) + 3}{24} = \frac{17}{24} \end{aligned}$$

3. Find 10 rational numbers between $\frac{-2}{11}$ and $\frac{9}{11}$.

Sol. The given rational numbers are $\frac{-2}{11}$ and $\frac{9}{11}$. Since the denominators of both the rational numbers are equal and positive. Finding 10 rational numbers between $\frac{-2}{11}$ and $\frac{9}{11}$ is similar to finding 10 integers between -2 and 9 . We know that $-2 < -1 < 0 < 1 < 2 < 3 < 4 < 5 < 6 < 7 < 8 < 9$

So, $\frac{-1}{11}, \frac{0}{11}, \frac{1}{11}, \frac{2}{11}, \frac{3}{11}, \frac{4}{11}, \frac{5}{11}, \frac{6}{11}, \frac{7}{11}, \frac{8}{11}$ are the desired 10 rational numbers between $\frac{-2}{11}$ and $\frac{9}{11}$.

4. What should be added to $\left(\frac{1}{2} + \frac{1}{3} - \frac{1}{5}\right)$ to get 3?

Sol. Sum of the numbers = 3

$$\text{One of the numbers} = \left(\frac{1}{2} + \frac{1}{3} - \frac{1}{5}\right) = \frac{15 + 10 - 6}{30} = \frac{19}{30}$$

The other number = Sum - one number

$$= 3 - \frac{19}{30} = \frac{90 - 19}{30} = \frac{71}{30}$$

5. The speed of car is $54\frac{1}{2}$ km per hour. Find the distance travelled in $\frac{7}{2}$ hours and $\frac{35}{2}$ minutes.

Sol. We first calculate the time in hour.

$$\frac{35}{2} \text{ minutes} = \frac{\cancel{35}^7}{2 \times \cancel{60}_{12}} = \frac{7}{24} \text{ hour.}$$

Therefore, total time (in hours) = $\frac{7}{2} + \frac{7}{24}$ hour = $\frac{84+7}{24} = \frac{91}{24}$ hours

Distance = Speed $\times$ time

Since, Speed = $54\frac{1}{2} = \frac{109}{2}$ km/hr

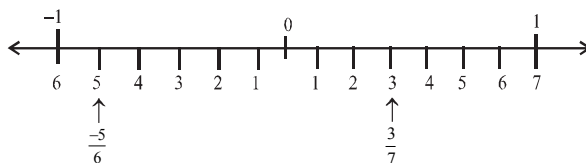
$$\Rightarrow \text{Distance} = \frac{109}{2} \times \frac{91}{24} \text{ km} = \frac{9919}{48} \text{ km}$$

6. Represent the the following on the number line.

(i) $\frac{3}{7}$

(ii) $-\frac{5}{6}$

Sol. (i) Since $\frac{3}{7}$ can not be expressed as a mixed fraction, divide the unit part in equal parts, where 3rd part will represent $\frac{3}{7}$ as shown.



(ii) Since $-\frac{5}{6}$ can not be expressed as a mixed fraction therefore, divide the unit part into 6 equal parts, where 5th part on the left of origin will represent $-\frac{5}{6}$ as shown above.

7. A farmer has one piece of rectangular land measuring $\frac{50000}{13}$ m length and $\frac{30000}{11}$ m breadth. If he distributes his land among his 5 children, how much area will each child get?

Sol. Length of land = $\frac{50000}{13}$ m

Breadth of land = $\frac{30000}{11}$ m

$$\text{Area of land} = \frac{50000}{13} \times \frac{30000}{11} = \frac{1500000000}{143} \text{ m}^2$$

$$\text{Area of one piece of land when it is distributed among 5 children} = \frac{1500000000}{143} \div 5$$

$$= \frac{1500000000}{143} \times \frac{1}{5} = \frac{300000000}{143} \text{ m}^2$$

8. The perimeter of a square is $\frac{24}{10}$ cm. Find the side of the square.

Sol. The perimeter of the square = $\frac{24}{10}$ cm

But the perimeter of square = $4 \times \text{side}$

$$4 \times \text{side} = \frac{24}{10} \text{ cm}$$

$$\text{Thus, side} = \frac{24}{10} \div 4 = \frac{24}{10} \times \frac{1}{4} = \frac{3}{5} \text{ cm}$$

The side of the square is $\frac{3}{5}$ cm.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- The product of two positive rational numbers is always
- The product of two negative rational numbers is always.....
- $\frac{1}{2} \times \left(\frac{3}{4} + \frac{-5}{12} \right) = \frac{1}{2} \times \dots + \dots \times \frac{-5}{12}$
- If a is reciprocal of b , then the reciprocal of b is
- The number 0 is the reciprocal of any number.
- A rational number between $\frac{1}{5}$ and $\frac{2}{5}$ is
- The denominator of a rational number cannot be
- If $\frac{-2}{5} = \frac{13}{x}$, then $x = \dots$
- For any rational number a ($a \neq 0$), $a \div (-a) = \dots$
- A rational number between $\frac{1}{2}$ and $\frac{-1}{3}$ is

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- The rational number $\frac{11}{15}$ lies on the left of zero on the number line.
- The rational numbers $\frac{1}{3}$ and $\frac{-5}{2}$ are on opposite sides of 0 on the number line.
- Every whole number is a rational number.
- If $\frac{x}{y}$ is a rational number, then y is always a whole number.
- Every integer is a rational number.
- All rational numbers can be represented on a number line.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | | |
|---|---|
| 1. Column-I | Column-II |
| (A) Distributive law | (p) If a & b are rational number, then $a + b$ is rational |
| (B) Commutative law | (q) If a & b are rational numbers, then $a + b = b + a$ |
| (C) Associative law | (r) If a , b & c are rational numbers, then $a + (b + c) = (a + b) + c$ |
| (D) Closure law | (s) If a , b & c are rational numbers, then $a \times (b + c) = ab + ac$ |
| 2. Column-I | Column-II |
| A. Distributive property of multiplication over addition is | (p) rational numbers |
| B. A rational number which lies between any two rational numbers a and b is | (q) For any three rational numbers a , b , and c ; we have $a(b + c) = ab + ac$ |
| C. All integers numbers are | (r) irrational |
| D. Square root of all positive prime numbers are | (s) $\frac{a + b}{2}$ |

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

- Express $\frac{4}{-14}$ rational number with positive denominator.
- Fill in blank : $\frac{-9}{14} + \dots = -1$
- Simplify : $\frac{-5}{9} \times \left(\frac{-10}{13} \right) \times \left(\frac{21}{11} \right) \times (-7)$
- The cost of $7\frac{2}{3}$ metres of rope is ₹ $12\frac{3}{4}$, find its cost per metre.
- Find three rational numbers between 0 and 0.2

Rational Numbers



6. (i) $\frac{1}{2} \times \left(\frac{3}{4} + \frac{-5}{12} \right) = \frac{1}{2} \times \dots + \dots \times \frac{-5}{12}$
- (ii) $\frac{-4}{5} \times \left(\frac{5}{7} + \frac{-8}{9} \right) = \left(\frac{-4}{5} \times \dots \right) + \frac{-4}{5} \times \frac{-8}{9}$
7. (i) $\frac{-7}{9} + \dots = 3$
- (ii) $\dots + \frac{15}{23} = 4$
8. Write the additive inverse of each of the following rational numbers :
- (i) $\frac{4}{9}$ (ii) $\frac{-13}{7}$
- (iii) $\frac{5}{-11}$ (iv) $\frac{-11}{-14}$

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- The product of two rational numbers is $\frac{63}{40}$. If one of the number is $\left(\frac{-7}{5} \right)$, find the other number.
- Evaluate : $\frac{11}{14} + \frac{19}{10} + \frac{-9}{5} + 0 + \frac{-7}{4}$
- If $\frac{3}{5}$ of a number exceeds its $\frac{2}{7}$ by 44, find the number.
- Find a rational number between $\frac{-2}{3}$ and $\frac{1}{4}$.
- Insert three rational numbers between $\frac{1}{3}$ and $\frac{1}{2}$.
- Represent $3\frac{2}{7}$ on the number line.
- Find : $\frac{-7}{4} + \frac{5}{3} + \frac{-5}{6} + \frac{1}{3} + \frac{-1}{2}$
- Simplify:

$$\left(\frac{13}{9} \times \frac{-15}{2} \right) + \left(\frac{7}{3} \times \frac{8}{5} \right) + \left(\frac{3}{5} \times \frac{1}{2} \right)$$
- Write any three rational numbers between -2 and 0.
- Add and express the sum as a mixed fraction :

(i) $\frac{-12}{5}$ and $\frac{43}{10}$ (ii) $\frac{101}{6}$ and $\frac{7}{8}$

11. Use the distributivity of multiplication of rational numbers over their addition to simplify :

(i) $\frac{3}{5} \times \left(\frac{35}{24} + \frac{10}{1} \right)$ (ii) $\frac{3}{4} \times \left(\frac{8}{9} - 40 \right)$

12. Simplify: $\frac{3}{8} + \frac{7}{2} + \frac{-3}{5} + \frac{9}{8} + \frac{-3}{2} + \frac{6}{5}$.

13. $\frac{1}{15} \div \left(\frac{4}{15} + \frac{1}{3} - \frac{3}{45} \right)$ is equal to

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

- Simplify $\left[\frac{3}{11} \times \frac{5}{6} \right] - \left[\frac{9}{12} \times \frac{4}{3} \right] + \left[\frac{5}{13} \times \frac{6}{15} \right]$
- Simplify $\left(\frac{1}{4} \times \frac{2}{7} \right) - \left(\frac{5}{14} \times \frac{-2}{3} \right) + \left(\frac{3}{7} \times \frac{9}{2} \right)$
- Find four rational numbers between $\frac{2}{3}$ and $\frac{4}{5}$
- For $x = \frac{1}{2}$ and $y = \frac{2}{3}$ verify that $-(x+y) = (-x) + (-y)$
- If $x = \frac{1}{2}, y = \frac{4}{3}, z = \frac{-3}{5}$, then verify
 $x \times (y-z) = (x \times y) - (x \times z)$
- By what number should we multiply $\frac{3}{-14}$, so that the product may be $\frac{5}{12}$.
- The value of $4 - \frac{5}{1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}}$ is
- Find ten rational numbers between $\frac{-2}{5}$ and $\frac{1}{2}$.
- Verify the property : $x \times (y+z) = x \times y + x \times z$ by taking:
 $x = \frac{-12}{5}, y = \frac{-15}{4}, z = \frac{8}{3}$

2 EXERCISE

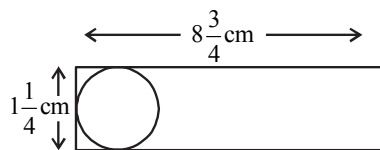
Text-Book Exercise :

- Find $\frac{3}{7} + \left(\frac{-6}{11}\right) + \left(\frac{-8}{21}\right) + \left(\frac{5}{22}\right)$
- Find $\frac{-4}{5} \times \frac{3}{7} \times \frac{15}{16} \times \left(\frac{-14}{9}\right)$
- Using appropriate properties, find.
 - $-\frac{2}{3} \times \frac{3}{5} + \frac{5}{2} - \frac{3}{5} \times \frac{1}{6}$
 - $\frac{2}{5} \times \left(-\frac{3}{7}\right) - \frac{1}{6} \times \frac{3}{2} + \frac{1}{14} \times \frac{2}{5}$
- Find the multiplicative inverse of the following.
 - $\frac{-5}{8} \times \frac{-3}{7}$
 - -1
- Multiply $\frac{6}{13}$ by the reciprocal of $\frac{-7}{16}$.
- Is $\frac{8}{9}$ the multiplicative inverse of $-1\frac{1}{8}$? Why or why not?
- Find any ten rational numbers between $\frac{-5}{6}$ and $\frac{5}{8}$.
- Represent $\frac{-2}{11}, \frac{-5}{11}, \frac{-9}{11}$ on the number line.
- Find five rational numbers between .
 - $\frac{2}{3}$ and $\frac{4}{5}$
 - $\frac{-3}{2}$ and $\frac{5}{3}$
 - $\frac{1}{4}$ and $\frac{1}{2}$
- Find ten rational numbers between $\frac{3}{5}$ and $\frac{3}{4}$.
- A farmer has a field of area $49\frac{4}{5}$ ha. He wants to divide it equally among his one son and two daughters. Find the area of each one's share (ha means hectare; 1 hectare = 10.000 m²)
- Let a, b, c, be the three rational numbers where $a = \frac{2}{3}, b = \frac{4}{5}$ and $c = -\frac{5}{6}$
 - $a + (b + c) = (a + b) + c$ (Associative property of addition)
 - $a \times (b \times c) = (a \times b) \times c$ (Associative property of multiplication)
- Identify the rational number which is different from the other three : $\frac{2}{3}, \frac{-4}{5}, \frac{1}{2}, \frac{1}{3}$. Explain your reasoning.
- The product of two rational numbers is -7 . If one of the number is -10 , find the other.
- Tell which property allows you to compute $\frac{1}{5} \times \left[\frac{5}{6} \times \frac{7}{9}\right]$ as $\left[\frac{1}{5} \times \frac{5}{6}\right] \times \frac{7}{9}$
- Verify the property $x \times (y \times z) = (x \times y) \times z$ of rational numbers by using $x = \frac{-2}{7}, y = \frac{-5}{6}$ and $z = \frac{1}{4}$
- Use the distributivity of multiplication of rational numbers over addition to simplify
 - $\frac{3}{5} \times \left[\frac{35}{24} + \frac{10}{1}\right]$
 - $\frac{-5}{4} \times \left[\frac{8}{5} + \frac{16}{15}\right]$
 - $\frac{2}{7} \times \left[\frac{7}{16} - \frac{21}{4}\right]$
 - $\frac{3}{4} \times \left[\frac{8}{9} - 40\right]$

Exemplar Questions :

- Find $\frac{4}{7} \times \frac{14}{3} \div \frac{2}{3}$.
- Using appropriate properties, find $\frac{2}{3} \times \left(\frac{-5}{7}\right) + \frac{7}{3} + \frac{2}{3} \times \left(\frac{-2}{7}\right)$.
- Let O, P and Z represent the numbers 0, 3 and -5 respectively on the number line. choose a point T between Z and O so that $ZT = TO$. Which rational number does T represent?
- Simplify
 - $\frac{32}{5} + \frac{23}{11} \times \frac{22}{15}$
 - $\frac{3}{7} \times \frac{28}{15} \div \frac{14}{5}$
 - $\frac{3}{7} + \frac{-2}{21} \times \frac{-5}{6}$
 - $\frac{7}{8} + \frac{1}{16} - \frac{1}{12}$

12. A $117\frac{1}{3}$ m long rope is cut into equal pieces measuring $7\frac{1}{3}$ m each. How many such small pieces are these?
13. Riya, Reena and Seema received a total of Rs. 2,016 as monthly allowance from their mother such that Seema gets $\frac{1}{2}$ of what Riya get and Reena gets $1\frac{2}{3}$ times Seema's share. How much money do the three sisters get individually?
14. On a winter day the temperature at a place in Himachal Pradesh was -16°C . Convert it in degree Fahrenheit ($^\circ\text{F}$) by using the formula. $\frac{C}{5} = \frac{F-32}{9}$
15. Shalini has to cut out circles of diameter $1\frac{1}{4}$ cm from an aluminium strip of dimensions $8\frac{3}{4}$ cm by $1\frac{1}{4}$ cm. How many full circles can Shalini cut?



HOTS Questions :

- Find the value of $\frac{(-2)}{3} \times \frac{3}{2} + \frac{5}{2} - \frac{3}{5} \times \frac{1}{6}$
- The speed of car is $54\frac{1}{2}$ km per hour. What is the distance travelled in $\frac{7}{2}$ hours and $\frac{35}{2}$ minutes?
- A mother and her two daughters got a room constructed for Rs. 60,000. The elder daughter contributes $\frac{3}{8}$ of her mother's contribution while the younger daughter contributes $\frac{1}{2}$ of her mother's share. How much do the three contribute individually?
- One fruit salad recipe requires $\frac{1}{2}$ cup of sugar. Another recipe for the same fruit salad requires 2 tablespoons of sugar. If 1 tablespoon is equivalent to $\frac{1}{16}$ cup, how much more sugar does the first recipe require?

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The standard form of $\frac{192}{-168}$ is
 - $\frac{-2}{3}$
 - $\frac{-8}{7}$
 - $\frac{-1}{7}$
 - $\frac{-6}{7}$
- The number which is subtracted from $\frac{27}{13}$ to get $\frac{-3}{7}$ is
 - $\frac{228}{91}$
 - $\frac{1}{91}$
 - $\frac{200}{91}$
 - $\frac{198}{91}$
- A rational number between $\frac{1}{4}$ and $\frac{1}{3}$ is
 - $\frac{7}{24}$
 - 0.29
 - $\frac{13}{48}$
 - All of these
- Which of the following statements is TRUE?
 - $\frac{5}{7} < \frac{7}{9} < \frac{9}{11} < \frac{11}{13}$
 - $\frac{11}{13} < \frac{9}{11} < \frac{7}{9} < \frac{5}{7}$
 - $\frac{5}{7} < \frac{11}{13} < \frac{7}{9} < \frac{9}{11}$
 - $\frac{5}{7} < \frac{9}{11} < \frac{11}{13} < \frac{7}{9}$
- If $\frac{3}{7} + x + \left(\frac{-8}{21}\right) + \frac{5}{22} = \frac{-125}{462}$, then x is
 - $\frac{6}{11}$
 - $\frac{-5}{11}$
 - $\frac{-6}{11}$
 - $\frac{5}{11}$

6. What number should be added to $\frac{7}{12}$ to get $\frac{4}{15}$?

- (a) $-\frac{19}{60}$ (b) $-\frac{11}{30}$
(c) $\frac{51}{60}$ (d) $\frac{1}{20}$

7. Choose the rational number which does not lie between rational numbers $-\frac{2}{5}$ and $-\frac{1}{5}$:

- (a) $-\frac{1}{4}$ (b) $-\frac{3}{10}$
(c) $\frac{3}{10}$ (d) $-\frac{7}{20}$

8. The product of two numbers is $-\frac{28}{27}$. If one of the numbers is $\left(-\frac{4}{9}\right)$, then the other number is ____.

- (a) $\frac{5}{2}$ (b) $\frac{7}{3}$
(c) $\frac{1}{3}$ (d) $\frac{2}{7}$

9. The sum of the additive inverse and multiplicative inverse of 2 is:

- (a) $\frac{3}{2}$ (b) $-\frac{3}{2}$
(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

10. Which of the following is false?

- (a) $\frac{-4}{-5} + 0 = \frac{4}{5}$
(b) $\frac{-5}{6} + \frac{-7}{12} = \frac{-7}{12} + \frac{-5}{6}$
(c) $\frac{8}{9} + \left(\frac{11}{3} + \frac{-2}{3}\right) = \left(\frac{8}{9} + \frac{11}{3}\right) + \frac{-2}{3}$
(d) $\frac{8}{9} - \frac{7}{11} = \frac{7}{11} - \frac{8}{9}$

11. Value of $13\frac{1}{2} - [4\frac{1}{2} - \{3 - (2 - \frac{1}{2})\}]$ will be -

- (a) $9\frac{1}{2}$ (b) $10\frac{1}{2}$
(c) $8\frac{1}{2}$ (d) $11\frac{1}{2}$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. Which of the following forms a pair of equivalent rational numbers?

- (a) $\frac{14}{35}$ and $\frac{21}{45}$ (b) $\frac{-12}{26}$ and $\frac{-18}{39}$
(c) $\frac{-3}{7}$ and $\frac{-21}{56}$ (d) $\frac{-7}{28}$ and $\frac{-5}{20}$

2. The value of $\left(-\frac{5}{9} \div \frac{2}{3}\right)$ is

- (a) $-\frac{5}{2}$ (b) $-\frac{5}{6}$
(c) $-\frac{10}{12}$ (d) $-\frac{6}{5}$

3. Which of the following rational numbers is in the standard form?

- (a) $\frac{-12}{26}$ (b) $\frac{-49}{91}$
(c) $\frac{-9}{16}$ (d) $\frac{-4}{15}$

4. The product of two number is $-\frac{16}{35}$. If one of the numbers is $-\frac{15}{14}$, find the other.

- (a) $\frac{-2}{5}$ (b) $\frac{-32}{-75}$
(c) $\frac{32}{75}$ (d) $-\frac{8}{3}$

5. $\left|\frac{2}{3} - \frac{3}{4}\right|$ is not equal to

- (a) $\frac{-7}{12}$ (b) $-\frac{1}{12}$
(c) $\frac{1}{12}$ (d) $\frac{17}{12}$

6. The additive inverse of $\frac{-6}{-17}$ is not equal to

- (a) $\frac{6}{17}$ (b) $\frac{-6}{17}$
(c) $\frac{-17}{-6}$ (d) $\frac{-17}{6}$

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE - I

Properties of Addition of Rational Numbers

(i) **Commutative property:** The addition of rational numbers

is commutative i.e., if $\frac{a}{b}$ and $\frac{c}{d}$ are any two rational numbers, then

$$\frac{a}{b} + \frac{c}{d} = \frac{c}{d} + \frac{a}{b}$$

(ii) **Associative Properties:** The addition of Rational numbers is associative i.e. if $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ are any three rational numbers, then

$$\frac{a}{b} + \left(\frac{c}{d} + \frac{e}{f} \right) = \left(\frac{a}{b} + \frac{c}{d} \right) + \frac{e}{f}$$

1. Using above properties of addition of rational numbers,

express the following as a rational numbers $\frac{3}{5} + \frac{-7}{6} + \frac{2}{5}$

- (a) $\frac{-1}{6}$ (b) $\frac{1}{30}$
(c) $\frac{1}{6}$ (d) $\frac{-1}{30}$

2. Using above properties of addition of rational numbers,

express the following as a rational number: $\frac{7}{3} + \frac{11}{2} + \frac{5}{3}$

- (a) $\frac{-19}{6}$ (b) $\frac{19}{2}$
(c) $\frac{19}{6}$ (d) $\frac{-19}{3}$

3. Using above properties of addition of rational numbers,

express the following as a rational number: $\frac{-5}{7} + 3 + \frac{2}{7}$

- (a) $\frac{-18}{7}$ (b) $\frac{24}{7}$
(c) $\frac{-24}{7}$ (d) $\frac{18}{7}$

PASSAGE - II

If $\frac{a}{b}$, $\frac{c}{d}$ and $\frac{e}{f}$ are any three rational numbers, then

$$\frac{a}{b} \times \left(\frac{c}{d} + \frac{e}{f} \right) = \frac{a}{b} \times \frac{c}{d} + \frac{a}{b} \times \frac{e}{f}$$

4. $\frac{2}{3} \times \frac{-7}{10} + \frac{-2}{3} \times \frac{8}{9} = ? \times \left[\frac{-7}{10} + ? \right]$

- (a) $\frac{2}{3}, \frac{8}{9}$ (b) $\frac{-2}{3}, \frac{-8}{9}$
(c) $\frac{2}{3}, \frac{-8}{9}$ (d) $\frac{-2}{3}, \frac{-8}{9}$

5. Name the property used above.

- (a) Commutativity of multiplication over addition
(b) Commutativity of addition over multiplication
(c) Distributivity of multiplication over addition
(d) Distributivity of addition over multiplication

6. $\frac{2}{5} \times \frac{-8}{9} + ? \times \frac{5}{9} = \frac{2}{5} \times [? + ?]$

- (a) $\frac{2}{5}, \frac{-8}{9}, \frac{5}{9}$ (b) $\frac{2}{5}, \frac{8}{9}, \frac{-5}{9}$
(c) $\frac{-2}{5}, \frac{-8}{9}, \frac{-5}{9}$ (d) $\frac{-2}{5}, \frac{-8}{9}, \frac{5}{9}$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
(b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
(c) If **Assertion** is **correct** but **Reason** is **incorrect**.
(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Zero is a rational number.

Reason : Each rational number is a quotient of any two integers, while its divisor should not be zero. Thus, a

number of the form $\frac{p}{q}$ where p and q are integers and $q \neq 0$ is a rational number.

2. **Assertion :** If x, y, z be rational numbers such that $x > y$ and $y > z$, then $x > z$

Reason : The sum of two rational numbers is always greater than third rational numbers.

3. **Assertion :** One of the rational between $\frac{1}{4}$ and $\frac{1}{3}$ is $\frac{7}{24}$

Reason : If x and y are any two rational numbers such that

$x < y$, then $\frac{1}{2}(x + y)$ is a rational number between x and y

such that $x < \frac{1}{2}(x + y) < y$.

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and four statements (p, q, r, s) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Match the following

Column-I	Column-II
(A) Associative law is satisfied by	(p) Addition of rational numnbers
(B) Commutative law is satisfied by	(q) Subtraction of rational numbers
(C) Closure law is satisfied by	(r) Multiplication of rational number
(D) Distributive law of multiplication is satisfied by	(s) Division of rational number

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- Verify the distributive property $x \times (y + z) = xy + xz$ where $x = \frac{-4}{3}, y = \frac{1}{2}, z = \frac{-7}{5}$ and write the denominator of $(x + y) \times z$.
- Find the sum of digits of numerator and denominator of reciprocal of rational number $\left(\frac{2}{5} + \frac{5}{4}\right)$.
- $\frac{1}{2}$ can't be expressed as rational number having denominator 5 beacuse 5 is not the multiple of ____.
- The product of two numbers is $\frac{45}{56}$. If one of them is $9/7$, then denominator of other number is ____.
- On simplifying $\left(\frac{-4}{3} \times \frac{12}{-5}\right) + \left(\frac{3}{7} \times \frac{21}{15}\right)$ we get $\frac{19}{k}$. Find the value of k .
- The sum of two rational numbers is $\frac{-3}{5}$. If one of the number is $\frac{-9}{20}$, than the other, number is $\frac{-k}{20}$. Find the value of k .

SOLUTIONS

**Brief Explanations
of
Selected Questions**

1 EXERCISE

Fill in the Blanks :

- Positive
- Positive
- $\frac{3}{4}, \frac{1}{2}$
- a
- not
- $\frac{3}{10}$
- Zero
- $\frac{-65}{2}$
- 1
- $\frac{1}{12}$

True / False :

- False : $\frac{11}{15}$ lies on the right of zero on the number line. (+ve)
rational numbers lies right of zero on the number line.
- True
- True
- False; If $\frac{x}{y}$ is a rational number, x and y must be integers.
- True
- True

Match the Columns :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (q); (C) $\rightarrow$ (r); (D) $\rightarrow$ (p)
- (A) $\rightarrow$ (q); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (r)

Very Short Answer Questions :

- $\frac{4}{-14} = \frac{4 \times (-1)}{-14 \times (-1)} = \frac{-4}{14}$
- $\frac{-9}{14} + x = -1$
 $\Rightarrow x = -1 + \frac{9}{14} = \frac{-14+9}{14} = \frac{-5}{14}$
- We have
 $\frac{-5}{9} \times \left(\frac{-10}{13}\right) \times \left(\frac{21}{11}\right) \times (-7)$
 $= \frac{(-5) \times (-10) \times 21 \times (-7)}{9 \times 13 \times 11} = \frac{-7350}{1287}$

- The cost of $\frac{23}{3}$ metres of rope = ₹ $\frac{51}{4}$

$$\text{So, the cost of 1 metre} = \frac{51}{4} \div \frac{23}{3}$$

$$= \frac{51}{4} \times \frac{3}{23} = \frac{153}{92} = ₹ 1 \frac{61}{92}$$

- Rational numbers between 0 and 0.2

$$\Rightarrow 0.1, 0.12, 0.13$$

$$\Rightarrow \frac{1}{10}, \frac{12}{100} \text{ and } \frac{13}{100}$$

- $\frac{1}{2} \times \frac{3}{4} + \frac{1}{2} \times \frac{-5}{12}$
 - $\frac{5}{7}$
- $\frac{34}{9}$
 - $\frac{77}{23}$
- $-\frac{4}{9}$
 - $\frac{13}{7}$
 - $\frac{5}{11}$
 - $-\frac{11}{4}$

Short Answer Questions :

- Let the other number be x.

$$\text{Now, } x \times \left(\frac{-7}{5}\right) = \frac{63}{40}$$

$$\Rightarrow \frac{-7x}{5} = \frac{63}{40} \Rightarrow -7x \times 40 = 63 \times 5$$

$$\Rightarrow -x = \frac{63 \times 5}{7 \times 40} \Rightarrow -x = \frac{9}{8}$$

$$\Rightarrow x = -\frac{9}{8}$$

$$\text{The other number} = -\frac{9}{8}$$

- $$\frac{11}{14} + \frac{19}{10} - \frac{9}{5} + \left(\frac{-7}{4}\right) - \left[\frac{11}{14} + \frac{19}{10}\right] + \left[\frac{-9}{5} - \frac{7}{4}\right]$$

$$= \left(\frac{55+133}{70}\right) + \left[\frac{-36-35}{20}\right] = \frac{188}{70} + \left(\frac{-71}{20}\right)$$

$$= \frac{376 + (-497)}{140} = \frac{-121}{140}$$

3. Let the number be x .

$$\frac{3}{5}x - \frac{2}{7}x = 44$$

$$\Rightarrow \frac{21x - 10x}{35} = 44 \Rightarrow \frac{11x}{35} = 44$$

$$\Rightarrow x = \frac{35 \times 44}{11} = 140$$

The number is 140.

4. The rational number $\left(\frac{-2}{3} + \frac{1}{4}\right) \div 2$ lies between

$$\frac{-2}{3} \text{ and } \frac{1}{4}$$

$$\Rightarrow \left(\frac{-2}{3} + \frac{1}{4}\right) = \frac{(-2) \times 4 + 3 \times 1}{12} = \frac{-8 + 3}{12} = \frac{-5}{12}$$

$$\Rightarrow \left(\frac{-2}{3} + \frac{1}{4}\right) \div 2 = \frac{-5}{12} \div 2 = \frac{-5}{12} \times \frac{1}{2} = \frac{-5}{24}$$

Thus, the required rational number is $\frac{-5}{24}$

$$\text{i.e., } \frac{-2}{3} < \frac{-5}{24} < \frac{1}{4}$$

5. A rational number between $\frac{1}{3}$ and $\frac{1}{2}$ is $\frac{1}{2}\left(\frac{1}{3} + \frac{1}{2}\right) = \frac{5}{12}$

$$\text{We have, } \frac{1}{3} < \frac{5}{12} < \frac{1}{2}$$

$$\text{A rational number between } \frac{1}{3} \text{ and } \frac{5}{12} \text{ is } \frac{1}{2}\left(\frac{1}{3} + \frac{5}{12}\right) = \frac{3}{8}$$

$$\text{A rational number between } \frac{5}{12} \text{ and } \frac{1}{2} \text{ is } \frac{1}{2}\left(\frac{5}{12} + \frac{1}{2}\right) = \frac{11}{24}$$

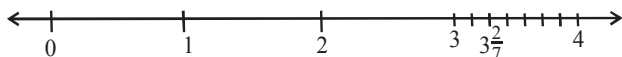
$$\text{Clearly, we have } \frac{1}{3} < \frac{3}{8} < \frac{5}{12} < \frac{11}{24} < \frac{1}{2}$$

$$\text{Hence, three rational numbers between } \frac{1}{3} \text{ and } \frac{1}{2} \text{ are } \frac{3}{8}, \frac{5}{12}$$

$$\text{and } \frac{11}{24}.$$

6. In order to represent $3\frac{2}{7}$ on the number line, take 3 unit lengths between 0 and 3 and divide the unit length between 3 and 4 into seven equal parts and take the end of 2nd part on it.

This point represents the rational number $3\frac{2}{7}$.



7. LCM of 4, 3, 6, 3 and 2 is 12

$$\begin{array}{r|rrrr} 2 & 4 & 3 & 6 & 2 \\ 2 & 2 & 3 & 3 & 1 \\ 3 & 1 & 3 & 3 & 1 \\ \hline & 1 & 1 & 1 & 1 \end{array}$$

$$\therefore \text{LCM of } 4, 3, 6, 3, 2 \text{ is } 2 \times 2 \times 3 = 12$$

$$\therefore \frac{-7}{4} + \frac{5}{3} + \frac{-5}{6} + \frac{1}{3} + \frac{-1}{2}$$

$$= \frac{((-7) \times 3) + (5 \times 4) + ((-5) \times 2) + (1 \times 4) + ((-1) \times 6)}{12}$$

$$= \frac{(-21) + 20 + (-10) + 4 + (-6)}{12}$$

$$= \frac{(-37) + 24}{12} = \frac{-13}{12}$$

$$8. \frac{-102}{15}$$

9. We can write

$$-2 = \frac{-2}{1} = \frac{-2 \times 5}{1 \times 5} = \frac{-10}{5} \text{ and } 0 = \frac{0}{5}$$

integers between -10 and 0 are $-9, -8, -7, -6, -5, \dots, -1$.

$$\therefore \frac{-9}{5}, \frac{-8}{5}, \frac{-7}{5}, \dots, \frac{-2}{5}, \frac{-1}{5} \text{ are rational numbers between } -2 \text{ and } 0.$$

We can choose any three of these rational numbers.

$$10. (i) 1\frac{9}{10} \quad (ii) 17\frac{17}{24}$$

$$11. (i) \frac{55}{8} \quad (ii) \frac{-88}{3}$$

$$12. \frac{3}{8} + \frac{7}{2} + \frac{(-3)}{5} + \frac{9}{8} + \frac{(-3)}{2} + \frac{6}{5}$$

$$\frac{15 + 140 - 24 + 45 - 60 + 48}{40}$$

$$\frac{248 - 84}{40} = \frac{164}{40} = \frac{41}{10}$$

$$\begin{aligned} 13. & \frac{1}{15} \div \left(\frac{4}{15} + \frac{1}{3} - \frac{3}{45}\right) \\ &= \frac{1}{15} \div \left(\frac{12 + 15 - 3}{45}\right) \\ &= \frac{1}{15} \div \left(\frac{24}{45}\right) \\ &= \frac{1}{15} \times \frac{45}{24} \\ &= \frac{1}{8} \end{aligned}$$

Long Answer Questions :

$$\begin{aligned}
 1. \quad & \left(\frac{3}{11} \times \frac{5}{6}\right) - \left(\frac{3}{4} \times \frac{4}{3}\right) + \left(\frac{5}{13} \times \frac{2}{5}\right) \\
 &= \left(\frac{1}{11} \times \frac{5}{2}\right) - 1 + \frac{2}{13} \\
 &= \frac{5}{22} - 1 + \frac{2}{13} \\
 &= \frac{65 - 286 + 44}{286} \text{ (L.C.M. of (13, 22) = 286)} \\
 &= \frac{109 - 286}{286} = \frac{-177}{286}
 \end{aligned}$$

$$\begin{aligned}
 2. \quad & \left(\frac{1}{4} \times \frac{2}{7}\right) - \left(\frac{5}{14} \times \frac{-2}{3}\right) + \left(\frac{3}{7} \times \frac{9}{2}\right) \\
 &= \left(\frac{1}{2} \times \frac{1}{7}\right) - \left(\frac{5}{7} \times \frac{-1}{3}\right) + \left(\frac{3}{7} \times \frac{9}{2}\right) \\
 &= \frac{1}{14} - \left(\frac{-5}{21}\right) + \left(\frac{27}{14}\right) = \frac{1}{14} + \frac{5}{21} + \frac{27}{14} \\
 &= \frac{3 + 10 + 81}{42} \text{ (L.C.M of (14, 21, 14) = 42)} \\
 &= \frac{13 + 81}{42} = \frac{94}{42} = \frac{47}{21}
 \end{aligned}$$

3. First we convert given rational numbers to rational numbers with the same denominator equal to the LCM of their denominators. The LCM of denominators 3 and 5 is 15.

$$\therefore \frac{2}{3} = \frac{2 \times 5}{3 \times 5} = \frac{10}{15} \text{ and } \frac{4}{5} = \frac{4 \times 3}{5 \times 3} = \frac{12}{15}$$

Between the numerators 10 and 12 of these equivalent rational numbers there is only one integer. So, we replace these numbers by equivalent rational numbers having a sufficiently large common denominator.

$$\frac{2}{3} = \frac{10}{15} = \frac{40}{60} \text{ and } \frac{4}{5} = \frac{12}{15} = \frac{48}{60}$$

Between 40 and 48 there are seven integers 41, 42, 43, 47.

Therefore, $\frac{41}{60}, \frac{42}{60}, \dots, \frac{47}{60}$ are seven rational numbers between

$\frac{40}{60} \left(= \frac{2}{3}\right)$ and $\frac{48}{60} \left(= \frac{4}{5}\right)$ We can take any four of these rational numbers.

$$\begin{aligned}
 4. \quad & \text{For } x = \frac{1}{2} \text{ and } y = \frac{2}{3} \\
 &= -(x + y) = -\left(\frac{1}{2} + \frac{2}{3}\right) = -\left(\frac{1}{2} \times \frac{3}{3} + \frac{2}{3} \times \frac{2}{2}\right) \\
 &= -\left(\frac{3}{6} + \frac{4}{6}\right) = \frac{-7}{6} \text{ and } \\
 &= (-x) + (-y) = \left(\frac{-1}{2}\right) + \left(\frac{-2}{3}\right) = \frac{-3-4}{6} = \frac{-7}{6} \\
 &\text{So, } -(x + y) = (-x) + (-y)
 \end{aligned}$$

$$5. \quad \text{L.H.S.} = x \times (y - z)$$

$$\begin{aligned}
 &= \frac{1}{2} \times \left[\frac{4}{3} - \left(\frac{-3}{5}\right)\right] = \frac{1}{2} \times \left[\frac{4}{3} + \frac{3}{5}\right] \\
 &= \frac{1}{2} \times \left[\frac{20+9}{15}\right] = \frac{1}{2} \times \frac{29}{15} = \frac{29}{30}
 \end{aligned}$$

$$\text{R.H.S.} = (x \times y) = \left(\frac{1}{2} \times \frac{4}{3}\right) = \frac{4}{6} = \frac{2}{3}$$

$$(x \times z) = \left[\frac{1}{2} \times \left(\frac{-3}{5}\right)\right] = \frac{-3}{10}$$

$$\begin{aligned}
 (x \times y) - (x \times z) &= \frac{2}{3} - \left(\frac{-3}{10}\right) = \frac{2}{3} + \frac{3}{10} \\
 &= \frac{20+9}{30} = \frac{29}{30}
 \end{aligned}$$

$$\therefore \text{L.H.S.} = \text{R.H.S.}$$

6. We have,

$$\text{Product of two numbers} = \frac{5}{12}, \text{ One number} = \frac{3}{-14}$$

$$\therefore \text{The other number } x = \frac{5}{12} \div \frac{3}{-14}$$

$$= \frac{5}{12} \times \frac{-14}{3}$$

$$= \frac{5 \times (-14)}{12 \times 3} = \frac{-(5 \times 14)}{12 \times 3} = \frac{-(5 \times 7)}{6 \times 3} = \frac{-35}{18}$$

$$7. \quad 4 - \frac{5}{1 + \frac{1}{3 + \frac{1}{2 + \frac{1}{4}}}} = 4 - \frac{5}{1 + \frac{1}{\frac{31}{9}}} = 4 - \frac{155}{40} = \frac{1}{8}$$

$$8. \quad \frac{-7}{20}, \frac{-6}{20}, \frac{-5}{20}, \dots, \frac{9}{20}$$

$$9. \quad x \times (y + z) = x \times y + x \times z$$

$$\text{When } x = \frac{-12}{5}, y = \frac{-15}{4}, z = \frac{8}{3}$$

$$\frac{-12}{5} \times \left[\frac{-15}{4} + \frac{8}{3}\right] = \left(\frac{-12}{5}\right) \times \left(\frac{-15}{4}\right) + \left(\frac{-12}{5}\right) \times \left(\frac{8}{3}\right)$$

$$\frac{-12}{5} \times \left[\frac{-13}{12}\right] = \left(\frac{180}{20}\right) + \left(\frac{-96}{15}\right)$$

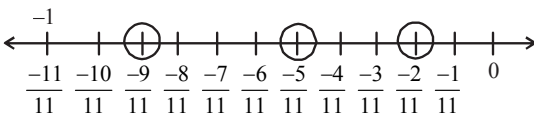
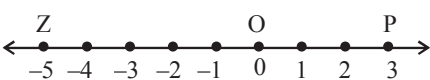
$$\frac{13}{5} = 9 - \frac{32}{5}$$

$$\frac{13}{15} = \frac{45-32}{5}$$

$$\frac{13}{5} = \frac{13}{5}$$

2 EXERCISE

Text-Book Exercise :

1. $\frac{3}{7} + \left(\frac{-6}{11}\right) + \left(\frac{-8}{21}\right) + \frac{5}{22}$
 $= \left[\frac{3}{7} + \left(\frac{-8}{21}\right)\right] + \left[\frac{-6}{11} + \frac{5}{22}\right]$
 (by using commutativity and associativity)
 $= \left[\frac{9+(-8)}{21}\right] + \left[\frac{-12+5}{22}\right]$
 (LCM of 7 and 21 is 21; LCM of 11 and 22 is 22)
 $= \frac{1}{21} + \left(\frac{-7}{22}\right) = \frac{22-147}{462} = \frac{-125}{462}$
2. $\frac{-4}{5} \times \frac{3}{7} \times \frac{15}{16} \times \left(\frac{-14}{9}\right)$
 $= \left(\frac{-4}{5} \times \frac{15}{16}\right) \times \left[\frac{3}{7} \times \left(\frac{-14}{9}\right)\right]$
 (Using commutativity and associativity)
 $= \frac{-3}{4} \times \left(\frac{-2}{3}\right) = \frac{1}{2}$
3. (i) 2 (ii) $\frac{-11}{28}$
4. (i) $\frac{56}{15}$ (ii) -1
5. $\frac{-96}{91}$
6. No, because the product is not 1.
7. We first convert $\frac{-5}{6}$ and $\frac{5}{8}$ to rational numbers with the same denominators.
 $\frac{-5 \times 4}{6 \times 4} = \frac{-20}{24}$ and $\frac{5 \times 3}{8 \times 3} = \frac{15}{24}$
 Thus we have $\frac{-19}{24}, \frac{-18}{24}, \frac{-17}{24}, \dots, \frac{14}{24}$ as the rational numbers between $\frac{-20}{24}$ and $\frac{15}{24}$. You can take any ten of these.

8. 

9. (i) $\frac{41}{60}, \frac{42}{60}, \frac{43}{60}, \frac{44}{60}, \frac{45}{60}$ (ii) $\frac{-8}{9}, \frac{-7}{6}, 0, \frac{1}{6}, \frac{2}{6}$

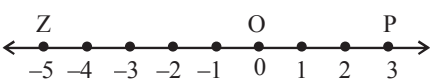
(iii) $\frac{9}{32}, \frac{10}{32}, \frac{11}{32}, \frac{12}{32}, \frac{13}{32}$

(There can be many more such rational numbers)

10. $\frac{97}{160}, \frac{98}{160}, \frac{99}{160}, \frac{100}{160}, \frac{101}{160}, \frac{102}{160}, \frac{103}{160}, \frac{104}{160}, \frac{105}{160}, \frac{106}{160}$

(There can be many more such rational numbers)

Exemplar Questions :

1. $\frac{4}{7} \times \frac{14}{3} \div \frac{2}{3} = \frac{4}{7} \times \left(\frac{14}{3} \times \frac{3}{2}\right) = \frac{4}{7} \times 7 = 4$
2. $\frac{2}{3} \times \left(\frac{-5}{7}\right) + \frac{7}{3} + \frac{2}{3} \times \left(\frac{-2}{7}\right) = \frac{-5}{7} \times \frac{2}{3} - \frac{2}{7} \times \frac{2}{3} + \frac{7}{3}$
 $= \left(\frac{-5}{7} - \frac{2}{7}\right) \times \frac{2}{3} + \frac{7}{3} = \frac{-7}{7} \times \frac{2}{3} + \frac{7}{3} = \frac{5}{3}$
3. 
 So, T is the mid-point of OZ, i.e.
 $T = \frac{0 + (-5)}{2} = \frac{-5}{2}$
4. $49\frac{4}{5} \text{ ha} = \frac{249}{5} \text{ ha}$
 Each share $= \frac{1}{3} \times \frac{249}{5} \text{ ha} = \frac{83}{5} \text{ ha} = 16\frac{3}{5} \text{ ha}$
5. (i) L.H.S $= a + (b + c)$
 $= \frac{2}{3} + \left[\frac{4}{5} + \left(\frac{-5}{6}\right)\right] = \frac{2}{3} + \left[\frac{24-25}{30}\right]$
 $= \frac{2}{3} + \left(\frac{-1}{30}\right) = \frac{20-1}{30} = \frac{19}{30}$
 R.H.S. of (i) $= (a + b) + c$
 $= \left(\frac{2}{3} + \frac{4}{5}\right) + \left(\frac{-5}{6}\right) = \left(\frac{10+12}{15}\right) + \left(\frac{-5}{6}\right)$
 $= \frac{22}{15} - \frac{5}{6} = \frac{44-25}{30} = \frac{19}{30}$
 So, $\frac{2}{3} + \left[\frac{4}{5} + \left(\frac{-5}{6}\right)\right] = \left(\frac{2}{3} + \frac{4}{5}\right) + \left(\frac{-5}{6}\right)$
 Hence verified.

(ii) L.H.S = $a \times (b \times c)$

$$= \frac{2}{3} \times \left[\frac{4}{5} \times \left(\frac{-5}{6} \right) \right]$$

$$= \frac{2}{3} \times \left(\frac{-20}{30} \right) = \frac{2}{3} \times \left(\frac{-2}{3} \right)$$

$$= \frac{2 \times (-2)}{3 \times 3} = \frac{-4}{9}$$

$$= (a \times b) \times c$$

R.H.S. = $(a \times b) \times c$

$$= \left(\frac{2}{3} \times \frac{4}{5} \right) \times \left(\frac{-5}{6} \right) = \frac{2 \times 4}{3 \times 5} \times \frac{-5}{6}$$

$$= \frac{8}{15} \times \left(\frac{-5}{6} \right) = \frac{8 \times (-5)}{15 \times 6} = \frac{-40}{90} = \frac{-4}{9}$$

So, $\frac{2}{3} \times \left[\frac{4}{5} \times \left(\frac{-5}{6} \right) \right] = \left[\frac{2}{3} \times \frac{4}{5} \right] \times \left(\frac{-5}{6} \right)$

6. $\frac{-4}{5}$ lies on the left side of zero while others lie on the right side of zero on the number line.

7. Let the other rational number be x

$$-10 \times x = -7$$

$$x = \frac{-7}{-10}, x = \frac{7}{10}$$

Check

$$-10 \times \frac{7}{10} = -7. \text{ Hence, the result is correct.}$$

8. Associative property

9. $x \times (y \times z) = (x \times y) \times z$

$$x = -\frac{2}{7}, y = -\frac{5}{6}, z = \frac{1}{4}$$

$$-\frac{2}{7} \left[\left(\frac{-5}{6} \right) \times \left(\frac{1}{4} \right) \right] = \left[\left(\frac{2}{7} \right) \times \left(\frac{-5}{6} \right) \right] \times \frac{1}{4}$$

$$-\frac{2}{7} \left(\frac{-5}{24} \right) = \left(\frac{10}{42} \right) \times \frac{1}{4}$$

$$\frac{10}{168} = \frac{10}{168}$$

(Hence verified).

10. (i) $6\frac{7}{8}$ (ii) $3\frac{1}{3}$

- (iii) $\frac{11}{8}$ or $1\frac{3}{8}$ (iv) $\frac{88}{3}$ or $29\frac{1}{3}$

11. (i) $\frac{142}{15}$ or $9\frac{7}{15}$ (ii) $\frac{2}{7}$

- (iii) $\frac{32}{63}$ (iv) $\frac{41}{48}$

12. 16 pieces.

13. ₹ 864, ₹ 720, ₹ 432

14. 3.2°F

15. 7

HOTS Questions :

1. We have $\frac{(-2)}{3} \times \frac{3}{2} + \frac{5}{2} - \frac{3}{5} \times \frac{1}{6}$
- $$= -\frac{1}{1} + \frac{5}{2} - \frac{3}{30} = \frac{-30 + 5 \times 15 - 3}{30}$$
- $$= \frac{-30 + 75 - 3}{30} = \frac{42}{30} = \frac{7}{5}$$

2. Speed = $\frac{\text{distance}}{\text{time}}$

$$1 \text{ min.} = \frac{1}{60} \text{ hrs.}$$

$$\frac{35}{2} \text{ min s.} = \frac{35}{2} \times \frac{1}{60} \text{ hrs} = \frac{35}{120} \text{ hrs.}$$

$$\therefore \text{ Total time taken} = \left(\frac{7}{2} + \frac{35}{120} \right) \text{ hrs.}$$

$$= \left(\frac{7 \times 60 + 35}{120} \right) \text{ hrs.}$$

$$= \frac{455}{120} \text{ hrs} = \frac{131}{24} \text{ hrs.}$$

$$\text{Speed} = \frac{109}{2} \text{ km/hr}$$

$$\text{Distance} = \frac{109}{2} \times \frac{131}{24} \text{ km} = \frac{14279}{48}$$

3. ₹ 32,000, ₹ 12,000, ₹ 16,000

4. $\frac{3}{8}$ cup

3 EXERCISE

Single Option Correct :

$$1. \quad (b) \quad \frac{192}{-168} = \frac{-192}{168} = \frac{-192 \div 4}{168 \div 4} \\ = \frac{-48}{42} = \frac{-48 \div 6}{42 \div 6} = \frac{-8}{7}$$

$$2. \quad (a) \quad \text{We have } \frac{27}{13} - x = \frac{-3}{7} \\ x = \frac{27}{13} + \frac{3}{7} \Rightarrow x = \frac{189 + 39}{91} \Rightarrow x = \frac{228}{91}$$

$$3. \quad (d) \quad \frac{1}{4} = 0.25, \frac{1}{3} = 0.33 \dots\dots$$

$$(a) \quad \frac{\frac{1}{4} + \frac{1}{3}}{2} = \frac{\frac{7}{12}}{2} = \frac{7}{24}$$

$$\frac{1}{4} < \frac{7}{24} < \frac{1}{3},$$

$$(b) \quad 0.25 < 0.29 < 0.33 \dots\dots$$

$$(c) \quad \frac{\frac{1}{4} + \frac{7}{24}}{2} = \frac{\frac{6+7}{24}}{2} = \frac{13}{48}$$

$\therefore$ all are correct.

$$4. \quad (a) \quad \text{L.C.M. of } 13, 11, 9, 7 = 117 \times 77 = 9009$$

$$\frac{11}{13} \times \frac{693}{693} = \frac{7623}{9009}, \frac{9}{11} \times \frac{819}{819} = \frac{7371}{9009}$$

$$\frac{7}{9} \times \frac{1001}{1001} = \frac{7007}{9009}, \frac{5}{7} \times \frac{1287}{1287} = \frac{6435}{9009}$$

$$\frac{6435}{9009} < \frac{7007}{9009} < \frac{7371}{9009} < \frac{7623}{9009} \text{ or } \frac{5}{7} < \frac{7}{9} < \frac{9}{11} < \frac{11}{13}$$

$$5. \quad (c) \quad \frac{3}{7} + x + \left(\frac{-8}{21}\right) + \left(\frac{5}{22}\right) = -\frac{125}{462}$$

$$\Rightarrow \frac{9 + 21x - 8}{21} = \frac{-125}{462} - \frac{5}{22}$$

$$\Rightarrow \frac{21x + 1}{21} = \frac{-(125 + 105)}{462}$$

$$\Rightarrow \frac{21x + 1}{21} = \frac{-(230)}{462} \Rightarrow \frac{21x + 1}{21} = \frac{-(230)}{462}$$

$$\Rightarrow 21x + 1 = \frac{-230}{22} \Rightarrow 21x = \frac{-230}{22} - 1$$

$$\Rightarrow 21x = \frac{-252}{22}$$

$$\Rightarrow x = \frac{-252}{22 \times 21} = \frac{-6}{11}$$

$$6. \quad (a) \quad \text{Let the number to be added be } x$$

$$\Rightarrow \frac{7}{12} + x = \frac{4}{15}$$

$$x = \frac{4}{15} - \frac{7}{12} = \frac{16 - 35}{60} = \frac{-19}{60}$$

$$7. \quad (c) \quad \frac{3}{10} > -\frac{2}{5} \text{ and } \frac{3}{10} > -\frac{1}{5}$$

$$8. \quad (b) \quad \text{Let us consider first rational number be } x.$$

$$\text{Now, from the equation, } \left(-\frac{4}{9}\right) \times x = \frac{-28}{27}$$

$$x = \left(-\frac{28}{27}\right) \div \left(-\frac{4}{9}\right) = \left(-\frac{28}{27}\right) \times \left(\frac{-9}{4}\right) = \frac{7}{3}$$

$$9. \quad (b) \quad -2 + \frac{1}{2} = \frac{-3}{2}$$

$$10. \quad (d) \quad \frac{8}{9} - \frac{7}{11} = \frac{88 - 63}{99} = \frac{25}{99}$$

$$\frac{7}{11} - \frac{8}{9} = \frac{63 - 88}{99} = \frac{-25}{99}$$

$$\therefore \frac{8}{9} - \frac{7}{11} \neq \frac{7}{11} - \frac{8}{9}$$

$$11. \quad (b) \quad 13\frac{1}{2} - \left[4\frac{1}{2} - \left\{3 - \left(2 - \frac{1}{2}\right)\right\}\right]$$

$$= \frac{27}{2} - \left[\frac{9}{2} - \left\{3 - \left(\frac{4-1}{2}\right)\right\}\right]$$

$$= \frac{27}{2} - \left[\frac{9}{2} - \left\{3 - \frac{3}{2}\right\}\right]$$

$$= \frac{27}{2} - \left[\frac{9}{2} - \left(\frac{6-3}{2}\right)\right]$$

$$= \frac{27}{2} - \left[\frac{9}{2} - \frac{3}{2}\right] = \frac{27}{2} - \left[\frac{6}{2}\right]$$

$$= \frac{21}{2} = 10\frac{1}{2}$$

More Than One Option Correct :

1. (b, d) Because $\frac{-12}{26} = \frac{-6}{13}$ is its simplest form

$$\frac{-6}{13} = \frac{-12}{26} = \frac{-18}{39} \text{ and } \frac{-7}{28} = \frac{-5}{20}$$

Because $\frac{-1}{4}$ is simplest form of $\frac{-7}{28}$ & $\frac{-5}{20}$

2. (b, c) $\left(\frac{-5}{9} \div \frac{2}{3}\right) = \frac{-5}{9} \times \frac{3}{2} = \frac{-5}{6} = -\frac{10}{12}$
 3. (c, d)
 4. (b, d) Let the other number be x

$$\frac{-15}{14} \times x = \frac{-16}{35}$$

$$x = \frac{-16}{35} \div \left(\frac{-15}{14}\right) \Rightarrow x = \frac{-16}{35} \times \frac{-14}{15}$$

$$x = \frac{-16}{5} \times \frac{-2}{15} = \frac{32}{75}$$

5. (a, b and d)
 6. (a, c and d)

Passage Based Questions :

1. (a) $\frac{3}{5} + \frac{-7}{6} + \frac{2}{5} = \frac{3}{5} + \frac{2}{5} + \frac{-7}{6}$ [Commutative Property]

$$= \left(\frac{3}{5} + \frac{2}{5}\right) + \frac{-7}{6} = \left(\frac{3+2}{5}\right) + \frac{-7}{6}$$

[Associative Property]

$$= 1 + \frac{-7}{6} = \frac{6+(-7)}{6} = \frac{-1}{6}$$

2. (b) $\frac{7}{3} + \frac{11}{2} + \frac{5}{3} = \frac{7}{3} + \frac{5}{3} + \frac{11}{2}$

[Commutative Properties]

$$= \left(\frac{7}{3} + \frac{5}{3}\right) + \frac{11}{2} \quad [\text{Associative Property}]$$

$$= \left(\frac{7+5}{3}\right) + \frac{11}{2} = 4 + \frac{11}{2} = \frac{8+11}{2} = \frac{19}{2}$$

3. (d) $\frac{-5}{7} + 3 + \frac{2}{7} = \frac{-5}{7} + \frac{2}{7} + 3$ [Commutative property]

$$= \left(\frac{-5}{7} + \frac{2}{7}\right) + 3 \quad [\text{Associative Property}]$$

$$= \frac{-5+2}{7} + 3 = \frac{-3}{7} + 3 = \frac{-3+21}{7} = \frac{18}{7}$$

4. (b) $\frac{2}{3} \times \frac{-7}{10} + \frac{-2}{3} \times \frac{8}{9} = \frac{2}{3} \times \left[\frac{-7}{10} + \left(\frac{-8}{9}\right)\right]$

5. (c) Distributivity of Multiplication over Addition.

6. (a) $\frac{2}{5} \times \frac{-8}{9} + \frac{2}{5} \times \frac{5}{9} = \frac{2}{5} \times \left(\frac{-8}{9} + \frac{5}{9}\right)$

Assertion & Reason :

1. (a) Assertion : True, Reason : True.
 2. (d) Assertion : True, Reason : False.
 3. (a) Assertion : True, Reason : True.

$$4 > 5 \Rightarrow \frac{1}{4} < \frac{1}{5} \Rightarrow \frac{1}{4} < \frac{9}{40} < \frac{1}{5}$$

Multiple Matching Questions :

1. (A) $\rightarrow$ (p, r); (B) $\rightarrow$ (p, r); (C) $\rightarrow$ (p, q, r, s); (D) $\rightarrow$ (p, r)

Integer Type Questions :

1. (6) $x \times (y+z) = xy + xz$

$$\text{L.H.S.} \quad x \times (y+z) = \frac{-4}{3} \times \left(\frac{1}{2} + \left(\frac{-7}{5}\right)\right)$$

$$= \frac{-4}{3} \times \left(\frac{1}{2} - \frac{7}{5}\right) = \frac{-4}{3} \times \left(\frac{5-14}{10}\right)$$

$$= \frac{-4}{3} \times \frac{-9}{10} = \frac{6}{5}$$

$$\text{R.H.S.} \quad xy + xz = \frac{-4}{3} \times \frac{1}{2} + \frac{-4}{3} \times \left(\frac{-7}{5}\right)$$

$$= \frac{-2}{3} + \frac{28}{15} = \frac{-10+28}{15} = \frac{18}{15} = \frac{6}{5}$$

$$\text{Denominator of } (x+y) \times z = \left(\frac{-4}{3} + \frac{1}{2}\right) \times \left(\frac{-7}{5}\right)$$

$$= \left(\frac{-8+3}{6}\right) \times \left(\frac{-7}{5}\right) = \frac{-5}{6} \times \frac{-7}{5} = \frac{7}{6}$$

$\therefore$ unit digit = 6

2. (8) $\left(\frac{2}{5} + \frac{5}{4}\right) = \frac{8+25}{20} = \frac{33}{20}$

Reciprocal of $\frac{33}{20} = \frac{20}{33}$

Sum of digits of numerator and denominator of $\frac{20}{33}$

$= 2 + 0 + 3 + 3 = 8$

3. (2) Because 5 is not the multiple of 2.

4. (5) Product of two numbers $= \frac{45}{56}$

One of them $= \frac{9}{7}$

Let other be x, $\frac{45}{56} \times x = \frac{9}{7}$

$x = \frac{9}{7} \div \frac{45}{56} = \frac{9}{7} \times \frac{56}{45}$

Other number $= 8/5$. Denominator = 5.

5. (5) $\left(\frac{-4}{3} \times \frac{12}{-5}\right) + \left(\frac{3}{7} \times \frac{21}{15}\right) = \frac{16}{5} + \frac{3}{5}$

$= \frac{19}{5}$

So, k = 5

6. (3) It is given that,

Sum of the numbers $= \frac{-3}{5}$ and, One of the numbers $\frac{-9}{20}$

Suppose the other rational number is x. Since the sum is

$\frac{-3}{5}$

$\therefore x + \left(\frac{-9}{20}\right) = \frac{-3}{5}$

$\Rightarrow x = \frac{-3}{5} - \left(\frac{-9}{20}\right)$

$\Rightarrow x = \frac{-3}{5} + \frac{9}{20} \quad \left[\because -\left(\frac{-9}{20}\right) = \frac{9}{20} \right]$

$\Rightarrow x = \frac{(-3) \times 4 + 9 \times 1}{20}$

$\Rightarrow x = \frac{(-3) \times 4 + 9 \times 1}{20} = \frac{-12 + 9}{20} = \frac{-3}{20}$

k = 3

Chapter

2

Linear Equations in One Variable

INTRODUCTION

A statement of equality of two algebraic expressions involving a variable is called an equation.

In this chapter, we learn how a mathematical problem is converted into an equation and it will be solved to find the solution of the problem. Solution of an equation can be find out algebraically.

There are many types of equations. But here we study only about Linear equation in one variable.

Forming the equation(s) and solving it, has great use in many branch of mathematics. It is used as a great tool in algebra and has its application in all spheres of life, be it Finance, Economics, Business, Marketing, Health, Sciences, etc.

VARIABLE OR UNKNOWN

In an equation, the symbol (letter) whose numerical value can vary is called a variable. Variables are usually denoted by letters x, y, z etc.

LINEAR EQUATION

When two algebraic expressions are connected by the sign of equality ($=$), they form an equation. e.g. $2x + 5 = 10$ or $2x + 4 = 6x + 2$ are examples of an equation. There 'x' is a variable.

Both the above example contain only one variable and the maximum power of variable is one. These type of equations are called linear equations.

Solution of a Linear Equation of One Variable

The value of variable which satisfy a given equation is called solution of the equation. The solution is also known as root of the equation. For e.g., for a linear equation: $2x - 8 = 0$, $x = 4$ is a solution or root, because by putting $x = 4$ in the equation, both sides of the equation become equal. In other words, $x = 2$ satisfies equation $2x - 8 = 0$.

NOTE: Solution of an equation is also known as root of the equation.

ILLUSTRATION : 1

Verify that $x = 3$ is a solution of equation $3x + 5 = 14$

SOLUTION :

If $x = 3$ is a solution of the given equation, then by putting $x = 3$ in the given equation will make both sides of equation equal

$\therefore$ Put $x = 3$ in the equation, we get

$$\text{L. H. S.} = 3 \times 3 + 5 = 9 + 5$$

$$= 14 = \text{R. H. S.}$$

Hence $x = 3$ is a solution of $3x + 5 = 14$

ILLUSTRATION : 2

Verify that $x = 2$ is a solution of $2x + 3 = 5x - 6$.

SOLUTION :

By putting $x = 2$, we get

$$\text{L. H. S.} = 2x + 3 = 2 \times 2 + 3 = 7$$

$$\text{R. H. S.} = 5x - 6 = 5 \times 2 - 6 = 4$$

So $x = 2$ is not a solution of $2x + 3 = 5x - 6$

Rules for Solving a Linear Equation in One Variable

While solving a linear equation, following rules are to be remembered.

- Addition/ subtraction of same quantity to/from both sides of an equation does not change the equality
- Multiplication by a same number on both sides of an equation does not change the equality
- Division by a same non-zero number on both sides of an equation does not change the equality.
- Any term may be transposed from one side of equality to another side by changing the sign of the term (from + to -, or from - to +),
- A multiplier may be removed from one side of an equation by making it a divisor on the other side.
- A divisor may be removed from one side of an equation by making it a multiplier on the other side.

SOLVING EQUATIONS HAVING VARIABLE TERMS ON ONE SIDE AND NUMBER(S) ON THE OTHER SIDE

Following examples will illustrate the method of solving linear equations in one variable having variable terms on one side and numbers on the other side.

ILLUSTRATION : 3

Solve the equation : $\frac{x}{5} + 11 = \frac{1}{15}$ and check the result.

SOLUTION :

We have, $\frac{x}{5} + 11 = \frac{1}{15} \Rightarrow \frac{x}{5} + 11 - 11 = \frac{1}{15} - 11$ (Subtracting 11 from both sides)

$$\Rightarrow \frac{x}{5} = \frac{1}{15} - 11 \Rightarrow \frac{x}{5} = -\frac{164}{15}$$

$$\Rightarrow 5 \times \frac{x}{5} = 5 \times -\frac{164}{15} \Rightarrow x = -\frac{164}{3}$$

Thus, $x = -\frac{164}{3}$ is the solution of the given equation.

Check

Substituting $x = -\frac{164}{3}$ in the given equation, we get

$$\text{L.H.S.} = \frac{x}{5} + 11 = \frac{-164}{3} \times \frac{1}{5} + 11 = \frac{-164}{15} + 11 = \frac{-164 + 165}{15} = \frac{1}{15}$$

$$\text{and, R.H.S.} = \frac{1}{15}$$

$$\therefore \text{L.H.S.} = \text{R.H.S. for } x = -\frac{164}{3}$$

Hence, $x = -\frac{164}{3}$ is the solution of the given equation.

ILLUSTRATION : 4

Solve: $\frac{1}{3}x - \frac{5}{2} = 6$

SOLUTION :

We have,

$$\frac{1}{3}x - \frac{5}{2} = 6$$

$$\Rightarrow \frac{1}{3}x - \frac{5}{2} + \frac{5}{2} = 6 + \frac{5}{2} \quad \left(\text{Adding } \frac{5}{2} \text{ on both sides} \right)$$

$$\Rightarrow \frac{1}{3}x = 6 + \frac{5}{2} \Rightarrow \frac{1}{3}x = \frac{12 + 5}{2}$$

$$\Rightarrow \frac{1}{3}x = \frac{17}{2} \Rightarrow 3 \times \frac{1}{3}x = 3 \times \frac{17}{2} \quad (\text{Multiplying both sides by 3})$$

$$\Rightarrow x = \frac{51}{2}$$

Thus, $x = \frac{51}{2}$ is the solution of the given equation.

NOTE: We can check or verify the solution obtained by solving and putting the solution obtained in the equation. If L.H.S. = R.H.S., then solution is verified.

SOLVING EQUATIONS HAVING THE VARIABLE ON BOTH SIDES

Sometimes both sides of an equation contains both, variable and constants. In such cases, we first transpose (shift) terms containing variable on R.H.S. to L.H.S. and constant terms on L.H.S. to R.H.S. While transposing, we change the sign of the term and then carry it to the other side. Following examples will illustrate this method of transposition.

ILLUSTRATION : 5

Solve : $x + 7 - \frac{8x}{3} = \frac{17}{6} - \frac{5x}{8}$

SOLUTION :

$$\frac{x}{1} + \frac{7}{1} - \frac{8x}{3} = \frac{17}{6} - \frac{5x}{8}$$

$$\Rightarrow \frac{x}{1} - \frac{8x}{3} + \frac{5x}{8} = \frac{17}{6} - \frac{7}{1} \quad \left[\text{Transposing } \frac{5x}{8} \text{ to L.H.S. and } -7 \text{ to R.H.S.} \right]$$

$$\Rightarrow 24 \left(x - \frac{8x}{3} + \frac{5x}{8} \right) = 24 \left(\frac{17}{6} - 7 \right) \quad [\text{Multiplying both sides by 24, the LCM of 3, 6 and 8}]$$

$$\Rightarrow 24x - 64x + 15x = 68 - 168$$

$$\Rightarrow -25x = -100$$

$$\Rightarrow x = \frac{-100}{-25} = 4 \quad [\text{Dividing both sides by } -25]$$

ILLUSTRATION : 6

Solve the following linear equation:

$$\frac{x}{2} - \frac{1}{5} = \frac{x}{3} + \frac{1}{4}$$

SOLUTION :

$$\frac{x}{2} - \frac{1}{5} = \frac{x}{3} + \frac{1}{4}$$

$$\Rightarrow \frac{x}{2} - \frac{x}{3} = \frac{1}{4} + \frac{1}{5} \quad \left(\begin{array}{l} \text{Transposing } \frac{x}{3} \text{ to LHS} \\ \text{and } -\frac{1}{5} \text{ to RHS} \end{array} \right)$$

$$\Rightarrow \frac{3x - 2x}{6} = \frac{5 + 4}{20}$$

$$\Rightarrow \frac{x}{6} = \frac{9}{20}$$

$$\Rightarrow x = \frac{9}{20} \times 6 \quad (\text{Multiplying both sides by 6})$$

$$\Rightarrow x = \frac{27}{10}$$

$$= \frac{9}{10} + \frac{1}{4} = \frac{18 + 5}{20} = \frac{23}{20}$$

Therefore, LHS = RHS

ILLUSTRATION : 7

Solve : $\frac{n}{2} - \frac{3n}{4} + \frac{5n}{6} = 21$

SOLUTION :

$$\frac{n}{2} - \frac{3n}{4} + \frac{5n}{6} = 21 \Rightarrow \frac{6n - 9n + 10n}{12} = 21 \quad (\text{LCM}(2, 4, 6) = 12)$$

$$\Rightarrow \frac{7n}{12} = 21 \Rightarrow n = 21 \times \frac{12}{7} = 36 \quad (\text{Multiplying both sides by } \frac{12}{7})$$

CROSS-MULTIPLICATION METHOD

Let equation be of the form : $\frac{ax+b}{cx+d} = \frac{m}{n}$.

Consider the equation $\frac{3x+5}{2x+1} = \frac{1}{3}$ where $a = 3, b = 5, c = 2, d = 1, m = 1$ and $n = 3$.

Equation is obtained directly by equating the product of numerator of L.H.S. and denominator of R.H.S. to the product of denominator of L.H.S. and numerator of R.H.S.

This can be exhibited as follows : $\frac{3x+5}{2x+1} = \frac{1}{3}$

This process of multiplying the numerator of L.H.S. with the denominator on R.H.S. and equating it to the product of the denominator on L.H.S. with the numerator on R.H.S. is called cross-multiplication.

So, we can convert an equation of the form $\frac{ax+b}{cx+d} = \frac{m}{n}$ to a linear equation $n(ax+b) = m(cx+d)$.

ILLUSTRATION : 8A

Solve : $\frac{5x-7}{3x} = 2$ and check your solution.

SOLUTION :

$$\frac{5x-7}{3x} = 2 \Rightarrow \frac{5x-7}{3x} = \frac{2}{1}$$

$$\Rightarrow (5x-7) = 2 \times 3x \Rightarrow 5x-7 = 6x$$

$$\Rightarrow 5x-6x = 7 \Rightarrow -x = 7 \text{ or } x = -7$$

Check: When $x = -7$

$$\text{L.H.S.} = \frac{5(-7)-7}{3(-7)} = \frac{-35-7}{-21} = \frac{-42}{-21} = 2 = \text{R.H.S.}$$

ILLUSTRATION : 9

Solve : $\frac{2x-1}{x+3} + \frac{1-2x}{x-3} = \frac{4-3x}{x^2-9}$

SOLUTION :

$$\frac{(2x-1)(x-3) + (1-2x)(x+3)}{x^2-9} = \frac{4-3x}{x^2-9}$$

[Using the identity $(a+b)(a-b) = a^2 - b^2$]

$$\Rightarrow \frac{[2x^2 - 6x - x + 3] + [x + 3 - 2x^2 - 6x]}{x^2-9} = \frac{4-3x}{(x^2-9)} \Rightarrow 2x^2 - 7x + 3 + 3 - 2x^2 - 5x = 4 - 3x$$

$$\Rightarrow -12x + 6 = 4 - 3x \Rightarrow -12x + 3x = 4 - 6$$

$$\Rightarrow -9x = -2 \Rightarrow x = \frac{2}{9}$$

WORD PROBLEMS IN THE SYSTEM OF SIMULTANEOUS LINEAR EQUATIONS

Working Rule :

- Make assumptions using one variable say x .
- Express the conditions the problem in symbolic language and form equation in term of x .
- Solve the equation and verify the result

ILLUSTRATION : 10

Mohan is three times as old as his son Manoj. Five years later Mohan shall be two and a half times as old as his son. Find present ages of Mohan and Manoj.

SOLUTION :

Let present ages of Manoj and Mohan are x and $3x$ years respectively.

5 years later, age of Mohan = $(3x + 5)$ years

5 years later, age of Manoj = $(x + 5)$ years

∴ According to question,

$$(3x + 5) = 2\frac{1}{2}(x + 5)$$

$$\Rightarrow (3x + 5) = \frac{5}{2}(x + 5)$$

$$\Rightarrow 2(3x + 5) = 5(x + 5)$$

$$\Rightarrow 6x + 10 = 5x + 25$$

$$\Rightarrow 6x - 5x = 25 - 10$$

$$\Rightarrow x = 15$$

∴ Present age of Mohan = 45 years and
present age of Manoj = 15 years.

ILLUSTRATION : 11

The denominator of a rational number is greater than its numerator by 8. If the numerator is increased by 17 and the denominator is decreased by 1, the number obtained is $\frac{3}{2}$. Find the rational number.

SOLUTION :

Let the rational number be $\frac{x}{x+8}$

According to the question,

$$\frac{x+17}{x+8-1} = \frac{3}{2}$$

$$\Rightarrow \frac{x+17}{x+7} = \frac{3}{2}$$

On cross multiplying, we get

$$2(x+17) = 3(x+7)$$

$$\Rightarrow 2x + 34 = 3x + 21$$

$$\Rightarrow 34 - 21 = 3x - 2x$$

$$\Rightarrow x = 13$$

$$\text{and } x + 8 = 13 + 8 = 21$$

Hence, the rational number is $\frac{13}{21}$.

MISCELLANEOUS SOLVED EXAMPLES

1. Solve: $0.16(5x - 2) = 0.4x + 7$

Sol. We have,

$$\begin{aligned} 0.16(5x - 2) &= 0.4x + 7 \\ \Rightarrow 0.8x - 0.32 &= 0.4x + 7 \\ \Rightarrow 0.8x - 0.4x &= 0.32 + 7 && [\text{Transposing } 0.4x \text{ to LHS and } -0.32 \text{ to RHS}] \\ \Rightarrow 0.4x &= 7.32 \\ \Rightarrow \frac{0.4x}{0.4} &= \frac{7.32}{0.4} && [\text{Dividing both sides by } 0.4] \\ \Rightarrow x &= \frac{732}{40} \Rightarrow x = \frac{183}{10} = 18.3 \end{aligned}$$

Hence, $x = 18.3$ is the solution of the given equation.

2. Solve for x : $2 + \frac{2x - 3}{2x + 3} = \frac{3x + 4}{x + 2}$

Sol. $\frac{2(2x + 3) + (2x - 3)}{2x + 3} = \frac{3x + 4}{x + 2} \Rightarrow \frac{4x + 6 + 2x - 3}{2x + 3} = \frac{3x + 4}{x + 2}$

$$\begin{aligned} \Rightarrow \frac{6x + 3}{2x + 3} &= \frac{3x + 4}{x + 2} \Rightarrow (6x + 3)(x + 2) = (3x + 4)(2x + 3) && (\text{by cross multiplication}) \\ \Rightarrow 6x^2 + 3x + 12x + 6 &= 6x^2 + 8x + 9x + 12 \Rightarrow 6x^2 + 15x + 6 = 6x^2 + 17x + 12 \\ \Rightarrow 6x^2 + 15x - 6x^2 - 17x &= 12 - 6 \Rightarrow -2x = 6 \quad (\text{on dividing both sides by } -2) \\ \Rightarrow x &= -3 \end{aligned}$$

3. Solve for y : $\frac{1}{2}(3y + 1) - \frac{1}{3}(5y + 2) = y - 1$

Sol. Given equation is $\frac{1}{2}(3y + 1) - \frac{1}{3}(5y + 2) = y - 1$

To remove fractions, we multiply the equation on both sides by LCM of 2 & 3 i.e.

$$\begin{aligned} 6 \left[\frac{1}{2}(3y + 1) - \frac{1}{3}(5y + 2) \right] &= 6(y - 1) \\ \Rightarrow 3(3y + 1) - 2(5y + 2) &= 6y - 6 \\ \Rightarrow 9y + 3 - 10y - 4 &= 6y - 6 \\ \Rightarrow 9y - 10y - 6y &= -6 + 4 - 3 \\ \Rightarrow -7y &= -5 \\ \Rightarrow y &= \frac{5}{7} \end{aligned}$$

4. Solve the equation : $\frac{2x-1}{3} - \frac{6x-2}{5} = \frac{1}{3}$

Sol. Given equation is $\frac{2x-1}{3} - \frac{6x-2}{5} = \frac{1}{3}$

Multiplying both sides by LCM of 3 and 5 i.e. 15

$$\Rightarrow 5(2x-1) - 3(6x-2) = 5 \Rightarrow 10x - 5 - 18x + 6 = 5$$

$$\Rightarrow -8x = 5 + 5 - 6 \Rightarrow -8x = 4$$

$$\Rightarrow x = -\frac{4}{8} = -\frac{1}{2}$$

$\therefore x = -\frac{1}{2}$ is the solution of given equation.

5. Solve for x : $\frac{6x-7}{2x+1} = \frac{3x+1}{x+5}$

Sol. $\frac{6x-7}{2x+1} = \frac{3x+1}{x+5}$, on cross multiplying, we get

$$\Rightarrow (6x-7)(x+5) = (3x+1)(2x+1)$$

$$\Rightarrow 6x^2 + 30x - 7x - 35 = 6x^2 + 3x + 2x + 1$$

$$\Rightarrow 23x - 3x - 2x = 35 + 1$$

$$\Rightarrow 18x = 36$$

$$\Rightarrow x = \frac{36}{18} \Rightarrow x = 2$$

$\therefore x = 2$ is the solution of given equation.

6. The sum of two numbers, one of which is $\frac{2}{3}$ times the other, is 50. Find two numbers.

Sol. Let one of the number = x

According to question; other number = $\frac{2}{3}x$

Now, it is given that $x + \frac{2}{3}x = 50$

$$\Rightarrow \frac{5}{3}x = 50$$

$$\Rightarrow x = \frac{50 \times 3}{5} = 30$$

$\therefore$ one number = 30

Other number = $30 \times \frac{2}{3} = 20$

7. The length of a rectangle is 4 cm more than the breadth and the perimeter is 11 cm more than the breadth.

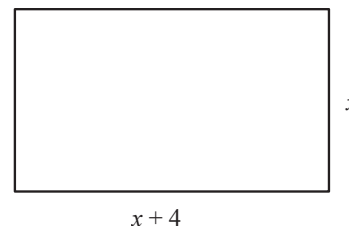
Find the length and breadth of the rectangle.

Sol. Let breadth of rectangle = x

$\therefore$ Length of rectangle = $(x+4)$ cm

Now, perimeter (P) of rectangle = $2(\text{length} + \text{breadth})$

$$= 2(x + x + 4) \text{ cm} = 2(2x + 4) = 4x + 8$$



According to question,

$$\text{Perimeter} = \text{Breadth} + 11$$

$$\Rightarrow 4x + 8 = x + 11$$

$$\Rightarrow 4x - x = 11 - 8 \Rightarrow 3x = 3$$

$$\Rightarrow \boxed{x = 1} = \text{Breadth}$$

$$\therefore \text{Length} = x + 4 = 1 + 4 = 5 \text{ cm}$$

$$\text{Perimeter} = \text{Breadth} + 11 = 1 + 11 = 12 \text{ cm}$$

8. A number consists of two digits. The digit at ten's place is two times the digit at the unit's place. The number formed by reversing the digits, is 27 less than the original number. Find the original number.

Sol. Let the digit at units place = x

$$\therefore \text{number at ten's place} = 2x$$

$$\therefore \text{Original number} = 10(2x) + x = (20x + x)$$

$$\text{and, number formed by reversing the digits} = 10x + 2x$$

$$\text{Now, new number} = \text{original number} - 27$$

$$10x + 2x = (20x + x) - 27$$

$$12x = 21x - 27, \Rightarrow 21x - 12x = 27$$

$$\Rightarrow 9x = 27, \Rightarrow x = \frac{27}{9} = 3$$

$$\therefore \text{Unit's digit of original number} = 3$$

$$\text{Ten's digit of original number} = 2x = 2 \times 3 = 6$$

$$\therefore \text{Original number} = 10(2x) + x = 10 \times 6 + 3 = 63$$

9. Divide 300 into two parts so that half of one part may be less than the other by 48.

Sol. Let one part of 300 be ' x ' so other part will be $(300 - x)$.

$$\text{Now, according to the question, } \frac{1}{2}x = (300 - x) - 48$$

$$\Rightarrow \frac{1}{2}x = 252 - x \quad \Rightarrow 504 - 2x = x$$

$$\Rightarrow 3x = 504 \quad \Rightarrow x = \frac{504}{3}$$

$$\Rightarrow x = 168$$

$$\therefore \text{other part} = 300 - 168 = 132$$

10. Saurabh has ₹ 34 in the form of fifty paise and twenty-five paise coins. If the number of 25 paise coins be twice the number of 50-paise coins, how many coins of each kind does he have?

Sol. Let the number of 50-paise coins be x . Then, number of 25-paise coins = $2x$

$$\therefore \text{Value of } x \text{ fifty-paise coins} = 50 \times x \text{ paise} = ₹ \frac{50 \times x}{100} = ₹ \frac{x}{2}$$

$$\text{Value of } 2x \text{ twenty-five paise coins} = 25 \times 2x \text{ paise} = 50x \text{ paise} = ₹ \frac{50x}{100} = ₹ \frac{x}{2}$$

$$\therefore \text{Total value of all coins} = ₹ \left(\frac{x}{2} + \frac{x}{2} \right) = ₹ x$$

$$\text{But the total value of the money is ₹ 34} \Rightarrow x = 34$$

$$\text{Hence, number of 50 paise coins} = 34 \text{ and number of 25 paise coins} = 68.$$

11. A boy covers a distance of 25 km in 4 hours partly on foot at the rate of 3.5 km/hr and partly on cycle at 9 km/hr. Find the distance covered on foot.

Sol. Let the distance covered on foot be x km.

Then, the distance covered on bicycle = $(25 - x)$ km.

Time taken to cover x km at 3.5 km/hr = $\left(\frac{x}{3.5}\right)$ hours and $(25 - x)$ km at 9 km/hr = $\left(\frac{25 - x}{9}\right)$ hours

$$\text{Time taken} = \frac{x}{3.5} + \frac{25 - x}{9} = 4$$

$$\Rightarrow \frac{2x}{7} + \frac{25 - x}{9} = 4 \Rightarrow \frac{18x + 175 - 7x}{63} = 4$$

$$\Rightarrow 11x + 175 = 252 \Rightarrow 11x = 77 \Rightarrow x = 7$$

$\therefore$ Distance covered on foot = 7 km.

12. A motor boat covers a certain distance downstream in a river in five hours. It covers the same distance upstream in five and a half hours. The speed of water is 1.5 km/hr. Find the speed of the boat in still water?

Sol. Let speed of boat in still water = x km/hr and speed of water = 1.5 km/hr

$\therefore$ Speed of boat downstream = $(x + 1.5)$ km/hr

$\therefore$ Distance covered downstream in 1 hour = $(x + 1.5)$ km

$\therefore$ Distance covered downstream in 5 hours = $5(x + 1.5) = 5x + 7.5$ km ... (i)

Speed of boat upstream = $(x - 1.5)$ km/hr

Distance covered upstream in 1 hour = $(x - 1.5)$ km

Distance covered upstream in $5\frac{1}{2}$ hours = $\frac{11}{2}(x - 1.5)$ km = $\left(\frac{11x}{2} - \frac{16.5}{2}\right)$ km ... (ii)

Since distance covered is same, we have

$$5x + 7.5 = \frac{11x}{2} - \frac{16.5}{2} \quad (\text{from (i) \& (ii)})$$

$$\Rightarrow 2(5x + 7.5) = 1(11x - 16.5) \Rightarrow 10x + 15.0 = 11x - 16.5$$

$$\Rightarrow 10x - 11x = -16.5 - 15$$

$$\Rightarrow -x = -31.5 \Rightarrow x = 31.5$$

Hence, speed of boat in still water = 31.5 km/hr.

13. The sum of two numbers is 2490. If 6.5% of one number is equal to 8.5% of the other, find the numbers.

Sol. Let the first number be x . And, Second number = $2490 - x$

$$\text{Now, } 6.5\% \text{ of the first number} = \frac{6.5}{100} \times x = \frac{65x}{1000}$$

$$\text{and, } 8.5\% \text{ of the second number} = \frac{8.5}{100} \times (2490 - x) = \frac{85}{1000} (2490 - x)$$

According to the question

$$\therefore \frac{65x}{1000} = \frac{85}{1000} (2490 - x)$$

$$\Rightarrow 65x = 85 (2490 - x) \quad [\text{Multiplying both sides by 1000}]$$

$$\Rightarrow 65x + 85x = 2490 \times 85$$

$$\Rightarrow x = \frac{2490 \times 85}{150} = 1411$$

$\therefore$ First number = 1411, Second number = $2490 - 1411 = 1079$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- If $\frac{1}{2} - x = \frac{1}{2}$, then $x =$ _____.
- If $5x = 5$, then $x =$ _____.
- If $x + a = b$, then $x =$ _____.
- If $\frac{x}{a} = b$, then $x =$ _____.
- If 5 less than a number is 72 then the number is _____.
- Solution of the equation $-\frac{3(p-1)}{10} = 2$ is _____.
- Solution of $0.18(5x-4) = 0.5x + 0.8$ is _____.
- An equation involving only _____ variable is called a linear equation.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- When a number is divided by 9, it is diminished by 56. Then, the number is 47.
- A number which satisfies the given equation is called solutions or root of the equation.
- Statements which are written in the form $ax + b < 0$, $ax + b \leq 0$, $ax + b > 0$ or $ax + b \geq 0$ are called linear in equations.
- The solution of the equation $2x + 5 = 0$ is $\frac{5}{2}$.
- An equation altered if we multiply or divide both sides of the equation by the same non-zero number.
- The two numbers whose sum is 28 and difference is 4 are 18 and 10.
- The solution of the equation $5x - 2(2x - 7) = 2(3x - 1) + \frac{7}{2}$ is $\frac{4}{5}$.
- When we divide 4500 into two parts such that 5% of the first part is equal to 10% of the second part, then the first part and second part is 3000, 1500.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.

Column-I	Column-II
1. (A) Solution of $2x - 3 = 7$ is	(p) -8
(B) Solution of $\frac{15}{4} - 7x = 9$ is	(q) 0.6
(C) Solution of $1.6 = \frac{y}{1.5}$ is	(r) $-\frac{3}{4}$
(D) Solution of $0.25(4t-3) = 0.05(10t-9)$ is	(s) 5
(E) Solution of $\frac{3x+4}{2-6x} = -\frac{2}{5}$ is	(t) 2.4

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- Find the value of 'p', if $x = 2$ and $y = 1$, is a solution of the equation $2x + 3y = p$
- When driven in town, a car runs x kilometres on each litre of petrol.
Find, in terms of x , the number of litres of petrol used when the car is driven 200 km in town.
- Find the solution of $\frac{2x+1}{3x-1} = \frac{3}{2}$.
- If the angles of a triangle are in the ratio 2 : 3 : 4, then find the difference between the greatest and smallest angles
- If $x = \frac{1}{2}$, then find the value of $x + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}$
- If $x - \frac{1}{x-2} = 2 - \frac{1}{x-2}$, then x is equal to ?
- Find three even consecutive integers whose sum is 90.
- Find the value of S in $\frac{1}{3} + S = \frac{2}{5}$.
- Find the value of d if $5 - \frac{2d+7}{9} = 0$ is
- Given that $-0.3k + 2.1 = 0.4k$, the value of $k = ?$

Short Answer Questions :

DIRECTIONS : Give answer in two to three sentences.

1. Solve the following equations :

(i) $\frac{2x-3}{2} + 1 = \frac{5x+2}{3}$

(ii) $\frac{x-5}{4} + 1 = \frac{3-2x}{7}$

(iii) $\frac{5x+3}{3} - \frac{4x-1}{4} = 2$

2. Solve the equation:

$$\frac{t}{2} + \frac{2t}{3} - \frac{3t}{4} = \frac{5t}{6} + \frac{11t}{12} + 4$$

3. If two-thirds of a number is greater than one-fourth of the number by 5, find the number.

4. A man left one-fourth of his money to his wife, two fifth to his son and donated balance amount of ₹ 210000 to a charitable fund. How much money did he have? How much money was received by his wife and son separately?

5. Three numbers are in the ratio 1:2:3 and their sum is 120. Find the numbers.

6. If the side of chess board is smaller than its perimeter by 24 cm, then find the side of the chess board.

7. A number is such that it is as much greater than 112 as it is less is than it. Find the number.

8. When seventh part of a number is increased by 2 and fifth part of the same number is decreased by 4, they become equal. Find the number.

9. Solve the following equation for given variable:

(i) $\frac{x}{10} + \frac{70-x}{2} = 19$ (ii) $8(x+40) = 1.5(2x+8)$.

10. Solve for x

(i) $12(3-x) = 48$

(ii) $2x + (x+1) + (x+2) = 103$

(iii) $\frac{x}{3} + 1 = \frac{7}{15}$.

11. Form equations and solve the following statements:

(i) Thrice a number equals 60. What is the number?

(ii) 60 subtracted from a number gives 52. What is the number?

(iii) Two numbers are in the ratio 5 : 8. If their sum is 130, then find the numbers.

12. Write the equations for the following statements:

(i) A number increased by 34 gives 86.

(ii) Twice a number equals 20.

(iii) Half of a number is 16.

(iv) Five times Romi's age equals 100 years.

(v) Eight times a number added to 2 gives 60.

13. Solve $\frac{3(x-5)}{7} = 12$.

14. Find the number if one-fourth of which is 8 more than 5.

15. (i) Seven times a number is 49. What is the number?

(ii) One and half times of a number is 300, find the number.

16. Solve the equation : $\frac{2x-1}{3} - \frac{6x-2}{5} = \frac{1}{3}$

17. $\frac{-4}{3}y = -\frac{3}{4}$, then y = ?

18. A number consists of two digits. The digit in the ten's place exceeds the digit in the unit's place by 4. The sum

of the digits is $\frac{1}{7}$ of the number. Find the number.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Solve for y : $\frac{1}{2}(3y+1) - \frac{1}{3}(5y+2) = y-1$

2. Find the value of n such that

$$\frac{2}{3}(4n-1) - \left(2n - \frac{1+n}{3}\right) = \frac{1}{3}n + \frac{4}{3}.$$

3. Solve, $\frac{x-4}{3} + \frac{2x-3}{35} = \frac{5x-32}{9} - \frac{x+9}{28}$

4. Solve the following:

$$\frac{1}{2}(x+1) + \frac{1}{3}(x-1) = \frac{5}{12}(x-2)$$

5. Find the solution for $\frac{2}{x+3} - \frac{4}{x-3} = \frac{-6}{x+3}$

6. The sum of three numbers is 98. The ratio of the first to the second is $\frac{2}{3}$ and the ratio of the second to the third is $\frac{5}{8}$.

Find the second number.

7. The difference between two natural numbers is 96. If the larger number is divided by the smaller one, then the quotient is 6 and the remainder is 11. Find the numbers.

8. A number consists of two digits. The digit at ten's place is two times the digit at the unit's place. The number formed by reversing the digits, is 27 less than the original number. Find the original number.
9. 300 is divided into two parts such that half of one part is less than the other by 48. Find the two parts.
10. One number is three times another. If the larger number is subtracted from 60, the result is 5 less than the smaller number subtracted from 55. Find the numbers.
11. Two trains, one travelling 15 km/hr faster than the other, leave the same station at the same time, one travelling east and the other west. At the end of 6 hours they are 570 km apart. What is the speed of each train?
12. The perimeter of a rectangle is 40cm. The ratio of its side is 2 : 3. Find its length & breadth.
13. A lady has 25 p and 50 p coins in her purse. If in all she has 40 coins totalling ₹.12.50, find the number of coins of each type she has.

2 EXERCISE

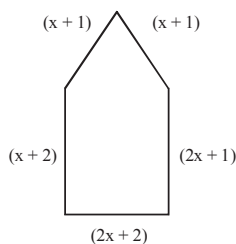
Text-Book Exercise :

1. $8x + 4 = 3(x - 1) + 7$
2. $x = \frac{4}{5}(x + 10)$
3. $\frac{2x}{3} + 1 = \frac{7x}{15} + 3$
4. $2y + \frac{5}{3} = \frac{26}{3} - y$
5. $3m = 5m - \frac{8}{5}$
6. $x + 7 - \frac{8x}{3} = \frac{17}{6} - \frac{5x}{2}$
7. $\frac{3t-2}{4} - \frac{2t+3}{3} = \frac{2}{3} - t$
8. $m - \frac{m-1}{2} = 1 - \frac{m-2}{3}$
9. $0.25(4f - 3) = 0.05(10f - 9)$
10. The sum of three consecutive multiples of 11 is 363. Find these multiples.
11. Deveshi has a total of ₹590 as currency notes in the denominations of ₹50, ₹20 and ₹10. The ratio of the number of ₹50 notes and ₹20 notes is 3:5. If she has a total of 25 notes, how many notes of each denomination she has?
12. Baichung's father is 26 years younger than Baichung's grandfather and 29 year solder than Baichung. The sum of the ages of all the three is 135 years. What is the age of each one of them?
13. A rational number is such that when you multiply it by $\frac{5}{2}$ and add $\frac{2}{3}$ to the product, you get $-\frac{7}{12}$. What is the number?
14. The organisers of an essay competition decide that a winner in the competition gets a prize of ₹100 and a participant who does not win gets a prize of ₹25. The total prize money distributed is ₹3,000. Find the number of winners, if the total number of participants is 63.
15. There is a narrow rectangular plot, reserved for a school, in Mahuli village. The length and breadth of the plot are in the ratio 11:4. At the rate ₹ 100 per metre it will cost the village panchayat ₹ 75000 to fence the plot. What are the dimensions of the plot?
16. Hasan buys two kinds of cloth materials for school uniforms, shirt material that costs him ₹50 per metre and trouser material that costs him ₹90 per metre. For every 3 metres of the shirt material, he buys 2 metres of the trouser material. He sells the materials at 12% and 10% profit respectively. His total sale is ₹36,600. How much trouser material did he buy?

Exemplar Questions :

1. A steamer goes downstream from one point to another in 7 hours. It covers the same distance upstream in 8 hours. If the speed of stream be 2 km/hr, find the speed of the steamer in still water and the distance between the ports.
2. Distance between two stations A and B is 690 km. Two cars start simultaneously from A and B towards each other, and the distance between them after 6 hours is 30 km. If the speed of one car is less than the other by 10 km/hr, find the speed of each car.
3. $1 - (x - 2) - [(x - 3) - (x - 1)] = 0$
4. $\frac{3t-2}{3} + \frac{2t+3}{2} = t + \frac{7}{6}$
5. $4(3p + 2) - 5(6p - 1) = 2(p - 8) - 6(7p - 4)$
6. ₹13500 are to be distributed among Salma, Kiran and Jenifer in such a way that Salma gets ₹1000 more than Kiran and Jenifer gets ₹500 more than Kiran. Find the money received by Jenifer.

7. The volume of water in a tank is twice of that in the other. If we draw out 25 litres from the first and add it to the other, the volumes of the water in each tank will be the same. Find the volume of water in each tank.
8. Find a number whose fifth part increased by 30 is equal to its fourth part decreased by 30.
9. Two equal sides of a triangle are each 4m less than three times the third side. Find the dimensions of the triangle, if its perimeter is 55m.
10. Anima left one-half of her property to her daughter, one-third to her son and donated the rest to an educational institute. If the donation was worth ₹1,00,000, how much money did Anima have?
11. A man was engaged as typist for the month of February in 2009. He was paid ₹500 per day but ₹100 per day were deducted for the days he remained absent. He received ₹9,100 as salary for the month. For how many days did he work?
12. For what value of x is the perimeter of shape 77 cm?



HOTS Questions :

1. A man has a certain number of 5-rupee coins, 2-rupee coins, 1-rupee coins and 50-paise coins. The number of 5-rupee coins is 7 less than 2-rupee coins, the number of 1-rupee coins is 15 more than 5-rupee coins and the number of 50-paise coins is twice the number of 2-rupee coins. If a man has ₹198 in all, find the number of coins of each denomination.

2. Advanced purchase tickets to an art exhibition cost ₹4, while tickets purchased at the door cost ₹6. If a total of 150 tickets were sold and ₹680 was collected, how many advanced purchase tickets were sold?

3. An alloy of silver and gold weighs 90 gm in air and 84 gm in a liquid. Assuming that silver loses one-tenth of its weight in the liquid and that gold loses one-nineteenth of its weight. Find the weight of each metal in the alloy.

4. Simplify the following linear equations, then solve them:

(i) $\frac{7y-2}{5y-1} = \frac{3+2y}{4+5y}$

(ii) $\frac{1}{a+2} + \frac{1}{a+1} = \frac{2}{a+10}$

(iii) $\frac{0.1}{2}x - 5\left(0.2x - \frac{2}{25}\right) = 3x$

5. $6.5x + \frac{19.5x - 32.5}{2} = 6.5x + 13 + \left(\frac{13x - 26}{2}\right)$

6. $(3x-8)(3x+2) - (4x-11)(2x+1) = (x-3)(x+7)$

7. Solve: $\frac{4x+17}{18} - \frac{13x-2}{17x-32} + \frac{x}{3} = \frac{7x}{12} - \frac{x+16}{36}$

8. Divide ₹1380 among Ahmed, John and Babita so that the amount Ahmed receives is 5 times as much as Babita's share and is 3 times as much as John's share.

9. The length of a rectangle exceeds its breadth by 4 cm. If length and breadth are each increased by 3 cm, the area of the new rectangle will be 81 cm² more than that of the given rectangle. Find the length and breadth of the given rectangle.

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. If $\frac{1}{x+1} + \frac{1}{x+4} = 0$, then $x =$

- (a) $2\frac{1}{2}$ (b) $-2\frac{1}{2}$
(c) 3 (d) -3

2. If $\frac{x}{pq} + \frac{x}{qr} + \frac{x}{pr} = p+q+r$, then $x =$

- (a) pqr (b) $\frac{pq}{r}$
(c) $\frac{p}{qr}$ (d) $\frac{q}{pr}$

3. The equation $\frac{12x+1}{4} = \frac{13x-1}{5} + 3$ is true for

- (a) $x = \frac{1}{8}$ (b) $x = 2$
(c) $x = \frac{5}{8}$ (d) $x = \frac{3}{4}$

4. The root of equation $\frac{4}{2z-3} = \frac{5}{3z-2}$ is
 (a) $-7/2$ (b) $7/2$
 (c) $3/2$ (d) $-3/2$
5. The sum of two numbers is 90 and the greater number exceeds thrice the smaller number by 14. The number is
 (a) 18, 72 (b) 19, 71
 (c) 20, 70 (d) 15, 75
6. A boat covers a certain distance downstream in 3 hours, and it covers the same distance upstream in 5 hours. If the speed of the boat in still water is 8 km/hr, then the speed of the stream is
 (a) 1 km/hr (b) 1.5 km/hr
 (c) 2 km/hr (d) 3 km/hr
7. Two numbers are in the ratio 5 : 3. If they differ by 18, then. The numbers are
 (a) 45, 27 (b) 25, 15
 (c) 35, 21 (d) 65, 39
8. The sum of three consecutive multiples of 8 is 888, then multiples are
 (a) 160, 168, 176 (b) 288, 296, 304
 (c) 320, 328, 336 (d) 264, 272, 280.
9. The ages of Rahul and Haroon are in the ratio 5 : 7. If four years later, the sum of their ages will be 56 years, then Rahul's present age is
 (a) 28 years (b) 26 years
 (c) 20 years (d) 18 years
10. A bag contains 50P, 25P and 10P coins in the ratio 2:3:4: amounting to ₹ 129. Find the number of coins of each type
 (a) 120, 180, 240 (b) 180, 150, 200
 (c) 200, 180, 120 (d) 180, 200, 140
11. Sum of the digits of a two digit number is 9. When we interchange the digits, it is found that the resulting two digit new number is greater than the original number by 27. Then the two digit number is
 (a) 63 (b) 36
 (c) 45 (d) 54
12. The denominator of a rational number is greater than its numerator by 8. If the numerator is increased by 17 and the denominator is decreased by 1, the number obtained is $\frac{3}{2}$. Then the rational number is
 (a) $\frac{13}{20}$ (b) $\frac{20}{13}$
 (c) $\frac{21}{13}$ (d) $\frac{13}{21}$
2. Divide 34 into two parts in such a way that $\left(\frac{4}{7}\right)^{\text{th}}$ of one part is equal to $\left(\frac{2}{5}\right)^{\text{th}}$ of the other.
 (a) 10 (b) 24
 (c) 14 (d) 20
3. The numerator of a fraction is 4 less than the denominator. If 1 is added to both its numerator and denominator, it becomes $1/2$. Which of the following is the multiple of the denominator?
 (a) 6 (b) 14
 (c) 7 (d) 10
4. Shikhaa has Piggy bank. It is full of one-rupee and fifty-paise coins. It contains 3 times as many fifty paise coins as one rupee coins. The total amount of the money in the bank is ₹ 35. How many coins of each kind are there in the bank?
 (a) 14 (b) 16
 (c) 48 (d) 42
5. The sum of two numbers is 45 and their ratio is 7 : 8. Find the numbers.
 (a) 21 (b) 24
 (c) 25 (d) 20
6. Two numbers are such that the ratio between them is 3:5. If each is increased by 10, the ratio between the new numbers so formed is 5:7. Find the original numbers.
 (a) 12 (b) 20
 (c) 25 (d) 15
7. The age of a man is same as his wife's age with the digits reversed. Then sum of their ages is 99 years and the man is 9 years older than his wife. The age of man and his wife is
 (a) 50 years (b) 45 years
 (c) 54 years (d) 44 years
8. Two angles in a triangle are in the ratio 4 : 5. If the sum of these angles is equal to the third angle, then the angles are
 (a) 180° (b) 40°
 (c) 50° (d) 90°
9. The sum of two numbers is 2490. If 6.5% of one number is equal to 8.5% of the other, the numbers are
 (a) 1311 (b) 1411
 (c) 1089 (d) 1079
10. The length of a rectangle is 6 m less than three times its breadth. The length and breadth of the rectangle if its perimeter is 148 m, is
 (a) 54m (b) 30m
 (c) 40m (d) 20m

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. One of the two digits of a two digit number is three times the other digit. If you interchange the digits of this 2-digit number and add the resulting number to the original number, you get 88. The original number is

Passage Based Questions :

DIRECTIONS : Study the given paragraph (s) and answer the following questions.

PASSAGE-I

Twenty years ago, my age was one-third of what it is now.

- My present age is
(a) 66 years (b) 30 years
(c) 33 years (d) 36 years
- My age twenty years ago is
(a) 40 years (b) 15 years
(c) 22 years (d) 10 years
- Ratio of my present age to age twenty years ago is
(a) 3 : 2 (b) 2 : 1
(c) 3 : 1 (d) 1 : 2

PASSAGE-II

A rectangle has a perimeter of 60 cm.

- If length and breadth of rectangle is $(k + 4)$ cm and $(3k - 2)$ cm respectively, then value of k is
(a) 7 (b) 9
(c) 12 (d) 29
- The length and breadth of the rectangle is
(a) 11, 7 (b) 11, 19
(c) 19, 20 (d) 23, 11
- If perimeter of rectangle is equal to perimeter of square, then side of square is
(a) 60 cm (b) 18 cm
(c) 225 cm (d) 15 cm

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- If **Assertion** is **correct** but **Reason** is **incorrect**.
- If **Assertion** is **incorrect** but **Reason** is **correct**.

- Assertion :** The length of rectangle exceeds its breadth by 4 cm. If length and breadth are each increased by 3 cm, the area of the new rectangle will be 81cm^2 more than that of given rectangle. Length and breadth are 14 and 10 cm respectively.

Reason : Area of rectangle = Length $\times$ Breadth.

- Assertion :** $\frac{4}{5}x + 4 = \frac{3}{4}x$ is a linear equation.

Reason : Four-fifth of a number is more than three fourth of the number by 4 is a statement of linear equation.

- Assertion :** $4x + \frac{8}{9} = \frac{6x-2}{8}$ is a linear equation.

Reason : Solution of the equation is -2 .

- Assertion :** $\frac{5}{3}x^{-1} + 11 = 1$ is a linear equation.

Reason : In a linear equation power of x cannot be negative.

- Assertion :** $\frac{7}{4}\sqrt{x} + 9x = 3$ is not a linear equation.

Reason : Linear equation involves only linear polynomials.

Multiple Matching Questions :

DIRECTIONS : Following question has statements (A, B, C, D) given in Column-I and statements (p, q, r, s) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- Find the value of x .

Column-I

- $8x + 4 = 3(x - 1) + 7$
- $\frac{3x-2}{4} - \frac{2x+3}{3} = \frac{2}{3} - x$
- $3(5x-7) - 2(9x-11) = 4(8x-13) - 17$
- $\frac{x+17}{x+7} = \frac{3}{2}$

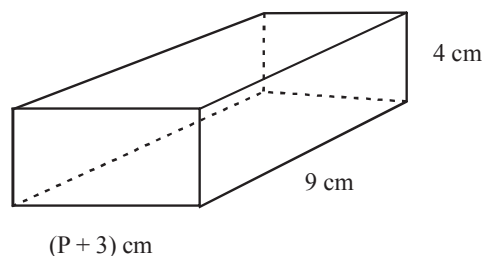
Column-II

- 2
- 13
- 0
- $0 < x < 4$

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- What must be subtracted from each term of the ratio $4 : 7$, so that the ratio becomes $2 : 5$?
- A's age is twice that of B, B's age is twice that of C and C's age is twice that of D. If the average of the ages of A, B, C and D is 15 years, then what is the present age of D?
- A man can row at 8 kmph in still water. If the river is running at 2 kmph, it takes him 48 minutes to row to a place and back. How far is the place?
- The figure shows a cuboid with a volume of 180cm^3 .



What is the value of p ?

- If $\frac{a(x-b)}{a-b} + \frac{b(x-a)}{b-a} = 1$, then $x =$

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- | | | |
|-----------------|-----------|----------|
| 1. 0 | 2. 1 | 3. $b-a$ |
| 4. $a \times b$ | 5. 77 | 6. -15 |
| 7. 3.8 | 8. linear | |

True/False :

- False
- True
- True
- False : The solution of the equation $2x + 5 = 0$ is $x = -\frac{5}{2}$.
- False : An equation remains unaltered if we multiply or divide both sides of the equation by the same non-zero number.
- False : Let one number be x
Other number = $28 - x$
 $\Rightarrow x - (28 - x) = 4 \Rightarrow x - 28 + x = 4$
 $\Rightarrow 2x = 28 + 4 \Rightarrow 2x = 32 \Rightarrow x = 16$
One number is 16
Other number is $(28 - x) = 28 - 16 = 12$.
- False : $5x - 2(2x - 7) = 2(3x - 1) + \frac{7}{2}$

$$\Rightarrow 5x - 4x + 14 = 6x - 2 + \frac{7}{2}$$

$$\Rightarrow x + 14 = 6x - 2 + \frac{7}{2}$$

$$\Rightarrow x - 6x = -2 + \frac{7}{2} - 14 \Rightarrow -5x = \frac{-4 + 7 - 28}{2}$$

$$\Rightarrow -5x = \frac{-25}{2} \Rightarrow x = \frac{5}{4}$$

- True : Let first part be x
Second part = $4500 - x$
Now, 5% of $x = 10\%$ of $(4500 - x)$

$$\frac{5}{100} \times x = \frac{10}{100} (4500 - x)$$

$$\Rightarrow 5x = 10(4500 - x)$$

$$\Rightarrow 5x = 45000 - 40x$$

$$15x = 45000 \Rightarrow x = \frac{45000}{15} = 3000$$

$$\text{First part} = x = 3000$$

$$\text{Second part} = (4500 - x) = 4500 - 3000 = 1500$$

Match the Columns :

- (A) $\rightarrow$ (s); B $\rightarrow$ (r); (C) $\rightarrow$ (t); (D) $\rightarrow$ (q); (E) $\rightarrow$ (p)

Very Short Answer Questions :

- $2(2) + 3(1) = p$
 $\Rightarrow p = 7$

- $\frac{200}{x}$

- $\Rightarrow \frac{2x+1}{3x-1} = \frac{3}{2}$
 $\Rightarrow 4x+2 = 9x-3$
 $\Rightarrow x = 1$

- $2x + 3x + 4x = 180^\circ$
 $9x = 180^\circ$
 $x = 20^\circ$

$$\text{Difference} = 4x - 2x = 2x = 2 \times 20^\circ = 40^\circ.$$

- $x + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}} = \frac{1}{2} + \frac{1}{1 + \frac{1}{1 + \frac{1}{\frac{1}{2}}}} = \frac{1}{2} + \frac{1}{1 + \frac{1}{1+2}}$
 $= \frac{1}{2} + \frac{1}{1 + \frac{1}{3}} = \frac{1}{2} + \frac{1}{\frac{4}{3}} = \frac{1}{2} + \frac{3}{4} = \frac{2+3}{4} = \frac{5}{4}$

- $x - \frac{1}{x-2} = 2 - \frac{1}{x-2}$

$$\Rightarrow x = 2 - \frac{1}{x-2} + \frac{1}{x-2} \Rightarrow x = 2$$

- The three consecutive even integers are $x, x+2, x+4$.
 $x + x + 2 + x + 4 = 90$
 $3x + 6 = 90$
 $x = 28$

$\therefore$ The required numbers are 28, 30 and 32.

- $\frac{1}{3} + S = \frac{2}{5}$

$$S = \frac{2}{5} - \frac{1}{3} \Rightarrow S = \frac{1}{15}.$$

9. $5 - \frac{2d+7}{9} = 0$
 $\Rightarrow 45 - 2d - 7 = 0$
 $\Rightarrow 38 = 2d \Rightarrow d = 19$
10. $-0.3k + 2.1 = 0.4k$
 $\Rightarrow 2.1 = 0.4k + 0.3k$
 $\Rightarrow k = \frac{2.1}{0.7} = 3$

Short Answer Questions :

1. (i) $x = -\frac{7}{4}$; (ii) $x = \frac{47}{15}$; (iii) $x = \frac{9}{8}$
2. $t = -3$
3. 12
4. Let the man had ₹ x with him.

Wife received = ₹ $\frac{x}{4}$ and

Son received = ₹ $\frac{2x}{5}$

$\therefore$ The amount left with the man = ₹ $\left(x - \frac{x}{4} - \frac{2x}{5}\right) = ₹ \frac{7x}{20}$

According to the question,

$$\frac{7x}{20} = 210000 \Rightarrow x = \frac{210000 \times 20}{7} = 6,00,000$$

Hence, Man has ₹ 6,00,00.

$$\text{Wife received} = 600000 \times \frac{1}{4} = ₹ 1,50,000$$

$$\text{Son received} = 600000 \times \frac{2}{5} = ₹ 2,40,000$$

5. Let the numbers be x , $2x$ and $3x$.
 According to the question,
 $x + 2x + 3x = 120$

$$6x = 120 \Rightarrow x = \frac{120}{6} = 20$$

Hence, numbers are 20, 40 and 60.

6. Side = 8 cm
7. If the number is x .

$$\text{then } x - 112 = 112 - x$$

$$\Rightarrow 2x = 224 \Rightarrow x = 112$$

8. Let the required number be x .

$$\text{Seventh part of a number} = \frac{x}{7}$$

When it is increased by 2, it becomes $\frac{x}{7} + 2$

$$\text{Fifth part of a number} = \frac{x}{5}$$

When it is decreased by 4, it becomes $\frac{x}{5} - 4$

$$\text{Thus, } \frac{x}{7} + 2 = \frac{x}{5} - 4 \Rightarrow x = 105$$

9. (i) $x = 40$.
 (ii) $x = -61.6$.
10. (i) $x = -1$
 (ii) $x = 25$
 (iii) $x = \frac{-8}{5}$.
11. (i) $3x = 60$, where $x = 20$.
 (ii) $x - 60 = 52$, where $x = 112$.
 (iii) $5x + 8x = 130$, where $x = 50$ and 80 .
12. (i) $x + 34 = 86$, where x is a number.
 (ii) $2x = 20$, where x is a number.
 (iii) $\frac{x}{2} = 16$, where x is a number.
 (iv) $5y = 100$, where y is Romi's age.
 (v) $8x + 2 = 60$, where x is a number.
13. $x = 33$.
14. 52
15. (i) The number is 7.
 (ii) The number is 200.

16. Given equation is $\frac{2x-1}{3} - \frac{6x-2}{5} = \frac{1}{3}$

Multiplying both sides by LCM of 3 and 5 i.e. 15

$$\Rightarrow 5(2x-1) - 3(6x-2) = 5$$

$$\Rightarrow 10x - 5 - 18x + 6 = 5$$

$$\Rightarrow -8x = 5 + 5 - 6 \Rightarrow -8x = 4$$

$$\Rightarrow x = -\frac{4}{8} = -\frac{1}{2}$$

$\therefore x = -\frac{1}{2}$ is the solution of given equation.

17. $-\frac{4}{3}y = -\frac{3}{4}$

$$y = \frac{-3}{4} \times \left(\frac{-3}{4}\right) = \frac{9}{16}$$

18. Let the digit in the unit's place = x
 Let the digit in the ten's place = $x + 4$
 $\therefore$ The number is
 $10(x+4) + x = 10x + 40 + x = 11x + 40$
 Sum of the digits = $x + x + 4 = 2x + 4$

$$\text{Given, } 2x + 4 = \frac{1}{7}(11x + 40)$$

$$\Rightarrow 14x + 28 = 11x + 40 \Rightarrow 3x = 12 \Rightarrow x = 4$$

$\therefore$ The number is $11 \times 4 + 40 = 44 + 40 = 84$.

Long Answer Questions :

1. Given equation is $\frac{1}{2}(3y+1) - \frac{1}{3}(5y+2) = y-1$

To remove fractions, we multiply the equation on both sides by LCM of 2 & 3 i.e.

$$6\left[\frac{1}{2}(3y+1) - \frac{1}{3}(5y+2)\right] = 6(y-1)$$

$$\Rightarrow 3(3y+1) - 2(5y+2) = 6y-6$$

$$\Rightarrow 9y+3-10y-4 = 6y-6$$

$$\Rightarrow 9y-10y-6y = -6+4-3$$

$$\Rightarrow -7y = -5$$

$$\Rightarrow y = \frac{5}{7}$$

2. $\frac{2}{3}(4n-1) - \left(2n - \frac{1+n}{3}\right) = \frac{1}{3}n + \frac{4}{3}$

$$\frac{2}{3}(4n-1) - \left(\frac{6n-1-n}{3}\right) = \frac{1}{3}n + \frac{4}{3}$$

$$2(4n-1) - (5n-1) = n+4$$

$$8n-2-5n+1 = n+4$$

$$(8n-5n) + (-2+1) = n+4$$

$$3n-1 = n+4$$

$$3n-n = 4+1$$

$$2n = 5$$

$$n = \frac{5}{2}$$

3. $\frac{x-4}{3} + \frac{2x-3}{35} = \frac{5x-32}{9} - \frac{x+9}{28}$

Multiplying throughout by 9, we have

$$3x-12 + \frac{18x-27}{35} = 5x-32 - \frac{9x+81}{28}$$

$$\text{transposing, } \frac{18x-27}{35} + \frac{9x+81}{28} = 2x-20$$

Now clear of fractions by multiplying by $5 \times 7 \times 4$ or 140

$$\text{thus } 72x-108+45x+405 = 280x-2800$$

$$\therefore 2800-108+405 = 280x-72x-45x$$

$$\therefore 3097 = 163x$$

$$\therefore x = 19$$

4. $\frac{1}{2}(x+1) + \frac{1}{3}(x-1) = \frac{5}{12}(x-2)$

$$\frac{x}{2} + \frac{1}{2} + \frac{x}{3} - \frac{1}{3} = \frac{5x}{12} - \frac{10}{12}$$

$$\frac{x}{2} + \frac{x}{3} - \frac{5x}{12} = \frac{-10}{12} + \frac{1}{3} - \frac{1}{2}$$

$$\frac{6x+4x-5x}{12} = \frac{-10+4-6}{12}$$

$$\frac{5x}{12} = \frac{-12}{12}$$

$$\frac{5x}{12} = -1$$

$$x = \frac{-12}{5}$$

5. $\frac{2}{x+3} - \frac{4}{x-3} = \frac{-6}{x+3}$

$$\Rightarrow \frac{2(x-3) - 4(x+3)}{(x+3)(x-3)} = \frac{-6}{x+3}$$

$$\Rightarrow \frac{2x-6-4x-12}{(x+3)(x-3)} = \frac{-6}{(x+3)}$$

$$\Rightarrow \frac{-2x-18}{x-3} = -6$$

$$\Rightarrow -2x-18 = -6x+18$$

$$\Rightarrow 4x = 36 \Rightarrow x = 9$$

6. Let three numbers are x_1, x_2 and x_3
Then

$$x_1 : x_2 = 2 : 3$$

$$x_2 : x_3 = 5 : 8$$

$$\Rightarrow x_1 : x_2 : x_3 = 10 : 15 : 24$$

$$\Rightarrow 10x + 15x + 24x = 98$$

$$49x = 98$$

$$x = 2$$

$$x_2 = 15x = 30$$

7. Let the two numbers are x and y and assume that $x > y$
Thus, $x - y = 96$... (i)

$$\text{and } x = 6y + 11$$

$$\text{But } x = 96 + y \text{ (} \because \text{ From (i))}$$

$$\therefore 96 + y = 6y + 11$$

$$\Rightarrow 96 - 11 = 5y$$

$$\Rightarrow 85 = 5y$$

$$\Rightarrow y = 17$$

$$\text{and } x = 96 + 17 = 113$$

8. Let the digit at units place = x

$$\therefore \text{ number at ten's place} = 2x$$

$$\therefore \text{ the original number}$$

$$= 10(2x) + x = (20x + x)$$

Now, number formed by reversing the digits = $10x + 2x$

Now, new number = original number - 27

$$10x + 2x = (20x + x) - 27$$

$$12x = 21x - 27, \Rightarrow 21x - 12x = 27$$

$$\Rightarrow 9x = 27, \Rightarrow x = \frac{27}{9} = 3$$

$\therefore$ Unit's digit of original number = 3

$$\text{Ten's digit of original number} = 2x = 2 \times 3 = 6$$

$$\therefore \text{Original number} = 10(2x) + x = 10 \times 6 + 3 = 63$$

9. Let one part of 300 be 'x' so other part will be $(300 - x)$.

$$\text{Now, according to the question, } \frac{1}{2}x = (300 - x) - 48$$

$$\Rightarrow \frac{1}{2}x = 252 - x \Rightarrow 504 - 2x = x$$

$$\Rightarrow 3x = 504 \Rightarrow x = \frac{504}{3}$$

$$\Rightarrow x = 168$$

$$\therefore \text{other part} = 300 - 168 = 132$$

10. Let the smaller number = x

$$\text{Larger number subtracted from 60} = 60 - 3x$$

$$\text{Smaller number subtracted from 55} = 55 - x$$

Solution is $x = 5$ (smaller no.)

11. Speed of fast train = $(x + 15)$ km/hr

$$\text{Distance travelled by slow train} = 6x$$

$$\text{Distance travelled by fast train} = 6(x + 15)$$

$$\text{Total distance} = 6x + 6(x + 15) = 570$$

$$\text{So, speed of fast train} = 55 \text{ km/hr}$$

12. Let length & breadth be $2x$ cm & $3x$ cm respectively.

According to problem,

$$2(x + y) = 40 \quad \dots\dots\dots (1)$$

$$\text{and } \frac{y}{x} = \frac{2}{3} \quad \dots\dots\dots (2)$$

$$\text{on solving, } x = 12, y = 8$$

$$\therefore \text{length be 12 cm. \& breadth be 8 cm.}$$

13. Let the lady has x coins of 25 p and $(40 - x)$ coins of 50 p.

Then acc. to problem

$$25x + 50(40 - x) = 1250 \quad \dots\dots\dots (2)$$

Solving for x, we get

$$x = 30 \text{ (25 P coins) \& } y = 10 \text{ (50 P coins)}$$

3. $x = 10$

4. $y = \frac{7}{3}$

5. $m = \frac{4}{5}$

6. $x = -5$

7. $t = 2$

8. $m = \frac{7}{5}$

9. $f = 0.6$

10. Take three consecutive multiples of 11 as x, $x + 11$ and $x + 22$.
Sum of these consecutive multiples of 11 = 363.

$$\therefore x + (x + 11) + (x + 22) = 363$$

$$\text{or } x + x + 11 + x + 22 = 363$$

$$\text{or } 3x + 33 = 363$$

$$\text{or } 3x = 363 - 33$$

$$\text{or } 3x = 330$$

$$\text{or } x = \frac{330}{3}$$

Hence, the three consecutive multiples are 110, 121, 132.

11. Let the number of ₹50 notes and ₹20 notes be $3x$ and $5x$, respectively.

But she has 25 notes in total.

$$\text{Therefore, the number of ₹10 notes} = 25 - (3x + 5x) = 25 - 8x$$

The amount she has

$$\text{from ₹50 notes : } 3x \times 50 = ₹150x$$

$$\text{from ₹20 notes : } 5x \times 20 = ₹100x$$

$$\text{from ₹10 notes : } (25 - 8x) \times 10 = ₹(250 - 80x)$$

$$\text{Hence the total money she has} = 150x + 100x + (250 - 80x) = ₹(170x + 250)$$

$$\text{now, } 170x + 250 = 590$$

$$\text{or } 170x = 590 - 250 = 340$$

$$\text{or } x = \frac{340}{170} = 2$$

$$\text{The number of ₹ 50 notes she has} = 3x$$

$$= 3 \times 2 = 6$$

$$\text{The number of ₹ 20 notes she has} = 5x = 5 \times 2 = 10$$

$$\text{The number of ₹ 10 notes she has} = 25 - 8x$$

$$= 25 - (8 \times 2) = 25 - 16 = 9$$

12. Baichung's age: 17 years; Baichung's father's age: 46 years; Baichung's grandfather's age = 72 years

13. $-\frac{1}{2}$

14. 19

15. Length = 275 m; breadth = 100 m

16. 200m

2 EXERCISE

Text-Book Exercise :

1. $x = 0$

2. $x = 40$

Exemplar Questions :

- Let speed of steamer in still water = x km/hr
Speed of stream = 2 km/hr
Speed downstream = $(x + 2)$ km/hr
Speed upstream = $(x - 2)$ km/hr
Distance covered in 7 hours while downstream = $7(x + 2)$
Distance covered in 8 hours while upstream = $8(x - 2)$
According to the condition,
 $7(x + 2) = 8(x - 2)$
 $7x + 14 = 8x - 16$
 $x = 30$ km/hr
Total Distance = $7(x + 2)$ km
 $= 7(30 + 2)$ km
 $= 7 \times 32$ km
 $= 224$ km
- Let speed of faster car = x km/hr
Then speed of other = $(x - 10)$ km/hr
Let 1st one start from A and other from B.
M and N be their position after 6 hours.

 $AM = 6x, BN = 6(x - 10)$
According to condition,
 $6x + 6x - 60 + 30 = 690$
 $12x = 690 + 30$
 $12x = 720$
 $x = 60$ km/hr
Speed of other car = 50 km/hr.
- $x = 5$
- $t = \frac{1}{3}$
- $P = \frac{-5}{22}$
- ₹4500
- 50l, 100l
- 1200
- 9m, 23m, 23m
- ₹3,00,000
- 20 days
- $x = 10$ cm

HOTS Questions :

- No. of 2-rupee coins = 25
No. of 5-rupee coins = 18
No. of 1-rupee coins = 33
No. of 50-paise coins = 50.
- Let x = no. of advanced purchase tickets sold.
No. of tickets sold at the door = $150 - x$

According to question,

$$4x + 6(150 - x) = 680$$

On solving, we get

$$x = 110$$

- Weight of silver = x gm

$$\text{Weight of gold} = (90 - x) \text{ gm}$$

$$\text{Weight lost by silver} = \frac{x}{10} \text{ gm}$$

$$\text{Weight lost by gold} = \frac{1}{19}(90 - x) \text{ gm}$$

$$\text{Given, } \frac{9x}{10} + \frac{18(90 - x)}{19} = 84$$

- (i) $y = \frac{1}{2}$ (ii) $a = \frac{-26}{17}$

$$(iii) \quad x = \frac{8}{79}$$

- $x = -2.8$

- $x = 4$

$$7. \quad \frac{4x + 17}{18} - \frac{13x - 2}{17x - 32} + \frac{x}{3} = \frac{7x}{12} - \frac{x + 16}{36}$$

$$\Rightarrow \frac{4x + 17}{18} - \frac{7x}{12} + \frac{x + 16}{36} + \frac{x}{3} = \frac{13x - 2}{17x - 32}$$

Multiplying both sides by 36 i.e., the LCM of 18, 12, 36 and 3, we get

$$36 \times \frac{4x + 17}{18} - 36 \times \frac{7x}{12} + 36 \times \frac{x + 16}{36}$$

$$+ 36 \times \frac{x}{3} = 36 \times \left(\frac{13x - 2}{17x - 32} \right)$$

$$\Rightarrow 2(4x + 17) - 3 \times 7x + x + 16 + 12x = 36 \times \left(\frac{13x - 2}{17x - 32} \right)$$

$$\Rightarrow 8x + 34 - 21x + x + 16 + 12x = 36 \times \left(\frac{13x - 2}{17x - 32} \right)$$

[By cross-multiplication]

$$\Rightarrow 50 = 36 \times \left(\frac{13x - 2}{17x - 32} \right)$$

$$\Rightarrow 50 \times (17x - 32) = 36(13x - 2)$$

$$\Rightarrow 850x - 468x = 1600 - 72$$

$$\Rightarrow 382x = 1528 \Rightarrow x = \frac{1528}{382} = 4$$

Hence, $x = 4$, is the solution of the given equation.

8. Let Babita's share be ₹ x . Then,
 Ahmed's share = ₹ $5x$
 $\therefore$ John's share = Total amount –
 (Babita's share + Ahmed's share)
 $= ₹ [1380 - (x + 5x)] = ₹ (1380 - 6x)$
 $\therefore 5x = 3(1380 - 6x)$
 $\Rightarrow 5x = 4140 - 18x$
 $\Rightarrow 5x + 18x = 4140$
 $\Rightarrow 23x = 4140$
 $\Rightarrow x = \frac{4140}{23} = 180$
 $\therefore$ Babita's share = ₹ 180, Ahmed's share
 $= ₹ (5 \times 180) = ₹ 900$
 John's share = ₹ $(1380 - 6 \times 180) = ₹ 300$
9. Let the breadth of the given rectangle be x cm. Then, Length
 $= (x + 4)$ cm
 $\therefore$ Area = Length $\times$ Breadth $= (x + 4)x = x^2 + 4x$.
 When length and breadth are each increased by 3 cm.
 New length = $(x + 4 + 3)$ cm $= (x + 7)$ cm,
 New breadth = $(x + 3)$ cm
 $\therefore$ Area of new rectangle = Length $\times$ Breadth
 $= (x + 7)(x + 3)$
 $= x(x + 3) + 7(x + 3)$
 $= x^2 + 3x + 7x + 21 = x^2 + 10x + 21$
 It is given that the area of new rectangle is 80 cm^2 more than
 the given rectangle.
 $\therefore x^2 + 10x + 21 = x^2 + 4x + 81$
 $\Rightarrow x^2 + 10x - x^2 - 4x = 81 - 21$
 $\Rightarrow 6x = 60$
 $\Rightarrow x = \frac{60}{6} = 10$
 Thus,
 Length of the given rectangle = $(x + 4)$ cm $= (10 + 4)$ cm
 $= 14$ cm

3 EXERCISE

Single Option Correct :

1. (b) 2. (a) 3. (d) 4. (a)
5. (b)
6. (c) Let the speed of the stream is x km/hr. Since the speed of the boat in still water is 8 km/hr., the speed of boat downstream = $(8 + x)$ km/hr. and the speed of boat upstream = $(8 - x)$ km/hr.
- Then the distance covered by the boat downstream in 8 hours = $3(8 + x)$ km [distance = time $\times$ speed]

The distance covered by the boat upstream in 5 hours
 $= 5(8 - x)$ km.

As the distances covered by the boat downstream and upstream are equal, we get

$$3(8+x)=5(8-x)$$

$$\Rightarrow 24 + 3x = 40 - 5x$$

$$\Rightarrow 8x = 16$$

$$\Rightarrow x = \frac{16}{8} = 2$$

$\therefore$ Speed of the stream = 2 km/hr

7. (a) 8. (b)
9. (c) Let the present ages of Rahul and Haroon be x and Y respectively.

$\therefore$ it is given $\frac{x}{y} = \frac{5}{7}$

$$\Rightarrow x = \frac{5}{7} y \quad \dots(1)$$

Four years later ages of Rahul of Haroon will be $(x + 4)$ and $(y + 4)$ respectively.

$$\therefore (x + 4) + (y + 4) = 56$$

$$\Rightarrow x + y = 48 \quad \dots(2)$$

From (1) and (2)

$$\Rightarrow \frac{5}{7}y + y = 48$$

$$\Rightarrow \frac{5y + 7y}{7} = 48$$

$$\Rightarrow \frac{12y}{7} = 48$$

$$\Rightarrow y = 28$$

(1) from (2)

$$x + 28 = 48$$

$$\Rightarrow x = 20$$

$\therefore$ Present age of Rahul = 20 years.

10. (a)
11. (b) Let the two digit number be xy
[i.e. digit at units place be 'y' and digit at tens place be x]
 $\therefore$ the two digit number can be expressed as
 $\Rightarrow 10x + y$
Number obtained by reversing the digits is
 $\Rightarrow 10y + x$
Now according to question
New number – Original number = 27
 $(10y + x) - (10x + y) = 27$
 $\therefore 9y - 9x = 27$
 $\Rightarrow y - x = 3$
 $\Rightarrow y = 3 + x \quad \dots(1)$
Also it is given that

$$x + y = 9 \quad \dots(2)$$

From (1) and (2)

$$\Rightarrow x + (3 + x) = 9$$

$$\Rightarrow 2x + 3 = 9$$

$$\Rightarrow 2x = 6$$

$$\Rightarrow x = \frac{6}{2} = 3$$

From (1) $y = 3 + 3 = 6$

$\therefore$ the original number $(xy) = 36$.

12. (d) Let the numerator of rational number = x

Let the denominator of rational number = y

$$\therefore \text{rational number} = \frac{x}{y}$$

Now according to question $y = x + 8$

$$\therefore \text{rational number} = \frac{x}{x+8} \quad \dots(1)$$

Further it is given that

$$\frac{x+17}{x+8-1} = \frac{3}{2}$$

$$\Rightarrow \frac{x+17}{x+7} = \frac{3}{2}$$

$$\Rightarrow 2x + 34 = 3x + 21$$

$$\Rightarrow 3x - 2x = 34 - 21$$

$$\Rightarrow x = 13$$

$$\therefore \text{the required rational no} = \frac{x}{x+8} = \frac{13}{13+8} = \frac{13}{21}$$

More than One Option Correct :

1. (b, c)
2. (c, d) Let one part be x . Then, other part is $(34 - x)$. It is given that

$$\left(\frac{4}{7}\right)^{\text{th}} \text{ of one part} = \left(\frac{2}{5}\right)^{\text{th}} \text{ of the other part}$$

$$\Rightarrow \frac{4}{7}x = \frac{2}{5}(34 - x)$$

$$\Rightarrow 20x = 14(34 - x)$$

[Multiplying both sides by 35, the LCM of 7 and 5]

$$\Rightarrow 20x + 14 \times 34 = 14x$$

$$\Rightarrow 20x + 14x = 14 \times 34$$

$$\Rightarrow 34x = 14 \times 34$$

$$\Rightarrow \frac{34x}{34} = \frac{14 \times 34}{34} \quad [\text{Dividing both sides by 34}]$$

$$\Rightarrow x = 14 \text{ and other part} = 34 - 14 = 20$$

3. (b, c) Let the denominator of the fraction be x . Then,
Numerator of the fraction = $x - 4$

$$\therefore \text{Fraction} = \frac{x-4}{x}$$

If 1 is added to both its numerator and denominator, the fraction becomes $\frac{1}{2}$.

$$\therefore \frac{x-4+1}{x+1} = \frac{1}{2}$$

$$\Rightarrow \frac{x-3}{x+1} = \frac{1}{2}$$

$$\Rightarrow 2(x-3) = x+1 \quad [\text{Using cross-multiplication}]$$

$$\Rightarrow 2x - 6 = x + 1$$

$$\Rightarrow 2x - x = 6 + 1$$

$$\Rightarrow x = 7$$

Putting $x = 7$ in (i), we get

$$\text{Fraction} = \frac{7-4}{7} = \frac{3}{7}. \text{ Hence, the required denominator is 7.}$$

4. (a, d) Let there be x one rupee coins in the bank. Then,

Number of 50-paise coins = $3x$

$\therefore$ Value of x one rupee coins = ₹ x

Value of $3x$ fifty-paise coins = $50 \times 3x$ paise

$$= 150x \text{ paise} = 150x \text{ paise} = ₹ \frac{150}{100}x = ₹ \frac{3x}{2}$$

$$\therefore \text{Total value of all the coins} = ₹ \left(x + \frac{3x}{2} \right)$$

But, the total amount of the money in the bank is given as ₹ 35.

$$\therefore x + \frac{3x}{2} = 35$$

$$\Rightarrow 2x + 3x = 70 \quad [\text{Multiplying both sides by 2}]$$

$$\Rightarrow 5x = 70 \Rightarrow \frac{5x}{5} = \frac{70}{5} \Rightarrow x = 14$$

$\therefore$ Number of one rupee coins = 14,

Number of 50 paise coins = $3x = 3 \times 14 = 42$.

5. (a, b) Let one of the numbers be x . Since the sum of the two numbers is 45.

Therefore, the other number will be $45 - x$.

It is given that the ratio of the numbers is 7 : 8.

$$\therefore \frac{x}{45-x} = \frac{7}{8}$$

$$\Rightarrow 8 \times x = 7 \times (45 - x)$$

$$\Rightarrow 8x = 315 - 7x$$

$$\Rightarrow 8x + 7x = 315$$

$$\Rightarrow 15x = 315$$

$$\Rightarrow x = \frac{315}{15} = 21$$

Thus, one number is 21 and, Other number = $45 - x = 45 - 21 = 24$

6. (c, d) Since the ratio between the numbers is 3 : 5. So, let the two numbers be $3x$ and $5x$.

If each number is increased by 10, the new numbers are $3x + 10$

It is given that the ratio between the new numbers is 5 : 7.

$$\therefore \frac{3x + 10}{5x + 10} = \frac{5}{7}$$

$$\Rightarrow 21x + 70 = 25x + 50$$

$$\Rightarrow 4x = 20$$

$$\Rightarrow x = 5$$

$\therefore$ Numbers are 15 and 25.

7. (b, c) Let Man's age = x years, Wife's age = $99 - x$

$$\Rightarrow x - 9 = 99 - x \Rightarrow x + x = 99 + 9$$

$$2x = 108 \Rightarrow x = 54$$

Man's age = 54 years

Wife's age = $99 - x = 45$ years

8. (b, c, d) Let the two angles be $4x$, $5x$

$4x + 5x =$ third angle [Given]

$$\Rightarrow 9x = \text{third angle}$$

$$\text{Now, } 4x + 5x + 9x = 180^\circ \Rightarrow 18x = 180^\circ$$

(Angle sum property of a triangle)

$$\Rightarrow x = 10^\circ, \text{ then the angles are } 4x = 40^\circ, 5x = 50^\circ \text{ and } 9x = 90^\circ.$$

9. (b, d) Let the first number be x . Then,

Second number = $2490 - x$

[$\therefore$ Sum of the numbers is given to be 2490]

Now, 6.5% of the first number = $\frac{6.5}{100} \times x = \frac{65x}{1000}$ and, 8.5% of the second number

$$= \frac{8.5}{100} \times (2490 - x) = \frac{85}{1000} (2490 - x)$$

It is given that 6.5% of the first number is equal to 8.5% of the other.

$$\therefore \frac{65x}{1000} - \frac{85}{1000} (2490 - x) \Rightarrow 65x = 85(2490 - x)$$

$$\Rightarrow 65x = 2490 \times 85 - 85x$$

$$\Rightarrow 65x + 85x = 2490 \times 85$$

$$\Rightarrow 150x = 2490 \times 85$$

$$\Rightarrow x = \frac{2490 \times 85}{150} \Rightarrow x = 1411$$

$\therefore$ First number = 1411

Second number = $2490 - 1411 = 1079$

10. (a, d) Let breadth of the rectangle = x metres

Length of the rectangle = $(3x - 6)$ metres

Perimeter = 148m

$$\Rightarrow 2(l + b) = 148 \Rightarrow 2(3x - 6 + x) = 148$$

$$\Rightarrow 2(4x - 6) = 148 \Rightarrow 4x - 6 = 74$$

$$\Rightarrow 4x = 74 + 6 \Rightarrow 4x = 80 \Rightarrow x = 20$$

Breadth of the rectangle = 20 metres

Length of the rectangle = $3 \times 20 - 6 = 54$ metres.

Passage Based Questions :

1. (b) Let my present age be x years.

20 years ago, my age = $(x - 20)$ yrs.

$$\therefore (x - 20) = \frac{1}{3}x$$

$$\Rightarrow x - \frac{1}{3}x = 20$$

$$\Rightarrow x = 30$$

2. (d) my age, 20 years ago = $30 - 20 = 10$ years.

3. (a) Ratio = $\frac{30}{20} = 3 : 2$

4. (a) Length of rectangle = $(k + 4)$ cm

Breadth of rectangle = $(3k - 2)$ cm

Perimeter of rectangle = $2(l + b) = 60$ cm

We have, $2[(k + 4) + (3k - 2)] = 60$

$$\Rightarrow 2[k + 3k + 4 - 2] = 60$$

$$\Rightarrow 2[4k + 2] = 60 \Rightarrow 8k + 4 = 60$$

$$8k = 60 - 4 \Rightarrow 8k = 56 \Rightarrow k = 7.$$

5. (b) Since, $k = 7$ (calculated in 1st part)

Length of the rectangle = $(k + 4)$ cm

$$= (7 + 4) \text{ cm} = 11 \text{ cm}$$

Breadth of the rectangle = $(3k - 2)$ cm

$$= (3 \times 7 - 2) = 21 - 2 = 19 \text{ cm}$$

6. (d) We are given that

Perimeter of the rectangle = Perimeter of the square

Perimeter of square = 60 cm

$$4 \times \text{side} = 60 \Rightarrow \text{Side} = 60 \div 4 = 15 \text{ cm.}$$

Assertion and Reason :

1. (a)

2. (b) **Assertion** : Correct, **Reason** : Correct; But Assertion is not the correct explanation of Reason.

3. (c) **Assertion** : Correct, **Reason** : Incorrect

$$\frac{4x + 8}{9} = \frac{6x - 2}{8}$$

$$\Rightarrow 8(4x+8)=9(6x-2)$$

$$\Rightarrow 32x+64=54x-18$$

$$\Rightarrow -22x=-82 \Rightarrow x=\frac{82}{22}=\frac{41}{11}$$

4. (c) **Assertion** : Correct, **Reason** : Incorrect.

$$\frac{3}{2}x^{-1}+9=1 \text{ is a linear equation because } \frac{3}{2x}+9=1,$$

Now multiplying the whole equation by x we get

$$\frac{3}{2}+9x=x \text{ which can be solved in terms of } x.$$

5. (c) **Assertion** : Correct, **Reason** : Incorrect

$$\frac{5}{2}\sqrt{x}+88x=9 \text{ is not a linear equation because the highest power of } x \text{ has to be } 1 \text{ and in this case } x \text{ has a power of } \frac{1}{2}.$$

Multiple Matching Questions :

1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (p, s); (C) $\rightarrow$ (p, s); (D) $\rightarrow$ (q)

Integer Type Questions :

1. (2) $\frac{4-x}{7-x} = \frac{2}{5}$

$$20-5x=14-2x$$

$$3x=6 \Rightarrow x=2$$

2. (4) Let Present age of D be x yrs

$\therefore$ Present age of A, B and C are $8x$, $4x$ and $2x$ respectively

$$\Rightarrow \frac{8x+4x+2x+x}{4}=15$$

$$\frac{15x}{4}=15$$

$$x=\frac{15 \times 4}{15}=4 \text{ yrs}$$

$\therefore$ D's Present age = 4 yrs

3. (3) Speed of the man in still water = 8 kmph

Speed of the river = 2 kmph

Downstream = $8+2=10$ kmph

Upstream = $8-2=6$ kmph

$$\frac{x}{10}+\frac{x}{6}=\frac{48}{60} \Rightarrow 8x=24$$

$$x=3 \text{ km.}$$

4. (2) Volume of cuboid = 180 cm^3

But volume = $l \times b \times h$

$$\Rightarrow 4 \times 9 \times (P+3)=180 \text{ cm}^3$$

$$\Rightarrow 36(P+3)=180$$

$$\Rightarrow P+3=180 \div 36$$

$$\Rightarrow P+3=5$$

$$\Rightarrow P=2$$

5. (1) $\frac{a(x-b)}{(a-b)}+\frac{b(x-a)}{(b-a)}=1$

$$\frac{a(x-b)-b(x-a)}{(a-b)}=1$$

$$\frac{ax-ab-[bx-ab]}{(a-b)}=1$$

$$ax-ab-bx+ab=a-b$$

$$(a-b)x=a-b$$

$$x=1$$

Chapter

3

Understanding Quadrilaterals

INTRODUCTION

A quadrilateral is a four sided polygon with four angles. There are many kinds of quadrilaterals. The most common types are the parallelogram, the rectangle, the square, the trapezoid and the rhombus.

In this chapter, we will learn about polygons, different types of quadrilaterals and the special cases, of parallelograms. We shall also learn the angle sum property of a quadrilateral and other polygon.

POLYGONS

A simple closed curve made up of only line segments is called a polygon.

We classify polygons according to the number of sides they have

No. of sides	Name of Polygon
3	Triangle
4	Quadrilateral
5	Pentagon
6	Hexagon
7	Heptagon
8	Octagon
9	Nonagon
10	Decagon

Classification of polygons

Polygons can be classified into regular and irregular polygons.

- (a) **Regular polygons** :- Polygons with all the sides and angles equal i.e regular polygons are equilateral and equiangular.
 (b) **Irregular polygons** :- Polygons in which all the sides or angles are not of the same measure.

Polygons can also be classified into convex and concave polygons.

Convex polygons are polygons in which all the line segments joining any two points of the polygon lie completely inside the polygon, otherwise the polygon is concave polygon.

Diagonals

The line segments obtained by joining two non-consecutive vertices of a polygon is called the diagonals of the polygon.

ANGLE SUM PROPERTY

Recall that the sum of three angles of triangles is 180° . We shall use this property to find the sum of four angles of a quadrilateral.

Consider a quadrilateral, ABCD.

Join the diagonal AC and name the angles so formed as $\angle 1, \angle 2, \angle 3, \angle 4, \angle 5$ and $\angle 6$.

The diagonal AC divides the quadrilateral ABCD in two triangles, namely, $\triangle ABC$ and $\triangle ACD$.

In $\triangle ABC$,

$$\angle 4 + \angle 5 + \angle 6 = 180^\circ \text{ (angle sum property of triangle)(i)}$$

Similarly, in $\triangle ACD$,

$$\angle 1 + \angle 2 + \angle 3 = 180^\circ \text{(ii)}$$

Adding (i) and (ii),

$$\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 = 180^\circ + 180^\circ$$

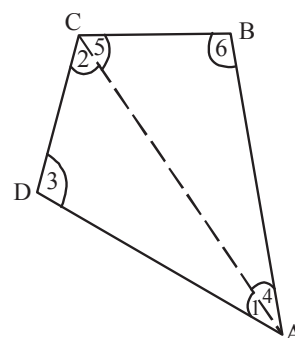
$$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 = 360^\circ$$

$$\Rightarrow \angle DAB + \angle ABC + \angle BCD + \angle CDA = 360^\circ$$

Thus the sum of all angles of a quadrilateral is 360° .

$\therefore$ Sum of all the interior angles of a polygon with ' n ' sides $= (n - 2) \times 180^\circ$

and Perimeter of polygon $= n \times a$ (a = side of regular polygon) and each interior angle of n -sided polygon $= \frac{(n-2)}{n} \times 180^\circ$.



No. of sides	Name of Polygon	Sum of all the interior angles = $(n - 2) \times 180$
3	Triangle	$(3 - 2) \times 180^\circ = 180^\circ$
4	Quadrilateral	$(4 - 2) \times 180^\circ = 360^\circ$
5	Pentagon	$(5 - 2) \times 180^\circ = 540^\circ$
6	Hexagon	$(6 - 2) \times 180^\circ = 720^\circ$
7	Heptagon	$(7 - 2) \times 180^\circ = 900^\circ$
8	Octagon	$(8 - 2) \times 180^\circ = 1080^\circ$
9	Nonagon	$(9 - 2) \times 180^\circ = 1260^\circ$
10	Decagon	$(10 - 2) \times 180^\circ = 1440^\circ$

ILLUSTRATION : 1

Find the number of sides of a regular polygon whose interior angle measures 108° .

SOLUTION :

Recall, the interior angle of a n -sided polygon = $\frac{(n-2)}{n} \times 180^\circ$

$$\Rightarrow 108^\circ = \frac{(n-2)}{n} \times 180^\circ$$

$$\Rightarrow \frac{(n-2)}{n} = \frac{6}{10}$$

By cross multiplication,

$$10n - 20 = 6n$$

$$\Rightarrow 4n = 20$$

$$\Rightarrow n = 5$$

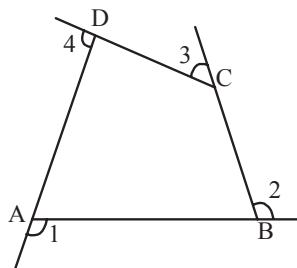
Thus, a regular pentagon has each interior angle of 108° .

SUM OF THE MEASURES OF THE THE EXTERIOR ANGLES OF POLYGON

If the sides of a quadrilateral are produced in order, the sum of four exterior angles so formed is 360° .

Let the side of a quadrilateral ABCD be produced in order as shown in figure forming exterior angles $\angle 1$, $\angle 2$, $\angle 3$ and $\angle 4$

Since $\angle 1$ and $\angle DAB$ forms a linear pair and the sum of the angles of a linear pair is 180°



$$\therefore \angle 1 + \angle DAB = 180^\circ \quad \dots (i)$$

Similarly, we have

$$\angle 2 + \angle CBA = 180^\circ \quad \dots (ii)$$

$$\angle 3 + \angle DCB = 180^\circ \quad \dots (iii)$$

$$\text{and } \angle 4 + \angle ADC = 180^\circ \quad \dots (iv)$$

Understanding Quadrilaterals

Adding (i), (ii), (iii) and (iv), we get

$$(\angle 1 + \angle 2 + \angle 3 + \angle 4) + (\angle DAB + \angle CBA + \angle DCB + \angle ADC) = 180^\circ + 180^\circ + 180^\circ$$

$$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 + 360^\circ = 720^\circ \quad [\because \angle DAB + \angle CBA + \angle DCB + \angle ADC = 360^\circ]$$

$$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 = 720^\circ - 360^\circ = 360^\circ$$

Thus, the sum of the measures of the external angles of any polygon is 360° .

NOTE: Each exterior angle of a regular polygon of n sides is equal to $\left(\frac{360}{n}\right)^\circ$

ILLUSTRATION : 2

The interior angle of a regular polygon is 135° , find number of sides of the polygon.

SOLUTION :

If interior angle of regular polygon = 135° then exterior angle = $180^\circ - 135^\circ = 45^\circ$

$$\therefore \text{no. of sides} = \frac{360^\circ}{\text{exterior angle}} = \frac{360^\circ}{45^\circ} = 8$$

$$\therefore \text{no. of sides} = 8$$

ILLUSTRATION : 3

Each interior angle of a regular polygon is 100° greater than each exterior angle. How many sides the polygon has ?

SOLUTION :

Let measure of each exterior angle is x .

Then measure of each interior angle = $(x + 100)$

$$\text{Now } x + (x + 100) = 180^\circ$$

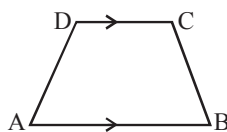
$$\Rightarrow 2x = 80^\circ \Rightarrow x = 40^\circ$$

$$\text{Now, no. of sides} = \frac{360^\circ}{\text{exterior angle}} = \frac{360^\circ}{40^\circ} = 9$$

DIFFERENT TYPES OF QUADRILATERALS

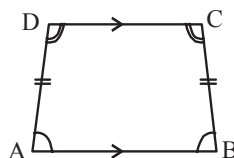
Trapezium

A quadrilateral in which one pair of opposite sides is parallel and the other pair of sides is non-parallel and not equal is called trapezium. In the given figure, ABCD is a trapezium in which $AB \parallel CD$ whereas AD and BC are its non-parallel sides.



Isosceles Trapezium

If the non-parallel sides of the trapezium (AD and BC as shown in figure) are equal, then it is called an isosceles trapezium. In the trapezium ABCD, $AB \parallel CD$; AD and BC are its non-parallel sides such that $AD = BC$. So, ABCD is an isosceles trapezium. In an isosceles trapezium, $\angle A = \angle B$ and $\angle C = \angle D$.



Isosceles Trapezium

NOTE: Non-parallel sides of a trapezium are called lateral sides.

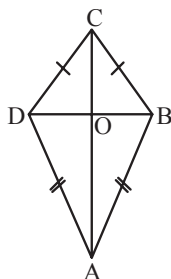
Kite

A quadrilateral in which two pairs of adjacent sides are equal is called a kite. It has unequal opposite sides.

In the adjoining figure, ABCD is a kite.

$AB = AD$ and $BC = CD$

Diagonals AC and BD intersect each other at right angles.

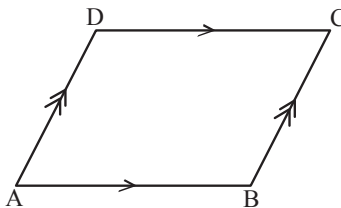


Parallelogram

A quadrilateral with both pair of opposite sides parallel is called a **parallelogram**.

In the adjoining figure ABCD, $AB \parallel CD$ and $AD \parallel BC$.

Thus, ABCD is a parallelogram.



Sides of Parallelogram

The opposite sides of a parallelogram are equal.

Consider a parallelogram ABCD.

Join the diagonal AC.

In $\triangle ADC$ and $\triangle ABC$,

$\angle 1 = \angle 2$ ($AB \parallel CD$, alternate interior angles).

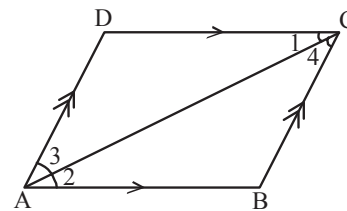
$AC = CA$ (common),

and $\angle 3 = \angle 4$ ($AD \parallel BC$, alternate interior angles).

By ASA congruency rule, $\triangle ADC \cong \triangle ABC$

By c.p.c.t.s., $AD = BC$ and $AB = CD$.

Thus, pair of opposite sides are equal.



Angles of Parallelogram

Opposite angles of a parallelogram are equal.

Let us consider a parallelogram ABCD in which $AB \parallel CD$ and $BC \parallel AD$.

Now, $AB \parallel DC$ and AD is a transversal

$\angle A + \angle D = 180^\circ$... (i)

[Sum of the measures of interior angles on the same side of a transversal is 180°]

Again, $AD \parallel BC$ and AB is a transversal.

$\therefore \angle A + \angle B = 180^\circ$... (ii)

From (i) and (ii), we have

$\therefore \angle D = \angle B$

Similarly, we can prove that $\angle A = \angle C$.

Thus, the adjacent angles of a parallelogram are supplementary and the opposite angles are equal.

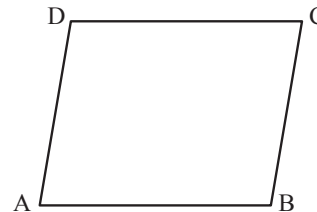


ILLUSTRATION 4:

Two adjacent angles of a parallelogram are equal. What is the measure of each?

SOLUTION:

Let the measure of each angle be x° .

Since the sum of any two consecutive angles of a parallelogram is 180° .

$$\therefore x^\circ + x^\circ = 180^\circ \Rightarrow 2x^\circ = 180^\circ$$

$$\Rightarrow x^\circ = \frac{180^\circ}{2} = 90^\circ$$

Hence, the measure of each angle is 90° .

Diagonals of a Parallelogram

The diagonals of parallelogram bisect each other.

Let the diagonals of parallelogram ABCD intersect at O.

In $\triangle AOB$ and $\triangle COD$,

$$\angle 1 = \angle 2 \quad [\text{alternate interior angles}].$$

$$AB = CD \quad [\text{opposite sides of a parallelogram}].$$

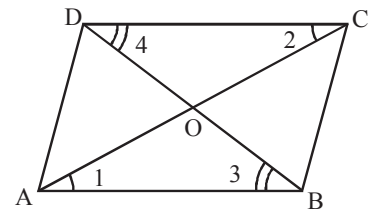
$$\text{and } \angle 3 = \angle 4 \quad [\text{alternative interior angles}].$$

$\therefore$ By ASA congruency rule,

$$\triangle AOB \cong \triangle COD$$

Thus, $OA = OC$ and $OB = OD$ (c.p.c.t.s.)

Therefore, diagonals of a parallelogram bisect each other.

**ILLUSTRATION : 5**

In the given kite, calculate x , y and z .

SOLUTION:

Both pairs of adjacent sides of a kite are equal, therefore,

$$AB = AD \text{ and } BC = CD$$

$$\text{In } \triangle ABD, \angle ABD = \angle ADB = 40^\circ \quad (\text{isos. } \triangle \text{ property})$$

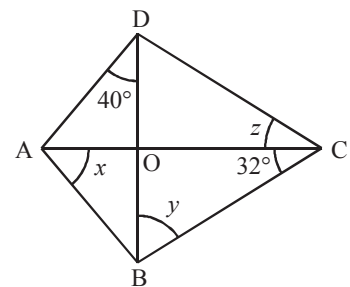
Since the diagonals of a kite intersect at rt. $\angle$ s, $\angle AOB = 90^\circ$.

$$\begin{aligned} \text{In } \triangle AOB, \quad x = \angle OAB &= 180^\circ - (90^\circ + 40^\circ) \\ &= 180^\circ - 130^\circ = 50^\circ \end{aligned}$$

$$\begin{aligned} \text{In } \triangle BOC, \quad y = \angle OBC &= 180^\circ - (90^\circ + 32^\circ) \\ &= 180^\circ - 122^\circ = 58^\circ \end{aligned}$$

Diagonals of kite bisect the angles at the vertices

$$z = 32^\circ$$

**ILLUSTRATION : 6**

PQRS is a parallelogram. Then find the value of y .

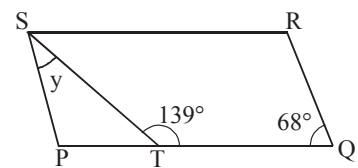
SOLUTION:

$$(a) \quad \angle PSR = \angle PQR = 68^\circ \quad (\text{opp. } \angle \text{s of a } \parallel \text{ gm are equal})$$

$$\angle PTS = 180^\circ - 139^\circ = 41^\circ \quad (\text{PTQ is a straight line})$$

$$\therefore \angle RST = \angle PTS = 41^\circ \quad (\text{SR} \parallel \text{PQ alt. } \angle \text{s are equal})$$

$$\therefore y = \angle PSR - \angle RST = 68^\circ - 41^\circ = 27^\circ$$

**SOME SPECIAL PARALLELOGRAMS****Rhombus**

A parallelogram having all sides equal is called a rhombus.

Properties of Rhombus

It has all properties of a parallelogram. In addition it has one more property as follows.

Diagonals of a rhombus bisect each other at right angles.

Let us consider a rhombus ABCD with diagonal AC and BC.

In $\triangle AOB$ and $\triangle BOC$, we have

$AB = BC$ [Sides of a rhombus]

$BO = BO$ [Common]

$AO = OC$ [Diagonals bisect each other]

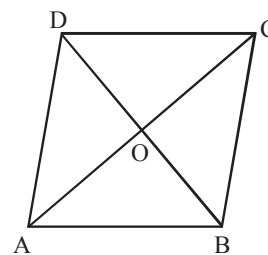
$\therefore \triangle AOB \cong \triangle BOC$ [SSS congruency]

Hence, $\angle AOB = \angle BOC$ [C.P.C.T.]

But, $\angle AOB + \angle BOC = 180^\circ$ [Linear Pair]

$\Rightarrow 2\angle AOB = 180^\circ$

$\therefore \angle AOB = \angle BOC = 90^\circ$

**Rectangle**

1. A rectangle is a quadrilateral with all right angles.

In a rectangle ABCD, $\angle A = \angle B = \angle C = \angle D = 90^\circ$

Consider, $\angle A = 90^\circ + 90^\circ = 180^\circ$

$\Rightarrow AB \parallel CD$

Similarly, $AD \parallel AB$

Since the pairs of opposite sides are parallel, ABCD is a parallelogram.

Thus,

A rectangle is a parallelogram with one of the angles as 90° .

2. The diagonals of a rectangle are equal in length.

Consider a rectangle ABCD, whose diagonals meet each other at O.

In $\triangle ABD$ and $\triangle DCA$

$DC = AB$ (opposite sides of rectangle)

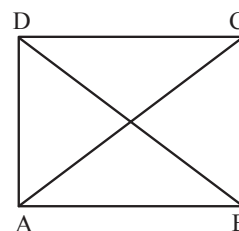
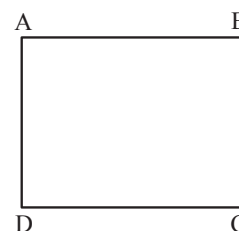
$\angle BAD = \angle CDA$ (each 90°)

$AD = DA$ (Common)

therefore $\triangle ABD \cong \triangle DCA$ by S.A.S., congruency rule.

By c.p.c.t.s. $AC = BD$

Thus, the diagonals of rectangle are equal in length.

**ILLUSTRATION : 7**

GHJK is a rectangle. $GH = HI$ and $\angle HKJ = 50^\circ$. HLK and GLI are straight lines. Find $\angle GLK$.

SOLUTION:

In $\triangle GHI$

$\angle HGI = \angle HIG$ [$\because GH = HI$] ... (i)

So by angle sum property

$\angle GHI + \angle HGI + \angle HIG = 180^\circ$

$\Rightarrow 90^\circ + \angle HGI + \angle HGI = 180^\circ$ [From (i) all the angles of rectangle are 90°]

$\Rightarrow \angle HGI = 45^\circ$

$\therefore \angle KGI = 90^\circ - \angle HGI = 90^\circ - 45^\circ$

$\angle KGI = 45^\circ$

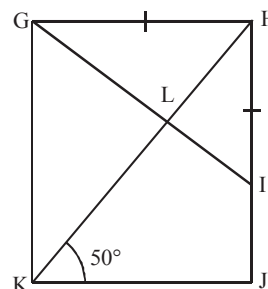
Now, $\angle GKL + \angle HKJ = 90^\circ$

$\Rightarrow \angle GKL = 90^\circ - 50^\circ = 40^\circ$

So, In $\triangle GLK$

$\angle GKL + \angle KGI + \angle GLK = 180^\circ$

$\Rightarrow \angle GLK = 180^\circ - 95^\circ = 85^\circ$

**Square**

A square is a rhombus with each of its angles as 90° .

Properties of A square

It has all properties of a rhombus. In addition, it has one more property as follows:

Diagonals of a square are equal.

Let us consider a square ABCD with diagonals AC and BD.

In $\triangle ABD$ and $\triangle BAC$, we have

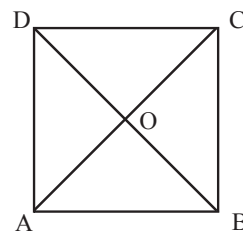
$AD = BC$ [Opposite sides of square]

$AB = AB$ [Common]

$\angle DAB = \angle ABC = 90^\circ$

$\therefore \angle ABD \cong \triangle BAC$ [S.A.S. congruency]

$AC = BD$ [C.P.C.T.]



NOTE: Square is a rhombus and rectangle at the same time.

ILLUSTRATION : 8

ABCD is a square, $BA = BQ$, QPC and BPD are straight lines and $\angle PBQ = 21^\circ$. Then, $\angle BAQ$ equals

SOLUTION:

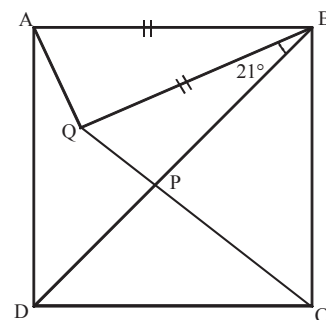
Since the diagonal of a square bisects the angle at the vertices, $\angle DBC = 45^\circ$,

$\therefore \angle ABQ = 90^\circ - (21^\circ + 45^\circ) = 90^\circ - 66^\circ = 24^\circ$

In $\triangle BAQ$, $BA = BQ$

$\Rightarrow \angle BQA = \angle BAQ$ (isos. $\triangle$ property)

$\therefore \angle BAQ = \frac{180^\circ - 24^\circ}{2} = \frac{156^\circ}{2} = 78^\circ$



Diagonal Properties of all Parallelograms

Sl. No.	Diagonal Properties	Type of Parallelogram			
		Parallelogram	Rectangle	Rhombus	Square
1	Diagonals bisect each other	✓	✓	✓	✓
2	Diagonals are equal	×	✓	×	✓
3	Diagonals are at 90° to each other	×	×	✓	✓

MISCELLANEOUS

SOLVED EXAMPLES

1. An angle is equal to one-third of its supplement. Find its measure.

Sol.

Two angles, the sum of whose measures is 180° are called supplementary angles.

Let the angle is x .

Its supplementary angle = $(180^\circ - x)$

$$\text{Now } x = \frac{180^\circ - x}{3} \text{ given}$$

$$\therefore 3x = 180^\circ - x$$

$$\Rightarrow 4x = 180^\circ$$

$$\Rightarrow x = 45^\circ$$

2. The complement of an angle exceeds the angle by 60° . Find the angle.

Sol.

Two angles, the sum of whose measures is 90° are called complementary angles.

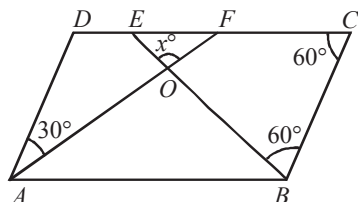
Let an angle is ' x ', its complementary angle will be $(90^\circ - x)$

$$90^\circ - x = x + 60^\circ \text{ (Given)}$$

$$\therefore 2x = 30^\circ$$

$$\Rightarrow x = 15^\circ$$

3. In the below given figure of parallelogram, find the measure of x° .



Sol.

$$\angle EOF = \angle AOB = x^\circ \text{ (Vertically opposite angles)}$$

In a parallelogram pair of opposite angles are equal.

$$\therefore \angle A = \angle C = 60^\circ \text{ and } \angle B = \angle D$$

Also in a parallelogram adjacent angles are supplementary.

$$\therefore \angle C + \angle D = 180^\circ, \angle A + \angle B = 180^\circ$$

$$\therefore \angle D = 180 - \angle C$$

$$\angle D = 180 - 60^\circ = 120^\circ$$

$$\Rightarrow \angle B = 120^\circ$$

$$\Rightarrow 60^\circ + \angle OBA = 120^\circ$$

$$\Rightarrow \angle OBA = 60^\circ$$

$$\text{Now, } \angle A = \angle C = 60^\circ$$

$$\Rightarrow 30^\circ + \angle OAB = 60^\circ$$

$$\Rightarrow \angle OAB = 30^\circ$$

$$\therefore \angle OAB = 30^\circ \text{ and } \angle OBA = 60^\circ$$

In $\triangle OAB$

$$\angle OAB + \angle OBA + \angle AOB = 180^\circ \text{ [Angle sum property]}$$

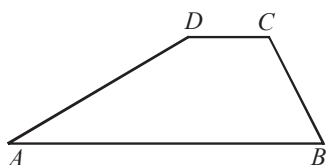
$$\therefore \angle AOB = 180^\circ - (\angle OAB + \angle OBA)$$

$$= 180 - (30 + 60)$$

$$\angle AOB = 180 - 90 = 90^\circ$$

$$\therefore \angle x = \angle AOB = 90^\circ$$

4. The measure of the angles of a quadrilateral $ABCD$ are respectively in the ratio 1 : 2 : 3 : 4. Find the type of quadrilateral $ABCD$



Sol.

Let the ratio factor be 'x'

$$\therefore \angle A = x, \angle B = 2x,$$

$$\angle C = 3x, \angle D = 4x$$

$$x + 2x + 3x + 4x = 360^\circ$$

$$10x = 360^\circ$$

$$x = 36^\circ$$

$$\therefore \angle A = 36^\circ, \angle B = 72^\circ, \angle C = 108^\circ, \angle D = 144^\circ$$

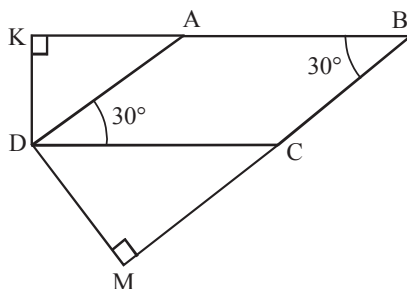
Now we see that

$$\angle A + \angle D = 180^\circ \text{ and } \angle B + \angle C = 180^\circ$$

It shows that $AB \parallel DC$

$\therefore ABCD$ is a trapezium.

5. $ABCD$ is a parallelogram. DK perpendicular to BA produced at K and DM perpendicular to BC produced at M . Find the measure of $\angle KDM$.



Sol.

$AD \parallel BC$ and BC is extended to BM

$$\therefore AD \parallel BM$$

So $\angle DCM = \angle ADC = 30^\circ$

And from $\triangle CDM$, $\angle CDM = 60^\circ$

Similarly BK is extended from BA

$$\therefore DC \parallel BK$$

$$\therefore \angle ADC = \angle DAK = 30^\circ$$

$$\therefore \text{From } \triangle ADK, \angle KDA = 60^\circ$$

$$\begin{aligned} \text{So } \angle KDM &= \angle KDA + \angle ADC + \angle CDM = 60 + 30 + 60 \\ \angle KDM &= 150^\circ \end{aligned}$$

6. Find the measure of the angle between the internal bisectors of any two adjacent angles of a parallelogram.

Sol.

Let $ABCD$ is a parallelogram and internal bisectors of adjacent angles A & B intersect at M . We are required to find $\angle AMB$.

Let $\angle AMB = x^\circ$.

$$\angle MAB = \frac{1}{2} \angle A, \quad \angle MBA = \frac{1}{2} \angle B$$

$$\angle MAB + \angle MBA = \frac{1}{2} [\angle A + \angle B]$$

$$\angle MAB + \angle MBA = \frac{1}{2} \times 180^\circ = 90^\circ \quad \dots\dots\dots (1)$$

(Adjacent angles of a parallelogram are supplementary)

Now from $\triangle AMB$

$$\angle MAB + \angle AMB + \angle MBA = 180^\circ$$

$$\angle AMB = 180^\circ - (\angle MAB + \angle MBA)$$

$$\angle AMB = 180^\circ - 90^\circ \quad \text{From (1)}$$

$$\therefore x^\circ = 90^\circ$$

$\therefore$ measure of angle between the internal bisectors of any two adjacent angles in a parallelogram is 90° .

7. $ABCD$ is a rhombus with $\angle ABC = 56^\circ$, find the value of $\angle ACD$ where AC is the diagonal of rhombus $ABCD$.

Sol.

$ABCD$ is a rhombus.

$$\therefore AB = BC$$

So, $\angle BAC = \angle BCA = x$ (let)

From $\triangle ABC$

$$x + x + 56^\circ = 180^\circ$$

$$2x + 180^\circ - 56^\circ$$

$$2x = 124$$

$$x = 62^\circ$$

Now $AD \parallel BC$ and AC is a transversal.

$$\therefore \angle BAC = \angle ACD = 62^\circ$$

$$\therefore \angle ACD = 62^\circ$$

(Alternate Interior Angles)

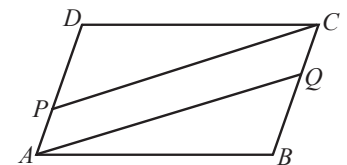
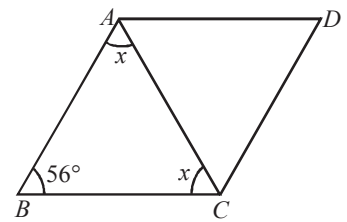
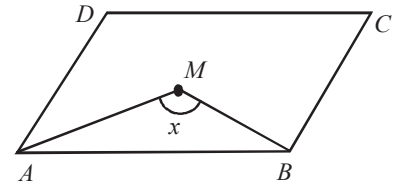
8. $ABCD$ is a parallelogram. P is a point on AD such that $AP = \frac{1}{3} AD$ and Q is a point on BC such that $CQ = \frac{1}{3} BC$. Then what kind of quadrilateral $AQCP$ is ?

Sol.

$ABCD$ is a parallelogram.

$$\therefore AD = BC \text{ and } AD \parallel BC$$

$$\therefore \frac{1}{3} AD = \frac{1}{3} BC$$



$$AP = CQ \text{ (Given)} \quad \dots (1)$$

AP lies on AD and CQ lies on BC .

$$\therefore AP \parallel CQ \quad \dots (2)$$

From (1) and (2)

$$AP = CQ \text{ and } AP \parallel CQ$$

$APCQ$ is a parallelogram. [Opposite sides are equal and parallel]

9. Four angles of a quadrilateral are in the ratio 3 : 5 : 7 : 9. Find the angles.

Sol. Suppose the measure of four angles are $3x$, $5x$, $7x$ and $9x$. By angle sum property we get

$$3x + 5x + 7x + 9x = 360^\circ$$

$$\Rightarrow (3 + 5 + 7 + 9)x = 360^\circ$$

$$\Rightarrow 24x = 360^\circ$$

$$\Rightarrow x = \frac{360^\circ}{24} = 15^\circ$$

Here, angle are:

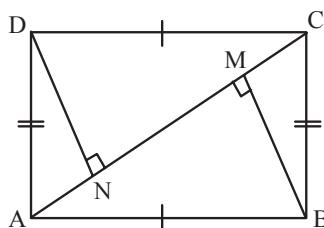
$$3x = 3 \times 15^\circ = 45^\circ$$

$$5x = 5 \times 15^\circ = 75^\circ$$

$$7x = 7 \times 15^\circ = 105^\circ$$

$$9x = 9 \times 15^\circ = 135^\circ$$

10. In a rectangle ABCD, DN and BM are perpendiculars drawn on the diagonal AC as shown in the figure. Prove that $DN = BM$.



Sol. In $\triangle AND$ and $\triangle CMB$, we have

$$AD = BC \quad (\text{opposite sides of rectangle } ABCD)$$

$$\angle DNA = \angle BMC \quad (90^\circ \text{ each})$$

$$\angle DAN = \angle BCM \quad (AD \parallel BC \text{ and } AC \text{ is transversal and these angles are alternate interior angles.})$$

Therefore, $\triangle AND \cong \triangle CMB$

$$\therefore DN = BM \quad (\text{c.p.c.t.})$$

Hence Proved.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- All regular polygons are _____.
- A rhombus is a parallelogram in which _____.
- A square is a rhombus in which _____.
- A rhombus has all its sides of _____.
- The diagonals of a rhombus bisect _____.
- If the diagonals of a parallelogram bisect each other at right angles, then it is a _____.
- If ABCD is a parallelogram, then $\angle A - \angle C = \dots\dots\dots$
- The perimeter of a parallelogram is 180 cm. One side exceeds the another by 10 cm. The adjacent sides of the parallelogram are.....
- ABCD is a parallelogram in which $\angle DAB = 75^\circ$ and $\angle DBC = 60^\circ$ then $\angle CDB = \dots\dots\dots$
- The diagonal of rectangle is thrice its smaller side. The ratio of its sides is _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- All squares are also rectangles.
- All rectangles are also squares.
- Some rectangles are also squares.
- All squares and rectangles are also parallelograms.
- If the diagonals of a rhombus are equal, its always a square.
- In a parallelohrum opposite angles are not equal.
- Every rhombus is a kite.
- The diagonals of a rectangle bisect each other at right angles.
- The diagonals of a square bisect each other at right angles.
- Opposite sides of a trapezium are parallel.

Match the Columns:

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | 1. | Column-I | Column-II |
|-----|--|--------------|
| (A) | A polygon bounded by ten line segments is called a | (p) pentagon |

- | | | |
|-----|--|-------------|
| (B) | A polygon bounded by eight line segments is called a | (q) hexagon |
| (C) | A polygon which is made by fivec line segments is called a | (r) decagon |
| (D) | A polygon bounded by six line segments is called a | (s) octagon |

2. Match the column:

Column-I

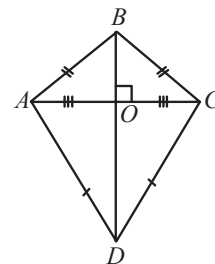
Column-II

- | | | |
|-----|--|-----------------|
| (A) | ABCDE is a regular pentagon. The bisector of $\angle A$ of the pentagon meets the side CD in M. Then $\angle AMC$ is | (p) 70° |
| (B) | The angles of a quadrilateral are respectively equal to 110° , 50° and 40° . The fourth angle is. | (q) 80° |
| (C) | In a quadrilateral ABCD, the bisector of $\angle A$ and $\angle B$ meet at point P. If $\angle C = 100^\circ$, $\angle D = 60^\circ$, the $m\angle APB$ is | (r) 90° |
| (D) | Three angles of a quadrilateral are equal: Fourth angle is of measure 150° . The measure of equal angles is | (s) 160° |

Very Short Answer Questions:

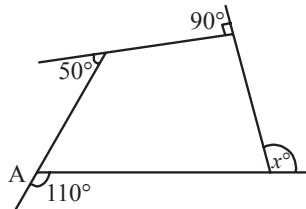
DIRECTIONS : Give answer in one word or one sentence.

- Write the angle-sum property of a quadrilateral.
- What is the name of the polygon which is bounded by three line segments.
- In which condition a polygon is called a regular polygon?
- The sides of a quadrilateral are produced in order. What is the sum of the four exterior angles?
- An isosceles trapezium has two parallel sides 60 m and 40 m. If the perimeter of this trapezium is 160 m, find the length of the other two sides.
- Which of the shapes have only one pair of opposite sides parallel?
- What is the sum of exterior angles in a convex polygon?
- Identify the picture.

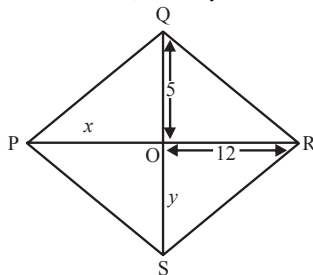


Understanding Quadrilaterals

9. In a quadrilateral PQRS; $\angle P = 70^\circ$, $\angle Q = 90^\circ$; $\angle R = 55^\circ$. Find the measure of $\angle S$. What kind of quadrilateral is it? Convex or concave.
10. Find measure x .



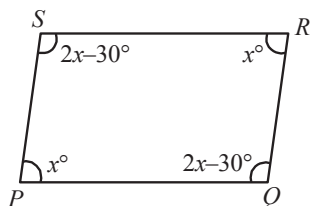
11. Let PQRS be a rhombus, find x , y .



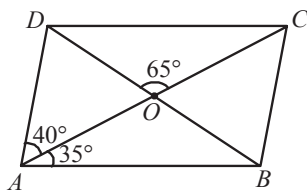
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- Find the sum of interior angles of
 - a polygon with 12 sides.
 - a polygon with 18 sides.
- Each interior angle of regular polygon is 144° . Find the interior angle of a regular polygon which has double the number of sides as the first polygon.
- Find the measure of each angle of a parallelogram, if one of its angles is 30° less than twice the smallest angle.

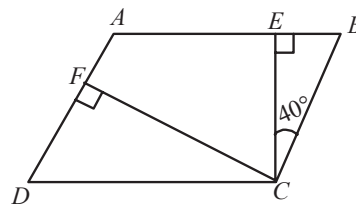


4. ABCD is a parallelogram in which $\angle DAO = 40^\circ$, $\angle BAO = 35^\circ$ and $\angle COD = 65^\circ$. Find $\angle ODC$.

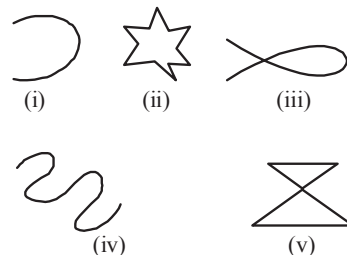


5. The angles of a quadrilateral are in the ratio 1 : 3 : 4 : 7. Find all the angles of the quadrilateral.

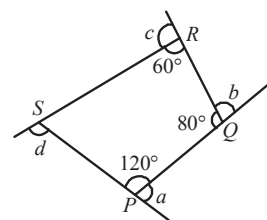
- Find the dimensions of a rectangle with perimeter 16 cm, whose long side is 3 times its short side.
- If the sum of the two angles of a quadrilateral is 180° then what is the sum of the remaining two angles?
- Find the number of sides of a polygon whose exterior and interior angles are in the ratio 1 : 5.
- Find the $\angle ADC$, $\angle ECF$ and $\angle EBC$ in the below given figure.



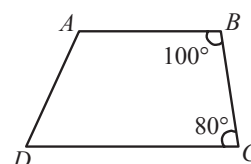
- The shorter side of a parallelogram is 4.8 cm and the longer side is half as much again as the shorter side. Find the perimeter of the parallelogram.
- The diagonals of a quadrilateral are of lengths 6 cm and 8 cm. If the diagonals bisect each other at right angles, what is the length of each side of the quadrilateral?
- Classify the following curves as open or closed.



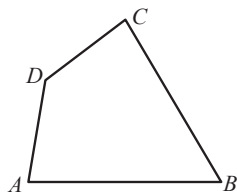
13. In the given figure, find $a + b + c + d$.



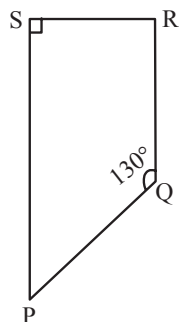
- Find the measure of an angle if seven times its complement is 10° less than three times its supplement.
- Explain how the following figure is a trapezium? Which of its two sides are parallel?



16. Study the figure and answer the following questions.



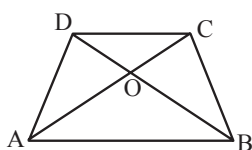
- Name a pair of adjacent sides.
 - How many pairs of opposite sides are there?
 - How many pairs of adjacent angles are there?
 - Name a pair of opposite angles?
 - How many pairs of opposite angles are there?
17. The adjacent figure PQRS is a trapezium in which $SP \parallel RQ$, find the measures of $\angle P$ and $\angle R$.



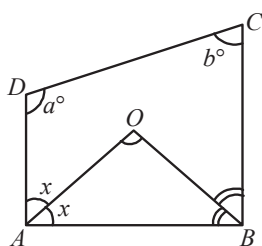
Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

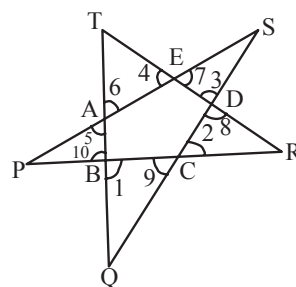
1. In Fig., $AB \parallel DC$. Find the value of x .



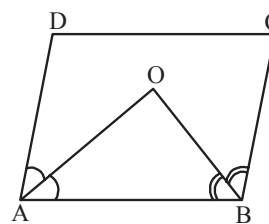
2. In the given figure, $ABCD$ is a quadrilateral and $\angle ADC = a^\circ$, $\angle BCD = b^\circ$. AO and BO are bisectors of $\angle DAB$ and $\angle ABC$ respectively meeting at O . Find $\angle AOB$ in terms of a° and b° .



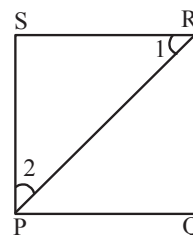
3. In a quadrilateral $ABCD$, AO and BO are the bisectors of $\angle A$ and $\angle B$ respectively. Prove that $\angle AOB = \frac{1}{2}(\angle C + \angle D)$.
4. The diagonals of a rectangle $ABCD$ meet at O . If $\angle BOC = 44^\circ$, find $\angle OAD$.
5. In Fig, determine $\angle P + \angle Q + \angle R + \angle S + \angle T$.



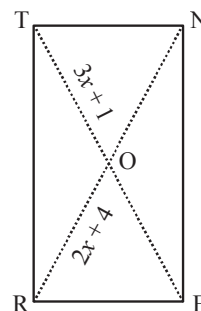
6. In a parallelogram $ABCD$, the bisectors of $\angle A$ and $\angle B$ meet at O . Find $\angle AOB$.



7. In Fig, PQRS is a square. Determine $\angle SRP$.



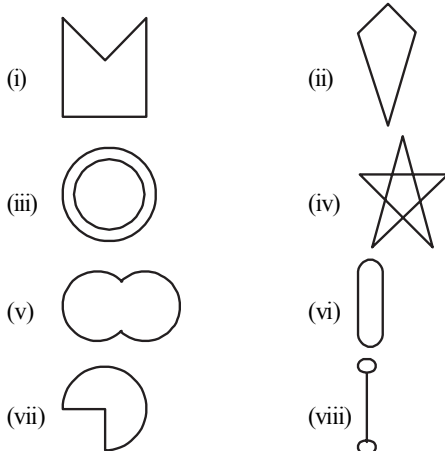
8. In figure, RENT is a rectangle. Its diagonals meet at O . Find x , if $OR = 2x + 4$ and $OT = 3x + 1$.



2 EXERCISE

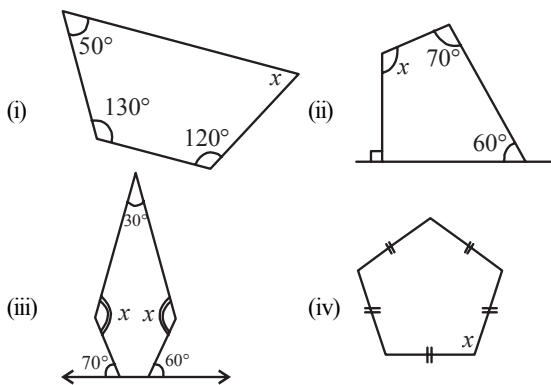
Text-Book Exercise :

- How many diagonals does each of the following have?
(i) A convex quadrilateral (ii) A regular hexagon
(iii) A triangle
- Given here are some figures.



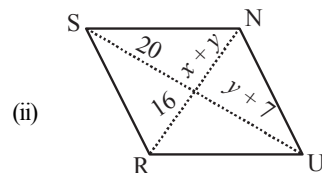
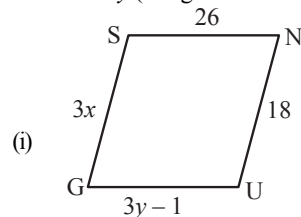
- Classify each of them on the basis of the following.
- Simple curve
 - Simple closed curve
 - Polygon
 - Convex polygon
 - Concave polygon

- Find the angle measure x in the following figures.

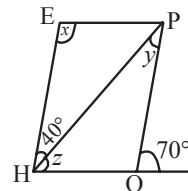


- Find the number of sides of a regular polygon whose each exterior angle has a measure of 45° .
- Find the measure of each exterior angle of a regular polygon of
(i) 9 sides (ii) 15 sides
- Can a quadrilateral ABCD be a parallelogram if
(i) $\angle D + \angle B = 180^\circ$?
(ii) $AB = DC = 8$ cm, $AD = 4$ cm and $BC = 4.4$ cm?
(iii) $\angle A = 70^\circ$ and $\angle C = 65^\circ$?

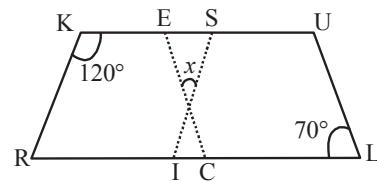
- The measure of two adjacent angles of a parallelogram are in the ratio 3 : 2. Find the measure of each of the angles of the parallelogram.
- Two adjacent angles of a parallelogram have equal measure. Find the measure of each of the angles of the parallelogram.
- The following figures GUNS and RUNS are parallelograms. Find x and y . (Length are in cm)



- The adjacent figure HOPE is a parallelogram. Find the angle measures x , y and z .

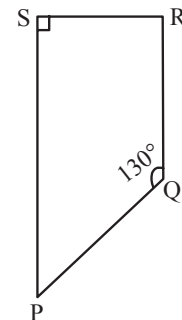


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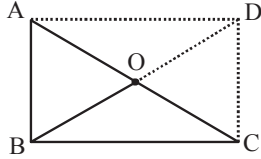


In the above figure both RISK and CLUE are parallelograms. Find the value of x .

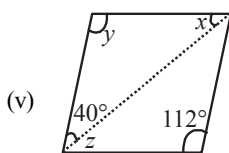
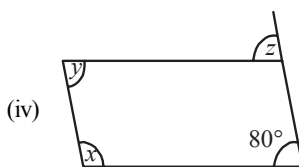
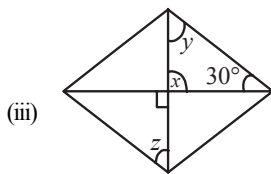
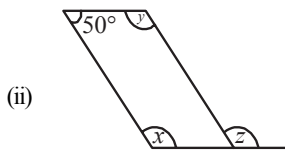
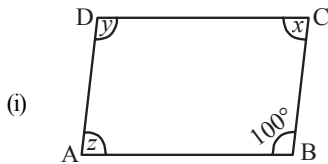
- Find the measure of $\angle P$ and $\angle S$ if $SP \parallel RQ$ in Fig. (If you find $m\angle R$, is there more than one method to find $m\angle P$?)



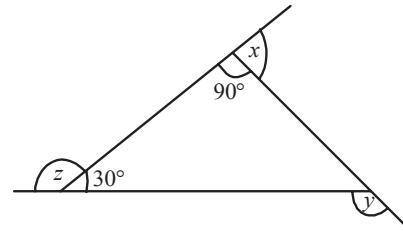
13. Name the quadrilaterals whose diagonals.
- bisect each other
 - are perpendicular bisectors of each other
 - are equal.
14. ABC is a right-angled triangle and O is the mid point of the side opposite to the right angle. Explain why O is equidistant from A, B and C. (The dotted lines are drawn additionally to help you).



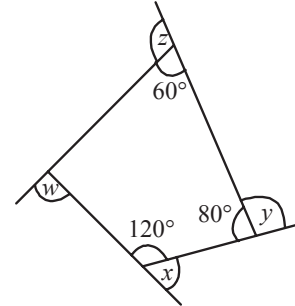
15. Consider the following parallelograms. Find the values of the unknown x , y , z .



16. How many sides does a regular polygon have if each of its interior angles is 165° ?
17. (i) What is the minimum interior angle possible for a regular polygon? Why?
- (ii) What is the maximum exterior angle possible for a regular polygon?
18. (i) Find $x + y + z$

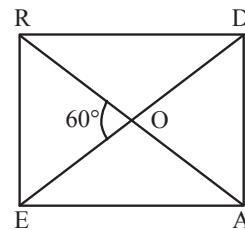


- (ii) Find $x + y + z + w$

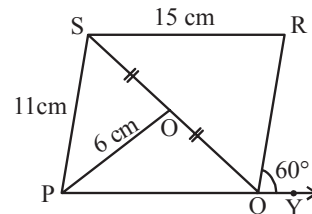


Exemplar Questions :

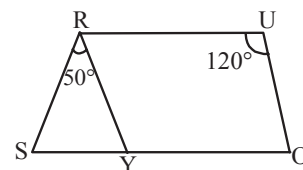
- The ratio between exterior angle and interior angle of a regular polygon is 1:5. Find the number of sides of the polygon.
- Two sticks each of length 7 cm are crossing each other such that they bisect each other at right angles. What shape is formed by joining their end points? Give reason.
- In rectangle READ, find $\angle EAR$, $\angle RAD$ and $\angle ROD$



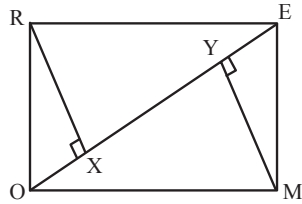
4. In parallelogram PQRS, O is the mid point of SQ. Find $\angle S$, $\angle R$, $\angle PQ$, $\angle QR$ and diagonal PR.



5. In the given parallelogram YOUR, $\angle RUO = 120^\circ$ and OY is extended to point S such that $\angle SRY = 50^\circ$. Find $\angle YSR$.

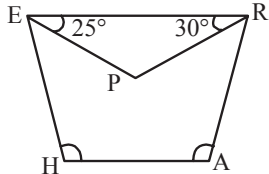


6. A rectangular MORE is shown below:

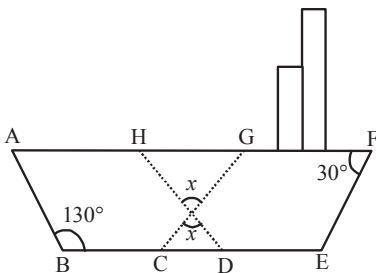


Answer the following questions by giving appropriate reason.

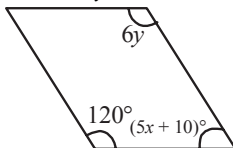
- (i) Is $RE = OM$? (ii) Is $\angle MYO = \angle RXE$?
 (iii) Is $\angle MOY = \angle REX$? (iv) Is $\triangle MYO \cong \triangle RXE$?
 (v) Is $MY = RX$?
7. In trapezium HARE, EP and RP are bisectors of $\angle E$ and $\angle R$ respectively. Find $\angle HAR$ and $\angle EHA$.



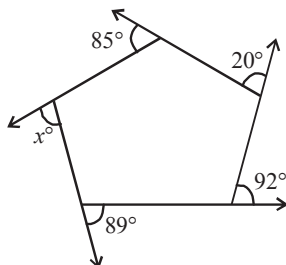
8. In the following figure of a ship, ABDH and CEFG are two parallelograms. Find the value of x .



9. Find the values of x and y in the following parallelogram.

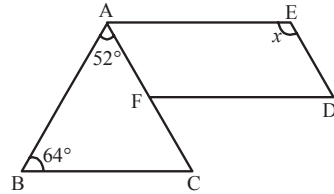


10. In the figure, find the value of x .

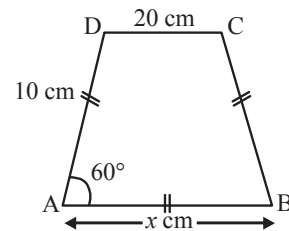


11. A line l is parallel to line m and a transversal p intersects them at X, Y respectively. Bisectors of interior angles at X and Y intersect at P and Q. Is PXQY a rectangle? Give reason.
12. ABCD is a parallelogram. Points P and Q are taken on the sides AB and AD respectively and the parallelogram PRQA is formed. If $\angle C = 45^\circ$, find $\angle R$.

13. In the following figure, $FD \parallel BC \parallel AE$ and $AC \parallel ED$. Find the value of x .



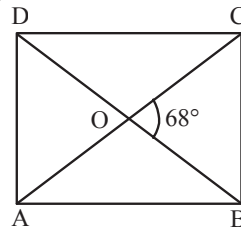
14. In the following figure, $AB \parallel DC$ and $AD = BC$. Find the value of x .



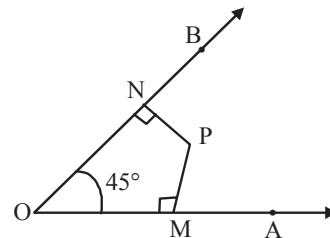
15. The angle between the two altitudes of a parallelogram through the vertex of an obtuse angle of the parallelogram is 45° . Find the angles of the parallelogram.

HOTS Questions :

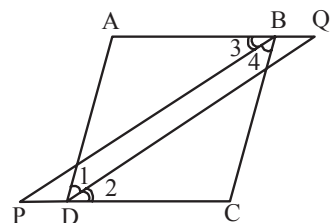
1. The diagonals of a rectangle ABCD intersect in O. If $\angle BOC = 68^\circ$, find $\angle ODA$.



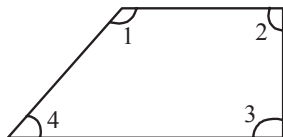
2. In Fig, find the measure of $\angle MPN$.



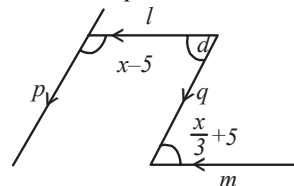
3. In Fig, bisectors of $\angle B$ and $\angle D$ of quadrilateral ABCD meet CD and AB produced at P and Q respectively. Prove that $\angle P + \angle Q = \frac{1}{2}(\angle ABC + \angle ADC)$.



4. Look at the angles in this quadrilateral. Which angle measure is closest to 48° ?



5. In the given figure lines p and q are parallel. Find value of x so that lines l and m be parallel.

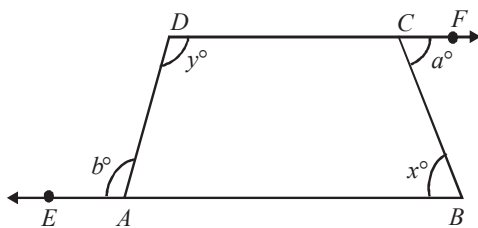


3 EXERCISE

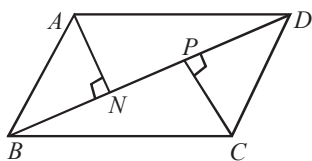
Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. The sides BA and DC of quadrilateral ABCD are produced as shown in the figure given below. Then $x + y$ is equal to

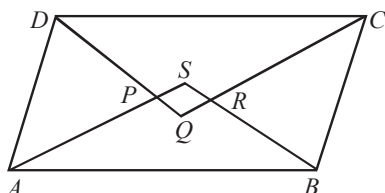


- (a) $a + b$ (b) $a - b$ (c) $\frac{a}{b}$ (d) $\frac{b}{a}$
2. In the figure given below AN and CP are perpendiculars to the diagonal BD of a parallelogram. Then



- (a) $AN = CP$ (b) $AN > CP$
(c) $AN < CP$ (d) None of these

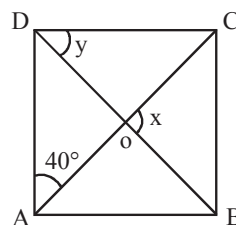
3. The bisectors of angles of a parallelogram enclose a



- (a) Rhombus (b) Rectangle
(c) Square (d) Parallelogram
4. The measures of the angles of a quadrilateral ABCD are respectively in the ratio 1 : 2 : 3 : 4. Then which of the following is true?

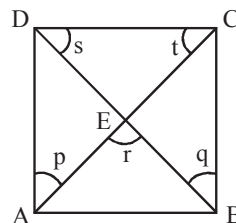
- (a) $AC = BD$ (b) $AD \parallel BC$
(c) $AB \parallel DC$ (d) None of these

5. In a quadrilateral ABCD, $\angle B = 90^\circ$ and $AD^2 = AB^2 + BC^2 + CD^2$ then $\angle ACD$ is equal to
(a) 90° (b) 60°
(c) 30° (d) None of these
6. ABCD is a rhombus whose one angle is 60° . The ratio of the lengths of its diagonals is
(a) $\sqrt{3} : 1$ (b) $\sqrt{2} : 1$
(c) 2 : 1 (d) 3 : 1
7. In the diagram, ABCD is a rhombus.



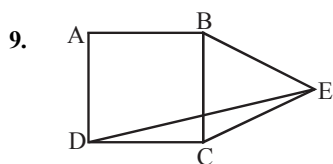
The value of $x - y$ is :

- (a) 50° (b) 40°
(c) 30° (d) 20°
8. In the diagram, ABCD is a rhombus. AEC and BED are straight lines.



$p + q + r + s = ?$

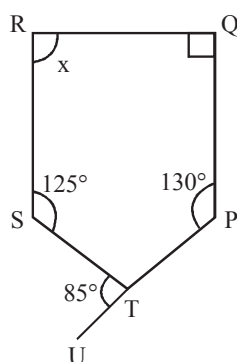
- (a) 200° (b) 270°
(c) 360° (d) 540°



In the above figure, ABCD is a square and BCE is an equilateral triangle, what is the measure of angle DEC?

- (a) 15° (b) 30°
(c) 20° (d) 45°

10. In the figure given below, PTU is a straight line.



What is the value of x ?

- (a) 100° (b) 110°
(c) 120° (d) 130°

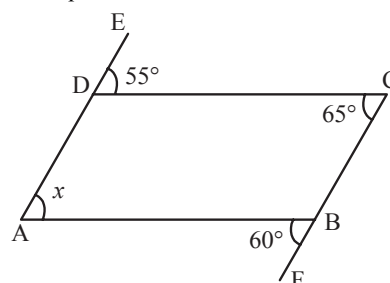
More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following properties are not true for a parallelogram?
 - Its diagonals are equal.
 - Its diagonals are perpendicular to each other.
 - The diagonals divide the figure into four congruent triangles.
 - Each diagonal divides a parallelogram into two congruent triangles.
- A parallelogram which has equal diagonals is a
 - Square
 - Rectangle
 - Rhombus
 - None

- Which of the following statement(s) is/are true?
 - A parallelogram in which two adjacent angles are equal is a rectangle.
 - A quadrilateral in which both pairs of opposite angles are equal is parallelogram.
 - In a parallelogram, the number of acute angles is zero or two.
 - None of these
- Which of the following is/are concave polygon?
 -
 -
 -
 -

- In which quadrilateral diagonals bisect each other?
 - Parallelogram
 - Rectangle
 - Square
 - Rhombus
- A mason has made a concrete slab. He needs it to be rectangular. In what way(s) can he make sure that it is rectangular?
 - By measuring each angle
 - By making opposite sides parallel
 - By measuring the lengths of the diagonals
 - All of these
- Which quadrilateral have four sides of equal length?
 - Rhombus
 - Rectangle
 - Parallelogram
 - Square
- If in the adjoining figure, ADE and CBF are straight lines, then x not equal to
 - 55°
 - 60°
 - 65°
 - 50°



- (a) 55° (b) 60°
(c) 65° (d) 50°

9. If two adjacent angles of a parallelogram are in the ratio 4 : 5, then the measure of the angles not equal to is
- (a) $80^\circ, 100^\circ$ (b) $160^\circ, 200^\circ$
 (c) $40^\circ, 50^\circ$ (d) $114^\circ, 36^\circ$

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

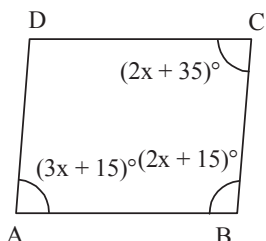
PASSAGE - I

The angles of a quadrilateral are in the ratio 3 : 5 : 7 : 9.

1. If measure of angles be $(3x)^\circ$, $(5x)^\circ$, $(7x)^\circ$ and $(9x)^\circ$, then the value of x is
- (a) 20 (b) 15
 (c) 25 (d) 10
2. The measure of all angles are
- (a) $45^\circ, 70^\circ, 105^\circ, 140^\circ$ (b) $40^\circ, 80^\circ, 105^\circ, 135^\circ$
 (c) $45^\circ, 75^\circ, 110^\circ, 130^\circ$ (d) $45^\circ, 75^\circ, 105^\circ, 135^\circ$
3. The sum of the least and greatest angle is
- (a) 175° (b) 180°
 (c) 170° (d) 185°

PASSAGE - II

Consider a parallelogram ABCD in the figure given below



4. The value of x is
- (a) 50 (b) 30
 (c) 20 (d) 40
5. The value of y is
- (a) 45° (b) 30
 (c) 20 (d) 40
6. The measure of $\angle A$ and $\angle B$ is
- (a) $115^\circ, 65^\circ$ (b) $100^\circ, 180^\circ$
 (c) $95^\circ, 85^\circ$ (d) $75^\circ, 105^\circ$

PASSAGE - III

A quadrilateral having one pair of opposite sides parallel and the other pair of opposite sides nonparallel is called a trapezium. If the two non-parallel sides of a trapezium are equal in length, then it is called an isosceles trapezium.

Based on the above passage answer the following questions:

7. The parallel sides of the trapezium are known as
- (a) bases (b) perpendiculars
 (c) hypotenuse (d) none

8. In the trapezium ABCD with $AB \parallel CD$, we have

- (a) $\angle A + \angle D = 90^\circ$
 (b) $\angle A + \angle D = 180^\circ$
 (c) $\angle A + \angle D = 360^\circ$
 (d) $\angle A + \angle D = 45^\circ$

9. In the isosceles trapezium PQRS with $PQ \parallel RS$ and $QR = PS$, which of the following is correct ?

- (a) $\angle P + \angle S = 180^\circ$ and $\angle Q + \angle R = 180^\circ$
 (b) $PS = QR$ and $PR = QS$
 (c) $\angle P = \angle Q$ and $\angle R = \angle S$
 (d) All of the above

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** A polygon bounded by four line segments is called a quadrilateral.

Reason : A polygon bounded by seven line segments is called a hexagon.

2. **Assertion :** A polygon having 16 sides is called 16 sided polygon or 16-gon.

Reason : A polygon bounded by n -line segments is called n -sides polygon or n -gon.

3. **Assertion :** In a regular polygon,

- (i) all sides are equal
 (ii) all interior angles are equal
 (iii) all exterior angles are equal.

Reason : A polygon is called regular polygon if all its sides as well as angles are equal.

4. **Assertion :** In a parallelogram if one angle is a right angle, then it is called a rectangle.

Reason : In a rectangle if all the sides are equal then it is called a square.

5. **Assertion :** The adjacent angles in a parallelogram are supplementary.

Reason : In a parallelogram, the adjacent angles are always equal.

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s, and t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1.	Column-I	Column-II
	(A) Parallelogram	(p) opposite sides are equal.
	(B) Rhombus	(q) Diagonals are equal.
	(C) Rectangle	(r) Diagonals bisect each other.
	(D) Square	(s) opposite angles are equal.
		(t) All sides are equal.

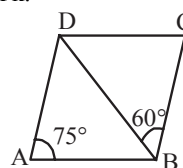
Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

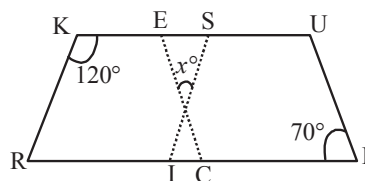
1. The ratio of two sides of a parallelogram is 4 : 3 and its perimeter is 56 m. If the sides of parallelogram be a and b .

Then $\frac{a+b}{7}$ is

2. In Fig, ABCD is a parallelogram in which $\angle DAB = 75^\circ$, $\angle DBC = 60^\circ$ then the value of $\angle CDB$ and $\angle ADB$ is $12k^\circ$. Find the value of k .



3. In the following figure RISK and CLUE are parallelograms. Then the measure of x is $10k^\circ$. Find the value of k .



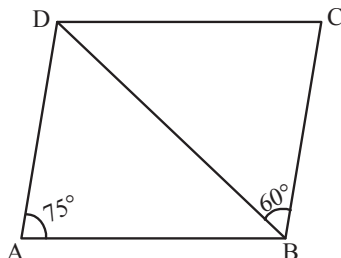
SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- convex
- adjacent sides are equal
- one angle is right angle
- equal length
- each other at right angles.
- rhombus
- In a parallelogram, opposite angles are equal.
So $\angle A = \angle C \Rightarrow \angle A - \angle C = 0^\circ$.
- Two sides are $x, x + 10$
Perimeter = 180 cm
i.e., $x + x + 10 + x + x + 10 = 180$
 $4x + 20 = 180$
 $x = 40$ cm.
The sides are 40 cm, 50 cm.



We have,

$$\angle ADB = \angle CBD = 60^\circ \quad (\because AD \parallel BC)$$

In $\triangle ADB$

$$\angle A + \angle D + \angle B = 180^\circ$$

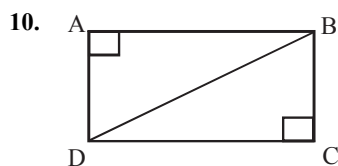
$$75^\circ + 60^\circ + \angle B = 180^\circ$$

$$\angle B = 45^\circ$$

$$\therefore \angle ABD = 45^\circ$$

$$\therefore \angle CDB = \angle ABD = 45^\circ$$

(Alternate interior angles)



Let the smaller side of rectangle be x .

Let the length be l .

$$\therefore \text{Diagonal} = 3x$$

Now, since the sides of rectangle are perpendicular,

$$(3x)^2 = l^2 + x^2$$

(By pythagoras theorem)

or

$$l^2 = 9x^2 - x^2 = 8x^2$$

$$l = 2\sqrt{2}x,$$

$$\therefore \text{ratio of sides} = 2\sqrt{2}x / x = 2\sqrt{2} : 1$$

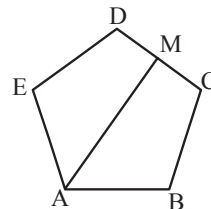
True / False :

- True
- False
- True
- True
- True
- False: Opposite angles are equal in a parallelogram.
- False: In case of rhombus both the diagonals bisect each other at right angles but in a kite only one of the diagonals bisects the other. So, every rhombus is not a kite.
- False
- True
- False

Match the Columns :

- (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q)
 - (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p)
- (A) We know that the measure of each interior angle of a regular pentagon is 108° .

$$\therefore \angle BAM = \frac{1}{2} (108^\circ) = 54^\circ$$



Since the sum of the angles of a quadrilateral is 360° .
Therefore, in quadrilateral ABCM, we have

$$\therefore \angle BAM + \angle ABC + \angle BCM + \angle AMC = 360^\circ$$

$$\Rightarrow 54^\circ + 108^\circ + 108^\circ + \angle AMC = 360^\circ$$

$$\Rightarrow \angle AMC = 90^\circ$$

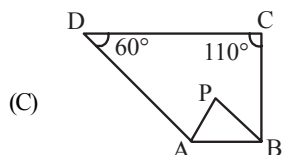
(B) Let the fourth angle be x .

$$x + 110^\circ + 50^\circ + 40^\circ = 360^\circ$$

[Sum of angles in a quadrilateral is 360°]

$$\Rightarrow x + 200^\circ = 360^\circ \Rightarrow x = 160^\circ$$

So, fourth angle = 160°



Since, sum of angles of a quadrilateral.

$$\text{So, } \angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$\Rightarrow \angle A + \angle B + 100^\circ + 60^\circ = 360^\circ$$

$$\Rightarrow \angle A + \angle B = 200^\circ$$

In $\triangle APB$, we have

$$\frac{1}{2} \angle A + \frac{1}{2} \angle B + \angle APB = 180^\circ$$

$$\Rightarrow \angle APB = 180^\circ - \frac{1}{2} (\angle A + \angle B)$$

$$\Rightarrow \angle APB = 180^\circ - \left(\frac{1}{2} \times 200 \right) = 80^\circ$$

(D) Let ABCD be a quadrilateral such that $\angle A = 50^\circ$. Let $\angle B = \angle C = \angle D = x^\circ$.

Now, by angle sum property of a quadrilateral

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$\Rightarrow 150^\circ + x + x + x = 360^\circ$$

$$\Rightarrow 3x + 150^\circ = 360^\circ$$

$$\Rightarrow 3x = 360^\circ - 150^\circ \Rightarrow x = 70^\circ$$

Hence, the measure of each of the remaining three equal angles is 70° .

Very Short Answer Questions :

- The sum of the four angles of any quadrilateral is always 360° or 4 right angles.
- Triangle
- A polygon is called regular polygon if all its sides as well as angles are equal.
- 360°
- 30 m each
- Trapezium
- 360°
- The given picture is a Kite.
- Since $\angle P + \angle Q + \angle R + \angle S = 360^\circ$
 $\Rightarrow 70^\circ + 90^\circ + 55^\circ + \angle S = 360^\circ$
 $\Rightarrow 215^\circ + \angle S = 360^\circ \Rightarrow \angle S = 360^\circ - 215^\circ = 145^\circ$
 $\therefore$ ABCD is convex quadrilateral.

$$10. x + 90^\circ + 50^\circ + 110^\circ = 360^\circ$$

$$\Rightarrow x + 250^\circ = 360^\circ \Rightarrow x = 110^\circ$$

11. Here, $x = 12, y = 5$ [Diagonals bisect each other]

Short Answer Questions :

- (i) Sum of interior angles of a polygon with 12 sides
 $= (2 \times 12 - 4) \text{ right angles}$
 $= 20 \text{ right angles}$
 $= 20 \times 90^\circ = 1800^\circ$
 (ii) $(2 \times 18 - 4) \times 90^\circ = 32 \times 90 = 2880$
- 162° .
- $70^\circ, 110^\circ, 70^\circ, 110^\circ$.
- Given, $\angle AOB = \angle COD = 65^\circ$
 In $\triangle AOB$, $\angle A + \angle O + \angle B = 180^\circ$
 $\Rightarrow \angle ABO = 80^\circ$
 $\angle ODC = \angle ABO = 80^\circ$ (Alternate interior angles)
- Required angles are $24^\circ, 72^\circ, 96^\circ$ and 168° .
 2 cm by 6 cm.
- 180° .
- Let no. of sides of polygon = n
 Exterior angle = x°
 Interior angle = $5x^\circ$
 $x = 30^\circ$ ($\because x + 5x = 180$)
 Sum of all exterior angle = $(30n)^\circ$
 Hence, $n = 12$
- $\angle EBC = 50^\circ, \angle ADC = 50^\circ, \angle FCE = 50^\circ$
- 24 cm.
- 5 cm.
- Open curve:
 (i), (iii), (iv)
 Close curve
 (ii), (v)
- Since, sum of measure of interior angles of a quadrilateral is 360° .
 $\therefore 120 + 80 + 60 + \angle RSP = 360^\circ$
 $\Rightarrow \angle RSP = 100$
 $a = \text{ext } \angle SPQ = 60^\circ$
 Similarly,
 $b = 100, c = 120, d = 80$
- Two angles are complementary when their sum is 90° and are supplementary when their sum is 180° .
 Use the condition
 $7(90^\circ - x) = 3(180^\circ - x) - 10$
 Required angle = 25°
- We have
 $\angle DCB + \angle ABC = 80 + 100 = 180$
 $\Rightarrow$ Both the angles are supplementary.
 $\Rightarrow DC \parallel AB$.
 Thus, ABCD is a trapezium in which $DC \parallel AB$.

16. (i)
- AB, BC

(ii) 2

(iii) 4

(iv) $\angle A, \angle C$ or $\angle B, \angle D$

(v) 2

17. Since PQRS is a trapezium in which
- $SP \parallel RQ$
- .

$$\therefore \angle P + \angle Q = 180^\circ \text{ and } \angle S + \angle R = 180^\circ$$

$$\Rightarrow \angle P + 130^\circ = 180^\circ \text{ and } 90^\circ \angle R = 180^\circ$$

$$\Rightarrow \angle P = 180^\circ - 130^\circ = 50^\circ \text{ and } \angle R = 180^\circ - 90^\circ = 90^\circ$$

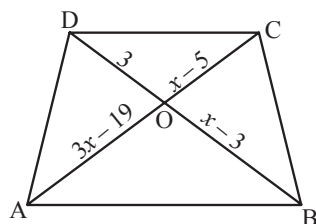
Long Answer Questions :

1. Since the diagonals of a trapezium divide each other proportionally.

$$\therefore \frac{AO}{OC} = \frac{BO}{OD} \quad \dots (iii)$$

$$\Rightarrow \frac{3x-19}{x-5} = \frac{x-3}{3}$$

$$\Rightarrow 3(3x-19) = (x-5)(x-3)$$



$$\Rightarrow 9x - 57 = x^2 - 8x + 15$$

$$\Rightarrow x^2 - 17x + 72 = 0$$

$$\Rightarrow (x-8)(x-9) = 0$$

$$\Rightarrow x-8=0, \text{ or } x-9=0 \Rightarrow x=8 \text{ or } x=9$$

2. Since ABCD is a quadrilateral therefore

$$\angle BAD + \angle ABC + \angle C + \angle D = 360^\circ$$

$$\text{Let } \angle OAB = x, \angle OBA = y$$

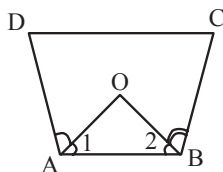
$$\therefore a + b + 2x + 2y = 360^\circ$$

$$\Rightarrow x + y = \frac{360 - (a + b)}{2}$$

Also, let $\angle AOB = Z$ and use sum of angles of triangle is 180° .

$$\text{Hence, } \angle AOB = \frac{a^\circ + b^\circ}{2}.$$

3. In
- $\triangle AOB$
- , we have



$$\angle AOB + \angle 1 + \angle 2 = 180^\circ$$

$$\Rightarrow \angle AOB = 180^\circ - (\angle 1 + \angle 2)$$

$$\Rightarrow \angle AOB = 180^\circ - \left(\frac{1}{2} \angle A + \frac{1}{2} \angle B \right)$$

$$\left[\angle 1 = \frac{1}{2} \angle A \text{ and } \angle 2 = \frac{1}{2} \angle B \right]$$

$$\Rightarrow \angle AOB = 180^\circ - \frac{1}{2} (\angle A + \angle B)$$

$$\Rightarrow \angle AOB = 180^\circ - \frac{1}{2} [360^\circ - (\angle C + \angle D)]$$

$$[\therefore \angle A + \angle B + \angle C + \angle D = 360^\circ]$$

$$\Rightarrow \angle AOB = 180^\circ - 180^\circ + \frac{1}{2} (\angle C + \angle D)$$

$$\Rightarrow \angle AOB = \frac{1}{2} (\angle C + \angle D)$$

4. We have,

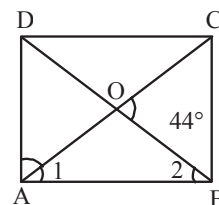
$$\angle BOC + \angle BOA = 180^\circ \quad [\text{Linear pair}]$$

$$\Rightarrow 44^\circ + \angle BOA = 180^\circ \Rightarrow \angle BOA = 136^\circ$$

Since diagonals of a rectangle are equal and they bisect each other. Therefore, in $\triangle OAB$, we have $OA = OB \Rightarrow \angle 1 = \angle 2$

$$[\therefore \text{Angles opposite to equal sides are equal}]$$

In $\triangle AOB$, we have



$$\angle 1 + \angle 2 + \angle BOA = 180^\circ$$

$$\Rightarrow 2\angle 1 + 136^\circ = 180^\circ \Rightarrow 2\angle 1 = 44^\circ$$

$$\Rightarrow \angle 1 = 22^\circ$$

Since each angle of a rectangle is a right angle.

$$\therefore \angle BAD = 90^\circ$$

$$\Rightarrow \angle 1 + \angle 3 = 90^\circ \Rightarrow 22^\circ + \angle 3 = 90^\circ$$

$$\Rightarrow \angle 3 = 68^\circ$$

$$\text{Hence, } \angle OAD = 68^\circ.$$

5. We know that the sum of the measures of the exterior angles of any polygon is equal
- 360°

$$\therefore \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 = 360^\circ$$

$$\text{and, } \angle 6 + \angle 7 + \angle 8 + \angle 9 + \angle 10 = 360^\circ$$

$$\Rightarrow \angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 + \angle 9 + \angle 10 = 720^\circ \dots (i)$$

The sum of the angles of a triangles is 180°

$$\therefore (\angle P + \angle 5 + \angle 10) + (\angle Q + \angle 1 + \angle 9) + \angle R + \angle 2 + \angle 8 + (\angle S + \angle 3 + \angle 7) + (\angle T + \angle 4 + \angle 6) = 5 \times 180^\circ = 900^\circ$$

$$\Rightarrow (\angle P + \angle Q + \angle R + \angle S + \angle T) + (\angle 1 + \angle 2 + \angle 3 + \angle 4 + \angle 5 + \angle 6 + \angle 7 + \angle 8 + \angle 9 + \angle 10) = 900^\circ$$

$$\Rightarrow (\angle P + \angle Q + \angle R + \angle S + \angle T) + 720^\circ = 900^\circ \text{ [Using (i)]}$$

$$\Rightarrow (\angle P + \angle Q + \angle R + \angle S + \angle T) = 180^\circ.$$

6. Since OA and OB are the bisectors of $\angle A$ and $\angle B$ respectively.

$$\therefore \angle OAB = \frac{1}{2} \angle A \text{ and } \angle OBA = \frac{1}{2} \angle B$$

In $\triangle AOB$, we have

$$\angle OAB + \angle AOB + \angle OBA = 180^\circ$$

$$\Rightarrow \frac{1}{2} \angle A + \angle AOB + \frac{1}{2} \angle B = 180^\circ$$

$$\Rightarrow \angle AOB = 180^\circ - \frac{1}{2} (\angle A + \angle B)$$

$$\Rightarrow \angle AOB = 180^\circ - \frac{1}{2} (180^\circ)$$

$$\Rightarrow \angle AOB = 180^\circ - 90^\circ = 90^\circ$$

[$\therefore \angle A$ and $\angle B$ are adjacent angles of parallelogram ABCD $\therefore \angle A + \angle B = 180^\circ$]

7. PQRS is a square.

$$\therefore PS = SR \text{ and } \angle PSR = 90^\circ$$

In $\triangle PSR$, we have

$$PS = SR$$

$$\Rightarrow \angle 1 = \angle 2 \quad [\therefore \text{Angles opp. to equal sides are equal}]$$

$$\text{But, } \angle 1 + \angle 2 + \angle PSR = 180^\circ$$

$$\therefore 2\angle 1 + 90^\circ = 180^\circ \quad [\therefore \angle PSR = 90^\circ]$$

$$\Rightarrow 2\angle 1 = 90^\circ$$

$$\Rightarrow \angle 1 = 45^\circ$$

8. Since diagonals of a rectangle bisect each other.

$$OR = \frac{1}{2} RN \text{ and } OT = \frac{1}{2} TE$$

Also, diagonals of a rectangle are equal.

$$\therefore RN = TE$$

$$\Rightarrow \frac{1}{2} RN = \frac{1}{2} TE$$

$$\Rightarrow OR = OT$$

$$\Rightarrow 2x + 4 = 3x + 1$$

$$\Rightarrow 3x - 2x = 4 - 1$$

$$\Rightarrow x = 3.$$

2 EXERCISE

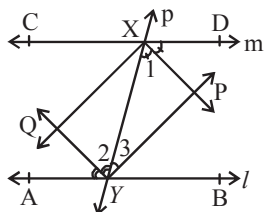
Text-Book Exercise :

- 2
 - 9
 - 0
- (i), (ii), (v), (vi), (vii)
 - (i), (ii), (v), (vi), (vii)
 - (i), (ii)
 - (ii)
 - (i)
- 60°
 - 140°
 - 140°
 - 108°
- Total measure of all exterior angles = 360°
Measure of each exterior angle = 45°

Therefore, the number of exterior angles = $\frac{360}{45} = 8$
The polygon has 8 sides.
- $\frac{360^\circ}{9} = 40^\circ$
 - $\frac{360^\circ}{15} = 24^\circ$
- Can be, but need not be.
 - No; (in a parallelogram, opposite sides are equal; but here, $AD \neq BC$).
 - No; (in a parallelogram, opposite angles are equal; but here, $\angle A \neq \angle C$).
- 108°; 72°;
- Each is a right angle.
- $x = 6; y = 9$
 - $x = 3; y = 13$;
- $x = 110^\circ; y = 40^\circ; z = 30^\circ$
- $x = 50^\circ$
- $\angle P = 50^\circ; \angle S = 90^\circ$
- Parallelogram; rhombus; square; rectangle.
 - Rhombus; square
 - Square; rectangle
- $\overline{AD} \parallel \overline{BC}; \overline{AB} \parallel \overline{DC}$. So, in parallelogram ABCD, the mid-point of diagonal $\overline{AC}$ is O.
- $x = 80^\circ; y = 100^\circ; z = 80^\circ$
 - $x = 130^\circ; y = 130^\circ; z = 130^\circ$
 - $x = 90^\circ; y = 60^\circ; z = 60^\circ$
 - $x = 100^\circ; y = 80^\circ; z = 80^\circ$
 - $y = 112^\circ; x = 28^\circ; z = 28^\circ$
- Number of sides = 24
- The equilateral triangle being a regular polygon of 3 sides has the least measure of an interior angle = 60°.
 - By (a), we can see that the greatest exterior angle is 120°.
- $x + y + z = 360^\circ$
 - $x + y + z + w = 360^\circ$

Exemplar Questions :

- 12
- Rhombus
- $30^\circ, 60^\circ, 120^\circ$
- $120^\circ, 60^\circ, 15 \text{ cm}, 11 \text{ cm}, 12 \text{ cm}, 52 \text{ cm}$
- 70°
- Yes, opposite sides of a rectangle are equal.
 - Yes, MY and RX are perpendicular to OE.
 - Yes, these are alternate interior angles.
 - Yes, $MYO \cong \Delta RXE$
- 120°
- 100°
- $x = 10^\circ, y = 20^\circ$
- 74°
- $l \parallel m$



$$\angle DXY = \angle XYA \quad [\text{alt int. } \angle S]$$

$$\frac{\angle DXY}{2} = \frac{\angle XYA}{2} \quad (\div 2)$$

$\angle 1 = \angle 2$ (XP and YQ are bisectors)

$$\therefore XP \parallel QY \quad (1)$$

$$\text{Similarly } XQ \parallel PY \quad (2)$$

From (1) and (2)

PXQY is a parallelogram

$$\angle DXY + \angle XYB = 180^\circ$$

$$\frac{\angle DXY}{2} + \frac{\angle XYB}{2} = \frac{180^\circ}{2} \quad (\text{Divide by 2})$$

$$\angle 1 + \angle 3 = 90^\circ$$

In ΔXYP

$$\angle 1 + \angle 3 + \angle P = 180^\circ$$

$$90^\circ + \angle P = 180^\circ$$

$$\angle P = 90^\circ \quad \dots (3)$$

From (1), (2) and (3), PXQY is a rectangle.

- 45°
- 116°
- 30 cm
- $135^\circ, 45^\circ, 135^\circ, 45^\circ$

HOTS Questions :

- We have.

$$\angle BOC = 68^\circ$$

$$\Rightarrow \angle AOD = 68^\circ \quad [\text{Vertically opposite angles}]$$

Since the diagonals of a rectangle are equal and they bisect each other.

$$\therefore OA = OD$$

Thus, in ΔAOD , we have

$$OA = OD \quad [\text{Prove above}]$$

$$\Rightarrow \angle ODA = \angle OAD \quad [\angle s \text{ opposite to equal sides}]$$

$$\text{But, } \angle ODA + \angle OAD + \angle AOD = 180^\circ$$

[By angle sum property in ΔAOD]

$$\Rightarrow 2\angle ODA + 68^\circ = 180^\circ$$

$$\Rightarrow 2\angle ODA = 180^\circ - 68^\circ$$

$$\Rightarrow 2\angle ODA = 112^\circ$$

$$\Rightarrow \angle ODA = \frac{112^\circ}{2} = 56^\circ$$

- 135°

In ΔPBC , we have

$$\therefore \angle P + \angle 4 + \angle C = 180^\circ$$

$$\Rightarrow \angle P + \frac{1}{2}\angle B + \angle C = 180^\circ \quad \dots (i)$$

In ΔQAD , we have

$$\angle Q + \angle A + \angle 1 = 180^\circ$$

$$\Rightarrow \angle Q + \angle A + \frac{1}{2}\angle D = 180^\circ \quad \dots (ii)$$

Adding (i) and (ii), we get

$$\angle P + \angle Q + \angle A + \angle C + \frac{1}{2}\angle B + \frac{1}{2}\angle D = 180^\circ + 180^\circ$$

$$\Rightarrow \angle P + \angle Q + \angle A + \angle C + \frac{1}{2}\angle D + \frac{1}{2}\angle D = 360^\circ$$

But, in quadrilateral ABCD, we have

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

From (i) and (ii), we have

$$\angle P + \angle Q + \angle A + \angle C + \frac{1}{2}(\angle B + \angle D) = \angle A + \angle B$$

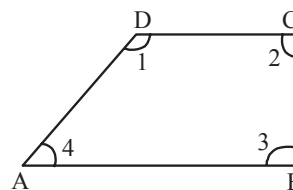
$$+ \angle C + \angle D$$

$$\Rightarrow \angle P + \angle Q = \frac{1}{2}(\angle B + \angle D)$$

$$\Rightarrow \angle P + \angle Q = \frac{1}{2}(\angle ABC + \angle ADC)$$

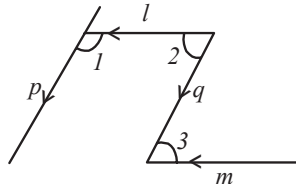
- In trapezium, $\angle 2 = \angle 3 = 90^\circ$

$\angle 4$ is acute angle, $\angle 1$ is obtuse angle



Hence $\angle 4 = 48^\circ$ is the acute angle.

5. $\angle 1 + \angle 2 = 180^\circ$ (Since $p \parallel q$)
 $x - 5 + \angle 2 = 180^\circ$... (1)
 $\angle 2 = \angle 3$ (Since $l \parallel m$)



$$\angle 2 = \frac{x}{3} + 5 \quad \dots (2)$$

put $\angle 2$ from (2) in (1), we get

$$x - 5 + \frac{x}{3} + 5 = 180^\circ$$

$$\frac{4x}{3} = 180^\circ \Rightarrow 4x = 180^\circ \times 3$$

$$4x = 540^\circ \Rightarrow x = 540^\circ \div 4 \Rightarrow x = 135^\circ$$

3 EXERCISE

Single Option Correct :

1. (a) In a quadrilateral sum of all internal angles = 360° .

$$\angle DAB = 180 - \angle DAE$$

$$\angle DAB = 180 - b$$

$$\angle BCD = 180 - \angle BCF$$

$$\angle BCD = 180 - a$$

$\therefore$ In quadrilateral $ABCD$

$$\angle DAB + \angle ABC + \angle BCD + \angle CDA = 360^\circ$$

$$180 - b + x + 180 - a + y = 360^\circ$$

$$\Rightarrow x + y = a + b$$

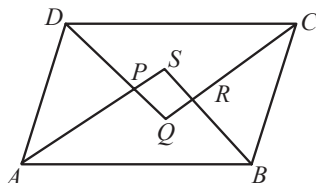
2. (a)

3. (b) $ABCD$ is a parallelogram.

$\therefore AB \parallel DC$ and AD cuts them

$$\therefore \angle A + \angle D = 180^\circ$$

$$\Rightarrow \frac{1}{2} \angle A + \frac{1}{2} \angle D = 90^\circ$$



$\therefore$ In $\triangle ADP$, $\angle DAP + \angle ADP = 90^\circ$

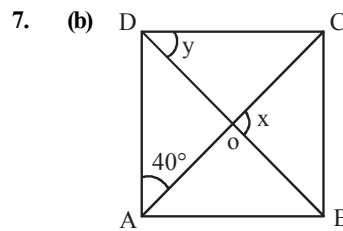
$$\therefore \angle DPA = 90^\circ$$

So $\angle SPQ = 90^\circ$ (Vertically opp. angles)

Similarly $\angle PQR = 90^\circ$, $\angle QRS = 90^\circ$ and $\angle RSP = 90^\circ$

$\therefore PQRS$ is a quadrilateral each of whose angles is 90° therefore $PQRS$ is a rectangle.

4. (c) 5. (a) 6. (a)



$$x = 90^\circ$$

($\because$ diagonals of rhombus bisect each other at right angles.)

Now, in $\triangle ADC$, $AD = DC$

(sides of rhombus)

$$\therefore \angle DAC = \angle ACD = 40^\circ$$

($\because$ equal sides have equal angles opposite to them.)

Now in $\triangle OCD$,

$$y = 180^\circ - (90^\circ + 40^\circ)$$

$$= 180^\circ - 130^\circ = 50^\circ$$

$$\therefore x - y = 90^\circ - 50^\circ = 40^\circ$$

8. (b) In rhombus $ABCD$,

$$\angle A + \angle B + \angle C + \angle D = 360^\circ$$

$$\Rightarrow \frac{\angle A}{2} + \frac{\angle B}{2} + \frac{\angle C}{2} + \frac{\angle D}{2} = \frac{360^\circ}{2}$$

$$p + q + s + t = 180 \quad \dots (1)$$

($\because$ diagonals of rhombus are angle bisectors)

$$\text{Now } \angle r = 90^\circ \quad \dots (2)$$

($\because$ diagonals of rhombus bisect the diagonals at right angles.)

Adding (1) and (2), we get

$$p + q + r + s + t = 180^\circ + 90^\circ$$

$$\therefore p + q + r + s + t = 270^\circ$$

9. (a) $\angle BCD = 90^\circ$ [angle of a square]

$$\angle BCE = 60^\circ$$
 [angle of an equilateral triangle]

$$\therefore \angle DCE = 90^\circ + 60^\circ = 150^\circ$$

$\triangle DCE$ is an isosceles triangle with $DC = EC$

$\therefore \angle CDE = \angle CED$ (angles opposite of equal sides)

$$\Rightarrow \angle EDC + \angle DEC + \angle DCE = 180^\circ$$

$$2\angle DEC = 180^\circ - \angle DCE$$

$$= 180^\circ - 150^\circ$$

$$\angle DEC = \frac{1}{2}(30^\circ) = 15^\circ$$

10. (a) $\angle STP = 180^\circ - 85^\circ = 95^\circ$

We have, sum of all interior angles of "n" sided polygon

$$= (n-2) \times 180^\circ = (5-2) \times 180^\circ = 3 \times 180^\circ = 540^\circ$$

$$\Rightarrow x + 90^\circ + 125^\circ + 130^\circ + 95^\circ = 540^\circ$$

$$\Rightarrow x + 440^\circ = 540^\circ$$

$$\Rightarrow x = 540^\circ - 440^\circ = 100^\circ$$

More Than One Option Correct :

1. (a, b, c) 2. (a, b) 3. (a, b, c)
4. (a, c) If one or more of the interior angles of a polygon is greater than 180° , it is called concave polygon.
5. (a, b, c, d) 6. (a, b, c, d)
7. (a, d) Rhombus and square are quadrilaterals that have four sides of equal lengths.
8. (d) 9. (b, d, c)

Passage Based Questions :

1. (b) Since the sum of all angles of a quadrilateral is 360° .
So, $3x + 5x + 7x + 9x = 360^\circ$
 $\Rightarrow 24x = 360^\circ \Rightarrow x = 15$
2. (d) The measure of all angles are
 $\angle 1 = 3x = 3 \times 15 = 45^\circ$
 $\angle 2 = 5x = 5 \times 15 = 75^\circ$
 $\angle 3 = 7x = 7 \times 15 = 105^\circ$
 $\angle 4 = 9x = 9 \times 15 = 135^\circ$
3. (b) The sum of the least and greater angle is
 $\angle 1 + \angle 4 = 45^\circ + 135^\circ = 180^\circ$
4. (c) $\angle A = \angle C$
[Opposite angles in a parallelogram are equal]
 $\Rightarrow (3x + 15^\circ) = (2x + 35^\circ)$
 $\Rightarrow 3x - 2x = 35 - 15 \Rightarrow x = 20$
5. (a) $\angle A + \angle B = 180^\circ$
[Sum of adjacent angles of a parallelogram are supplementary]
 $\Rightarrow (3x + 15^\circ) + (2y + 15^\circ) = 180^\circ$
 $\Rightarrow 3x + 15 + 2y + 15 = 180^\circ$
 $\Rightarrow 3x + 2y + 30 = 180^\circ$
 $\Rightarrow 3x + 2y = 150^\circ$
Substituting the value of $x = 20$ in (i), we get
 $\Rightarrow 3 \times 20 + 2y = 150^\circ$
 $\Rightarrow 60 + 2y = 150^\circ \Rightarrow 2y = 150 - 60$
 $\Rightarrow 2y = 90 \Rightarrow y = 45$
6. (d) Measure of $\angle A = (3x + 15)^\circ$
 $= 3 \times 20 + 15 = 75^\circ$

$$\text{Measure of } \angle B = (2y + 15)^\circ = 2 \times 45 + 15 = 105^\circ$$

7. (a) 8. (b) 9. (d)

Assertion & Reason :

1. (c) **Assertion** is correct and **Reason** is Incorrect.
A polygon bounded by seven line segments is called a heptagon.
2. (a) **Assertion** and **Reason** both are correct.
Reason is the correct explanation of **Assertion**
3. (a) Both Statements are correct and **Reason** is the correct explanation of **Assertion**.
4. (b) **Assertion** and **Reason** both are correct but **Reason** is not the correct explanation of **Assertion**.
5. (c) Adjacent angles are not equal in a parallelogram.

Multiple Matching Questions :

1. (A) $\rightarrow$ (p, r, s); (B) $\rightarrow$ (t, r); (C) $\rightarrow$ (p, q, r, s); (D) $\rightarrow$ (q, r, s, t)

Integer Type Questions :

1. (4) The ratio of two sides of a parallelogram is $4x$ and $3x$.
Perimeter = 2 (length + breadth)
 $\rightarrow 56 = 2(4x + 3x)$
 $\rightarrow 56 = 14x \rightarrow x = 4$.
Sides of parallelogram is $4 \times 4 = 16$ m and $4 \times 3 = 12$ m

Now, value of $\frac{a+b}{7} = \frac{16+12}{7} = 4$ m
2. (5) Since opposite angles of a parallelogram are equal.
 $\therefore \angle BCD = \angle DAB = 75^\circ$
In $\triangle BCD$, we have
 $\angle DBC + \angle BCD + \angle CDB = 180^\circ$
 $\Rightarrow 60^\circ + 75^\circ + \angle CDB = 180^\circ$
 $\Rightarrow \angle CDB = 180^\circ - (60^\circ + 75^\circ) = 45^\circ$
Since $\angle ADB$ and $\angle CBD$ are alternate interior angles.
 $\therefore \angle ADB = \angle CBD = 60^\circ = 12 \times 5 \Rightarrow k = 5$
[$\therefore \angle CBD = 60^\circ$]
3. (5) $x = 50^\circ$
 $\Rightarrow k = 5$

Chapter

4

Practical Geometry

INTRODUCTION

Familiarity with the step-by-step methods used for constructing geometric figures and knowing related definition of terms help you understand the practical applications of geometric construction to problem solving. Simplified- or preferred methods of geometric construction, as well as alternate methods, are valuable knowledge factors when used with drafting instruments to create accurate drawings. Geometric construction applies equally to computer- generated drawings as it does to more traditional instrumental drawings using triangles, compasses and protractors.

It is widely used by civil engineers, designers.

CONSTRUCTION OF QUADRILATERALS

A quadrilateral has ten parts in all, four sides, four angles and two diagonals. It is not necessary to know all the parts to construct a quadrilateral. If we know five parts out of ten, we will be able to construct a quadrilateral.

Following are the cases, we will consider for construction of quadrilaterals.

- When 4 sides and one diagonal are given
- When three sides and two diagonals are given.
- When four sides and one angle are given.
- When three sides and two included angles are given.
- When two adjacent sides and three angles are given.
- When other special properties are given.

When the Length of Four Sides and a Diagonal are Given

ILLUSTRATION : 1

Construct a quadrilateral ABCD with $AB = 4.5$ cm, $BC = 4$ cm, $CD = 6.5$ cm, $DA = 3$ cm and $BD = 6.5$ m.

SOLUTION :

Steps of construction :

- Draw $CD = 6.5$ cm.
- With C as centre draw an arc with radius 4 cm and with D as centre draw an arc with radius 3 cm.
- Again with D as centre and radius 6.5 cm, which cut the previously drawn arc from C at B.
- With B as centre and radius equal to 4.5 cm draw arc to cut the previously drawn arc through D at A.
- Join AB, BC, AD and BD. ABCD is the required quadrilateral with given dimensions.

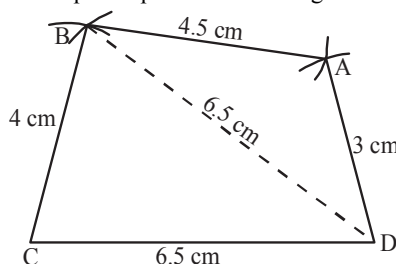
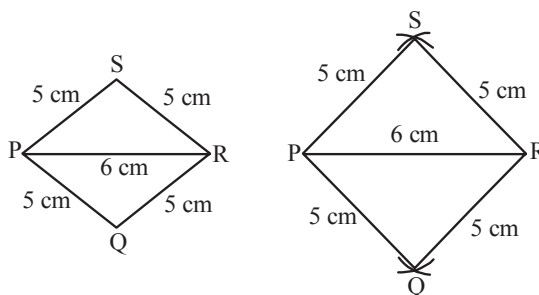


ILLUSTRATION : 2

Construct a rhombus with side 5 cm and the diagonal as 6 cm.

SOLUTION :



Steps of construction :

- Draw the rough sketch of rhombus and mark the measurements.
- Draw $PR = 6$ cm.
- With P as centre and radius 5 cm, draw arcs on both sides of PR.
- With R as centre radius 5 cm draw another arcs intersecting the previously drawn arcs at S and Q.
- Join PS, PQ, QR and PR.
PQRS is the required rhombus.

When Three Sides and Two Diagonals are Given

When four sides and a diagonal are given, we draw the diagonal as the base and then draw the triangle with the available data above and below the diagonal to complete a quadrilateral. In this case we shall use the same method of constructing the triangle first to complete the quadrilateral.

ILLUSTRATION : 3

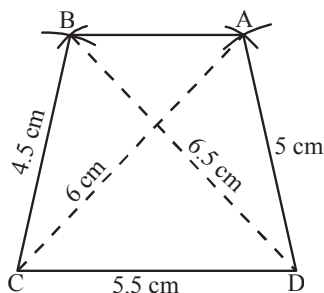
Construct a quadrilateral ABCD in which $BC = 4.5$ cm, $CA = 6$ cm, $AD = 5$ cm, $CD = 5.5$ cm and $BD = 6.5$ cm. Write its steps of construction also.

SOLUTION :

Steps of construction :

- (i) Draw $CD = 5.5$ cm.
- (ii) With C as centre and radius 4.5 cm, draw an arc above CD.
- (iii) With D as centre and radius 6.5 cm, draw an arc cutting the arc drawn in step (ii) at B.
- (iv) With D as centre and radius 5 cm draw an arc above CD.
- (v) With C as centre and radius 6 cm, draw an arc cutting the arc shown in step (iv) at A.
- (vi) Join AB, BC and DA

Thus ABCD is the required quadrilateral.



When Four Sides and One Angle are Given

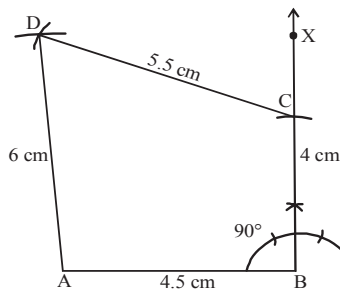
ILLUSTRATION : 4

Construct a quadrilateral ABCD in which $AB = 4.5$ cm, $BC = 4$ cm, $CD = 5.5$ cm, $AD = 6$ cm and $\angle B = 90^\circ$.

SOLUTION :

Steps of construction :

- (i) Draw $AB = 4.5$ cm
- (ii) Draw a ray BX making an angle 90° with AB, and mark $BC = 4$ cm on BX.
- (iii) By taking A as centre and radius 6 cm, draw an arc.
- (iv) With C as centre and radius 5.5 draw an arc to cut the arc drawn in step (iii) at D.
- (v) Join CD and DA.
- (vi) ABCD is the required quadrilateral.



When Three Sides and Two Included Angles are Given

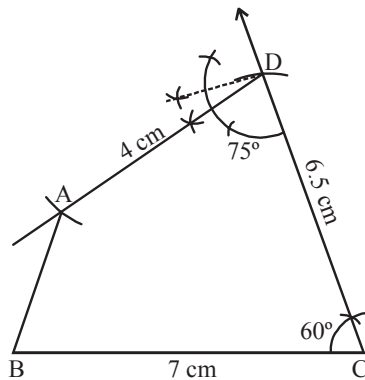
ILLUSTRATION : 5

Construct a quadrilateral ABCD, if $BC = 7$ cm, $CD = 6.5$ cm, $DA = 4$ cm, $\angle C = 60^\circ$ and $\angle D = 75^\circ$.

SOLUTION :

Steps of construction :

- (i) Draw $BC = 7$ cm
- (ii) Draw a ray CD such that $\angle BCD = 60^\circ$ and cut off $CD = 6.5$ cm
- (iii) At D , construct an angle of 75° and cut off $DA = 4$ cm.
- (iv) Join AB .
- (v) ABCD is the required quadrilateral.



When Two Adjacent Sides and Three Angles are Given

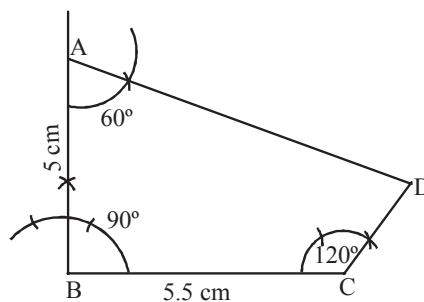
ILLUSTRATION : 6

Construct quadrilateral ABCD in which $AB = 5$ cm, $BC = 5.5$ cm, $\angle A = 60^\circ$, $\angle B = 90^\circ$ and $\angle C = 120^\circ$

SOLUTION :

Steps of construction :

- (i) Draw $BC = 5.5$ cm
- (ii) At B , construct an angle of 90° and mark $BA = 5$ cm.
- (iii) At C , construct an angle such that $\angle BCD = 120^\circ$.
- (iv) At A , construct an angle 60° with a ray to intersect the ray given in step (iii) at point D .
- (v) Join AD and DC .
- (vi) ABCD is the required quadrilateral.



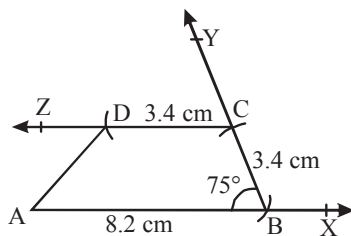
SOME SPECIAL CASES**Construction of Special Types of Quadrilaterals**

For the construction of the quadrilaterals like parallelogram, rhombus, trapezium, etc., we can use their special features, such as

- (i) opposite sides of a parallelogram are equal,
- (ii) opposite angles of a parallelogram are equal,
- (iii) diagonals of a parallelogram bisect each other,
- (iv) diagonals of a rhombus bisect each other at right angles,
- (v) a pair of opposite sides of a trapezium are parallel,
- (vi) opposite sides of a rhombus are equal,
- (vii) opposite angles of a rhombus are equal.

ILLUSTRATION : 7

Construct a trapezium ABCD in which $AB \parallel CD$, $AB = 8.2$ cm, $BC = 3.4$ cm, $CD = 3.4$ cm and $\angle B = 75^\circ$.

SOLUTION :**Steps for Construction :**

- (i) Draw $\overline{AX}$ and cut $AB = 8.2$ cm on it.
- (ii) At B, draw a ray $\overline{BY}$ such that $\angle ABY = 75^\circ$.
- (iii) With centre B and radius equal to 3.4 cm draw an arc cutting $\overline{BY}$ at C.
- (iv) Through C draw $\overline{CZ}$ such that $\angle BCZ = 180^\circ - 75^\circ = 105^\circ$.
- (v) With centre C and radius equal to 3.4 cm, draw an arc cutting $\overline{CZ}$ at D. Join A and D.
Thus, ABCD is the required trapezium.

MISCELLANEOUS

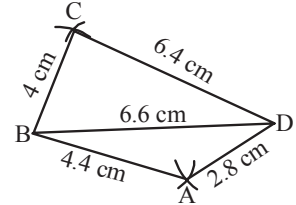
SOLVED EXAMPLES

1. Construct a quadrilateral ABCD in which $AB = 4.4$ cm, $BC = 4$ cm, $CD = 6.4$ cm, $DA = 2.8$ cm and $BD = 6.6$ cm.

Sol. Steps of Construction :

- Draw $BD = 6.6$ cm.
- With B as centre and radius $BC = 4$ cm, draw an arc.
- With D as centre and radius $CD = 6.4$ cm, draw an arc, to intersect the arc drawn in step (ii) at C.
- With B as centre and radius $BA = 4.4$ cm, draw an arc on the side of BD opposite to that of C.
- With D as centre and radius $AD = 2.8$ cm, draw another arc to intersect the arc drawn in step (iv) at A.
- Join BA, DA, BC and CD.

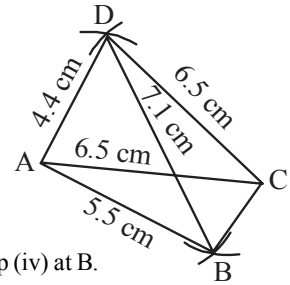
The quadrilateral ABCD so obtained is the required quadrilateral.



2. Construct a quadrilateral ABCD in which $AB = 5.5$ cm, $AD = 4.4$ cm, $CD = 6.5$ cm, $AC = 6.5$ cm and $BD = 7.1$ cm.

Sol. Steps of Construction :

- Draw $AC = 6.5$ cm.
- With A as centre and radius $AD = 4.4$ cm, draw an arc.
- With C as centre and radius $CD = 6.5$ cm, draw an arc to intersect the arc drawn in step (ii) at D.
- With A as centre and radius $AB = 5.5$ cm, draw an arc on the side of AC opposite to that of D.
- With D as centre and radius $BD = 7.1$ cm, draw another arc to intersect the arc drawn in step (iv) at B.
- Join AD, CD, AB and CB to obtain the required quadrilateral.

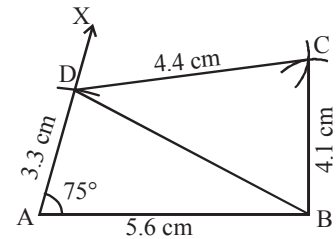


3. Construct a quadrilateral ABCD given $AB = 5.6$ cm, $BC = 4.1$ cm, $CD = 4.4$ cm, $AD = 3.3$ cm and $\angle A = 75^\circ$.

Sol. Steps of Construction :

- Draw $AB = 5.6$ cm.
- Construct $\angle BAX = 75^\circ$.
- With A as centre and radius $AD = 3.3$ cm, cut off $AD = 3.3$ cm along AX.
- Join BD.
- With D as centre and radius $DC = 4.4$ cm, draw an arc.
- With B as centre and radius $BC = 4.1$ cm, draw an arc to cut the arc drawn in step (v) at C.

Join BC, CD to obtain the required quadrilateral ABCD.

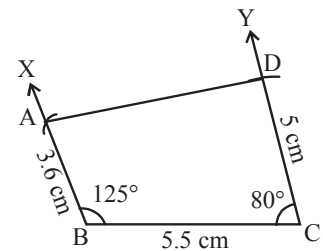


4. Construct a quadrilateral ABCD in which $AB = 3.6$ cm, $BC = 5.5$ cm, $CD = 5$ cm, $\angle B = 125^\circ$ and $\angle C = 80^\circ$.

Sol. Steps of Construction :

- Draw $BC = 5.5$ cm.
- At B construct $\angle XBC = 125^\circ$.
- At C construct $\angle YCB = 80^\circ$, such that X and Y are on the same side of BC.
- With B as centre and radius $AB = 3.6$ cm, draw an arc to intersect BX at A.
- With C as centre and radius $CD = 5$ cm, draw an arc to intersect CY at D.
- Join AD.

The quadrilateral ABCD so obtained is the required quadrilateral.



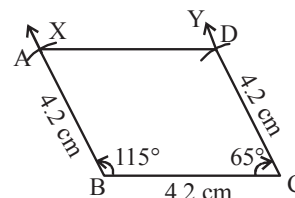
5. Construct a rhombus with side 4.2 cm and one of its angles equal to 65° .

Sol. Clearly, the adjacent angle $= (180^\circ - 65^\circ) = 115^\circ$. So, we may proceed according to the steps given below.

Steps of Construction :

- Draw $BC = 4.2$ cm.
- Make $\angle CBX = 115^\circ$ and $\angle BCY = 65^\circ$.
- Cut off $BA = 4.2$ cm along BX and $CD = 4.2$ cm along CY .
- Join AD .

Then, $ABCD$ is the required rhombus.



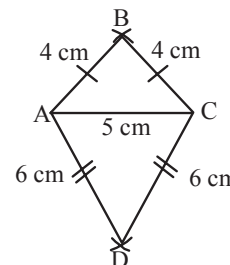
6. Construct a kite $ABCD$ in which $AB = 4$ cm, $AD = 6$ cm and $AC = 5$ cm.

Sol. We know that a kite has two pairs of equal adjacent sides.

Steps of Construction :

- Draw $AC = 5$ cm.
- Cut off $AB = CB = 4$ cm at point B .
- Similarly, cut off $AD = CD = 6$ cm at point D .
- Join AB , CB , AD and CD .

Then, $ABCD$ is the required kite.



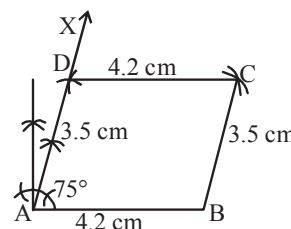
7. Construct a parallelogram $ABCD$ where $AB = 4.2$ cm, $AD = 3.5$ cm and $\angle A = 75^\circ$.

Sol. We first draw a rough sketch of the parallelogram $ABCD$

Steps of Construction :

- Draw a line segment $AB = 4.2$ cm
- At A draw an angle $\angle BAX = 75^\circ$.
- From AX cut off $AD = 3.5$ cm
- With D as centre and radius ($DC = AB$) equal to 4.2 cm, draw an arc. Again with B as centre and radius ($BC = 3.5$ cm), draw another arc intersecting the first arc at C .
- Join BC and DC .

Then, $ABCD$ is the required parallelogram.



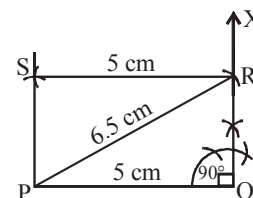
8. Construct a rectangle $PQRS$ given that $PQ = 5$ cm and the diagonal $PR = 6.5$ cm.

Sol. We first draw a rough sketch of the rectangle $PQRS$.

Steps of Construction :

- Draw a line segment $PQ = 5$ cm.
- At Q , draw $QX \perp PQ$. With P as centre and radius equal to 6.5 cm, draw an arc cutting QX at R . Join PR .
- With R as centre and radius $= 5$ cm, draw an arc.
- With P as centre and radius equal to QR draw another arc cutting the first arc at S . Join PS and RS .

Then, $PQRS$ is the required rectangle.

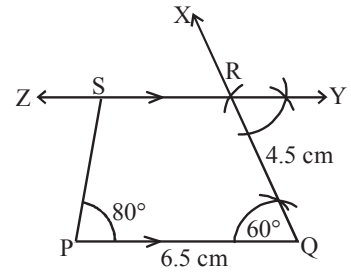


9. Construct a trapezium PQRS in which $PQ \parallel SR$, $PQ = 6.5$ cm, $QR = 4.5$ cm, $\angle P = 80^\circ$ and $\angle Q = 60^\circ$.

Sol. Steps of Construction :

- (i) Draw a line segment $PQ = 6.5$ cm.
- (ii) Make $\angle PQX = 60^\circ$.
- (iii) Cut off $QR = 4.5$ cm on QX .
- (iv) Make $\angle QRY = \angle PQX$ (Alternative interior angle)
- (v) Draw a line YZ passing through R parallel to PQ .
- (vi) At P draw an angle such that $\angle QPS = 80^\circ$

Thus, PQRS is the required trapezium.



1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. A quadrilateral can be constructed uniquely if its three sides and _____ angle are given.
2. _____ measurements can determine a quadrilateral uniquely.
3. Data about five parts of a quadrilateral in order to be sufficient must also satisfy _____ inequality and _____ of a triangle, wherever applicable.
4. If only one diagonal of a quadrilateral bisects the other, then the quadrilateral is known as _____.
5. To construct a quadrilateral uniquely, it is necessary to have the knowledge of a least _____ independent elements.
6. A quadrilateral can be constructed uniquely if its _____ sides and one diagonal are given

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. For constructing a unique parallelogram lengths of only two sides should be given.
2. A quadrilateral can be drawn if only measures of four sides are given.
3. A quadrilateral can have all four angles as obtuse.
4. A quadrilateral can be drawn when all the four angles and one side is given.
5. A quadrilateral can be drawn if three sides and two diagonals are given.
6. A parallelogram can be constructed uniquely if both diagonals and the angle between them is given.
7. A rhombus can be constructed uniquely if both diagonals are given.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

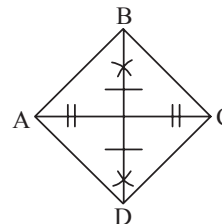
- | 1. | Column-I
(Given Data) | Column-II
(Constructed figure) |
|-----|--|-----------------------------------|
| (A) | Given $AB = 6.6$ cm,
$AC = 3.5$ cm, the altitude
AL from $A = 2.6$ cm,
opposite sides are parallel
and equal, then $ABCD$ is a | (p) square |

- | | | |
|-----|---|-------------------|
| (B) | A quadrilateral in which each side measure 6.2 cm and each angle equal to 90° is a | (q) trapezium |
| (C) | Given $AB = 6$ cm, $BC = 4.1$ cm, $CD = 3.3$ cm, $\angle B = 75^\circ$ and $DC \parallel AB$, then $ABCD$ is a | (r) Rectangle |
| (D) | $QR = 3.6$ cm, diagonal $PR = 6$ cm and each angle equal to 90° , then $PQRS$ is a | (s) Parallelogram |

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Construct a square each of whose side length is 5 cm.
2. Construct a square LMNO whose diagonal is 6 cm.
3. The construction of which figure is shown in the diagram?

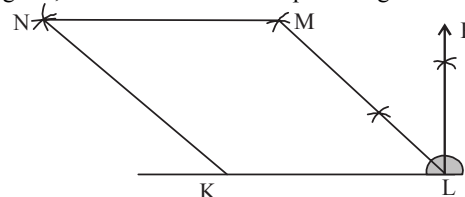


4. If $AB = 7.3$ cm, one angle is 60° and diagonal bisect each other at 90° , then which figure can be constructed?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Construct a quadrilateral $ABCD$ in which $AB = 3.8$ cm, $BC = 5$ cm, $CD = 5$ cm, $DA = 5.5$ cm and $AC = 8$ cm.
2. Construct a quadrilateral $ABCD$ in which $AB = 5$ cm, $AC = 6$ cm, $AD = 8$ cm and $\angle ABC = \angle ACD = 90^\circ$.
3. Construct a quadrilateral $ABCD$ in which $AB = 4$ cm, $BC = 3.4$ cm, $AD = 2.8$ cm, diagonal $AC = 4.6$ cm and diagonal $BD = 4.2$ cm.
4. In the diagram, $KLMN$ is a constructed parallelogram.



Find the value of $\angle KLM$.

Long Answer Questions :**DIRECTIONS :** Give answer in four to five sentences.

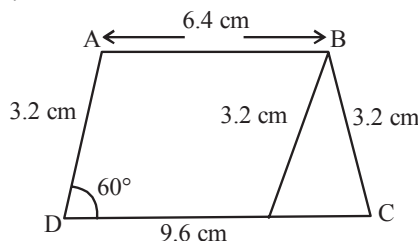
- Construct a quadrilateral $ABCD$ in which $AB = 4$ cm, $\angle A = 75^\circ$, $BC = 2.8$ cm, $CD = 3.3$ cm and $AD = 2.5$ cm
- Construct a quadrilateral $ABCD$ in which $AB = 4.5$ cm, $BC = 3$ cm, $CD = 3.2$ cm, $AD = 2.5$ cm and the diagonal $AC = 5$ cm.
- Construct a parallelogram $ABCD$, where $AB = 4.5$ cm, $BC = 3$ cm and $\angle B = 105^\circ$.
- Construct a rectangle $PQRS$ with $PQ = 5.5$ cm and $QR = 3$ cm.
- Construct a rectangle $PQRS$ with $PQ = 6$ cm and diagonal $PR = 6.6$ cm.

2

EXERCISE

Text-Book Exercise :

- Construct the Quadrilateral JUMP, where
 $JU = 3.5$ cm
 $UM = 4$ cm
 $MP = 5$ cm
 $PJ = 4.5$ cm
 $PU = 6.5$ cm
- Construct a quadrilateral $ABCD$, given that $BC = 4.5$ cm, $AD = 5.5$ cm, $CD = 5$ cm the diagonal $AC = 5.5$ cm and diagonal $BD = 7$ cm.
- Construct the Parallelogram HEAR, where
 $HE = 5$ cm
 $EA = 6$ cm
 $\angle R = 85^\circ$
- Construct a quadrilateral $ABCD$, where $AB = 4$ cm, $BC = 5$ cm, $CD = 6.5$ cm and $\angle B = 105^\circ$ and $\angle C = 80^\circ$.
- Is it possible to construct a rhombus $ABCD$ where $AC = 6$ cm and $BD = 7$ cm? Justify your answer.
- A rhombus whose diagonals are 5.2 cm and 6.4 cm long.
- A parallelogram $OKAY$ where $OK = 5.5$ cm and $KA = 4.2$ cm. Is it unique?
- Construct the Quadrilateral DEAR, where
 $DE = 4$ cm
 $EA = 5$ cm
 $AR = 4.5$ cm
 $\angle E = 60^\circ$
 $\angle A = 90^\circ$
- Construct a rectangle whose side is 3 cm and a diagonal equal to 5 cm.
- Construct a quadrilateral $NEWS$ in which $NE = 7$ cm, $EW = 6$ cm, $\angle N = 60^\circ$, $\angle E = 110^\circ$ and $\angle S = 85^\circ$.
- Is it possible to construct a quadrilateral $ROAM$ in which $RO = 4$ cm, $OA = 5$ cm, $\angle O = 120^\circ$, $\angle R = 105^\circ$ and $\angle A = 135^\circ$? If not, why?
- Construct a trapezium $ABCD$ where $AB \parallel CD$, $AD = BC = 3.2$ cm, $AB = 6.4$ cm and $CD = 9.6$ cm. Measure $\angle B$ and $\angle A$.



[Hint : Difference of two parallel sides given an equilateral triangle.]

- Construct a rhombus $PAIR$, given that $PA = 6$ cm and angle $\angle A = 110^\circ$.
- Is it possible to construct a quadrilateral $ABCD$ in which $AB = 3$ cm, $BC = 4$ cm, $CD = 5.4$ cm, $DA = 5.9$ and diagonal $AC = 8$ cm? If not, why?

HOTS Questions :

- Construct a quadrilateral $ABCD$ given $AB = 5.3$ cm, $AD = 2.9$ cm, $\angle A = 70^\circ$, $\angle B = 95^\circ$, $\angle C = 85^\circ$.
- Construct a quadrilateral $BDEF$, where $DE = 4.5$ cm, $EF = 3.5$ cm, $FB = 6.5$ cm, $\angle F = 50^\circ$ and $\angle E = 100^\circ$.
- Construct, if possible, a quadrilateral $ABCD$ given $AB = 6$ cm, $BC = 3.7$ cm, $CD = 5.7$ cm, $AD = 5.5$ and $BD = 6.1$ cm. Give reason for not being able to construct it, if you cannot.

Exemplar Questions :

- Construct a quadrilateral $ABCD$ in which $AB = AD = 5$ cm, $BC = CD = 7$ cm and $BD = 6$ cm. What type of quadrilateral is this?
- Construct a trapezium $ABCD$ in which $AB \parallel DC$, $\angle A = 105^\circ$, $AD = 3$ cm, $AB = 4$ cm and $CD = 8$ cm.

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- To construct a parallelogram, the minimum number of measurements required is :
(a) 2 (b) 3
(c) 4 (d) 1
- To construct a convex quadrilateral, which of the following cases is not correct?
(a) When the lengths of four sides and one diagonal are given.
(b) When the lengths of three sides and the two diagonals are given.
(c) When the lengths of four sides and one angle are given.
(d) When the lengths of two sides and two included angles are given.
- Which of the following is true for constructing a quadrilateral?
(a) 5 sides and 3 diagonals
(b) 3 sides only
(c) 3 angles and their 2 included sides
(d) 2 diagonals only
- Given $AB = 3$ cm, $BC = 5$ cm, $AC = 9$ cm, $AD = 6$ cm, $CD = 2$ cm. Which of the following is true about the construction of a quadrilateral?
(a) It is possible to draw the quadrilateral.
(b) It is not possible to draw the quadrilateral since $AD + DC < AC$.
(c) It is possible to draw the quadrilateral since $AD + DC < AC$.
(d) None of these
- If $AB = 5$ cm, $BC = 4$ cm, $\angle B = 90^\circ$ and opposite sides are equal, then which of the following figure can be constructed?
(a) Square (b) Rectangle
(c) trapezium (d) Rhombus
- To construct a kite which of the following is necessary?
(a) Two adjacent unequal sides and included diagonal.
(b) Two adjacent equal sides and included diagonal.
(c) Opposite sides length.
(d) None of these.

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- To construct a quadrilateral ABCD, which of the following parts is necessary?
(a) Length of AB
(b) Length of BC
(c) Measure of $\angle A$, $\angle B$ and $\angle C$
(d) Both (a) and (b)
- Which of the given properties of a parallelogram are necessary to construct it?
(a) Opposite sides of a parallelogram are equal should be given.
(b) Opposite angles of a parallelogram are equal should be given.
(c) diagonals of parallelogram bisect each other should be given.
(d) both (a) and (b)
- Which of the following options is/ are true about a trapezium to construct it?
(a) one pair of opposite sides of a trapezium should be parallel and unequal should be given
(b) two non – parallel sides length should be given
(c) two equal opposite angles should be given
(d) all of these
- Given below are the steps of construction of a quadrilateral ABCD, where $AB = 3.5$ cm, $BC = 6.5$ cm, $\angle A = 75^\circ$, $\angle B = 105^\circ$ and $\angle C = 120^\circ$. Which of the following is a correct step?
(a) Step I : Draw $AB = 3.5$ cm.
(b) Step II : Draw $\angle XAB = 75^\circ$ at A and $\angle ABY = 105^\circ$ at B.
(c) Step III : With B as centre and radius $BC = 6.5$ cm, draw an arc to intersect BY at C.
(d) Step IV : At C draw $\angle ADC = 120^\circ$ such that CZ meets AX at D.
- Given $AB = 3$ cm, $BC = 5$ cm, $AC = 9$ cm, $AD = 6$ cm, $CD = 2$ cm. Which of the following is false about the construction of a quadrilateral?
(a) It is possible to draw the quadrilateral.
(b) It is not possible to draw the quadrilateral since $AD + DC < AC$.
(c) It is possible to draw the quadrilateral since $AD + DC < AC$.
(d) None of these

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE

Given a quadrilateral ABCD in which $BC = 5$ cm, $\angle BCD = 120^\circ$, $CD = 4.8$ cm and opposite sides are parallel and equal.

- What is the name of the quadrilateral which can be drawn from the given data?
 (a) square (b) Rhombus
 (c) Parallelogram (d) Rectangle
- If $\angle BCD = 120^\circ$, then $\angle BAD =$
 (a) 60° (b) 120°
 (c) 180° (d) none of these
- If $BC = 5$ cm, then $AD =$
 (a) 5 cm (b) 4.8 cm
 (c) 6 cm (d) none of these

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- If **Assertion** is **correct** but **Reason** is **incorrect**.
- If **Assertion** is **incorrect** but **Reason** is **correct**.

- Assertion :** A quadrilateral can be constructed if at least any five independent elements are given.

Reason : Data about the five parts of a quadrilateral in order to be sufficient must also satisfy (i) the triangle inequality and (ii) angle sum property of a triangle, wherever applicable.

- Assertion :** In all cases of quadrilateral, it is convenient and helpful to draw rough sketch of the quadrilateral and indicate the data on it.

Reason : This suggests the steps of construction.

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- To construct a quadrilateral, if 3 angles are given then how many included sides are required.
- To construct a quadrilateral, if 2 diagonals are given, then how many sides are required.
- The number of measurements required to construct a quadrilateral is _____.
- The minimum number of dimensions needed to construct a rectangle is _____.
- The minimum number of measurements needed to construct a square is _____.

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

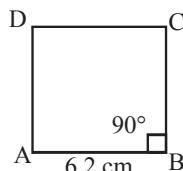
1. 2 included
2. 5
3. triangle, angle sum property.
4. Kite
5. Five

True / False :

1. False.
2. False.
3. False.
4. False.
5. True.
6. True.
7. True.

Match the Columns :

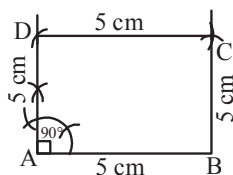
1. (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)
- (A) Hence, one side, one diagonal and an altitude from, A is given. Also, opposite sides are parallel and equal. So, a parallelogram can be constructed.
- (B) Here each side measurement and each angle equal to 90° are given. So, in this cases a square can be constructed.



- (C) Here, AB, BC, CD, DB and DC $\parallel$ AB are given and by using these values we can construct a trapezium.
- (D) Here, side QR, diagonal PR and each angle equal to 90° . So, using these measurements we can construct a rectangle.

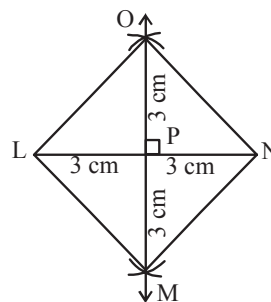
Very Short Answer Questions :

1. Steps of Construction :



- (i) Draw a line segment AB = 5 cm.
- (ii) With A as centre, draw an angle of 90° at point A.
- (iii) Cut off AD equal to 5 cm. With B as centre and radius 5 cm; draw an arc.
- (iv) With D as centre and same radius cut off previously drawn arc at C.
- (v) Join D to C and C to B.

2. We know that the diagonals of the square bisect each other at right angles and are equal.



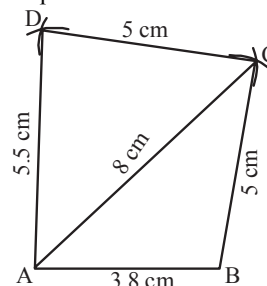
Steps of Construction :

- (i) Draw a line segment LN = 6 cm
- (ii) Draw a line OM, its right bisector, meeting LN at P.
- (iii) Cut off PO and PM each equal to 3 cm.
- (iv) Join LM, MN, NO and OL.
Then, LMNO is the required square.
3. Since diagonals are bisecting each other at right angle, it is a rhombus.
4. Here, one side, one angle 60° and diagonal bisect each other at 90° is given and we know that diagonals of a rhombus bisect each other at right angle. So, a rhombus can be drawn since one angle is 60° .

Short Answer Questions :

1. Steps of Construction :

- (i) Draw AB = 3.8 cm
- (ii) With A as centre and radius equal to 8 cm draw an arc.
- (iii) With B as centre and radius equal to 5 cm draw another arc, cutting the previous arc at C.
- (iv) Join B to C.
- (v) With A as centre and radius equal to 5.5 cm, draw an arc.
- (vi) With C as centre and radius equal to 5 cm, draw another arc, cutting the previous arc at D.

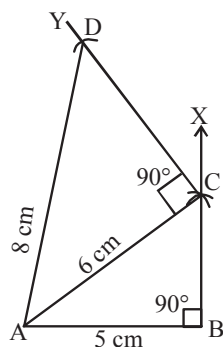


- (vii) Join AD and CD.

Then, ABCD is the required quadrilateral.

2. Steps of Construction :

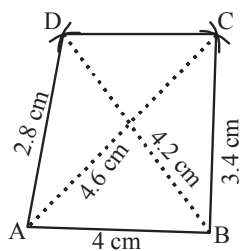
- (i) Draw $AB = 5$ cm
- (ii) Make $\angle ABX = 90^\circ$
- (iii) With A as centre and radius equal to 6 cm cut an arc on BX at C. Join AC.



- (iv) Again make $\angle ACY = 90^\circ$.
 - (v) With A as centre and radius equal to 8 cm cut an arc on CY at D. Join AD.
- Thus, ABCD is the required quadrilateral.

3. Steps of Construction :

- (i) Draw $AB = 4$ cm
- (ii) With A as centre and radius equal to 4.6 cm, draw an arc.
- (iii) With B as centre and radius equal to 3.4 cm, draw another arc, cutting the previous arc at C.



- (iv) Join B to C.
 - (v) With A as centre and radius equal to 2.8 cm draw an arc.
 - (vi) With B as centre and radius equal to 4.2 cm draw another arc cutting the previous arc at point D.
 - (vii) Join C to D and A to D.
- Thus, ABCD is the required quadrilateral.

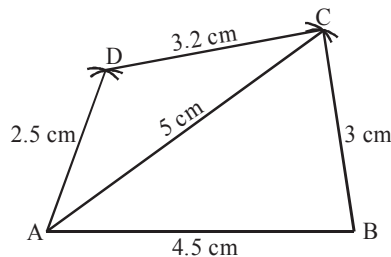
4. $\angle KLP = 90^\circ$

LM is an angular bisector of $\angle KLP$

$$\therefore \angle KLM = \frac{1}{2} \times 90^\circ = 45^\circ$$

Long Answer Questions :

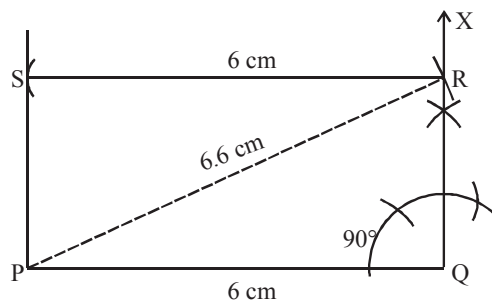
2.



Steps of construction:

- (i) Draw $AB = 4.5$ cm.
- (ii) With B as centre draw an arc with radius 3 cm and with A as centre with radius 5 cm draw another arc cutting previously drawn arc at C.
- (iii) Join B to C.
- (iv) With C as centre and radius 3.2 cm draw an arc and with A as centre and radius 2.5 cm draw another arc cutting previously drawn arc at D.
- (v) Join A to D and C to D.
- (vi) ABCD is the required quadrilateral.

5.



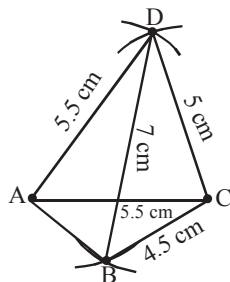
Steps of construction:

- (i) Draw $PQ = 6$ cm.
- (ii) At Q, draw a ray QX such that $\angle PQX = 90^\circ$
- (iii) With P as centre and radius 6.6 cm draw an arc, cutting QX at R.
- (iv) With R as centre and radius 6 cm draw an arc.
- (v) With P as centre and radius equal to QR draw another arc, cutting previously drawn arc at S.
- (vi) Join PS and RS.
- (vii) PQRS is the required rectangle.

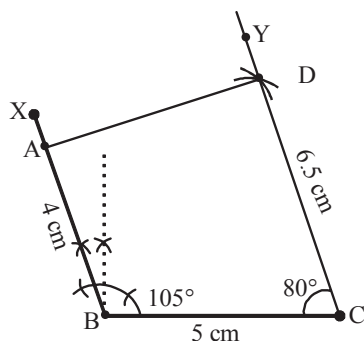
2 EXERCISE

Text-Book Exercise :

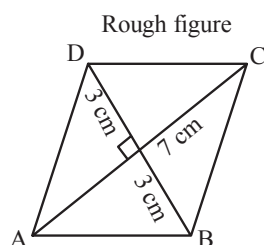
2.



4.



6. Only two (diagonal) measurements of the rhombus are given. However, since it is a rhombus, we can find more help from its properties.



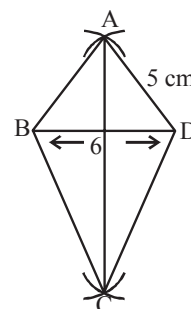
7. So, first draw $AC = 7$ cm and then construct its perpendicular bisector. Let them meet at O . Cut off 3 cm lengths on either side of the drawn bisector. you now get B and D . Join AB , BC , CD and AD .

It is not a unique parallelogram. There can be endless parallelogram with those dimensions.

Exemplar Questions :

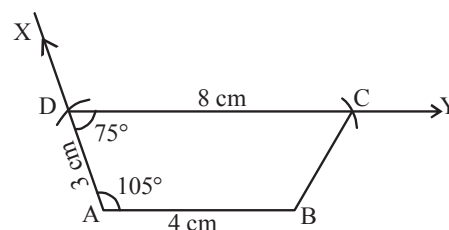
- Draw a line segment $BD = 6$ cm. Taking B and D as centres and 5 cm radius, draw arcs to intersect at the point A .
 - Taking B and D as centres and 7 cm radius, draw arcs in the opposite side of A intersect at the point C .
 - Join AB , AD and BC , DC .

$ABCD$ is the required quadrilateral. It is a kite.

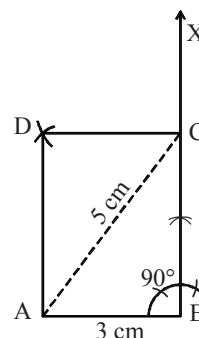


- $\angle A + \angle D = 180^\circ$
 $105^\circ + \angle D = 180^\circ$
 $\angle D = 75^\circ$

Steps of construction:



- Draw $AB = 4$ cm
 - Draw $\overline{AX}$ such that $\angle BAX = 105^\circ$
 - Mark a point D on AX such that $AD = 3$ cm
 - Draw $\overline{DY}$ such that $\angle ADY = 75^\circ$
 - Mark a point C such that $CD = 8$ cm
 - Join BC . $ABCD$ is the required trapezium.
3. Diagonals of a rectangle are equal.

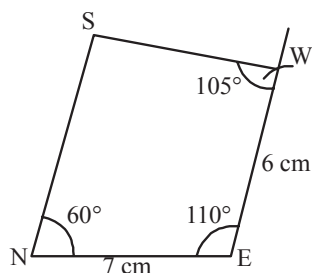


$AC = BD = 5$ cm

Steps of construction:

- Draw $AB = 3$ cm
- Draw a ray BX such that $\angle ABX = 90^\circ$
- Draw an arc such that $AC = 5$ cm
- With B as centre m draw an arc of radius 5 cm. With C as centre draw another arc of radius 3 cm which intersect first arc at a point D .
- Join CD and AD .

4.



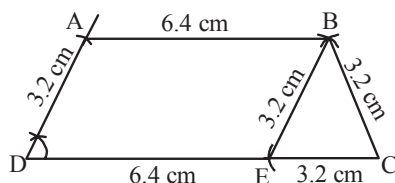
$$\begin{aligned}\text{Fourth angle} &= 360^\circ - (60^\circ + 110^\circ + 85^\circ) \\ &= 360^\circ - 255^\circ \\ &= 105^\circ\end{aligned}$$

5. No,

$$\angle O + \angle R + \angle A = 120^\circ + 105^\circ + 135^\circ = 360^\circ$$

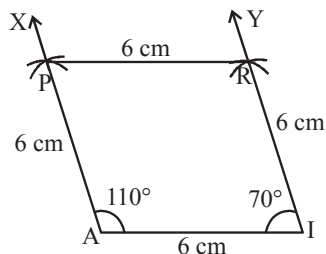
The sum of four angles of a quadrilateral must be 360° .

6.



$\triangle BEC$ is an equilateral triangle
 $\angle A = 120^\circ$, $\angle B = 60^\circ$

7.



Since in a rhombus, all sides are equal so, $PA = AI = IR = RP = 6$ cm

Also, rhombus is a parallelogram

so, adjacent angle, $\angle I = 180^\circ - 110^\circ = 70^\circ$

Steps of Construction :

- (i) Draw $AI = 6$ cm
- (ii) Draw ray $\overrightarrow{AX}$ such that $\angle IAX = 110^\circ$ and draw $\overrightarrow{IY}$ such that $\angle AIY = 70^\circ$.
- (iii) With A and I as centres and radius 6 cm draw arcs intersecting AX and IY at P and R respectively.
- (iv) Join PR.

Thus, PAIR is the required rhombus.

8. No,

In a Δ , sum of two sides always is greater than the third side.

$$AB + BC \not> AC$$

HOTS Questions :

1. We know that the sum of the angles of a quadrilateral is 360° .

$$\therefore \angle A + \angle B + \angle C + \angle D = 360^\circ$$

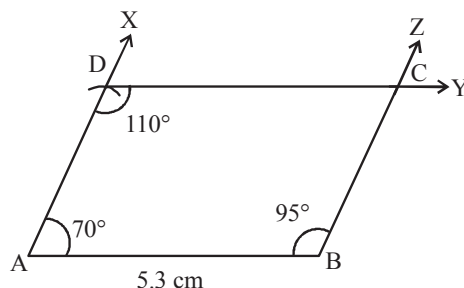
$$\Rightarrow 70^\circ + 95^\circ + 85^\circ + \angle D = 360^\circ$$

$$\Rightarrow 250^\circ + \angle D = 360^\circ$$

$$\Rightarrow \angle D = 360^\circ - 250^\circ = 110^\circ$$

Steps of Construction :

- (i) Draw $AB = 5.3$ cm.
- (ii) At A draw $\angle XAB = 70^\circ$.
- (iii) With A as centre and radius 2.9 cm draw an arc to cut AX at D.
- (iv) At D draw $\angle ADY = 110^\circ$.
- (v) At B draw $\angle ZBA = 95^\circ$ so that ZB intersects YD at C.

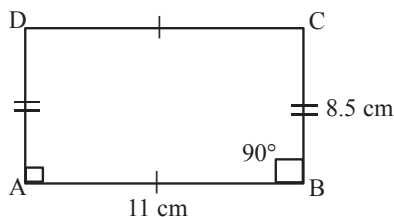


The quadrilateral ABCD so obtained is the required quadrilateral.

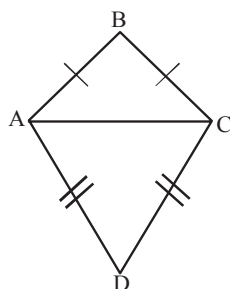
3 EXERCISE

Single Option Correct :

1. (b) To construct a parallelogram minimum 3 measurements are required.
2. (d) Only four measurements are given, which are not sufficient to construct a convex quadrilateral.
3. (c) 3 angles and their 2 included sides. There is only 4 sides and 2 diagonals. So option (1) is not valid. Only 3 sides are not sufficient to draw a quadrilateral. So, option (2) is also invalid. Only 2 diagonals are not sufficient to draw a quadrilateral. So, option (4) is also not valid.
4. (b) Here, it is given that $AD + DC < AC$ which does not satisfy triangle inequality and hence quadrilateral cannot be drawn.
5. (b) Here, two adjacent sides, one right angle and opposite sides are equal as given. So, a rectangle can be constructed.



6. (a) two adjacent unequal sides and included diagonal means, two unequal sides and diagonal from common vertex. i.e., sides AB, AD and diagonal AC are given.



More Than One Option Correct :

- (a, b, c) : We known that to construct a quadrilateral, measure of atleast five parts is necessary.
In case of quadrilateral ABCD, if length of AB, length of BC and measure of $\angle A$, $\angle B$ and $\angle C$ is required. So, option (a), (b) and (c) are correct.
- (a, b, d) : We know that a parallelogram can be constructed if its opposite sides are given, opposite angles are given, length of its diagonals are given or at least one angle is given.
Here , option (a), (b) and (d) fulfilled the requirement.
- (a, b) : A trapezium can be constructed if length of opposite unequal parallel sides are given and non – parallel sides are given.
So, options (a), (b) are correct. Option (c) is not correct, since opposite angles of a trapezium are not equal.
- (a, b, c)
- (a, b) Here, it is given that $AD + DC < AC$ which does not

satisfy triangle inequality and hence quadrilateral cannot be drawn.

Passage Based Questions :

- (c) Here, $BC = 5$ cm, $\angle BCD = 120^\circ$, $CD = 4.8$ cm and opposite sides are parallel and equal. By using these value we can construct a parallelogram.
- (b) Here, $\angle BCD = 120^\circ$,
 $\therefore \angle BAD = \angle BCD = 120^\circ$,
(opposite angles of a parallelogram are equal)
- (a) If $BC = 5$ cm then $AD = BC = 5$ cm
(Given as opposite sides are equal)

Assertion & Reason :

- (a) **Assertion** : correct, **Reason** : correct
Reason is correct explannation of Assertion since sides and angles of the quadrilateral must satisfy the triangle inequality and angle sum property of a triangle.
- (a) **Assrtion** : correct, **Reason** : correct
Reason is the correct explanation of Assertion since by using rough sketch we feel easy in writing steps of construction.

Integer Type Questions :

- (2) Since 3 angle + 2 included sides = 5 independent elements which are required to construct a quadrilateral.
- (3) Since, at least 5 independent elements are necessary to construct a quadrilateral. Therefore three sides are required.
- (3) Minimum number of measurement required to construct a unique quadrilateral is five.
- (2) Two dimensions are sufficient to construct a rectangle.
- (1) Only 1 measurement is sufficient to construct a square.

Chapter

5

Data Handling

INTRODUCTION

Data from current and past events is often a useful tool in business, economics, finance, social and medical sciences. Gathering data is the first step in developing mathematical models that can be used to analyse a situation and predict future performance. Graphical representations of data are commonly used in all spheres of life. Some of the commonly encountered graphs are 'production of wheat from 1971 to 2010,' 'Over-wise run-rate maintained by India in a 20-20 Cricket match,' etc.

Statistical models have become increasingly useful in manufacturing, agriculture, marketing, social sciences and in all types of research.

Statistics has applications to almost every aspect of modern life. The digital age is creating a wealth of data that needs to be summarized, visualized, and analyzed whether we want to know executives' earnings per year, movies' earnings at the box office on a given weekend, or the sales within sectors of the home and leisure industry.

Life is full of uncertainty so to identify the clear trends or future prospects of a product, a system or any change we can't just depend on the collected data or the statistical results. Here probability comes in which enables us to deal with uncertainty.

DATA

The information collected in the form of numerical figures is called data.

Ungrouped or Raw Data

The data which is collected for specific purpose and put as it is (without any arrangement) is called raw data. Each entry in raw data is known as an observation.

For example : runs scored by batsmen in a T-20 cricket match are :

17, 20, 15, 42, 25, 17, 23, 18, 5, 15

By observation we can find the lowest and highest scores.

Range

The difference between the highest and lowest values of given data is called range. In the above case,

Highest Score = 42, Lowest score = 5

$\therefore$ Range = Highest Score – Lowest score

Range = $42 - 5 = 37$

Frequency : The number of times a particular observation occurs is called its frequency. In the above example, frequency of 15 and 17 is two and for other scores the frequency is 1.

Discrete Frequency Distribution Table

Consider the marks obtained by 20 students in mathematics annual examination of class VIII (Marks are out of 100) :

60, 70, 59, 60, 50, 58, 62, 56, 59, 59, 58, 70, 58, 62, 50, 58, 58, 50, 62, 56

To make the data easily understandable we write it in the following table.

Marks obtained	50	56	58	59	60	62	70
No. of students (frequency)	3	2	5	3	2	3	2

The above table is called on **Discrete a Frequency Distribution Table**.

Grouped Frequency Distribution Table

When our data is large e.g. we are given marks out of 100 of 60 students. To make the data more sensible, we condense the data into groups like 50 – 55, 56 – 60,, 65 – 70.

These groupings are called **Classes** or **Class-intervals**. Size of class is called class - size or class width. In one case, class - size is 5. In each of the class least number is called lower class limit and greatest number is called upper class limit e.g. in 50 – 55, lower class limit is 50 and upper class limit is 55. The given data can now be arranged in tabular form as given in the adjoining table.

Marks Obtained	No. of Students (frequency)
50 – 55	3
56 – 60	12
61 – 65	3
66 – 70	2
Total	20

This table is called Grouped Frequency Distribution Table. From this, we can observe that more than 60% marks are obtained by $3 + 2 = 5$ students out of 20 students.

Inclusive and Exclusive Grouped Frequency Distribution

Frequency distribution in which groups are of the form 1 – 9, 10 – 19, 20 – 29... in which the upper limit of one class interval does not coincide with the lower limit of the next class interval is called inclusive grouped frequency distribution. In this type of distribution

both the lower limit and upper limit belongs to that class only e.g. for inclusive class 1 – 9 both 1 and 9 belongs to class 1 – 9. Actually the inclusive class 1 – 9 includes equal and greater than ($1 - 0.5 = 0.5$) but less than ($9 + 0.5 = 9.5$).

There is another type of frequency distribution like 10 – 20, 20 – 30, 30 – 40.... in which upper limit of one class interval is same as the lower class limit of succeeding class interval. This is called exclusive grouped frequency distribution. In this type of grouped frequency distribution the upper limit of class interval belongs to succeeding class.

An inclusive frequency distribution can be converted to exclusive frequency distribution in the following way :

- Find the difference between the upper limit of a class and the lower limit of its succeeding class.
- Add half of this difference to each of the upper limits and subtract the same from each of the lower limits

Consider frequency distribution in the given table.

1 – 9, 10 – 19, 20 – 29...

Lower limit of 10 – 19 = 10

Upper limit of 1 – 9 = 9

Difference = $10 - 9 = 1$

Half the difference = $1/2 = 0.5$

$\therefore$ New class interval for 1 – 9 is $(1 - 0.5) - (9 + 0.5)$

i.e. 0.5 – 9.5 similarly other intervals can be formed.

Inclusive Class Interval	Exclusive Class Interval
1 – 9	0.5 – 9.5
10 – 19	9.5 – 19.5
20 – 29	19.5 – 29.5

Class Marks

Class marks are the mid points of the class intervals. To find the class-mark of a class interval, we find the sum of upper limit and lower limit of a class and divide it by 2.

$$\therefore \text{Class Mark} = \frac{\text{Lower limit} + \text{Upper limit}}{2}$$

For a class interval 10 – 20,

$$\text{Class mark} = \frac{10 + 20}{2} = 15$$

GRAPHICAL REPRESENTATION OF DATA

To make the data useful and attractive, we represent them by means of diagrams and graphs. The representation of quantitative (numerical) data suitably through chart and diagram is known as graphical representation of data.

I. Pictograph

Pictorial representation of numerical data, using picture symbol, is known as pictograph of the data. Most business and industrial concerns show their product by depicting pictures.

Steps involved in drawing a pictograph

- Allot a symbol for a unit
- The symbol used must be simple, clear and self explanatory
- The quantity that each symbol represent must be clearly indicated
- All symbols used in the problem should be of uniform size.

ILLUSTRATION : 1

Read the following numerical data about the total number of students scored more than 90% aggregate in annual examination in classes I to V of a school with 120 students in each class having 4 sections and with 30 students in each section.

Class	I	II	III	IV	V
No. of students scored more than 90% aggregate	10	8	8	7	9

SOLUTION :

The pictograph of the above data is shown in following **table**, where one student is represented by one Δ .

I	$\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta$	10
II	$\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta$	8
III	$\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta$	8
IV	$\Delta\Delta\Delta\Delta\Delta\Delta\Delta$	7
V	$\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta\Delta$	9

II. Bar Graph

Representation of numerical data through pictograph is both tedious and time-consuming. In some situations we may have to draw part pictorial symbol which may not convey some given numbers correctly. Therefore instead of picture symbols we draw horizontal or vertical bars (rectangles) to represent the given data. Such a representation is called bar graph.

Bar graphs are the simple most popular method of showing data related to one variable. Bars are drawn, of uniform width with equal spaces in between them, on X-axis. On Y-axis height of bar (rectangle) is shown.

ILLUSTRATION : 2

The expenditure (in 10 lakhs of rupees) on health by ABC Pvt Ltd. During five years is given below:

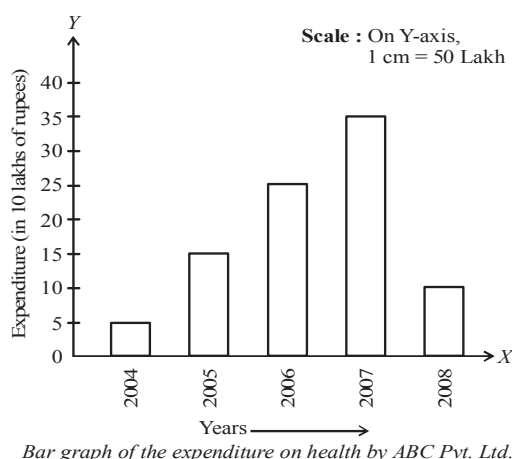
Year	Expenditure (in 10 lakh of rupees)
2004	5
2005	15
2006	25
2007	35
2008	10

Construct a bar graph to represent the above data.

SOLUTION :

- On a graph paper, draw two perpendicular lines to represent X-axis (horizontal) and Y-axis vertical.
- Along horizontal axis we write and mark the information 'Year'. Along vertical axis we write and mark Expenditure (in 10 lakhs of rupees)
- Choose a suitable scale to show height of the bars along vertical axis.
1 cm = 50 lakhs
- Calculate heights of various bars
Year 2004 = 1 cm
Year 2005 = 3 cm
Year 2006 = 5 cm
Year 2007 = 7 cm
Year 2008 = 2 cm

The required bar graph is shown below :



NOTE : The breadth of rectangles in a bar graph are uniform.

III. Histograms

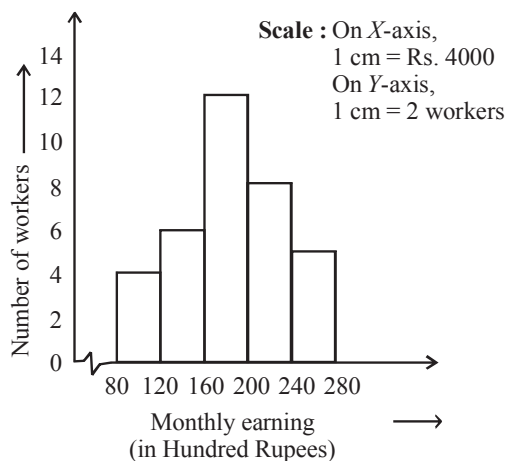
A histogram is very similar to bar graph. However, unlike a bargraph, there is no gap between the bars of a histogram. Histograms are used for exclusive (or continuous) class intervals. In a histogram, height of bars are proportional to the frequencies represented.

ILLUSTRATION : 3

Draw a histogram to represent the following data of the earnings of workers:

Monthly Earnings (in hundred Rupees)	No. of Workers
80 – 120	4
120 – 160	6
160 – 200	12
200 – 240	8
240 – 280	5

SOLUTION :



- On the horizontal axis, mark the class intervals with a uniform scale.
- On the vertical axis mark the frequencies, with a uniform scale.
- Construct rectangles with class intervals as bases and the corresponding frequencies as heights

In the illustration, the class sizes are same (40), so the rectangles will have the same width and the height of each rectangle will be proportional to frequency (no. of workers).

Since the first class interval is started from 80 not from 0, hence we use a 'kink' or a break on the X-axis.

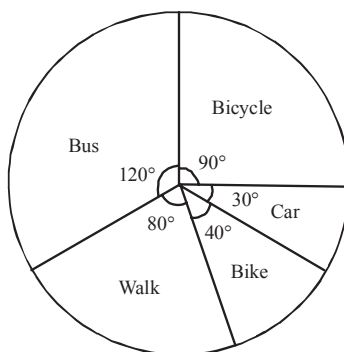
IV. Pie Chart or Circle Graph

A pie chart is a way of showing how something is shared or divided. It shows the relationship between a whole and its parts.

In pie charts, entire circle of appropriate radius represent total allocation. The circle is divided into sectors. The size of each sector is proportional to the information it represents.

$$\text{Central angle} = \frac{\text{Frequency}}{\text{Total Frequency}} \times 360^\circ$$

Following figure is the Pie chart which shows how 720 students usually come to school.



The whole group of 720 students is represented by a circle. The total angle at the centre of the circle is 360° .

Here, the angle of 120° at the centre corresponds to students who come to school by bus.

Since angle of 360° at the centre corresponds to whole group of 720° students.

$$\therefore 1^\circ \text{ corresponds to } \frac{720}{360} = 2 \text{ students}$$

$$\therefore 120^\circ \text{ corresponds to } \frac{720}{360} \times 120 = 240 \text{ students}$$

Similarly, students coming by bike are $\frac{720}{360} \times 40 = 80$

Students walking to school are $\frac{720}{360} \times 80 = 160$ and so on.

NOTE : We use a pie chart when we want to display data as parts of a whole.

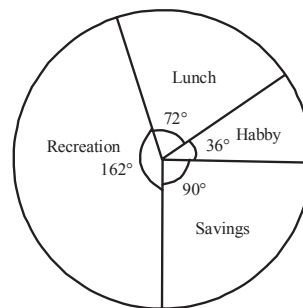
ILLUSTRATION : 4

The way Mr. Goel spends his allowance is given below. Draw the pie chart.

Item	Percent
Hobby	10%
Lunch	20%
Recreation	45%
Savings	25%
Total	100%

SOLUTION :

Item	%	Fractional Part	Central Angle
Hobby	10%	$\frac{10}{100} = \frac{1}{10}$	$\frac{1}{10} \times 360^\circ = 36^\circ$
Lunch	20%	$\frac{20}{100} = \frac{1}{5}$	$\frac{1}{5} \times 360^\circ = 72^\circ$
Recreation	45%	$\frac{45}{100} = \frac{9}{20}$	$\frac{9}{20} \times 360^\circ = 162^\circ$
Savings	25%	$\frac{25}{100} = \frac{1}{4}$	$\frac{1}{4} \times 360^\circ = 90^\circ$



1. Draw a circle of convenient radius.
2. Divide it into sectors of central angles 36° , 72° , 162° and 90° .
3. Write the items in the corresponding sectors.

The obtained figure is the required pie chart.

PROBABILITY

In our day to day life when we are not sure of happening of an event then we use the words probably, chance, possible etc. for example

- (a) Probably India may win the cricket match against Australia this time.
- (b) Probably it may rain today.
- (c) Chances of suresh passing the examination are very low.

In all the above statements probably, chances, etc convey the sense of certainty about the occurrence of any particular event. It appears that there cannot be any exact measurement for these uncertainties. But in Mathematics, we have methods for calculating the degree of certainty of an event in numerical value under certain conditions.

Some Basic Terms

Random Experiments : An experiment whose outcome has to be among a set of events that are completely known but whose exact outcome is unknown is called a random experiment, e.g. throwing of a dice, tossing of a coin. etc.

Sample Space : It is the set of all possible outcomes of a random experiment. e.g. when a coin is tossed. Sample space is (Head, Tail). Sample space when a dice is thrown is $\{1, 2, 3, 4, 5, 6\}$

Event : The set representing the desired outcome of a random experiment is called the event, e.g. getting head in a toss of coin is an event.

Trial : Performing an experiment once is called a trial.

Equally Likely Events: The results of a random experiment are said to be equally likely if the different outcomes have the same or equal chances of occurrence. For example, when a dice is thrown, each of the faces having 1,2,3,4,5 or 6 numbers are equally likely to appear. Once the event : getting a head and the event : setting a tail are mutually exclusive events.

Favourable Outcome(s) : The outcome(s) which ensure the occurrence of an event are called favourable outcome(s) to that event. For example, the favourable outcomes to the occurrence of an even number when a die is thrown are 2, 4 or 6.

Probability : If an event A can happen, then the probability of the happening of the event A,

$$P(A) = \frac{\text{Number of favourable outcomes}}{\text{Total number of possible outcomes}}$$

For example: On tossing a coin, there are two possibilities either head may come up or tail may come up. Therefore, total no. of possible outcomes = 2.

The number of favourable outcomes for getting a head = 1

$$\therefore \text{probability of getting a head } P(H) = \frac{1}{2}$$

$$\text{Similarly, probability of getting a tail } P(T) = \frac{1}{2}$$

CONNECTING TOPIC

Cumulative Frequency Table : The cumulative frequency corresponding to a class is the sum of all the frequencies upto and including that class.

For example : consider marks obtained by 45 students of class VIII in an assessment is given below :

Marks	0	5	10	14	16	18	20
No. of students Frequency	1	4	8	9	12	6	5

From the table we conclude that

1 student secured zero marks

4 students secured 5 marks

8 students secured 10 marks and so on.

Now, how many students secured 10 marks or less

To answer this question we have to add all the students who secured 10 or less marks i.e. $(1 + 4 + 8) = 13$ students.

13 is termed as cumulative frequency of marks 10. Similarly cumulative frequency of marks 16 is $(1 + 4 + 8 + 9 + 12) = 34$.

Types of cumulative frequency table : There are two types of cumulative frequency table :

(a) Less than type

(b) More than type

ILLUSTRATION : 5

Construct cumulative frequency distribution (less than type and more than type) from the following data

Marks obtained	0 – 20	20 – 40	40 – 60	60 – 80	80 – 100
No. of students	2	7	11	18	12

SOLUTION :

(a) **Less Than Type Cumulative Frequency Distribution Table.**

Marks Obtained	No of Students (Cumulative Freq.)
Less than 20	2
Less than 40	$2 + 7 = 9$
Less than 60	$2 + 7 + 11 = 20$
Less than 80	$2 + 7 + 11 + 18 = 38$
Less than 100	$2 + 7 + 11 + 18 + 12 = 50$

Here the numbers 20, 40, 60 are all upper limits of given class intervals.

(b) **More Than Type Cumulative Frequency Distribution table.**

Marks Obtained	No of Students (Cumulative Freq.)
More than or equal to 0	$12 + 18 + 11 + 7 + 2 = 50$
More than or equal to 20	$12 + 18 + 11 + 7 = 48$
More than or equal to 40	$12 + 18 + 11 = 41$
More than or equal to 60	$12 + 18 = 30$
More than or equal to 80	$12 = 12$

Here the numbers 0, 20, 40, 60, 80 are all lower limits of given class intervals.

Variate	TallyMarks	Frequency
15		4
16		6
18		6
20		6
24		5
25		5
27		3
28		3
29		1
30		1
Total		40

4. The data on the mode of transport used by 720 students are given below:

Mode of transport	Bus	Cycle	Train	Car	Scooter
Number of students	120	180	240	80	100

Represent the above data by a pie chart.

Sol. Total number of students = 720

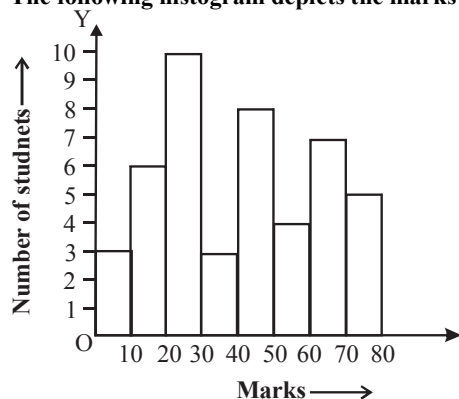
Central angle for a mode of transport

$$= \left(\frac{\text{number of students using that mode}}{\text{total number of students}} \times 360 \right)^\circ$$

Calculation of central angles

Mode of transport	Number of students	Central angle
Bus	120	$\left(\frac{120}{720} \times 360 \right)^\circ = 60^\circ$
Cycle	180	$\left(\frac{180}{720} \times 360 \right)^\circ = 90^\circ$
Train	240	$\left(\frac{240}{720} \times 360 \right)^\circ = 120^\circ$
Car	80	$\left(\frac{80}{720} \times 360 \right)^\circ = 40^\circ$
Scooter	100	$\left(\frac{100}{720} \times 360 \right)^\circ = 50^\circ$

5. The following histogram depicts the marks obtained by 45 students of a class:



Look at the histogram and answer the following questions:

- What is the class size?
- How many students obtained less than 10 marks?
- How many students obtained 30 or more marks but less than 40?
- What is the interval of highest marks and how many students are there in this interval?
- If passing marks are 30, what is the number of failures?

Sol. (i) The class intervals are 0 – 10, 10 – 20, 20 – 30, ..., 70 – 80. So class size = 10.

(ii) Number of students getting less than 10 marks = 3

(iii) Number of students getting 30 or more marks but less than 40 marks = 3.

(iv) Clearly, 70 – 80 is the interval of highest marks. The height of the corresponding bar in the histogram is 5 units. Thus, 5 students are there in the interval of highest marks.

(v) The failures are the students who obtained marks in the intervals 0 – 10, 10 – 20 and 20 – 30.

$\therefore$ Number of failures = (3 + 6 + 10) = 19

6. Find the range of the data 143, 148, 135, 150, 128, 139, 149, 146, 151 and 153.

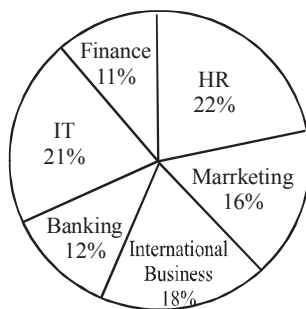
Sol. Maximum value of the given data = 153

Minimum value of the given data = 128

∴ Range = Maximum value – minimum value
 Range = 153 – 128
 Range = 25

7. Study the pie-chart carefully to answer the following questions.

Percentage of students studying different specialisations in a management institute
 Total number of students = 4500



Ratio of men to women in each specialisation

Specialisation	Men	Women
HR	2	3
Marketing	7	5
International Business	5	1
Banking	17	10
IT	1	2
Finance	5	4

- (i) What is the total number of men studying in the institute?
 (a) 1665 (b) 2421 (c) 2025 (d) 1954
- (ii) The number of women studying Marketing is approximately what per cent of the total number of students in the institute?
 (a) 5 (b) 12 (c) 15 (d) 7
- (iii) What is the total number of men studying Banking and International Business?
 (a) 1125 (b) 297 (c) 1015 (d) 594
- (iv) The number of women studying Finance is what per cent of the number of men studying the same?
 (a) 80 (b) 65 (c) 95 (d) 70

Sol.

- (i) (b) It is obvious from above.

Specialisation		Total number of students	Male	Female
1	HR	990	396	594
2	Marketing	720	420	300
3	International Business	810	675	135
4	Banking	540	340	200
5	IT	945	315	630
6	Finance	495	275	220
		4500	2421	2079

- (ii) (d) The required per cent = $\frac{300}{4500} \times 100 = 6.6\% = 7\%$
- (iii) (c) The required number of men = $675 + 340 = 1015$
- (iv) (a) The required per cent = $\frac{220}{275} \times 100 = 80\%$

8. The distribution of weights (in kg) of 50 students in a class is as following :

Weight	35 – 38	38 – 41	41 – 44	44 – 47	47 – 50
No. of students	3	13	19	10	5

- Find the lower limit of the class 44 – 47.
- Find the upper limit of the class 38 – 41.
- Find the size of the class intervals.
- How many students weight less than 41 kg ?
- How many students weight less than 47 kg ?
- Find the mid value of the class intervals 41 – 44 and 47 – 50.

Sol. (a) The lower limit of the class 44 – 47 is 44 kg.
 (b) The upper limit of the class 38 – 41 is 41 kg.
 (c) The size of the class intervals is 3 kg. Since the difference of the lower and the upper limits is 3 everywhere.
 (d) Cumulative Frequency table.

Class intervals	f	Cf
35 – 38	3	3
38 – 41	13	3+13=16
41 – 44	19	16+19=35
44 – 47	10	35+10=45
47 – 50	5	45+5=50

No. of students whose weigh less than 41 kg is 16.

(e) No. of students whose weigh less than 47 kg is 45.

(f) The mid value of 41 – 44 is $\frac{41+44}{2} = 42.5$

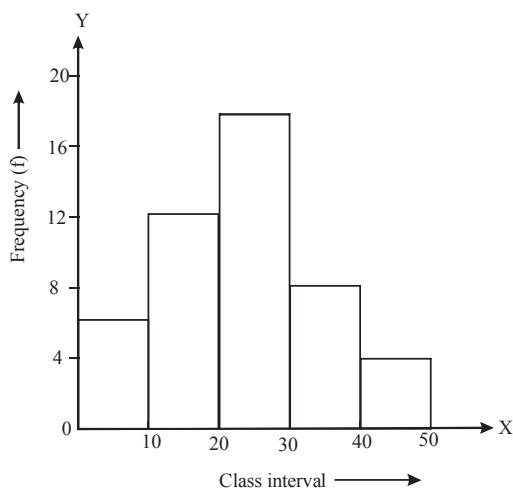
The mid value of 47 – 50 is $\frac{47+50}{2} = 48.5$

9. Represent the following frequency distribution by a histogram.

Class Intervals	0 – 10	10 – 20	20 – 30	30 – 40	40 – 50
Frequency (f)	6	12	18	8	4

Sol. Scaling the x-axis as 1 cm = 10 and the y-axis as 1 cm = 4.

The diagram as shown below is the required histogram.



10. Find the probability that a number selected at random from the numbers 1 to 25 is not a prime number when each of the given number is equally likely to be selected. Also find the probability of not getting a prime number.

Sol. Prime number between 1 to 25 = {2, 3, 5, 7, 11, 13, 17, 19, 23}
 Total number of prime numbers = 9
 $\therefore$ Number of favourable outcomes = 16
 Total no. of outcomes = 25

$$\therefore \text{probability} = \frac{\text{Total number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{16}{25}$$

11. Find the probability of getting a number less than 5 in a single throw of dice.

Sol. Total number of outcomes = 6
 $\therefore$ Number of favourable outcomes = 4 {1, 2, 3, 4}

$$\therefore \text{Probability} = \frac{4}{6} = \frac{2}{3}$$

12. A card is drawn at random from a well shuffled pack of 52 cards. Find the probability that the card drawn is neither a red card nor a queen.

Sol. Total number of outcomes = 52
 There are 26 red cards including 2 red queens and two more queens are there. Thus we have
 $26 + 2 = 28$ cards
 $\therefore$ Total number of favourable outcomes = $(52 - 28) = 24$ cards.

$$\therefore \text{Probability} = \frac{24}{52} = \frac{6}{13}$$

13. A bag contains 5 black, 7 red and 3 white balls. A ball is drawn from the bag at random. Find the probability that the ball drawn is

(i) red (ii) not black.

Sol. Number of black balls = 5 ; Number of red balls = 7; Number of white balls = 3
 $\therefore$ Total number of balls in the bag = $5 + 7 + 3 = 15$
 $\therefore$ Total number of outcomes = 15
 (i) Total number of favourable outcomes = 7

$$\therefore \text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{7}{15}$$

(ii) Total number of favourable outcomes = 10

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{10}{15} = \frac{2}{3}$$

Alternatively,

$$P(\text{Not Black}) = 1 - P(\text{Black})$$

$$P(\text{Not Black}) = 1 - \frac{5}{15} = 1 - \frac{1}{3} = \frac{2}{3}$$

14. A card is drawn from a well shuffled deck of playing cards. Find the probability of drawing

(a) A face card (b) A red face card.

Sol. Total no. of cards = 52.
 $\therefore$ total number of possible outcomes = 52
 (a) Number of face cards (Jack, Queen, King and Ace) in a deck of cards = $4 \times 4 = 16$
 $\therefore$ no. of favourable outcomes = 16

$$\therefore \text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{16}{52} = \frac{4}{13}$$

(b) $\therefore$ Number of favourable outcomes = $4 + 4 = 8$

$$\text{Probability} = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}} = \frac{8}{12} = \frac{2}{3}$$

SOLVED EXAMPLES BASED ON CONNECTING TOPICS

15. Prepare a Less Than Type frequency distribution table for the following data.

Class Interval	0-8	8-16	16-24	24-32	32-40	40-48
Frequency	8	12	20	16	8	10

Sol.

Less Than	Cummulative Frequency
Less than 8	8
Less than 16	$8 + 12 = 20$
Less than 24	$8 + 12 + 20 = 40$
Less than 32	$8 + 12 + 20 + 16 = 56$
Less than 40	$8 + 12 + 20 + 16 + 8 = 64$
Less than 48	$8 + 12 + 20 + 16 + 8 + 10 = 74$

16. The frequency distribution of scores obtained by 230 candidates in a medical entrance test is as follows:

Scores	400-450	450-500	500-550	550-600	600-650	650-700	700-750	750-800
No. of Candidate	20	35	40	32	24	27	18	34

Prepare a more Than Type frequency distribution table.

Sol.

Scores	No. of Candidates
More than or equal to 400	230
More than or equal to 450	$230 - 20 = 210$
More than or equal to 500	$210 - 35 = 175$
More than or equal to 550	$175 - 40 = 135$
More than or equal to 600	$135 - 32 = 103$
More than or equal to 650	$103 - 24 = 79$
More than or equal to 700	$79 - 27 = 52$
More than or equal to 750	$52 - 18 = 34$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. Range of the data 32, 46, 80, 56, 78, 21 is
2. Difference between upper limit and lower limit of a class is called class
3. Sum of central angles in a pie chart is
4. Histogram is a bar graph with no between the bars.
5. Each outcome or a collection of outcomes in an experiment makes an
6. An experiment whose outcomes cannot be predicted exactly in advance is called a experiment.
7. When a dice is rolled, the six possible outcomes are
8. The total number of outcomes when a coin is tossed is
9. Tally marks are used to find
10. An operation which can produce some well defined outcomes is called an

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. In a pie chart a whole circle is divided into sectors.
2. On throwing a dice once, the probability of occurrence of a composite number is $\frac{1}{2}$.
3. In a throw of a dice, the probability of getting an even number is the same as that of getting an odd number.
4. In a pie chart two or more central angles can be equal.
5. In 65 throws of a dice, 5 is obtained 22 times. Now, in a random throw of a dice, the probability of getting 5 is $\frac{22}{65}$.
6. A dice has 6 faces.
7. In a dice, the one face shows 4, then its opposite face shows 2.
8. In toss of a coin, all possible outcomes are head and tail.
9. In a histogram, for each class a rectangle is constructed with base as size of the class.
10. An experiment whose outcome is sure is called a certain experiment.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1.	Column-I	Column-II
	(A) Sum of all central angles in	(p) $\frac{1}{6}$
	(B) Chance of being Friday tomorrow, if today is Saturday	(q) 1
	(C) Probability of getting 2 when a die is thrown	(r) No
	(D) Probability of getting 9 when a die is thrown	(s) 360°
	(E) Probability that a number less than 7 will appear on the top if a die is thrown	(t) 0
2.	Column-I	Column-II
	(A) Raw data can be presented through	(p) circle graph
	(B) Histogram is a type of	(q) class - interval
	(C) Data can also presented using	(r) observation
	(D) A group into which the raw frequency data is condensed is called as distribution	(s) grouped data
	(E) Each numerical figure in a data is called an	(t) Bar diagram

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Name the three types of graphs to represent the statistical data.
2. Give two example of a random experiment.
3. Find the probability that a vowel selected at random in English language is an "i".

Data Handling



4. In a study of number of accidents per day, the observations for 30 days were obtained as follows:
4, 3, 5, 6, 4, 3, 2, 5, 4, 2, 6, 2, 1, 2, 2
0, 5, 4, 6, 1, 3, 0, 5, 3, 6, 1, 5, 5, 2, 6.
Prepare a frequency table.
5. The marks obtained by Kunal in his annual examination are shown below:

Subject	Hindi	English	Mathematics	Science	Social studies
Marks Obtained	63	75	90	72	58

Draw a bar graph to represent the above data.

6. It is known that a box of 600 electric bulbs contains 12 defective bulbs. One bulb is taken out at random from this box. What is the probability that it is a non-defective bulb?

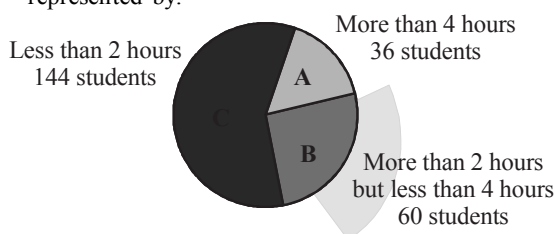
Short Answer Questions :

DIRECTIONS : Give answer in two to three sentences.

- Define the equally likely events.
- What is the probability of an event?
- Draw a pie-chart of the data given below.
The time spent by a child during a day.
Sleep – 8 hours
School – 6 hours
Home work – 4 hours
Play – 4 hours
Others – 2 hours
- For the following frequency distribution, draw a histogram.

Class-interval	Frequency
0 - 5	2
5 - 10	7
10 - 15	18
15 - 20	10
20 - 25	8
25 - 30	5

- What is the difference between Bar graph and Histogram?
- In a survey, 240 parents of a middle school were asked about the number of hours spent by their children on their study (including homework) in a day. The data obtained is represented by:



From this pie chart,

- Find the proportion of each sector (i.e., part: whole)
- How many students spend more than 2 hours and less than 4 hours doing home work?

- How many students put less than 4 hours doing home work?
 - Convert the number of students in each sector in terms of percentage.
7. A bag contains 5 red balls, 8 white balls, 4 green balls and 7 black balls. If one ball is drawn at random, find the probability that it is:
- black
 - red
 - not green
8. The frequency distribution of weights (in kg) of 40 persons of a locality is given below:

Weights (in kg)	40 - 45	45 - 50	50 - 55	55 - 60	60 - 65
Frequency	4	12	13	6	5

- What is the upper limit of the fourth class interval?
- Find the class marks of all the classes.
- What is the class size of each class interval?
- Which class interval has the highest frequency?

Long Answer Questions :

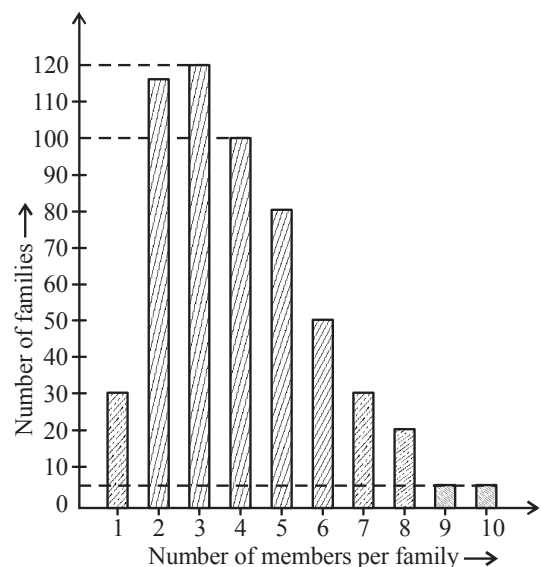
DIRECTIONS : Give answer in four to five sentences.

- How will you proceed to draw a Histogram?
- The education level of the males in U.P. in 1984 is given below.

Education level	Middle	Secondary	Intermediate	Graduate and above
No. of males	91	58	35	33

Draw a bar graph for the above data.

- Study the bar graph and answer the following questions:



Bar graph of the number of families with different number of members per family in a locality

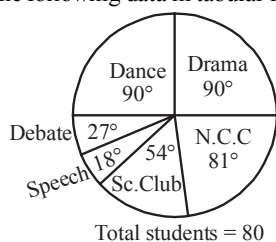
- What information is given by the bar graph?
- How many number of families having 4 members?
- How many members per family are there in the maximum number of families?

4. The number of cars sold of a company, in different months in a city are given below:

Months	Jan	April	June	Aug	Sep	Dec
No. of cars sold	200	500	250	300	400	450

Represent the above information by a pictograph.

5. Represent the following data in tabular form:



6. Represent the following data by a pie-diagram

Items	Expenditure	
	Family A	Family B
Food	4000	6400
Clothing	2500	4800
Rent	1500	3200
Education	400	1000
Miscellaneous	1600	600
Total	10,000	16000

2 EXERCISE

Text-Book Exercise :

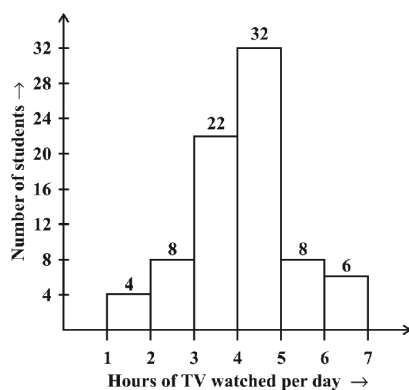
1. The shoppers who come to a departmental store are marked as: man (M), woman (W), boy (B) or girl (G). The following list gives the shoppers who came during the first hour in the morning:
 W W G B W W M G G M M W W W G B M W B G G
 M W W M M W W
 W M W B W G M W W W W G W M M W W M W G W
 M G W M M B G G W

Make a frequency distribution table using tally marks. Draw a bar graph to illustrate it.

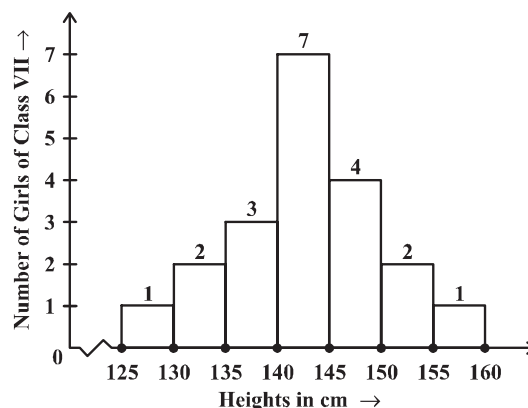
2. The number of hours for which students of a particular class watched television during holidays is shown through the given graph.

Answer the following.

- For how many hours did the maximum number of students watch TV?
- How many students watched TV for less than 4 hours?
- How many students spent more than 5 hours in watching TV?



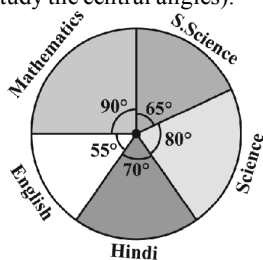
3. Observe the histogram and answer the questions given below.



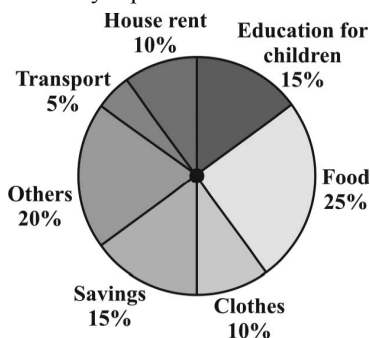
- What information is being given by the histogram?
 - Which group contains maximum girls?
 - How many girls have a height of 145 cms and more?
 - If we divide the girls into the following three categories, how many would there be in each?
 150 cm and more — Group A
 140 cm to less than 150 cm — Group B
 Less than 140 cm — Group C
4. A group of 360 people were asked to vote for their favourite season from the three seasons rainy, winter and summer.
- Which season got the most votes?
 - Find the central angle of each sector.
 - Draw a pie chart to show this information.

Season		No. of votes
Summer		90
Rainy		120
Winter		150

5. The adjoining pie chart gives the marks scored in an examination by a student in Hindi, English, Mathematics, Social Science and Science. If the total marks obtained by the students were 540, answer the following questions.
- In which subject did the student score 105 marks?
- (Hint: for 540 marks, the central angle = 360° . So, for 105 marks, what is the central angle?)
- How many more marks were obtained by the student in Mathematics than in Hindi?
 - Examine whether the sum of the marks obtained in Social Science and Mathematics is more than that in Science and Hindi.
- (Hint: Just study the central angles).



6. Adjoining pie chart gives the expenditure (in percentage) on various items and savings of a family during a month.
- On which item, the expenditure was maximum?
 - Expenditure on which item is equal to the total savings of the family?
 - If the monthly savings of the family is ₹ 3000, what is the monthly expenditure on clothes?



7. When a die is thrown, list the outcomes of an event of getting
- (a) a prime number (b) not a prime number.
 - (a) a number greater than 5 (b) a number not greater than 5.

8. Numbers 1 to 10 are written on ten separate slips (one number on one slip), kept in a box and mixed well. One slip is chosen from the box without looking into it. What is the probability of.
- getting a number 6?
 - getting a number less than 6?
 - getting a number greater than 6?
 - getting a 1-digit number?
9. A bag has 4 red balls and 2 yellow balls. (The balls are identical in all respects other than colour). A ball is drawn from the bag without looking into the bag.
- What is probability of getting a red ball? Is it more or less than getting a yellow ball?

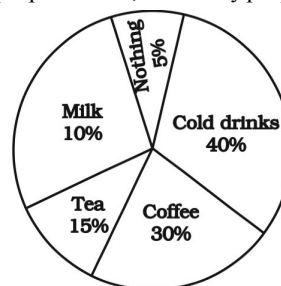
Exemplar Questions :

1. For the development of basic infrastructure in a district, a project of ₹ 108 crore approved by Development Bank is as follows:

Item Head	Road	Electricity	Drinking Water	Sewerage
Amount in crore (₹)	43.2	16.2	27.00	21.6

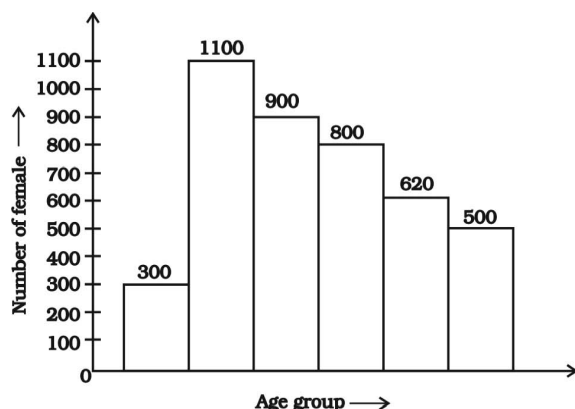
Draw a pie chart for this data.

2. A survey was carried out to find the favourite beverage preferred by a certain group of young people. The following pie chart shows the findings of this survey.
- From this pie chart answer the following:
- Which type of beverage is liked by the maximum number of people.
 - If 45 people like tea, how many people were surveyed?



3. In a hypothetical sample of 20 people, the amount of money (in thousands of rupees) with each was found to be as follows:
- 114, 108, 100, 98, 101, 109, 117, 119, 126, 131, 136, 143, 156, 169, 182, 195, 207, 219, 235, 118.
- Draw a histogram of the frequency distribution, taking one of the class intervals as 50 – 100.
4. The below histogram shows the number of literate females in the age group of 10 to 40 years in a town.
- Write the class intervals assuming all the classes are of equal width.
 - What is the class width?

- (iii) In which age group are literate females the least?
 (iv) In which age group is the number of literate females the highest?

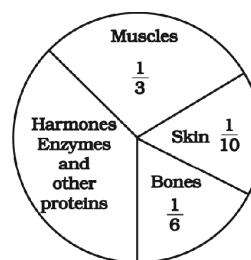


5. Construct a frequency distribution table for the following weights (in grams) of 35 mangoes, using the equal class intervals, one of them is 40 – 45 (45 not included).
 30, 40, 45, 32, 43, 50, 55, 62, 70, 70, 61, 62, 53, 52, 50, 42, 35, 37, 53, 55, 65, 70, 73, 74, 45, 46, 58, 59, 60, 62, 74, 34, 35, 70, 68.
- (i) How many classes are there in the frequency distribution table?
 (ii) Which weight group has the highest frequency?
6. A dice is rolled once. What is the probability that the number on top will be
- (i) Odd (ii) Greater than 5
 (iii) A multiple of 3 (iv) Less than 1
 (v) A factor of 36 (vi) A factor of 6
7. Given below is a frequency distribution table. Read it and answer the questions that follow:

Class Interval	Frequency
10 – 20	5
20 – 30	10
30 – 40	4
40 – 50	15
50 – 60	12

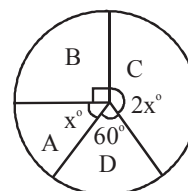
- (i) What is the lower limit of the second class interval?
 (ii) What is the upper limit of the last class interval?
 (iii) What is the frequency of the third class?
 (iv) Which interval has a frequency of 10?
 (v) Which interval has the lowest frequency?
 (vi) What is the class size?
8. A glass jar contains 6 red, 5 green, 4 blue and 5 yellow marbles of same size. Hari takes out a marble from the jar at random. What is the probability that the chosen marble is of red colour?
9. Numbers 1 to 5 are written on separate slips, i.e one number on one slip and put in a box. Wahida pick a slip from the box without looking at it. What is the probability that the slip bears an odd number?

10. A coin is tossed 200 times and head appeared 120 times. The probability of getting a head in this experiment is
11. What is the probability of choosing a vowel from the alphabets?
12. In a school only, 3 out of 5 students can participate in a competition.
 What is the probability of the students who do not make it to the competition?
13. The following pie chart represents the distribution of proteins in parts of a human body. What is the ratio of distribution of proteins in the muscles to that of proteins in the bones?



HOTS Questions :

1. The pie chart shows the number of participants from four schools A, B, C and D taking part in a tennis tournament. Given that there are 12 participants from school D. Find the number of participants from school C.

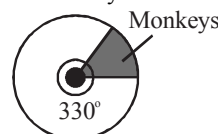


2. In a survey of 500 ladies, it was found that 180 like coffee while rest of the ladies dislike it. From these ladies, one is chosen at random. What is the probability that the chosen lady dislikes coffee?
3. Pooja spends different hours of a working day as follows:

Activity	Number of hours
School	8
Coaching	3
Play	2
Sleep	8
Other	3

What is the difference in central angles for sleep and play in the pie chart?

4. The circle graph represents a total of 240 animals. The shaded area represents the number of monkeys. How many of the animals are monkeys?



3 EXERCISE

Single Option correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- What is the probability that a leap year selected at random will contain 53 sundays?
 - $\frac{1}{2}$
 - $\frac{2}{7}$
 - $\frac{5}{7}$
 - $\frac{6}{7}$
- An unbiased dice is tossed, The probability of getting a multiple of 3 is
 - $\frac{2}{3}$
 - $\frac{1}{3}$
 - $\frac{1}{6}$
 - $\frac{1}{2}$
- The probability of getting not more than 7 in rolling of a dice is
 - $\frac{1}{7}$
 - $\frac{1}{4}$
 - $\frac{1}{2}$
 - 1
- The class size of an interval 10 – 20 is
 - 10
 - 5
 - 20
 - 15
- If  represents 7 boxes of balls, what does  stand for?
 - 30 balls
 - 35 balls
 - 35 boxes of balls
 - 70 boxes of balls
- The heights in cm of 10 students in a class are 134, 138, 142, 136, 129, 144, 137, 138, 142, 140. The range of the above data is
 - 12
 - 10
 - 15
 - 20

- The population of four towns A, B, C and D as on 2011 are as follows:

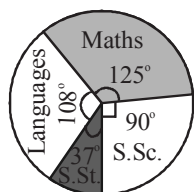
Town	Population
A	6863
B	519
C	12185
D	1755

What is the most appropriate diagram to present the above data?

- Pie chart
 - Bar chart
 - Histogram
 - Line graph
- From the following frequency table, find out how many students failed if the pass marks are 40.

Mark	0–19	20–39	40–49	50–59	60–89	90–100
Number of students	8	6	15	13	18	7

- 29
 - 7
 - 8
 - 14
- From the following table, the percentage of the families with less than 3 children is
- | | | | | | |
|--------------------|---|---|----|---|---|
| Number of children | 0 | 1 | 2 | 3 | 4 |
| Number of families | 5 | 8 | 15 | 8 | 4 |
- 70%
 - 60%
 - 54%
 - 45%
- The lower limit of 45 – 50 is
 - 45
 - 50
 - 5
 - 47.5
 - The probability of drawing an ace from a deck of cards is
 - $\frac{1}{4}$
 - $\frac{15}{12}$
 - $\frac{4}{52}$
 - $\frac{1}{52}$
 - Today is Sunday. What is the chance that tomorrow will be Monday?
 - No
 - Sure
 - Poor
 - Even
 - The adjoining pie chart shows the interests of the students of a school in different subjects. How many students like languages if there are 1800 students in the school?



- (a) 450 (b) 540
(c) 480 (d) 390

More than One Option correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d). Out of which ONE OR MORE may be correct.

- Which of the following experiments have equally likely outcomes?
 - A driver attempts to start a car. The car starts or does not start.
 - A player attempts to shoot a basketball. She/he shoots or misses the shot.
 - A trial is made to answer a true-false question. The answer is right or wrong.
 - A baby is born. It is a boy or a girl.
- In a test, the marks obtained by 15 students are 43, 73, 44, 93, 54, 64, 53, 24, 84, 40, 93, 33, 34, 74, 44. The probability that a pupil chosen at random passed the test, if the passing marks are 40 is
 - 8/15
 - 4/5
 - 7/15
 - 48/60
- An electronic machine chooses random numbers from 1 to 30. What is the probability that the number chosen is an even number?
 - 15/30
 - 1/10
 - 1/6
 - 1/2
- In a single throw of two dice, what is the probability of getting a total of 11.
 - 1/9
 - 1/18
 - 1/12
 - 2/36
- What is the probability that a card drawn at random from a pack of playing cards is a king?
 - 1/13
 - 2/13
 - 3/13
 - 4/52
- A bag contains 4 blue, 5 red and 7 green balls. If a ball is drawn at random, what is the probability that it is blue?
 - 4/16
 - 1/4
 - 1/256
 - 1/64

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the questions given below each passage.

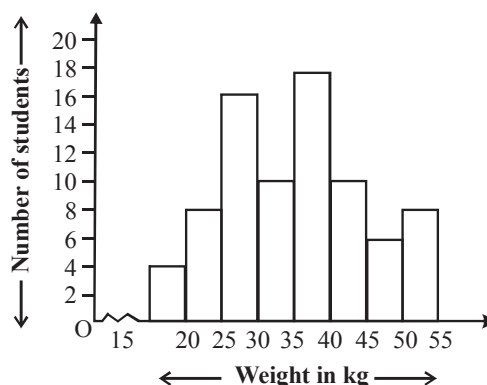
PASSAGE-I

The weights of new born babies (in kg) in a hospital on a particular day are as follows : 2.3, 2.2, 2.1, 2.7, 2.6, 3.0, 2.5, 2.9, 2.8, 3.1, 2.5, 2.8, 2.7, 2.9, 2.4.

- The range is
 - 2.1 kg
 - 0.5 kg
 - 1.0 kg
 - 1.5 kg
- How many baby weights are below 2.5 kg?
 - 4
 - 3
 - 2
 - 1
- How many babies weigh 2.8 kg?
 - 1
 - 2
 - 3
 - 4

PASSAGE - II

Given below is the histogram showing weight (in kg) of the students of VIII class in a school.



Study the histogram and answer the following questions:

- How many students have been observed?
 - 20
 - 55
 - 40
 - 80
- What is the class size?
 - 15
 - 10
 - 5
 - 55
- How many students weight less than 35 kg?
 - 38
 - 24
 - 16
 - 18

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.

(c) If **Assertion** is **correct** but **Reason** is **incorrect**.

(d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion:** The marks in maths of 7 students are as follows: 53, 61, 78, 68, 62, 8, 48. Here, range = $78 - 8 = 70$.

Reason: Range is defined as the difference between the highest and lowest values of observations in a data.

2. **Assertion:** Weighing of an apple is an example of random experiment.

Reason: A random experiment is that in which outcomes may differ each time when we perform an experiment.

3. **Assertion:** The probability of getting an odd number when a dice is thrown is $\frac{1}{2}$.

Reason: Probability of an event

$$= \frac{\text{Number of times the event occur in trials}}{\text{total number of trials}}$$

Multiple Matching Questions :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and five statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

- | 1. | Column-I | Column-II |
|-----|---|--|
| (A) | A bag contains 10 red and 8 white balls. One ball is drawn at random. The probability that the ball drawn is white is | (p) $\frac{1}{5}$ |
| (B) | The probability that a number selected from the numbers 1, 2, 3, ...15 is a multiple of 4 is | (q) $\frac{21}{26}$ |
| (C) | A bag contains 5 white and 7 blue balls. One ball is drawn at random. The probability of getting a white ball is | (r) $\frac{5}{12}$ |
| (D) | If we put 21 consonants and 5 vowels in a bag the probability of getting a constant is | (s) $\frac{4}{9}$
(t) $\frac{10}{24}$ |

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. A die is thrown 24 times. If number 4 come up 12 times, the probability of number '4' is $\frac{1}{k}$. Then the value of k is
2. The following data gives the number (in ten thousands) of electrified villages in India in some years

Year	1961	1971	1981	1982	1983	1984
Number of electrified villages (in ten thousands)	15	16	19	24	29	37

The unit digit of the range is.

3. The area cultivated under different crops in the year 1982 is given as follows:

Crop	Rice	Pulses	Wheat	Other cereals	Oil - Seeds	Sugarcane	Cotton
Area (in ten lakhs hectares)	38	23	23	40	17	4	8

If range = r^2 , then r equals _____.

4. Getting a number less than 1 when a die is thrown.

4

ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Prepare “Less than” and “More than” frequency distribution table for the following data.

Marks Obtained	50-60	60-70	70-80	80-90	90-100
No. of Students (Cumulative frequency)	4	8	12	6	6

2. The marks obtained by 400 students in medical entrance exam are given in the following table.

Marks	400-450	450-500	500-550	550-600	600-650	650-700	700-750	750-800
No. of Examiners	30	45	60	52	54	67	45	47

Prepare “Less than” and “more than” frequency distribution table.

3. The following table gives production yield per hectare wheat of 100 farms of a village.

Production yield (in kg/he)	Number of farms
50-55	2
55-60	8
60-65	12
65-70	24
70-75	38
75-80	16

Change the distribution to a more than type distribution.

4. During the medical check-up of 35 students of a class, their weights were recorded as follows:

Weight (in kg)	Number of students
Less than 38	0
Less than 40	3
Less than 42	5
Less than 44	9
Less than 46	14
Less than 48	28
Less than 50	32
Less than 52	35

Give a frequency table for this data.

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- 59
- width
- 360°
- gaps
- event
- Random
- 1, 2, 3, 4, 5, 6
- 2
- frequency
- experiment

True / False :

- True
- False
- True
- True
- True
- True
- False
- True
- True
- True

Match the Column :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (r); (C) $\rightarrow$ (p); (D) $\rightarrow$ (t); (E) $\rightarrow$ (q)
- (A) $\rightarrow$ (s); (B) $\rightarrow$ (t); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q); (E) $\rightarrow$ (r)

Very Short Answer Questions :

- (i) Bar graphs (ii) Histogram
(iii) Frequency polygon
- (i) Tossing a coin (ii) Rolling an unbiased dice
- Total no. of vowels = 5

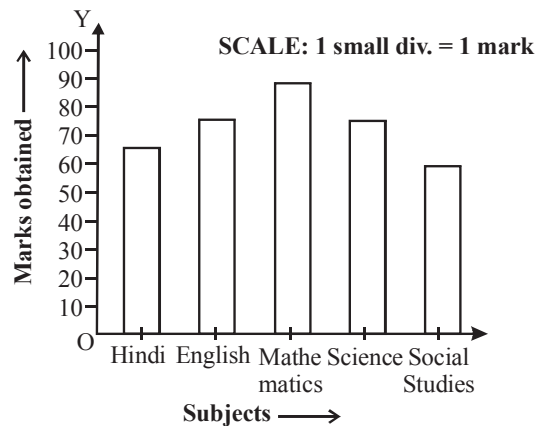
$$\therefore \text{Required Prob} = \frac{1}{5}$$

- Arranging the data in ascending order, we get
0, 0, 1, 1, 1, 2, 2, 2, 2, 2, 3, 3, 3, 3
4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 6, 6, 6, 6, 6
Now, we represent the above data in the tabular form as shown below:

Frequency Table

No. of accidents	Tally Mark	Frequency
0		2
1		3
2		6
3		4
4		4
5		6
6		5
Total		30

5.



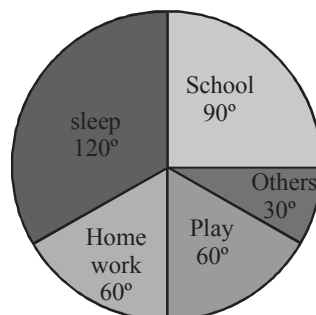
- Out of 600 electric bulbs one bulb can be chosen in 600 ways.
 $\therefore$ Total number of elementary events: 600
There are 588 (= 600 - 12) non-defective bulbs out of which one bulb can be chosen in 588 ways.
 $\therefore$ Favourable number of elementary events = 588
Hence, $P(\text{Getting a non-defective bulb})$
$$= \frac{588}{600} = \frac{49}{50} = 0.98.$$

Short Answer Questions :

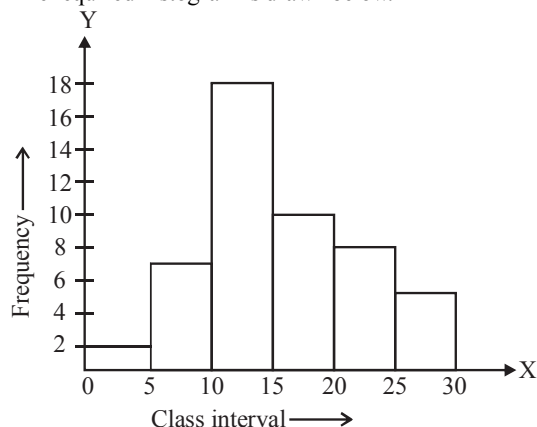
- Two or more events are said to be equally likely if one of them cannot be expected in preference to the other.
- Let we have n equally likely, mutually exclusive and exhaustive outcomes.
Then, The probability of occurring an event E with m favourable outcomes is given by

$$P(E) = \frac{\text{Number of favourable outcomes to } E}{\text{Total number of possible outcome}} = \frac{m}{n}$$

3.



4. The required histogram is drawn below.



6. (i) Proportion of students spend more than 4 hours

$$= \frac{36}{240} = \frac{3}{20}$$

Proportion of students spend more than 2 hours but

$$\text{less than 4 hours} = \frac{60}{240} = \frac{1}{4}$$

Proportion of students spend less than 2 hours

$$= \frac{144}{240} = \frac{3}{5}$$

(ii) Number of students = 60

(iii) Number of students = $144 + 60 = 204$

(iv) % of students in sector A = $\frac{36}{240} \times 100 = 15\%$

$$\text{\% of students in sector B} = \frac{60}{240} \times 100 = 25\%$$

$$\text{\% of students in sector C} = \frac{144}{240} \times 100 = 60\%$$

7. Total number of balls in the bag = $5 + 8 + 4 + 7 = 24$

$\therefore$ Total number of outcomes = 24

(i) There are 7 black balls in the bag.

$\therefore$ Favourable number of outcomes = 7

$$\text{Hence, } P(\text{Getting a black ball}) = \frac{7}{24}$$

(ii) There are 5 red balls in the bag.

$\therefore$ Favourable number of outcomes = 5

$$\text{Hence, } P(\text{Getting a red ball}) = \frac{5}{24}$$

(iii) There are $5 + 8 + 7 = 20$ balls which are not green.

$\therefore$ Favourable number of outcomes = 20

$$\text{Hence, } P(\text{Not Getting a green ball}) = \frac{20}{24} = \frac{5}{6}$$

8. (i) Upper limit of $55 - 60 = 60$

$$\text{(ii) Class marks of } 40 - 45 = \frac{40 + 45}{2} = \frac{85}{2} = 42.5$$

$$\text{Class marks of } 45 - 50 = \frac{45 + 45}{2} = \frac{95}{2} = 47.5$$

$$\text{Class marks of } 50 - 55 = \frac{50 + 55}{2} = \frac{105}{2} = 52.5$$

$$\text{Class marks of } 55 - 60 = \frac{55 + 60}{2} = \frac{115}{2} = 57.5$$

$$\text{Class marks of } 60 - 65 = \frac{60 + 65}{2} = \frac{125}{2} = 62.5$$

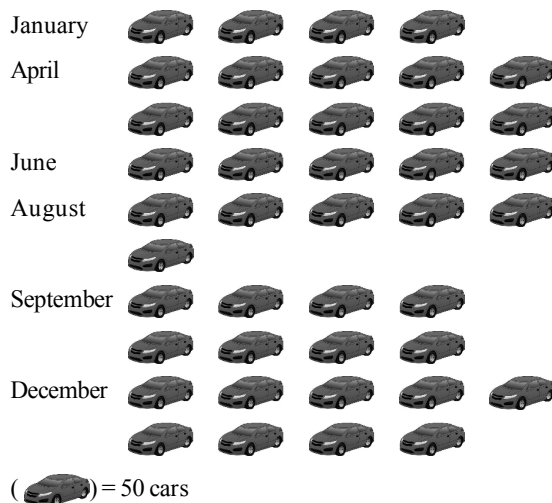
(iii) Class size = 5

(iv) $50 - 55$.

Long Answer Questions :

- To draw a histogram, we use the following steps:
 - On the horizontal axis (X-axis), mark the class intervals on a uniform scale.
 - On the Y-axis, mark the frequencies also on a uniform scale.
 - Construct rectangles with class-intervals as bases and the corresponding frequencies as heights.
- The bar graph represents the number of families with different number of members per family in a locality.
 - 100
 - We observe that the height of the rectangle at 3 on the horizontal axis is maximum and its top corresponds to 120 on the vertical axis.
So, maximum number of families is 120 and members per family are 3.

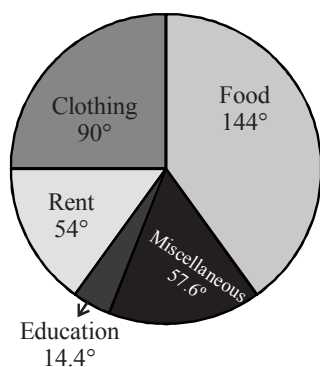
4. Months No. of cars sold



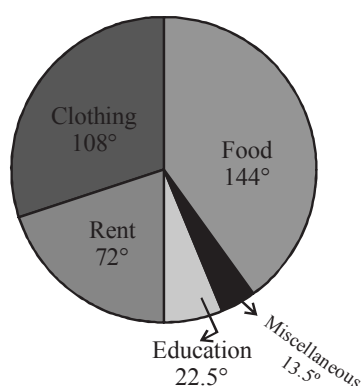
5.

Item	No. of Students
Drama	20
Dance	20
Debate	6
Speech	4
Sc. Club	12
N.C.C	18
Total	80

6. Family A



Family B



2 EXERCISE

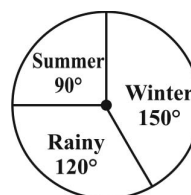
Text Book Exercise :

1.

Shopper	Tally marks	Number
W		28
M		15
B		5
G		12
	Total	60

2. (i) 4 - 5 hours (ii) 34 (iii) 14
3. (i) The given histogram shows the height of girls of class VII.
- (ii) Maximum no. of girls have (140–145) cm height.
- (iii) $4 + 2 + 1 = 7$
- (iv) Group A $\rightarrow 2 + 1 = 3$
 Group B $\rightarrow 7 + 4 = 11$
 Group C $\rightarrow 42 + 3 = 6$
4. (i) Winter (ii) Winter - 150° , Rainy - 120° , Summer - 90°

(iii)



5. (i) Hindi
 (ii) 30 marks
 (iii) Yes
6. (i) Expenditure is maximum on food.
 (ii) Expenditure on Education of children is the same (i.e., 15%) as the savings of the family.
 (iii) 15% represents ₹ 3000

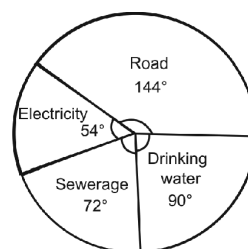
Therefore, 10% represents ₹ $\frac{3000}{15} \times 10$
 $= ₹ 2000$

7. Outcomes of an event of getting
 (i) (a) 2, 3, 5 (b) 1, 4, 6
 (ii) (a) 6 (b) 1, 2, 3, 4, 5
8. (i) $\frac{1}{10}$ (ii) $\frac{1}{2}$
 (iii) $\frac{2}{5}$ (iv) $\frac{9}{10}$
9. There are in all $(4 + 2) = 6$ outcomes of the event. Getting a red ball consists of 4 outcomes.

Therefore, the probability of getting a red ball is $\frac{4}{6} = \frac{2}{3}$. In the same way the probability of getting a yellow ball $\frac{2}{6} = \frac{1}{3}$. Therefore, the probability of getting a red ball is more than that of getting a yellow ball.

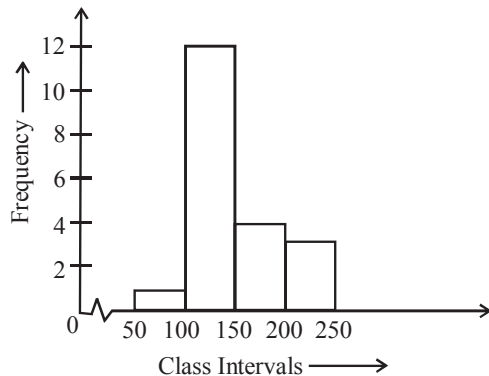
Exemplar Question :

1.



2. (i) Cold drinks (ii) 300

3.



4. (i) 10-15, 15-20, 20-25, 25-30, 30-35, 35-40
 (ii) 5 (iii) 10-15
 (iv) 15-20

5.

Class interval	Tally marks	Frequency
30-35		3
35-40		3
40-45		3
45-50		3
50-55		5
55-60		4
60-65		5
65-70		2
70-75		7
	Total	35

- (i) 9 (ii) (70-75) g
 6. (i) $\frac{1}{2}$ (ii) $\frac{1}{6}$ (iii) $\frac{2}{6}$ or $\frac{1}{3}$ (iv) 0
 (v) $\frac{5}{6}$ (vi) $\frac{4}{6}$ or $\frac{2}{3}$
 7. (i) 20 (ii) 60 (iii) 4 (iv) 20-30
 (v) 30-40 (vi) 10
 8. $\frac{3}{10}$ 9. $\frac{3}{5}$ 10. $\frac{3}{5}$ 11. $\frac{5}{26}$
 12. 0.4 13. 2:1

HOTS Questions :

1. Number of participants in school D = 12
 Central Angle = $\frac{\text{Frequency of variable}}{\text{Total frequency}} \times 360^\circ$
 $\Rightarrow 60^\circ = \frac{12}{\text{Total Number of students}} \times 360^\circ$
 $\Rightarrow \text{Total Number of students} = \frac{12 \times 360^\circ}{60^\circ} = 72$

$$\text{Total angle} = A + B + C + D = 360^\circ$$

$$\Rightarrow x^\circ + 90^\circ + 2x^\circ + 60^\circ = 360^\circ$$

$$\Rightarrow 3x^\circ + 150^\circ = 360^\circ$$

$$\Rightarrow 3x = 210^\circ$$

$$\Rightarrow x = 70^\circ$$

$$\angle A = 70^\circ \text{ and } \angle C = 70^\circ \times 2 = 140^\circ$$

Number of participants from school C

$$= \text{Central angle} \times \frac{\text{Total number of participants}}{360^\circ}$$

$$= \frac{140^\circ \times 72}{360^\circ} = 28$$

2. Number of ladies who like coffee = 180

Total number of ladies = 500

Required probability that chosen lady dislike coffee

$$= 1 - \frac{180}{500} = 1 - \frac{9}{25} = \frac{25-9}{25} = \frac{16}{25}$$

3. Central angle = $\frac{\text{Value of frequency}}{\text{Total frequency}} \times 360^\circ$

Central angle for sleep

$$= \frac{8}{8+3+2+8+3} \times 360^\circ$$

$$= \frac{8}{24} \times 360^\circ = 120^\circ$$

$$\text{Central angle for play} = \frac{2}{24} \times 360^\circ = 30^\circ$$

Difference between central angle for sleeps and play
 $= 120^\circ - 30^\circ = 90^\circ$

4. Number of monkeys = $\frac{30^\circ \times 240}{360} = 20$

3 EXERCISE

Single Option Correct :

1. (b) In a leap year there are 366 days.
 366 days = 52 weeks and 2 days.
 These remaining two days can be
 (a) Sunday, Monday
 (b) Monday, Tuesday
 (c) Tuesday, Wednesday
 (d) Wednesday, Thursday
 (e) Thursday, Friday
 (f) Friday, Saturday
 (g) Saturday, Sunday
 $\therefore$ Total no. of cases (a to g) = 7
 No. of favourable cases (a, g) = 2
 $\therefore$ Required probability = $\frac{2}{7}$

2. (b) 3. (d) 4. (a) 5. (c)
 6. (c) 7. (b) 8. (d) 9. (a)
 10. (a) 11. (c) 12. (b) 13. (b)

More Than One Option Correct :

1. (c, d)
 2. (b, d) Marks obtained by 15 students are 43, 73, 44, 93, 54, 64, 53, 24, 84, 40, 93, 33, 34, 74, 44. Required Probability

$$= \frac{\text{Number of favourable outcomes}}{\text{Number of total possible outcomes}}$$
 Number of possible outcomes = 15
 Number of favourable outcomes = 12

$$\text{Required probability} = \frac{12}{15} = \frac{4}{5}$$
 3. (a, d) Number of favourable outcomes = 15(2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30)
 Number of possible outcomes = 30

$$\text{Required Probability} = \frac{15}{30} = \frac{1}{2}$$
 4. (b, d) Number of possible outcomes = 36 [(1, 1), ..., (6, 6)]
 Number of favourable outcomes = 2 [(5, 6), (6, 5)]

$$\therefore \text{Required Probability} = \frac{2}{36} = \frac{1}{18}$$
 5. (a) Number of total possible outcomes = 52
 Number of favourable outcomes = 4

$$\therefore \text{Required Probability} = \frac{4}{52} = \frac{1}{13}$$
 6. (a, b) Number of total possible outcomes = (4 + 5 + 7) = 16
 Number of favourable outcomes = 4

$$\therefore \text{Required Probability} = \frac{4}{16} = \frac{1}{4}$$

Passage Based Questions :

1. (c) Range = Highest observation – lowest observation = 3.1 – 2.1 = 1
 For answering (2) and (3), we prepare frequency table of weight below 2.5 kg and weight 2.8 kg
 2. (a) Number of babies weighing below 2.5 kg
 $= 1 + 1 + 1 + 1 = 4$
 3. (b) Number of babies weighing 2.8 kg = 2
 4. (d)

Class Interval	Number of students
15 – 20	4
20 – 25	8
25 – 30	16
30 – 35	10
35 – 40	18
40 – 45	10
45 – 50	6
50 – 55	8
Total	80

Number of students observed
 $= (4 + 8 + 16 + \dots + 8) = 80$

5. (c) Class – size = upper limit – lower limit
 $= 20 - 15 = 5$
 6. (a) Number of students, weighing less than 35 kg
 $= 4 + 8 + 16 + 10 = 12 + 26 = 38$

Assertion Reason :

1. (a) Assertion is true and reason is true and reason is the correct explanation of assertion.
 2. (d) Assertion is false, reason is true.
 3. (c) Assertion is true, Reason is false.
 Resulting outcomes when a dice is rolled
 $= 6 (1, 2, 3, 4, 5, 6)$
 Number of odd outcomes = (1, 3, 5)
 Probability of getting an odd number $= \frac{3}{6} = \frac{1}{2}$
 and Probability of an event
 $= \frac{\text{Number of favourable outcomes}}{\text{Number of total possible outcomes}}$

Multiple Matching Questions :

1. (A) $\rightarrow$ (s), (B) $\rightarrow$ (p), (C) $\rightarrow$ (r), (t), (D) $\rightarrow$ (q)

Integer Type Questions :

1. (2) a die is thrown 24 times
 $\therefore$ Number of possible outcomes = 24
 Number 4 come up 12 times
 $\therefore$ Number of favourable outcomes = 12

$$\text{Required Probability} = \frac{12}{24} = \frac{1}{2}$$
 Denominator of Probability = 2
 2. (2) Range = Highest observation – Lowest observation
 $= 37 - 15 = 22$
 Unit digit of Range is 2.
 3. (6) Range = Highest observation – Lowest observation
 $= 40 - 4 = 36, r^2 = 36$
 $\Rightarrow r = 6$
 4. (0) When a die is thrown, then number of possible outcomes = 6 (1, 2, 3, 4, 5, 6)
 Number of possible outcomes less than 1 are 0.
 $\therefore$ Required Probability = $\frac{0}{6}$

4 ADVANCED EXERCISE
 BASED ON CONNECTING TOPICS

Marks Obtained	No. of Students (Cumulative frequency)
More than or equal to 50	36
More than or equal to 60	$36 - 4 = 32$
More than or equal to 70	$32 - 8 = 24$
More than or equal to 80	$24 - 12 = 12$
More than or equal to 90	$12 - 6 = 6$

Marks Obtained	No. of Students (Cumulative frequency)
Less than 60	4
Less than 70	12
Less than 80	24
Less than 90	30
Less than 100	36

2. “Less than” cumulative frequency table:

Marks Obtained	No. of Examinees
Less than 450	30
Less than 500	75
Less than 550	135
Less than 600	187
Less than 650	241
Less than 700	308
Less than 750	353
Less than 800	400

“More than” cumulative frequency table:

Marks Obtained	No. of Examinees
400 and more	400
450 and more	370
500 and more	325
550 and more	265
600 and more	213
650 and more	159
700 and more	92
750 and more	47

- 3.

Production	Cumulative frequency
More than or equal to 50	100
More than or equal to 55	98 (100 – 2)
More than or equal to 60	90 (98 – 8)
More than or equal to 65	78 (90 – 12)
More than or equal to 70	54 (78 – 24)
More than or equal to 75	16 (54 – 38)

- 4.

Weight (in kg)	Number of students (Frequency) f_i
36-38	0 = 0
38-40	3 - 0 = 3
40-42	5 - 3 = 2
42-44	9 - 5 = 4
44-46	14 - 9 = 5
46-48	28 - 14 = 14
48-50	32 - 28 = 4
50-52	35 - 32 = 3

Chapter

6

Square and Square Roots

INTRODUCTION

To find square and square roots of different numbers is an algebraic process.

If a number is multiplied by itself, the number thus obtained is the square of the number.

And the number itself is the square root of the squared number.

This chapter has a direct practical use in

- (i) Calculating the area of a square when length of its one side is given :

Area of the square = $(\text{side})^2 = \text{side} \times \text{side}$

- (ii) Calculating the length of a side of square whose area is given :

Length of a side = Square root of the area

SQUARE

Let p and q are natural numbers such that $p = q^2$ then we say ' p ' is the square of number ' q ', e.g., $9 = 3^2$, so 9 is square of 3 and we call 9 as a perfect square number. Table below shows number and their squares from 1 to 10.

Number	1	2	3	4	5	6	7	8	9	10
Square	1	4	9	16	25	36	49	64	81	100

Properties of Square Numbers

By careful examination of table, we can write the following properties of the square numbers.

- (1) No square number ends in 2, 3, 7 or 8 i.e. unit place digit in a square number can never be 2, 3, 7 or 8.
- (2) A square number end with either 0, 1, 4, 5, 6 or 9. But it does not mean that all numbers that end with 0, 1, 4, 5, 6 or 9 are perfect square numbers.
- (3) If a number is even, then its square is also even. If a number is odd then its square is also odd.
- (4) No. of zeroes at the end of a perfect square number is always even. In other words, we can say that numbers ending with odd number of zeroes are never perfect squares.
- (5) A perfect square leaves a remainder 0 or 1 when divided by 3 but all numbers which leave remainder 0 or 1 when divided by 3 need not be a perfect square.
- (6) If p is a square number (perfect square), then $2p$ will never be a square number. 4 is a square number, $2 \times 4 = 8$ is not a square number.
- (7) For every natural number ' n ' the sum of first ' n ' odd natural numbers $= n^2$
e.g. $1 + 3 + 5 + 7 + 9 = 25 = 5^2$

NOTE: A square number is never negative.

METHODS OF FINDING SQUARES OF A NUMBER

(a) Column Method

This method of finding square of a two digit number is based on the identity $(a + b)^2 = a^2 + 2ab + b^2$.

- For a given number ab (two digit number), where ' b ' is units place digit and ' a ' is tens place digit, we calculate a^2 , $2ab$ and b^2 and arrange them in 3 columns as shown below

Column I	Column II	Column III
a^2	$2ab$	b^2

- We underline the units place digit of b^2 and add tens place digit (if present) to $2ab$ under column II.
- Underline the units place digit of column II and add remaining digit (if present) to a^2 under column I.
- Underline the digits of column I.
- The underlined digits give the required square.

For Example : To calculate square of 46 with column method. Here $a = 4$, $b = 6$. We write $a^2 = 16$, $2ab = 48$ and $b^2 = 36$ under columns I, II and III as follows:

Column I	Column II	Column III
16	48	36

- Underline 6 in column III and add 3 to 48 under column II as follows:

Column I	Column II	Column III
16	48	36
	$+3$	
	<u>51</u>	

- Underline 1 in column II and add 5 to 16 under column I as follows:

Column I	Column II	Column III
16	48	36
$+5$	$+3$	
<u>21</u>	<u>51</u>	

- Underline the number in column I as follows:

Column I	Column II	Column III
16	48	36
$+5$	$+3$	
<u>21</u>	<u>51</u>	

$\therefore$ The required square is 2116.

Square of 68

$$\begin{array}{r|l} 68 \times 68 & \\ \hline 36 & 96 \\ + 10 & + 6 \\ \hline 46 & 102 \end{array}$$

$$\therefore 68^2 = 4624$$

Square of 92

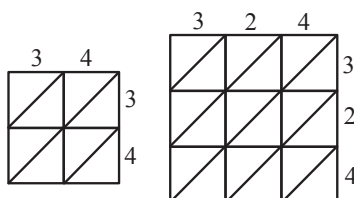
$$\begin{array}{r|l} 92 \times 92 & \\ \hline 81 & 36 \\ + 3 & 4 \\ \hline 84 & \end{array}$$

$$\therefore 92^2 = 8464$$

(b) Diagonal Method

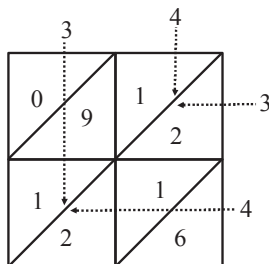
As the number of digits increases, we may use another method called Diagonal Method. The method is illustrated with following examples:

Here we first form a square. We divide the square into sub-squares and draw some diagonals and write digits of the numbers to be squared as shown below:

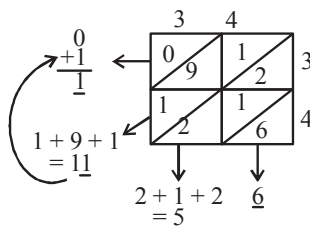


If the number to be squared is a two digit number, we divide our square into 4-sub squares, if number of digits in the given number is 3, then we divide the square into 9 sub-squares.

After arranging the digits as shown before, multiply each digit on the top with each digit on the right and write the product in the corresponding sub-square. If the product is a single digit number, write it below the diagonal and if the product is a two digit number, write tens place digit above the diagonal and units place digit below the diagonal.

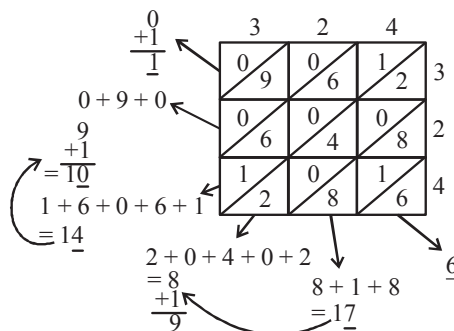


Now, starting from the lowest diagonal, sum the digits along the diagonal. Underline the units digit of the sum and carry over the tens digit if any to the sum of the digits of the diagonal just above. The units digits so underlined together with all the digits in the sum above the top most diagonal gives the square of the number given.



$$\therefore 34^2 = 1156$$

Example : Find value of $(324)^2$ using diagonal method.



$$\therefore (324)^2 = 104976$$

(c) Squaring a Number by Yavadunam Method

This method is used for numbers which are near to a base (some power of 10 e.g. 10, 100, 1000 etc.).

This method is based on one of the vedic mathematics formula “Yavadunam Tavdunikritya Vargamcha Yojayet” which means - whatever the extent of deficiency of a number from base, lesson it to the same extent and set up the square of the deficiency.

Although this sutra tells about the deficiency, it can also be applied to the excess. This excess is to be added to the number followed by setting up the square of the excess.

Example : If we need to find square of 98 and 104 using ‘Yavadunam method’, first we observe that both the numbers are near 100, so base in this case is 100. Now 98 is less than 100 by 2, so the deficiency in this case is 2. 104 is more than 100 by 4 so excess in this case is 4.

Now the square can be calculated in two steps.

First step- The LHS of answer is number plus deviation. The deviation can be positive or negative depending on whether the number is more or less than the base respectively.

Second step - The RHS of the answer being square of the deviation and contains the **same number of digits as the number of zeros in the base.**

This can be done by putting sufficient number of zeroes if required to the left of RHS of the answer.

Example : $(98)^2 = \text{LHS / RHS}$
 Base = 100 (No of zeros = 2)
 LHS = $98 - 2 = 96$
 RHS = $(2)^2 = 4 = 04$ [digits to be equal to no. of zeros of base]
 $\therefore (98)^2 = 9604$

Example : $(104)^2 = (104 + 4) / (4)^2$
 $(104)^2 = 10816$

Example : $(1002)^2 = (1002 + 2) / (2)^2$ [Base = 1000]
 $= 1004 / 4 = 1004 / 004$
 $= 1004004$

Example : $(9999)^2 = (9999 - 1) / (1)^2$ [Base = 10000]
 $= 9998 / 1 = 9998 / 0001$
 $(9999)^2 = 99980001$

SQUARE ROOTS

If $p = q^2$ where p and q are integers, then we say that q is the square root of p . For example $9 = 3^2$, therefore 3 is the square root of 9, similarly 7 is the square root of 49 and 12 is the square root of 144. We can say that if p is a perfect square, then its square root is an integer and if p is not a perfect square, then it does not have an integral square root.

Symbolically, square root of a positive number ‘ n ’ is written as $\sqrt{n}$ or $\sqrt[n]{n}$ or $(n)^{1/2}$. Therefore

$$\sqrt{16} = 4 \text{ or } \sqrt[4]{16} = 4 \text{ or } (16)^{1/2} = 4.$$

METHODS OF FINDING SQUARE ROOTS

(a) Method of Successive Subtraction

We have seen that sum of first 'n' odd natural numbers is ' n^2 ' i.e.

$$1 + 3 + 5 + 7 + \dots + (2n - 1) = n^2$$

The above property can be used to find square root of a perfect square number.

Suppose we want to calculate square root of a number p. We start subtracting odd numbers 1, 3, 5, 7,etc from p.

Now, if p is a perfect square number, then we will get zero after some steps of subtraction. At this stage, the number of times we performed the subtraction process is the square root of given number p.

NOTE: The method of successive subtraction is suitable for small numbers only.

ILLUSTRATION : 1

Calculate square root of 64.

SOLUTION :

- | | | |
|--------------------|-------------------|--------------------|
| (a) $64 - 1 = 63$ | (b) $63 - 3 = 60$ | (c) $60 - 5 = 55$ |
| (d) $55 - 7 = 48$ | (e) $48 - 9 = 39$ | (f) $39 - 11 = 28$ |
| (g) $28 - 13 = 15$ | (h) $15 - 15 = 0$ | |

Total number of subtraction = 8

$\therefore$ Square root of 64, i.e. $\sqrt{64} = 8$.

(b) Prime Factorization Method

Let 'p' be the number whose square root is to be find

Steps

- Write all the prime factors of 'p', and pair them such that primes in each pair are equal.
- Choose one prime from each pair.
- Multiply all primes selected in step (ii)
- Product so obtained is the square root of 'p'.

ILLUSTRATION : 2

Find square root of 4096.

SOLUTION :

2	4096
2	2048
2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2

$$\therefore 4096 = (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (2 \times 2)$$

$$\sqrt{4096} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$$

This method of calculation of square root is efficient only if the given number has small prime factors.

(c) Division Method

The number of digits can be determined by placing bars on every pair of digits starting from units digit. If the number of digits is odd, then the left most single digit will have a bar on it. The number of bars give the number of digits in the square root of the number. For example: square root of $\overline{20} \overline{25}$ will have 2 digits whereas $\sqrt{2 \overline{72} \overline{25}}$ has 3 digits.

Steps of Division Method:

- (i) Place a bar over every pair of digits starting from the units digit.
- (ii) Find the largest number whose square is less than or equal to the number under the left most bar.
- (iii) Take this number as the divisor and number under left most bar as dividend. Divide them to get the remainder. You will see that in this step the divisor and quotient are same.
- (iv) Write down the number under next bar at the right side of the remainder. This is our new dividend.
- (v) New divisor is obtained by adding the quotient in the divisor obtained in step (iii) and putting a suitable digit at the right of it. The digit is chosen in such a way that its product with new divisor is equal or just less than new dividend.

Repeat steps (iv) and (v) till all bars have been considered. The final quotient is the square root of the given number.

ILLUSTRATION : 3

Find square root of 106929.

SOLUTION :

Square root of $\overline{10} \ \overline{69} \ \overline{29}$ will have 3 digits. As 3 is the largest digit whose square is less than 10 (number under left most bar). Here 10 is our dividend and 3 is our divisor and quotient.

$$\begin{array}{r}
 327 \\
 3 \overline{) 10 \ 69 \ 29} \\
 \underline{9} \\
 62 \\
 62 \overline{) 1 \ 69} \\
 \underline{1 \ 24} \\
 647 \\
 647 \overline{) 45 \ 29} \\
 \underline{45 \ 29} \\
 0
 \end{array}$$

$$\therefore \sqrt{106929} = 327$$

ILLUSTRATION : 4

Find square root of 11664.

SOLUTION :

$$\begin{array}{r}
 108 \\
 1 \overline{) 11 \ 66 \ 64} \\
 \underline{1} \\
 20 \\
 20 \overline{) 16} \\
 \underline{16} \\
 208 \\
 208 \overline{) 16 \ 64} \\
 \underline{16 \ 64} \\
 0
 \end{array}$$

$$\therefore \sqrt{11664} = 108$$

SQUARE ROOT OF RATIONAL NUMBERS WHOSE NUMERATORS AND DENOMINATORS ARE PERFECT SQUARES.

We will use the following rules to calculate square root:

- (i) $\sqrt{\frac{p}{q}} = \frac{\sqrt{p}}{\sqrt{q}}$, where $q \neq 0$
- (ii) If p and q are positive numbers, then

$$\sqrt{pq} = \sqrt{p} \times \sqrt{q}$$

ILLUSTRATION : 5

Find square root of $\frac{144}{625}$.

SOLUTION :

$$\sqrt{\frac{144}{625}} = \frac{\sqrt{144}}{\sqrt{625}}$$

$$\sqrt{144} = \sqrt{2 \times 2 \times 2 \times 2 \times 3 \times 3}$$

$$\begin{aligned}
 &= \sqrt{2^2 \times 2^2 \times 3^2} \\
 &= 2 \times 2 \times 3 = 12 \\
 \sqrt{625} &= \sqrt{5 \times 5 \times 5 \times 5} = 5 \times 5 = 25 \\
 \therefore \sqrt{\frac{144}{625}} &= \frac{12}{25}
 \end{aligned}$$

ILLUSTRATION : 6

Find square root of $3\frac{4}{64}$.

SOLUTION :

$$\begin{aligned}
 3\frac{4}{64} &= \frac{196}{64} \\
 \sqrt{\frac{196}{64}} &= \frac{\sqrt{196}}{\sqrt{64}} = \frac{14}{8} = \frac{7}{4}
 \end{aligned}$$

NOTE: All square numbers end with 0, 1, 4, 5, 6 or 9 at unit's place.

SQUARE ROOT OF PERFECT SQUARE DECIMAL NUMBER BY DIVISION METHOD

As we have seen that the square root of these kinds of numbers can be found by first converting them into rational number. However by using division method we can find the square root directly. Follow the steps explained below:

- (1) Place the bar on integral part (from left side of decimal) of the number in usual manner.
- (2) Place bars on decimal part (from right side of decimal) on every pair of digits.
- (3) Apply division method and find square root.
- (4) Place the decimal point in the quotient as soon as the integral part is exhausted.

ILLUSTRATION : 7

Find square root of 52.8529.

SOLUTION :

$$\begin{array}{r}
 7.27 \\
 7 \overline{) 52.85 \overline{) 29}} \\
 \underline{49} \\
 142 \overline{) 385} \\
 \underline{284} \\
 1447 \overline{) 10129} \\
 \underline{10129} \\
 0
 \end{array}$$

$$\therefore \sqrt{52.8529} = 7.27$$

ILLUSTRATION : 8

Find square root of 0.000169.

SOLUTION :

$$\begin{array}{r}
 0.013 \\
 1 \overline{) 0.00 \overline{) 0169}} \\
 \underline{1} \\
 23 \overline{) 069} \\
 \underline{69} \\
 0
 \end{array}$$

$$\therefore \sqrt{0.000169} = 0.013$$

SQUARE ROOT OF NUMBERS WHICH ARE NOT PERFECT SQUARES

Division method can also be applied for finding square root of numbers which are not perfect square numbers. Method is explained with the following illustrative examples.

ILLUSTRATION : 9

Find square root of 3 upto 3 decimal places.

SOLUTION :

$$\begin{array}{r}
 1.732 \\
 1 \overline{) 3.00\ 00\ 00} \\
 \underline{1} \\
 200 \\
 \underline{189} \\
 1100 \\
 \underline{1029} \\
 7100 \\
 \underline{6924} \\
 176
 \end{array}$$

Here, we have added 3 pairs of zeros after decimal. One pair each for 1 digit after decimal point.

$\therefore \sqrt{3} = 1.732$ upto three decimal places.

ILLUSTRATION : 10

Find square root of $5\frac{2}{15}$ upto 3 decimal places.

SOLUTION :

$$5\frac{2}{15} = 5.133333 \text{ (approx.)}$$

$$\begin{array}{r}
 2.265 \\
 2 \overline{) 5.13\ 33\ 33} \\
 \underline{4} \\
 113 \\
 \underline{84} \\
 2933 \\
 \underline{2676} \\
 4525 \\
 \underline{4525} \\
 03108
 \end{array}$$

$$\therefore \sqrt{5\frac{2}{15}} = \sqrt{5.133333} = 2.265 \text{ (approx.) up to 3 decimal places.}$$

PYTHAGOREAN TRIPLETS

A Pythagorean triplet is a set of three natural numbers which form the sides of a right angled triangle.

For every natural number $m > 1$;

$(2m, m^2 - 1, m^2 + 1)$ form a Pythagorean Triplet.

ILLUSTRATION : 11

Find the Pythagorean triplet whose one member is 6.

SOLUTION :

$$\begin{array}{ll}
 \text{Put} & 2m = 6 \\
 \Rightarrow & m = 3 \\
 \text{also,} & m^2 - 1 = (3)^2 - 1 = 8 \\
 \text{and} & m^2 + 1 = (3)^2 + 1 = 10 \\
 \therefore & (6, 8, 10) \text{ is required Pythagorean triplet.}
 \end{array}$$

ILLUSTRATION : 12

Find the Pythagorean triplet whose smallest member is 3.

SOLUTION :

$$\begin{array}{ll}
 \text{Put} & m^2 - 1 = 3 \\
 \Rightarrow & m^2 = 3 + 1 = 4 \\
 \Rightarrow & m = 2 \\
 \text{also,} & 2m = 2(2) = 4 \\
 \text{and} & m^2 + 1 = (2)^2 + 1 = 5 \\
 \therefore & (3, 4, 5) \text{ is the required Pythagorean triplet.}
 \end{array}$$

MISCELLANEOUS

SOLVED EXAMPLES

1. Without adding, find the sum,

(a) $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19$

(b) $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23$

Sol. We know that sum of first n odd natural numbers is n^2

$\therefore$ (a) Given sum is the sum of first 10 odd natural numbers

$\therefore 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = (10)^2 = 100.$

(b) Given sum is the sum of the first 12 odd natural numbers.

$\therefore 1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 = (12)^2 = 144.$

2. Write pythagorean triplet whose one number is

(a) 8

(b) 12

Sol. A triplet of three natural numbers p, q, r is called pythagorean triplet if $p^2 + q^2 = r^2$ and is written as (p, q, r) .

For any natural number ' k ' > 1 , $(2k, k^2-1, k^2+1)$ is a pythagorean triplet.

(a) Now if $2k = 8$

then $k = 4$

$k^2 - 1 = 16 - 1 = 15$

$k^2 + 1 = 16 + 1 = 17$

$\therefore$ Pythagorean triplet is $(8, 15, 17)$

(b) $2k = 12 \Rightarrow k = 6$

$k^2 - 1 = 36 - 1 = 35$

$k^2 + 1 = 36 + 1 = 37$

$\therefore$ Pythagorean triplet is $(6, 35, 37)$.

3. Find the smallest number by which 252 must be multiplied so that the product becomes a perfect square. Also find the square root of the perfect square so obtained.

Sol. Writing 252 as product of its prime factors, we get

$252 = 2 \times 2 \times 3 \times 3 \times 7$

We find that prime factors 2 and 3 occur in pairs but prime factor 7 occurs alone.

Therefore 252 must be multiplied with 7 to get a perfect square number

$\therefore$ New number $= 252 \times 7 = 1764$

Now $1764 = (2 \times 2) \times (3 \times 3) \times (7 \times 7)$

$\therefore \sqrt{1764} = 2 \times 3 \times 7 = 42.$

4. Find the smallest number by which 15552 must be divided so that it becomes a perfect square. Also find the square root of new perfect square number.

Sol. By writing 15552 as product of its prime factors we get

$15552 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3$

$= (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3) \times 3$

Here we find that prime factors 2 occur in pair but one of the prime factor 3 occurs alone.

$\therefore$ If we divide 15552 by 3, we get a perfect square.

$\therefore$ New Number $= \frac{15552}{3} = 5184$

Now $5184 = (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3)$

$\therefore \sqrt{5184} = 2 \times 2 \times 2 \times 3 \times 3 = 8 \times 9 = 72$

5. Find the smallest square number which is divisible by each of the number 4, 6 and 12.

Sol. We know that smallest number which is divisible by each of the number 4, 6, 12 is L C M (4, 6, 12)

Now L C M (4, 6, 12) is 12.

By resolving 12 into its prime factors we find $12 = (2 \times 2) \times 3$, Since 3 occurs alone

$\therefore$ To make 12 a perfect square we have to multiply it with 3.

$\therefore$ Smallest square number divisible by 4, 6, 12 is 36.

6. The product of two numbers is 1575 and their quotient is $\frac{9}{7}$. Find the numbers.

Sol. Let one of the two numbers be K. As the product of numbers is 1575, the other number will be $\frac{1575}{K}$.

$$\text{Quotient of numbers} = \frac{9}{7}$$

$$\therefore \frac{K}{\frac{1575}{K}} = \frac{9}{7} \Rightarrow \frac{K^2}{1575} = \frac{9}{7} \quad \therefore K^2 = \frac{9 \times 1575}{7} = 9 \times 225$$

$$K^2 = 3 \times 3 \times 15 \times 15 = (3 \times 3) \times (3 \times 3) \times (5 \times 5)$$

$$\therefore K = 3 \times 3 \times 5 = 45$$

$$\therefore \text{other number is } \frac{1575}{45} = 35$$

$\therefore$ The required numbers are 45 and 35.

7. Find the least number which must be subtracted from 18315 to make it a perfect square. Also find the square root of the resulting number.

Sol. First we apply the long division method of finding square root on 18315.

$$\begin{array}{r} 135 \\ 1 \overline{) 18315} \\ \underline{1} \\ 23 \\ \underline{265} \\ 1415 \\ \underline{1325} \\ 90 \end{array}$$

Here we find that remainder in the last step is 90, it means 18315 is 90 more than the square of 135. So we must subtract 90 from 18315 to make it a perfect square.

$$\therefore \text{Perfect square number } 18315 - 90 = 18225 \text{ and } \sqrt{18225} = 135$$

8. Find the least number of four digits which is a perfect square.

Sol. Least number of 4 digits is 1000, it is not perfect square.

First we apply long division method on 1000.

$$\begin{array}{r} 31 \\ 3 \overline{) 1000} \\ \underline{9} \\ 61 \\ \underline{61} \\ 39 \end{array}$$

This clearly shows that $(31)^2 < 1000$ and next square number is, $(32)^2 > 1000$

So, smallest number of 4 - digits which is a perfect square is $(1000 + 24) = 1024$

9. 7396 students are sitting in an auditorium in such a manner that there are as many students in a row as there are rows in the auditorium. How many rows are there in the auditorium?

Sol. Let number of students sitting in a row = 'x'

$\therefore$ Number of rows in auditorium = 'x'

$\therefore$ Number of students sitting in auditorium = $x \times x = x^2$

$$\text{Now } x^2 = 7396$$

$$\therefore x = \sqrt{7396}$$

$$\begin{array}{r} 86 \\ 8 \overline{) 7396} \\ \underline{64} \\ 166 \\ \underline{166} \\ 00 \end{array}$$

$$\therefore x = \sqrt{7396} = 86$$

Number of rows in auditorium = 86.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. Square numbers can only have..... number of zeros at the end.
2. The number of zeroes at the end of the square of a number is the number of zeroes at the end of the number.
3. The square root of a 4-digit or a 3 digit number is a digit number.
4. Square of a number between 0 and 1 is than the number itself.
5. If 'a' is a square root of 'b' then 'b' is of 'a'.
6. A number whose square root is exact is called a
7. The number of zeroes at the end of a perfect square is always
8. A perfect square leaves a remainder 0 or 1, by dividing with

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. If $7^2 = 49$ and $0.7^2 = 0.49$ then $0.007^2 = 0.000049$.
2. The square of a proper fraction is always greater than itself.
3. The square root of a prime number can be obtained approximately but not exactly.
4. The square of an odd number is always odd.
5. A number ending in an odd number of zeros is never a perfect square.
6. Square of a prime number is always prime.
7. The sum of two square numbers is a square number.
8. The difference of two square number is a square number.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

1. Without actually finding the squares match the Column I with Column II.

Column-I

- (A) $18^2 - 17^2$
(B) $31^2 - 30^2$
(C) $51^2 - 50^2$

Column-II

- (p) 101
(q) 89
(r) 39

(D) $45^2 - 44^2$ (s) 35

(E) $20^2 - 19^2$ (t) 61

2. Column-I

A. If $\sqrt{(75.24 + x)} = 8.71$
then the value of x is

B. If $\sqrt{0.04 \times 0.4 \times a}$
 $= 0.4 \times 0.04 \times \sqrt{b}$
then the value of $\frac{a}{b}$ is

C. If $\sqrt{256} \div \sqrt{x} = 2$ then the
value of x is

D. If $\sqrt{\frac{x}{169}} = \frac{54}{39}$ then
the value of x is

Column-II

(p) 324

(q) 64

(r) 0.016

(s) .6241

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. How do we find the square root of a fraction ?
2. Is 3 the square of any rational number ?
3. Evaluate: $\sqrt{10 + \sqrt{36}}$.
4. Without adding find the sum :
 $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19$
5. Express 49 as the sum of 7 odd natural numbers.
6. How many natural numbers lie between squares of 12 and 13.
7. An odd number when multiplied by itself gives 2401. Find the number.

8. Find the square root of $\frac{m^{n^2} n^{m^2} a^{m+n}}{(m+n)^{(m+n)^2}}$.
9. If x and y are two positive numbers, then
Put = or $\neq$ in the given boxes.

(i) $\sqrt{x} \times \sqrt{y}$ ☐ $\sqrt{xy}$

(ii) $\sqrt{\frac{x}{y}}$ ☐ $\frac{\sqrt{x}}{\sqrt{y}}, y \neq 0$

(iii) $\sqrt{x+y}$ ☐ $\sqrt{x} + \sqrt{y}$

(iv) $\sqrt{x-y}$ ☐ $\sqrt{x} - \sqrt{y}$

10. If $(2 \times 9)(2 \times 9)$ is 324, then find the value of $\sqrt{324}$.
11. Find the value of $100^2 - 99^2$.
12. Find the square of the number 509 using the identity $(a + b)^2 = a^2 + 2ab + b^2$.
13. Find the least square number exactly divisible by each of the number 6, 5, 10 and 20.
14. Evaluate: $\sqrt{1.96}$.
15. Find the value of $\sqrt{45} \times \sqrt{20}$.

Short Answer Questions :

DIRECTIONS : Give answer in two to three sentences.

1. Find the square root of 298116 by prime factorization.
2. Find the value of
(i) $\sqrt{99} \times \sqrt{396}$ (ii) $\sqrt{147} \times \sqrt{243}$
3. Find the square root of
(i) 480249 (ii) 0.00008281
4. If $(y - 4)^2 = 16$ then find the value of y .
5. Find the square root of $6\frac{19}{25}$.
6. Find the least square number which is exactly divisible by 8, 9 and 10.
7. Find the square root of 2.9 upto two decimal places.
8. Find the square of 42 without actual multiplication.
9. Find the square root of $\frac{81b^2a^4}{36x^2y^6}$.
10. If 1156 students in a school are to form a square pattern on the field for the mass drill function on the sports day, how many students will form each side of the square?
11. A PT teacher wants to arrange maximum possible number of 6000 students in a field such that the number of rows is equal to the number of columns. Find the number of rows if 71 were left out after the arrangement.

12. Find the square root of 54756 by long division method.
13. Given that $\sqrt{2} = 1.414$ and $\sqrt{5} = 2.236$, evaluate each of the following:
(i) $\sqrt{\frac{36}{5}}$ (ii) $\sqrt{\frac{625}{98}}$
14. Find the greatest number of five digits which is a perfect square.
15. Find the smallest number that must be added to 1780 to make it a perfect square.

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Find the square root of each of the following by prime factorization.
(i) 586756 (ii) 3013696
2. Find the square root of the following numbers by division method.
(i) 12321 (ii) 96721 (iii) 10609
3. Find the value of
(i) $\sqrt{146.8944}$ (ii) $\sqrt{964.288809}$
4. The product of two numbers is 1575 and their quotient is $\frac{9}{7}$. Find the numbers.
5. The area of a square field is $101\frac{1}{400}$ square metres. Find the length of one side of the field.
6. Find the square root of 237.615 correct to three places of decimal.
7. A general wishing to arrange his men, who were 335250 in number in the form of a square found that there were 9 men left over. How many were there in each row?

2

EXERCISE

Text-Book Exercise :

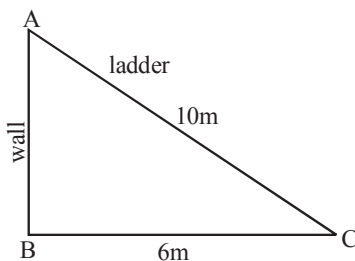
1. Observe the following pattern and find the missing digits.
 $11^2 = 121$
 $101^2 = 10201$
 $1001^2 = 1002001$
 $100001^2 = 1 \dots\dots 2 \dots\dots 1$
 $10000001^2 = \dots\dots\dots$

2. Without adding, find the sum.
 (i) $1 + 3 + 5 + 7 + 9$
 (ii) $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19$
 (iii) $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23$
3. How many numbers lie between squares of the following numbers?
 (i) 12 and 13 (ii) 25 and 26 (iii) 99 and 100
4. Find the square of the following numbers without actual multiplication.
 (i) 39 (ii) 42

- Write a Pythagorean triplet whose smallest member is 8.
- Find the square root of 6400.
- Is 2352 a perfect square? If not, find the smallest multiple of 2352 which is a perfect square. Find the square root of the new number.
- Find the greatest 4-digit number which is a perfect square.
- Find the least number that must be added to 1300 so as to get a perfect square. Also find the square root of the perfect square.
- Area of a square plot is 2304 m^2 . Find the side of the square plot.
- In a right triangle ABC, $\angle B = 90^\circ$.
 - If $AB = 6 \text{ cm}$, $BC = 8 \text{ cm}$, find AC
 - If $AC = 13 \text{ cm}$, $BC = 5 \text{ cm}$, find AB
- A gardener has 1000 plants. He wants to plant these in such a way that the number of rows and the number of columns remain same. Find the minimum number of plants he needs more for this.

Exemplar Questions :

- Using distributive law, find the square of 43.
- A ladder 10m long rests against a vertical wall. If the foot of the ladder is 6m away from the wall and the ladder just reaches the top of the wall, how high is the wall?



- Find the length of a diagonal of a rectangle with dimensions 20m by 15m.
- The area of a rectangular field whose length is twice its breadth is 2450 m^2 . Find the perimeter of the field.
- A decimal number is multiplied by itself. If the product is 51.84, find the number.
- Find three numbers in the ratio 2:3:5, the sum of whose squares is 608.
- The perimeters of two squares are 40 and 96 metres respectively.
Find the perimeter of another square equal in area to the sum of the first two squares.
- 13 and 31 is a strange pair of numbers such that their squares 169 and 961 are also mirror images of each other. Can you find two other such pairs?

HOTS Questions :

- If $(6.453)^2 = 41.64$ then find the value of $(64.53)^2$.
- Two thirds of the area of a square is equal to 48 cm^2 . Another square is $\frac{1}{8}$ the area of the original square. Find the area of the second square and the length of its side.
- Observe the following pattern and use it to complete the given statements.

$$15^2 = 1 \times 2 \text{ (hundred)} + 5^2 = 225$$

$$25^2 = 2 \times 3 \text{ (hundred)} + 5^2 = 625$$

$$35^2 = 3 \times 4 \text{ (hundred)} + 5^2 = 1225$$

$$45^2 = 4 \times 5 \text{ (hundred)} + 5^2 = 2025$$
 - $55^2 = 5 \times 6 \times (\text{_____}) + 25 = \text{_____}$
 - $65^2 = 6 \times \text{_____} \times (\text{hundred}) + \text{_____} = 4225$
 - $75^2 = \text{_____} \times \text{_____} \times (\text{hundred}) + 25 = 5625$
- Find the square root of
 - 75724804
 - 36663025
 - 4036081
 - 146894400
- 5929 students are sitting in an auditorium in such a manner that there are as many students in a row as there are rows in the auditorium. How many rows are there in the auditorium?
- The area of a square field is 60025 m^2 . A man cycles along its boundary at 18 km/hr . In how much time will he return at the starting point?
- Simplify:
 - $\frac{\sqrt{59.29} - \sqrt{5.29}}{\sqrt{59.29} + \sqrt{5.29}}$
 - $\frac{\sqrt{0.2304} + \sqrt{0.1764}}{\sqrt{0.2304} - \sqrt{0.1764}}$
- If $\sqrt{\frac{125a^6b^4c^2}{5a^4b^2}} = x$ then find $\frac{x^2}{abc}$.
- Find the square root of $x^{m^2-n^2} \cdot x^{n^2+2mn} \cdot x^{n^2}$
- If $\sqrt{5^n} = 125$ then find the value of $5^{\frac{n}{4}}$.
- If $\sqrt{2^n} = 1024$, then find the value of $3^{\frac{n}{4}-4}$.

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The smallest number by which 136 must be multiplied so that it becomes a perfect square is
(a) 2 (b) 17
(c) 34 (d) None of these
- The smallest number by which 3888 must be divided so that the resulting number is a perfect square is
(a) 2 (b) 6
(c) 3 (d) None of these.
- Class 'X -A' of a school collected ₹ 1024 as contribution to AP flood relief fund. If each student contributed as one student in the class. The number of students in the class are
(a) 24 (b) 32
(c) Can't say (d) None of these.
- The least square number exactly divisible by 4, 6, 10, 15 is
(a) 400 (b) 100
(c) 25 (d) 900
- The least 6 digit number which is perfect square is
(a) 100000 (b) 100144
(c) 100489 (d) 100225
- The least number to be subtracted from 24136 to make it a perfect square
(a) 155 (b) 111
(c) 156 (d) None of these.
- What must be added to 24136 to make it a perfect square?
(a) 100 (b) 200
(c) 111 (d) None of these.
- Area of a square field is 22500 m^2 . A man cycles along its boundary at 15 km/hr . The time will be taken by a man to return to starting point, is
(a) 2 min 24 sec. (b) 3 min 12 sec.
(c) 4 mins. (d) None of these.
- A gardener arranges plants in rows to form a square. He finds that in doing so 15 plants are left out. If the total number of plants are 3984, the number of plants in each row are,
(a) 62 (b) 63
(c) 64 (d) None of these.
- A number is multiplied by $\frac{1}{3}$ times itself and 18 is added to the result. If the final result is 2901, the number is
(a) 54 (b) 93
(c) 83 (d) 84
- A four digit perfect square whose first two digits and last two digits taken separately are also perfect squares, is:
(a) 6481 (b) 4925
(c) 3625 (d) 1681
- The smallest natural number which when added to the difference of squares of 16 and 12 gives a perfect square, is
(a) 9 (b) 8
(c) 5 (d) 3
- If $\left(\sqrt{6+\sqrt{6+\sqrt{6+\dots}}}\right) = P$, then value of P is
(a) 6 (b) 4
(c) 2 (d) 3
- You have a rectangular frame that is 40 cm by 60 cm. Can you put a square picture that has an area of 800 cm^2 completely inside the frame?
(a) Yes (b) No
(c) Can't say (d) Data insufficient
- Pooja's rectangular living room is 12 m by 5.5 m. She has a square rug that covers $\frac{3}{7}$ the area of the floor. The side length of square rug is
(a) 28.3m (b) 6.2 m
(c) 5.3m (d) None of these

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following triplets is/are Pythagorean?
(a) (8, 15, 17) (b) (18, 80, 82)
(c) (14, 48, 51) (d) (12, 35, 38)
- Which of the following number is/are squares of even numbers ?
(a) 121 (b) 256
(c) 324 (d) 5476
- Which of the following is/are correct?
(a) The number of digits in a square number is even.
(b) The sum of two square numbers is a square number.
(c) There is no square number between 50 and 60.
(d) There are fourteen square number upto 200.
- Which of the following is/are incorrect?
(a) The square of a prime number is prime.
(b) No square number is negative.
(c) The product of two square numbers is a square number.
(d) The difference of two square numbers is a square number.

5. Which one is/are correct?
 - (a) The square root of 7744 by prime factorization is 88.
 - (b) 3600 is the smallest square number, divisible by each one of the numbers 8, 9 and 10.
 - (c) Possible unit digits of the square root of 657666025 is 5.
 - (d) The square root of 390625 has three digits.
6. Which of the following statements is/are incorrect?
 - (a) $(5.01)^2$ lies between 16 and 25.
 - (b) $0.1 < (0.12)^2 < 0.2$
 - (c) $144 < (12.1)^2 < 169$
 - (d) $(26.1)^2$ lies between 400 and 900.
7. Which of the following is/are incorrect for $\sqrt{9+4}$?
 - (a) $\sqrt{9} + \sqrt{4}$
 - (b) $3 + 2$
 - (c) $\sqrt{13}$
 - (d) $\sqrt{3+2}$
8. Which of the following is/are correct for $\sqrt{a^2 \times b^2 \times c^2}$?
 - (a) $\sqrt{a^2} \times \sqrt{b^2} \times \sqrt{c^2}$
 - (b) $a \times b \times c$
 - (c) $\sqrt{(abc)^2}$
 - (d) None of these.
9. Given numbers are perfect squares. Which of them are the squares of odd numbers?
 - (a) 1681
 - (b) 2116
 - (c) 1089
 - (d) 1764

Passage Based Questions :

DIRECTIONS : Study the given passage(s) and answer the questions given below each passage.

PASSAGE - I

The sum of first n odd natural numbers is n^2 . If a number is a square number, it has to be the sum of consecutive odd numbers starting from 1. This means that if a number is not equal to the sum of successive odd natural numbers starting with 1, then it is not a perfect square.

1. The sum $1 + 3 + 5 + 7 + 9$ is
 - (a) 25
 - (b) 26
 - (c) 24
 - (d) 27
2. 121 is the sum of n odd natural numbers. Where n is
 - (a) 10
 - (b) 11
 - (c) 9
 - (d) None of these

PASSAGE - II

The area of two square fields are respectively 256 and 625 square metres.

3. The side of two square fields respectively are
 - (a) 16, 25
 - (b) 25, 16
 - (c) 17, 25
 - (d) 16, 5
4. The ratio of their sides are
 - (a) 17 : 25
 - (b) 25 : 16
 - (c) 16 : 25
 - (d) 16 : 5
5. If both the squares are made into a new square field, then the area will be
 - (a) 811 m^2
 - (b) 1681 m^2
 - (c) 891 m^2
 - (d) 440 m^2

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 - (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 - (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 - (d) If **Assertion** is **incorrect** but **Reason** is **correct**.
1. **Assertion :** 50 natural numbers lie between squares of 25 and 26.
Reason : There are $2n$ non-perfect square numbers between two consecutive square numbers n^2 and $(n+1)^2$.
 2. **Assertion :** $\frac{5}{4}$ is always a rational.
Reason : If a and b are perfect squares, then $\sqrt{\frac{a}{b}}$ is always a rational number.
 3. **Assertion :** In a right triangle ABC , $\angle B = 90^\circ$. If $AB = 6 \text{ cm}$, $BC = 8 \text{ cm}$ then $AC = 10 \text{ cm}$.
Reason : Pythagorus theorem says,
 $(\text{Perpendicular})^2 + (\text{Base})^2 = (\text{Hypotenuse})^2$
 4. **Assertion :** $136^2 - 135^2 = 136 + 135 = 271$
Reason : The difference of square of two consecutive natural numbers is equal to their difference.
 5. **Assertion :** (8, 15, 17) are Pythagorean triplets.
Reason : For any natural number $m > 1$, we have $2m^2, m^2 - 1, m^2 + 1$ as pythagorean triplets.

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. The smallest number by which 3888 must be divided so that the resulting number is a perfect square is
2. The value of $\sqrt{388 + \sqrt{127 + \sqrt{289}}}$ is $5(K)$ then $K = ?$
3. The value of $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$ is:
4. The perfect square number out of 2, 3, 4 and 5 is
5. If $\sqrt{1 + \frac{27}{169}} = 1 + \frac{x}{13}$, then $x = \underline{\hspace{2cm}}$
6. The value of $\sqrt{(-1)} \cdot \sqrt{(-1)}$ is:
7. How many pairs of natural numbers are there so that difference of their squares is 60?
8. $\frac{\sqrt{32} + \sqrt{48}}{\sqrt{8} + \sqrt{12}} = ?$
9. What will be the number of zeroes in the squares of the number 100?

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- even
- twice
- two
- decimal, less
- square
- perfect square
- even
- three

True/False :

- True
- False, the square of a proper fraction is always less than itself.
- True
- True.
- True.
- False.
- False, because $1^2 + 2^2 = 5$, which is not a square number.
- False, because $5^2 - 2^2 = 21$, which is not a square number.

Match The Columns :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (t); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q); (E) $\rightarrow$ (r)
- (A) $\rightarrow$ (s); (B) $\rightarrow$ (r); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p)

$$(A) \quad 75.24 + x = 8.71 \times 8.71 \Rightarrow x = .6241$$

$$(B) \quad \sqrt{0.016a} = 0.016 \times \sqrt{b}$$

$$\Rightarrow \sqrt{\frac{a}{b}} = \sqrt{0.016} \Rightarrow \frac{a}{b} = 0.016$$

$$(C) \quad \frac{\sqrt{256}}{\sqrt{x}} = 2 \Rightarrow 16 = 2\sqrt{x} \Rightarrow x = 64$$

$$(D) \quad \sqrt{\frac{x}{169}} = \frac{54}{39} \Rightarrow x = \frac{54}{39} \times \frac{54}{39} \times 169 = 324$$

Very Short Answer Questions :

- Square root of a fraction

$$= \frac{\text{square root of its numerator}}{\text{square root of its denominator}}$$
- No, 3 is not the square of any rational number.
So, it is not a perfect square.
- $\sqrt{10 + \sqrt{36}} = \sqrt{10 + 6} = \sqrt{16} = 4$
- $10^2 = 100$
- $49 = 1 + 3 + 5 + 7 + 9 + 11 + 13$
- $12 \times 2 = 24$ natural numbers.
- Required number $= \sqrt{2401} = 49$

$$8. \quad \frac{\frac{n^2}{m^2} \cdot \frac{m^2}{n^2} \cdot \frac{m+n}{a^2}}{\frac{(m+n)^2}{(m+n)^2}}$$

$$9. \quad (i) \quad \sqrt{xy} \quad \boxed{=} \quad \sqrt{xy}$$

$$(ii) \quad \sqrt{\frac{x}{y}} \quad \boxed{=} \quad \sqrt{\frac{x}{y}}$$

$$(iii) \quad \sqrt{x+y} \quad \boxed{\neq} \quad \sqrt{x} + \sqrt{y}$$

$$(iv) \quad \sqrt{x-y} \quad \boxed{\neq} \quad \sqrt{x} - \sqrt{y}$$

$$10. \quad \sqrt{324} = 2 \times 9 = 18$$

$$11. \quad \text{Here, } m^2 - n^2 = (m+n)(m-n) \\ \text{So, } 100^2 - 99^2 = (100+99)(100-99) = 199$$

$$12. \quad \text{We have,} \\ 509^2 = (500)^2 + 2 \times 500 \times 9 + 9^2 \\ = 250000 + 9000 + 81 = 259081$$

- The smallest number divisible by each one of the numbers 6, 5, 10 and 20 is their L.C.M

$$\begin{array}{r|l} 2 & 6, 5, 10, 20 \\ 5 & 3, 5, 5, 10 \\ \hline & 3, 1, 1, 2 \end{array}$$

LCM. of 6, 5, 10 and 20 is 60.

$$60 = (2 \times 2) \times 3 \times 5$$

Grouping the factors into pairs of equal factors, we see that three is no factor to form pairs with 3 and 5. Here the smallest square number divisible by 6, 5, 10, 20 is $60 \times 15 = 900$.

- Using the division method we may find the square root of the given number as shown below.

$$\begin{array}{r} 14 \\ 1 \overline{) 196} \\ \underline{-1} \\ 24 \\ \underline{-96} \\ 0 \end{array}$$

$$\therefore \sqrt{1.96} = 1.4$$

- We have

$$\begin{aligned} \sqrt{45} \times \sqrt{20} &= \sqrt{45} \times \sqrt{20} = \sqrt{3 \times 3 \times 5 \times 2 \times 2 \times 5} \\ &= \sqrt{3 \times 3 \times 2 \times 2 \times 5 \times 5} = (3 \times 2 \times 5) = 30. \end{aligned}$$

Short Answer Questions :

1. $\sqrt{298116} = 2 \times 3 \times 7 \times 13 = 546$

2. (i) $\sqrt{99} \times \sqrt{396} = \sqrt{99 \times 396}$
 $= \sqrt{3 \times 3 \times 11 \times 2 \times 2 \times 3 \times 3 \times 11}$
 $= 3 \times 3 \times 2 \times 11 = 198$

(ii) $\sqrt{147} \times \sqrt{243} = 189$

3. (i) $\sqrt{480249} = 693$

(ii) $\sqrt{0.00008281} = 0.0091$

4. 8

5. $6\frac{19}{25} = \frac{169}{25}$

$\therefore \sqrt{6\frac{19}{25}} = \sqrt{\frac{169}{25}} = \frac{13}{5}$ or $2\frac{3}{5}$

6. L.C.M. of 8, 9 and 10 is 360 which is not a perfect square. To make it perfect square, it must be multiplied by $2 \times 5 = 10$.

7. 1.70

8. $42 = (40 + 2)^2$

Use $(a + b)^2 = a^2 + b^2 + 2ab$
 $= (40)^2 + 2^2 + 2(40)(2) = 1764$

9. $\frac{3ba^2}{2xy^3}$

10. The number of students on each side $= \sqrt{1156} = 34$

11. Number of students = 6000
 Number of left out students = 71
 Number of students which can be arranged in a perfect square = $6000 - 71 = 5929$
 Because number of rows and columns are equal. So, number of rows = $\sqrt{5929}$

$5929 = 7 \times 7 \times 11 \times 11$

$\therefore \sqrt{5929} = \sqrt{7 \times 7 \times 11 \times 11} = 7 \times 11 = 77$

Hence, number of rows = 77.

12.
$$\begin{array}{r} 234 \\ 2 \overline{) 54756} \\ \underline{-4} \\ 43 \quad 147 \\ \underline{-129} \\ 464 \quad 1856 \\ \underline{-1856} \\ 0 \end{array}$$

 $\therefore \sqrt{54756} = 234$

13. (i) $\sqrt{\frac{36}{5}} = \frac{\sqrt{36}}{\sqrt{5}} = \frac{6}{\sqrt{5}} = \frac{6}{\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{6\sqrt{5}}{5}$
 $\Rightarrow \sqrt{\frac{36}{5}} = \frac{6 \times 2.236}{5} = \frac{13.416}{5} = 2.6832$

(ii) $\sqrt{\frac{625}{98}} = \frac{\sqrt{625}}{\sqrt{98}} = \frac{\sqrt{25 \times 25}}{\sqrt{49 \times 2}} = \frac{25}{7\sqrt{2}}$
 $= \frac{25}{7\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} = \frac{25\sqrt{2}}{7 \times 2} = \frac{25\sqrt{2}}{14}$
 $\Rightarrow \sqrt{\frac{625}{98}} = \frac{25 \times 1.414}{14} = \frac{35.35}{14} = 2.525$

14. Greatest number of 5 digit = 99999

$$\begin{array}{r} 316 \\ 3 \overline{) 99999} \\ \underline{-9} \\ 61 \quad 99 \\ \underline{-61} \\ 626 \quad 3899 \\ \underline{-3756} \\ 143 \end{array}$$

$\therefore$ Required number = $(99999 - 143) = 99856$.

15.
$$\begin{array}{r} 42 \\ 4 \overline{) 1780} \\ \underline{-16} \\ 82 \quad 180 \\ \underline{-164} \\ 16 \end{array}$$

$\therefore$ Number to be added
 $= (43)^2 - 1780 = 1849 - 1780 = 69$.

Long Answer Questions :

- (i) 766 (ii) 1736
- (i) 111 (ii) 311 (iii) 103
- (i) 12.12 (ii) 31.053

4. Let one of the two numbers be a
 The product of numbers is 1575.

$\therefore$ Other number = $\frac{1575}{a}$

It is given that the quotient of the two number is $\frac{9}{7}$.

$\therefore \frac{a}{\frac{1575}{a}} = \frac{9}{7}$

$$\Rightarrow \frac{a^2}{1575} = \frac{9}{7} \Rightarrow a^2 = \frac{9 \times 1575}{7}$$

$$\Rightarrow a^2 = 2025 \Rightarrow a^2 = (5 \times 5) \times (3 \times 3)^2$$

$$\Rightarrow a = 5 \times 3^2 = 45$$

Hence, the numbers are 45 and $\frac{1575}{45} = 35$.

5. Let the length on one side of the square field to x metres. Then,
Area of the field = x^2

$$\text{But, area of the field} = 101\frac{1}{400}\text{m}^2 = \frac{40401}{400}\text{m}^2$$

$$\therefore x^2 = \frac{40401}{400}$$

$$\Rightarrow x = \sqrt{\frac{40401}{400}} = \frac{\sqrt{40401}}{\sqrt{400}}$$

Square roots of 40401 and 400 separately, given below:

$\begin{array}{r} 201 \\ 2 \overline{) 40401} \\ \underline{4} \\ 40 \\ \underline{40} \\ 0 \end{array}$	$\begin{array}{r} 20 \\ 2 \overline{) 400} \\ \underline{4} \\ 40 \\ \underline{40} \\ 0 \end{array}$
--	---

Thus, $\sqrt{40401} = 201$ and $\sqrt{400} = 20$

$$\therefore x = \frac{\sqrt{40401}}{\sqrt{400}} = \frac{201}{20}$$

Hence, the length of one side of the field is $\frac{201}{20}$ metres.

6. We affix 5 zeros to the right of decimal part.

$$237.615 = 237.6150000$$

15.4147	
1	$\overline{) 237.6150000}$
25	$\underline{1}$
304	137
3081	$\underline{125}$
30824	1261
308287	$\underline{1216}$
	4550
	$\underline{3081}$
	146900
	$\underline{123296}$
	2360400
	$\underline{2158009}$
	2020391

$$\therefore \sqrt{237.615} = 15.4147 \text{ upto four places of decimal}$$

$$\Rightarrow \sqrt{237.615} = 15.415 \text{ correct to three places of decimal.}$$

7. Let there were ' a ' men in each row. Then,
Number of rows = a
 $\therefore$ Total number of men in a rows = $a \times a = a^2$
It is given that after arranging men in the form of a square, 9 men were left over.
 $\therefore$ Total number of men = $a^2 + 9$
But the total number of men was 335250
 $\Rightarrow a^2 = 335250 - 9 \Rightarrow a^2 = 335241$
 $\Rightarrow a = \sqrt{335241}$ [By prime factorization]
 $\Rightarrow a = \sqrt{3 \times 3 \times 193 \times 193}$ [Taking one factor from each pair]
 $\Rightarrow a = 3 \times 193 \Rightarrow a = 579$.
Hence, there were 579 men in each row.

2 EXERCISE

Text-Book Exercise :

- 10000200001, 100000020000001
- (i) 25 (ii) 100 (iii) 144
- (i) 24 (ii) 50 (iii) 198
- (i) $39^2 = (30+9)^2 = 30(30+9) + 9(30+9)$
 $= 30^2 + 30 \times 9 + 9 \times 30 + 9^2$
 $= 900 + 270 + 270 + 81 = 1521$
 (ii) $42^2 = (40+2)^2 = 40(40+2) + 2(40+2)$
 $= 40^2 + 40 \times 2 + 2 \times 40 + 2^2$
 $= 1600 + 80 + 80 + 4 = 1764$
- We can get Pythagorean triplets by using general form $2m$, $m^2 - 1$, $m^2 + 1$.
 Let us first take $m^2 - 1 = 8$
 So, $m^2 = 8 + 1 = 9$
 which gives $m = 3$
 Therefore, $2m = 6$ and $m^2 + 1 = 10$
 The triplet is thus 6, 8, 10. But 8 is not the smallest member of this.
 So, let us try $2m = 8$
 then $m = 4$
 We get $m^2 - 1 = 16 - 1 = 15$
 and $m^2 + 1 = 16 + 1 = 17$
 The triplet is 8, 15, 17 with 8 as the smallest member.
- Write $6400 = \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{2 \times 2} \times \underline{5 \times 5}$
 Therefore $\sqrt{6400} = 2 \times 2 \times 2 \times 2 \times 5 = 80$

2	6400
2	3200
2	1600
2	800
2	400
2	200
2	100
2	50
5	25
	5

7. We have $2352 = 2 \times 2 \times 2 \times 2 \times 3 \times 7 \times 7$
 As the prime factor 3 has no pair, 2352 is not a perfect square.
 If 3 gets a pair then the number will become perfect square.
 So, we multiply 2352 by 3 to get,
 $2352 \times 3 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 7 \times 7$

2	2352
2	1176
2	588
2	294
3	147
7	49
	7

Now each prime factor is in a pair. Therefore, $2352 \times 3 = 7056$ is a perfect square.
 Thus the required smallest multiple of 2352 is 7056 which is a perfect square.

And, $\sqrt{7056} = 2 \times 2 \times 3 \times 7 = 84$

8. Greatest number of 4-digits = 9999. We find $\sqrt{9999}$ by long division method. The remainder is 198. This shows 99^2 is less than 9999 by 198.

	99
9	9999
	-81
189	1899
	-1701
	198

Therefore, the required perfect square is $9999 - 198 = 9801$.

And, $\sqrt{9801} = 99$

9. We find $\sqrt{1300}$ by long division method. The remainder is 4. This shows that $36^2 < 1300$.

	36
3	1300
	-9
66	400
	-396
	4

Next perfect square number is $37^2 = 1369$.

Hence, the number to be added is $37^2 - 1300$
 $= 1369 - 1300 = 69$.

10. Area of square plot = 2304 m²

	48
4	2304
	-16
88	704
	704
	0

Therefore, side of the square plot = $\sqrt{2304}$ m

We find that, $\sqrt{2304} = 48$

Thus, the side of the square plot is 48 m.

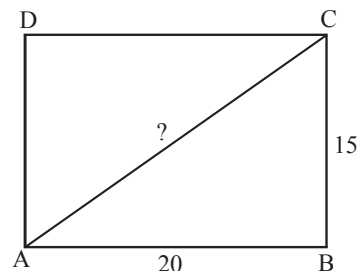
11. (a) 10 cm (b) 12 cm
 12. 24 plants

Exemplar Questions :

1. $43 = 40 + 3$
 So $43^2 = (40 + 3)^2 = (40 + 3)(40 + 3) = 40(40 + 3) + 3(40 + 3)$
 $= 40 \times 40 + 40 \times 3 + 3 \times 40 + 3 \times 3$
 $= 1600 + 240 + 9$
 $= 1849$
 So, $43^2 = 1849$
2. Let AC be the ladder.
 Therefore, AC = 10m
 Let BC be the distance between the foot of the ladder and the wall.
 Therefore, BC = 6m
 $\triangle ABC$ forms a right angled triangle, right angled at B.
 By Pythagoras theorem,
 $AC^2 = AB^2 + BC^2$
 $10^2 = AB^2 + 6^2$
 or $AB^2 = 10^2 - 6^2 = 100 - 36 = 64$
 or $AB = \sqrt{64} = 8$ m
 Hence, the wall is 8m high.
3. Using Pythagoras theorem, we have
 Length of diagonal of the rectangle

$$= \sqrt{(l^2 + b^2)} \text{ units} = \sqrt{(20^2 + 15^2)} \text{ m}$$

$$= \sqrt{400 + 225} \text{ m} = \sqrt{625} \text{ m} = 25 \text{ m}$$



- Hence, the length of diagonal is 25 m.
4. Let the breadth of the field be x metres. Then length of the field is 2x metres.
 Therefore, area of the rectangular field = length $\times$ breadth
 $= (2x)(x) = (2x^2) \text{ m}^2$
 Given that area is 2450 m².
 Therefore, $2x^2 = 2450$
 $x^2 = \frac{2450}{2}$
 $x = \sqrt{1225}$ or $x = 35$ m
 Hence, breadth = 35m and length $35 \times 2 = 70$ m
 Perimeter of the field = $2(l + b)$
 $= 2(70 + 35) \text{ m} = 2 \times 105 \text{ m} = 210 \text{ m}$

5. 7.2
7. 104
6. 8, 12, 20
8. 12, 21, 102, 201

HOTS Questions :

- $(64.53)^2 = 4164$
- Let area of original square = a^2
 $\therefore \frac{2}{3}a^2 = 48 \Rightarrow a^2 = 72$
 Area of second square = b^2
 Thus, $\frac{1}{8}a^2 = b^2 \Rightarrow b^2 = 9 \text{ cm}^2$
- (i) hundred, 3025 (ii) 7, 25
(iii) 7, 8
- (i) 8702 (ii) 6055
(iii) 2009 (iv) 12120
- Hint : Rows = $\sqrt{5929} = 77$
- 3 min 16 sec.
- (i) 0.54
 (ii) $2 \overline{) 23 \text{ } 04}$ $2 \overline{) 17 \text{ } 64}$
 $\begin{array}{r} 48 \\ 2 \overline{) 23 \text{ } 04} \\ \underline{16} \\ 704 \\ \underline{-704} \\ 0 \end{array}$ $\begin{array}{r} 42 \\ 2 \overline{) 17 \text{ } 64} \\ \underline{16} \\ 164 \\ \underline{-164} \\ 0 \end{array}$
 $\therefore \sqrt{2304} = 48$ and $\sqrt{1764} = 42$
 $\sqrt{0.2304} = \frac{48}{100}$ and $\sqrt{0.1764} = \frac{42}{100}$
 $\therefore$ Given expression = 15
- Consider $\sqrt{\frac{125a^6b^4c^2}{5a^4b^2}} = x$
 $\Rightarrow \sqrt{25a^2b^2c^2} = x$
 $\Rightarrow 5abc = x \Rightarrow x^2 = 25a^2b^2c^2$
 $\therefore \frac{x^2}{abc} = \frac{25a^2b^2c^2}{abc} = 25abc$
- Consider $x^{m^2-n^2} \cdot x^{n^2+2mn} \cdot x^{n^2}$
 $= x^{m^2-n^2+n^2+2mn+n^2}$
 $= x^{m^2+2mn+n^2} = x^{(m+n)^2}$
 So, required square root = $x^{\frac{(m+n)^2}{2}}$
- Given $\sqrt[n]{5^n} = 125$
 $\Rightarrow 5^{\frac{n}{2}} = 5 \times 5 \times 5 = 5^3$
 $\Rightarrow n = 6$
 $\therefore 5^{\frac{n}{6}} = 5^2$ ($\because 6^{\text{th}}$ root of $64 = 2$)
 $= 25$
- $n = 20$
 Required Ans is 9.

3 EXERCISE**Single Option Correct :**

- (c) Resolve 136 into prime factors and make group of two of each prime factor
 $136 = 2 \times 2 \times 2 \times 17$
 $136 = (2 \times 2) \times 2 \times 17$
 We find that 2 and 17 doesn't appear in group of two. So, 136 has to be multiplied with 34 to make it a perfect square.
 $\therefore 136 \times 2 \times 17 = (2 \times 2) \times (2 \times 2) \times (17 \times 17)$
 $\Rightarrow 4624 = (2 \times 2) \times (2 \times 2) \times (17 \times 17)$
 $\Rightarrow$ Required square root = $2 \times 2 \times 17 = 68$
- (c) Resolving 3888 into its prime factors, we find that
 $3888 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3$
 $3888 = (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3) \times 3$
 Here we find that prime factor 3 is appearing alone.
 So, if we divide 3888 by 3, we will get a perfect square number
 $\frac{3888}{3} = (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3)$
 $\therefore 1296 = (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3)$
 $\therefore \sqrt{1296} = 2 \times 2 \times 3 \times 3 = 36$
- (b) Let no. of students in the class = x
 $\therefore$ Contribution of each student = ₹ x
 $\therefore x \times x = 1024$
 $\Rightarrow x^2 = 1024$
 $\Rightarrow x^2 = (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (2 \times 2) \times (2 \times 2)$
 $\Rightarrow x = 2 \times 2 \times 2 \times 2 \times 2 = 32$
 $\therefore$ no. of students in the class = 32
- (d) Least number which is divisible by 4, 6, 10, 15 is LCM (4, 6, 10, 15)
 $\text{LCM}(4, 6, 10, 15) = 2 \times 3 \times 5 \times 2$
 $\text{LCM}(4, 6, 10, 15) = 60$
 $60 = 2 \times 2 \times 3 \times 5$
 Here we find that 3 and 5 occurs alone. So, if we multiply 60 by $3 \times 5 = 15$, we get a perfect square no.
 $\therefore 60 \times 15 = 900$
 900 is the least square no. which is divisible by 4, 6, 10, 15.
- (c) Least six digit number is 100000, which is not a perfect square because it has odd number of zeroes.
 First let us extract the square number hidden in it.

$$\begin{array}{r} 2 \overline{) 136} \\ \underline{2} 68 \\ 2 \overline{) 68} \\ \underline{2} 34 \\ 17 \overline{) 34} \\ \underline{17} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 3888} \\ \underline{2} 1944 \\ 2 \overline{) 1944} \\ \underline{2} 972 \\ 2 \overline{) 972} \\ \underline{2} 486 \\ 3 \overline{) 486} \\ \underline{3} 243 \\ 3 \overline{) 243} \\ \underline{3} 81 \\ 3 \overline{) 81} \\ \underline{3} 27 \\ 3 \overline{) 27} \\ \underline{3} 9 \\ 3 \overline{) 9} \\ \underline{3} 3 \\ 3 \overline{) 3} \\ \underline{3} \\ 1 \end{array}$$

$$\begin{array}{r} 2 \overline{) 4, 6, 10, 15} \\ \underline{3} 2, 3, 5, 15 \\ 5 \overline{) 2, 3, 5, 15} \\ \underline{2} 1, 1, 1 \\ 2 \overline{) 1, 1, 1} \\ \underline{1} 1, 1, 1 \end{array}$$

$$\begin{array}{r} 316 \\ 3 \overline{) 10\ 00\ 00} \\ \underline{9} \\ 61 \\ 61 \\ \underline{626} \\ 3900 \\ 3756 \\ \underline{154} \end{array}$$

- ∴ We find that $100000 > (316)^2$ by 154
Next square number $(317)^2 > 100000$
∴ $(316)^2 < 100000 < (317)^2$
∴ If we add $(317)^2 - 100000 = 489$ to 100000
We get least six digit perfect sq. no.
∴ Least four digit perfect square no. is 100489.

6. (b) Let us extract the square root from 24136.

$$\begin{array}{r} 155 \\ 1 \overline{) 2\ 41\ 36} \\ \underline{1} \\ 25 \\ 25 \\ \underline{305} \\ 1636 \\ 1525 \\ \underline{111} \end{array}$$

- ∴ 24136, is 111 more than $(155)^2$. So if we Subtract 111 from 24136, we will get a perfect sq. number.

7. (b)

8. (a) Let the side of square field = 'a' m

$$\therefore \text{Area of square field} = a^2 \text{ sq. m}$$

$$a^2 = 22500 \text{ m}^2$$

$$\Rightarrow a = 150 \text{ m}$$

$$\text{Speed of cycling} = 15 \text{ km/hr}$$

$$= \frac{15 \times 1000}{60 \times 60} = \frac{25}{6} \text{ m/s.}$$

$$\text{Now, total distance to be covered along the boundary} \\ = 4 \times 150 = 600 \text{ m}$$

$$\therefore \frac{25}{6} \text{ m is covered in 1 sec.}$$

$$\therefore 600 \text{ m is covered in } \frac{600}{25} \times 6 = 144 \text{ sec} \\ = 2 \text{ min } 24 \text{ sec.}$$

9. (b) Gardener arranges $(3984 - 15) = 3969$ plants in different rows to form a square.

$$\text{Let no. of plants in each row be 'x'}$$

$$\therefore x \times x = 3969$$

$$x^2 = 3969 \Rightarrow x = 63$$

10. (b) Let the required number be 'x'

$$\text{Now, as per question.}$$

$$x \times \frac{1}{3}x + 18 = 2901$$

$$\Rightarrow \frac{x^2}{3} = 2901 - 18 = 2883 \Rightarrow x^2 = 8649$$

$$x^2 = (3 \times 3) \times (31 \times 31)$$

$$\Rightarrow x = 3 \times 31 = 93$$

11. (d)

12. (a)

13. (d) Given $P = \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$ (1)

$$P^2 = 6 + \sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$$

$$P^2 = 6 + P \quad \text{from (1)}$$

$$\Rightarrow P^2 - P - 6 = 0$$

$$\Rightarrow P^2 - 3P + 2P - 6 = 0$$

$$\Rightarrow P(P - 3) + 2(P - 3) = 0$$

$$\Rightarrow (P - 3)(P + 2) = 0$$

$$\Rightarrow P = 3 \text{ or } P = -2$$

From (1) it is clear that P cannot be negative

$$\therefore P = 3$$

14. (a)

15. (d)

More Than One Option Correct :

1. (a, b)

Pythagorean triplet is given as $(2m, m^2 - 1, m^2 + 1)$.

2. (b, c, d)

3. (c, d)

4. (a, d)

5. (a, b, c, d)

$$(a) \sqrt{7744} = 2 \times 2 \times 2 \times 11 = 88$$

(b) The smallest number divisible by each one of the numbers 8, 9, 10 is their L.C.M..

$$\text{Their L.C.M.} = 360$$

But to make 360 a perfect square, we must multiply it by 10.

- (c) True

$$(d) \overline{36} \overline{06} \overline{25}$$

Since, we have 3 bars over pair of digits.

$$\therefore \sqrt{390625} \text{ has 3 digits.}$$

6. (b, c, d)

7. (a, b, d)

8. (a, b, c)

9. (a, c)

1681 and 1089 are the squares of odd numbers.

Passage Based Questions :

1. (a) $5^2 = 25$

2. (b) 11

3. (a) Area of 1st square field = 256 m^2

$$\text{Area of 2nd square field} = 625 \text{ m}^2$$

Side of 1st and 2nd square field are 16 m and 25 m

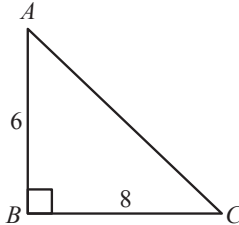
$$\begin{array}{r} 16 \\ 1 \overline{) 256} \\ \underline{1} \\ 26 \\ 156 \\ \underline{156} \\ 0 \end{array}$$

$$\begin{array}{r} 25 \\ 2 \overline{) 625} \\ \underline{4} \\ 45 \\ 225 \\ \underline{225} \\ 0 \end{array}$$

4. (c) Ratio of their sides = 16 : 25

5. (b) Side of the new square field
 $= 16 + 25 = 41$ m
 Area of new square field $= (41)^2 = 1681$ m²

Assertion & Reason :

1. (a) Between 25^2 and 26^2 there are $2 \times 25 = 50$ natural numbers.
 2. (a) Assertion is true. Because 25 and 16 are perfect squares.
 Reason is the correct explanation for Assertion.
 3. (a) Assertion
 $(AC)^2 = 6^2 + 8^2$
 $= 36 + 64 = 100$
 $AC = 10$
- 
4. (c) For every natural number n ,
 $(n+1)^2 = n^2 + (n+1) + n$
i.e., the difference of squares of two consecutive natural numbers is equal to their sum.
 So, $136^2 - 135^2 = 136 + 135 = 271$.
 5. (a) For any natural number $m > 1$,
 $(2m, m^2 - 1, m^2 + 1)$ is a pythagorean triplet.
 Let $m = 4$
 $\Rightarrow 2m = 8, m^2 - 1 = 15, m^2 + 1 = 17$.

Integer Type Questions :

1. (3) Resolving 3888 into its prime factors, we find that
 $3888 = 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3$
 $3888 = (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3) \times 3$
- | | |
|---|------|
| 2 | 3888 |
| 2 | 1944 |
| 2 | 972 |
| 2 | 486 |
| 3 | 243 |
| 3 | 81 |
| 3 | 27 |
| 3 | 9 |
| 3 | 3 |
| 3 | 1 |

Here we find that prime factor 3 is appearing alone.
 So, if we divide 3888 by 3, we will get a perfect square number

$$\frac{3888}{3} = (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3)$$

$$\therefore 1296 = (2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (3 \times 3)$$

$$\therefore \sqrt{1296} = 2 \times 2 \times 3 \times 3 = 36$$

2. (4) $\sqrt{388 + \sqrt{127 + \sqrt{289}}}$
 $= \sqrt{388 + \sqrt{127 + 17}} \quad [\because \sqrt{289} = 17]$
 $= \sqrt{388 + \sqrt{144}} \quad [\because \sqrt{144} = 12]$
 $= \sqrt{388 + 12} = \sqrt{400}$
 $= 20 \quad [\because \sqrt{400} = 20] \quad K = \frac{20}{5} = 4$
3. (4) $\sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + \sqrt{225}}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{154 + 15}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + \sqrt{169}}}}$
 $= \sqrt{10 + \sqrt{25 + \sqrt{108 + 13}}} = \sqrt{10 + \sqrt{25 + \sqrt{121}}}$
 $= \sqrt{10 + \sqrt{25 + 11}} = \sqrt{10 + \sqrt{36}} = \sqrt{10 + 6} = \sqrt{16} = 4$
4. (4) $4 = 2 \times 2 = 2^2$.
5. (1) $\sqrt{1 + \frac{27}{169}} = \sqrt{\frac{196}{169}} = \frac{14}{13} = 1\frac{1}{13} = 1 + \frac{1}{13}$
 $\therefore x = 1$
6. (1) We define $\sqrt{-a}$ as the quantity which is being multiplied by itself (or) squared become equal to $-a$.
 So $\sqrt{-a} \times \sqrt{-a} = (\sqrt{-a})^2 = -a$
 $\therefore \sqrt{-1} \times \sqrt{-1} = -1$
7. (3) Let numbers be x and y then $x^2 - y^2 = 60$
 $(x - y)(x + y) = 60$
 $5 \times 12 = 60$
 $2 \times 30 = 60$
 $4 \times 15 = 60$
 $1 \times 60 = 60$
 $3 \times 20 = 60$
 $6 \times 10 = 60$
 Here, pairs "2 and 30" and "6 and 10" satisfying the given condition.
8. (2) $\frac{\sqrt{32} + \sqrt{48}}{\sqrt{8} + \sqrt{12}} = \frac{\sqrt{16 \times 2} + \sqrt{16 \times 3}}{\sqrt{4 \times 2} + \sqrt{4 \times 3}}$
 $\frac{4\sqrt{2} + 4\sqrt{3}}{2\sqrt{2} + 2\sqrt{3}} = \frac{4(\sqrt{2} + \sqrt{3})}{2(\sqrt{2} + \sqrt{3})} = 2$
9. (2) Number of zeroes at the end of the number $100 = 2$
 $\therefore$ Number of zeroes at the end of the square of the number $100 = 2 \times 2 = 4$

Chapter

7

Cube and Cube Roots

INTRODUCTION

If a number multiply twice by itself the number thus obtained is the cube of the number.

And the number itself is the cube root of the cubed number.

This chapter has a direct practical use in

- (i) Calculating the cubes of non-zero numbers
- (ii) Finding the relation between unit digit of a number and its cube.
- (iii) Calculating the cube roots using different methods.

CUBES

If 'n' is a non-zero number, then $n \times n \times n$, written as n^3 is called cube of n or n cubed.

For example, $3 \times 3 \times 3 = 27$ is cube of 3 or 3 cubed.

A number 'n' is a perfect cube number if there is an integer 'm' such that $n = m \times m \times m$.

For example, 64 is a perfect cube number as 4 is an integer such that $4 \times 4 \times 4 = 64$.

Test of Perfect Cube

"In a prime factorization of perfect cube, every prime number occurs three times or a multiple of three times.

For Example :

(i) $64 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 2^6$

(ii) $729 = 9 \times 9 \times 9 = 3 \times 3 \times 3 \times 3 \times 3 \times 3 = 3^6$

ILLUSTRATION : 1

Examine (i) 243 (ii) 21636 are perfect cubes.

SOLUTION :

(i) $243 = (3 \times 3 \times 3) \times 3 \times 3$

Here occurrence of 3 is not in multiple of '3', so 243 is not a perfect cube number.

(ii) $21636 = (2 \times 2 \times 2) \times (3 \times 3 \times 3) \times 11 \times 11$

In 21636, prime number 11 does not occur in group of three, so 21636 is not a perfect cube number.

Properties of Cubes of Numbers

1. If a number is negative, then its cube is also negative.

Example : (i) $(-2)^3 = (-2) \times (-2) \times (-2) = -8$

(ii) $(-5)^3 = (-5) \times (-5) \times (-5) = -125$

This shows that negative numbers can also be perfect cubes.

2. Cube of an even number is also even.

Example : (i) $(2)^3 = 2 \times 2 \times 2 = 8$

(ii) $(6)^3 = 6 \times 6 \times 6 = 216$.

3. Cube of an odd number is also odd.

Example : (i) $(3)^3 = 3 \times 3 \times 3 = 27$

(ii) $(7)^3 = 7 \times 7 \times 7 = 343$.

Relation Between Unit Digit of a Number and its Perfect Cube

Observe the adjoining table-I.

It can be observed from the table that if the unit digit of a number is 1, 4, 5, 6, 9, then unit digit of cube is also 1, 4, 5, 6, 9. If the unit digit of number is 2 or 8, then the unit digit in cube is 8 or 2 respectively.

Table-I

x	1	2	3	4	5	6	7	8	9
x^3	1	8	27	64	125	216	343	512	729

3 & 7 form similar pair as 2 and 8.

Example :

- (i) Unit place digit of cube of 41 is 1 because unit place digit of 41 is 1 .
 (ii) Unit place digit of cube of 76 is 6 because unit place digit of 76 is 6 .
 (iii) Unit place digit of cube of 92 is 8 because unit place digit of 92 is 2.

CUBE OF TWO DIGIT POSITIVE NUMBER (SHORT-CUT METHOD)

Let ab is a two digit number. To find cube of ab using short-cut method, we use the identity $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$ and form 4 columns $a^3 / 3a^2b / 3ab^2 / b^3$. This method is same as finding square by column method. We retain the unit digit after adding numbers in each column and add the remaining digit to the next column.

ILLUSTRATION : 2

Find value of $(25)^3$.

SOLUTION :

Take $a = 2$, $b = 5$

a^3	$3a^2b$	$3ab^2$	b^3	
8	$3 \times 4 \times 5$	$3 \times 2 \times 25$	125	
8	60	150	125	
8	60	150		
+ 7	16	+ 12		
15	76	162		
15	6	2	5	

$\therefore (25)^3 = 15625$

CUBE ROOTS

If x is a perfect cube such that $x = y^3$, then y is called cube root of ' x '. For example $8 = 2^3$, so 2 is cube root of 8, 6 is cube root of 216. It is written as $6 = \sqrt[3]{216}$.

The symbol $\sqrt[3]{}$ is used to represent cube root.

Cube Root by Successive Subtraction

If we subtract 1, 7, 19, 37, 61, 91 from a given number, till we get zero, then the number of subtractions will give the cube root of the given number. The numbers 1, 7, 19, 37 are obtained by putting $m = 1, 2, 3, 4, \dots$ in

$$[1 + m \times (m-1) \times 3]$$

Example: $64 - 1 = 63$, $63 - 7 = 56$, $56 - 19 = 37$, $37 - 37 = 0$.

Here number of subtractions till we get zero is 4. Therefore $\sqrt[3]{64} = 4$

Cube Root by Prime Factorization

We know that in a perfect cube primes, occurs in multiples of three. We can find cube root by following steps.

- Find prime factors of given number.
- Group the factors in triples such that all three factors in each triple are the same.
- If any prime factor is left ungrouped, then the given number is not a perfect cube.
- If no factor is left ungrouped, choose one factor from each group and take the product. The product is the cube root of given number.

ILLUSTRATION : 3

Find cube root of 32768.

SOLUTION :

$$32768 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)$$

$$\therefore \sqrt[3]{32768} = 2 \times 2 \times 2 \times 2 \times 2 = 32$$

2	32768
2	16384
2	8192
2	4096
2	2048
2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

Relation Between Unit Digit of a Perfect Cube and Unit Digit of its Cube Root

Table-2 shows the relationship between the unit digit of a perfect cube and unit digit of its cube root.

Table - 2

Unit digit of a perfect cube no.	1	2	3	4	5	6	7	8	9	0
Unit digit of cube root	1	8	7	4	5	6	3	2	9	0

Using Table -1 (given before) and Table-2 we can calculate cube roots upto 2- digits very easily, however by suitably extending Table -1, we can calculate cube root upto any number of digits.

Steps :

- (1) Put a slash before the last three digits of given perfect cube number. 103823 can represent as 103 / 823, 1331 can represent as 1 / 331.
- (2) The number on right side of slash will give us the unit place digit of cube root and that on left hand side will give rest of the digits of cube root.
- (3) The unit place digit of cube root can be found by looking at unit place digit of right side of given number and comparing it with table-2. For example, cube root of 287496 will have 6 at units place in its cube root because the number itself has 6 in its place.
- (4) To find the left hand part of the answer, we take the number which lies to the left of the slash. Now we need to find two perfect cubes between which the left hand part lies. For example in 287496, left part 287 lies between 216 (cube of 6) and 343 (cube of 7).
- (5) Now, out of the numbers 6 and 7 obtained above, we take the smaller number and put it on the left of unit digit obtained in step - (3). Thus root of 287496, which is a perfect cube, is 66.

Thus, by simple observation, we will be able to find cube root of given perfect cube number.

ILLUSTRATION : 4

Find cube root of 1157625.

SOLUTION :

- (a) Represent 1157625 as 1157 / 625.
- (b) Number ends with 5, so the cube root ends with 5 (Table-2)
- (c) Left part 1157 lies between 1000 (cube of 10) and 1331 (cube of 11).
- (d) Smaller number is 10. So, cube root of 1157625 is 105.

Cube Root of Negative Numbers

Cube root of a product of two perfect cubes is the product of cube roots of perfect cubes.

$$\sqrt[3]{p \times q} = \sqrt[3]{p} \times \sqrt[3]{q}$$

$$\sqrt[3]{64 \times 125} = \sqrt[3]{64} \times \sqrt[3]{125}$$

By using the above result, we can find cube root of negative integers.

$$\text{We have, } \sqrt[3]{-n} = \sqrt[3]{-1} \times \sqrt[3]{n}$$

$$\therefore \sqrt[3]{-n} = -\sqrt[3]{n} \quad \left[\sqrt[3]{-1} = -1 \text{ because } (-1)^3 = -1 \right]$$

ILLUSTRATION : 5

Find cube root of

(i) -64

(ii) -1331

SOLUTION :

(i) $\sqrt[3]{-64} = -\sqrt[3]{64} = -4$

(ii) $\sqrt[3]{-1331} = -\sqrt[3]{1331} = -11.$

Cube Root of Rational Numbers

If p and q , where $q \neq 0$, are perfect cubes, then $\sqrt[3]{\frac{p}{q}} = \frac{\sqrt[3]{p}}{\sqrt[3]{q}}$. Here $\sqrt[3]{p}$ and $\sqrt[3]{q}$ are integers and $\sqrt[3]{q} \neq 0$. Therefore $\frac{\sqrt[3]{p}}{\sqrt[3]{q}}$ is a rational number.

Example: (i) $\sqrt[3]{\frac{125}{64}} = \frac{\sqrt[3]{125}}{\sqrt[3]{64}} = \frac{5}{4}$ (ii) $\sqrt[3]{\frac{-729}{1331}} = -\frac{\sqrt[3]{729}}{\sqrt[3]{1331}} = -\frac{9}{11}$

MISCELLANEOUS SOLVED EXAMPLES

1. Find the smallest number that must be subtracted from

(a) 132

(b) 348 so as to make them perfect cube.

Sol. (a) By successive subtraction method we have

$$132 - 1 = 131; 131 - 7 = 124; 124 - 19 = 105;$$

$$105 - 37 = 68; 68 - 61 = 7$$

Now we cannot subtract 91 from 7 as $7 < 91$ so if we subtract 7 from 132, we get a perfect cube number.

$$\therefore 132 - 7 = 125 = (5)^3.$$

(b) By successive subtraction method, we have

$$348 - 1 = 347; 347 - 7 = 340; 340 - 19 = 321;$$

$$321 - 37 = 284; 284 - 61 = 223; 223 - 91 = 132;$$

$$132 - 127 = 5$$

Now we cannot subtract 169 from 5 as $5 < 169$. So if we subtract 5 from 348, we get a perfect cube number.

$$\therefore 348 - 5 = 343 = (7)^3.$$

2. Find cube root of

(a) 250047 and

(b) -13824 by prime factorization method.

Sol. In a perfect cube, prime factors occurs in triplets.

$$(a) 250047 = (3 \times 3 \times 3) \times (3 \times 3 \times 3) \times (7 \times 7 \times 7)$$

$$\therefore \sqrt[3]{250047} = 3 \times 3 \times 7$$

$$\sqrt[3]{250047} = 63$$

3	250047
3	83349
3	27783
3	9261
3	3087
3	1029
7	343
7	49
7	7
	1

$$(b) \sqrt[3]{-13824} = -\sqrt[3]{13824}$$

$$13824 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (3 \times 3 \times 3)$$

$$\therefore \sqrt[3]{13824} = 2 \times 2 \times 2 \times 3 = 24$$

$$\therefore -\sqrt[3]{13824} = -24$$

2	13824
2	6912
2	3456
2	1728
2	864
2	432
2	216
2	108
2	54
3	27
3	9
3	3
	1

3. Divide the number 13122 by the smallest number so that the quotient is a perfect cube. Also find the cube root of the quotient.

Sol. Resolving 13122 into its prime factors, we get

$$13122 = 2 \times (3 \times 3 \times 3) \times (3 \times 3 \times 3) \times 3 \times 3$$

In a perfect cube, each prime factor occurs thrice.

So the smallest number by which 13122 must be divided to make it a perfect cube is $2 \times 3 \times 3 = 18$

$$\therefore \text{New number} = \frac{13122}{18} = 729$$

$$\text{Now } \sqrt[3]{729} = \sqrt[3]{(3 \times 3 \times 3) \times (3 \times 3 \times 3)} = 3 \times 3 = 9.$$

4. Three numbers are in the ratio 1: 2: 3. The sum of their cubes is 98784. Find the numbers.

Sol. Let the ratio of numbers = x

$\therefore$ Numbers are x, 2x, 3x.

$$\text{Now } x^3 + (2x)^3 + (3x)^3 = 98784$$

$$\Rightarrow 36x^3 = 98784$$

$$\Rightarrow x^3 = \frac{98784}{36} = 2744$$

$$\Rightarrow x = \sqrt[3]{14 \times 14 \times 14} = 14$$

$\therefore$ Numbers are 14, 28 and 42.

5. Find the value of $\sqrt[3]{0.000027}$.

$$\text{Sol. } \sqrt[3]{0.000027} = (0.000027)^{1/3} = \left(\frac{27}{1000000}\right)^{1/3} = \frac{(27)^{1/3}}{(1000000)^{1/3}} = \frac{(3 \times 3 \times 3)^{1/3}}{(100 \times 100 \times 100)^{1/3}} = \frac{3}{100} = 0.03.$$

6. The volume of a cubical box is 32.768 cubic metres. Find the length of a side of the box.

Sol. Let the side of cubical box be 'a' metre. The volume of cubical box is a^3 cubic metre.

$$\therefore a^3 = 32.768 \Rightarrow a = \sqrt[3]{32.768} \Rightarrow a = \sqrt[3]{\frac{32768}{1000}}$$

$$32768 = (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2)$$

$$\therefore \sqrt[3]{32768} = 2 \times 2 \times 2 \times 2 \times 2 = 32$$

$$\therefore \sqrt[3]{1000} = \sqrt[3]{10 \times 10 \times 10} = 10$$

$$\therefore a = \sqrt[3]{\frac{32768}{1000}} = \frac{\sqrt[3]{32768}}{\sqrt[3]{1000}} = \frac{32}{10} = 3.2$$

$\therefore$ side of cubical box = 3.2 m.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- Numbers obtained when a number is multiplied by itself three times are known as
- The smallest number by which 81 should be divided to make it a perfect cube is
- If a number ends in two 9's then its cube ends in number of 9's.
- The cube of a number is that number raised to power
- Cube of even numbers are always
- Cubes of odd numbers are always
- The cube root of x is denoted by
- Cubes of negative integers are always

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- When a number is multiplied by itself, the product is said to be cube of that number.
- The cube root of 125 is 25.
- If x is a non-zero number, then $x \times x \times x$ is written as x^3 .
- Square root is the inverse operation of cube root.
- If in the prime factorization of any number each factor appears three times, then the number is a perfect cube.
- The symbol $\sqrt[3]{}$ denotes cube root and $\sqrt{}$ denotes square root.
- Cube of a natural number which is a multiple of 3 is also a multiple of 27.
- Cube root of a negative number can be positive.
- No cube can end with exactly two zeros.
- For an integer a , a^3 is always greater than a^2 .

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.

1. Column-I (Numbers)

- A. 121
B. 25
C. 67
D. 68
E. 52

2. Column-I

- A. $\sqrt[3]{125 \times 27} = 3 \times \dots$
B. $\sqrt[3]{8 \times \dots} = 8$
C. $\sqrt[3]{1728} = 4 \times \dots$
D. $\sqrt[3]{\dots} = \sqrt[3]{4} \times \sqrt[3]{5} \times \sqrt[3]{6}$
E. $\sqrt[3]{480} = \sqrt[3]{3} \times 2 \times \sqrt[3]{\dots}$
F. $\sqrt[3]{\frac{729}{1331}} = \frac{9}{\dots}$

Column-II (Digit in the unit place for the cube)

- (p) 3
(q) 8
(r) 2
(s) 1
(t) 5

Column-II

- (p) 120
(q) 20
(r) 64
(s) 11
(t) 3
(u) 5

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- Find $\sqrt[3]{101 + \sqrt[3]{13824}}$.
- If $a = 2b$ and $b = 4c$ then find the value of $\sqrt[3]{\frac{a^2}{16bc}}$.
- The value of $(1.5)^3$ is
- The cube root of 1.331 is
- If $36a^4 = a^6$, then a^3 is equal to
- $\sqrt[3]{-3\frac{3}{8}} = \dots$
- Find the value of $\left[(5^2 + 12^2)^{\frac{1}{2}} \right]^3$.
- Find the value of $\sqrt[3]{\frac{0.027}{0.008}} - \sqrt{\frac{0.09}{0.04}} - 1$
- If $y = \sqrt[3]{2\frac{93}{125}}$, then find the value of y .

10. Given that $\sqrt[3]{x} = -6$, find the value of x .
11. Calculate the exact value of $\sqrt[3]{8 \times 10^6}$.
12. $9 + \sqrt[3]{-125} = \underline{\hspace{2cm}}$.
13. The value of $(27 \times 2744)^{1/3}$ is $\underline{\hspace{2cm}}$.
14. $\sqrt[3]{(156 + x)} = 12$, then find the value of x .
15. $\sqrt[3]{0.125} + 3 = \underline{\hspace{2cm}}$
8. If the product of two numbers is 21 and their difference is 4, then find the ratio of the sum of their cubes to the difference of their cubes.
9. Find the cube root of -13824 .
10. In a five digit number $1b\ 6a3$, 'a' is the greatest single digit perfect cube and twice of it exceeds 'b' by 7. Then find the sum of the number and its cube root.
11. What is the value of expression $2(3^3 - 6) + 2$?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. Prove that if a number is doubled, then its cube is eight times the cube of the given number.
2. Given that $y = -8$. Find the value of $\sqrt[3]{y}$.
3. Evaluate :
 - (i) $125 \sqrt[3]{a^6} - \sqrt[3]{125a^6}$
 - (ii) $\sqrt[3]{700 \times 2 \times 49 \times 5}$
4. Consider the following pattern.
 $2^3 - 1^3 = 1 + 2 \times 1 \times 3$
 $3^3 - 2^3 = 1 + 3 \times 2 \times 3$
 $4^3 - 3^3 = 1 + 4 \times 3 \times 3$
 Using the above pattern find the value of $7^3 - 6^3$
5. Find the value of $\sqrt[3]{288} \times \sqrt[3]{432} \times \sqrt[3]{648}$.
6. If $\sqrt[3]{4 \frac{12}{125}} = x$, then the value of x is $\underline{\hspace{2cm}}$.
7. The three numbers are in the ratio 2 : 3 : 4. The sum of their cubes is 33957. Find the numbers.

$$12. \sqrt[3]{\frac{a^6 \times b^3 \times c^{21}}{c^9 \times a^{12}}} = ?$$

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Find the smallest number which when multiplied with 137592 will make the product a perfect cube. Further find the cube root of the product.
2. Show that -17576 is a perfect cube. Also, find the number whose cube is -17576 .
3. The value of $(4096)^{\left(\frac{1}{\sqrt{3}}\right)^{(\sqrt{2})^2}}$ is $\underline{\hspace{2cm}}$.
4. Show that $\frac{-216}{42875}$ is the cube of a rational number. Also, find that rational number.
5. Divide the number 26244 by the smallest number so that the quotient is a perfect cube. Also, find the cube root of the quotient.
6. The volume of a cubical box is 32.768 cubic metres. Find the length of a side of the box.

2

EXERCISE

Text-Book Exercise :

1. Is 68600 a perfect cube? If not, find the smallest number by which 68600 must be multiplied to get a perfect cube.
2. Find the smallest number by which each of the following numbers must be multiplied to obtain a perfect cube.
 - (i) 256 (ii) 675 (iii) 100
3. Find the smallest number by which each of the following numbers must be divided to obtain a perfect cube.
 - (i) 81 (ii) 135 (iii) 704
4. Parikshit makes a cuboid of plasticine of sides 5 cm, 2 cm, 5 cm. How many such cuboids will he need to form a cube?
5. Find the cube root of 13824 by prime factorisation method.
6. Find the cube root of 17576 through estimation.

Exemplar Questions :

1. Is 9720 a perfect cube? If not, find the smallest number by which it should be divided to get a perfect cube.
2. Find the length of each side of a cube if its volume is 512 cm^3 .
3. Three numbers are in the ratio 2:3:4. The sum of their cubes is 0.334125. Find the numbers.
4. Evaluate: $\sqrt[3]{27} + \sqrt[3]{0.008} + \sqrt[3]{0.064}$
5. $\left\{ \left[6^2 + (8^2)^{\frac{1}{2}} \right] \right\}^3 = ?$

HOTS Questions :

- Evaluate: $\frac{\sqrt[3]{15625 \times 216}}{\sqrt[3]{3375}}$
- Evaluate: $\sqrt[3]{216 \times (-343)}$.
- Find the volume of cube whose surface area is 150 m^2 .
- For a big icecream of volume 2744 cm^3 , Mukti wants to make a box. What should be the edge of box so that the block can be put into it?
- Consider the following pattern:
 $2^3 - 1^3 = 1 + 2 \times 1 \times 3$
 $3^3 - 2^3 = 1 + 3 \times 2 \times 3$
 $4^3 - 3^3 = 1 + 4 \times 3 \times 3$
 Find the value of $12^3 - 11^3$, using the above

- Find the value of $\sqrt{2\frac{1}{4}} \times \left(1\frac{1}{3}\right)^2 + 1 \div \sqrt[3]{3\frac{3}{8}}$.
- What number will come in place of question mark?
 $(\sqrt{8} \times \sqrt{8})^{\frac{1}{2}} + (9)^{\frac{1}{2}} = (?)^3 + \sqrt{8} - 340$
- Find the value of $\frac{(2.3)^3 - 0.027}{(2.3)^2 + 0.69 + 0.09}$
- $\sqrt[3]{\sqrt{0.000064}}$ equals _____.

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The smallest number which when multiplied with 7200 will make the product a perfect cube, is
 (a) 10 (b) 20
 (c) 30 (d) None of these.
- The smallest number by which 16384 must be divided so that quotient is a perfect cube, is
 (a) 2 (b) 4
 (c) 12 (d) None of these.
- If the sum of cubes of digits of a number is equal to the number itself, the number is called 'Armstrong Number'. then the Armstrong Number is
 (a) 367 (b) 470
 (c) 153 (d) None of these.
- The perfect cube nearest to 2750 is
 (a) 2749 (b) 2747
 (c) 2744 (d) 2754
- Value of $\sqrt[3]{392} \times \sqrt[3]{448}$ is
 (a) 50 (b) 52
 (c) 54 (d) 56
- A $8 \times 6 \times 4 \text{ cm}^3$ metallic cuboid is melted. The minimum volume of molten metal which should be added to mould it into a cube whose edge is 'x' where 'x' is an integer, is
 (a) 20 cm^3 (b) 21 cm^3
 (c) 23 cm^3 (d) 24 cm^3
- The volumes of two cubes are in the ratio 343 : 1331, the ratio of their edges, is
 (a) 7:10 (b) 7 : 11
 (c) 7 : 12 (d) None of these.
- The cube of a number is 8 times the cube of another number. If the sum of the cubes of numbers is 243, the difference of the numbers is,
 (a) 3 (b) 4
 (c) 6 (d) None of these
- The square of a natural number when subtracted from its cube results in 48. The number is
 (a) 6 (b) 5
 (c) 4 (d) 8
- $\frac{16}{9} \times \left(-1\frac{1}{2}\right)^3 = \dots\dots\dots$
 (a) -12 (b) -6 (c) $-\frac{8}{3}$ (d) $\frac{8}{9}$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d), out of which ONE or MORE may be correct.

- Which of the following numbers are the cube of a negative whole number ?
 (a) -64 (b) -2197
 (c) -1056 (d) -3888
- Which of the following is/are correct statement(s) ?
 (a) 8640 is not a perfect cube.
 (b) If a divides b , then a^3 divides b^3 .
 (c) If a and b are integers such that $a^2 > b^2$, then $a^3 > b^3$.
 (d) None of these

3. Which of the following statements are correct?

- (a) $m \times m \times m = 3m$
 (b) $x^3 = n \Rightarrow x = n^{\frac{1}{3}}$
 (c) $\sqrt[3]{a} + \sqrt[3]{b} = \sqrt[3]{a+b}$
 (d) $\sqrt[3]{a} \div \sqrt[3]{b} = \sqrt[3]{a \div b}$

4. Which of the following numbers are perfect /complete cube?

- (a) 64 (b) 216
 (c) 343 (d) 256

5. Which of the following numbers are cubes of rational numbers?

- (a) $\frac{27}{64}$ (b) $\frac{125}{128}$
 (c) 0.001331 (d) 0.04

6. $(-216 \times 1728)^{1/3} = \dots\dots\dots$

- (a) -6×-12 (b) -72
 (c) -6×12 (d) -27

7. The value of $\sqrt[3]{-125 \times (-1000)}$ is $\dots\dots\dots$

- (a) $(5 \times 10)^{1/3}$ (b) $(5^3 \times 10^3)^{1/3}$
 (c) $(5 \times 10)^3$ (d) $((5 \times 10)^3)^{1/3}$

8. Find the value of $\sqrt[3]{2\frac{10}{27}} + (2^2 - 3^2)$.

- (a) $\frac{-15}{4}$ (b) $\left(\frac{-15}{4}\right)^{-1}$
 (c) $\left(\frac{1}{15}\right)^{-1}$ (d) $\frac{-4}{15}$

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE-I

Let $a = \frac{m}{n}$ be a rational number (m, n are non-zero integers such that $n \neq \pm 1$) other than an integer, then the cube of a is defined as $a^3 = a \times a \times a$ or $\left(\frac{m}{n}\right)^3 = \frac{m}{n} \times \frac{m}{n} \times \frac{m}{n} = \frac{m^3}{n^3}$

1. Show that $\frac{-216}{42875}$ is the cube of a rational number.

2. Show that 0.001728 is the cube of a rational number.

3. Which of the following numbers are cubes of rational numbers?

- (a) $\frac{27}{64}$ (b) $\frac{125}{128}$
 (c) 0.001331 (d) 0.04

PASSAGE - II

We have for any positive integer x , $-x$ is a negative integer such that

$$(-x)^3 = (-x) \times (-x) \times (-x) = -x^3$$

$$\therefore \sqrt[3]{-x^3} = -x \Rightarrow \sqrt[3]{-x^3} = -\sqrt[3]{x^3}$$

Thus, the cube root of a negative perfect cube is negative of the cube root of its absolute value. In other words, to find the cube root of a negative perfect cube, we find the cube root of its absolute value and multiply it by -1 .

4. Find the cube root of -226981 and -175616 .

- (a) $-51, -46$ (b) $-61, -56$
 (c) $-59, -96$ (d) None of these

5. Cube root of -125×-3375 is

- (a) 57 (b) 75
 (c) 15 (d) 51

6. Cube root of $\frac{-3375}{-2744}$ is

- (a) $\frac{14}{15}$ (b) $\frac{-14}{15}$
 (c) $\frac{15}{14}$ (d) $\frac{-15}{14}$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** Cube root of 64000 is 40 and 64000 is a perfect cube.

Reason : No cube can end with exactly two zeros.

2. **Assertion :** $14^3 = 2744, 24^3 = 13822$

Reason : The digits of number end with 4, then cubes of the number ends with same digit 4.

3. **Assertion :** $\sqrt[3]{-4^3} = \sqrt[3]{\left(\frac{1}{4}\right)^3}$

Reason : For any positive integer x , $-x$ is a negative integer such that $\sqrt[3]{-x^3} = -x$.

4. **Assertion :** We know that
 $9 \div 3 = 3$ and $9^3 \div 3^3 = 27$
 $8 \div 2 = 4$ and $8^3 \div 2^3 = 64$
Reason : If a divides b , then a^3 divides b^3 .

Multiple Matching Questions :

DIRECTIONS : Following question has five statements (A, B, C, D and E) given in Column I and five statements (p, q, r, s and t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column II.

- | 1. | Column-I | Column-II |
|----|---|----------------------------|
| A. | The smallest number which must be subtracted from 1730 to make it a perfect cube, is | (p) an even number |
| B. | If $x^y z = 2^8$ then $x + y + z$ is equal to, (x, y and z are positive integers) | (q) 7 |
| C. | If $a = 2b, b = 4c$ then $\sqrt[3]{\frac{a^4}{32b^2c^2}} =$ | (r) 2 |
| D. | If single digit natural numbers satisfying $x^2 + y^2 = 145$, then the value of $x + y$ is | (s) 17 |
| E. | If a and b are whole numbers such that $b^a = 512, b > a, 1 < a < 4$ then $\sqrt[4]{b} = ?$ | (t) Smallest prime number. |

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- The product $864 \times n$ is a perfect cube. What is the smallest possible value of 'n'?
- If $72k$ is a perfect cube, then find the value of k .
- If $\sqrt[3]{\frac{a^6 \times b^3 \times c^{21}}{c^9 \times a^{12}}} = \frac{bc^k}{a^{k/2}}$, then $k =$ _____.
- $\sqrt[3]{0.125} + \sqrt[3]{0.729} = n \times 0.7$. Find n .
- The cube of a number ending in 3, ends in _____.
- The cube root of the number 10648 is $k\sqrt[3]{121}$. Find k .

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- cube numbers
- 3
- two
- 3, cubes of $a = a \times a \times a = a^3$. So it raised to power 3.
- even, Example: $2^3 = 2 \times 2 \times 2 = 8$.
- odd, Example: $3^3 = 3 \times 3 \times 3 = 27$.
- $\sqrt[3]{x}$
- negative, Example: $\sqrt[3]{(-1)} = \sqrt[3]{-1 \times -1 \times -1} = -1$.

True/False :

- False.
- False, the cube root of 125 is 5.
- True.
- False, square root is the inverse operation of square.
- True.
- True.
- True: Let the number be $3m$.
Cube of $3m = (3m)^3 = 3m \times 3m \times 3m = 27m^3$
 $\therefore$ cube is multiple of 27.
- False : Cube root of a negative number is always negative.
- True
- False: For negative integers, a^3 is less than a^2
For example $(-2)^3 = -8$
 $(-2)^2 = 4$ and $(-2)^3 < (-2)^2$.

Match The Columns :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (t); (C) $\rightarrow$ (p); (D) $\rightarrow$ (r); (E) $\rightarrow$ (q)
- (A) $\rightarrow$ (u); (B) $\rightarrow$ (r); (C) $\rightarrow$ (t); (D) $\rightarrow$ (p); (E) $\rightarrow$ (q); (F) $\rightarrow$ (s)

Very Short Answer Questions :

- $\sqrt[3]{101+24} = \sqrt[3]{125} = 5$
- $\sqrt[3]{\frac{a^2}{16bc}} = \sqrt[3]{\frac{(2b)^2}{16bc}} = 1$
- $(1.5)^3 = 3.375$
- $\sqrt[3]{1.331} = \sqrt[3]{(1.1)^3} = 1.1$
- $36a^4 = a^6$
On taking square root of both sides, we get
 $6a^2 = a^3$

$$6. \quad \sqrt[3]{-3\frac{3}{8}} = \sqrt[3]{\frac{-27}{8}} = \sqrt[3]{\frac{-3^3}{2^3}} = \frac{-3}{2}$$

$$7. \quad [(5^2 + 12^2)^{1/2}]^3 \\ = [(25 + 144)^{1/2}]^3 \\ = [(169)^{1/2}]^3 = (13)^3 = 2197$$

$$8. \quad \sqrt[3]{\frac{0.027}{0.008}} - \sqrt{\frac{0.09}{0.04}} - 1 \\ = \frac{0.3}{0.2} - \frac{0.3}{0.2} - 1 = -1$$

$$9. \quad y = \sqrt[3]{2\frac{93}{125}} = \sqrt[3]{\frac{343}{125}} = \frac{7}{5}$$

$$10. \quad \sqrt[3]{x} = -6 \\ x = (-6)^3 = (-6) \times (-6) \times (-6) = -216$$

$$11. \quad \sqrt[3]{8 \times 10^6} \\ = \sqrt[3]{2 \times 2 \times 2 \times 10 \times 10 \times 10 \times 10 \times 10 \times 10} \\ = 2 \times 10 \times 10 = 200.$$

$$12. \quad 9 + \sqrt[3]{-125} = 9 - 5 = 4$$

$$13. \quad (27 \times 2744)^{1/3} = 3 \times 14 = 42$$

$$14. \quad \sqrt[3]{156 + x} = 12 \\ \Rightarrow 156 + x = 1728 \\ \Rightarrow x = 1728 - 156 = 1572$$

$$15. \quad \sqrt[3]{0.125} + 3 = 0.5 + 3 = 3.5$$

Short Answer Questions :

- Let the given number be 'a'.
Let b be the double of 'a'
i.e. $b = 2a$
Consider, $b^3 = b \times b \times b$
 $= 8 \times a \times a \times a = 8a^3$
 $\Rightarrow b^3 = 8 \times (\text{cube of } a)$
- $\sqrt[3]{-8} = (-2)$
- (i) $125\sqrt[3]{a^6} - \sqrt[3]{125a^6}$
 $= 125a^2 - 5a^2 = 120a^2$
(ii) $\sqrt[3]{700 \times 2 \times 49 \times 5}$
 $= \sqrt[3]{7 \times 2 \times 2 \times 2 \times 5 \times 5 \times 7 \times 7 \times 5}$
 $= 7 \times 2 \times 5 = 70$

2	4096
2	2048
2	1024
2	512
2	256
2	128
2	64
2	32
2	16
2	8
2	4
	2

$$\Rightarrow \sqrt[3]{4056} = 2 \times 2 \times 2 \times 2 = 16$$

4. Resolving 216 and 42875 into prime factors, we get

$$216 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \text{ and } 42875 = 5 \times 5 \times 5 \times 7 \times 7 \times 7$$

2	216	5	42875
2	108	5	8575
2	54	5	1715
3	27	7	343
3	9	7	49
3	3	7	7
	1		1

Clearly, 216 and 42875 can be grouped into triples of equal factors.

$$\therefore 216 = (2 \times 3)^3 \text{ and } 42875 = (5 \times 7)^3$$

$$\Rightarrow 216 = 6^3 \quad \text{and } 42875 = 35^3$$

$$\text{Now, } \frac{-216}{42875} = \frac{(-6)^3}{35^3} = \left(\frac{(-6)}{35}\right)^3$$

Thus, $\frac{-216}{42875}$ is the cube of a rational number $\frac{-6}{35}$.

5. On resolving 26244 into prime factors, we get

$$26244 = 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3$$

Grouping the factors in triples of equal factors, we get

$$26244 = 2 \times 2 \times (3 \times 3 \times 3) \times (3 \times 3 \times 3) \times 3 \times 3$$

Therefore, the smallest number by which the given number must be divided in order that the quotient is a perfect cube is $2 \times 2 \times 3 \times 3 = 36$

Also, the quotient = $\frac{26244}{36} = 729$

$$\therefore 729 = (3 \times 3 \times 3) \times (3 \times 3 \times 3)$$

and $\sqrt[3]{729} = 3 \times 3 = 9$

6. Let the length of a side of the box be x metres. Then, its volume is x^3 cubic metres. But, the volume is given as 32.768 cubic metres.

$$\therefore x^3 = 32.768$$

$$\Rightarrow x = \sqrt[3]{32.768} = \sqrt[3]{\frac{32768}{1000}} = \frac{\sqrt[3]{32768}}{\sqrt[3]{1000}}$$

Now, resolving 32768 and 1000 in prime factors, we have

[illegible]

and, $1000 = 2 \times 2 \times 2 \times 5 \times 5 \times 5$

$$\therefore \sqrt[3]{32768} = \underbrace{2 \times 2 \times 2}_{\text{3 times}} \times \underbrace{2 \times 2 \times 2}_{\text{3 times}} = 32 \text{ and}$$

$$\sqrt[3]{1000} = 2 \times 5 = 10$$

$$\text{Hence, } x = \frac{\sqrt[3]{32768}}{\sqrt[3]{1000}} = \frac{32}{10} = 3.2 \text{ metres}$$

2 EXERCISE

Text-Book Exercise:

1. We have, $68600 = 2 \times 2 \times 2 \times 5 \times 5 \times 7 \times 7 \times 7$. In this factorisation, we find that there is no triplet of 5. So, 68600 is not a perfect cube. To make it a perfect cube we multiply it by 5.
- Thus, $68600 \times 5 = 2 \times 2 \times 2 \times 5 \times 5 \times 5 \times 7 \times 7 \times 7$
 $= 343000$, which is a perfect cube of 70.
2. (i) $256 = \underline{2 \times 2 \times 2} \times \underline{2 \times 2 \times 2} \times 2 \times 2$
 $\therefore$ The smallest number by which 256 must be multiplied to obtain a perfect cube = 2.
- (ii) $675 = 5 \times 5 \times \underline{3 \times 3 \times 3}$
 $\therefore$ The smallest number by which 675 must be multiplied to obtain a perfect cube = 5.
- (iii) $100 = 2 \times 2 \times 5 \times 5$
 $\therefore$ The smallest number by which 100 must be multiplied to obtain a perfect cube = $2 \times 5 = 10$
3. (i) $81 = \underline{3 \times 3 \times 3} \times 3$
 $\therefore$ The smallest number by which 81 must be divided to obtain a perfect cube = 3
- (ii) $135 = 5 \times \underline{3 \times 3 \times 3}$
 $\therefore$ The smallest number by which 135 must be divided to obtain a perfect cube = 5.
- (iii) $704 = \underline{2 \times 2 \times 2} \times \underline{2 \times 2 \times 2} \times 11$
 $\therefore$ The smallest number by which 704 must be divided to obtain a perfect cube = 11.
4. Volume of a cuboid = $5 \times 2 \times 5 \text{ cm}^3$.
- Since there is only one 2 and only two 5 in the prime factorisation, so, we need $2 \times 2 \times 5$, i.e., 20 to make a perfect cube. Therefore, we need 20 such cuboids to make a cube.

5. $13824 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$
 $= 2^3 \times 2^3 \times 2^3 \times 3^3$.

Therefore, $\sqrt[3]{13824} = 2 \times 2 \times 2 \times 3 = 24$

6. The given number is 17576.

Step 1 Form groups of three starting from the rightmost digit of 17576.

17 576. In this case one group i.e., 576 has three digits whereas 17 has only two digits.

Step 2 Take 576.

The digit 6 is at its one's place.

We take the one's place of the required cube root as 6.

Step 3 Take the other group, i.e., 17.

Cube of 2 is 8 and cube of 3 is 27. 17 lies between 8 and 27. The smaller number among 2 and 3 is 2.

The one's place of 2 is 2 itself. Take 2 as ten's place of the cube root of 17576.

Thus, $\sqrt[3]{17576} = 26$ (Check it!)

Exemplar Questions :

1. We have, $9720 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3 \times 3 \times 5$

9720 is not a perfect cube

Thus, the smallest number by which it should be divided to get a perfect cube $= 3 \times 3 \times 5 = 45$

2. Volume of cube $= 512 \text{ cm}^3$

or $(\text{side})^3 = 512$

$\Rightarrow \text{side} = \sqrt[3]{512}$

$\Rightarrow \text{side} = 8 \text{ cm}$

3. Let the numbers be $2x$, $3x$ and $4x$.

$\therefore (2x)^3 + (3x)^3 + (4x)^3 = 0.334125$

$\Rightarrow 8x^3 + 27x^3 + 64x^3 = 0.334125$

$\Rightarrow 99x^3 = 0.334125$

$\Rightarrow x^3 = \frac{0.334125}{99} = 0.003375$

$\Rightarrow x = 0.15$

$\therefore$ Required numbers are $2(0.15)$, $3(0.15)$ and $4(0.15)$.
 i.e. 0.3, 0.45 and 0.6.

4. $\sqrt[3]{27} + \sqrt[3]{0.008} + \sqrt[3]{0.064} = 3 + 0.2 + 0.4 = 3.6$

5. $\left\{ \left[6^2 + (8^2)^{\frac{1}{2}} \right]^3 \right\} = \{36 + 8\}^3 = (44)^3 = 85184$

HOTS Questions :

1. We have: $\frac{\sqrt[3]{15625} \times \sqrt[3]{216}}{\sqrt[3]{3375}}$
 $= \frac{\sqrt[3]{25 \times 25 \times 25} \times \sqrt[3]{6 \times 6 \times 6}}{\sqrt[3]{15 \times 15 \times 15}}$
 $= \frac{25 \times 6}{15} = 10$

2. We have: $\sqrt[3]{216 \times (-343)}$

$= \sqrt[3]{216} \times \sqrt[3]{-343} = \sqrt[3]{216} \times -\sqrt[3]{343}$

$= -\sqrt[3]{6 \times 6 \times 6} \times \sqrt[3]{(7) \times (7) \times (7)}$

$= -[6 \times (7)] = -42$.

3. Let the length of each edge of the given cube x metres. Then,

Surface area $= 150 \text{ m}^2$

$\Rightarrow 6x^2 = 150 \Rightarrow x^2 = \frac{150}{6} = 25 \text{ m}^2$

$\Rightarrow x = \sqrt{25} = \sqrt{5 \times 5} = 5 \text{ m}$

$\therefore$ Volume of the cube $= x^3 \text{ m}^3$
 $= 5^3 \text{ m}^3$
 $= (5 \times 5 \times 5) \text{ m}^3$
 $= 125 \text{ m}^3$

4. Volume of box $= 2744 \text{ cm}^3$

Let edge of box $= x \text{ cm}$

$\therefore x^3 = 2744$

$\Rightarrow x = \sqrt[3]{2744}$

$\Rightarrow x = \sqrt[3]{2 \times 2 \times 2 \times 7 \times 7 \times 7} = 2 \times 7 = 14$

$\therefore$ Edge of box is 14 cm.

5. $12^3 - 11^3 = 1 + 12 \times 11 \times 3 = 1 + 396 = 397$.

6. $\sqrt[3]{2\frac{1}{4} \times \left(1\frac{1}{3}\right)^2} + 1 \div \sqrt[3]{3\frac{3}{8}}$
 $= \sqrt[3]{\frac{9}{4} \times \left(\frac{4}{3}\right)^2} + 1 \div \sqrt[3]{\frac{27}{8}} = \frac{3}{2} \times \left(\frac{4}{3}\right)^2 + 1 \div \left(\frac{3}{2}\right)$
 $= \frac{3}{2} \times \frac{16}{9} + 1 \div \frac{3}{2} = \frac{8}{3} + 1 \times \frac{2}{3}$
 $= \frac{8}{3} + \frac{2}{3} = \frac{8+2}{3} = \frac{10}{3} = 3\frac{1}{3}$

7. Let the number be x

$(\sqrt{8} \times \sqrt{8})^{\frac{1}{2}} + (9)^{\frac{1}{2}} = (x)^3 + \sqrt{8} - 340$

$\Rightarrow (8)^{\frac{1}{2}} + (3^2)^{\frac{1}{2}} = (x)^3 + (8)^{\frac{1}{2}} - 340$

$\Rightarrow 3 + 340 = (x)^3$

$\Rightarrow x^3 = 343$

$\Rightarrow x = 7$

8. Given expression

$\frac{(2.3)^3 - (0.3)^3}{(2.3)^2 + (2.3)(0.3) + (0.3)^2}$ is in the form of $\frac{a^3 - b^3}{a^2 + ab + b^2}$

Now, $a^3 - b^3 = \frac{(a-b)(a^2 + ab + b^2)}{a^2 + ab + b^2} = a - b$

$\therefore a - b = 2.3 - 0.3 = 2$

9. $\sqrt[3]{\sqrt{0.000064}} = \sqrt[3]{\sqrt{\frac{64}{1000000}}}$

$= \sqrt[3]{\frac{8}{1000}} = \frac{2}{10} = 0.2$

2	2744
2	1372
2	686
7	343
7	49
7	7
	1

3 EXERCISE

Single Option Correct :

- (c) Expressing 7200 as its prime factors
 $7200 = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 5 \times 5$
 $7200 = (2 \times 2 \times 2) \times (2 \times 2) \times (3 \times 3) \times (5 \times 5)$
 We find that prime factors 2, 3 & 5 appear in groups of two, so to make the given no. perfect cube, we must multiply it with $2 \times 3 \times 5 = 30$
 $\therefore 7200 \times 30 =$
 $(2 \times 2 \times 2) \times (3 \times 3 \times 3) \times (2 \times 2 \times 2) \times (5 \times 5 \times 5)$
 is a perfect cube.
- (b) We have, $16384 = 2^{14}$
 $\therefore$ To make 16384 perfect cube, we must divide it with $2 \times 2 = 4$.
- (c) According to definition of Armstrong no.
 $1^3 + 5^3 + 3^3 = 1 + 125 + 27$
 $= 126 + 27 = 153$
 $\therefore$ 153 is Armstrong no.
- (c) 2744, which is a cube of 14.
- (d) $\sqrt[3]{392} \times \sqrt[3]{448} = \sqrt[3]{2 \times 2 \times 2 \times 7 \times 7}$
 $\times \sqrt[3]{2 \times 2 \times 2 \times 2 \times 2 \times 7}$
 $= \sqrt[3]{(2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (2 \times 2 \times 2) \times (7 \times 7 \times 7)}$
 $= 2 \times 2 \times 2 \times 7 \quad [\because \sqrt[3]{p} \times \sqrt[3]{q} = \sqrt[3]{pq}]$
 $= 56$
- (d) Volume cuboid $= 8 \times 6 \times 4 = 192 \text{ cm}^3$
 $\therefore$ Nearest perfect cube to 192 = 216
 Thus, 24 cm^3 should be added to 192 cm^3 to mould it into a cube whose edge is 6 cm where 6 is an integer.
- (b) $\frac{V_1}{V_2} = \frac{343}{1331} \Rightarrow \frac{(S_1)^3}{(S_2)^3} = \frac{343}{1331}$
 $\Rightarrow \frac{S_1}{S_2} = \frac{\sqrt[3]{343}}{\sqrt[3]{1331}} = \frac{7}{11}$
 $\Rightarrow$ Ratio of their edges = 7 : 11
- (a) Let the two numbers are x & y , $x > y$
 $\therefore x^3 = 8y^3 \quad \dots(1)$
 and $x^3 + y^3 = 243 \quad \dots(2)$
 $\Rightarrow 8y^3 + y^3 = 243 \quad \text{From (1)}$
 $\Rightarrow 9y^3 = 243$
 $\Rightarrow y^3 = 27 \Rightarrow y = 3$
 $x = 6 \quad \text{From (1)}$
 $\therefore$ Difference $(x - y) = 6 - 3 = 3$

- (c) Let the natural number be 'x'.

$$\therefore x^3 - x^2 = 48$$

$$\Rightarrow x^2(x - 1) = 48$$

$$\Rightarrow 4^2(4 - 1) = 48$$

$$\therefore x = 4$$

- (b) $\frac{16}{9} \times \left(\frac{-3}{2}\right)^3 = \frac{16}{9} \times \frac{-3}{2} \times \frac{-3}{2} \times \frac{-3}{2}$
 $= 2 \times (-3) = (-6)$

More Than One Option Correct :

- (a, b)
 $-64 = -4 \times -4 \times -4$
 $-2197 = -13 \times -13 \times -13$
- (a, b)
- (b, d)
- (a, b, c)
 $64 = 4 \times 4 \times 4$
 $216 = 6 \times 6 \times 6$
 $343 = 7 \times 7 \times 7$
- (a, c): $\frac{27}{64} = \left(\frac{3}{4}\right)^3$, $\frac{1331}{1000000} = \left(\frac{11}{10}\right)^3$
 $\frac{4}{100} = \left(\frac{2}{10}\right)^2$, $\frac{125}{128} = \frac{5^3}{2^7}$
 $\therefore$ Only $\frac{27}{64}$ and 0.001331 are cubes of rational numbers.
- (b, c): $(-216 \times 1728)^{1/3} = \sqrt[3]{-216} \times \sqrt[3]{1728}$
 $= -6 \times 12 = -72$
- (b, d): $\sqrt[3]{(-125) \times (-1000)} = \sqrt[3]{-125} \times \sqrt[3]{-1000}$
 $= (-5) \times (-10) = 50$
- (b, d): $\sqrt[3]{2 \frac{10}{27} \div (2^2 - 3^2)} = \sqrt[3]{\frac{64}{27} \div (2^2 - 3^2)}$
 $= \frac{4}{3} \div (4 - 9) = \frac{4}{3} \div (-5) = \frac{4}{3} \times \left(\frac{-1}{5}\right) = -\frac{4}{15}$

Passage Based Questions :

PASSAGE-I

- $\frac{-216}{42875} = \frac{(-6) \times (-6) \times (-6)}{35 \times 35 \times 35}$
 $= \left(\frac{-6}{35}\right)^3$
- $0.001728 = \frac{1728}{1000000} = \left(\frac{12}{10}\right)^3$

3. (a,c) $\frac{27}{64} = \left(\frac{3}{4}\right)^3$

and $0.001331 = \frac{1331}{1000000} = \left(\frac{11}{10}\right)^3$

PASSAGE-II

4. (b) $\sqrt[3]{-226981} = -61$ and $\sqrt[3]{-175616} = -56$

5. (b) $\sqrt[3]{-125 \times -3375}$
 $= \sqrt[3]{(-5)^3 \times (-15)^3}$
 $= (-5) \times (-15) = 75$

6. (c) $\sqrt[3]{\frac{-3375}{-2744}} = \frac{\sqrt[3]{-3375}}{\sqrt[3]{-2744}} = \frac{15}{14} = \frac{15}{14}$

Assertion & Reason :

1. (a) Assertion and Reason both are correct.
 Reason is the correct explanation of Assertion.

2. (d) $14^3 = 2744$, $24^3 = 13824$
 $\therefore$ Assertion is incorrect but reason is correct.

3. (d) $\sqrt[3]{(-4)^3} = \sqrt[3]{-4 \times -4 \times -4}$
 $= -4 \neq \sqrt[3]{\left(\frac{1}{4}\right)^3} = \frac{1}{4}$
 $\therefore$ Assertion is incorrect.
 $\sqrt[3]{-x^3} = -x$ is always true for all integers.
 $\therefore$ Reason is correct.

4. (a) a divides b
 $\Rightarrow b = ak$, for some k , $b^3 = a^3 k^3$
 $\therefore a^3$ divides b^3 .

Multiple Matching Questions :

- (A) $\rightarrow$ (p); (B) $\rightarrow$ (q); (C) $\rightarrow$ (r, t); (D) $\rightarrow$ (s); (E) $\rightarrow$ (r, t)
 A. Since 1728 is the perfect cube nearest to 1730.
 $\therefore$ Required no. = $1730 - 1728 = 2$
 B. Since x , y and z are positive integers.
 Hence, $x^{y^z} = 2^8$
 $\Rightarrow x^{y^z} = 2^{2^3} \Rightarrow x + y + z = 2 + 2 + 3 = 7$
 C. Since $a = 2b$, $b = 4c$
 $\Rightarrow a = 2(4c)$, $b = 4c \Rightarrow a = 8c$, $b = 4c$

So, $\sqrt[3]{\frac{a^4}{32b^2c^2}} = \sqrt[3]{\frac{(8c)^4}{32(4c)^2 c^2}} = \sqrt[3]{8} = 2$

D. $x = 8$, $y = 9$

$x^2 + y^2 = (8)^2 + (9)^2 = 64 + 81 = 145$

So, $x + y = 17$

E. Let $b = 8$

$\therefore 8^3 = 512$

So, $a = 3$, $b = 8$

$\sqrt[3]{b} = \sqrt[3]{8} = 2$

Integer Type Questions :

- (2) $864 \times n$ is a perfect cube.
 $864 = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3$
 $\Rightarrow n = 2$
- (3) $72 = 2 \times 2 \times 2 \times 3 \times 3$
 If $k = 3$ then
 $72k = 72 \times 3 = 216 = 6^3$ is a perfect cube.

3. (4) We have $\sqrt[3]{\frac{a^6 \times b^3 \times c^{21}}{c^9 \times a^{12}}}$
 $= \sqrt[3]{a^{6-12} \times b^3 \times c^{21-9}} = \sqrt[3]{\frac{b^3 \times c^{12}}{a^6}}$
 $= \frac{(b^3)^{1/3} (c^{12})^{1/3}}{(a^6)^{1/3}} = \frac{bc^4}{a^2} = \frac{bc^4}{a^{4/2}}$
 Hence $k = 4$.

4. (2) $\sqrt[3]{\frac{125}{1000}} + \sqrt[3]{\frac{729}{1000}} = \sqrt[3]{\frac{5 \times 5 \times 5}{10 \times 10 \times 10}} + \sqrt[3]{\frac{9 \times 9 \times 9}{10 \times 10 \times 10}}$
 $= \frac{5}{10} + \frac{9}{10} = \frac{14}{10} = 1.4 = 2 \times 0.7$
 Hence $n = 2$.

5. (7) If unit's place digit of number is 3, then unit's place digit of its cube is 7.

6. (2) On resolving 10648 into prime factors, we get
 $10648 = 2 \times 2 \times 2 \times 11 \times 11 \times 11$
 $\sqrt[3]{10648} = 2 \times 11 = 22 = 2 \times 11 = 2 \times \sqrt{121}$
 $\Rightarrow k = 2$

Chapter

8

Comparing Quantities

INTRODUCTION

A shopkeeper sells 100 copies of a book in a month at its maximum retail price (MRP). He offered a discount of 20% on the MRP of the book in the next month and was able to sell 150 copies of the same book. Thus by offering a discount he has been able to fetch more sales, which would definitely make a difference to his profit/loss. Most people take out a loan for a big purchase, such as a car, a major appliance, or a house. Loan payments must be accurately determined, and it may take some work to find the "best deal." It is important for both businesspersons and consumers to understand commercial mathematics in order to make sound financial decisions. In this chapter we would deal with ratios, percentages, interests, profits / losses, discounts, sales tax, VAT, etc.

RATIO

A ratio is a comparison of two numbers (quantities) by division. The ratio of a to b is written as $a : b$ or $\frac{a}{b}$.

A ratio is a number, so to find the ratio of two quantities they must be expressed in the same units.

ILLUSTRATION : 1

Find the

(i) ratio of ₹ 20 to ₹ 80

(ii) ratio of 3 km to 600 m

SOLUTION :

(i) 20 : 80 or 1 : 4

(ii) 3000m : 600m or 5 : 1

Simplest Form of a Ratio

The ratio of two or more quantities is said to be in simplest form, if the HCF of the quantities is 1.

If the HCF of the quantities is not 1, then each quantity of the ratio is divided by the HCF to get the ratio in simplest form.

ILLUSTRATION : 2

Express 81 : 36 in simplest form.

SOLUTION :

HCF of 81 & 36 is 9. So we divide 81 & 36 by 9

$$\therefore 81 : 36 = \frac{81}{9} : \frac{36}{9} = 9 : 4$$

$\therefore$ Simplest form of 81 : 36 is 9 : 4

NOTE: We should always express the ratio in the lowest or standard form.

PERCENTAGE

The word percent means per hundred or for every hundred. The symbol % is used for the term percent. Thus 20 percent written as 20% and it means 20 out of 100. This can also be written as $\frac{20}{100}$.

Thus percent is a fraction whose denominator is 100 and numerator of this fraction is called rate percent.

$$\therefore \frac{20}{100} = 20\% ; \frac{a}{100} = a\%$$

Conversion of Percent into Fraction

To convert a percent into fraction, divide it by 100 and remove the % sign.

ILLUSTRATION : 3

Convert 15% into fraction.

SOLUTION :

$$15\% = \frac{15}{100} = \frac{3}{20}$$

Conversion of Fraction into Percent

To convert any fraction to percent, multiply it by 100 and put the % sign.

ILLUSTRATION : 4

Convert $\frac{2}{5}$ into percent.

SOLUTION :

$$\left(\frac{2}{5} \times 100\right)\% = 40\%$$

Conversion of Percent into Ratio

First change the percent into fraction by dividing it by 100 and remove the % sign, finally reduce the obtained fraction into its simplest form.

ILLUSTRATION : 5

Convert 35%, 0.65% and 4.5% into ratio.

SOLUTION :

$$35\% = \frac{35}{100} = \frac{7}{20} = 7 : 20$$

$$0.65\% = \frac{0.65}{100} = \frac{65}{10000} = \frac{13}{2000} = 13 : 2000$$

$$4.5\% = \frac{4.5}{100} = \frac{45}{1000} = \frac{9}{200} = 9 : 200$$

Conversion of Ratio into Percent

$$x : y = \frac{x}{y} = \left(\frac{x}{y} \times 100 \right) \%$$

ILLUSTRATION : 6

Convert 3 : 4 to percent.

SOLUTION :

$$3 : 4 = \frac{3}{4} = \left(\frac{3}{4} \times 100 \right) \% = 75\%$$

Conversion of Percent into Decimal

To convert the percent to decimal, first change it to fraction by dividing it by 100 and remove the % sign, finally put the decimal point accordingly.

ILLUSTRATION : 7

Convert 18% to decimal.

SOLUTION :

$$18\% = \frac{18}{100} = 0.18$$

Conversion of Decimal to Percent

First change the decimal into fraction and then multiply by 100 and put the % sign

ILLUSTRATION : 8

Convert 1.5 and 0.25 to percent.

SOLUTION :

$$1.5 = \frac{15}{10} = \left(\frac{15}{10} \times 100 \right) \% = 150\%$$

$$0.25 = \frac{25}{100} = \left(\frac{25}{100} \times 100 \right) \% = 25\%$$

Percentage of a Number

To find percentage of a given number, multiply the given number by required percentage.

ILLUSTRATION : 9

Find 20% of 400.

SOLUTION :Let the required value is x

$$\therefore x = 20\% \text{ of } 400$$

$$x = \frac{20}{100} \times 400 = 80$$

$$\therefore 20\% \text{ of } 400 \text{ is } 80.$$

Original Number From its Percent**ILLUSTRATION : 10**

Find the number whose 20% is 60.

SOLUTION :Let the required number be ' x '.Now, 20% of x is 60

$$\Rightarrow 20\% \text{ of } x = 60$$

$$\Rightarrow \frac{20}{100} \times x = 60$$

$$\Rightarrow x = \frac{60 \times 100}{20} = 300$$

$$\therefore \text{The required number is } 300.$$

Percentage of One Quantity of AnotherIf $x\%$ of a is b , then $\frac{x}{100} \times a = b$

$$\Rightarrow x = \frac{100b}{a}$$

ILLUSTRATION : 11

What percent of 120 is 30?

SOLUTION :Let $x\%$ of 120 = 30

$$\therefore \frac{x}{100} \times 120 = 30$$

$$\Rightarrow x = \frac{30 \times 100}{120} = 25$$

$$\therefore 25\% \text{ of } 120 \text{ is } 30.$$

Finding the Increase or Decrease Percent

- (i) If there is an increase in any number or quantity from A to B , then percentage increase

$$= \left(\frac{\text{increase in number (or quantity)}}{\text{original number (or quantity)}} \times 100 \right) \%$$

- (ii) If there is a decrease in any number or quantity from A to B , then percentage decrease

$$= \left(\frac{\text{decrease in number (or quantity)}}{\text{original number (or quantity)}} \times 100 \right) \%$$

ILLUSTRATION : 12

The price of a scooter which was Rs. 34000 last year increased by 20% this year. What is the price now?

SOLUTION :

Let original price be ₹ 100

Increase in price = ₹ 20.

∴ Increased price = ₹ 100 + ₹ 20 = ₹ 120

If original price is ₹ 100, then increased price = ₹ 120

If original price is ₹ 34000, increased price

$$= ₹ \frac{120}{100} \times 34000 = ₹ 40800.$$

PROFIT AND LOSS

Business transactions have become common feature of daily life. When we purchase and sale some items, we either make profit or loss. Profit and loss are part of every business transactions. Let us first investigate them.

Cost Price : The Price at which a person buys an item is the Cost Price (C.P.) of that item. In other words, amount paid or expenses in purchasing or producing an item is called its cost price (C.P.).

Selling Price : The Price at which a person sells an item is the selling price (S.P.) of item. In other words, amount received when an item sold is called selling price (S.P.) of that item to the seller.

Overheads : All the expenses that a buyer paid on transportation or repair of purchased item are called overheads. All over heads are added to basic Cost Price (C.P.) of that item.

Profit : When a person sells an item at a price which is more than its cost price we say that person has made a profit or gain. The difference (S.P – C.P) is called profit or gain.

Thus Profit or Gain = S.P – C.P, if S.P > C.P.

Loss : When a person sells an item at a price which is less than its cost price, we say that the person has suffered loss. The difference (C.P – S.P) is called loss.

Thus Loss = C.P – S.P, if S.P < C.P.

Basic Formulae :

Case-I : If S.P > C.P, then there is a profit.

$$\therefore \text{Profit} = \text{S.P} - \text{C.P}$$

$$\text{S.P} = \text{C.P} + \text{Profit}$$

$$\text{Profit \%} = \frac{\text{S.P} - \text{C.P}}{\text{C.P}} \times 100 = \frac{\text{Profit} \times 100}{\text{C.P}}$$

Case-II : If SP < CP, then there is a loss.

$$\therefore \text{Loss} = \text{CP} - \text{SP}$$

$$\text{SP} = \text{CP} - \text{Loss}$$

$$\text{Loss \%} = \frac{\text{CP} - \text{SP}}{\text{CP}} \times 100 = \frac{\text{Loss} \times 100}{\text{C.P.}}$$

ILLUSTRATION : 13

Rishi bought a wrist watch for ₹ 2200 and sold it for ₹ 1980. Find his loss and loss percent.

SOLUTION :

We have,

C.P. of watch = ₹ 2200

S.P. of watch = ₹ 1980

Since S.P. < C.P. So, there is loss given by

$$\text{Loss} = \text{C.P.} - \text{S.P.}$$

$$= ₹ 2200 - ₹ 1980 = ₹ 220$$

$$\text{Now, Loss \%} = \left(\frac{\text{Loss}}{\text{C.P.}} \times 100 \right) \% = \left(\frac{220}{2200} \times 100 \right) \% = 10\%$$

Hence, Loss = ₹ 220 and Loss % = 10%

Discount

Sometimes to increase the sales or dispose off the old stock, dealer or seller offers his goods at reduced prices. This reduction in price offered by dealer/seller is called discount.

Marked Price (M.P.) : The printed price or the tagged price of an article is called its marked price (M.P.). It is also called list price. Discount is always calculated on M.P. of the article.

Now,

$$\text{Selling Price (S.P.)} = \text{M.P.} - \text{Discount}$$

$$\therefore \text{Discount} = \text{M.P.} - \text{S.P.}$$

$$\% \text{ Discount} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\% \text{ Discount} = \frac{\text{MP} - \text{SP}}{\text{M.P.}} \times 100$$

$$\text{Or } \text{S.P.} = \text{M.P.} \left(\frac{100 - \text{Discount \%}}{100} \right)$$

ILLUSTRATION : 14

At a clearance sale, all goods are on sale at 45% discount. If I buy a skirt marked Rs 600, how much would I need to pay?

SOLUTION :

We have,

$$\text{M.P.} = ₹ 600, \text{Discount} = 45\%$$

$$\therefore \text{Discount} = 45\% \text{ of } ₹ 600 = \left(₹ \frac{45}{100} \times 600 \right) = ₹ 270$$

$$\therefore \text{S.P.} = \text{M.P.} - \text{Discount}$$

$$\Rightarrow \text{S.P.} = ₹ 600 - ₹ 270 = ₹ 330$$

Thus, the amount I need to pay is ₹ 330.

Sales Tax

Government lays some charges upon persons or organizations to provide various facilities (such as roads, transport etc. to general public). These charges are called taxes. Some taxes are excise duty, luxury tax, sales tax etc.

Sales tax is charged from persons or organisations involved in the business of selling, purchasing of various items. It is charged as certain percentage of selling price.

$$\text{Sales Tax} = \frac{\text{Rate of sales tax}}{100} \times \text{S.P.}$$

$$\text{Total billing amount} = \text{selling price (SP)} + \text{Sales Tax.}$$

ILLUSTRATION : 15

The cost of a table is ₹ 800. The Sales tax charged is 4%. Find the bill amount.

SOLUTION :

$$\text{Selling Price (SP) of table} = ₹ 800.$$

$$\text{Sales Tax} = 4\% \text{ of SP}$$

$$= \frac{4}{100} \times 800 = ₹ 32$$

$$\therefore \text{Billing amount} = ₹ (800 + 32) = ₹ 832$$

Value Added Tax (VAT)

The government spend a lot of money to provide various types of facilities for the public. To meet these expenditures, the government imposes different types of taxes. Value added tax (VAT) is one of these. VAT is calculated on selling price. Thus if discount is given, VAT is calculated on discounted S.P.

ILLUSTRATION : 16

Suresh purchased a T.V. set for ₹ 15000 including VAT of 10%. Find the cost of TV set before VAT was added.

SOLUTION :

$$\text{Let the cost of TV set without VAT} = ₹ x.$$

$$\text{Now, } x + \text{VAT} = 15000$$

$$\Rightarrow x + x \times \frac{10}{100} = 15000$$

$$\Rightarrow \frac{11x}{10} = 15000$$

$$\Rightarrow x = \frac{15000 \times 10}{11} = 13636.3636$$

$\therefore$ Cost of TV set before VAT is ₹ 13636.36

SIMPLE INTEREST

If any agency (an individual, a firm or a bank etc.) borrow some money from any other agency, then the first agency is called the *borrower* and the second agency is called the *lender*. The borrowed money is called the *principal*.

If the borrower has to pay some additional money together with the borrowed money for the benefit of using borrowed money for a certain time period called *loan period*, then this additional money is called the *interest* and the principal together with the interest is called the *amount* (i.e. Amount = Principal + Interest)

When we deposit money in a bank, we earn interest, Interest is calculated according to an agreement which specifies the rate of interest, Generally the rate of interest is taken as “per cent per annum” which means “per ₹ 100 per year”, for example, a rate of 10% per annum means ₹ 10 on ₹ 100 for 1 year.

Simple Interest (S.I.) : If the principal remains the same for whole loan period, then the interest is called the simple interest.

$$\text{S.I.} = \frac{PRT}{100}$$

Where P = Principal, R = Rate of interest in percent per annum, T = Loan period (or whole time period) (in years)

ILLUSTRATION : 17

At what rate percent by simple interest, will a sum of money double itself in 5 years 4 months ?

SOLUTION :

Let $P = ₹ x$

Amount $A = ₹ 2x$

$$\therefore \text{S.I.} = A - P = ₹ 2x - ₹ x = ₹ x$$

$$T = 5 \text{ years } 4 \text{ months} = 5 \frac{4}{12} \text{ years} = 5 \frac{1}{3} \text{ years} = \frac{16}{3} \text{ years}$$

Let R be the rate percent per annum.

$$\text{Using } R = \frac{\text{S.I.} \times 100}{P \times T}, \text{ We get } R = \frac{x \times 100}{x \times \frac{16}{3}} = \frac{300}{16} = 18.75.$$

Hence required rate = 18.75% p.a.

COMPOUND INTEREST (C.I.)

If the borrower and the lender agree to fix up a certain interval of time (a year, a half year or a quarter of a year etc.) called conversion period, so that the amount (= principal + interest) at the end of an conversion period becomes the principal for the next conversion period, then the total interest over all the loan period calculated in this way is called the compound interest.

NOTE: The main difference between the simple interest and the compound interest is that the principal in the case of simple interest remains constant throughout the loan period where as in the case of compound interest, the principal changes periodically (i.e. after each conversion period) throughout the loan period.

(i) **Computation of Compound Interest when interest is Compounded annually :**

$$A = P \left(1 + \frac{r}{100} \right)^n$$

$\therefore \text{C.I.} = A - P = P \left[\left(1 + \frac{r}{100} \right)^n - 1 \right]$, where A is the amount, P is the principal, r is the rate of interest in percent per conversion period and n is the number of conversion periods in the whole loan period.

ILLUSTRATION : 18

Roohi deposited ₹ 7,000 with a finance company for 3 years at an interest of 15% per annum compounded annually. What is the compound interest and the amount that Roohi gets after 3 years?

SOLUTION :

Principal, $P = ₹ 7,000$, $n = 3$ years, $r = 15\%$ per annum

$$A = P \left(1 + \frac{r}{100} \right)^n = 7,000 \left(1 + \frac{15}{100} \right)^3 = 7,000 \left(\frac{115}{100} \right)^3$$

$$= 7,000 \times \frac{115}{100} \times \frac{115}{100} \times \frac{115}{100} = 10,646.125 = ₹ 10,646.13 = ₹ 10,646 \text{ (approx.)}$$

$$\text{Compound interest} = A - P = 10,646 - 7,000 = ₹ 3,646$$

(ii) Computation of compound interest when interest is compounded k times every year :

If r be the rate of interest in percent per year, then the rate of interest in percent per conversion period is $\frac{r}{k}$.

If n be the number of years in the whole loan period (or whole time period), then the number of conversion period is nk .

$$\therefore A = P \left(1 + \frac{r}{100k} \right)^{nk}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{100k} \right)^{nk} - 1 \right]$$

(a) In case of interest compounded half-yearly, $k = 2$

$$\therefore A = P \left(1 + \frac{r}{200} \right)^{2n}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{200} \right)^{2n} - 1 \right]$$

(b) In case of interest compounded quarterly, $k = 4$

$$\therefore A = P \left(1 + \frac{r}{400} \right)^{4n}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{400} \right)^{4n} - 1 \right]$$

(c) In case of interest compounded monthly, $k = 12$

$$\therefore A = P \left(1 + \frac{r}{1200} \right)^{12n}$$

$$\text{and C.I.} = P \left[\left(1 + \frac{r}{1200} \right)^{12n} - 1 \right]$$

ILLUSTRATION : 19

Find the compound interest on ₹ 1000 at the rate of 10% per annum for 18 months when interest is compounded half-yearly.

SOLUTION :

Here, $P = ₹ 1000$, $R = 10\%$ per annum and, $n = \frac{18}{12}$ years $= \frac{3}{2}$ years

$$\begin{aligned}\therefore \text{Amount after 18 months} &= P \left(1 + \frac{R}{200} \right)^{2n} \\ &= ₹ 1000 \times \left(1 + \frac{10}{200} \right)^{2 \times \frac{3}{2}} = ₹ 1000 \times \left(1 + \frac{1}{20} \right)^3 \\ &= ₹ 1000 \times \left(\frac{21}{20} \right)^3 = ₹ 1000 \times \frac{21}{20} \times \frac{21}{20} \times \frac{21}{20} \\ &= ₹ 1157.63\end{aligned}$$

Hence, Compound interest = Amount – Principal

$$= ₹ 1157.63 - ₹ 1000 = ₹ 157.63$$

(iii) **Computation of Compound Interest when Interest is Compounded annually but rate of interest in percent being different for different years :**

$$A = P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \dots \left(1 + \frac{R_n}{100} \right),$$

where $R_1, R_2, \dots, R_n$ are rate of interest in percent per year for different years.

ILLUSTRATION : 20

Ram Singh bought a refrigerator for ₹ 4000 on credit. The rate of interest for the first year is 5% and of the second year is 15%. How much will it cost him if he pays the amount after two years?

SOLUTION :

Here, $P = ₹ 4000$, $R_1 = 5\%$ per annum and $R_2 = 15\%$ per annum.

$$\begin{aligned}\therefore \text{Amount after 2 years} &= P \left(1 + \frac{R_1}{100} \right) \left(1 + \frac{R_2}{100} \right) \\ &= ₹ 4000 \times \left(1 + \frac{5}{100} \right) \left(1 + \frac{15}{100} \right) \\ &= ₹ 4000 \times \left(1 + \frac{1}{20} \right) \left(1 + \frac{3}{20} \right) \\ &= ₹ 4000 \times \frac{21}{20} \times \frac{23}{20} = ₹ 4830\end{aligned}$$

Thus, the refrigerator will cost ₹ 4830 to Ram Singh.

(iv) If P be the value of an article (or population of a town or country etc.) at a certain time and $R\%$ per annum is the rate of depreciation, then the value A at the end of n years is given by,

$$A = P \left(1 - \frac{R}{100} \right)^n$$

ILLUSTRATION : 21

The population of a town 2 years ago was 62500. Due to migration to cities, it decreases every year at the rate of 4% per annum. Find its present population.

SOLUTION :

We have,

Population two years ago = 62500

Rate of decrease of population = 4% per annum.

$$\begin{aligned}\therefore \text{Present population} &= 62500 \times \left(1 - \frac{4}{100}\right)^2 \\ &= 62500 \times \left(1 - \frac{1}{25}\right)^2 = 62500 \times \left(\frac{24}{25}\right)^2 \\ &= 62500 \times \frac{24}{25} \times \frac{24}{25} = 57600.\end{aligned}$$

Hence, present population = 57600.

- (v) If P be the population of a country (or value of an article etc.) at a certain time, which increases at the rate $R_1\%$ per year for first n_1 years and decreases at the rate of $R_2\%$ per year for next n_2 years, then the population at the end of $(n_1 + n_2)$ years is given by

$$A = P \left(1 + \frac{R_1}{100}\right)^{n_1} \cdot \left(1 - \frac{R_2}{100}\right)^{n_2}$$

This formula can be extended for more than 2 different periods and rates.

ILLUSTRATION : 22

10000 workers were employed to construct a river bridge in four years. At the end of first year, 10% workers were retrenched. At the end of the second year, 5% of the workers at that time were retrenched. However to complete the project in time, the number of workers was increased by 10% at the end of the third year. How many workers were working during the fourth year?

SOLUTION :

We have,

Initial number of workers = 10000

Reduction of workers at the end of first year = 10%

Reduction of workers at the end of second year = 5%

Increase of workers at the end of third year = 10%

$\therefore$ Number of workers working during the fourth year

$$\begin{aligned}&= 10000 \left(1 - \frac{10}{100}\right) \left(1 - \frac{5}{100}\right) \left(1 + \frac{10}{100}\right) \\ &= 10000 \times \frac{9}{10} \times \frac{19}{20} \times \frac{11}{10} = 9405.\end{aligned}$$

Hence, the number of workers working during the fourth year was 9405.

MISCELLANEOUS

SOLVED EXAMPLES

1. Divide ₹ 3000 among A, B, C in the ratio 2 : 3 : 5.

Sol: A's share = $\left(\frac{2}{2+3+5}\right) \times 3000 = \frac{2}{10} \times 3000 = ₹ 600$

B's share = $\left(\frac{3}{2+3+5}\right) \times 3000 = \frac{3}{10} \times 3000 = ₹ 900$

C's share = $\left(\frac{5}{2+3+5}\right) \times 3000 = \frac{5}{10} \times 3000 = ₹ 1500$

2. If $a : b = 2 : 3$, find $(3a + 4b) : (4a - b)$.

Sol: Since $\frac{a}{b} = \frac{2}{3}$

$$\therefore \frac{3a+4b}{4a-b} = \frac{3\left(\frac{a}{b}\right)+4}{4\left(\frac{a}{b}\right)-1} \quad [\text{dividing by 'b'}]$$

$$= \frac{3 \times \frac{2}{3} + 4}{4 \times \frac{2}{3} - 1} = \frac{2+4}{\frac{5}{3}} = \frac{18}{5}$$

3. A bag contains coins of ₹ 5, ₹ 2 and ₹ 1 in the ratio 3 : 7 : 4, amounting to ₹ 1980. Find the number of coins of each type.

Sol: Ratio of coins = 3 : 7 : 4

$\therefore$ number of coins of ₹ 5, ₹ 2 and ₹ 1 are $3x$, $7x$ and $4x$.

Now, $5 \times (3x) + 2 \times (7x) + 1 \times (4x) = 1980$

$\Rightarrow 33x = 1980$

$\therefore x = \frac{1980}{33} = 60 \Rightarrow x = 60$

$\therefore$ No of coins of ₹ 5 = $60 \times 3 = 180$

No of coins of ₹ 2 = $60 \times 7 = 420$

No of coins of ₹ 1 = $60 \times 4 = 240$

4. A mixture contains alcohol and water in the ratio 3 : 2. If 4 litres of water is added to the mixture, the ratio becomes 3 : 4. Find the quantity of water in the given mixture.

Sol: Since ratio of alcohol and water be 3 : 2.

$\therefore$ Quantity of alcohol and water are $3x$ and $2x$ respectively

Now $\frac{3x}{2x+4} = \frac{3}{4}$

$\Rightarrow 12x = 6x + 12 \Rightarrow 6x = 12 \Rightarrow x = 2$

$\therefore$ Quantity of water in given mixture = $2 \times 2 = 4$ liters.

5. What must be subtracted from each term of the ratio 3 : 7, so that the ratio becomes 2 : 5 ?

Sol: Let the required number to be subtracted is 'x'.

$$\therefore \frac{3-x}{7-x} = \frac{2}{5}$$

$$\Rightarrow 15 - 5x = 14 - 2x \Rightarrow 3x = 1 \Rightarrow x = \frac{1}{3}$$

$\therefore$ The required number to be subtracted is $\frac{1}{3}$.

6. If $\frac{a}{x-y} = \frac{b}{y-z} = \frac{c}{z-x}$, then prove that $a + b + c = 0$

Sol: Let $\frac{a}{x-y} = \frac{b}{y-z} = \frac{c}{z-x} = K$

$$\therefore a = K(x-y), b = K(y-z), c = K(z-x)$$

$$\text{Now } a + b + c = K(x-y) + K(y-z) + K(z-x)$$

$$= K(x - y + y - z + z - x)$$

$$a + b + c = K \times 0 = 0$$

$$\Rightarrow a + b + c = 0$$

7. The sum of the present ages of A, B and C is 90 years. Six years ago, their ages were in the ratio 1 : 2 : 3. What is the present age of B.

Sol: 6 years ago the sum of ages of A, B, C = $90 - 6 \times 3 = 72$ years.

$$\therefore x + 2x + 3x = 72$$

$$\Rightarrow 6x = 72$$

$$\Rightarrow x = 12$$

$$\text{6 years ago age of 'B' } = 2 \times 12 = 24 \text{ years}$$

$$\text{B's present age} = 24 + 6 = 30 \text{ years.}$$

8. If $\frac{x}{a} = \frac{y}{b} = \frac{z}{c}$, then prove that each ratio is equal to $\left(\frac{2ax + 5by + 7cz}{2a^2 + 5b^2 + 7c^2} \right)$

Sol: Let $\frac{x}{a} = \frac{y}{b} = \frac{z}{c} = K$

$$\Rightarrow x = aK, y = bK, z = cK$$

$$\text{Now } \left(\frac{2ax + 5by + 7cz}{2a^2 + 5b^2 + 7c^2} \right) = \frac{2a \times aK + 5b \times bK + 7c \times cK}{2a^2 + 5b^2 + 7c^2} = K \left(\frac{2a^2 + 5b^2 + 7c^2}{2a^2 + 5b^2 + 7c^2} \right) = K$$

9. Two numbers are in the ratio 9 : 14. If the larger number is 55 more than the smaller number, find the numbers.

Sol: Since the ratio of given numbers = 9 : 14.

$\therefore$ We can suppose the number be 9x and 14x

$$\text{According to question } 14x = 9x + 55$$

$$\Rightarrow 5x = 55$$

$$\Rightarrow x = 11$$

$$\therefore \text{Numbers are } 9 \times 11, 14 \times 11 = 99, 154$$

10. 25% of a number is 20. What is 40% of that number ?

Sol: Let 20 is 25% of x

$$\therefore \frac{25}{100} \times x = 20$$

$$\Rightarrow x = 80$$

$$\text{Now } 40\% \text{ of } 80 = \frac{40}{100} \times 80$$

$$40\% \text{ of } 80 = 32$$

11. $P_1\%$ of number N_1 is equal to $P_2\%$ of number N_2 . Find what percent of N_1 is N_2 ?

Sol: We are required to find $\frac{N_2}{N_1} \times 100\%$

$$\text{Given } \frac{P_1}{100} \times N_1 = \frac{P_2}{100} \times N_2$$

$$\Rightarrow \frac{N_2}{N_1} = \frac{P_1}{P_2}$$

$$\therefore \frac{N_2}{N_1} \times 100\% = \frac{P_1}{P_2} \times 100\%$$

12. The side of a square increases by $A\%$, then find by what percent does its area increases ?

Sol: Let the side of square be ' x ' units

$$\therefore \text{Area of square} = x \cdot x = x^2 \text{ units.}$$

$$\text{After increase, side of square is } \left(x + \frac{Ax}{100}\right) \text{ units} = x \left(1 + \frac{A}{100}\right) \text{ units}$$

$$\text{New area of square} = x^2 \left(1 + \frac{A}{100}\right)^2 \text{ units}$$

$$\text{Now, increase in area} = x^2 \left(1 + \frac{A}{100}\right)^2 - x^2$$

$$= x^2 \left[\left(1 + \frac{A}{100}\right)^2 - 1 \right]$$

$$= x^2 \left[\frac{A}{100} \left(2 + \frac{A}{100}\right) \right] \quad [\text{using } p^2 - q^2 = (p + q)(p - q)]$$

$$\% \text{ increase in area} = \frac{x^2 \left[\frac{A}{100} \left(2 + \frac{A}{100}\right) \right]}{x^2} \times 100\%$$

$$\% \text{ increase in area} = \left(2A + \frac{A^2}{100}\right) \%$$

13. When N is reduced by 4, it becomes 80% of itself. What is the value of N ?

Sol: $N - 4 = 80\% \text{ of } N$

$$\Rightarrow N - 4 = \frac{80}{100} N$$

$$\Rightarrow N - 4 = \frac{4}{5} N \Rightarrow N - \frac{4}{5} N = 4 \Rightarrow \frac{N}{5} = 4 \Rightarrow N = 20$$

14. If 10% of an electricity bill is deducted, ₹ 45 is still to be paid. How much was the bill?

Sol: Let the bill amount was ₹ ' x '

$$\therefore x - 10\% \text{ of } x = 45$$

$$\Rightarrow x - 10 \times \frac{x}{100} = 45$$

$$\Rightarrow \frac{9x}{10} = 45$$

$$\Rightarrow x = ₹ 50$$

15. If 120 is 20% of a number, then what will be 120% of that number?

Sol: Let the number be ' x '

$$\therefore 20\% \text{ of } x = 120$$

$$\Rightarrow 20 \times \frac{x}{100} = 120$$

$$\Rightarrow x = 600$$

$$\text{Now } 120\% \text{ of } 600 = 120 \times \frac{600}{100} = 720.$$

16. When 75% of a number is added to 75 the result is the same number. Find the number.

Sol: Let the required number be x

$$\text{Now } 75\% \text{ of } x + 75 = x$$

$$\Rightarrow \frac{75x}{100} + 75 = x \Rightarrow x - \frac{3}{4}x = 75$$

$$\Rightarrow \frac{x}{4} = 75 \Rightarrow x = 300.$$

17. If two numbers are respectively 30% and 40% more than a third number, what percent is the first to the second?

Sol: Let third number be 100.

$$\therefore \text{First and second numbers are 130 and 140}$$

$$\text{Now \% of first to second} = \frac{130}{140} \times 100\% = \frac{650}{7} = 92\frac{6}{7}\%$$

18. Naresh secured 50% marks in Hindi, 60% in English and 70% in Maths as well as Science. What were his total marks, if the maximum marks obtainable in each of these 4 subjects was 50?

Sol: Total marks = 50% of 50 + 60% of 50 + 70% of 50 + 70% of 50

$$= 50 \times \frac{50}{100} + \frac{60}{100} \times 50 + \frac{70}{100} \times 50 + \frac{70}{100} \times 50$$

$$= 25 + 30 + 35 + 35$$

$$\text{Total marks} = 125$$

19. The salary of a person was reduced by 20% . By what percent should his reduced salary be raised so as to bring it at par with his original salary ?

Sol: Let the original salary = ₹ 'x'

$$\therefore \text{New salary (20\% reduced)} = x - \frac{20}{100}x = \frac{4}{5}x$$

Let, y % increase in salary bring the reduced salary at par with original salary

$$\frac{4}{5}x + \frac{4}{5}x \times \frac{y}{100} = x$$

$$\Rightarrow \frac{4}{5}x \left(1 + \frac{y}{100}\right) = x$$

$$\Rightarrow 1 + \frac{y}{100} = \frac{5}{4}$$

$$\Rightarrow \frac{y}{100} = \frac{5}{4} - 1 = \frac{1}{4}$$

$$\Rightarrow y = 25$$

$\therefore$ 25% increase in reduced salary bring the salary at par with original salary.

20. 300 gm of sugar solution has 40% sugar in it. How much sugar should be added to make it 50% in the solution ?

Sol: Quantity of sugar = $300 \times \frac{40}{100} = 120$ gm

Let 'x' gm sugar is added to make it 50% in the solution

$$\text{Now } \frac{120 + x}{300 + x} = \frac{50}{100}$$

$$\Rightarrow 240 + 2x = 300 + x$$

$$\Rightarrow 2x - x = 60$$

$$\Rightarrow x = 60 \text{ gm.}$$

21. If 60% of people of a certain village are illiterate, 28% of literate population are women. If the number of literate men be 4320, then find the total population of the village.

Sol: Let the total population of the village be 'x'.

Population of illiterate persons = 60% of x

$$\text{Population of literate persons} = 40\% \text{ of } x = \frac{40}{100}x = \frac{2}{5}x$$

$$\text{Now, population of literate women} = 28\% \text{ of } \frac{2}{5}x$$

$$\text{and population of literate men} = 72\% \text{ of } \frac{2}{5}x$$

$$4320 = \frac{72}{100} \times \frac{2}{5}x$$

$$15000 = x$$

22. Imran's salary was decreased by 20% and subsequently increased by 20% . How much percent does he lose?

Sol: Let Imran's original salary = ₹100

$$\text{After 20\% decrease, his salary} = 100 - \frac{20}{100} \times 100 = ₹ 80$$

$$\text{After 20\% increase salary} = 80 + 80 \times \frac{20}{100} = 96$$

$\therefore$ Total decrease = 4%.

23. Mr. Sahu buys a TV Set for ₹ 4500 . For how much should he sell so that there is a gain of 8%.

Sol: Given CP = ₹ 4500, gain % = 8%

Let Selling Price (S.P.) be ₹ 'x'

As gain % is given, So SP > CP.

Now SP = CP + Profit (Gain)

$$\therefore \text{SP} = 4500 + 8\% \text{ of CP}$$

$$x = 4500 + 8\% \text{ of } 4500$$

$$\therefore x = 4500 + \frac{8}{100} \times 4500$$

$$\Rightarrow x = 4500 + 360 = ₹ 4860$$

$$\therefore \text{S.P of TV Set} = ₹ 4860$$

24. By selling a washing machine for ₹ 7200, Rajesh loses 10%. Find the Cost Price of the washing machine.

Sol: Let cost price of washing machine be ₹ 'x'

Given,

$$\text{SP} = ₹ 7200, \text{ Loss \%} = 10\%$$

As % loss given, so CP > SP

$$\Rightarrow \text{SP} = \text{CP} - \text{Loss}$$

$$\Rightarrow 7200 = x - 10\% \text{ of } x$$

$$\Rightarrow x - \frac{10}{100}x = 7200$$

$$\Rightarrow \frac{9x}{10} = 7200$$

$$\Rightarrow x = \frac{7200 \times 10}{9} = 8000$$

$$\therefore \text{Cost price of washing machine} = ₹ 8000$$

25. By selling a radio for ₹ 1536, Ajay lost 20% . What percent shall he gain or loss by selling it for ₹ 2000 ?

Sol: Let cost price (CP) of radio = ₹ x .

Given that if selling price (SP) is ₹ 1536, loss % = 20%

$$\text{SP} = \text{CP} - \text{loss}$$

$$1536 = x - \frac{20}{100} \times x = \frac{5x - x}{5} = \frac{4x}{5}$$

$$x = \frac{1536 \times 5}{4} = 384 \times 5 = 1920$$

$$\therefore \text{Cost Price} = ₹ 1920.$$

$$\text{By Selling radio for ₹ 2000, Profit made by Ajay} = ₹ 2000 - 1920 = ₹ 80$$

$$\therefore \text{Profit \%} = \frac{\text{Profit}}{\text{CP}} \times 100 = \frac{80}{1920} \times 100\%$$

$$\text{Profit \%} = \frac{25}{6} = 4\frac{1}{6}\%.$$

26. Two books were sold at ₹ 850 each. On one a gain of 15% is made and on the other, a loss of 15%. How much loss % or gain % is made in the whole transaction?

Sol: Total cost of selling two books = ₹ 850 + ₹ 850 = ₹ 1700

Let cost price of book with 15% gain = ₹ x_1

Let cost price of book with 15% loss = ₹ x_2

∴ $SP = CP + \text{Profit}$ [for book with 15% Profit]

$$\Rightarrow 850 = x_1 + \frac{15}{100} \times x_1 = \frac{20x_1 + 3x_1}{20}$$

$$850 = \frac{23x_1}{20} \Rightarrow x_1 = \frac{850 \times 20}{23} \quad \dots(1)$$

For book with 15% loss.

$SP = CP - \text{loss}$

$$850 = x_2 - \frac{15}{100}x_2 = \frac{20x_2 - 3x_2}{20} = \frac{17x_2}{20}$$

$$\Rightarrow x_2 = \frac{850 \times 20}{17} \quad \dots(2)$$

$$\text{Total cost price of two books} = x_1 + x_2 = \frac{850 \times 20}{23} + \frac{850 \times 20}{17} = \frac{850 \times 20 \times (23 + 17)}{23 \times 17} = \frac{850 \times 20 \times 40}{23 \times 17}$$

Total cost price = ₹ 1739.13

∴ $CP > SP$ for whole transaction.

$$\therefore \% \text{ Loss} = \frac{\text{Loss}}{\text{CP}} \times 100 = \frac{1739.13 - 1700}{1739.13} \times 100$$

$$\% \text{ Loss} = 2.25\%$$

27. A dealer prices an article at 20% more than the cost price and allows a discount of 10% on it. Find the gain percent.

Sol: Let the cost price of an article = ₹ ' x '

Marked Price (M.P.) = $CP + 20\%$ of CP .

$$M.P. = x + \frac{20}{100} \times x = \frac{6x}{5}$$

Discount % = 10%

S.P. = $MP - 10\%$ of MP

$$S.P. = \frac{6x}{5} - \frac{10}{100} \times \frac{6x}{5} = \frac{30x}{25} - \frac{3x}{25} = \frac{27x}{25}$$

$$\text{gain \%} = \frac{SP - CP}{CP} \times 100 = \frac{\frac{27}{25}x - x}{x} \times 100\% = \frac{2x}{25x} \times 100\%$$

$$\text{gain \%} = 8\%$$

28. A garment dealer allows his customers 10% discount on marked price of the goods and still makes a profit of 25%. Find the cost price of a shirt if it is marked at ₹ 1250.

Sol: Marked Price (MP) = ₹ 1250, Discount = 10%, Profit = 25%

Let Cost Price (C.P.) = ₹ ' x '

Selling Price (S.P.) = $M.P. - \text{Discount}$

$$SP = 1250 - 10\% \text{ of } 1250$$

$$SP = 1250 - 125 = ₹ 1125$$

$$\text{Now, \% Profit} = \frac{\text{SP} - \text{CP}}{\text{CP}} \times 100$$

$$25 = \left(\frac{1125 - x}{x} \right) \times 100$$

$$\Rightarrow x = 1125 \times 4 - 4x$$

$$\Rightarrow 5x = 1125 \times 4$$

$$\Rightarrow x = \frac{1125 \times 4}{5} = 225 \times 4 = ₹ 900$$

$$\therefore x = ₹ 900$$

$$\text{C.P.} = ₹ 900$$

29. Lussy deposited ₹ 7500 in a bank which pays him 12% interest per annum compounded quarterly. What is the amount which she receives after 9 months?

Sol: Here, $P = ₹ 7500$, $R = 12\%$ per annum and $n = 9$ months $= \frac{9}{12}$ year $= \frac{3}{4}$ year.

$$\therefore \text{Amount after 9 months} = P \left(1 + \frac{R}{400} \right)^{4n} = ₹ 7500 \times \left(1 + \frac{12}{400} \right)^{4 \times \frac{3}{4}}$$

$$= ₹ 7500 \times \left(1 + \frac{3}{100} \right)^3$$

$$= ₹ 7500 \times \frac{103}{100} \times \frac{103}{100} \times \frac{103}{100}$$

$$= ₹ 8195.45$$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- A ratio is a comparison by division of two quantities with the _____ unit.
- Ratio has _____ units as it is a number.
- The word percent means _____.
- _____ % of 50 is 19.
- _____ is calculated on the sale value by applying the rate of tax as applicable.
- _____ is the price at which the article is purchased
- _____ price is the price at which the article is sold.
- _____ = selling price – cost price
- _____ = cost price – selling price
- Real cost = cost price + _____?

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- The word cent means century i.e. hundred.
- To convert a fraction or a decimal or a ratio into a percentage, divide by 100.
- 15 % of 300 is 54.
- The sum of the amount and the money borrowed is called the simple interest.
- The deduction made on the marked price is called discount.
- The profit percent is the profit that would be obtained for a S.P. of ₹ 100.
- A ratio with its second term 100 is also called a percent.
- A student buys a pen for ₹ 90 and sells it for ₹ 100, the gain percent is $\frac{100}{9}\%$.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D,...) in column-I have to be matched with statements (p, q, r, s,...) in column-II.

- | | | |
|-----|---|------------------|
| 1. | Column-I | Column-II |
| (A) | 1 cm to 2 m | (p) 1 : 1000 |
| (B) | 1 min to 1 h | (q) 24 : 1 |
| (C) | 1 m to 1 km | (r) 1 : 365 |
| (D) | 1 day to 1 h | (s) 1 : 200 |
| (E) | 1 day to 1 year | (t) 1 : 60 |
| 2. | Column-I | Column-II |
| (A) | The number whose $6\frac{1}{4}\%$ is 2 is | (p) 8560 |
| (B) | Marked price of a book is ₹ 30. It is sold at a discount of 15%. The discount allowed on the book in ₹ is | (q) 650 |
| (C) | Rohan purchased a house for ₹ 6500 and sold it on the same day for ₹ 7150. His gain in ₹ is | (r) 32 |
| (D) | A shopkeeper purchased goods worth ₹ 8000. He sold them at a profit of 7%. His selling price in ₹ is | (s) 4.50 |

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

- Replace each box with $>$, $<$ or $=$ to make each statement true.
 - 60% of 120 60% of 12
 - 25% of 150 35% of 120
- What is the value of $x\%$ of $(y\%$ of $x)$?
- A number when increased by 3% becomes 20.6. Then what is the number?
- Find 90% of a number whose 500% is 200.
- Increase 48 by $12\frac{1}{2}\%$.
- If 486 is 54%, what percent is 639?
- Given ratios are all equivalent to 4:3 for each, tell by which numbers both terms were multiplied.

(i) 12 : 9	(ii) 8 : 6
(iii) 16 : 12	(iv) 24 : 18

Comparing Quantities

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8. The ratio in each part are equivalent, find the missing term.

(i) $4:5 = \square:25$ (ii) $2:3 = 12:\square$ (iii) $15:25 = \square:5$

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

1. A new car costs ₹ 360000. Its price depreciates at the rate of 10% a year during the first two years and at the rate of 20% a year thereafter. What will be the price of the car after 3 years ?
2. The bacteria in a culture grows by 10% in the first hour, decreases by 10% in the second hour and again increases by 10% in the third hour. If the original count of the bacteria in a sample is 10000, find the bacteria count at the end of 3 hours.
3. Divide ₹ 8000 into two parts so that the SI on the first part for 5 years at 12% per annum is equal to SI on the second part for 2 years at 18% per annum?
4. Ritu visits a departmental store and purchase the following articles:

	Item	VAT
(i)	1 rain coat for ₹ 300	10%
(ii)	1 pair of shoes for ₹ 460	9%
(iii)	Food articles for ₹ 450	5%
(iv)	Clothes for ₹ 800	1%

Calculate the total amount of the bill.

5. In an examination, a student obtained 365 marks in total. If he obtained 73% marks then find the maximum marks.
6. If 7% of 500 is subtracted from 8% of a number, then the result is 5. Find the number.
7. The price of a box of soap powder increased from ₹ 2.50 to ₹ 2.65. Find the percentage increase in the price.
8. If 33% of a number is more than 23% of 40 by 0.7, then find the number.
9. The selling price, with VAT, on a cooking range is ₹ 19610. What is the original price of the cooking range? The VAT is 6%.
10. Express using a percent.
 - (i) On average, 10 out of every 100 people are left handed.
 - (ii) Jasmine got 86 out of 100 in her math test.
 - (iii) 35 of the world's 100 highest mountains are in Pakistan.

11. The ratio of the length of the human body to the length of the head is about 8:1. What is the length of the head of a person who is 168 cm tall?

12. A property decreases in value every year at the rate of $6\frac{1}{4}$ percent on its value at the beginning of that year. If its value at the end of 3 years be ₹ 21093.75, what was it worth at the beginning of these three years ?
13. A winning candidate got 520000 votes which represented 52% of the electorate. His only opponent secured 25% votes of the electorate. Find the total number of the electorate and the number of persons who did not cast their votes.
14. A man spends 15% of his income on rent, 44% on food and 10% on entertainment. He saves 10% and spends the rest on education. If his savings per month is ₹ 2500, find his income. Also find how much he spends on each.
15. The price of a T.V. set inclusive sales tax of 9% is ₹ 13407. Find its marked price. If the sales tax is increased to 13%, how much more does the customer pay for the TV?
16. In what time will ₹ 64000 amount to ₹ 68921 at 5% per annum, interest being compounded half-yearly?

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Find the compound interest on ₹ 15,625 for 9 months at 16% per annum, compounded quarterly.
2. Devendra borrowed ₹ 16000 at 17.5% per annum simple interest. He lent it to Inder at the same rate, the same day but compounded annually. What does he gain at the end of 2 years?
3. Determine the rate per cent per annum if ₹ 25,000 amounts to 26,010 in 6 months, interest being compounded quarterly.
4. The population of a town 2,50,000 three years ago. If it had increased by 3% , 4% and 6% in the last three years, find the present population of the town.
5. If the simple interest on a certain sum of money for 3 yrs at 5% is ₹ 150, find the corresponding CI.
6. 24000 blood donors were registered with a charitable hospital. The number of donors increased at the rate of 5% every six month. Find the time period at the end of which the total number of blood donors becomes 27783.

2 EXERCISE

Text-Book Exercise :

- Find the ratio of the following.
 - Speed of a cycle 15 km per hour to the speed of scooter 30 km per hour.
 - 5 m to 10 km
 - 50 paise to ₹ 5
- Convert the following ratios to percentages.
 - 3 : 4
 - 2 : 3
- A football team won 10 matches out of the total number of matches they played. If their win percentage was 40, then how many matches did they play in all?
- If 60% people in a city like cricket, 30% like football and the remaining like other games, then what per cent of the people like other games? If the total number of people are 50 lakh, find the exact number who like each type of game.
- A shopkeeper purchased 200 bulbs for ₹ 10 each. However 5 bulbs were fused and had to be thrown away. The remaining were sold at ₹ 12 each. Find the gain or loss %.
- Meenu bought two fans for ₹ 1200 each. She sold one at a loss of 5% and the other at a profit of 10%. Find the selling price of each. Also find out the total profit or loss.
- The cost of a pair of roller skates at a shop was ₹ 450. The sales tax charged was 5%. Find the bill amount.
- Waheeda bought an air cooler for ₹ 3300 including a tax of 10%. Find the price of the air cooler before VAT was added.
- The cost of an article was ₹ 15,500. ₹ 450 were spent on its repairs. If it is sold for a profit of 15%, find the selling price of the article.
- A VCR and TV were bought for ₹ 8,000 each. The shopkeeper made a loss of 4% on the VCR and a profit of 8% on the TV. Find the gain or loss percent on the whole transaction.
- I purchased a hair-dryer for ₹ 5,400 including 8% VAT. Find the price before VAT was added.
- A TV was bought at a price of ₹ 21,000. After one year the value of the TV was depreciated by 5% (Depreciation means reduction of value due to use and age of the item). Find the value of the TV after one year.

Exemplar Questions :

- Find the compound interest on Rs 48,000 for one year at 8% per annum when compounded half yearly.
- Vishakha offers a discount of 20% on all the items at her shop and still makes a profit of 12%. What is the cost price of an article marked at ₹ 280?
- In a factory, women are 35% of all the workers, the rest of the workers being men. The number of men exceeds that of women by 252. Find the total number of workers in the factory.
- Three bags contain 64.2 kg of sugar. The second bag contains $\frac{4}{5}$ of the contents of the first and the third contains $45\frac{1}{2}\%$ of what there is in the second bag. How much sugar is there in each bag?
- In 2007–08, the number of students appeared for Class X examination was 1,05,332 and in 2008–09, the number was 1,16,054. If 88,151 students pass the examination in 2007–08 and 103804 students in 2008–09. What is the increase or decrease in pass % in Class X result?
- A lady bought an air-conditioner for Rs 15,200 and spent Rs 300 and Rs 500 on its transportation and repair respectively. At what price should she sell it to make a gain of 15%?
- In the year 2001, the number of malaria patients admitted in the hospitals of a state was 4,375. Every year this number decreases by 8%. Find the number of patients in 2003.

HOTS Questions :

- Two friends Tanmay and Naman invested a total of ₹ 60000 together at compound interest at 12.5% per annum in such a way that Tanmay gets the same amount after two years what Naman gets after five years. Find the Tanmay's investment which is more than that of Naman
- A man bought two T.V sets for ₹ 42500. He sold one at a loss of 10% and other at a profit of 10%. If the selling price of each T.V set is same, determine the C.P of each set.
- A number is increased by 9% and the same number is then decreased by 9%. The difference between these two new numbers is 36. What is the number?
- A man borrowed ₹ 24000 from two money lenders. For one loan he paid 15% annum and for the other 18% per annum. At the end of one year, he paid simple interest of ₹ 4050. How much did he borrow at each rate?

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The ratio of three numbers is 3 : 4 : 5 and sum of their squares is 1250. The sum of the numbers is
(a) 30 (b) 50
(c) 60 (d) 90
- An amount of ₹735 was divided between A, B and C. If each of them had received ₹25 less, their shares would have been in the ratio of 1 : 3 : 2. The money received by 'C' was
(a) ₹195 (b) ₹220
(c) ₹225 (d) ₹245
- A man loses 12.5% of his money and after spending 70% of the remainder, has ₹210 left. At first the man had
(a) ₹720 (b) ₹600
(c) ₹800 (d) ₹880
- A cycle shop allows a discount of 25% on the marked price and earns a profit of 20% on the cost price. Its marked price on which shop earns ₹40 is
(a) ₹300 (b) ₹320
(c) ₹280 (d) ₹340
- A Salesman bought 500 m of electric wire at 75 paise per meter. The sold 60% of it at a profit of 8%. At what gain percent should he sell the remainder so as to gain 12% on the whole transaction.
(a) ₹8% (b) ₹11%
(c) ₹15% (d) ₹18%
- A person sells two watches for ₹500 each. On one he loses 10% and on the other he gained 10%. His gain or loss % is
(a) 1.5% gain (b) 1.5% loss
(c) 1% loss (d) 1% gain
- A sells a bicycle to B at a profit of 20%. B sells it to C at a profit of 25%. If C pays ₹225 to it, the cost price of the bicycle for A is
(a) ₹115 (b) ₹130
(c) ₹150 (d) ₹140
- A shopkeeper allows a discount of 10% on the marked price of an item but charges a sales tax of 8% on discounted price. If a customer pays ₹680.40 as the price of the item including sales tax. The marked price of the item is
(a) ₹650 (b) ₹680
(c) ₹700 (d) ₹725
- Two shopkeepers A and B sell machines at the same list price. The first (A) allows two successive discounts of 30% and 16% and the second (B) 20% and 26%. Which discount series is more advantageous to the customer?
(a) Discount offered by 'A'
(b) Discount offered by 'B'
(c) Both are equal
(d) Can't be determined.
- A shopkeeper sold sarees at ₹266 each after giving 5% discount on labelled price. Had he not given the discount, he would have earned a profit of 12% on the cost price. The cost price of each saree was
(a) ₹200 (b) ₹225
(c) ₹250 (d) ₹240
- Hemant purchased 120 rims of paper at ₹80 per rim. He spent ₹280 on transportation, paid octroi at the rate of 40 paise per rim and paid ₹72 to the coolie. If he wants to have a gain of 8%, the selling price per rim must be
(a) ₹85 (b) ₹88
(c) ₹90 (d) ₹95
- The population of a village increases @ 5% p.a.. If present population is 8000, after how many years the population will be 9261?
(a) 2 years (b) 3 years
(c) $3\frac{1}{2}$ years (d) 4 years
- Population of a town increases at a certain rate per cent per annum. Present population of the town is 3600 and in 5 years' time it becomes 4800. How much will it be in 10 years time?
(a) 5000 (b) 6000
(c) 6400 (d) 7000
- In what time will ₹72 become ₹81 at $6\frac{1}{4}\%$ p.a. SI?
(a) $1\frac{1}{2}$ year (b) $2\frac{1}{2}$ years
(c) 2 years (d) None of these
- Compound interest on ₹25000 at 20% p.a. for $2\frac{1}{2}$ years, if interest is compounded annually, is
(a) ₹39600 (b) ₹14600
(c) ₹37500 (d) ₹12500
- A man invested ₹16000 at compound interest for 3 years, interest compounded annually. If he got ₹18522 at the end of 3 years, then the rate of interest is
(a) 4% (b) 5%
(c) 6% (d) 7%

17. The difference between CI and SI on ₹ 8000 for 3 yrs at 2.5% p.a. is
 (a) ₹ 15.125 (b) ₹ 10.125
 (c) ₹ 18.125 (d) ₹ 19.125
18. The value of a machine depreciates at the rate of 10% after every year. Its purchase price if at the end of two years its value is ₹ 11340 was
 (a) 12000 (b) 14000
 (c) 13400 (d) None of these
19. Compound interest on ₹ 10000 for 3 years if rate of interest is 5%, 10% and 20% for first, second and third years respectively, is
 (a) ₹ 3310 (b) ₹ 3500
 (c) ₹ 3860 (d) ₹ 3980

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. The selling price of 12 articles is equal to the cost price of 15 articles. The gain per cent is
 (a) $6\frac{2}{3}\%$ (b) $22\frac{12}{4}\%$
 (c) 25% (d) 80%
2. A discount series of 10%, 20% and 40% is equal to a signal discount of (approx.)
 (a) 50% (b) 56.8%
 (c) 57% (d) 70.28%
3. An umbrella marked at ₹ 80 is sold for ₹ 68. The rate of discount is
 (a) 15% (b) $12\frac{12}{4}\%$
 (c) $17\frac{11}{17}\%$ (d) 20%
4. The simple interest on a certain sum for 3 years at 14% per annum is ₹ 235.20. The sum is
 (a) ₹ $\frac{100 \times 235.20}{3 \times 12}$ (b) ₹ $\frac{100 \times 235.20}{3 \times 14}$
 (c) ₹ 560 (d) ₹ 720
5. A sum amounts to ₹ 1352 in 2 years at 4% compound interest. The sum is
 (a) ₹ 1250 (b) ₹ 25×50
 (c) ₹ 1260 (d) ₹ 63×20
6. The compound interest on ₹ 2800 for $1\frac{1}{2}$ years at 10% per annum, is
 (a) ₹ 441.35 (b) ₹ (3234 – 2800)
 (c) ₹ 434 (d) ₹ (3200 – 3000)
7. Simple interest on a sum at 4% per annum for 2 years is ₹ 80. The compound interest on the same, sum for the same period is
 (a) ₹ 81.60 (b) ₹ 160
 (c) ₹ 120/2 (d) ₹ 163.2/2

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE - I

If V_0 is the value of an article at certain time which increases at the rate of R_1 for first n_1 years and decreases at the rate of R_2 for next n_2 years, then the value of the article V at the end of $(n_1 + n_2)$ years is given by

$$V = V_0 \left(1 + \frac{R_1}{100}\right)^{n_1} \left(1 - \frac{R_2}{100}\right)^{n_2}$$

1. The production of an article of a company in 2002 was 10000. Due to increase in demand, the company increased its production by 20% in the next 2 years. After 2 years due to decrease in the demand, the company decreased its production by 10% in the next year, then the production after 3 years is
 (a) 12950 (b) 12000
 (c) 12900 (d) 12960
2. Population of Chhapra district was 200000 on the last day of the year 2005. During next year is increased by 5% but due to an epidemic it decreased by 2% in the following year, then the population at the end of the year 2007 is
 (a) 205000 (b) 205800
 (c) 206000 (d) 205700
3. Sandeep started a factory with an initial investment of ₹ 500000. In the next year, he incurred a loss of 20%. However, during the second year, he earned a profit of 10% and in the third year he earned a profit of 15%, then his net profit for the entire period of three years is
 (a) ₹ 5000 (b) ₹ 6000
 (c) ₹ 50000 (d) ₹ 60000

PASSAGE - II

The number of listeners to a radio station in 1995 was 350000. In 1996, there were 392000 listeners. Each year, the radio station invites its listeners to vote for their favourite pieces of music. In 1995, 18% of the listeners voted. In 1997, the number of listeners was 460000 and 91000 of these voted. The number who voted in 1997 was 40% more than in 1996.

4. Find the percentage increase in the number of listeners from 1995 to 1996.
 (a) 10% (b) 12%
 (c) 8% (d) 14%
5. Calculate the number who voted in 1995.
 (a) 63000 (b) 62000
 (c) 65000 (d) 61000
6. Find the percentage of listeners who voted in 1997.
 (a) 17% (b) 20%
 (c) 19% (d) 18%
7. Calculate the number who voted in 1996.
 (a) 62000 (b) 63000
 (c) 65000 (d) 61000

PASSAGE - III

Increase = Increase % of original value and Decrease = Decrease % of original value.

8. The salary of a bank clerk was increased by 7%. If his present salary is ₹ 8025, his salary before increment is
 (a) ₹ 7500 (b) ₹ 8500
 (c) ₹ 7000 (d) ₹ 8000
9. The value of a machine depreciates every year by 5%. If the present value is ₹ 40,000, the value after a year would be
 (a) ₹ 380 (b) ₹ 30,000
 (c) ₹ 3800 (d) ₹ 38,000
10. The population of a town increases by 10% annually. If the present population is 60,000, the population after a year is
 (a) 6600 (b) 66,000
 (c) 660 (d) 60,000
11. The value of a machine depreciates 10% every year. If its present value is ₹ 38700, the value a year ago is
 (a) ₹ 4300 (b) ₹ 430
 (c) ₹ 43000 (d) ₹ 40300

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
- (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. In a test on percent applications Priyanka answered 28 of the 35 questions correctly.

Assertion : She answered 80% of the questions correctly.

Reason : She answered 20% of the questions correctly.

2. **Assertion :** An article is sold at ₹ 1425 at a loss of 5%. It's C.P is ₹ 1500.

Reason : If the shopkeeper has to make a 10% profit in question given in statement 1 then the S.P should be ₹ 6150.

3. **Assertion :** 0.5% of a is 3 $\Rightarrow \frac{0.5}{100} = 3a$

Reason : $p\% \text{ of } x = \frac{P}{100} \times x$, where x = number and p = required percent.

4. **Assertion :** Rekha bought a saree for ₹ 1240 and sold it for ₹ 1147. Her loss is ₹ 93.

Reason : If the selling price of an article is less than the list price, then the difference between the cost price and the selling price is called profit.

5. **Assertion :** The discount percent when M.P. = ₹ 900 and S.P. = ₹ 873 is 3%

Reason : Discount % is calculated using the formula

$$\text{Discount \%} = \frac{\text{Discount}}{\text{SP}} \times 100$$

Multiple Matching Question :

DIRECTIONS : Following question has three statements (A, B and C) given in Column I and six statements (p, q, r, s, t and u) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1.	Column-I	Column-II
(A)	Percentage is	(p) a fraction whose denominator is 100.
(B)	Selling price is	(q) $\frac{\text{Principal} \times \text{rate} \times \text{time}}{100}$
(C)	Simple interest is	(r) $\left(\frac{\text{One quantity}}{\text{Other quantity}} \times 100 \right) \%$
		(s) $\left(1 + \frac{\text{Profit}}{100} \right) \times \text{C.P.}$
		(t) C.P. – Loss
		(u) Amount–Principal

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. The population of a city reduced from 25000 to 24500. What is the percentage of reduction of population?
2. A sum of money invested at compound interest amounts to ₹800 in 3 year and to ₹840 in 4 years. What is the rate of interest per annum?
3. 72 % of 25 students are good at Mathematics. How many are not good at it?
4. If 50% of $(x - y) = 30\%$ of $(x + y)$, then $x = ky$. Find k.
5. Shobha's Mathematics Test had 75 problems i.e., 10 arithmetic, 30 algebra and 35 geometry problems. Although she answered 70% of the arithmetic, 40% of the algebra and 60% of the geometry problems correctly, she did not pass the test because in algebra she got less than 60% of the problems. How many more questions she would have needed to answer correctly to score a 60% passing grade?
6. In what time will ₹ 1000 amount to ₹ 1331 at 10% per annum compounded annually?

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- Same
- no.
- per hundred.
- 38%
- VAT
- Cost price
- Selling
- Profit
- Loss
- overhead charges

True / False :

- True
- False, multiply by 100 and put the sign%.
- False, $15\% \text{ of } 300 = 300 \times \frac{15}{100} = 3 \times 15 = 45$
- False; the difference between the amount and the money borrowed is called the compound interest for that period.
- True
- False; profit % = $\frac{\text{Profit}}{\text{C.P.}} \times 100$
- True
- True; C.P. = ₹ 90; S.P. = ₹ 100
Profit = S.P. - C.P. = 100 - 90 = ₹ 10

$$\text{Profit \%} = \frac{\text{Profit}}{\text{C.P.}} \times 100 = \frac{10}{90} \times 100 = \frac{100}{9} \%$$

Match the Columns :

- (A) → (s); (B) → (t); (C) → (p); (D) → (q); (E) → (r)
- (A) → (r); (B) → (s); (C) → (q); (D) → (p)

(A) Let the number be a

$$6\frac{1}{4}\% \text{ of } a \text{ is } 2$$

$$\Rightarrow \frac{25}{4}\% \times a = 2 \Rightarrow \frac{25}{4} \times \frac{1}{100} \times a = 2$$

$$\Rightarrow a = 2 \times 4 \times 4 = 32$$

(B) M.P. = ₹ 30, Discount % = 15%, Discount = ?

$$\text{Discount \%} = \frac{\text{Discount}}{\text{M.P.}} \times 100$$

$$\Rightarrow 15 = \frac{\text{Discount}}{30} \times 100$$

$$\Rightarrow \text{Discount} = \frac{45}{10} = ₹ 4.50$$

(C) C.P. of the house = ₹ 6500

S.P. of the house = ₹ 7150

$$\text{Gain} = \text{S.P.} - \text{C.P.} = 7150 - 6500 = ₹ 650$$

(D) Cost price of the goods = ₹ 8000

Profit % = 7

S.P. = ?

$$\text{S.P.} = \left(\frac{100 + \text{profit \%}}{100} \right) \times \text{C.P.}$$

$$= \left(\frac{100 + 7}{100} \right) \times 8000$$

$$= \frac{107}{100} \times 8000 = 107 \times 80 = ₹ 8560$$

Very Short Answer Questions :

- (i) 60% of 120 > 60% of 12
(ii) 25% of 150 < 35% of 120.

$$2. \quad y\% \text{ of } x = \frac{x \times y}{100} = \frac{xy}{100}$$

$$x\% \text{ of } (y\% \text{ of } x) = \frac{xy}{100} \times \frac{x}{100} = \frac{x^2y}{10000}$$

3. 20

4. Required number = 36.

5. 54

6. 71%

7. (i) 3

(ii) 2

(iii) 4

(iv) 6

8. (i) 20

(ii) 18

(iii) 3

Short Answer Questions :

1. We have,

Cost of the car = ₹ 360000

Rate of depreciation in first two years = 10% per annum.

Rate of depreciation in the third year = 20%

∴ Price of the car after 3 years

$$= ₹ \left\{ 360000 \times \left(1 - \frac{10}{100} \right) \times \left(1 - \frac{10}{100} \right) \times \left(1 - \frac{20}{100} \right) \right\}$$

$$= ₹ \left\{ 360000 \times \left(1 - \frac{1}{10} \right) \times \left(1 - \frac{1}{10} \right) \times \left(1 - \frac{1}{5} \right) \right\}$$

$$= ₹ \left\{ 360000 \times \frac{9}{10} \times \frac{9}{10} \times \frac{4}{5} \right\} = ₹ 233280$$

Hence, the price of the car after 3 years = ₹ 233280.

2. We have,

P = Original count of bacteria = 10000

∴ Bacteria count after 3 hours

$$= 10000 \times \left(1 + \frac{10}{100} \right) \times \left(1 + \frac{10}{100} \right) \times \left(1 + \frac{10}{100} \right)$$

$$= 10000 \times \frac{11}{10} \times \frac{11}{10} \times \frac{11}{10}$$

$$= 10 \times 11 \times 9 \times 11 = 10890.$$

3. Let the first part be x .

Second part = $8000 - x$

We have

$$\frac{x \times 5 \times 12}{100} = \frac{(8000 - x) \times 2 \times 18}{100}$$

$$5x = 24000 - 3x$$

$$8x = 24000$$

$$x = 3000$$

∴ First part = ₹ 3000.

Second part = ₹ 5000.

4. C.P = 300, VAT = 10%

Amount paid = 300 + 30

Similarly we can find the paid amount of other articles.

Hence, total amount of the bill = 2111.90.

5. Maximum marks = x

73% of x = 365

Thus, maximum marks = 500.

6. $\frac{8x}{100} - \frac{500 \times 7}{100} = 5 \Rightarrow x = 500$

7. $\left(\frac{265 - 250}{250} \right) \times 100 = 6\%$

8. Let the required number be x

According to the question,

$$\frac{33x}{100} = \frac{23}{100} \times 40 + 0.7$$

on solving, we get $x = 30$.

9. $x \left(1 + \frac{6}{100} \right) = 19610$

$$\Rightarrow x = 18500$$

Original price = ₹ 18500

10. (i) On average, 10 % people are left handed.

(ii) Jasmine got 86 % in her math test.

(iii) 35 % of the world's highest mountains are in Pakistan.

11. $\frac{\text{Length of human body}}{\text{Length of head}} = \frac{8}{1}$

$$\Rightarrow \text{Length of head} = \frac{168}{8} = 21 \text{ cm}$$

12. $P \left(1 - \frac{6.25}{100} \right)^3 = 21093 \frac{3}{4} \Rightarrow P \times \left(\frac{15}{16} \right)^3 = \frac{84375}{4}$

$$P = \frac{84375}{4} \times \frac{16 \times 16 \times 16}{15 \times 15 \times 15} = ₹ 25600$$

13. Total no. of electorate = x

52 % of $x = 520000 \Rightarrow x = 1000000$

the total no. of votes secured by two candidates

= 77% of 1000000

The number of persons who didn't cast vote = 23% of 1000000 = 230000

14. Total income = ₹ 25000

Entertainment = ₹ 2500

Rent = ₹ 3750

Food = ₹ 11000

Education = ₹ 5250

15. Marked price = x

$$\text{Sales tax} = \frac{9x}{100}$$

$$\text{Price of TV} = \frac{109x}{100}$$

$$\frac{109x}{100} = 13407 \Rightarrow x = 12300$$

Customer will pay when sales tax is 13%

$$\frac{113}{100} \times 12300 = 13899$$

∴ Customer has to pay ₹ (13899 - 13407) more i.e. ₹ 492

16. Here, Principal $P = ₹ 64000$, Amount $A = ₹ 68921$,
rate $R = 5\%$ per annum.
Since the interest is compounded half-yearly.

$$\therefore A = P \left(1 + \frac{R}{200} \right)^{2n}, \text{ where } n \text{ is the number of years.}$$

$$\Rightarrow 68921 = 64000 \left(1 + \frac{5}{200} \right)^{2n}$$

$$\Rightarrow \frac{68921}{64000} = \left(\frac{41}{40} \right)^{2n}$$

$$\Rightarrow \left(\frac{41}{40} \right)^3 = \left(\frac{41}{40} \right)^{2n} \Rightarrow 2n = 3.$$

$$\Rightarrow n = \frac{3}{2} \text{ years} = 1\frac{1}{2} \text{ years}$$

Long Answer Questions :

1. Here, Principal (P) = ₹ 15625
Rate (r) = 16% p.a. = 4% per quarter
Time (n) = 9 months = 3 quarters

$$\text{Now, Amount, } (A) = P \left(1 + \frac{r}{100} \right)^n$$

$$= ₹ 15625 \left(1 + \frac{4}{100} \right)^3$$

$$= ₹ 15625 \left(1 + \frac{1}{25} \right)^3$$

$$= ₹ 15625 \left(\frac{26}{25} \right)^3$$

$$= ₹ 15625 \times \frac{26}{25} \times \frac{26}{25} \times \frac{26}{25}$$

$$= ₹ 26 \times 26 \times 26 = ₹ 17576$$

Since, Compound Interest = Amount – Principal

$$= ₹ 17576 - ₹ 15625$$

$$= ₹ 1,951$$

2. $P = ₹ 16,000$, $R = 17.5\%$ p.a., $T = 2$ years

$$\text{S.I.} = \frac{P \times R \times T}{100} = ₹ \frac{16,000 \times 17.5 \times 2}{100} = ₹ 5600.$$

When compounded annually : $P = 16000$, $R = 17.5\%$

$$\text{Interest at the end of first year} = \frac{16000 \times 17.5 \times 1}{100}$$

$$= ₹ 2800.$$

$$\therefore \text{Principal for the second year} = ₹ 16000 + ₹ 2800$$

$$= ₹ 18800.$$

$$\text{Interest at the end of second year} = \frac{18800 \times 17.5 \times 1}{100}$$

$$= ₹ 3290.$$

$$\therefore \text{Compound interest for two years} = ₹ 2800 + ₹ 3290$$

$$= ₹ 6090.$$

$$\text{Gain of Devendra} = \text{C.I.} - \text{S.I.} = ₹ 6090 - ₹ 5600$$

$$= ₹ 490.$$

3. Here, $n = 2$ [$\because$ 6 months = 2 quarters]

Now, $A = P \left(1 + \frac{r}{100} \right)^n$, where r is the rate per cent per quarter.

$$\therefore 26010 = 25000 \left(1 + \frac{r}{100} \right)^2$$

$$\text{or } \left(1 + \frac{r}{100} \right)^2 = \frac{26010}{25000} = \frac{2601}{2500} = \left(\frac{51}{50} \right)^2$$

$$\text{or } \left(1 + \frac{r}{100} \right) = \frac{51}{50}$$

$$\text{or } \frac{r}{100} = \frac{51}{50} - 1 = \frac{51 - 50}{50} = \frac{1}{50}$$

$$\text{or } r = \frac{1}{50} \times 100 = 2\%$$

Hence, the required rate is 8% p.a.

4. Population after $(t_1 + t_2 + t_3 + \dots)$ years

$$= \left(1 + \frac{R_1}{100} \right)^{t_1} \times \left(1 + \frac{R_2}{100} \right)^{t_2} \times \left(1 + \frac{R_3}{100} \right)^{t_3} \dots$$

If p = population three years ago,

then population after three years becomes present population

Let x be the present populations

$$\Rightarrow x = 2,50,000 \left(1 + \frac{3}{100} \right) \left(1 + \frac{4}{100} \right) \left(1 + \frac{6}{100} \right)$$

$$\Rightarrow x = 2,50,000 \times \frac{103}{100} \times \frac{104}{100} \times \frac{106}{100} = 2,83,868$$

Hence, the present population of the town is 2,83,868.

5. Whenever the relationship between CI and SI is asked for 3 yrs of time, we use the formula:

$$\text{S.I.} = \frac{rt}{100 \left[\left(1 + \frac{r}{100} \right)^t - 1 \right]} \times \text{C.I.}$$

$$150 = \frac{5 \times 3}{100 \left[\left(1 + \frac{r}{100} \right)^3 - 1 \right]} \times \text{C.I.}$$

$$\therefore \text{C.I.} = \frac{150 \times 100 \left[\frac{9261 - 8000}{8000} \right]}{5 \times 3}$$

$$= \frac{150 \times 100 \times 1261}{5 \times 3 \times 8000} = \frac{1261}{8} = ₹ 157.62$$

6. We have,
 P = Initial number of donors = 24000, A = Final number of donors = 27783
 R = Rate of increase = 5% every six month = 10% p.a.

Let the total time be n years. Then,

$$A = P \left(1 + \frac{R}{200} \right)^{2n}$$

$$\Rightarrow 27783 = 24000 \left(1 + \frac{10}{200} \right)^{2n}$$

$$\Rightarrow 27783 = 24000 \left(\frac{21}{20} \right)^{2n}$$

$$\Rightarrow \frac{27783}{24000} = \left(\frac{21}{20} \right)^{2n} \Rightarrow \left(\frac{21}{20} \right)^3 = \left(\frac{21}{20} \right)^{2n} \Rightarrow 2n$$

$$= 3 \Rightarrow n = \frac{3}{2}$$

2 EXERCISE

Text-Book Exercise :

- (i) 1 : 2
 (ii) 1 : 2000
 (iii) 1 : 10
- (i) 75%
 (ii) $66\frac{2}{3}\%$
- 40% of total no. of matches = 10
 $\Rightarrow$ Total no. of matches = $\frac{10 \times 100}{40} = 25$ matches
- Percentage of people like other games = $[100 - (60 + 30)]$
 = 10 %, cricket $\rightarrow$ 60% of 50 lakh = 30 lakh;
 football $\rightarrow$ 30% of 50 lakh = 15 lakh;
 other games $\rightarrow$ 10% of 50 lakh = 5 lakh
- Cost price of 200 bulbs = ₹ 200 $\times$ 10 = ₹ 2000
 5 bulbs were fused. Hence, number of bulbs left = 200 - 5 = 195
 The SP of 195 bulbs = ₹ 195 $\times$ 12 = ₹ 2340
 $\therefore$ Profit = ₹ 2340 - ₹ 2000 = ₹ 340
 and Profit % = $\frac{340}{2000} \times 100 = 17\%$.

6. Overall CP of each fan = ₹ 1200. One is sold at a loss of 5%.
 This means if CP is ₹ 100, SP is ₹ 95.

$$\text{Therefore, when CP is Rs. 1200, then SP} = ₹ \frac{95}{100} \times 1200$$

$$= ₹ 1140$$

Also second fan is sold at a profit of 10%.

It means, if CP is ₹ 100, SP is ₹ 110.

$$\text{Therefore, when CP is ₹ 1200, then SP} = ₹ \frac{110}{100} \times 1200$$

$$= ₹ 1320$$

Total CP = ₹ 1200 + ₹ 1200 = ₹ 2400

Total SP = ₹ 1140 + ₹ 1320 = ₹ 2460

Since total SP > total CP,

$\therefore$ profit = ₹ (2460 - 2400) = ₹ 60

7. On ₹ 100, the tax paid was ₹ 5.

$$\text{On ₹ 450, the tax paid would be} = ₹ \frac{5}{100} \times 450$$

= ₹ 22.50

Bill amount = Cost of item + Sales tax = ₹ 450 + ₹ 22.50

= ₹ 472.50

8. If the price without VAT is ₹ 100 then price including VAT is ₹ 110.

Now, when price including VAT is ₹ 110, original price is ₹ 100.

Hence when price including tax is ₹ 3300, the original price

$$= ₹ \frac{100}{110} \times 3300 = ₹ 3000$$

9. ₹ 18,342.50

10. Gain of 2%

11. $\therefore$ When price including VAT is ₹ 108, original price = ₹ 100

$\therefore$ When price including VAT is ₹ 5,400,

$$\text{original price} = ₹ \frac{100}{108} \times 5,400$$

= ₹ 5,000

Hence, the price before VAT was added is = ₹ 5,000.

12. Principal = ₹ 21,000

Reduction = 5% of ₹ 21000 per year

$$= ₹ \frac{21000 \times 5 \times 1}{100} = ₹ 1050$$

Value at the end of 1 year = ₹ 21000 - ₹ 1050 = ₹ 19,950

Alternate method: We may directly get this as follows:

$$\text{Value at the end of 1 year} = ₹ 21000 \left(1 - \frac{5}{100} \right)$$

$$= ₹ 21000 \times \frac{19}{20} = ₹ 19,950$$

Exemplar Questions :

- $$A = P \left(1 + \frac{R}{200} \right)^{2n}$$

$$= 48,000 \left(1 + \frac{8}{200} \right)^2$$

$$= 48,000 \times \frac{26}{25} \times \frac{26}{25} = ₹ 51,916.80$$

$$\therefore \text{Compound Interest} = A - P$$

$$= ₹ (51,916.80 - 48,000)$$

$$= ₹ 3,916.80$$
- Discount = 20% of ₹280

$$= \frac{20}{100} \times 280 = ₹ 56$$

So selling price = ₹(280 - 56) = ₹224

Let the cost price be ₹ 100

Profit = 12% of ₹100

$$= ₹ 12$$

So selling price = ₹(100 + 12) = ₹112

If the selling price is ₹112, cost price = ₹100

If the selling price is ₹ 224, cost price

$$= ₹ \left(\frac{100}{112} \times 224 \right) = ₹ 200$$
- 840
- 29.67 kg, 23.73 kg, 10.79 kg or 10.8 kg (approx.)
- Increase 5.76
- ₹ 18,400
- 3703

HOTS Questions :

- Let Tanmay's investment = P

Then, Naman's investment = $60000 - P$

After 2 years Tanmay got $A = P \left(1 + \frac{12.5}{100} \right)^2$

After 5 years Naman got A_1

$$= (60000 - P) \left(1 + \frac{12.5}{100} \right)^5$$

$$\therefore P \left(1 + \frac{12.5}{100} \right)^2 = (60000 - P) \left(1 + \frac{12.5}{100} \right)^5$$

$$\text{or } P = (60000 - P) \times \frac{9}{8} \times \frac{9}{8} \times \frac{9}{8}$$

$$\Rightarrow \frac{512P}{729} = 60000 - P$$

$$\text{or } \frac{512P}{729} + P = 60000$$

$$\Rightarrow P = \frac{60000 \times 729}{1241} = ₹ 35246$$

Hence Naman's investment = ₹ 24754

Difference = $35246 - 24754 = ₹ 10492$

- C.P of first T.V = x

C.P of second T.V = $42500 - x$

$$\text{Loss on first T.V} = \frac{x}{10}$$

$$\text{S.P of first T.V} = \frac{9x}{10}$$

$$\text{Gain on 2nd T.V} = \frac{42500 - x}{10}$$

$$\text{S.P of second T.V} = \text{C.P} + \text{Gain}$$

$$= (42500 - x) \times \frac{11}{10}$$

Given, S.P of each T.V is same.

$$\therefore \frac{9x}{10} = (42500 - x) \times \frac{11}{10}$$

$$\Rightarrow x = 23375.$$

CP of first TV = ₹23375

CP of second TV = ₹18625

- Let required number = x

According to the question,

$$\left(x + \frac{9x}{100} \right) - \left(x - \frac{9x}{100} \right) = 36$$

$$\Rightarrow x = 200$$

- Let sum borrowed at 15% = x

Sum borrowed at 18% = $24000 - x$

$$\therefore \frac{x \times 15 \times 1}{100} + \frac{(24000 - x) \times 18 \times 1}{100} = 4050$$

$$\Rightarrow x = ₹ 9000$$

$$\therefore \text{Money borrowed at 18\%} = ₹ 15000$$

3 EXERCISE

Single Option Correct :

- (c) $(3x)^2 + (4x)^2 + (5x)^2 = 1250$

$$\Rightarrow x^2 + 25 \Rightarrow x = 5$$

$$\therefore 3x + 4x + 5x = 12x = 60$$
- (b)
- (c) Let the man at first had ₹ x

Money lost by man = 12.5% of x

$$= \frac{25}{2} \times \frac{1}{100} \times x = \frac{x}{8}$$

$$\therefore \text{Remaining money} = x - \frac{x}{8} = \frac{7x}{8}$$

$$\text{Money spent} = 70\% \text{ of } \frac{7x}{8}$$

$$= \frac{70}{100} \times \frac{7x}{8} = \frac{49}{80}x$$

$$\text{Money left with man} = \frac{7x}{8} - \frac{49}{80}x = \frac{21}{80}x$$

According to question

$$\text{Money left} = \frac{21}{80}x = 210$$

$$\Rightarrow x = \frac{210 \times 80}{21} = 800$$

$\therefore$ At first man had ₹ 800.

4. (b) Let marked price = ₹ x

$$\therefore \text{selling price (S.P)} = x - \frac{25}{100}x$$

$$\text{SP} = ₹ \frac{3}{4}x$$

Let cost price (CP) = ₹ y

Profit = 20%

$$\therefore \frac{20}{100}y = 40$$

$$\Rightarrow y = 200$$

$$\therefore \text{selling price (SP)} = 200 + 40 = ₹ 240$$

$$\therefore \frac{3}{4}x = 240 \Rightarrow x = \frac{240 \times 4}{3} = ₹ 320$$

5. (d)

6. (c) Let cost of two watches are ₹ x and ₹ y respectively

$$\therefore \text{total cost price} = ₹ (x + y)$$

$$\text{Selling price with 10\% profit} = x + \frac{10}{100}x = \frac{11}{10}x$$

$$\therefore \frac{11}{10}x = 500 \Rightarrow x = \frac{5000}{11}$$

$$\text{Selling price with 10\% loss} = y - \frac{10}{100}y = \frac{9}{10}y$$

$$\frac{9}{10}y = 500 \Rightarrow y = \frac{5000}{9}$$

$$\therefore \text{total cost price } x + y = \frac{5000}{11} + \frac{5000}{9}$$

$$= ₹ \frac{5000 \times 20}{99} > 1000$$

$$\therefore x + y > 1000 \text{ [total selling price]}$$

So there is a loss

$$\therefore \text{Loss} = \frac{5000 \times 20}{99} - 1000 = ₹ 1000 \times \frac{1}{99}$$

$$\% \text{ Loss} = \left(\frac{1000 \times \frac{1}{99}}{5000 \times \frac{20}{99}} \times 100 \right) \% = 1\%$$

7. (c) Let CP for A = ₹ x

$$\therefore \text{CP for B} = ₹ 1.2x$$

$$\text{and CP for C} = ₹ 1.5x$$

$$\therefore 1.5x = 225$$

$$\Rightarrow x = \frac{225}{1.5} = ₹ 150$$

$$\therefore \text{CP for A} = ₹ 150$$

8. (c)

9. (a) Discount offered by shopkeeper A is more advantageous to the customer.

10. (c)

11. (c) Cost price to Hemant = $(120 \times 80 + 280 + 0.4 \times 120 + 72)$
= ₹ 10,000

Selling price = cost price + 8% gain

$$= 10000 + \frac{8}{100} \times 10000$$

$$= ₹ 10800 \text{ [for 120 Rims]}$$

$$\therefore \text{Selling cost / Rim} = \frac{10800}{120} = ₹ 90.$$

12. (b) Let population become 9261 in 'x' years.

$$\frac{\text{Amount}}{\text{Principal}} = \left(1 + \frac{\text{Rate}}{100} \right)^{\text{Time}}$$

$$\therefore \frac{9261}{8000} = \left(\frac{21}{20} \right)^x$$

$$\therefore \left(\frac{21}{20} \right)^3 = \left(\frac{21}{20} \right)^x$$

$$\therefore \text{Time} = 3 \text{ years}$$

13. (c) Let rate of increase in population = $r\%$ p.a.

$$\text{Then } 4800 = 3600 \left(1 + \frac{r}{100} \right)^5$$

$$\therefore \left(1 + \frac{r}{100} \right)^5 = \frac{4800}{3600} = \frac{4}{3}$$

Population in the next 5 years will become

$$4800 \times \frac{4}{3} = 6400.$$

14. (c) $A = P \left(1 + \frac{TR}{100} \right)$

$$81 = 72 \left(1 + \frac{T \times \frac{25}{4}}{100} \right)$$

$$\frac{16+T}{16} = \frac{81}{72}$$

$$16+T=18$$

$$T=2 \text{ years.}$$

$$\begin{aligned} 15. \quad (b) \quad \text{Amount} &= 25000 \times \left(1 + \frac{20}{100}\right)^2 \times \left(1 + \frac{10}{100}\right)^1 \\ &= 25000 \times \left(\frac{6}{5}\right)^2 \times \frac{11}{10} = 39600 \\ \therefore \text{Compound interest} &= 39600 - 25000 = 14600. \end{aligned}$$

$$\begin{aligned} 16. \quad (b) \quad (1+r)^3 &= \frac{18522}{16000} = \frac{9261}{8000} = \left(\frac{21}{20}\right)^3 = \left(1 + \frac{1}{20}\right)^3 \\ \therefore \text{Rate of interest} &= \frac{1}{20} = 5\% \end{aligned}$$

$$\begin{aligned} 17. \quad (a) \quad \text{CI} - \text{SI} &= P \left[1 + \frac{R}{100}\right]^3 - P - \left(\frac{P \times R \times 3}{100}\right) \\ &= \frac{PR^2(R+300)}{(100)^3} \\ &= \frac{8000 \times 2.5 \times 2.5(300+2.5)}{100 \times 100 \times 100} \\ &= \frac{8 \times 25 \times 25 \times 3025}{100 \times 100 \times 100} = \frac{121}{8} = ₹ 15.125 \end{aligned}$$

$$\begin{aligned} 18. \quad (b) \quad \text{Let the purchase price be } P. \\ P(1 - 10/100)^2 &= 11340 \Rightarrow P(9/10)^2 = 11340 \\ P &= ₹ 14000. \end{aligned}$$

$$\begin{aligned} 19. \quad (c) \quad \text{Amount after 3 years} \\ &= \text{Principal} \times \frac{100+R_1}{100} \times \frac{100+R_2}{100} \times \frac{100+R_3}{100} \\ &= ₹ 10000 \times \frac{21}{20} \times \frac{11}{10} \times \frac{6}{5} = ₹ 13860 \\ \therefore \text{Compound interest} &= ₹ 13860 - ₹ 10000 = ₹ 3860 \end{aligned}$$

More Than One Option Correct :

$$\begin{aligned} 1. \quad (b,c) \quad \text{Let C.P. of each article} &= ₹ 1. \\ \text{Then, C.P. of 12 articles} &= ₹ 12. \\ \text{S.P. of 12 articles} &= \text{C.P. of 15 articles} = ₹ 15. \\ \therefore \text{Gain} &= \left(\frac{3}{12} \times 100\right)\% = 25\% \end{aligned}$$

$$\begin{aligned} 2. \quad (b,c) \quad \text{Let original price} &= ₹ 100. \\ \text{Price after first discount} &= ₹ 90. \\ \text{Price after second discount} \\ &= ₹ \left(\frac{80}{100} \times 90\right) = ₹ 72 \\ \text{Price after third discount} \\ &= ₹ \left(\frac{60}{100} \times 72\right) = ₹ 43.20 \end{aligned}$$

$$\therefore \text{Single discount} = (100 - 43.20) = 56.8\% \text{ or } 57\%$$

$$3. \quad (a,b) \quad \text{Discount} = \left(\frac{12}{80} \times 100\right)\% = 15\%$$

$$4. \quad (b,c) \quad \text{Sum} = ₹ \left(\frac{100 \times 235.20}{3 \times 14}\right) = ₹ 560.$$

$$5. \quad (a,b) \quad \text{Let the sum be } P. \text{ Then,}$$

$$1352 = P \left(1 + \frac{4}{100}\right)^2 \Rightarrow 1352 = P \times \frac{26}{25} \times \frac{26}{25}$$

$$\Rightarrow P = \frac{1352 \times 25 \times 25}{26 \times 26} = 1250$$

$$\therefore \text{Principal} = ₹ 1250.$$

$$\begin{aligned} 6. \quad (b,c) \quad \text{Amount} &= ₹ \left[2800 \left(1 + \frac{10}{100}\right) \left(1 + \frac{5}{100}\right)\right] \\ &= ₹ \left[2800 \times \frac{11}{10} \times \frac{21}{20}\right] = ₹ 3234 \end{aligned}$$

$$\therefore \text{C.I.} = ₹ (3234 - 2800) = ₹ 434$$

$$7. \quad (a,d) \quad \text{Principal} = ₹ \left(\frac{100 \times 80}{4 \times 2}\right) = ₹ 1000$$

$$\therefore \text{C.I.} = ₹ \left[\left\{1000 \times \left(1 + \frac{4}{100}\right)^2 - 1000\right\}\right] = ₹ 81.60$$

Passage Based Questions :

$$1. \quad (d) \quad \text{Production of the article after 3 years}$$

$$= 10000 \left(1 + \frac{20}{100}\right)^2 \left(1 - \frac{10}{100}\right)$$

$$= 10000 \times \frac{36}{25} \times \frac{9}{10} = 12960$$

$$2. \quad (b) \quad \text{Population of Chhapra district}$$

$$= 200000 \left(1 + \frac{5}{100}\right) \left(1 - \frac{2}{100}\right)$$

$$= 200000 \times \frac{21}{20} \times \frac{49}{50} = 205800$$

$$3. \quad (b) \quad \text{Amount after three years}$$

$$= 500000 \left(1 - \frac{20}{100}\right) \times \left(1 + \frac{10}{100}\right) \left(1 + \frac{15}{100}\right)$$

$$= 500000 \times \frac{4}{5} \times \frac{11}{10} \times \frac{23}{20} = 506000$$

$$\begin{aligned} \text{Net profit} &= ₹ (506000 - 500000) \\ &= ₹ 6000 \end{aligned}$$

4. (b) $\left(\frac{392000 - 350000}{350000} \right) \times 100 = 12\%$
5. (a) $350,000 \times \frac{18}{100} = 63000$
6. (b) $\frac{91000}{460000} \times 100 = 19.78\% = 20\%$
7. (c) $91000 \times \frac{100}{140} = 65,000$
8. (a) Let the original salary = ₹ x
Present salary = ₹ 8025
Increase in salary = 7% of original salary
= 7% of ₹ x
According to the question,
$$x + \frac{7}{100} \times x = 8025$$

$$\Rightarrow \frac{107}{100} x = 8025 \Rightarrow x = \frac{8025 \times 100}{107} = ₹ 7500$$
9. (d) Present value of the machine = ₹ 40,000.
Decrease in value = 5% of ₹ 40000
$$= \frac{5}{100} \times 40000 = ₹ 2000.$$

Thus, value of machine after a year
= ₹ 40000 - ₹ 2000 = ₹ 38000
10. (b) Original (present) population = 60,000
Increase in population = 10% of original
$$= \frac{10}{100} \times 60000 = 6000$$

Thus, population after a year = 60,000 + 6000 = 66000.
11. (c) Let the value year ago = x
$$x - \frac{10}{100} x = ₹ 38700$$

$$\Rightarrow \frac{90}{100} x = ₹ 38700$$

$$\Rightarrow x = \frac{38700 \times 100}{90} = ₹ 43000$$

Assertion & Reason :

1. (c) Required percent of the questions, She answered
$$= \frac{28}{35} \times 100$$
 only correctly. Assertion is correct.
2. (c) Assertion :
SP = ₹ 1425, loss = 5%
$$C.P = \frac{SP \times 100}{100 - \text{loss}} = \frac{1425 \times 100}{95} = 1500$$

Reason :
$$\text{Profit} = 10\%, S.P = \frac{1500 \times 110}{100} = 1650$$

Reason is false.

3. (d) Assertion : False, Reason : True
Since 0.5% of a = 3
$$\Rightarrow \frac{0.5}{100} \times a = 3 \quad (\text{since } p\% \text{ of } x = \frac{p}{100} \times x)$$
4. (c) Assertion : True, Reason : False
Cost price of a saree = ₹ 1240
Selling price of a saree = ₹ 1147
So, there is loss since C.P. > S.P.
Loss = C.P. - S.P. = 1240 - 1147 = ₹ 93
5. (c) Assertion : True, Reason : False
Since, Discount % = $\left(\frac{M.P. - S.P.}{M.P.} \right) \times 100$
$$= \left(\frac{900 - 873}{900} \right) \times 100$$

$$= \left(\frac{27}{900} \times 100 \right) = 3\%$$

Multiple Matching Questions :

1. (A) → (p,r); (B) → (s, t); (C) → (q, u)

Integer Type Questions :

1. (2) Decrease in population = 25000 - 24500 = 500
percentage decrease = $\frac{500}{25000} \times 100 = 2\%$
2. (5) Let the Principal = ₹ x
Now, $A = P \left(1 + \frac{R}{100} \right)^n$
$$\Rightarrow 800 = x \left(1 + \frac{R}{100} \right)^3 \quad \dots(i)$$

and $840 = x \left(1 + \frac{R}{100} \right)^4 \quad \dots(ii)$
Dividing (i) and (ii), we get
$$\frac{840}{800} = \left(1 + \frac{R}{100} \right)$$

$$\Rightarrow \frac{21}{20} - 1 = \frac{R}{100} \Rightarrow \frac{100}{20} = R \Rightarrow R = 5\%$$
3. (7) We have, number of students who are good at
Mathematics = 72% of 25 = $\frac{72}{100} \times 25 = 18$

∴ Number of students who are not good at Mathematics
 $= 25 - 18 = 7$.

4. (4) 50% of $(x - y) = 30\%$ of $(x + y)$

$$\Rightarrow \frac{50}{100}(x - y) = \frac{30}{100}(x + y)$$

$$\Rightarrow 5(x - y) = 3(x + y) \Leftrightarrow 2x = 8y \Leftrightarrow x = 4y.$$

$$\Rightarrow k = 4$$

5. (5) Number of questions attempted correctly
 $= (70\% \text{ of } 10 + 40\% \text{ of } 30 + 60\% \text{ of } 35)$
 $= (7 + 12 + 21) = 40.$

Question to be answered correctly for 60% grade
 $= 60\% \text{ of } 75 = 45.$

∴ Required number of questions $= (45 - 40) = 5.$

6. (3) Let the required time be n years.

Then, amount

$$= ₹ \left\{ 1000 \times \left(1 + \frac{10}{100} \right)^n \right\}$$

$$= ₹ \left\{ 1000 \times \left(\frac{11}{10} \right)^n \right\}$$

$$\therefore 1000 \times \left(\frac{11}{10} \right)^n = 1331 \quad [\because \text{amount} = ₹ 1331 \text{ (given)}]$$

$$\Rightarrow \left(\frac{11}{10} \right)^n = \frac{1331}{1000} = \frac{11 \times 11 \times 11}{10 \times 10 \times 10} = \left(\frac{11}{10} \right)^3$$

$$\Rightarrow \left(\frac{11}{10} \right)^n = \left(\frac{11}{10} \right)^3$$

$$\Rightarrow n = 3,$$

Hence, the required time is 3 years.

Chapter

9

Algebraic Expressions And Identities

INTRODUCTION

Algebra is the branch of mathematics concerning the study of the rules of operations and relations. Elementary algebra is the most basic form of algebra. It is taught to students who are presumed to have no knowledge of mathematics beyond the basic principles of arithmetic. In arithmetic, only numbers and their arithmetical operations (such as $+$, $-$, $\times$, $\div$) occur. In algebra, numbers are often denoted by symbols (such as a , x , or y). This is useful because:

- It allows the general formulation of arithmetical laws
- It allows the formulation of functional relationships. For example : "If you sell x tickets, then your profit will be ₹($3x - 10$) , or $f(x) = 3x - 10$, where f is the function, and x is the number to which the function is applied.

ALGEBRAIC EXPRESSIONS

Algebra can be considered as generalisation of arithmetic, where we use letters in place of numbers, which allows to write rules and form in general way.

Variable : A quantity that does not have a fixed numerical value is called a variable. It is represented by the letters like $x, y, \dots$

Constant : A symbol which has a fixed numerical value is called a **constant**.

Algebraic expressions : When variables and constants are combined with the help of mathematical operations of addition, subtraction, multiplication and division, we get an algebraic expression. For example, $3x + 7$, $15y - 23$ are algebraic expression.

Terms: Look at the expression $(3x + 7)$. This is formed by first forming $3x$ as product of 3 and x and then adding 7 to the product. Such parts of an expression which are formed first and then added are called **terms**.

Consider $(9y^2 - 8x)$, here we can say that $9y^2$ and $-8x$ are two terms of given expressions.

Factors of a term : Now we know that an expression consist of terms. $(9y^2 - 8x)$ has two terms $9y^2$ and $(-8x)$. The term $9y^2$ is a product of 9, y and y . Here we say that 9, y and y are factors of term $9y^2$. A term is represented as product of its factors.

For term $(-8x)$, -8 and x are factors.

Coefficients of a term : We know that any term of an expression can be expressed as product of its factors. These factors are numeric or variables. The numerical factor is called numerical **coefficient** or **coefficient** of the term.

In $9y^2$, 9 is the coefficient of the term. It is also called coefficient of y^2 . In $-10y^2z^2$, -10 is the coefficient of y^2z^2 .

If the coefficient of any term is $+1$, we omit it.

$1y^2$ can be written as y^2 , $1xy$ is written as xy coefficient (-1) is indicated by minus $(-)$ sign, $(-1)y^2$ is written as $-y^2$, $(-1)y^2z^2$ as $-y^2z^2$ etc.

For example: In the following expressions identify the terms, factors and coefficients.

$$(4x + 3y), 3xy^2 - 4x, 3p^2q + 7pq - 8pq^2$$

Expression	Terms	Factors	Coefficients
$4x + 3y$	$4x$	$4, x$	4
	$3y$	$3, y$	3
$3xy^2 - 4x$	$3xy^2$	$3, x, y, y$	3
	$-4x$	$-4, x$	-4
$3p^2q + 7pq - 8pq^2$	$3p^2q$	$3p, p, q$	3
	$7pq$	$7, p, q$	7
	$-8pq^2$	$-8, p, q, q$	-8

Like and Unlike Terms: In any algebraic expression, terms which have same variable(s) factor(s) are called like terms. Terms which have different variable(s) factors are called unlike terms. Take example of expression $3y^2 + 2x - 2y^2 + 5$, in this expression, factors of $3y^2$ are 3, y and y , factors of $-2y^2$ are -2 , y and y . Thus their variables factors are same so $3y^2$ and $-2y^2$ are like terms whereas $3y^2$ and $2x$ are unlike terms because their variable factors are different, similarly $2x$ and 5 are also unlike terms.

Types of Algebraic Expressions:

- An algebraic expression that has only one term is called a **monomial**. For example, x , $4x^2y$, $-3p^2q^2$ etc.
- An algebraic expression that contains two unlike terms is called a **binomial**. For example, $x + y$, $3x + 4y$, $x - 10$, $-y - 5$. etc.
- An algebraic expression that contains three unlike terms is called a **trinomial**. For example, $a + b + 5$, $x^2 - y^2 + 6$, $x^2y + xy^2 + xy$ etc. $2y + 10 - y$ is not a trinomial as $2y$ and $-y$ are like terms.

ADDITION OF ALGEBRAIC EXPRESSIONS

To add two algebraic expressions, we collect different groups of like terms and find the sum of like terms in each group.

ILLUSTRATION : 1

Add; $7x^2 - 4x + 5$, $-3x^2 + 2x - 1$ and $5x^2 - x + 9$.

SOLUTION :

$$\begin{aligned}
 & (7x^2 - 4x + 5) + (-3x^2 + 2x - 1) + (5x^2 - x + 9) \\
 &= 7x^2 - 4x + 5 - 3x^2 + 2x - 1 + 5x^2 - x + 9 \\
 &= 7x^2 - 3x^2 + 5x^2 - 4x + 2x - x + 5 - 1 + 9 \\
 &= (7 - 3 + 5)x^2 + (-4 + 2 - 1)x + (5 - 1 + 9) \\
 &= 9x^2 - 3x + 13
 \end{aligned}$$

SUBTRACTION OF ALGEBRAIC EXPRESSIONS

In order to subtract an algebraic expression from another, we change the signs (from $-$ to $+$ or $+$ to $-$) of all the terms of the expression which is to be subtracted and then the two expressions are added.

ILLUSTRATION : 2

Subtract $(2x^3 - 4x^2 + 3x + 5)$ from $(4x^3 + x^2 + x + 6)$

SOLUTION :

$$\begin{aligned} & (4x^3 + x^2 + x + 6) - (2x^3 - 4x^2 + 3x + 5) \\ &= 4x^3 + x^2 + x + 6 - 2x^3 + 4x^2 - 3x - 5 \\ &= 2x^3 + 5x^2 - 2x + 1 \end{aligned}$$

MULTIPLICATION OF ALGEBRAIC EXPRESSIONS

In the multiplication of algebraic expressions, we shall be using the following rules:

- (1) The product of two factors with like signs is positive and the product of two factors with unlike signs is negative.
 - (i) $(+) \times (+) = +$
 - (ii) $(-) \times (-) = +$
 - (iii) $(+) \times (-) = -$
 - (iv) $(-) \times (+) = -$
- (2) If x is any variable and a, b are positive integers, then

$$x^a \times x^b = x^{a+b}$$
 For example: $x^4 \times x^3 = x^{4+3} = x^7$

MULTIPLICATION OF TWO OR MORE MONOMIALS

While multiplying two or more monomials, we have the following two rules:

- (1) Multiply the coefficients of given monomials to find the coefficient of the product of monomials.
- (2) Multiply the variable parts in the given monomials to find the variable part of the product of monomials.

ILLUSTRATION : 3

Multiply : (i) $3x^2y^2z$ by $4xy^2z^2$ (ii) $\frac{-8}{5}x^2yz^3$ by $\frac{-3}{4}xy^2z$

SOLUTION :

$$\begin{aligned} \text{(i)} \quad & 3x^2y^2z \times 4xy^2z^2 = (3 \times 4) \times (x^2 \times x) \times (y^2 \times y^2) \times (z \times z^2) = 12x^3y^4z^3 \\ \text{(ii)} \quad & \frac{-8}{5}x^2yz^3 \times \frac{-3}{4}xy^2z = \left(\frac{-8}{5} \times \frac{-3}{4} \right) \times (x^2 \times x) \times (y \times y^2) \times (z^3 \times z) = \frac{6}{5}x^3y^3z^4 \end{aligned}$$

ILLUSTRATION : 4

Find the product of $4s^2t$, $3s^3t^3$, $2st^4$ and (-2)

SOLUTION :

$$\begin{aligned} & (4s^2t) \times (3s^3t^3) \times (2st^4) \times (-2) \\ &= (4 \times 3 \times 2 \times (-2)) \times (s^2 \times s^3 \times s \times t \times t^3 \times t^4) \\ &= -48s^6t^8 \end{aligned}$$

MULTIPLICATION OF A MONOMIAL BY A BINOMIAL

To multiply a binomial by a monomial, we shall use the following properties:

- (i) $P \times (Q + R) = (P \times Q) + (P \times R)$
- (ii) $P \times (Q - R) = (P \times Q) - (P \times R)$

ILLUSTRATION : 5

Multiply xz by $(x^2 + y^2)$

SOLUTION :

$$xz \times (x^2 + y^2) = (xz \times x^2) + (xz \times y^2) = x^3z + xzy^2$$

ILLUSTRATION : 6

Multiply $\frac{-3a^2b}{5}$ by $\left(\frac{2a}{3} - b\right)$

SOLUTION :

$$\left(\frac{-3a^2b}{5} \times \frac{2a}{3}\right) - \left(\frac{-3a^2b}{5} \times b\right) = \frac{-2a^3b}{5} - \left(\frac{-3a^2b^2}{5}\right) = \frac{-2a^3b}{5} + \frac{3a^2b^2}{5} = \frac{3a^2b^2}{5} - \frac{2a^3b}{5}$$

MULTIPLICATION OF A MONOMIAL BY A TRINOMIAL

To multiply a monomial by a trinomial, we use the following property:

$$P \times (Q + R + S) = (P \times Q) + (P \times R) + (P \times S)$$

ILLUSTRATION : 7

Multiply $2z$ ($z - x - y$)

SOLUTION :

$$2z(z - x - y) = (2z \times z) - (2z \times x) - (2z \times y) = 2z^2 - 2xz - 2zy$$

MULTIPLICATION OF TWO BINOMIALS

In order to multiply two binomials, we will use the distributive property of multiplication.

Consider $(a + b)(c + d)$.

Step 1 : Apply distributive property and we get; $a(c + d) + b(c + d)$

Step 2 : Multiply further to get : $ac + ad + bc + bd$.

ILLUSTRATION : 8

Multiply $(2x + 3y)$ and $(4x - 5y)$

SOLUTION :

$$(2x + 3y)(4x - 5y) = 2x(4x - 5y) + 3y(4x - 5y) = 8x^2 - 10xy + 12xy - 15y^2 = 8x^2 + 2xy - 15y^2$$

ALGEBRAIC IDENTITIES

Consider the equality $(x + 2)(x + 3) = x^2 + 5x + 6$

Let us evaluate both sides of this equality for some value of variable x say $x = 4$

$\therefore$ For $x = 4$

$$\text{LHS} \Rightarrow (x + 2)(x + 3) = (4 + 2)(4 + 3) = 6 \times 7 = 42$$

$$\text{RHS} \Rightarrow (4)^2 + 5 \times 4 + 6 = 16 + 20 + 6 = 42$$

So for $x = 4$

LHS = RHS

Let us calculate LHS & RHS for $x = -3$

$$\text{LHS} \Rightarrow (-3 + 2)(-3 + 3) = 0$$

$$\text{RHS} \Rightarrow (-3)^2 + 5(-3) + 6 = 9 - 15 + 6 = 0$$

$\therefore$ for $x = -3$, LHS = RHS

If we take any value of variable x , we find that LHS = RHS

Such an equality which is true for every value of variable present in it is called an identity thus $(x + 2)(x + 3) = x^2 + 5x + 6$ is an identity.

Identities differ from equations in the following manners. An equation is a statement of equality of two algebraic expression involving one or more variable. An equation is true for certain values of the variable.

For example:

$$\begin{aligned} 4x + 3 &= x - 3 \dots\dots\dots(1) \\ \Rightarrow 3x &= -6 \Rightarrow x = -2 \end{aligned}$$

Thus eqⁿ (1) is true only for $x = -2$, no other value of x satisfy eqⁿ (1).

Standard Identities:

$$(i) (a+b)^2 = a^2 + 2ab + b^2$$

$$(ii) (a-b)^2 = a^2 - 2ab + b^2$$

$$(iii) (a^2 - b^2) = (a+b)(a-b)$$

$$(iv) (x+a)(x+b) = x^2 + (a+b)x + ab$$

ILLUSTRATION : 9

Simplify the following:

$$(i) (2a+b)^2$$

$$(ii) (3x-4y)^2$$

SOLUTION :

$$(i) (2a+b)^2 = (2a)^2 + (b)^2 + 2(2a)(b) \\ = 4a^2 + b^2 + 4ab$$

$$(ii) (3x-4y)^2 = (3x)^2 + (4y)^2 - 2(3x)(4y) \\ = 9x^2 + 16y^2 - 24xy$$

CONNECTING TOPIC

Some More Identities

We have dealt with identities involving squares. Now we will see how to handle identities involving cubes.

$$(i) (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$(ii) (a+b)^3 = a^3 + b^3 + 3ab(a+b)$$

$$(iii) (a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

ILLUSTRATION : 10

Simplify the following:

$$(i) (2a-5b)^3$$

$$(ii) (3x+4y-z)^2$$

$$(iii) (28)^3 + (-15)^3 + (-13)^3$$

SOLUTION :

$$(i) (2a-5b)^3$$

$$\text{Here, } (a-b)^3 = a^3 - b^3 - 3ab(a-b)$$

$$\therefore (2a-5b)^3 = (2a)^3 - (5b)^3 - 3(2a)(5b)(2a-5b)$$

$$= 8a^3 - 125b^3 - 30ab(2a-5b)$$

$$= 8a^3 - 125b^3 - 60a^2b + 150ab^2$$

$$(ii) (3a+4y-z)^2$$

$$\text{Here, } (a+b+c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$$

$$(3x+4y-z)^2 = (3x)^2 + (4y)^2 + (-z)^2 + 2(3x)(4y) + 2(4y)(-z) + 2(-z)(3x)$$

$$= 9x^2 + 16y^2 + z^2 + 24xy - 8yz - 6xz$$

$$(iii) (28)^3 + (-15)^3 + (-13)^3$$

$$\text{Let } a = 28, b = -15, c = -13$$

$$\text{Here we see that } a+b+c = 28 + (-15) + (-13) = 0$$

$$\therefore a^3 + b^3 + c^3 = 3abc$$

$$\therefore (28)^3 + (-15)^3 + (-13)^3 = 3(28)(-15)(-13)$$

$$(28)^3 + (-15)^3 + (-13)^3 = 16380$$

MISCELLANEOUS

SOLVED EXAMPLES

1. Evaluate: $\left(\frac{x}{2} + \frac{y}{3}\right)\left(\frac{x}{2} + \frac{y}{3}\right)$

$$\begin{aligned}\text{Sol. } \left(\frac{x}{2} + \frac{y}{3}\right)\left(\frac{x}{2} + \frac{y}{3}\right) &= \left(\frac{x}{2} + \frac{y}{3}\right)^2 = \left(\frac{x}{2}\right)^2 + 2\left(\frac{x}{2}\right)\left(\frac{y}{3}\right) + \left(\frac{y}{3}\right)^2 \\ &\Rightarrow \left(\frac{x}{2} + \frac{y}{3}\right)\left(\frac{x}{2} + \frac{y}{3}\right) = \frac{x^2}{4} + \frac{xy}{3} + \frac{y^2}{9}\end{aligned}$$

2. Evaluate: $(2x^2 - 5y^2)^2$

$$\begin{aligned}\text{Sol. } (2x^2 - 5y^2)^2 &= (2x^2)^2 - 2(2x^2)(5y^2) + (5y^2)^2 \\ (2x^2 - 5y^2)^2 &= 4x^4 - 20x^2y^2 + 25y^4\end{aligned}$$

3. Evaluate: $(1 - 6x)(1 + 6x) + (2 + x)(2 - x)$

$$\begin{aligned}\text{Sol. } (1 - 6x)(1 + 6x) + (2 + x)(2 - x) &= \{1^2 - (6x)^2\} + \{2^2 - x^2\} = 1 - 36x^2 + 4 - x^2 \\ (1 - 6x)(1 + 6x) + (2 + x)(2 - x) &= 5 - 37x^2\end{aligned}$$

4. Evaluate: (a) $(204)^2$ (b) $(148)^2$

$$\begin{aligned}\text{Sol. (a) } (204)^2 &= (200 + 4)^2 = 200^2 + 2 \times 200 \times 4 + 4^2 \\ &= (204)^2 = 41616\end{aligned}$$

$$\begin{aligned}\text{(b) } (148)^2 &= (148)^2 = (150 - 2)^2 = (150)^2 - 2 \times 150 \times (148)^2 = (150 - 2)^2 = (150)^2 - 2 \times 150 \times 2 + 2^2 \\ &= 22500 - 600 + 4 \\ (148)^2 &= 21904\end{aligned}$$

5. If $x + y = 9$ and $xy = 16$, find (i) $x^2 + y^2$ (ii) $(x - y)^2$

$$\text{Sol. (i) } (x + y) = 9$$

$$\Rightarrow (x + y)^2 = 9^2 = 81 \text{ [By squaring both sides]}$$

$$\Rightarrow x^2 + 2xy + y^2 = 81$$

$$\Rightarrow x^2 + y^2 + 2 \times 16 = 81$$

$$\Rightarrow x^2 + y^2 = 81 - 32$$

$$\Rightarrow x^2 + y^2 = 49$$

$$\text{(ii) } (x - y)^2 = x^2 - 2xy + y^2$$

$$= x^2 + y^2 - 2xy = 49 - 2 \times 16$$

$$= 49 - 32$$

$$(x - y)^2 = 17$$

6. If $\left(x - \frac{1}{x}\right) = 9$, find $\left(x + \frac{1}{x}\right)^2$.

Sol. Given $x - \frac{1}{x} = 9$

$$\left(x - \frac{1}{x}\right)^2 = 81 \quad [\text{By squaring both sides}]$$

$$x^2 - 2x \cdot \frac{1}{x} + \frac{1}{x^2} = 81$$

$$\Rightarrow x^2 + \frac{1}{x^2} - 2 = 81$$

$$x^2 + \frac{1}{x^2} = 83 \quad \dots\dots\dots (1)$$

$$\text{Now, } \left(x + \frac{1}{x}\right)^2 = x^2 + 2 \cdot \cancel{x} \cdot \frac{1}{\cancel{x}} + \frac{1}{x^2} = x^2 + \frac{1}{x^2} + 2$$

$$= 83 + 2 \quad [\text{from (1)}]$$

$$\left(x + \frac{1}{x}\right)^2 = 85$$

7. Evaluate the following products without direct multiplication. (a) 103×107 (b) 104×96

Sol. (a) $103 \times 107 = (100 + 3)(100 + 7) = (100)^2 + (3 + 7) \times 100 + 3 \times 7 \quad \left[\text{using } (x+a)(x+b) = x^2 + (a+b)x + ab \right]$

$$= 10000 + 1000 + 21 = 11021$$

(b) $104 \times 96 = (100 + 4)(100 - 4)$

$$= 100^2 - 4^2 \quad [\text{using } (x+a)(x-a) = x^2 - a^2]$$

$$= 10000 - 16 = 9984$$

8. Simplify:

(i) $(2x-3)(x^3-2x^2+5x-7)$

(ii) $(3x-2)(2x-3) + (5x-3)(x+1)$

Sol. (i) $(2x-3)(x^3-2x^2+5x-7)$

$$= 2x(x^3-2x^2+5x-7) - 3(x^3-2x^2+5x-7)$$

$$= 2x^4 - 4x^3 + 10x^2 - 14x - 3x^3 + 6x^2 - 15x + 21$$

$$= 2x^4 - 4x^3 - 3x^3 + 10x^2 + 6x^2 - 14x - 15x + 21$$

$$= 2x^4 - 7x^3 + 16x^2 - 29x + 21$$

(ii) $(3x-2)(2x-3) + (5x-3)(x+1)$

$$= [3x(2x-3) - 2(2x-3)] + [5x(x+1) - 3(x+1)]$$

$$= (6x^2 - 9x - 4x + 6) + (5x^2 + 5x - 3x - 3)$$

$$= 6x^2 - 13x + 6 + 5x^2 + 2x - 3$$

$$= 11x^2 - 11x + 3$$

SOLVED EXAMPLES BASED ON CONNECTING TOPICS

9. Expand : $(2x - y + z)^2$

$$\text{Sol. } (2x - y + z)^2 = (2x)^2 + (-y)^2 + (z)^2 + 2(2x)(-y) + 2(-y)(z) + 2(2x)(z)$$

$$\left[\text{using } (a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca \right]$$

$$\therefore (2x - y + z)^2 = 4x^2 + y^2 + z^2 - 4xy - 2yz + 4zx$$

10. Evaluate: 103^3

$$\text{Sol. } (103)^3 = (100 + 3)^3$$

$$\text{Here, } a = 100, b = 3$$

$$\text{Using } (a + b)^3 = a^3 + 3ab(a + b) + b^3$$

$$= (100)^3 + 3 \times 100 \times 3(100 + 3) + (3)^3 = 1000000 + 900 \times 103 + 27$$

$$= 1092727$$

11. $(x - 2y)^3 + (2y - 3z)^3 + (3z - x)^3$ is

$$\text{Sol. Let } x - 2y = a, 2y - 3z = b \text{ and } 3z - x = c$$

$$\therefore a + b + c = x - 2y + 2y - 3z + 3z - x = 0$$

$$\Rightarrow a^3 + b^3 + c^3 = 3abc$$

$$\text{Hence } (x - 2y)^3 + (2y - 3z)^3 + (3z - x)^3$$

$$= 3(x - 2y)(2y - 3z)(3z - x)$$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- _____ are a combination of terms connected by the operations of addition, subtraction, multiplication or division.
- The numerical factor of the term is called _____.
- In expression $2x^2 + 4x$, the coefficient of x^2 is _____ and coefficient of x is _____.
- $5x + 4y$ is an expression having _____ term.
- An algebraic expression is called a _____ if there is only one term in it.
- Trinomial is an algebraic expression with _____ terms.
- $(-5ab^2c) \times (3a^3bc^2d) = \underline{\hspace{2cm}}$.
- Each term in an algebraic expression is a product of one or more numbers, numerical. These numbers are called the _____ of that term.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- In the term $5ab$, 5, a and b are the factors of this term.
- $(3y^2 + 3xyz) - (2x^2 - 3y^2 + 4z^2 - xyz)$ is $-2x^2 + y^2 - z^2 + xyz$.
- A constant term contains only variables.
- Only like terms can be added or subtracted.
- The terms of the expression having no literal factors is called a constant term.
- $3xy, 2xy, 9yx, 6xy$ are examples of like terms.
- An expression which contains two terms is called binomial.
- The product of $-\frac{6}{7}xy^3$ and $-\frac{4}{3}x^2y$ is $+\frac{8}{7}x^3y^4$.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | 1. Column-I | Column-II |
|---|--------------------------|
| (A) $\left(\frac{2}{3}a^2b\right)\left(\frac{-9}{4}ab^2\right)$ | (p) $\frac{-4}{9}p^5q^5$ |
| (B) $(-pq)(-2.3p^2q^2)$
$(-0.1p^2q)$ | (q) $0.225a^4b^4c$ |

- (C) $(-1.5a^2b)(0.3ab^2)$
 $(-0.5abc)$
- (D) $\left(\frac{-3}{7}p^3q^2\right)\left(\frac{-14}{9}pq^2\right)$
 $\left(\frac{-2}{3}pq\right)$
- (r) $\frac{-3}{2}a^3b^3$
- (s) $-0.23p^5q^4$

2. Column-I (Expression)

- (A) $(2x^2 + 3y^2)(-2x^2 - 3y^2)$
- (B) $(x^3 + 5)(x - 5) + 5$
- (C) $6x^2 + (x + 3y)(3 - 2y)$
- (D) $(x + 3y)(2x + 3y) + (x + 5y)$
 $+ (x - y)$

Column-II (Solution)

- (p) $-4x^4 - 12x^2y^2 - 9y^4$
- (q) $6x^2 + 3x - 2xy + 9y - 6y^2$
- (r) $2x^2 + 9xy + 9y^2 + 2x + 4y$
- (s) $x^4 - 5x^3 + 5x - 20$

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

- Find the product of $\frac{-3}{4}x^4y^5z^8$ and $\frac{-16}{9}xy^2z^3$.
- Multiply
 - $(a^2b^2 - 2ab + 4)$ by $(a + 2)$
 - $(x^2 + y^2 + z^2)$ by $(x + y + z)$
- Find the volume of the rectangular boxes with following length, breadth and height:

	Length	Breadth	Height
(i)	$2ax$	$3by$	$5cz$
(ii)	m^2n	n^2p	p^2m
(iii)	$2q$	$4q^2$	$8q^3$
- Multiply : $\{2m + (-n)\}$ by $\{-3m + (-5)\}$
- If $x^2 + \frac{1}{x^2} = 53$, find the value of $x - \frac{1}{x}$
- $\left(p + \frac{1}{3}\right)\left(p - \frac{5}{3}\right) = \underline{\hspace{2cm}}$
- The sum of two binomials is $5x^2 - 6x$. If one of the binomial is $3x^2 - 2x$. What is the other binomial?
- What is the value of $3(x^2 - 4x)$ when $x = 5$?
- Simplify each of the following expressions:
 - $12x^2 - 12x(x - 1) + x(7 + 3x)$
 - $pq(p - q) - 6p(p + q) + 2pq(2p - 2q)$
- The value of $16.1^2 - 8.9^2$ is _____.

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- Subtract
 - $a + b + c$ from $a^2 + b^2 + c^2$
 - $\frac{2}{3}a^2b - \frac{1}{9}ab^2 + \frac{2}{5}ab - 1$ from $-a^2b + 2$.
- Add :
 - $a^2 + b^2 - 3c^2$,
 $3b^2 + c^2$,
 $\frac{2}{3}c^2 - \frac{4}{5}b^2$ and $a^2 + b^2 + c^2$
 - $3ax - 4ax^2 + 1, 5 + 9ax^2 - 4ax$.
- Determine the literal part in the product of the monomials
 $\frac{187}{395}x^3y^2z^7, \frac{-2691}{1297}x^7y$ and $\frac{39251}{3931}xy^2z^3$
- Evaluate :
 - $(-3.5a^2b) \times (2ab^2) + (-7ab) \times (-a^2b^2)$
 - $(5ab) \times (-2ab^2) \times (-2a) + (2a^2b^2) \times \left(\frac{-1}{2}a\right) \times (-4b)$
- The sides of a rectangle are given by $x^2 + 3y^2$ and $x^3 - y^2$. Find its perimeter.
- The sides of a triangle are given by :
 $3x^2 - y^2, 4x^2 - 7xy + 4y^2, -3x^2 + 7xy + 8y^2$. Find its perimeter.
- Answer the following questions :-
 - How much smaller is $2x^3 - 2x + 5$ than $7 + 5x^2 + 3x$?
 - How much larger is $8x^2 - 9y^2$ than $3x^2 - 2y^2$?
 - What should be taken away from $3x^2 + 4x + 1$ to get $2x^2 - 4x + 35$?
- Subtract $2x^2 - 4y^2 + 3xy - 5$ from the sum of $-2y^2 + 5x^2 + 9$ and $7xy - 3x^2 + 11$.
- If $a = 2p - 5q + 7, b = -3p + 2q - 1$ and $c = p - q - 3$ then find.
 - $b - a - c$
 - $a - b + c$
 In terms of p and q
- Identify terms, factors and coefficients for each of the following expressions.
 - $8x^2y - 5zy$
 - $\frac{x}{3} - \frac{y}{2} + 1$
 - $0.5x^2 - 3y^2$
 - $z^2 - z + 1$
- Which of the following are monomials, binomials, or Trinomials?
 - 3
 - $x + y$
 - $x^3 + y^2 + 3$
 - $x^2 + y^2 + z^2 + 10$
- Simplify the following using identities
 - $\frac{58^2 - 42^2}{16}$
 - $1.73 \times 1.73 - 0.27 \times 0.27$

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

- If $a = \frac{3}{2}x^2 - \frac{1}{4}y^2 + \frac{5}{3}xy - 1$ and
 $b = \frac{-1}{2}x^2 + \frac{5}{3}y^2 - xy + \frac{5}{3}$ then calculate $a + b$ and $a - b$.
- Find the value of $\left(\frac{-5}{2}x^3y^2\right) \times \left(\frac{2}{-25}xy^2\right)$.
- Simplify and state the degree of the polynomial.
 - $6m(2mn - n^2) - 3(m^2n - 19mn^2) + 6n(-5mn - 7m^2)$
 - $4x(5x^2 - 2xy + y^2) - 9x(2x^2 + 6y^2 - xy)$
 $+ 3x(x^2 - 5xy - y^2)$
 - $5s(3s^2 - 7s + 2) + 2s(5s^2 + s - 6) + 9s(2s^2 + 6s - 4)$
- Evaluate 19.8×20.2
- The expression
$$\frac{bx(a^2x^2 + 2a^2y^2 + b^2y^2) + ay(a^2x^2 + 2b^2x^2 + b^2y^2)}{(ax + by)^2}$$
- By how much is the sum of $(a - 5)(a - 2)$ and $(a^2 + 6a + 7)$ is greater than $-a^2 - 2a + 1$?
- $x^2 + \frac{1}{x^2} = 25$, find the values of each of the following :
 - $x + \frac{1}{x}$
 - $x - \frac{1}{x}$
- Find the value of x , if
 - $6x = 23^2 - 17^2$
 - $4x = 98^2 - 88^2$
- Show that:
 - $(3x + 7)^2 - 84x = (3x - 7)^2$
 - $(9p - 5q)^2 + 180pq = (9p + 5q)^2$
 - $\left(\frac{4}{3}m - \frac{3}{4}n\right)^2 + 2mn = \frac{16}{9}m^2 + \frac{9}{16}n^2$
 - $(4pq + 3q)^2 - (4pq - 3q)^2 = 48pq^2$
 - $(a - b)(a + b) + (b - c)(b + c) + (c - a)(c + a) = 0$
- Using identities, evaluate:
 - 71^2
 - 99^2
 - 998^2
 - 5.2^2
 - 78×82
 - 8.9^2
 - 1.05×9.5
- Using $(x + a)(x + b) = x^2 + (a + b)x + ab$, find
 - 103×104
 - 5.1×5.2
 - 103×98
 - 9.7×9.8

2 EXERCISE

Text Book Exercise

- Add: $7xy + 5yz - 3zx$, $4yz + 9zx - 4y$, $-3xz + 5x - 2xy$.
- Subtract $5x^2 - 4y^2 + 6y - 3$ from $7x^2 - 4xy + 8y^2 + 5x - 3y$.
- Find the volume for given dimensions :

	length	breadth	height
(i)	$2ax$	$3by$	$5cz$
(ii)	m^2n	n^2p	p^2m
(iii)	$2q$	$4q^2$	$8q^3$

- Simplify the expressions and evaluate them as directed:

- $x(x - 3) + 2$ for $x = 1$,
- $3y(2y - 7) - 3(y - 4) - 63$ for $y = -2$

- Add:

- $5m(3 - m)$ and $6m^2 - 13m$
- $4y(3y^2 + 5y - 7)$ and $2(y^3 - 4y^2 + 5)$

- Subtract $3pq(p - q)$ from $2pq(p + q)$.

- Simplify $(a + b)(2a - 3b + c) - (2a - 3b)c$.

- (i) $(a + b)(c - d) + (a - b)(c + d) + 2(ac + bd)$

- $(x + y)(2x + y) + (x + 2y)(x - y)$

- $(x + y)(x^2 - xy + y^2)$

- $(1.5x - 4y)(1.5x + 4y + 3) - 4.5x + 12y$

- $(a + b + c)(a + b - c)$

- Simplify:

- $(a^2 - b^2)^2$

- $(7m - 8n)^2 + (7m + 8n)^2$

- $(2.5p - 1.5q)^2 - (1.5p - 2.5q)^2$

- $(ab + bc)^2 - 2ab^2c$

- Show that:

- $(3x + 7)^2 - 84x = (3x - 7)^2$

- $(9p - 5q)^2 + 180pq = (9p + 5q)^2$

- $\left(\frac{4}{3}m - \frac{3}{4}n\right)^2 + 2mn = \frac{16}{9}m^2 + \frac{9}{16}n^2$

- $(4pq + 3q)^2 - (4pq - 3q)^2 = 48pq^2$

- $(a - b)(a + b) + (b - c)(b + c) + (c - a)(c + a) = 0$

- Using $a^2 - b^2 = (a + b)(a - b)$, find

- $51^2 - 49^2$

- $(1.02)^2 - (0.98)^2$

- $153^2 - 147^2$

- $12.1^2 - 7.9^2$

- Using $(x + a)(x + b) = x^2 + (a + b)x + ab$, find

- 103×104

- 5.1×5.2

- 103×98

- 9.7×9.8

Exemplar Questions

- Simplify: $-pqr(p^2 + q^2 + r^2)$

- Simplify: $(px + qy)(ax - by)$

- Evaluate using suitable identities: 497×505

- Evaluate using suitable identities: 2.07×1.93

- Find the value of x , if $10000x = (9982)^2 - (18)^2$

- By using suitable identity, evaluate $x^2 + \frac{1}{x^2}$, if $x + \frac{1}{x} = 5$.

- Simplify: $(a - b)(a^2 + b^2 + ab) - (a + b)(a^2 + b^2 - ab)$

- Simplify: (i) $(x^2 + y^2)(x^2 - y^2)$

- $(a^2 + b^2)^2$

- If $x - y = 13$ and $xy = 28$, then find $x^2 + y^2$.

- If $m - n = 16$ and $m^2 + n^2 = 400$, then find mn .

- What should be added to $4c(-a + b + c)$ to obtain $3a(a + b + c) - 2b(a - b + c)$?

- Find the value of a , if

$$8a = 35^2 - 27^2$$

HOTS Questions :

- How much should $3xy - 4a^2 + 5b^2 + 2$ be increased to get $-5 + 4a^2 + 2b^2 - 7xy$?

- find the value of $(5a^6) \times (-10ab^2) \times (-2.1a^2b^3)$ for $a = 1$ and $b = \frac{1}{2}$.

- Simplify: (i) $15a^2 - 6a(a - 2) + a(3 + 7a)$

- $x^2(1 - 3y^2) + x(xy^2 - 2x) - 3y(y - 4x^2y)$

- Simplify: $(3x - 2)(x - 1)(3x + 5)$

- Simplify:

$$2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = (a - b)^2 + (b - c)^2 + (c - a)^2$$

3 EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The expanded form of $(x + y)^2$ is a _____.
(a) monomial (b) binomial
(c) trinomial (d) cubic
- The product of $(x^2 + 3x + 5)$ and $(x^2 - 1)$ is
(a) $x^4 + 3x^3 - 4x^2 - 3x - 5$
(b) $x^4 + 3x^3 + 4x^2 - 3x - 5$
(c) $x^4 + 3x^3 + 4x^2 + 3x - 5$
(d) None of these
- Evaluate : $(2x^2 - 5y^2)^2$
(a) $4x^2 + 20xy + 25y$
(b) $4x^2 + 20x^2y^2 + 25y^4$
(c) $4x^4 - 20x^2y^2 + 25y^4$
(d) $(4x^4 - 25y^4)(4x^4 + 25y^4)$
- Subtract : $3a - 3b + c$ from $4a + 5b - 3c$
(a) $2a + 3b + c$ (b) $a + 2b + 3c$
(c) $a + 8b - 4c$ (d) $-5x^2 + 5x + 5$
- Find the product of $5m^2n$, $-3mnp$ and $-5n^2p$.
(a) $75m^3n^3p^2$ (b) $75m^3n^4p^2$
(c) $75mn^3p^3$ (d) $75m^3np^2$
- Simplify $3x^2y^2(5x^2 - 4xy + 6y^2)$
(a) $15x^4y^2 - 12x^3y^3 + 18y^4x^2$
(b) $15x^2y^2 - 12x^3y^3 + 18y^2x^2$
(c) $15x^2y^3 - 12x^2y^2 + 6y^2y^2$
(d) $15xy^3 - 12x^3y^3 + 18x^2y^4$
- What must be subtracted from $3a^2 - 6ab - 3b^2 - 1$ to get $4a^2 - 7ab - 4b^2 + 1$?
(a) $-a^2 + ab + b^2 - 2$ (b) $a^2 + ab + b^2 + 2$
(c) $a^2 - ab - b^2 + 2$ (d) $a^2 - ab - 4b^2 - 2$
- The value of the product $\left(3 + \frac{5}{x}\right)\left(9 - \frac{15}{x} + \frac{25}{x^2}\right)$ at $x = 1$ is
(a) 150 (b) 148
(c) 152 (d) None of these
- The value of $(0.9)^2 - (0.6)^2$ is
(a) 0.54 (b) -0.54
(c) -0.45 (d) 0.45
- The value of $(a + b)^2 + (a - b)^2$ is:
(a) $2a + 2b$ (b) $2a - 2b$
(c) $2a^2 + 2b^2$ (d) $2a^2 - 2b^2$
- The value of $(a + b)^2 - (a - b)^2$ is:
(a) $4ab$ (b) $-4ab$
(c) $2a^2 + 2b^2$ (d) $2a^2 - 2b^2$
- Subtract $4p^2 - 2q + 7r^3 - 3$ from $3q + 7p^2 - 2r^3 + 4$.
(a) $p^2 + 2q + 5r^3 + 1$ (b) $11p^2 + q + 5r^3 + 1$
(c) $-3p^2 - 5q + 9r^3 - 7$ (d) $3p^2 + 5q - 9r^3 + 7$
- Find the product of $3a^2 - 4ax + x^2$ by $5x^2 - 2ax$
(a) $14a^3x + 23a^2x^2 - 22ax^3 + 5x^4$
(b) $14a^2x + 23a^3x^2 - 22ax^3 + 5x^4$
(c) $16a^3x + 25a^2x^2 - 22x^3 + 5x^4$
(d) $-6a^3x + 23a^2x^2 - 22ax^3 + 5x^4$
- The sum of two binomials is $5x^2 - 6x$. If one of the binomial is $3x^2 - 2x$. What is the other binomial?
(a) $2x^2 - 4x$ (b) $2x^2 - 8x$
(c) $8x^2 - 4x$ (d) $8x^2 - 8x$
- The cost of a notebook is ₹ $3a^2 - 4ab + 6b^2$. How much does $5a^2b^2$ notebooks cost?
(a) ₹ $15a^4b^2 - 20a^3b^3 + 30a^2b^4$
(b) ₹ $15a^4b^2 + 20a^3b^3 + 30a^2b^4$
(c) ₹ $15a^4b^2 - 20a^3b^3 - 30a^2b^4$
(d) ₹ $15a^4b^2 + 20a^3b^3 - 30a^2b^4$
- Simplify : $(0.8m + 1.1n)(0.8m - 1.1n)$
(a) $0.64m^2 + 1.12n^2$ (b) $0.64m^2 + 1.21n^2$
(c) $0.64m^2 - 1.21n^2$ (d) $0.64m^2 - 11.2n^2$
- By how much is the sum of $(a - 5)$ $(a - 2)$ and $a^2 + 6a + 7$ is greater than $-a^2 - 2a + 1$?
(a) $a^2 - a + 16$ (b) $a^2 - a - 16$
(c) $3a^2 + a + 16$ (d) $2a^2 + a + 16$

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following expressions are binomials and trinomials?
 (a) $\frac{3}{x^2}, 5x$ (b) $x + \frac{2}{x}, x^2 + 2x - 5$
 (c) $x^2 + 2x - 5, 9x^3 + 5x$ (d) $\frac{7x}{5} - \frac{9}{8}, 3x^2 - 4x + 5$
- Which of the following terms are like terms?
 (a) $8ab, -9ab^2$ (b) $3x^2y, -4yx^2$
 (c) $8xy^2, -11x^3y$ (d) $\frac{7}{4}xy, -\frac{5}{3}xy$
- Which of the following terms are unlike terms?
 (a) $\frac{3}{4}a^2bx, \frac{3}{4}ab^2x, \frac{3}{4}abx^2$
 (b) $-9xy, 5$
 (c) $-9x^2, -10x^2, 5x^2$
 (d) None of these
- In the term $\frac{25}{3}a^2bc^3$, which of the following is/are correct?
 (a) Coefficient of $a^2 = \frac{25}{3}bc^3$
 (b) Numerical coefficient = $\frac{25}{3}$
 (c) Coefficient of $c^3 = \frac{25}{3}a^2b$
 (d) Coefficient of $a^2bc^3 = \frac{25}{3}$
- The value of $23^2 - 22^2$ is
 (a) 45 (b) 55
 (c) 23×22 (d) $(23 + 22)(23 - 22)$
- The value of $(60 - 4)(60 - 6)$ is
 (a) 3204 (b) 3024
 (c) $3600 + 600 - 24$ (d) $3600 - 600 + 24$

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE - I

$-\frac{11}{3} - 9x^2y + 6y^2x^3 - \frac{5}{6}x^2y^2 + 15x^2y^2$ is an algebraic expression.

- Write the like terms given in the expression
 (a) $-9x^2y, -\frac{5}{6}x^2y^2$ (b) $-\frac{5}{6}x^2y^2, 15x^2y^2$
 (c) $6y^2x^3, 9x^2y$ (d) $6y^2x^3, -\frac{5}{6}x^2y^2$

- Which is the constant term?

- (a) $-\frac{11}{3}$ (b) $-\frac{5}{6}$
 (c) -9 (d) 6

- What is the coefficient of xy^2 in the term $-\frac{5}{6}x^2y^2$?

- (a) $-\frac{5}{6}$ (b) $\frac{5}{6}$
 (c) $-\frac{5}{6}x$ (d) $\frac{5}{6}x$

PASSAGE - II

$$a^2 - b^2 = (a + b)(a - b)$$

- The value of x if $25x = 536^2 - 136^2$ is
 (a) 10752 (b) 10572
 (c) 10257 (d) 10725
- $(10000)^2 - (9999)^2$ is
 (a) 1999 (b) 19999
 (c) 199999 (d) 999
- $(4000)^2 + (9000)^2 - (3999)^2 - (8999)^2$ equals
 (a) 9989 (b) 24997
 (c) 43998 (d) 25998

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- If **Assertion** is **correct** but **Reason** is **incorrect**.
- If **Assertion** is **incorrect** but **Reason** is **correct**.

- Assertion :** In the expression $3x^2 + 7y^2 - 2xy + 4x^2 + 8xy + 9y^2$, $3x^2, 4x^2$ are like terms, $-2xy, 8xy$ are like terms and $7y^2, 9y^2$ are like terms.
Reason : When the terms have same literal factors they are called unlike terms.
- Assertion :** In an algebraic expression $3x - 5$, 3 and -5 are called constants.
Reason : Constants does not have a fixed value.
- Assertion :** $99 \times 101 = (100 - 1)(100 + 1) = 9999$.
Reason : $(a - b)(a + b) = a^2 - b^2$
- Assertion :** $-8abc + 4bac$ is a binomial.
Reason : If an expression contains two terms it is called a binomial.

Multiple Matching Question :

DIRECTIONS : Following question has three statements (A, B and C) given in Column-I and five statements (p, q, r, s and t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Column-I	Column-II
(A) Monomials	(p) $\sqrt{6ab} - \frac{a^2b}{5}$
(B) Binomials	(q) $a^2 + 2ab + b^2$
(C) Trinomials	(r) p^2q^3y
	(s) $3xy + 4x^2y$
	(t) $\frac{3}{4}xy$

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- Find the value of $(25a^2 + 16b^2 + 9 + 40ab - 24b - 30a)$ at $a = -1$ and $b = 2$.
- The value of $25x^2 + 16y^2 + 40xy$ at $x = 1$ and $y = -1$ is _____.
- $3x - \frac{1}{2}x + \frac{3}{2}x - 4x =$ _____.
- The units digit of 34×36 is _____.
- The value of the product $(4a^2 + 3b)(4a^2 + 3b)$ at $a = 1$, $b = 2$ is k. The value of $k \div 100$ is _____.
- Find the value of $\frac{(997 + 496)^2 - (997 - 496)^2}{997 \times 496}$

4**ADVANCED EXERCISE
BASED ON CONNECTING TOPICS**

DIRECTIONS (Qs. 1-4) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- If $a + b + c = 15$ and $a^2 + b^2 + c^2 = 77$, then the value of $ab + bc + ca$ is
(a) 32 (b) 28
(c) 74 (d) 70
- If $x - \frac{1}{x} = 7$, then the value of $x^3 - \frac{1}{x^3}$ is
(a) 333 (b) 243
(c) 364 (d) None
- Value of $(25)^3 + (-10)^3 + (-15)^3$ is
(a) 11250 (b) -11250
(c) 12150 (d) -12150
- Value of $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$ is
(a) $a^3 + b^3 + c^3 - 3abc$ (b) $a^3 + b^3$
(c) $a^3 - b^3$ (d) $a^4 - b^4$

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- Algebraic expressions.
- coefficient.
- 2, 4
- two
- monomial
- three
- $-15a^4b^3c^3d$.
- factors.

True / False :

- True
- False, $-2x^2 + 6y^2 - 4z^2 + 4xyz$.
- False, A constant term contains only a number.
- True
- True, 2 is a constant as it has no variable.
- True, it has same literal factors.
- True
- True, $-\frac{6}{7}xy^3 \times \left(-\frac{4}{3}x^2y\right) = \frac{8}{7}x^3y^4$

Match the Columns :

- (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p)
- (A) $\rightarrow$ (p); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)
 (A) $(2x^2 + 3y^2)(-2x^2 - 3y^2)$
 $= -(2x^2 + 3y^2)(2x^2 + 3y^2)$
 $= -(2x^2 + 3y^2)^2$
 $= -[(2x^2)^2 + (3y^2)^2 + 2(2x^2)(3y^2)]$
 $= -[4x^4 + 9y^4 + 12x^2y^2]$
 $= -4x^4 - 12x^2y^2 - 9y^4$
 (B) $(x^3 + 5)(x - 5) + 5$
 $= x^3(x - 5) + 5(x - 5) + 5$
 $= x^4 - 5x^3 + 5x - 25 + 5$
 $= x^4 - 5x^3 + 5x - 20$
 (C) $6x^2 + (x + 3y)(3 - 2y)$
 $= 6x^2 + x(3 - 2y) + 3y(3 - 2y)$
 $= 6x^2 + 3x - 2xy + 9y - 6y^2$
 (D) $(x + 3y)(2x + 3y) + (x + 5y) + (x - y)$
 $= x(2x + 3y) + 3y(2x + 3y) + x + 5y + x - y$
 $= 2x^2 + 3xy + 6yx + 9y^2 + 2x + 4y$
 $= 2x^2 + 9xy + 9y^2 + 2x + 4y$

Very Short Answer Questions :

- $\left(-\frac{3}{4}x^4y^5z^8\right) \times \left(-\frac{16}{9}xy^2z^3\right) = \frac{4}{3}x^5y^7z^{11}$
- (i) $(a^2b^2 - 2ab + 4) \times (a + 2)$
 $= a^3b^2 - 2a^2b + 4a + 2a^2b^2 - 4ab + 8$

$$(ii) (x^2 + y^2 + z^2) \times (x + y + z) \\ = x^3 + y^3 + z^3 + x^2y + x^2z + xy^2 + y^2z + xz^2 + yz^2$$

- Volume $= l \times b \times h$

$$(i) l = 2ax, b = 3by, h = 5cz \\ V = 2ax \times 3by \times 5cz = 30abcxyz \\ (ii) l = m^2n, b = n^2p, h = p^2m \\ V = m^2n \times n^2p \times p^2m \\ V = m^3n^3p^3$$

$$(iii) l = 2q, b = 4q^2, h = 8q^3 \\ V = l \times b \times h = 2q \times 4q^2 \times 8q^3 = 64q^6.$$

- $(2m + (-n)) \times (-3m + (-5))$
 $= 2m(-3m - 5) - n(-3m - 5)$
 $= -6m^2 - 10m + 3mn + 5n$

$$5. \left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2 = 53 - 2 = 51 \\ \Rightarrow x - \frac{1}{x} = \sqrt{51}$$

$$6. \left(p + \frac{1}{3}\right)\left(p - \frac{5}{3}\right) = p^2 - \frac{5p}{3} + \frac{p}{3} - \frac{5}{9} \\ = p^2 - \frac{4p}{3} - \frac{5}{9}$$

- Let the other binomial be $b(x)$
 $3x^2 - 2x + b(x) = 5x^2 - 6x$
 $\Rightarrow b(x) = 5x^2 - 6x - 3x^2 + 2x$
 $= 2x^2 - 4x$

$$8. 3(x^2 - 4x) = 3x^2 - 12x = 3(5)^2 - 12(5) \\ = 75 - 60 = 15$$

- (i) $12x^2 - 12x(x - 1) + x(7 + 3x)$
 $= 12x^2 - 12x^2 + 12x + 7x + 3x^2$
 $= 19x + 3x^2$
 (ii) $pq(p - q) - 6p(p + q) + 2pq(2p - 2q)$
 $= p^2q - pq^2 - 6p^2 - 6pq + 4p^2q - 4pq^2$
 $= 5p^2q - 5pq^2 - 6p^2 - 6pq$

$$10. (16.1)^2 - (8.9)^2 = [16.1 + 8.9][16.1 - 8.9] \\ [\text{Using the identity } a^2 - b^2 = (a + b)(a - b)] \\ = 25 \times 7.2 = 180$$

Short Answer Questions :

- (i) $a^2 + b^2 + c^2 - a - b - c$
 (ii) $\frac{1}{9}ab^2 - \frac{5}{3}a^2b - \frac{2}{5}ab + 3$

$$2. \quad (i) \quad (a^2 + b^2 - 3c^2) + (3b^2 + c^2) + \left(\frac{2}{3}c^2 - \frac{4}{5}b^2\right) + (a^2 + b^2 + c^2)$$

$$= 2a^2 + \frac{21}{5}b^2 - \frac{1}{3}c^2$$

$$(ii) \quad (3ax - 4ax^2 + 1) + (5 + 9ax^2 - 4ax) = 5ax^2 - ax + 6$$

$$3. \quad x^{11}y^5z^{10}$$

$$4. \quad (i) \quad -7a^3b^3 + 7a^3b^3 = 0$$

$$(ii) \quad 20a^3b^3 + 4a^3b^3 = 24a^3b^3$$

$$5. \quad \text{Perimeter} = 2[(x^2 + 3y^2) + (x^3 - y^2)] = 2x^3 + 2x^2 + 4y^2$$

$$6. \quad \text{Perimeter} = (3x^2 - y^2) + (4x^2 - 7xy + 4y^2) + (-3x^2 + 7xy + 8y^2) = 4x^2 + 11y^2$$

$$7. \quad (i) \quad 7 + 5x^{2+3x} - (2x^3 - 2x + 5) = -2x^3 + 5x^2 + 5x + 2$$

$$(ii) \quad (8x^2 - 9y^2) - (3x^2 - 2y^2) = 5x^2 - 7y^2$$

$$(iii) \quad (3x^2 + 4x + 1) - (2x^2 - 4x + 35) = x^2 + 8x - 34$$

$$8. \quad 2y^2 + 25 + 4xy$$

$$9. \quad (i) \quad -6p + 8q - 5 \quad (ii) \quad 6p - 8q + 5$$

10.	Expression	Terms	Factors	Coefficients
(i)	$8x^2y - 5zy$	$8x^2y$ $-5zy$	$8, x, y$ $-5, z, y$	8 3
(ii)	$\frac{x}{3} - \frac{y}{z} + 1$	$\frac{x}{3}$ $-\frac{y}{z}$ 1	$\frac{1}{3}, x$ $-\frac{1}{z}, y$ 1	$\frac{1}{3}$ $-\frac{1}{2}$ 1
(iii)	$0.5x^2 - 3y^2$	$0.5x^2$ $-3y^2$	$0.5, x$ $-3, y$	0.5 -3
(iv)	$z^2 - z + 1$	z^2 $-z$ 1	$1, z$ $-1, z$ 1	1 -1 1

11. (i) Monomial (ii) Binomial
(iii) Trinomial (iv) Polynomial

$$12. \quad (i) \quad \frac{58^2 - 42^2}{16} = \frac{(58-42)(58+42)}{16}$$

$$= \frac{16 \times 100}{16} = 100$$

$$(ii) \quad 1.73 \times 1.73 - 0.27 \times 0.27 \\ = (1.73)^2 - (0.27)^2 \\ = (1.73 + 0.27)(1.73 - 0.27) \\ = 2 \times 1.46 = 2.92$$

Long Answer Questions :

$$1. \quad a + b = \frac{3}{2}x^2 - \frac{1}{4}y^2 + \frac{5}{3}xy - 1 + \frac{-1}{2}x^2 + \frac{5}{3}y^2 - xy + \frac{5}{3} \\ = x^2 + \frac{17}{12}y^2 + \frac{2}{3}xy + \frac{2}{3}$$

$$a - b = \frac{3}{2}x^2 - \frac{1}{4}y^2 + \frac{5}{3}xy - 1 + \frac{1}{2}x^2 - \frac{5}{3}y^2 + xy - \frac{5}{3} \\ = 2x^2 - \frac{23}{12}y^2 + \frac{8}{3}xy - \frac{8}{3}$$

$$2. \quad \text{Product} = \frac{1}{5}x^4y^4$$

$$3. \quad (a) \quad 21mn^2 - 33m^2n, \text{ degree} = 3$$

$$(b) \quad 5x^3 - 14x^2y - 53xy^2, \text{ degree} = 3$$

$$(c) \quad 43s^3 + 21s^2 - 38s, \text{ degree} = 3$$

$$4. \quad 399.96$$

$$5. \quad \frac{bx(a^2x^2 + 2a^2y^2 + b^2y^2) + ay(a^2x^2 + 2b^2x^2 + b^2y^2)}{(ax + by)^2}$$

$$= \frac{a^2x^2(ay + bx) + 2abxy(ay + bx) + b^2y^2(bx + ay)}{(ax + by)^2}$$

$$= \frac{(ay + bx)(a^2x^2 + 2abxy + b^2y^2)}{(ax + by)^2}$$

$$= \frac{(ay + bx)(ax + by)^2}{(ax + by)^2} = ay + bx$$

$$6. \quad \text{Sum of } (a-5)(a-2) \text{ and } (a^2 + 6a + 7) \text{ is} \\ (a^2 - 2a - 5a + 10) + (a^2 + 6a + 7) \\ (a^2 - 7a + 10) + a^2 + 6a + 7 \\ 2a^2 - a + 17$$

Required number

$$= 2a^2 - a + 17 + a^2 + 2a - 1 \\ = 3a^2 + a + 16$$

$$7. \quad (i) \quad x^2 + \frac{1}{x^2} = 25$$

$$\left(x + \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} + 2 = 25 + 2 = 27$$

$$\Rightarrow x + \frac{1}{x} = \sqrt{27} = 3\sqrt{3}$$

$$(ii) \quad x^2 + \frac{1}{x^2} = 25 \text{ (given)}$$

$$\left(x - \frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2 = 25 - 2 = 23$$

$$\Rightarrow x - \frac{1}{x} = \sqrt{23}$$

$$8. \quad (i) \quad 6x = 23^2 - 17^2$$

$$[\text{Using } a^2 - b^2 = (a - b)(a + b)]$$

$$6x = (23 - 17)(23 + 17)$$

$$6x = 6 \times 40 = 240 \Rightarrow x = 40$$

$$(ii) \quad 4x = 98^2 - 88^2 = (98 - 88)(98 + 88)$$

$$\Rightarrow 4x = (10) \times 186$$

$$\Rightarrow x = \frac{10 \times 186}{4} = 465$$

9. (i) $(3x+7)^2 - 84x = (3x-7)^2$
 L.H.S. $= (3x+7)^2 - 84x$
 $= \{(3x)^2 + 2(3x)(7) + (7)^2\} - 84x$
 $= (9x^2 + 42x + 49) - 84x$
 $= 9x^2 + (42x - 84x) + 49$

Combining the like terms

$$= 9x^2 - 42x + 49 \quad \dots(1)$$

$$\text{R.H.S.} = (3x-7)^2$$

$$= (3x)^2 - 2(3x)(7) + (7)^2$$

$$= 9x^2 - 42x + 49 \quad \dots(2)$$

From equations (1) and (2),

$$(3x-7)^2 - 84x = (3x-7)^2$$

(ii) $(9p-5q)^2 + 180pq = (9p+5q)^2$
 L.H.S. $= (9p-5q)^2 + 180pq$
 $= \{(9p)^2 - 2(9p)(5q) + (5q)^2\} + 180pq$
 $= (81p^2 - 90pq + 25q^2) + 180pq$
 $= 81p^2 + (180pq - 90pq) + 25q^2$

Combining the like terms

$$= 81p^2 + 90pq + 25q^2 \quad \dots(1)$$

$$\text{R.H.S.} = (9p+5q)^2$$

$$= (9p)^2 + 2(9p)(5q) + (5q)^2$$

$$= 81p^2 + 90pq + 25q^2 \quad \dots(2)$$

From equations (1) and (2),

$$(9p-5q)^2 + 180pq = (9p+5q)^2$$

(iii) $\left(\frac{4}{3}m - \frac{3}{4}n\right)^2 + 2mn = \frac{16}{9}m^2 + \frac{9}{16}n^2$

$$\text{L.H.S.} = \left(\frac{4}{3}m - \frac{3}{4}n\right)^2 + 2mn$$

$$= \left(\frac{4}{3}m\right)^2 - 2\left(\frac{4}{3}m\right)\left(\frac{3}{4}n\right) + \left(\frac{3}{4}n\right)^2 + 2mn$$

$$= \frac{16}{9}m^2 - 2mn + \frac{9}{16}n^2 + 2mn$$

$$= \frac{16}{9}m^2 + (2mn - 2mn) + \frac{9}{16}n^2$$

Combining the like terms

$$= \frac{16}{9}m^2 + \frac{9}{16}n^2 = \text{R.H.S.}$$

(iv) $(4pq+3q)^2 - (4pq-3q)^2 = 48pq^2$

$$\text{L.H.S.} = (4pq+3q)^2 - (4pq-3q)^2$$

$$= \{(4pq)^2 + 2(4pq)(3q) + (3q)^2\}$$

$$- \{(4pq)^2 - 2(4pq)(3q) + (3q)^2\}$$

$$= 16p^2q^2 + 24pq^2 + 9q^2 - (16p^2q^2 - 24pq^2 - 9q^2)$$

$$= 16p^2q^2 + 24pq^2 + 9q^2 - 16p^2q^2 + 24pq^2 - 9q^2$$

$$= (16p^2q^2 - 16p^2q^2) + (24pq^2 + 24pq^2) + (9q^2 - 9q^2)$$

Combining the like terms

$$= 48pq^2 = \text{R.H.S.}$$

(v) $(a-b)(a+b) + (b-c)(b+c) + (c-a)(c+a) = 0$

$$\text{L.H.S.} = (a-b)(a+b) + (b-c)(b+c)$$

$$+ (c-a)(c+a)$$

$$= a^2 - b^2 + b^2 - c^2 + c^2 - a^2$$

Using identity III

$$= (a^2 - a^2) + (b^2 - b^2) + (c^2 - c^2)$$

Combining the like terms

$$= 0 = \text{R.H.S.}$$

10. (i) 71^2

$$71^2 = (70+1)^2$$

$$= (70)^2 + 2(70)(1) + (1)^2$$

$$= 4900 + 140 + 1 = 5041$$

(ii) 99^2

$$99^2 = (100-1)^2$$

$$= (100)^2 - 2(100)(1) + (1)^2$$

$$= 10000 - 200 + 1 = 9801$$

(iii) 998^2

$$998^2 = (1000-2)^2$$

$$= (1000)^2 - 2(1000)(2) + (2)^2$$

$$= 1000000 - 4000 + 4 = 996004$$

(iv) 5.2^2

$$5.2^2 = (5+0.2)^2$$

$$= (5)^2 + 2(5)(0.2) + (0.2)^2$$

$$= 25 + 2 + 0.04 = 27.04$$

(vi) 78×82

$$78 \times 82 = (80-2) \times (80+2)$$

$$= (80)^2 - (2)^2$$

$$= 6400 - 4 = 6396$$

(vii) 8.9^2

$$8.9^2 = (9-0.1)^2$$

$$= (9)^2 - 2(9)(0.1) + (0.1)^2$$

$$= 81 - 1.8 + 0.01 = 79.21$$

(viii) 1.05×9.5

$$1.05 \times 9.5 = \frac{1}{10} \times 10.5 \times 9.5$$

$$= \frac{1}{10} (10+0.5) \times (10-0.5)$$

$$= \frac{1}{10} \times \{(10)^2 - (0.5)^2\}$$

$$= \frac{1}{10} \times (100 - 0.25)$$

$$= \frac{1}{10} \times 99.75 = 9.975$$

11. (i) 103×104

$$103 \times 104 = (100+3) \times (100+4)$$

$$= (100)^2 + (3+4)(100) + (3)(4)$$

$$= 10000 + 700 + 12 = 10712$$

(ii) 5.1×5.2

$$5.1 \times 5.2 = (5+0.1) \times (5+0.2)$$

$$= (5)^2 + (0.1+0.2)(5) + (0.1)(0.2)$$

$$= 25 + 1.5 + 0.02$$

$$= 26.52$$

(iii) 103×98

$$103 \times 98 = (100+3) \times (100-2)$$

$$= (100+3) \times \{(100) - (-2)\}$$

$$\begin{aligned}
 &= (100)^2 + \{3 + (-2)\} (100) + (3)(-2) \\
 &= 10000 + 100 - 6 = 10094 \\
 \text{(iv)} \quad &9.7 \times 9.8 \\
 &9.7 \times 9.8 = (10 - 0.3)(10 - 0.2) \\
 &= (10)^2 - (0.3 + 0.2)(10) + (0.3)(0.2) \\
 &= 100 - 5 + 0.06 = 95.06
 \end{aligned}$$

2 EXERCISE

Text-Book Exercise :

1. Writing the three expressions in separate rows, with like terms one below the other, we have

$$\begin{array}{r}
 7xy + 5yz - 3zx \\
 + \quad 4yz + 9zx - 4y \\
 + \quad -2xy - 3zx + 5x \quad \text{(Note: } xz \text{ is same as } zx) \\
 \hline
 5xy + 9yz + 3zx + 5x - 4y
 \end{array}$$

Thus, the sum of the expressions is $5xy + 9yz + 3zx + 5x - 4y$.

$$\begin{array}{r}
 7x^2 - 4xy + 8y^2 + 5x - 3y \\
 5x^2 - 4y^2 + 6y - 3 \\
 \hline
 (-) \quad (+) \quad (-) \quad (+) \\
 2x^2 - 4xy + 12y^2 + 5x - 9y + 3
 \end{array}$$

3. Volume = length $\times$ breadth $\times$ height

$$\begin{aligned}
 \text{Hence, for (i) volume} &= (2ax) \times (3by) \times (5cz) \\
 &= 2 \times 3 \times 5 \times (ax) \times (by) \times (cz) = 30abcxyz \\
 \text{for (ii) volume} &= m2n \times n2p \times p2m \\
 &= (m^2 \times m) \times (n \times n^2) \times (p \times p^2) = m^3n^3p^3 \\
 \text{for (iii) volume} &= 2q \times 4q^2 \times 8q^3 \\
 &= 2 \times 4 \times 8 \times q \times q^2 \times q^3 = 64q^6
 \end{aligned}$$

4. (i) $x(x-3) + 2 = x^2 - 3x + 2$
 For $x = 1$, $x^2 - 3x + 2 = (1)^2 - 3(1) + 2$
 $= 1 - 3 + 2 = 3 - 3 = 0$
- (ii) $3y(2y-7) - 3(y-4) - 63 = 6y^2 - 21y - 3y + 12 - 63$
 $= 6y^2 - 24y - 51$
 For $y = -2$, $6y^2 - 24y - 51 = 6(-2)^2 - 24(-2) - 51$
 $= 6 \times 4 + 24 \times 2 - 51$
 $= 24 + 48 - 51 = 72 - 51 = 21$
5. (i) First expression $= 5m(3-m) = (5m \times 3) - (5m \times m)$
 $= 15m - 5m^2$
 Now adding the second expression to it, $15m - 5m^2$
 $+ 6m^2 - 13m = m^2 + 2m$
- (ii) The first expression $= 4y(3y^2 + 5y - 7) = (4y \times 3y^2)$
 $+ (4y \times 5y) + (4y \times (-7))$
 $= 12y^3 + 20y^2 - 28y$
 The second expression $= 2(y^3 - 4y^2 + 5) = 2y^3 + 2$
 $\times (-4y^2) + 2 \times 5$
 $= 2y^3 - 8y^2 + 10$

Adding the two expressions,

$$\begin{array}{r}
 12y^3 + 20y^2 - 28y \\
 + \quad 2y^3 - 8y^2 + 10 \\
 \hline
 14y^3 + 12y^2 - 28y + 10
 \end{array}$$

6. We have $3pq(p-q) = 3p^2q - 3pq^2$ and
 $2pq(p+q) = 2p^2q + 2pq^2$

$$\begin{array}{r}
 2p^2q + 2pq^2 \\
 3p^2q - 3pq^2
 \end{array}$$

$$\text{Subtracting, } \begin{array}{r} - \quad + \\ -p^2q + 5pq^2 \end{array}$$

7. We have
 $(a+b)(2a-3b+c) = a(2a-3b+c) + b(2a-3b+c)$
 $= 2a^2 - 3ab + ac + 2ab - 3b^2 + bc$
 $= 2a^2 - ab - 3b^2 + bc + ac$
 (Note, $-3ab$ and $2ab$ are like terms)
 and $(2a-3b)c = 2ac - 3bc$
 Therefore,
 $(a+b)(2a-3b+c) - (2a-3b)c = 2a^2 - ab - 3b^2 + bc + ac$
 $- (2ac - 3bc)$

$$\begin{aligned}
 &= 2a^2 - ab - 3b^2 + bc + ac - 2ac + 3bc \\
 &= 2a^2 - ab - 3b^2 + (bc + 3bc) + (ac - 2ac) \\
 &= 2a^2 - 3b^2 - ab + 4bc - ac
 \end{aligned}$$

8. (i) $4ac$ (ii) $3x^2 + 4xy - y^2$ (iii) $x^3 + y^3$
 (iv) $2.25x^2 - 16y^2$ (v) $a^2 + b^2 - c^2 + 2ab$
9. (i) $a^4 - 2a^2b^2 + b^4$ (ii) $98m^2 + 128n^2$
 (iii) $4p^2 - 4q^2$ (vi) $a^2b^2 + b^2c^2$
10. (i) L.H.S. $= (3x+7)^2 - 84x$
 $= \{(3x)^2 + 2(3x)(7) + (7)^2\} - 84x$
 $= (9x^2 + 42x + 49) - 84x$
 $= 9x^2 + (42x - 84x) + 49$

Combining the like terms

$$= 9x^2 - 42x + 49 \quad \dots(1)$$

$$\begin{aligned}
 \text{R.H.S.} &= (3x-7)^2 \\
 &= (3x)^2 - 2(3x)(7) + (7)^2 \\
 &= 9x^2 - 42x + 49 \quad \dots(2)
 \end{aligned}$$

From equation (1) and (2),

$$(3x+7)^2 - 84x = (3x-7)^2$$

- (ii) L.H.S. $= (9p-5q)^2 + 180pq$
 $= \{(9p)^2 - 2(9p)(5q) + (5q)^2\} + 180pq$
 $= (81p^2 - 90pq + 25q^2) + 180pq$
 $= 81p^2 + (180pq - 90pq) + 25q^2$
 Combining the like terms
 $= 81p^2 + 90pq + 25q^2$
 R.H.S. $= (9p+5q)^2$
 $= (9p)^2 + 2(9p)(5q) + (5q)^2$
 $= 81p^2 + 90pq + 25q^2 \quad \dots(2)$

From eqns. (1) and (2),

$$(9p - 5q)^2 + 180pq = (9p + 5q)^2$$

$$\begin{aligned} \text{(iii) L.H.S.} &= \left(\frac{4}{3}m - \frac{3}{4}n\right)^2 + 2mn \\ &= \left(\frac{4}{3}m\right)^2 - 2\left(\frac{4}{3}m\right)\left(\frac{3}{4}n\right) + \left(\frac{3}{4}n\right)^2 + 2mn \\ &= \frac{16}{9}m^2 - 2mn + \frac{9}{16}n^2 + 2mn \\ &= \frac{16}{9}m^2 + (2mn - 2mn) + \frac{9}{16}n^2 \end{aligned}$$

Combining the like terms

$$= \frac{16}{9}m^2 + \frac{9}{16}n^2 = \text{R.H.S.}$$

$$\begin{aligned} \text{(iv) L.H.S.} &= (4pq + 3q)^2 - (4pq - 3q)^2 \\ &= \{(4pq)^2 + 2(4pq)(3q) + (3q)^2\} \\ &\quad - \{(4pq)^2 - 2(4pq)(3q) + (3q)^2\} \\ &= (16p^2q^2 + 24pq^2 + 9q^2) - (16p^2q^2 - 24pq^2 + 9q^2) \\ &= (16p^2q^2 + 24pq^2 + 9q^2) - 16p^2q^2 + 24pq^2 - 9q^2 \\ &= (16p^2q^2 - 16p^2q^2) + (24pq^2 + 24pq^2) + (9q^2 - 9q^2) \end{aligned}$$

Combining the like terms

$$= 48pq^2 = \text{R.H.S.}$$

$$\begin{aligned} \text{(v) L.H.S.} &= (a - b)(a + b) + (b - c)(b + c) + (c - a)(c + a) \\ &= a^2 - b^2 + b^2 - c^2 + c^2 - a^2 \quad \text{Using identity III} \\ &= (a^2 - a^2) + (b^2 - b^2) + (c^2 - c^2) \end{aligned}$$

Combining the like terms

$$= 0 = \text{R.H.S.}$$

11. (i) 200 (ii) 0.08 (iii) 1800 (iv) 84
12. (i) 10712 (ii) 26.52 (iii) 10094 (iv) 95.06

Exemplar Questions :

- $-pqr(p^2 + q^2 + r^2)$
 $= -(pqr) \times p^2 - (pqr) \times q^2 - (pqr) \times r^2$
 $= -p^3qr - pq^3r - pqr^3$
- $(px + qy)(ax - by)$
 $= px(ax - by) + qy(ax - by)$
 $= apx^2 - pbxy + aqxy - qby^2$
- $497 \times 505 = (500 - 3)(500 + 5)$
 $= 500^2 + (-3 + 5) \times 500 + (-3)(5)$
 $\quad \quad \quad [\text{using } (x + a)(x + b) = x^2 + (a + b)x + ab]$
 $= 250000 + 1000 - 15 = 250985$
- $2.07 \times 1.93 = (2 + 0.07)(2 - 0.07)$
 $= 2^2 - (0.07)^2 = 3.9951$
- R.H.S. $= (9982)^2 - (18)^2$
 $= (9982 + 18)(9982 - 18)$
 $\quad \quad \quad [\text{Since } (a^2 - b^2) = (a + b)(a - b)]$
 $= (10000) \times (9964)$

$$\text{L.H.S.} = (10000) \times x$$

Comparing L.H.S and R.H.S, we get

$$10000x = 10000 \times 9964$$

$$\text{or } x = \frac{10000 \times 9964}{10000} = 9964$$

$$6. \text{ Given that } x + \frac{1}{x} = 1$$

$$\text{So, } \left(x + \frac{1}{x}\right)^2 = 25$$

$$\text{Now, } \left(x + \frac{1}{x}\right)^2 = x^2 + 2 \times x \times \frac{1}{x} + \left(\frac{1}{x}\right)^2 \quad [\text{Using Identity}]$$

$$(a + b)^2 = a^2 + 2ab + b^2, \text{ with } a = x \text{ and } b = \frac{1}{x}$$

$$= x^2 + 2 + \left(\frac{1}{x^2}\right) \Rightarrow x^2 + \frac{1}{x^2} = 25 - 2 = 23$$

$$7. -2b^3 [\text{Using identity } a^3 - b^3 = (a - b)(a^2 + b^2 + ab)]$$

$$8. \text{ (i) } x^4 - y^4 \quad \text{(ii) } a^4 + 2a^2b^2 + b^4$$

$$9. 225$$

$$10. 72$$

$$11. 3a^2 + ab \quad 7ac + 2b^2 - 6bc - 4c^2$$

$$12. 62$$

HOTS Questions :

$$1. 4a^2 + 2b^2 - 7xy - 5$$

$$\begin{array}{r} -4a^2 + 5b^2 + 3xy + 2 \\ + \quad - \quad - \quad - \\ \hline 8a^2 - 3b^2 - 10xy - 7 \end{array}$$

$$2. \text{ We have,}$$

$$\begin{aligned} &(5a^6) \times (-10ab^2) \times (-2.1a^2b^3) \\ &= (5 \times -10 \times -2.1) \times (a^6 \times a \times a^2 \times b^2 \times b^3) \\ &= \left(5 \times -10 \times -\frac{21}{10}\right) \times (a^6 \times a \times a^2 \times b^2 \times b^3) \\ &= 105a^{6+1+2}b^{2+3} = 105a^9b^5 \end{aligned}$$

$$\text{Putting } a = 1 \text{ and } b = \frac{1}{2}, \text{ we have}$$

$$105a^9b^5 = 1045 \times (1)^9 \times \left(\frac{1}{2}\right)^5 = 105 \times 1 \times \frac{1}{32} = \frac{105}{32}$$

$$3. \text{ (i) We have,}$$

$$\begin{aligned} &15a^2 - 6a(a - 2) + a(3 + 7a) \\ &= 15a^2 - 6a^2 + 12a + 3a + 7a^2 \\ &= 15a^2 - 6a^2 + 7a^2 + 12a + 3 \\ &= 16a^2 + 15a \end{aligned}$$

$$\text{(ii) We have,}$$

$$\begin{aligned} &x^2(1 - 3y^2) + x(xy^2 - 2x) - 3y(y - 4x^2y) \\ &= x^2 \times 1 - 3y^2 \times x^2 + x \times xy^2 - x \times 2x - 3y \times y + 3y \times 4x^2y \\ &= x^2 - 3x^2y^2 + x^2y^2 - 2x^2 - 3y^2 + 12x^2y^2 \\ &= (x^2 - 2x^2) + (-3x^2y^2 + x^2y^2 + 12x^2y^2) - 3y^2 \\ &= -x^2 + 10x^2y^2 - 3y^2 \end{aligned}$$

4. We have,
 $(3x-2)(x-1)(3x+5)$
 $= \{(3x-2)(x-1)\} \times (3x+5)$
 $= \{3x(x-1)-2(x-1)\} \times (3x+5)$
 $= (3x^2-3x-2x+2) \times (3x+5)$
 $= (3x^2-5x+2) \times (3x+5)$
 $= 3x^2 \times (3x+5) - 5x(3x+5) + 2 \times (3x+5)$
 $= (9x^3+15x^2) + (-15x^2-25x) + (6x+10)$
 $= 9x^3+15x^2-15x^2-25x+6x+10$
 $= 9x^3-19x+10$
5. We have,
 $\text{LHS} = 2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca$
 $= (a^2 - 2ab + b^2) + (b^2 - 2bc + c^2) + (c^2 - 2ca + a^2)$
 $\quad \quad \quad [\text{Re-arranging the terms}]$
 $= (a-b)^2 + (b-c)^2 + (c-a)^2$
 $= \text{R.H.S.}$
Hence, $2a^2 + 2b^2 + 2c^2 - 2ab - 2bc - 2ca = (a-b)^2 + (b-c)^2 + (c-a)^2$

3 EXERCISE

Single Option Correct :

- (c) $(x+y)^2 = x^2 + 2xy + y^2$
 $\therefore$ Expanded form of $(x+y)^2$ is a trinomial
- (b) $(x^2+3x+5) \times (x^2-1)$
 $= (x^2 \times x^2) + (3x \times x^2) + (5 \times x^2)$
 $\quad + [x^2 \times (-1)] + [3x \times (-1)] + [5 \times (-1)]$
 $= x^4 + 3x^3 + 5x^2 - x^2 - 3x - 5$
 $= x^4 + 3x^3 + 4x^2 - 3x - 5$
- (c) $(2x^2 - 5y^2)^2 = (2x^2)^2 - 2(2x^2)(5y^2) + (5y^2)^2$
 $(2x^2 - 5y^2)^2 = 4x^4 - 20x^2y^2 + 25y^4$
- (c) $\frac{4a+5b-3c}{\pm 3a \mp 3b \pm c}$
 $\frac{a+8b-4c}{a+8b-4c}$
- (b) $5m^2n \times (-3mnp) \times (-5n^2p) = 75m^3n^4p^2$
- (a) $3x^2y^2(5x^2-4xy+6y^2)$
 $= 15x^4y^2 - 12x^3y^3 + 18y^4x^2$
- (a) $3a^2 - 6ab - 3b^2 - 1$
 $\frac{\pm 4a^2 \mp 7ab \mp 4b^2 \pm 1}{-a^2 + ab + b^2 - 2}$
- (c) $\left(3 + \frac{5}{x}\right)\left(9 - \frac{15}{x} + \frac{25}{x^2}\right) = 3^3 + \left(\frac{5}{x}\right)^3$
 $= 27 + \frac{125}{x^3}$
At $x = 1$,
 $27 + \frac{125}{x^3} = 27 + \frac{125}{1^3} = 27 + 125 = 152$
- (d) $(0.9)^2 - (0.6)^2 = (0.9+0.6)(0.9-0.6)$
 $= (1.5)(0.3)$
 $= 0.45$
- (a) $(a+b)^2 + (a-b)^2$
 $= a^2 + b^2 + \cancel{2ab} + a^2 + b^2 - \cancel{2ab}$
 $= 2a^2 + 2b^2$
- (a) $(a+b)^2 - (a-b)^2$
 $= a^2 + b^2 + 2ab - (a^2 + b^2 - 2ab)$
 $= \cancel{a^2 + b^2} + 2ab - \cancel{a^2 + b^2} + 2ab$
 $= 4ab$
- (d) $3q + 7p^2 - 2r^3 + 4 - [4p^2 - 2q + 7r^3 - 3]$
 $= 3q + 7p^2 - 2r^3 + 4 - 4p^2 + 2q - 7r^3 + 3$
 $= 3p^2 - 9r^3 + 5q + 7$
- (d) $(3a^2 - 4ax + x^2) \times (5x^2 - 2ax)$
 $= (3a^2 - 4ax + x^2) \times 5x^2 - (3a^2 - 4ax + x^2) \times 2ax$
 $= 15a^2x^2 - 20ax^3 + 5x^4 - 6a^3x + 8a^2x^2 - 2ax^3$
 $= 23a^2x^2 - 22ax^3 + 5x^4 - 6a^3x$
- (a) Other binomial $= 5x^2 - 6x - (3x^2 - 2x)$
 $= 5x^2 - 6x - 3x^2 + 2x$
 $= 2x^2 - 4x$
- (a) Cost of a notebook $= ₹ 3a^2 - 4ab + 6b^2$
Cost of $5a^2b^2$ notebooks
 $= ₹ [3a^2 - 4ab + 6b^2] (5a^2b^2)$
 $= ₹ (15a^4b^2 - 20a^3b^3 + 30a^2b^4)$
- (c) $(0.8m + 1.1n)(0.8m - 1.1n)$
 $= (0.8m)^2 - (1.1n)^2$
 $= 0.64m^2 - 1.21n^2$
[Using identity $a^2 - b^2 = (a+b)(a-b)$]
- (c) Sum of $(a-5)(a-2) + (a^2+6a+7)$
 $= a^2 - 2a - 5a + 10 + a^2 + 6a + 7$
 $= 2a^2 - 7a + 6a + 10 + 7$
 $= 2a^2 - a + 17$
 $\therefore 2a^2 - a + 17 - (-a^2 - 2a + 1)$
 $= 2a^2 - a + 17 + a^2 + 2a - 1$
 $= 3a^2 + a + 16$

More Than One Option Correct :

- (c, d)
- (b, d)
- (a, b)
- (a, b, c, d)
- (a, d) : $23^2 - 22^2 = (23+22)(23-22) = 45$
[Using identity $(a^2 - b^2) = (a+b)(a-b)$]
- (b, d) $(60-4)(60-6)$
 $= [60+(-4)][60+(-6)]$
[Using identity : $(x+a)(x+b) = x^2 + (a+b)x + ab$]
 $= (60)^2 + (-4-6) \times 60 + (-4) \times (-6)$
 $= 3600 - 600 + 24 = 3024$

Passage Based Questions :

- (b) Yes, it is a polynomial in x and y
degree = 5.
- (a) Constant term = $\frac{-11}{3}$
- (c) $\frac{-5}{6}x$
- (a) $25x = (536 - 136)(536 + 136) = (400)(672)$
 $\Rightarrow 25x = 268800 \Rightarrow x = 10752$
- (b) $(10000)^2 - (9999)^2$
 $(10000 + 9999)(10000 - 9999) = 19999$
- (d) $(4000)^2 - (3999)^2 + (9000)^2 - (8999)^2$
 $= (4000 + 3999)(4000 - 3999) + (9000 - 8999)(9000 + 8999)$
 $= 7999 + 17999 = 25998$

Assertion & Reason :

- (c) Assertion is true but Reason is false. When the terms have same literal factors they are called like terms.
- (c) Constant terms are terms of the expression having no literal factor. It has fixed numerical value.
- (a) $99 \times 101 = (100 - 1)(100 + 1) = (100)^2 - 1^2$
 $= 10000 - 1 = 9999$
[Using identity : $a^2 - b^2 = (a + b)(a - b)$]
- (d) $-8abc + 4bac = -8abc + 4abc = -4abc$ is a monomial.

Multiple Matching Questions :

- (A) $\rightarrow (r, t)$; (B) $\rightarrow (p, s)$; (C) $\rightarrow (q)$

Integer Type Questions :

- (0) Using the identity
 $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$
 $25a^2 + 16b^2 + 9 + 40ab - 24b - 30a$
 $= (5a)^2 + (4b)^2 + (-3)^2 + 2 \times 5a \times 4b + 2 \times 4b \times (-3) + 2 \times 5a \times (-3)$
 $= (5a + 4b - 3)^2$
 $\therefore$ Required value $= (5 \times (-1) + 4 \times 2 - 3)^2$
 $= (-5 + 8 - 3)^2 = (-8 + 8)^2 = 0$
- (1) $25x^2 + 16y^2 + 40xy$
At $x = 1$ and $y = -1$
 $25(1)^2 + 16(-1)^2 + 40(1)(-1)$
 $= 25 + 16 - 40 = 1$.
- (0) $3x - \frac{1}{2}x + \frac{3}{2}x - 4x = -x + \frac{2x}{2} = -x + x = 0$
- (4) $36 \times 34 = (30 + 6)(30 + 4)$
[Using $(x + a)(x + b) = x^2 + (a + b)x + ab$]

- (1) $(4a^2 + 3b)(4a^2 + 3b) = (4a^2 + 3b)^2$
 $= 16a^4 + 9b^2 + 24a^2b$
When $a = 1, b = 2$
 $16(1)^4 + 9(2)^2 + 24 \times (1)^2 \times 2$
 $16 + 36 + 48 = 100$
 $k \div 100 = 100 \div 100 = 1$
- (4) $\frac{(997 + 496)^2 - (997 - 496)^2}{997 \times 496}$
 $\frac{(997)^2 + (496)^2 + 2 \times 997 \times 496 - (997)^2 - (496)^2 + 2 \times 997 \times 496}{997 \times 496}$
 $= \frac{4 \times 997 \times 496}{997 \times 496} = 4$

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

- (c) $a + b + c = 15$
 $\Rightarrow (a + b + c)^2 = 15^2$
 $\Rightarrow a^2 + b^2 + c^2 + 2(ab + bc + ca) = 225$
 $\Rightarrow 77 + 2(ab + bc + ca) = 225 \Rightarrow ab + bc + ca = 74$
- (c) Given $x - \frac{1}{x} = 7$
Cubing both the sides
 $\left(x - \frac{1}{x}\right)^3 = 7^3$
 $\Rightarrow x^3 - 3 \cdot x \cdot \frac{1}{x} \left(x - \frac{1}{x}\right) - \left(\frac{1}{x}\right)^3 = 343$
 $\Rightarrow x^3 - 3(7) - \left(\frac{1}{x}\right)^3 = 343$
 $\Rightarrow x^3 - \frac{1}{x^3} = 343 + 21$
 $\Rightarrow x^3 - \frac{1}{x^3} = 364$
- (a) If $a + b + c = 0$, then
 $a^3 + b^3 + c^3 = 3abc$
 $(25)^3 + (-10)^3 + (-15)^3$
 $= 3(25)(-10)(-15)$
 $= 11250$
- (a) $(a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$
 $= a^3 + b^3 + c^3 - 3abc$

Chapter

10

Visualising Solid-Shapes

INTRODUCTION

In our daily use we encounter lot of things of different shapes and sizes.

For example : Mobile phones, Dice, Pipes, Balls, Gas Cylinder, Pencil, Pen, Shoeboxes, Marbles, Oranges, Ice-cream cones, Traffic cones, Funnels etc.

Can you identify the various kinds of shapes from the above list and sort them out. This chapter will enable you to see how the dimensions of rectangular prisms, cylinders, and other solid shapes can be drastically different.

Solid things aren't what they look like from a distance. If you see a cylindrical pipe standing vertically from a distance, it looks like a rectangle to you but when you approach the pipe, you find it to be cylindrical. In the present chapter, we will study about different solid shapes and how they look when viewed from different angles.

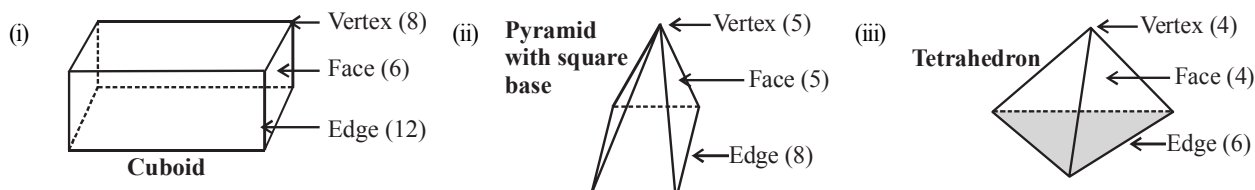
SOLIDS

Anything that occupies space and has three dimensions viz, length, breadth and height (or depth) is called a solid or a three-dimensional figure.

Wherever you are, whether in your sweet home, your classroom, in a train or in an aeroplane, you come across various solid figures such as books, pencils, houses, bricks, refrigerators, footballs, cricket bats, tables etc.

Parts of a Solid

Faces, Edges and Vertices are the different parts of solids. Lets have a look at the following solids to understand them more.



- (a) Face is each flat surface of the solid.
- (b) Vertex is the point (corner) where three faces of a solid meet.
- (c) Edge is the line segment that is intersection of two faces.

Now based on the above definitions, we will find out how many vertices, edges and faces each of the solids have.

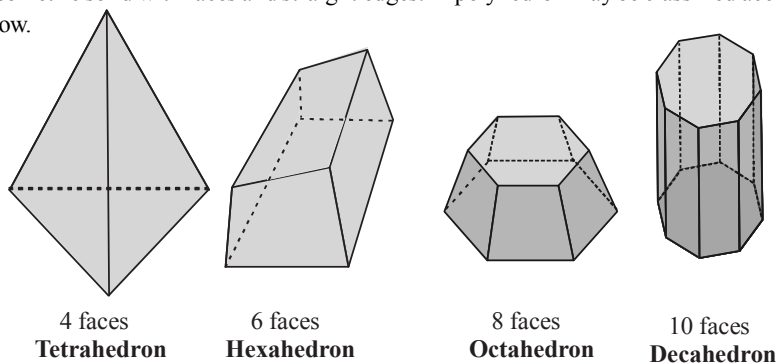
VARIOUS TYPES OF SOLID

I. Some Common Types of Solids:

Solid	Cube	Cuboid	Sphere	Cylinder	Cone
Vertices	8	8	0	0	1
Edges	12	12	0	2 Curved edges	2 Curved edges
Faces	6 flat	6 flat	1 Curved face	3(2 flat, 1 curved)	2(1 flat, 1 curved)

II. Polyhedrons

A polyhedron is a geometric solid with faces and straight edges. A polyhedron may be classified according to the number of its faces, as shown below.



NON-POLYHEDRON

Non-polyhedra are the solids that have curved surfaces, or a mix of both flat and curved surfaces. These solids are not made of polygons.

Example: Spheres, Cylinders and Cones.



Sphere



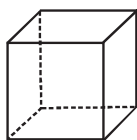
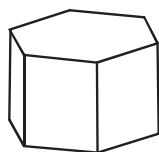
Cylinder



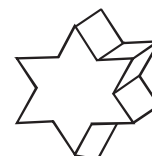
Cone

CONVEX POLYHEDRON

The diagonal of a polyhedron is an imaginary line that connects the vertices through the body of the solid figure. A polyhedron is classified as convex if all the diagonals lie entirely inside the polyhedron.



These are **convex polyhedrons**



These are **not convex or concave polyhedrons**

PRISMS AND PYRAMIDS

Two important members of polyhedron family are prisms and pyramids. So, let us know about these two polyhedrons.

I. Prisms

A prism is a solid formed by joining two congruent plane shapes together with straight line. The two congruent shapes called the bases, are parallel and determine the name of the prism.

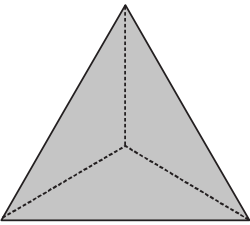
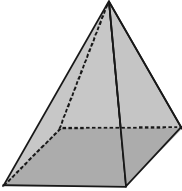
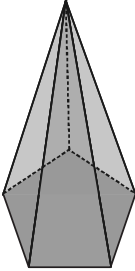
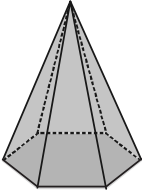
For example : A rectangular prism is formed by joining two identical rectangles together with straight lines.

Solids					
Vertices	8	8	6	10	12
Edges	12	12	9	15	18
Faces	6 (All rectangles)	6 (All squares)	5 [2 triangles 3 rectangles]	7 [2 Pentagons 5 rectangles]	8 [2 hexagon 6 rectangles]

II. Pyramids

A pyramid is any three-dimensional solid where the upper surfaces are triangular and converge at one point. It has one base (usually a polygon). A pyramid is named according to the shape of its base.

For example : A Pyramid with pentagonal base is a pentagonal pyramid.

Solids				
	Triangular Pyramid	Rectangular Pyramid	Pentagonal Pyramid	Hexagonal Pyramid
Vertices	4	5	6	7
Edges	6	8	10	12
Faces	4 (All Triangles)	5 [4 Triangles 1 Rectangle]	6 [5 Triangles 1 Pentagon]	7 [6 Triangles 1 Hexagon]

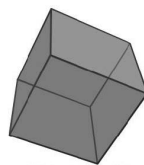
PLATONIC SOLIDS

These solids are also known as regular polyhedrons. In a regular polyhedron, the same number of polygons meet at each vertex. Only five regular polyhedrons exist: the tetrahedron (four triangular faces), the cube (six square faces), the octahedron (8 triangular faces), the dodecahedron (12 pentagonal faces), and the icosahedron (20 triangular faces).



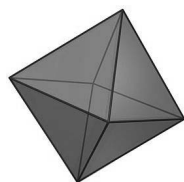
Tetrahedron

- 3 triangles meet at each vertex
- 4 Faces
- 4 Vertices
- 6 Edges



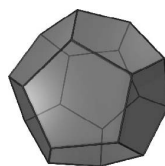
Cube

- 3 squares meet at each vertex
- 6 Faces
- 8 Vertices
- 12 Edges



Octahedron

- 4 triangles meet at each vertex
- 8 Faces
- 6 Vertices
- 12 Edges



Dodecahedron

- 3 pentagons meet at each vertex
- 12 Faces
- 20 Vertices
- 30 Edges



Icosahedron

- 5 triangles meet at each vertex
- 20 Faces
- 12 Vertices
- 30 Edges

EULER'S RULE

As you have already studied, the closed space figures have faces, vertices and edges. There is some relationship among the number of faces, vertices and edges.

Number of faces + Number of vertices = Number of edges + 2

or $F + V = E + 2$

or $F + V - E = 2$

This simple relationship among the number of vertices (v), the number of faces (f) and the number of edges (E) is known as **Euler's formula**. This rule was discovered by **Leonhard Euler**, a famous mathematician.

The following table verifies the Euler's Formula for the different solids.

Name of Solid	Faces	Vertices	Edges	$F + V - E = 2$
Cube	6	8	12	$6 + 8 - 12 = 2$
Pentagonal Prism	7	10	15	$7 + 10 - 15 = 2$
Tetrahedron	4	4	6	$4 + 4 - 6 = 2$
Hexagonal Pyramid	7	7	12	$7 + 7 - 12 = 2$
Triangular Prism	5	6	9	$5 + 6 - 9 = 2$

ILLUSTRATION : 1

Find the number of vertices in a polyhedron with 10 faces and 20 edges.

SOLUTION :

Here, $F = 10$, $E = 20$ and $V = ?$

Using Euler's formula: $E = V + F - 2$

$$\Rightarrow 20 = V + 10 - 2 \Rightarrow V = 20 - 8$$

$$\Rightarrow V = 12$$

Thus the given solid will have 12 vertices.

DIFFERENT VIEWS OF A SOLID

3-dimensional objects can look different from different positions. Thus they can be drawn from different perspectives.

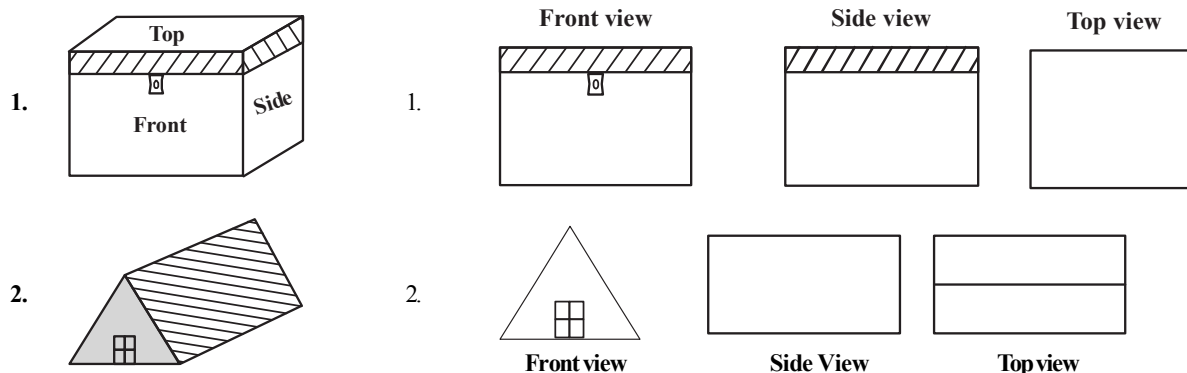
For example, the following objects when viewed from different positions look like:

(1) Top view

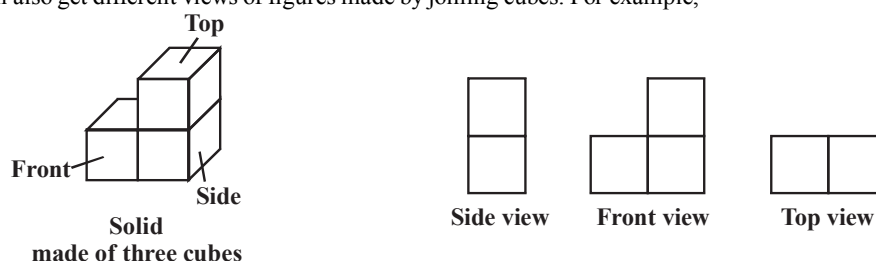
(2) Side view

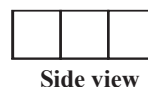
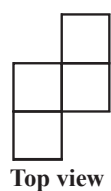
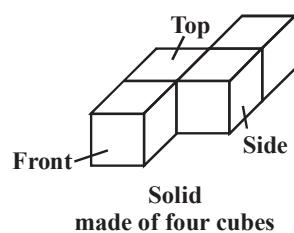
(3) Front view

We shall understand it by a few examples :



We can also get different views of figures made by joining cubes. For example,

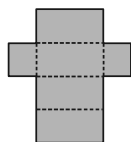




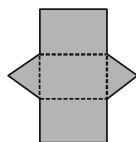
NETS OF SOLIDS

A net of a solid is the outline of its faces joined together from which a model of the solid can be made.

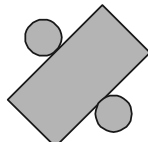
Nets of some of the solid figures are shown below.



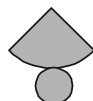
Cuboid



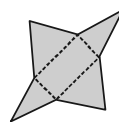
Triangular prism



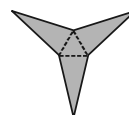
Cylinder



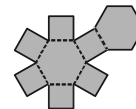
Cone



Rectangular pyramid



Tetrahedron



Hexagonal prism

MISCELLANEOUS

SOLVED EXAMPLES

1. Using Euler's formula, find the unknown

	(i)	(ii)	(iii)	(iv)
Faces	8	y	20	6
Vertices	x	5	14	w
Edges	12	9	z	12

Sol. From Euler's formula,

$$F + V = E + 2$$

(i) Here, $F = 8$, $V = x$, $E = 12$

$$\therefore 8 + x = 12 + 2$$

$$\Rightarrow x = 14 - 8 = 6$$

(iii) Here, $F = 20$, $V = 14$, $E = z$

$$\therefore 20 + 14 = z + 2$$

$$\Rightarrow z = 32$$

(ii) Here, $F = y$, $V = 5$, $E = 9$

$$\therefore y + 5 = 9 + 2$$

$$\Rightarrow y = 11 - 5 = 6$$

(iv) Here, $E = 6$, $V = w$, $F = 12$

$$\therefore 6 + w = 12 + 2$$

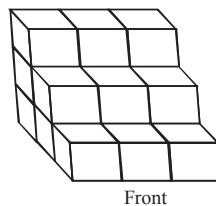
$$\Rightarrow w = 14 - 6 = 8$$

2. Draw rough sketches of the

(i) front - view

(ii) side - view and

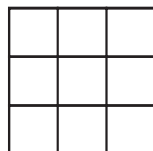
(iii) top - view for the given figure



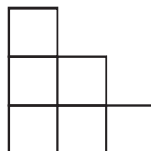
Front

Sol. As per given figure

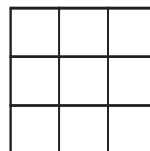
(i) Front - view



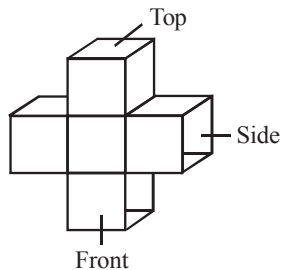
(ii) side - view



(iii) Top - view



3. Draw rough sketches of the top, side and front view of the following figure.



Front

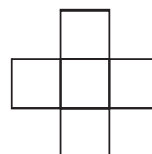
Sol.



Side - view



Top - view



Front - view

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. Solids are shown on paper by their _____ representations.
2. If all corners of a polygon are joined to a point not lying in its plane, we get a _____.
3. The side faces of a pyramid form its _____.
4. The end on which a prism may be supposed to stand is called _____ of the prism.
5. The perpendicular distance between the ends of a prism is called the _____ of the prism.
6. The straight line joining the centres of the ends of a prism is called the _____ of the prism.
7. A pyramid is called a quadrilateral pyramid if its base is _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

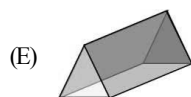
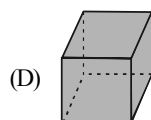
1. The length of a prism is the portion of the axis that lies between the perpendicular ends.
2. All faces other than the bases of a prism are known as its lateral faces.
3. The lines of perpendiculars of the lateral faces of a prism are called the lateral edges of the prism.
4. A prism is called a regular prism if ends are regular polygons.
5. A prism is called a right prism if its lateral edges are parallel to its ends.
6. The cuboid can cast a shadow in the shape of a square.
7. A triangular pyramid have 4 faces, 4 vertices and 6 edges.
8. A square pyramid has square base and triangles on the four lateral sides.
9. All the nets of a cube has 5 square polygon.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D, ...) in column-I have to be matched with statements (p, q, r, s, ...) in column-II.

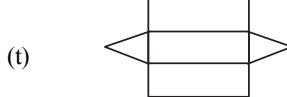
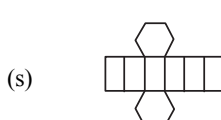
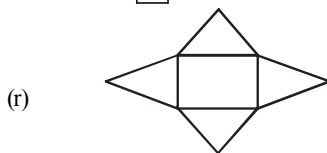
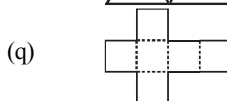
1. Column-I

Figure of prisms (with given edges)



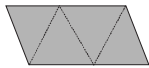
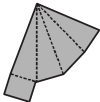
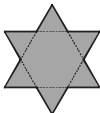
Column-II

Nets with area of faces

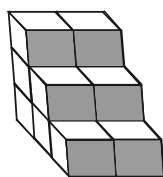


Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

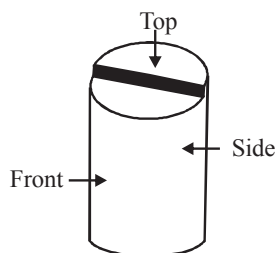
- Verify the Euler's formula for a Octahedron.
- How many equal triangular faces does a regular pentagonal pyramid have?
- Name the solids that have :
 - 4 faces
 - 1 curved surface
 - 6 faces
 - 5 faces and 5 vertices
 - 5 triangular faces
 - 6 rectangular faces and 2 hexagonal faces.
- Name the different plane shapes needed to draw the net of:
 - a cube
 - a triangular prism
 - a triangular pyramid
 - a cylinder
- Name the solid that would be formed by each net.
 - 
 - 
 - 
- Define the "net of the polyhedron".
- Is it possible to have more than one net for a polyhedron?
- Draw the correct picture of the given net.
 

- Name the solid having no flat faces and no vertices.
- Give the base-design for the given solid.



Front

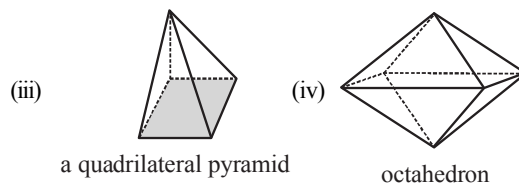
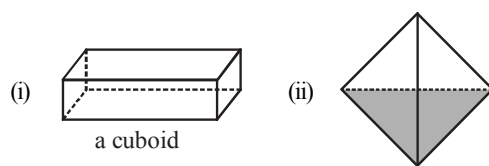
- Draw the top, side and front views of the following figure.



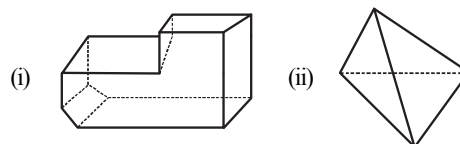
Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

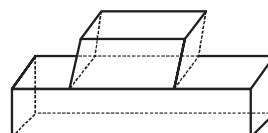
- Define the below given terms which are related to pyramid.
 - Vertex
 - Height
 - Axis
 - Lateral edges
 - Lateral faces
- Draw nets for each of the following polyhedron



- Verify the Euler formula for the given figure.



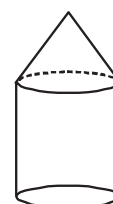
- Draw the top view, front view and side view of the given figure.



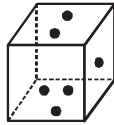
- Using Euler's formula, find the unknown.

Faces	x	10	40	12	8
Vertices	12	y	24	w	16
Edges	24	18	z	24	v

- Draw the net pattern for the given solid.



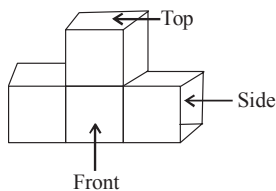
7. Draw the top view of the dice.



8. Draw the front view of the given figure.



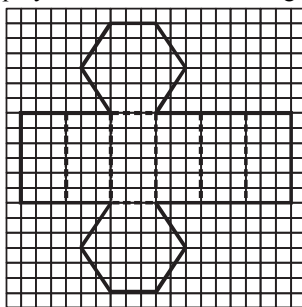
9. Draw the top view of the figure made up of cubes joined together



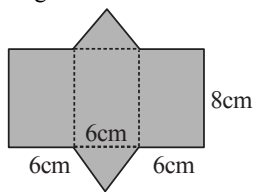
10. Find the number of vertices in a polyhedron which has 30 edges and 12 faces.
11. Using Euler's formula, find the unknown:

Faces	4	25
Vertices	?	12
Edges	7	?

12. Name the polyhedron that has following net pattern.



13. The below given net is made up of two equilateral triangles and three rectangles.

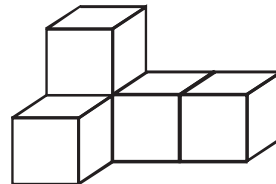


- (i) Name the solid it represents.
- (ii) How many faces, edges and vertices does this solid have?

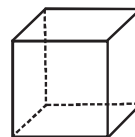
Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

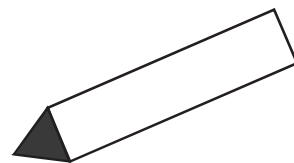
1. Draw the front-view, side-view and top-view of the following arrangement of cubes.



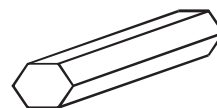
2. Draw rough sketches of nets for the following solids.



- (i) A cube



- (ii) A triangular prism



- (iii) A hexagonal prism

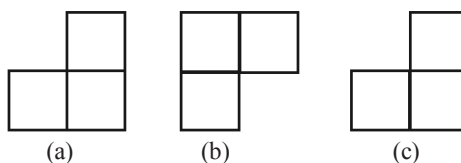
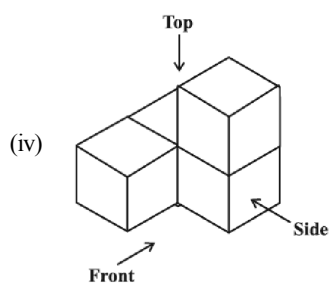
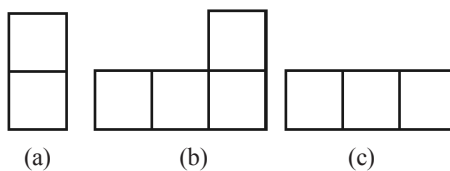
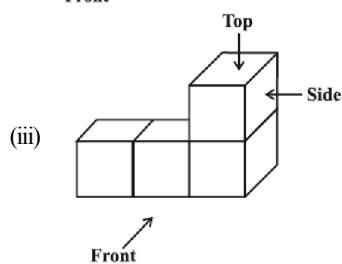
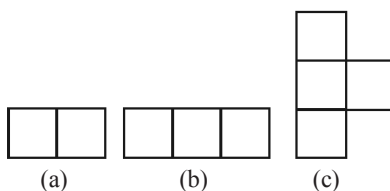
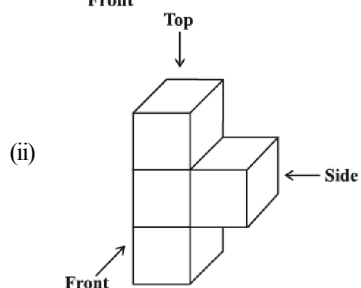
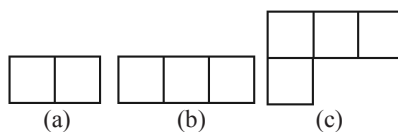
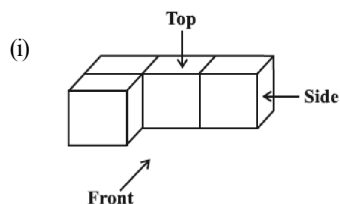


- (iv) A pentagonal pyramid

2 EXERCISE

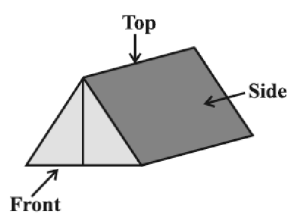
Text-Book Exercise :

1. For each given below solid, identify the top view, front view and side view.

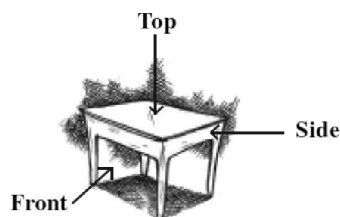


2. Draw the front view, side view and top view of the given objects.

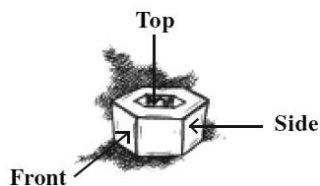
- (i) A military tent



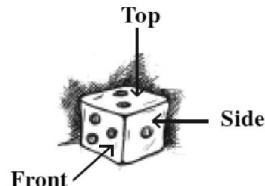
- (ii) A table



(iii) A nut



(iv) A dice



3. Can a polyhedron have for its faces

(i) 3 triangles?

(ii) 4 triangles?

(iii) a square and four triangles?

4. Which are prisms among the following?

(i)



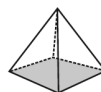
A nail

(ii)



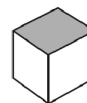
Unsharpened pencil

(iii)



A table weight

(iv)



A box

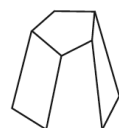
5. (i) How are prisms and cylinders alike?

(ii) How are pyramids and cones alike?

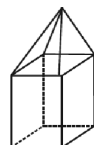
6. Is a square prism same as a cube? Explain.

7. Verify Euler's formula for these solids.

(i)



(ii)



8. Using Euler's formula find the unknown.

Faces	?	5	20
Vertices	6	?	12
Edges	12	9	?

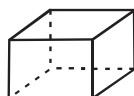
9. Can a polyhedron have 10 faces, 20 edges and 15 vertices?

Exemplar Questions :

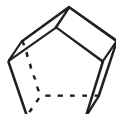
1. Name the following polyhedrons and verify the Euler's formula for each of them.



(i)



(ii)



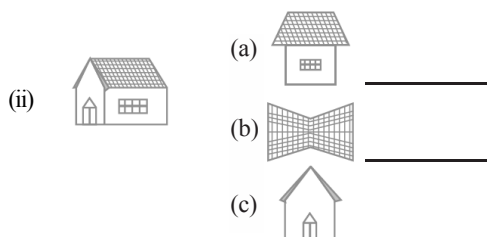
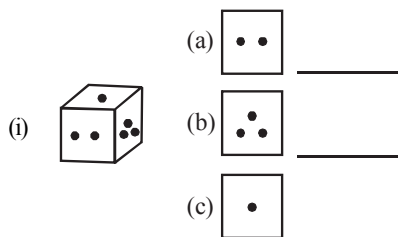
(iii)

2. In a solid if $F = V = 5$, then find the number of edges in the shape.

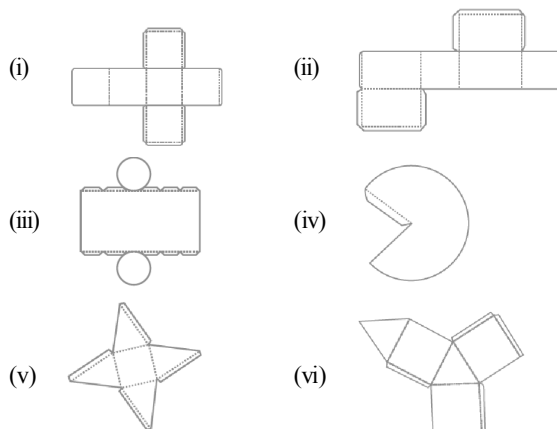
3. Complete the table given below:

S.No.	Solid	Number of faces (F)	Number of vertices (V)	Number of edges (E)	$F + V = E + 2$
(i)	Cuboid				
(ii)	Square Pyramid				
(iii)	Cube				
(iv)	Heptagonal Prism				

4. For each of the following solids, identify the front, side and top views and write it in the space provided.

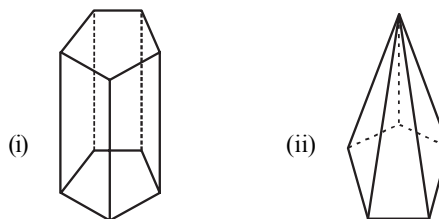


5. Identify the nets given below and mention the name of the corresponding solid.



HOTS Questions :

1. Name a geometric solid that is not polyhedra. Explain why not?
2. How are a pyramid and a cone alike ? How are they different?
3. Can a polyhedron have 12 faces, 22 edges and 17 vertices?
4. Can a polyhedron have 14. faces, 20 edges and 8 vertices ?
5. Verify Euler's formula for the given figures.

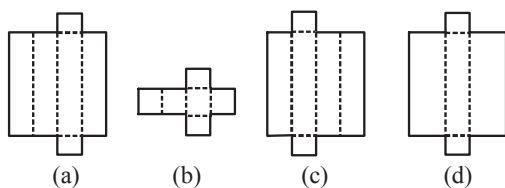


3 EXERCISE

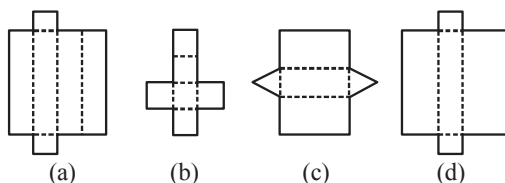
Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

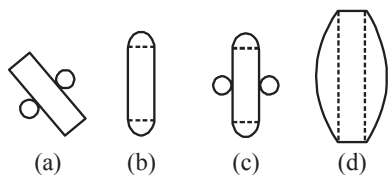
1. Which of these nets matches that of a cube?



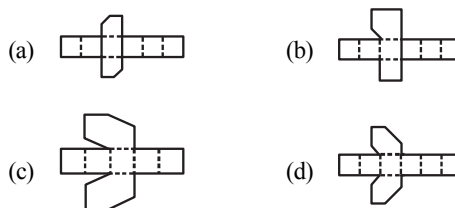
2. Which of these nets matches that of a cuboid?



3. Which of these nets matches that of a cylinder?

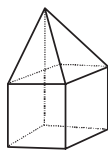


4. Which of the following picture is the correct for the given net ?

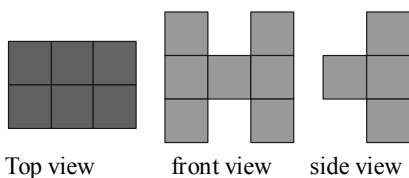


5. In which of the following solids has the least number of vertices
(a) Cone (b) Cylinder
(c) Cube (d) Pyramid
6. Which of the following is a solid ?
(a) Triangle (b) Cone
(c) Rhombus (d) Circle
7. How many faces a cube has?
(a) 6 (b) 8
(c) 5 (d) 4
8. Which of the following is/are true about triangular Pyramid?
(a) It has 4 faces (b) It has 7 edges
(c) It has 5 vertices (d) None of these

9. How many equal triangular faces does a regular pentagonal pyramid have?
 (a) 4 (b) 5
 (c) 6 (d) 3
10. A polyhedron has 11 faces and 18 vertices. Find the number of edges.
 (a) 15 (b) 27
 (c) 33 (d) 21
11. How many faces does the solid in the given figure have?



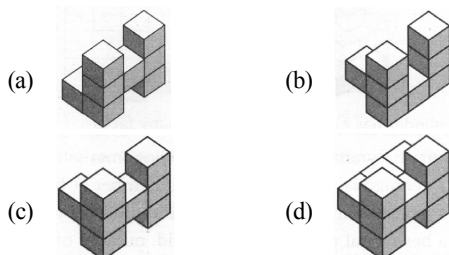
- (a) 5 (b) 7
 (c) 8 (d) 9
12. A solid has forty faces and sixty edges. Find the number of vertices of the solid.
 (a) 20 (b) 22
 (c) 24 (d) 26
13. Which of the given geometric solids has the maximum number of vertices?
 (a) Cone (b) Cylinder
 (c) Cuboid (d) Pyramid
14. If $F = n$ and $E = n + 1$, then using Euler's formula, the value of V , is:
 (a) 3 (b) 2
 (c) 1 (d) 0
15. The adjacent figure shows 3 different views of a three-dimensional figure made from cubes. Which could be a drawing of the figure?



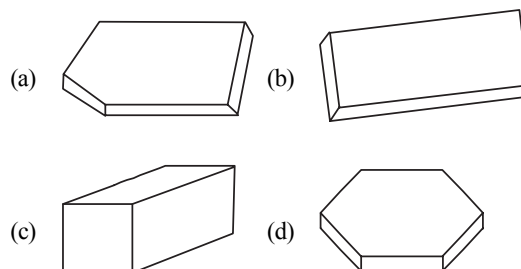
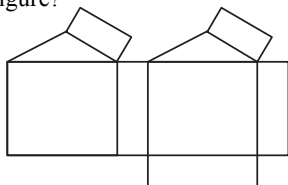
Top view

front view

side view



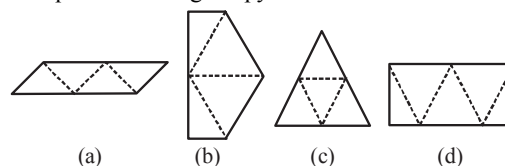
16. Which of the following solids can be formed from the net in the given figure?



More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

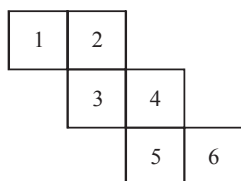
1. Which of the following(s) represents the Euler's formula?
 (a) $F + V - E = 2$
 (b) $F + V = 2 + E$
 (c) $F + V - 2 = E$
 (d) $E + F = V$
2. Given below are 4 nets. Which of them is the correct net of an equilateral triangular pyramid?



Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.
1. **Assertion :** A cube, a cuboid, a pyramid, a prism and a tetrahedron are convex polyhedrons.
Reason : For any polyhedron, $F + V = 2 + E$
2. **Assertion :** A right prism is called a right triangular prism if its ends are triangles.
Reason : A triangular prism is called a right triangular prism if its lateral edges are perpendicular to its ends.
3. **Assertion :** The following net be a net for a perfect die.

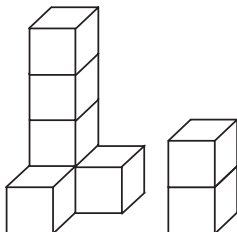


Reason: The total number of dots on the opposite faces of a perfect die is equal to seven.

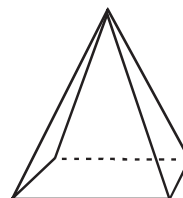
Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

1. Count the number of cubes in the given shapes.



2. If $F = 4$ and $V = 3$, then find the value of E (Use Euler's formula)
3. A square pyramid is shown in the figure. Total number of edges, vertices and lateral faces are denoted by E , V and F . Find $F + V - E$.



4. How many number of edges are there in a triangular prism?
5. How many number of vertices are there in a pentagonal pyramid?
6. In the Euler's formula $V + F = E + 2$, if $V = 5$, $F = 4$, then find the value of E ?
7. How many number of triangular faces are there in a Octahedron?

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- Two dimensional.
- Pyramid
- lateral surface
- base
- height
- Axis
- quadrilateral

True / False :

- False, It should lie between the parallel ends.
- True
- False, the lines of intersection of the lateral faces of a prism are called lateral edges.
- True
- False, A prism is a right prism if its lateral edges are $\perp$ to its ends (bases).
- False, A cuboid always casts a rectangular shadow.
- True, Triangular pyramid has 4 faces, 4 vertices and 6 edges.
- True, pyramids are named after their bases and the other faces are triangular.
- False, Since a cube has 6 square faces.
 $\therefore$ Net must contain 6 square polygons.

Match the Columns :

- (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q); (E) $\rightarrow$ (t)

Very Short Answer Questions :

- Euler's Formula :
Number of faces + number of vertices = Number of edges + 2
L.H.S = $8 + 6 = 14$
R.H.S = $12 + 2 = 14$
 $\therefore$ L.H.S. = R.H.S.
Hence Euler's Formula is verified.
- 5 faces
- Triangular Pyramid
 - Sphere
 - Cuboid
 - Square Based Pyramid
 - Pentagonal Pyramid
 - Hexagonal Prism
- Square
 - Rectangle, Triangle
 - Triangle
 - Circle, rectangle
- Triangular Pyramid
 - Square Based Pyramid
 - Hexagonal Pyramid
- A pattern that folds to form a polyhedron is called the net of the polyhedron.

- Yes

-

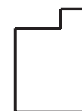


- A sphere (Ball)

- Base – Design of the given solid is

3	3
2	2
1	1

-



Top view Front view Side view

Short Answer Questions :

- Vertex = The common vertex of the triangular faces of a pyramid is called the vertex of the pyramid.
 - Height = The height of a pyramid is the length of the perpendicular from the vertex to the base.
 - Axis = The axis of a pyramid is the straight line joining the vertex to the central point of the base.
 - Lateral edges : The edges through the vertex of a pyramid are known as its lateral edges.
 - Lateral faces : The side faces of a pyramid are known as its lateral faces.
- -
 -
 -
- Euler Formula :
 - Number of faces + Number of Vertices = Number of edges + 2
L.H.S = $9 + 14 = 23$
R.H.S = $21 + 2 = 23$
 $\therefore$ L.H.S = R.H.S
Hence Euler Formula is verified.

Euler Formula :

(ii) Number of faces + Number of Vertices
= Number of edges + 2

$$\text{L.H.S} = 4 + 4 = 8$$

$$\text{R.H.S} = 6 + 2 = 8$$

$$\therefore \text{L.H.S} = \text{R.H.S}$$

Hence Euler Formula is verified.

4. Top view : 

Front view : 

Side view : 

5. Euler's Formula :
Number of faces + Number of Vertices
= Number of edges + 2

or

$$F + V = E + 2$$

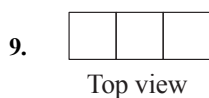
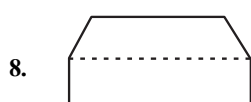
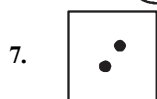
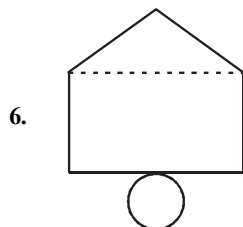
(a) $x + 12 = 24 + 2 \Rightarrow x = 26 - 12 = 14$

(b) $10 + y = 18 + 2 \Rightarrow y = 10$

(c) $40 + 24 = z + 2 \Rightarrow z = 62$

(d) $12 + w = 24 + 2 \Rightarrow w = 14$

(e) $8 + 16 = V + 2 \Rightarrow V = 22$



10. For any polyhedron,
 $F + V - E = 2$
Here, $F = 12$, $V = ?$, $E = 30$
Using above formula,
 $12 + V - 30 = 2$
 $V - 18 = 2$

$$V = 2 + 18$$

$$V = 20$$

11. By using Euler's formula, we have

$$\Rightarrow F + V = E + 2$$

$$\Rightarrow 4 + V = 7 + 2$$

$$\Rightarrow V = 9 - 4 = 5$$

Hence, the required number of vertices is 5.

Also, $E + 2 = F + V$

$$\Rightarrow E + 2 = 25 + 12 \Rightarrow E = 37 - 2 = 35$$

Hence, the required number of edges is 35.

12. Hexagonal pyramid

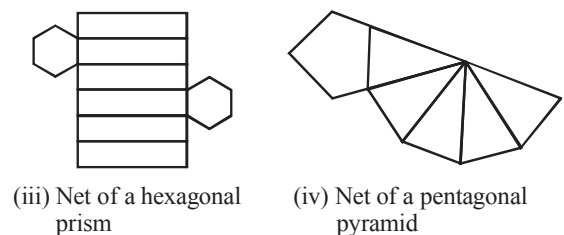
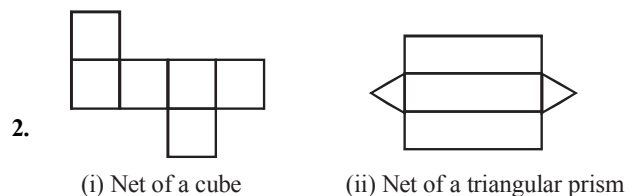
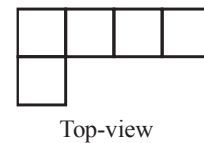
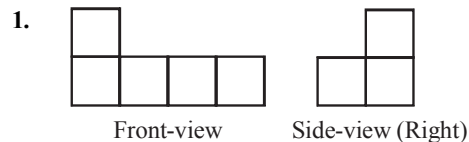
13. (i) Triangular Prism

(ii) No. of faces = 5

No. of edges = 9

No. of vertices = 6

Long Answer Questions :

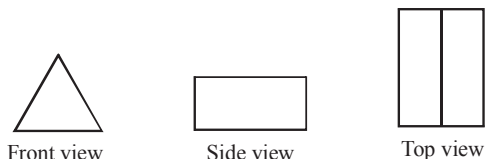


2 EXERCISE

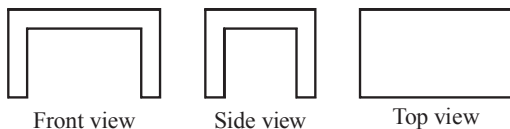
Text-Book Exercise :

1. (i) (a) Side; (b) Front; (c) Top
(ii) (a) Top; (b) Side; (c) Front
(iii) (a) Side; (b) Front; (c) Top
(iv) (a) Front; (b) Top; (c) Side

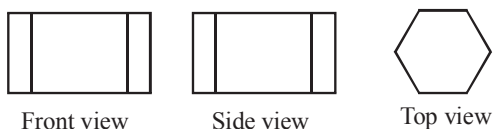
2. (i)



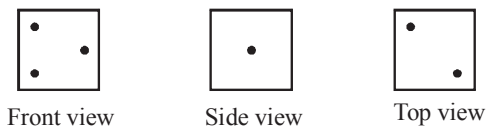
(ii)



(iii)



(iv)



3. (i) No; (ii) Yes; (iii) Yes
 4. Only (ii) and (iv)
 5. (i) A prism becomes a cylinder as the number of sides of its base becomes larger and larger.
 (ii) A pyramid becomes a cone as the number of sides of its base becomes larger and larger.

6. No. It can be a cuboid also

7. (i) $F = 7$; $V = 10$; $E = 15$
 $F + V = 7 + 10 = 17$
 $E + 2 = 15 + 2 = 17$

So, $F + V = E + 2$

Hence, Euler's formula verified.

- (ii) $F = 9$; $V = 9$; $E = 16$
 $F + V = 9 + 9 = 18$
 $E + 2 = 16 + 2 = 18$

So, $F + V = E + 2$

Hence, Euler's formula is verified.

8. (i) $F + V = E + 2$
 $\Rightarrow F + 6 = 12 + 2$
 $\Rightarrow F + 6 = 14$
 $\Rightarrow F = 14 - 6 = 8$
 (ii) $F + V = E + 2$
 $\Rightarrow 5 + V = 9 + 2$
 $\Rightarrow 5 + V = 11$
 $\Rightarrow V = 11 - 5 = 6$
 (iii) $F + V = E + 2$
 $\Rightarrow 20 + 12 = E + 2$
 $\Rightarrow 32 = E + 2$
 $\Rightarrow E = 32 - 2$
 $\Rightarrow E = 30$

9. Here, $F = 10$; $E = 20$; $V = 15$
 So, $F + V = 10 + 15 = 25$
 $E + 2 = 20 + 2 = 22$
 $\therefore F + V \neq E + 2$
 $\therefore$ Such a polyhedron is not possible.

Exemplar Questions :

- | S.No. | Polyhedron | F | V | F + V | E | F + V - E |
|-------|------------------|---|----|-------|----|-----------|
| (i) | Tetrahedron | 4 | 4 | 8 | 6 | 2 |
| (ii) | Cuboid | 6 | 8 | 14 | 12 | 2 |
| (iii) | Pentagonal prism | 7 | 10 | 17 | 15 | 2 |
2. (8)
 3. (i) 6, 8, 12, 14, 14
 (ii) 5, 5, 8, 10, 10
 (iii) 6, 8, 12, 14, 14
 (iv) 9, 14, 21, 23, 23
 4. (i) (a) Front View
 (b) Side View
 (c) Top view
 (ii) (a) side view
 (b) Top view
 (c) Front view
 5. (i) Cube
 (ii) Cuboid
 (iii) Cylinder
 (iv) Cone
 (v) Square Pyramid
 (vi) Triangular Prism

HOTS Questions :

1. A sphere has only one surface (which is curved). Hence sphere is not a polyhedra.
 2. A pyramid and a cone are alike in the sense that
 (i) Base of both pyramid and cone are plane surface.
 (ii) Lateral surface of both pyramid and cone converge at a point.
 A pyramid and a cone are different in the sense that
 (i) Base of the pyramid is always a polygon but base of the cone is always a circle.
 (ii) Lateral surfaces of a pyramid are always triangular but lateral surface of the cone is always curved.
 3. Euler's Formula for polyhedron :
 (No. of faces) + (No. of vertices) = (No. of edges) + 2
 L.H.S. = $12 + 17 = 29$
 R.H.S. = $22 + 2 = 24$
 $\therefore$ L.H.S. $\neq$ R.H.S.
 Thus Euler's formula for given number of faces, edges and vertices is not satisfied.
 Therefore there can not be polyhedra of the given number of faces, edges and vertices.

- $$\begin{aligned} 40 + V - 60 &= 2 \\ -20 + V &= 2 \\ V &= 2 + 20 \\ V &= 22. \end{aligned}$$

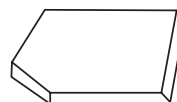
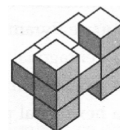
- 13. (c)** Cuboid
It has 8 vertices
Incorrect options
(1) Cone
It has only 1 (one) vertex.
(2) Cylinder
It has zero vertices.
(4) Pyramid
Here, no. of vertices depends on the shape of its base.
- 14. (a)** $F = n$
 $E = n + 1$
Using Eulers Formula,
 $F + V - E = 2$
 $V = E + 2 - F = n + 1 + 2 - n = 3$

Single Option Correct :

1. (b)
2. (a)
3. (a)
4. (d)
5. (b) Cylinder has 0 vertex
6. (b)
7. (a)
8. (a) Since, triangular Pyramid has triangular base.



9. (b) 5 faces
10. (b) Faces = 11
Vertices = 18
Using Euler's Formula,
 $F + V - E = 2$
 $E = F + V - 2$
 $= 11 + 18 - 2$
 $= 29 - 2 = 27$
11. (d) 9 faces.
(4 triangular face, 4 square faces and 1 square base)
12. (b) For any polyhedron,
 $F + V - E = 2$
Given,
 $F = 40$
 $E = 60$
 $V = ?$



- 16. (a)**

More than One Option Correct :

1. (a, b, c)
2. (a, c)

Assertion & Reason :

1. (c) Assertion is correct
Reason : $F + V - E = 2$
Where F = number of faces
 V = number of vertices
 E = number of edges
Hence, Reason is incorrect.
2. (a) Both Assertion & Reason are correct.
3. (d) Assertion : incorrect; Reason : correct
In a perfect die the sum of dots on opposite faces is seven.
The total numbers of dots on the opposite faces is not equal to seven i.e., $6 + 3 = 9 \neq 7$
 $4 + 1 = 5 \neq 7$
Hence, the given net is not a net of a perfect die.

Integer Type Questions :

1. (8) cubes

2. (5) $\therefore F = 4$ and $V = 3$

We have Euler formula as:

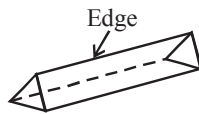
$$F + V = E + 2$$

$$\Rightarrow 4 + 3 = E + 2$$

$$\Rightarrow E = 5$$

3. (2) A square pyramid has 5 faces, 8 edges and 5 vertices.

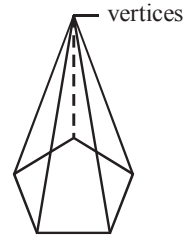
$$\text{So, } 5 + 5 - 8 = 2$$



4. (9)

Triangular prism
No. of edges = 9

5. (6)



Pentagonal pyramid
no of vertices = 6

6. (7) Euler's formula is given by

$$V + F = E + 2$$

$$\Rightarrow 5 + 4 = E + 2 \Rightarrow 9 = E + 2 \Rightarrow E = 9 - 2 = 7$$

7. (8) No of triangular faces = 8

Chapter

11

Mensuration

INTRODUCTION

In the broadest sense, mensuration is all about the process of measurement. Mensuration is based on the use of algebraic equations and geometric calculations to provide measurement data regarding the width, depth and volume of a given object or group of objects.

For example : You want to develop a house with 3 bedrooms, 1 hall and 1 kitchen on your plot of area equivalent to 1800 sqft. How would you decide on the sizes of the rooms, kitchens etc.?

You need to buy a water tank, to be kept on the roof of your house, in order to suffice your daily water requirements. What shape or volume you would like to go for?

The answers to all the above questions can be found out with the help of mensuration. In the present chapter, we will be studying all this and much more.

In our previous classes we have learnt how to find the area of a rectangle, a square, a triangle or a parallelogram. In this chapter, we shall learn to find the areas of some other simple. Closed figures such as rhombus, trapezium, quadrilateral etc.

1. Area of rectangle with sides l cm and b cm is given by $A = l \times b \text{ cm}^2$
2. Area of a square of side x cm is given by

$$A = x \times x = x^2 \text{ cm}^2$$

3. Area of a triangle of base b and altitude h cm is

$$A = \frac{1}{2} \times b \times h \text{ cm}^2$$

4. Area of a parallelogram of base b cm and height h cm is

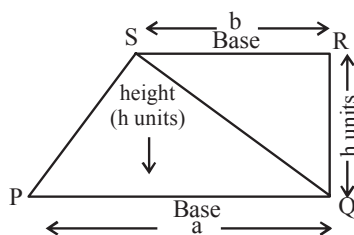
$$A = b \times h \text{ cm}^2$$

5. Area of circle $= \pi r^2 \text{ cm}^2$ where 'r' is the radius of circle.

AREA OF TRAPEZIUM

Trapezium is a quadrilateral which has only one pair of opposite sides as parallel. The parallel sides (PQ and RS in the figure) are called the **bases** of the trapezium and the perpendicular distance between them is called the **height** of the trapezium. Each diagonal divides the trapezium into two triangles.

In the given figure,



$$\text{Area of } \Delta PQS = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times PQ \times h = \frac{1}{2} \times a \times h$$

$$\text{Area of } \Delta QRS = \frac{1}{2} \times \text{base} \times \text{height}$$

$$= \frac{1}{2} \times RS \times h = \frac{1}{2} \times b \times h$$

$$\therefore \text{Area of trapezium PQRS} = \text{area of } \Delta PQS + \text{area of } \Delta QRS$$

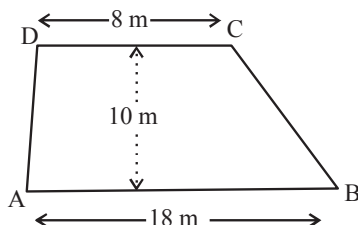
$$= \left(\frac{1}{2} \times a \times h \right) + \left(\frac{1}{2} \times b \times h \right)$$

$$= \frac{1}{2} \times h \times (a + b)$$

$$\text{Thus, the area of a trapezium} = \frac{1}{2} \times (\text{sum of the parallel sides}) \times \text{altitude}$$

ILLUSTRATION : 1

The parallel sides of a trapezium are 18 m and 8 m respectively and the distance between them is 10 m. What is the area of the trapezium?

SOLUTION :

Since, area of a trapezium $= \frac{1}{2} \times (\text{sum of the parallel sides}) \times \text{altitude}$

Here, the parallel sides are 18 m and 8 m. Altitude is 10 m.

$$\begin{aligned} \therefore \text{Area of trapezium ABCD} &= \frac{1}{2} \times (18 + 8) \times 10 \text{ m}^2 \\ &= \frac{1}{2} \times 26 \times 10 \text{ m}^2 \\ &= 13 \times 10 \text{ m}^2 = 130 \text{ m}^2 \end{aligned}$$

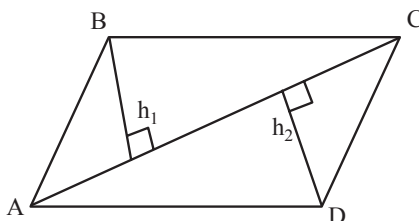
Hence, area of the trapezium is 130 m^2 .

AREA OF A GENERAL QUADRILATERAL

A general quadrilateral ABCD can be split into the triangles by drawing one of its diagonals.

Area of quadrilateral ABCD = (Area of $\triangle ABC$) + (Area of $\triangle ADC$)

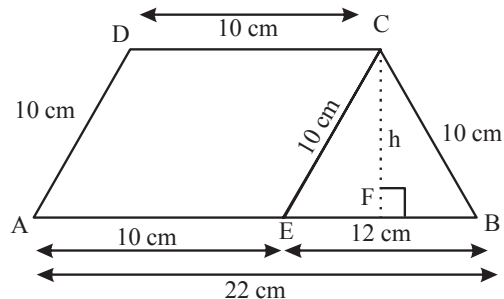
$$\begin{aligned} &= \left(\frac{1}{2} AC \times h_1 \right) + \left(\frac{1}{2} AC \times h_2 \right) \\ &= \frac{1}{2} AC \times (h_1 + h_2) \\ &= \frac{1}{2} d (h_1 + h_2) \text{ where } d \text{ is length of the diagonal } AC. \end{aligned}$$

**ILLUSTRATION : 2**

The parallel sides of an isosceles trapezium are 22 cm and 10 cm and the length of the non-parallel side is 10 cm. Find the area of the trapezium.

SOLUTION :

Let ABCD be a trapezium with



$$AB = 22 \text{ cm}, CD = 10 \text{ cm}$$

$$AD = BC = 10 \text{ cm}$$

Draw $CE \parallel AD$ and $CF \perp EB$

$$AB = AE + EB$$

$$\Rightarrow 22 = 10 + EB$$

$$\Rightarrow EB = 12 \text{ cm}$$

Since $CE \parallel AD$ and $AB \parallel CD$, AECD is a parallelogram

$$\therefore CE = 10 \text{ cm}$$

$\triangle CEB$ is an isosceles triangle as $CE = BC = 10 \text{ cm}$

$$\Rightarrow CF \text{ bisects } EB \text{ i.e. } EF = FB = \frac{1}{2} EB$$

$$\Rightarrow FB = 6 \text{ cm.}$$

In right $\triangle CFB$,

$$BC^2 = CF^2 + FB^2 \text{ (Pythagoras theorem)}$$

$$\Rightarrow (10)^2 = h^2 + (6)^2$$

$$\Rightarrow h^2 = 100 - 36$$

$$\Rightarrow h^2 = 64$$

$$\Rightarrow h = 8 \text{ cm.}$$

$$\text{Area of trapezium } ABCD = \frac{1}{2}(b_1 + b_2) \times h$$

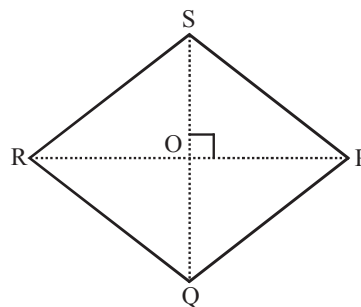
$$= \frac{1}{2}(22 + 10) \times 8 = \frac{1}{2} \times 32 \times 8$$

$$= 128 \text{ cm}^2.$$

AREA OF RHOMBUS

A rhombus is quadrilateral with all sides equal and the diagonals of rhombus bisect each other at right angles.

Again to find the area of a rhombus we shall split it into triangles by drawing the diagonals.



Consider a rhombus PQRS, join its diagonals PR and SQ. Let them intersect at a point O.

Since the diagonals of a rhombus bisect at right angles, $\angle SOP = 90^\circ$

Area of rhombus (PQRS) = Area ($\triangle SRP$) + Area ($\triangle PQR$)

$$\begin{aligned}
 &= \left(\frac{1}{2} \times PR \times SO \right) + \left(\frac{1}{2} \times PR \times OQ \right) \\
 &= \frac{1}{2} \times PR \times (SO + OQ) \\
 &= \frac{1}{2} \times PR \times SQ
 \end{aligned}$$

Area of rhombus = $\frac{1}{2} \times (d_1 \times d_2)$ where d_1 and d_2 are the diagonals.

ILLUSTRATION : 3

The area of rhombus is 28 cm^2 . If one of its diagonals is 7 cm , find the other.

SOLUTION :

We know area of rhombus = $\frac{1}{2} \times d_1 \times d_2$

$$\Rightarrow 28 = \frac{1}{2} \times 7 \times d_2 \Rightarrow d_2 = \frac{28 \times 2}{7}$$

$$\Rightarrow d_2 = 8 \text{ cm}$$

Thus, the other diagonals is 8 cm .

ILLUSTRATION : 4

Find the area of rhombus whose diagonals d_1 and d_2 are 4 cm & 6 cm respectively.

SOLUTION :

$$\text{Area of rhombus} = \frac{1}{2} \times (d_1 \times d_2)$$

$$= \frac{1}{2} \times (4 \times 6)$$

$$\Rightarrow \text{Area} = 12 \text{ cm}^2.$$

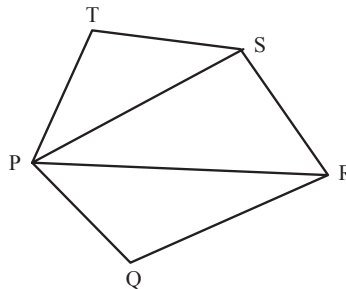
AREA OF POLYGON

We have already studied what a polygon is, it is a closed figure bounded by three or more straight lines. A polygon with 5 sides is called a pentagon and that with 6 sides is called a hexagon.

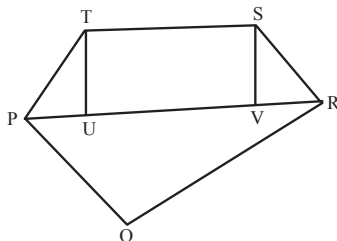
We know how to find area of a triangle. We also know how to find area of quadrilateral by splitting it into triangles. Similar methods can be used to find the area of a polygon. Consider following two cases.

(A) By drawing two diagonals PR and PS, the pentagon PQRST is divided into three triangles.

Area of pentagon PQRST = Area of $\triangle PQR$ + Area of $\triangle PRS$ + Area of $\triangle PST$.



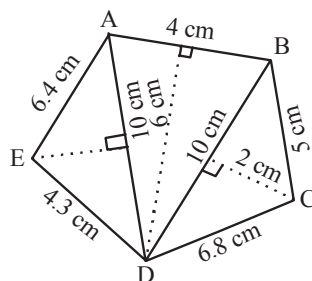
- (B) By drawing diagonal PR and two perpendiculars TU and SV on this diagonal, Pentagon PQRST is divided into four parts. This method is known as field.



Area of pentagon PQRST = Area of $\triangle PQT$ + Area of trapezium (TUVS) + Area of $\triangle SVR$ + Area of $\triangle PQR$.

ILLUSTRATION : 5

Find of area of the polygon given below :



SOLUTION :

The given polygon is divided into three triangle, namely, $\triangle ADE$, $\triangle ADB$, $\triangle BDC$.

Area of polygon ABCDE = area of $\triangle ADE$ + area of $\triangle ADB$ + area of $\triangle BDC$

$$\begin{aligned} &= \frac{1}{2} \times 2.5 \times 10 + \frac{1}{2} \times 6 \times 4 + \frac{1}{2} \times 10 \times 2 \\ &= 12.5 + 12 + 10 \\ &= 34.5 \text{ cm}^2. \end{aligned}$$

SOLID SHAPES

Solid : A solid is a figure which has three dimensions namely length, breadth (or width) and height (or thickness). The plane surfaces that bind it are called faces and the lines where faces meet are called edges.

The area of the plane surface that bind the solid is called its surface area.

The amount of space any solid figure occupy in three dimensional space is called its volume.

SURFACE AREA OF CUBOID, CUBE, CYLINDER

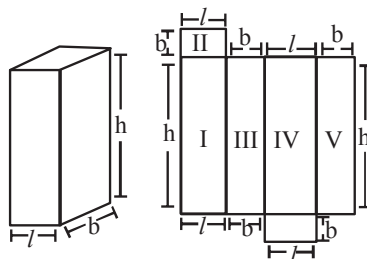
The total surface area of solid is the sum of the areas of its faces.

Cuboid

Surface Area of a Cuboid

The net of a cuboid is shown below :

The total surface area of a cuboid = area I + area II + area III + area IV + area V + area VI



$$= h \times l + b \times l + b \times h + l \times h + b \times h + l \times b$$

So, total surface area = $2(h \times l + b \times h + b \times l) = 2(lb + bh + hl)$

where h , l and b are the height, length and width of the cuboid respectively.

Lateral Surface Area of a Cuboid

The side walls (the faces excluding the top and bottom) is the lateral surface of the cuboid,

$$\therefore \text{Lateral surface area} = 2(h \times l + b \times h) = 2h(l + b)$$

Cube

In this case, its length breadth and height are the same.

So, surface area of cube = $2(\text{side} \times \text{side} + \text{side} \times \text{side} + \text{side} \times \text{side})$
 $= 2[3(\text{side})^2] = 6a^2$, where a is the side of the cube.

Lateral surface area of cube = $2(a + a) \times a = 4a^2$.

Cylinder

Let us consider a right circular cylinder of radius r and height h .

Cut it along a line on the surface parallel to axis (OO') of the cylinder and flatten this surface on a plane surface.

We will obtain a rectangle, whose length is equal to the

circumference of the base of the right circular cylinder and breadth is equal to the height of the cylinder.

Area of curved surface = Area of rectangle

= Circumference of base $\times$ height

$$= 2\pi r \times h = 2\pi rh.$$

Total surface area = Curved surface area + $2 \times$ Area of one circular end

$$= 2\pi rh + 2\pi r^2 = 2\pi r(r + h)$$

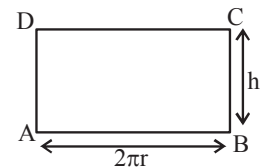
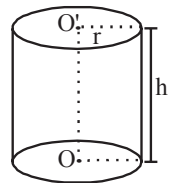
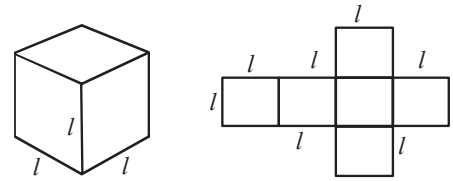


ILLUSTRATION : 6

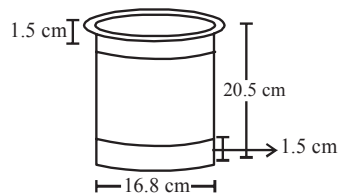
A company package its milk powder in cylindrical containers whose base has a diameter of 16.8 cm and height 20.5 cm. Company places a label around the curved surface of the container. If the label is placed 1.5 cm from the top and the bottom, what is the surface area of the label?

SOLUTION :

Clearly, surface area of the label is equal to the curved surface area of a cylinder of base radius $r = \frac{16.8}{2} \text{ cm} = 8.4 \text{ cm}$ and height $h = (20.5 - 1.5 - 1.5) \text{ cm} = 17.5 \text{ cm}$.

$$\therefore \text{Surface area of the label} = 2\pi rh = 2 \times \frac{22}{7} \times 8.4 \times 17.5 \text{ cm}^2$$

$$= 2 \times 22 \times 1.2 \times 17.5 \text{ cm}^2 = 924 \text{ cm}^2.$$



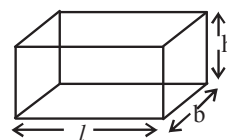
VOLUME

All three dimensional figures occupy some space. The measure of the space occupied by a solid is termed as volume.

Volume of a Cuboid

$$\text{Volume of a cuboid} = \text{Area of the base} \times \text{height}$$

$$= l \times b \times h$$



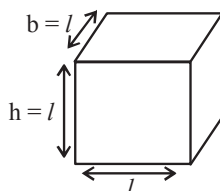
Volume of a Cube

A cube is a special type of cuboid where length, breadth and height are equal.

In a cube, $l = b = h$

Thus volume of cube $= l \times l \times l$

$\Rightarrow$ Volume of a cube $= l^3$.



Volume of a Cylinder

A cylinder is similar to a cuboid as both have got a top and a base which are congruent and parallel to each other. Also, their lateral surface is perpendicular to the base.

Thus, calculating the volume of cylinder is identical to that of a cuboid.

We know, volume of a cuboid $= \text{area of base} \times \text{height} = l \times b \times h = lbh$

Therefore, volume of cylinder $= \text{area of base} \times \text{height}$
 $= \text{area of circular cross section} \times \text{height}$
 $= \pi r^2 \times h$

$\Rightarrow$ volume of cylinder $= \pi r^2 h$

ILLUSTRATION : 7

A rectangular sheet of paper $44 \text{ cm} \times 18 \text{ cm}$ is rolled along its length and a cylinder is formed. Find the volume of the cylinder.
 (Use $\pi = 22/7$)

SOLUTION :

When the rectangular sheet is rolled along its length, we find that the length of the sheet forms the circumference of its base and breadth of the sheet becomes the height of the cylinder.

Let r cm be the radius of the base and h cm be the height. Then, $h = 18$ cm.

Now, circumference of the base = Length of the sheet

$\Rightarrow$ Circumference $= 44$ cm

$\Rightarrow 2\pi r = 44$

$\Rightarrow 2 \times \frac{22}{7} \times r = 44$

$\Rightarrow r = 7$ cm

$\therefore$ Volume of the cylinder $= \pi r^2 h \text{ cm}^3$

$$= \frac{22}{7} \times (7)^2 \times 18 \text{ cm}^3 = 2772 \text{ cm}^3$$

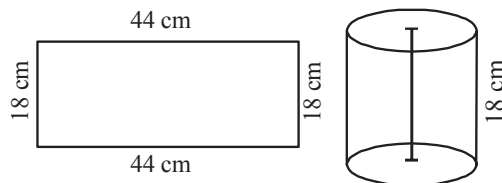


ILLUSTRATION : 8

How many 3 metre cubes can be cut from a cuboid measuring $18 \text{ m} \times 12 \text{ m} \times 9 \text{ m}$?

SOLUTION :

We have,

Edge of the each cube $= 3$ m

$\therefore$ Volume of each cube $= (\text{edge})^3 = (3^3) \text{ m}^3 = 27 \text{ m}^3$

Volume of the cuboid $= (18 \times 12 \times 9) \text{ m}^3 = 1944 \text{ m}^3$

$\therefore$ Number of cubes $= \frac{\text{Volume of the cuboid}}{\text{Volume of each cube}} = \frac{1944}{27} = 72$

CONNECTING TOPIC

AREA RELATED TO A CIRCLE

Circle

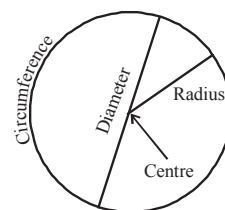
A circle is a path in a plane travelled by a point which moves in such a way that its distance from a fixed point is always constant. The fixed point is called centre of circle and fixed distance is called radius of the circle.

Circumference or perimeter of circle of radius ' r ', is

$$c = 2\pi r = \pi d \quad (2r = d, d = \text{diameter})$$

$$\begin{aligned} \text{Area of circle of radius 'r'} &= \pi r^2 = \frac{\pi d^2}{4} = \frac{(\pi d)^2}{4\pi} = \frac{c^2}{4\pi} \quad [\because c = \text{circumference}] \\ &= \frac{c^2}{2.2\pi r} \times r \end{aligned}$$

$$\text{Area of circle} = \frac{c^2}{2c} \times r = \frac{1}{2} \times c \times r$$

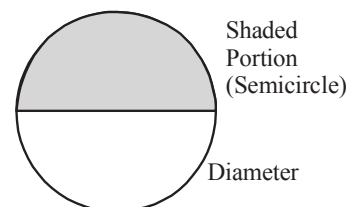


Semi-Circle

A semicircle is a figure enclosed by a diameter and part of circumference of the circle cut off by it.

$$\text{Area of semicircle of radius 'r'} = \frac{\pi r^2}{2}$$

$$\text{Circumference of semicircle of radius 'r'} = \pi r + 2r = r(\pi + 2)$$



Sector of Circle

Sector is the portion of a circle enclosed by two radii and arc cut by two radii of the circle. OACB is a sector of circle.

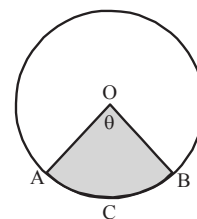
Let radius of circle = ' r '

Circumference ($2\pi r$) makes an angle 360° at the centre.

$$\therefore \text{Length of arc ACB (which make angle } \theta \text{ at centre)} = 2\pi r \times \frac{\theta}{360}$$

$$\text{Area of sector OACB} = \frac{1}{2} \times \text{arc (ACB)} \times \text{radius} = \frac{1}{2} \times \left(2\pi r \times \frac{\theta}{360^\circ} \right) \times r$$

$$\text{Area of sector OACB} = (\pi r^2) \times \frac{\theta}{360^\circ}$$



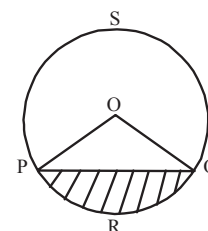
Segment

A segment of a circle is a region enclosed by a chord and an arc which it cuts off by the chord of the circle.

Any chord of a circle which is not a diameter (such as PQ) divides the circle into two segments, one greater (major segment) and one less (minor segment) than a semi-circle.

$$\text{Area of Segment PQR (Minor Segment)} = \text{Area of sector OPRQ} - \text{Area of } \triangle OPQ$$

$$\text{Area of Segment PSQ (Major Segment)} = \text{Area of Circle} - \text{Area of Segment PQR}$$



MISCELLANEOUS

SOLVED EXAMPLES

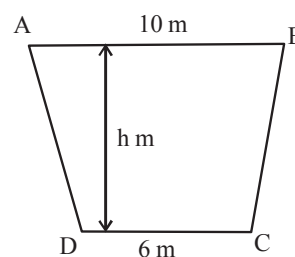
1. The cross section of a canal is trapezium in shape. If the canal is 10 m wide at the top and 6 m wide at the bottom and area of cross section is 800 m^2 , find the depth of the canal.

Sol. Let height of trapezium cross section ABCD = h 'm'

$$\text{Area of trapezium} = \frac{1}{2} \times (AB + CD) \times h \text{ m}^2,$$

$$800 = \frac{1}{2} \times (10 + 6) \times h$$

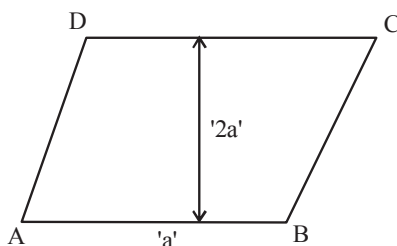
$$\Rightarrow h = \frac{800 \times 2}{16} \text{ m} = 100 \text{ m}$$



2. The area of a parallelogram is 338 m^2 . If its altitude is twice the corresponding base, determine the base and the altitude.

Sol. Let base of parallelogram = 'a' m

$\therefore$ Corresponding altitude = '2a' m



Area of parallelogram ABCD = AB $\times$ corresponding altitude

$$338 = a \times 2a$$

$$\Rightarrow a^2 = 169$$

$$\Rightarrow a = 13 \text{ m}$$

Corresponding altitude of parallelogram = $13 \times 2 = 26 \text{ m}$.

3. The diagonals of a rhombus are 72 cm and 96 cm. Find the area, side and height of the rhombus.

Sol. Let each side of rhombus = 'a' cm

Given AC = 72 cm, BD = 96 cm

Let height of rhombus

CE = 'h' cm

From $\triangle AOB$

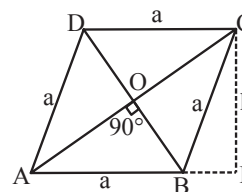
$$AB^2 = AO^2 + BO^2$$

$$a^2 = \left(\frac{AC}{2}\right)^2 + \left(\frac{BD}{2}\right)^2$$

$$\Rightarrow a^2 = \left(\frac{72}{2}\right)^2 + \left(\frac{96}{2}\right)^2$$

$$a = \frac{1}{2} \sqrt{72^2 + 96^2} = \frac{1}{2} \times 120 = 60 \text{ cm}$$

$\therefore$ side of rhombus = 60 cm.



$$\text{Area of rhombus} = \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 72 \times 96 = 36 \times 96 = 3456 \text{ cm}^2$$

$$\text{Also, area of rhombus} = AB \times CE = a \times h = 60h = 3456 \text{ cm}^2$$

$$\Rightarrow h = \frac{3456}{60} \text{ cm}$$

$$h = 57.6 \text{ cm}$$

4. The length of the floor of a rectangular hall is 10 m more than its breadth. If 34 carpets of size 6 m × 4 m are required to cover the floor of the hall, then find the length and breadth of the hall.

Sol. Let length and breadth of rectangular hall be l and b . It is given $l = (b + 10)$

Area of 34 carpets = Area of floor of hall

$$34 \times 6 \times 4 = b \times (b + 10)$$

$$\Rightarrow b \times (b + 10) = 24 \times (24 + 10) \quad [\because 34 = 24 + 10]$$

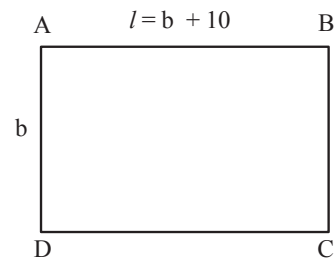
From above $b = 24$ m

and $l = b + 10$

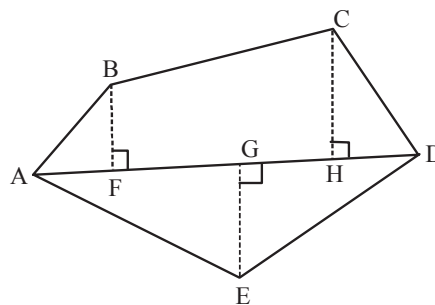
$$l = 24 + 10 = 34 \text{ m}$$

$\therefore$ Length of rectangular hall = 34 m

Breadth of rectangular hall = 24 m



5. Find the area of the pentagon ABCDE shown below if $AD = 8$ cm, $AH = 6$ cm, $AG = 4$ cm, $AF = 3$ cm, $BF = 2$ cm, $CH = 3$ cm, and $EG = 2.5$ cm.



Sol. Area of pentagon ABCDE = Area of $\triangle AFB$ + Area of trapezium BCHF + Area of $\triangle CHD$ + Area of $\triangle AED$.

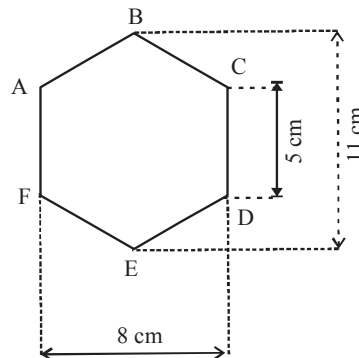
$$= \frac{1}{2} \times (AF \times BF) + \frac{1}{2} (BF + CH) \times (AH - AF) + \frac{1}{2} CH \times (AD - AH) + \frac{1}{2} (AD \times EG)$$

$$= \frac{1}{2} (3 \times 2) + \frac{1}{2} (2 + 3) \times (6 - 3) + \frac{1}{2} \times 3 \times (8 - 6) + \frac{1}{2} (8 \times 2.5)$$

$$= 3 + \frac{15}{2} + 3 + 10 = \frac{6 + 15 + 6 + 20}{2} = \frac{47}{2} = 23.5 \text{ cm}^2$$

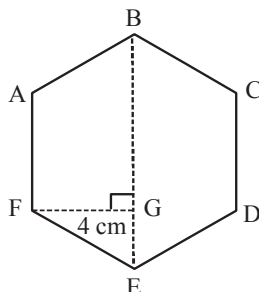
$\therefore$ Area of Pentagon = 23.5 cm^2

6. Figure shows a regular hexagon ABCDEF of side 5 cm. Find its area in two different ways.



Sol. Method - I : We join BE, it will divide the hexagon into two trapeziums of equal areas.

$$\therefore \text{Area of hexagon ABCDEF} = \text{Area of Trapezium ABEF} + \text{Area of Trapezium BCDE} = 2 \times (\text{Area of Trapezium ABEF})$$

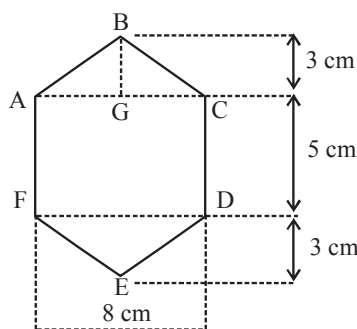


$$\text{Area of hexagon ABCDEF} = 2 \times \left(\frac{1}{2} \times (AF + BE) \times FG \right) = 2 \times \frac{1}{2} (5 + 11) \times 4$$

$$\text{Area of hexagon ABCDEF} = 64 \text{ cm}^2$$

Method-II Here we join AC and FD. The hexagon is now divided into two triangles ABC and FED of equal areas and a rectangle ACDF

$$\text{Area of hexagon ABCDEF} = 2 \times \text{Area } \triangle ABC + \text{Area of rectangle ACDF}$$



$$= 2 \times \frac{1}{2} \times AC \times BG + AF \times FD = 8 \times 3 + 5 \times 8 = 24 + 40 = 64 \text{ cm}^2$$

7. A circular plot covers an area of 154 m². How much wire is required for fencing the plot ?

Sol. Let radius of circular plot = 'r' m

$$\therefore \text{Area of circular plot} = \pi r^2 = 154 \text{ (Given)}$$

$$r^2 = \frac{154 \times 7}{22} = 49 \Rightarrow r = 7 \text{ m}$$

$$\therefore \text{Length of wire required for fencing} = \text{circumference of circular plot} = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44 \text{ m}$$

8. Find the length of a rope by which a buffalo must be tethered in order that she may be able to graze an area of 9856 m².

Sol. Let the length of rope be 'ℓ' m.

This length will act as radius of circle in which the buffalo move to graze.

$$\therefore \text{Area of circle (where buffalo is able to graze)} = \pi \ell^2$$

$$9856 = \frac{22}{7} \times \ell^2 \Rightarrow \ell = \sqrt{\frac{9856 \times 7}{22}}, \ell = \sqrt{448 \times 7} = \sqrt{64 \times 7 \times 7} = \sqrt{(8)^2 \times (7)^2} = 7 \times 8 = 56 \text{ m}$$

9. How many revolutions will a wheel make in travelling 528 m if its diameter measures 0.7 m?

Sol. Let the no. of revolutions made by wheel = 'n'
 Distance travelled in one revolution = circumference of wheel
 $= \pi \times d$ (d = diameter of wheel)
 $= \frac{22}{7} \times 0.7 = 2.2 \text{ m}$
 Now,
 No. of revolutions $\times$ Distance traveled in one revolution = Total distance travelled
 $\Rightarrow$ No. of revolution
 $= \frac{\text{Total distance travelled}}{\text{Distance travelled in one revolution}} = \frac{528}{2.2} = 240 \text{ revolutions}$

10. The surface area of a cube is 216 sq. cm. Find its volume.

Sol. Let each side of cube be 'a'
 $\therefore$ surface area of cube $= 6a^2$
 $6a^2 = 216$
 $\Rightarrow a^2 = 36$
 $\Rightarrow a = 6 \text{ cm}$
 Volume of cube $= a^3 \text{ cm}^3 = (6)^3 = 216 \text{ cm}^3$

11. On a rainy day 60 mm of rain falls, find how many cubic metres of water falls on 3 hectares of ground on that day.

Sol. Height (h) of rain water fall $= 60 \text{ mm} = \frac{60}{1000} \text{ m}$
 Area of ground (A) $= 3 \text{ hectares} = 3 \times 10000 \text{ m}^2$ (1 hectare $= 10000 \text{ m}^2$)
 $\therefore$ Volume of water fall $= \text{Area of ground} \times \text{height} = 3 \times 10000 \times \frac{60}{1000} = 30 \times 60 = 1800 \text{ m}^3$

12. Volume of two cubes are in the ratio 1 : 27, find the ratio of their surface areas.

Sol. Let sides of two cubes be a_1 and a_2 respectively.

$$\text{Volume of first cube } (V_1) = a_1^3$$

$$\text{Volume of second cube } (V_2) = a_2^3$$

$$\therefore \frac{V_1}{V_2} = \frac{a_1^3}{a_2^3} = \left(\frac{a_1}{a_2} \right)^3 = \frac{1}{27}$$

$$\therefore \frac{a_1}{a_2} = \frac{1}{3} \quad \dots (1)$$

$$\text{Now, surface area of first cube } (S_1) = 6a_1^2$$

$$\text{surface area of second cube } (S_2) = 6a_2^2$$

$$\text{Now, } \frac{S_1}{S_2} = \frac{6a_1^2}{6a_2^2} = \left(\frac{a_1}{a_2} \right)^2$$

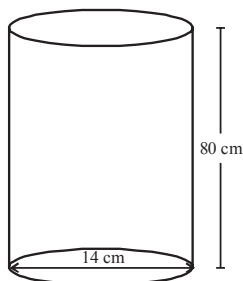
$$\frac{S_1}{S_2} = \left(\frac{1}{3} \right)^2$$

$$\frac{S_1}{S_2} = \frac{1}{9}$$

$\therefore$ Ratio of surface area $= 1 : 9$.

13. The height of a cylinder is 80 cm and diameter of the base is 14 cm. Find its curved surface, total surface and volume.

Sol. Height (h) = 80 cm, Diameter of base = 14 cm



$$\therefore \text{radius of base (r)} = \frac{14}{2} = 7 \text{ cm}$$

$$\therefore \text{curved surface area of cylinder} = 2\pi rh = 2 \times \frac{22}{7} \times 7 \times 80 = 44 \times 80 = 3520 \text{ sq. cm}$$

Total surface area (TSA) = C.S.A + Area of two ends

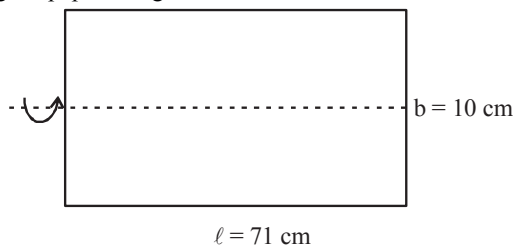
$$= 2\pi rh + 2\pi r^2 = 2\pi r(h + r) = 2 \times \frac{22}{7} \times 7 \times (80 + 7) = 2 \times 22 \times 87$$

T.S.A of cylinder = 3828 sq. cm

$$\text{Volume (V) of cylinder} = \pi r^2 h = \frac{22}{7} \times (7)^2 \times 80 = 22 \times 7 \times 80 = 154 \times 80 = 12320 \text{ cubic units.}$$

14. A rectangular piece of paper is 71 cm long and 10 cm wide. A cylinder is formed by rolling the paper along its breadth. Find the volume of the cylinder.

Sol. As the cylinder is made by rolling the paper along its breadth.



$\therefore$ Circumference of base of cylinder = Width of paper

$$\Rightarrow 2\pi r = 10 \quad (r = \text{radius of base of cylinder})$$

$$\Rightarrow r = \frac{5}{\pi} \text{ cm}$$

Now, length of paper = height of cylinder = 71 cm

$$\therefore \text{Volume of cylinder (V)} = \pi r^2 \ell = \pi \times \frac{25}{\pi \times \pi} \times 71 = \frac{25 \times 7 \times 71}{22}$$

Volume of cylinder (V) = 564.78 cubic cm.

15. Two circular cylinders of equal volume have their heights in the ratio 9 : 16. Find the ratio of their radii.

Sol. Let the radius and height of two cylinders are r_1 , h_1 and r_2 , h_2 respectively. Let v_1 and v_2 are their volumes.

$$\text{Now, } v_1 = \pi r_1^2 h_1 \text{ and } v_2 = \pi r_2^2 h_2$$

It is given that $v_1 = v_2$

$$\therefore \pi r_1^2 h_1 = \pi r_2^2 h_2$$

$$\frac{r_1^2}{r_2^2} = \frac{h_2}{h_1}$$

$$\Rightarrow \frac{r_1}{r_2} = \sqrt{\frac{h_2}{h_1}} = \sqrt{\frac{16}{9}} = \frac{4}{3}$$

$$\therefore r_1 : r_2 = 4 : 3$$

16. How many coins 1.4 cm in diameter and 0.4 cm thick is to be melted to form a right circular cylinder of height 16 cm and diameter 3.5 cm?

Sol. Let the number of required coins = 'n'

$$\therefore n \times \text{volume of one coin } (v_{co}) = \text{volume of cylinder } (v_{cy})$$

$$\text{Now, } v_{co} = \pi r_{co}^2 \times h_{co} \quad \left[\begin{array}{l} r_{co} = \text{radius of coin} \\ h_{co} = \text{height/thickness of coin} \end{array} \right]$$

$$= \pi \times \frac{1.4}{2} \times \frac{1.4}{2} \times 0.4$$

$$v_{co} = 0.196 \pi \text{ cubic cm}$$

$$v_{cy} = \pi r_{cy}^2 h_{cy} \quad \left[\begin{array}{l} r_{cy} = \text{radius of cylinder} \\ h_{cy} = \text{height of cylinder} \end{array} \right]$$

$$v_{cy} = \pi \times \frac{3.5}{2} \times \frac{3.5}{2} \times 16$$

$$\therefore n \times 0.196 \pi = 49 \pi$$

$$n = \frac{49}{0.196} = 250$$

Hence, number of coins = 250

SOLVED EXAMPLES BASED ON CONNECTING TOPICS

17. If a piece of wire 20 cm long is bent into an arc of a circle subtending an angle of 60° at the centre. Find the radius of the circle.

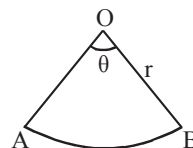
Sol: Length of wire = length of arc

Let radius of arc of circle = 'r' cm

Length of arc making angle ' θ ' at the centre

$$= 2\pi r \times \frac{\theta}{360} \quad [\theta = \text{central angle}]$$

$$20 = 2 \times \frac{22}{7} \times r \times \frac{60}{360} \Rightarrow r = \frac{20 \times 7 \times 6}{2 \times 22} \Rightarrow r = \frac{210}{11} \text{ cm.}$$



18. Find the area of a sector of a circle whose radius is 10 cm and the length of the arc is 13 cm.

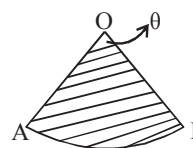
Sol: Length of arc which make angle ' θ ' at the centre = $2\pi r \times \frac{\theta}{360^\circ}$

r = radius of the circle

Area of sector made by above arc

$$= \pi r^2 \times \frac{\theta}{360^\circ} = \frac{1}{2} \left(2\pi r \times \frac{\theta}{360^\circ} \right) \times r$$

$$= \frac{1}{2} \times (\text{length of arc}) \times r = \frac{1}{2} \times 13 \times 10 = 65 \text{ cm}^2$$



1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- _____ is any figure bounded by four straight lines.
- Solids are _____ dimensional figures that have _____, _____ and _____.
- An edge forms when two _____ meet.
- A cube is a special type of _____.
- A regular polygon of n sides can be divided into _____ triangles of equal areas.
- In a _____, if all the dimensions are equal then the solid is called a _____.
- All the edges of a cube are equal in _____.
- The _____ of a closed plane figure is the length of its boundary.
- Each face of the cuboid is _____.
- If each side of a cube is doubled, its volume becomes _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- The face of a solid is two-dimensional in shape.
- If a solid has a curved surface then it has no faces.
- Volume of the solid is measured in cubic units.
- Area is the length of the boundary of a closed figure.
- Area is the total surface covered by a closed figure.
- The parallel sides are called bases of the trapezium.
- πr^2 is the area of a circle.
- πr is the circumference of a circle.
- The surface area of a chalk box whose length, breadth and height are 6 cm, 5 cm and 4 cm respectively is 148 cm^2 .
- The diameter of a cylinder is 14 cm and its height is 12 cm. The volume of the cylinder is 848 cm^3 .

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- Match the following columns :

Column-I	Column-II
(A) The area of triangle (in sq. m ²) whose base is 24 cm and whose altitude is 1.5 dm is	(p) 11
(B) The diameter of a wheel is 1.26 m. How far will it travel in 500 revolutions in metres?	(q) 34650
(C) The height (in cm) of a cuboid where volume is 275 cm^3 and base area is 25 cm^2 is	(r) 180
(D) The circumference of the base of the cylinder is 132 cm and its height is 25 cm. The volume of the cylinder (in cm ³) is	(s) 1980

- Match the following columns :

Column-I	Column-II
(A) The lateral surface area of the cube is	(p) $2\pi(r + h)$
(B) Area of circular base of cylinder is	(q) $l \times b \times h$
(C) Volume of a cuboid is	(r) $4P$
(D) Total surface area of the cylinder is	(s) πr^2

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

- Find the volume of a cylindrical can of height 21 cm and base radius 8 cm.
- What is the difference between Volume and Capacity?
- Find the volume in cu. dm of each of the cubes whose side is 1.2 m
- Find the volume of the drum if radius is 4.2 m and height is 3.5 m. (Take $\pi = 3.14$)
- How many paving stones each measuring $2.5 \text{ m} \times 2 \text{ m}$ are required to pave a rectangular courtyard 30 m long and 16.5 m wide?
- What will be the ratio of the circumference to the diameter of the circle if its original radius is tripled?

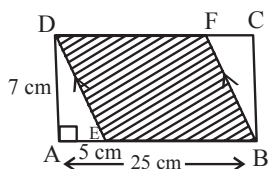
7. Do the following conversions :

- (i) 1 litre = _____ ml
- (ii) 5 litres = _____ dm^3
- (iii) 1 cm^3 = _____ ml
- (iv) 1 m^3 = _____ cm^3 = _____ l
- (v) 1 l = _____ cm^3

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- How do we find the “Area of a Polygon” ?
- Define a term ‘trapezium’.
- Two sides of a triangle are 85 m and 154 m, and the Perimeter is 324 m. Find the Area of the triangle.
- The parallel sides of a trapezium are in the ratio 3 : 5. If the distance between the parallel sides is 8 cm and its area is 128 cm^2 , Find the lengths of the parallel sides.
- There is cuboidal log of wood of size 2.5 m by 50 cm by 50 cm. Assuming that there is no wastage, it is cut into cubes of side 25 cm. Find the number of such cubes.
- A godown has dimension 7 m $\times$ 4.5 m $\times$ 2 m. How many cartons of dimension 70 cm $\times$ 22.5 cm $\times$ 40 cm can be stored in it ?
- Water in a canal, 6 m wide and 1.5 m deep, is flowing with a speed of 10 km/h. How much area will it irrigate in 30 minutes, if 8 cm of standing water is needed ?
- Find the area of the shaded regions :

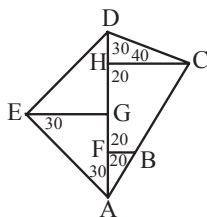


- A solid cube is cut into two cuboids of equal volumes. Find the ratio of the total surface area of the given cubes and that of one of the cuboids.
- Eight identical cuboidal wooden blocks are stacked one on top of the other. The total volume of the solid so formed is 128 cm^3 . If the height of each block is 1 cm and the base is a square, find the dimension of each block.
- If the length of a rectangle increases by 10% and the breadth of the rectangle decreases by 12% then find the % change in area.

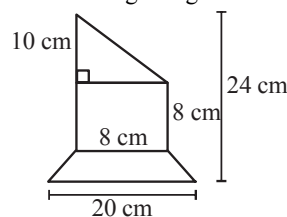
Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

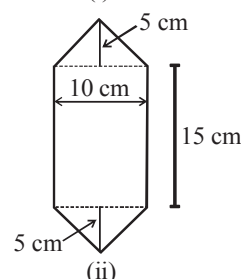
- Find the area of the following field. All dimensions are in metres.



- Find the areas of the figures given below:

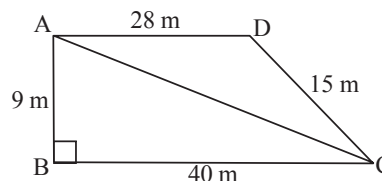


(i)

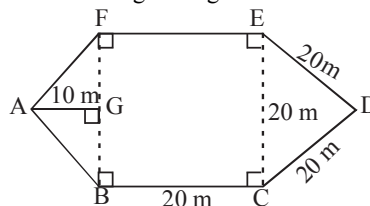


(ii)

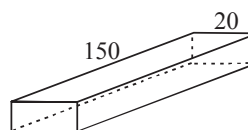
- Students of a school stage a rally for cleanliness campaign. They walked through the lanes in two groups. One group walked through the lanes AB, BC and CA; while the other through AC, CD and DA (see figure). Then they cleaned the area enclosed within their lanes. If $AB = 9 \text{ m}$, $BC = 40 \text{ m}$, $CD = 15 \text{ m}$, $DA = 28 \text{ m}$ and $\angle B = 90^\circ$, which group cleaned more area and by how much ? Find the total area cleaned by the students.



- How many cubic metres of earth must be dug out to sink a well 7 metres deep and diameter 28 m ? Find the cost of plastering its curved surface at ₹ 3.50 per sq. m.
- Find the area of the given figure:

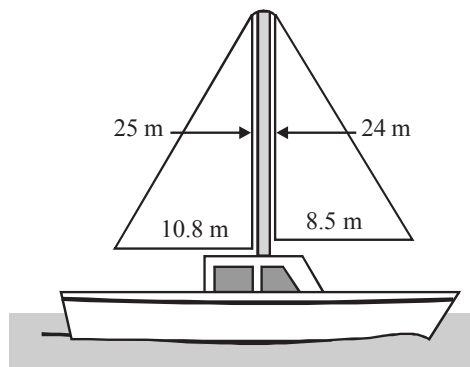


- A water tank built by a municipality of a town to supply water to its 25000 inhabitants at 125 litres per day per person is 40 m long and 31.25 m broad. The tank, when it is full, can supply water for two days to the inhabitants of the town. Find the depth of the tank.



7.

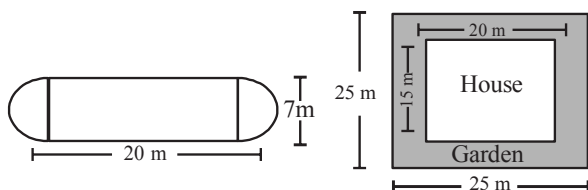
- (i) Given Diagram shows a rectangular trough of length 150 cm and width 20 cm.
The trough was completely filled with 48000 cm^3 of water from the tank.
Calculate the depth of the trough.
- (ii) After the trough had been filled, water started to leak from the tank. In 2 hours 30 minutes it was found that $20,000 \text{ cm}^3$ ran out of the tank.
Calculate the rate at which the level of water in the tank was falling.
Express your answer in centimeters per hour.
(The value of π is 3.142 correct to three decimal places).
8. The sailboat has 2 sails. What is the total area of material in the sails?



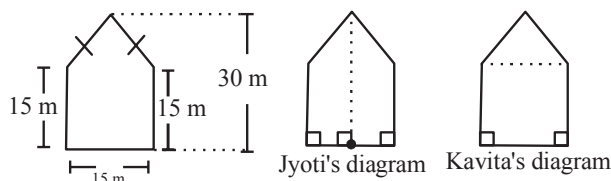
2 EXERCISE

Text-Book Exercise :

1. The shape of a garden is rectangular in the middle and semi circular at the ends as shown in the diagram. Find the area and the perimeter of the garden [Length of rectangle is $20 - (3.5 + 3.5)$ metres.]

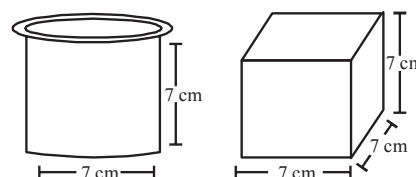


2. A flooring tile has the shape of a parallelogram whose base is 24 cm and the corresponding height is 10 cm. How many such tiles are required to cover a floor of area 1080 m^2 ? (If required you can split the tiles in whatever way you want to fill up the corners).
3. The area of a trapezium is 34 cm^2 and the length of one of the parallel sides is 10 cm and height is 4 cm. Find the length of the other parallel side.
4. The floor of a building consists of 3000 tiles which are rhombus shaped and each of its diagonals are 45 cm and 30 cm in length. Find the total cost of polishing the floor, if the cost per m^2 is ₹ 4.
5. There is a pentagonal shaped park as shown in the figure. For finding its area Jyoti and Kavita divided it in two different ways.

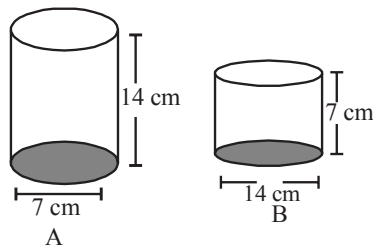


Find the area of this park using both ways. Can you suggest some other way of finding its area?

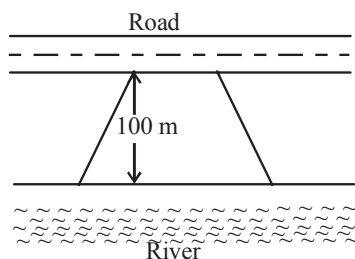
6. The internal measures of a cuboidal room are $12 \text{ m} \times 8 \text{ m} \times 4 \text{ m}$. Find the total cost of white washing all four walls of a room, if the cost of white washing is ₹ 5 per m^2 . What will be the cost of white washing if the ceiling of the room is also white washed.
7. In a building there are 24 cylindrical pillars. The radius of each pillar is 28 cm and height is 4 m. Find the total cost of painting the curved surface area of all pillars at the rate of ₹ 8 per m^2 .
8. Daniel is painting the walls and ceiling of a cuboidal hall with length, breadth and height of 15 m, 10 m and 7 m respectively. From each can of paint 100 m^2 of area is painted. How many cans of paint will she need to paint the room?



9. The lateral surface area of a hollow cylinder is 4224 cm^2 . It is cut along its height and formed a rectangular sheet of width 33 cm. Find the perimeter of rectangular sheet.
10. A rectangular paper of width 14 cm is rolled along its width and a cylinder of radius 20 cm is formed. Find the volume of the cylinder. (Take $\frac{22}{7}$ for π)
11. Diameter of cylinder A is 7 cm, and the height is 14 cm. Diameter of cylinder B is 14 cm and height is 7 cm. Without doing any calculations can you suggest whose volume is greater? Verify it by finding the volume of both the cylinders. Check whether the cylinder with greater volume also has greater surface area?



12. A milk tank is in the form of cylinder whose radius is 1.5 m and length is 7 m. Find the quantity of milk in litres that can be stored in the tank?
13. Mohan wants to buy a trapezium shaped field. Its side along the river is parallel to and twice the side along the road. If the area of this field is 10500 m^2 and the perpendicular distance between the two parallel sides is 100 m, find the length of the side along the river.

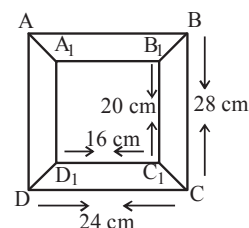


Exemplar Questions :

1. The walls and ceiling of a room are to be plastered. the length, breadth and height of the room are 4.5 m, 3 m and 350 cm respectively. Find the cost of plastering at the rate of ₹ 8 per m^2 .
2. The area of a trapezium with equal non – parallel sides is 168 m^2 . If the lengths of the parallel sides are 36 m and 20 m, find the length of the non – parallel sides.
3. There is a circular pond and a footpath runs along its boundary. A person walks around it, exactly once keeping close to the edge. if his step is 66 cm long and he takes exactly 400 steps to go around the pond, find the diameter of the pond.
4. The thickness of a hollow metallic cylinder is 2 cm. It is 70 cm long with outer radius of 14 cm. Find the volume of the metal used in making the cylinder, assuming that it is open at both the ends. Also find its weight if the metal weighs 8 g per cm^3 .
5. A carpenter makes a box which has a volume of $13,400 \text{ cm}^3$. The base has an area of 670 cm^2 . What is the height of the box?
6. A wooden box (including the lid) has external dimensions 40 cm by 34 cm by 30 cm. If the wood is 1 cm thick, how many cm^3 of wood is used in it?
7. A truck carrying 7.8 m^3 concrete arrives at a job site. A platform of width 5 m and height 2 m is being constructed at the site. Find the length of the platform, constructed from the amount of concrete on the truck?
8. A housing society consisting of 5,500 people needs 100 L of water per person per day. The cylindrical supply tank is 7 m high and has a diameter 10 m. For how many days will the water in the tank last for the society?
9. External dimensions of a closed wooden box are in the ratio 5 : 4 : 3. If the cost of painting its outer surface at the rate of ₹ 5 per dm^2 is ₹ 11,750, find the dimensions of the box.
10. Four times the area of the curved surface of a cylinder is equal to 6 times the sum of the areas of its bases. If its height is 12 cm, find its curved surface area.
11. A rectangular examination hall having seats for 500 candidates has to be built so as to allow 4 cubic metres of air and 0.5 square metres of floor area per candidate. If the length of hall be 25 m, find the height and breadth of the hall.

HOTS Questions :

1. The diameter of a wheel is 98 cm. How many times would the wheel rotate to cover 7.7 km? Find also the distance covered by the wheel in 500 revolutions.
2. A rope to which a calf is tied is increased in length from 12 metres to 23 metres. Find the area of additional ground which it can graze.
3. The length of a rectangle is increased by 33.33%. By what percent should the width be decreased to maintain the same area?
4. Diagram shown in Fig. of the adjacent of the picture frame has outer dimensions $24 \text{ cm} \times 28 \text{ cm}$ and inner dimensions $16 \text{ cm} \times 20 \text{ cm}$. Find the area of each section of the frame, if the width of each section is same.



5. A lead pencil consists of a cylinder of wood with a solid cylinder of graphite filled into it. The diameter of the pencil is 7 mm, the diameter of the graphite is 1 mm and the length of the pencil is 10 cm. Calculate the weight of the whole pencil, if the specific gravity of the wood is

0.7 gm/cm³ and that of the graphite is 2.1 gm/cm³.

6. If V is the volume of cuboid of dimensions a, b, c and S is

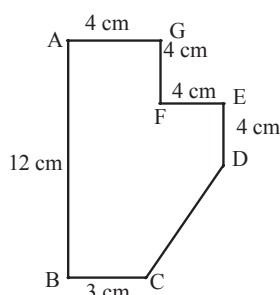
its surface area, then prove that $\frac{1}{V} = \frac{2}{S} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right)$.

3 EXERCISE

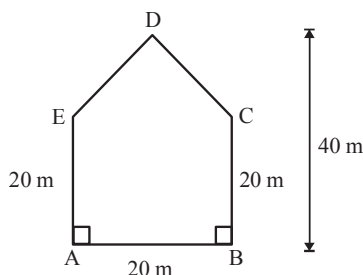
Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Two adjacent sides of a parallelogram are 60 m and 40 m and one of the diagonals is 80 m long. The area (in sqm) of the parallelogram is
(a) $800\sqrt{15}$ (b) $700\sqrt{15}$
(c) $600\sqrt{15}$ (d) $650\sqrt{15}$
- A triangle and parallelogram have the same base and same area. If the sides of the triangle are 26 cm, 28 cm and 30 cm, and the parallelogram stands on the base 28 cm, then the height of the parallelogram is
(a) 15 cm (b) 12 cm
(c) 16 cm (d) 14 cm
- Area of the field shown in the figure given below is:

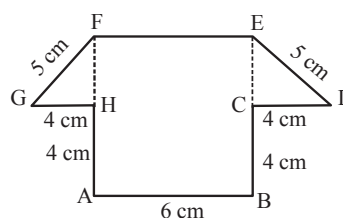


- (a) 68 cm² (b) 70 cm²
(c) 72 cm² (d) 75 cm²
4. Area of pentagonal park shown below is



- (a) 600 m² (b) 800 m²
(c) 450 m² (d) 700 m²

5. Area of the following shape is



- (a) 50 cm² (b) 54 cm²
(c) 60 cm² (d) 64 cm²
- A room of size 6.75 m long and 5.75 m wide is to be paved with square tiles. The minimum number of square tiles required is
(a) 630 (b) 430
(c) 621 (d) 421
 - Cost of levelling a square field at 15 paise per sq. meter is ₹ 135. The cost of fencing the same field if the unit cost of fencing is at 50 paise per meter ?
(a) ₹ 100 (b) ₹ 200
(c) ₹ 60 (d) ₹ 150
 - The area of a trapezium whose parallel sides are 77 cm and 60 cm and the other sides are 25 cm and 26 cm, is
(a) 1630 cm² (b) 1654 cm²
(c) 1644 cm² (d) 1674 cm²
 - The breadth of a room is twice its height and one half its length and the volume of the room is 512 cubic m. The height of the room is
(a) 6 m (b) 8 m
(c) 4 m (d) 10 m
 - A metallic sheet is of rectangular shape with measurements 48 cm × 36 cm. From each one of its corners a square of 8 cm is cut off. An open box is made of the remaining sheet. The volume of the box is
(a) 5020 cm³ (b) 1250 cm³
(c) 2150 cm³ (d) 5120 cm³
 - The amount of concrete required to build a concrete cylindrical pillar whose base has a perimeter of 8.8 m and whose curved surface area is 17.6 m², is
(a) 10 cu.m (b) 9.8 cu.m
(c) 12.5 cu.m (d) 12.32 cu.m

12. If the area of the three adjacent faces of a cuboidal box are 120 cm^2 , 72 cm^2 and 60 cm^2 respectively. The volume of the box is
 (a) 720 cm^3 (b) 780 cm^3
 (c) 728 cm^3 (d) 798 cm^3
13. Three cubes with sides in the ratio 3 : 4 : 5 are melted to form a single cube whose diagonal is $12\sqrt{3} \text{ cm}$. The sides of the cube are
 (a) 2 cm, 3 cm, 6 cm (b) 3 cm, 4 cm, 5 cm,
 (c) 6 cm, 8 cm, 10 cm (d) 4 cm, 6 cm, 8 cm
14. A field is 200 m long and 150 m broad. There is a plot 50 m long and 40 m broad, near the field. The plot is dug 7 m deep and the earth taken out is spread evenly on the field. By how many meters is the level of the field raised?
 (a) 0.25 m (b) 0.42 m
 (c) 0.49 m (d) 0.47 m
15. A two meter wide road is to be constructed surrounding a square plot of area 144 sq. m. If the unit cost of construction is ₹ 250 per sq.m., The total cost of construction is
 (a) ₹ 20,000 (b) ₹ 25,000
 (c) ₹ 28,000 (d) ₹ 30,000

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

1. Which of the following is/are related to cuboid?
 (a) A match box (b) A brick
 (c) A room (d) A kite
2. Which one of the following is/are correct?
 (a) Diameter of a circle = $2 \times$ radius
 (b) Area of a ring or circular track of radii R and $r = \pi(R^2 - r^2)$
 (c) The distance between the two parallel sides is called the altitude of the trapezium.
 (d) Perimeter of trapezium = Product of lengths of its sides
3. Which one is/are correct?
 (a) Total surface area of cuboid is $2(lb + bh + hl)$
 (b) Total surface area of a cube is $6l^2$
 (c) Area of four walls = $2h(l + b)$
 (d) Area of four walls = Height $\times$ Perimeter of the room
4. A Circular Cylinder can be separated into
 (a) circular end at the bottom
 (b) curved surface
 (c) circular end at the top
 (d) None of these
5. Cubes and Cuboids are the examples of
 (a) regular solids
 (b) rectangular solids
 (c) two-dimensional solids
 (d) None of these
6. The edges of a cuboid are in the ratio 1 : 2 : 3 and its surface area is 88 cm^2 . The length, breadth and height of cuboid is
 (a) 2 (b) 4
 (c) 6 (d) 8

7. Find the volume, curved surface area and the total surface area of a cylinder with diameter of base 7 cm and height 40 cm.
 (a) 1540 cm^3 (b) 880 cm^2
 (c) 957 cm^2 (d) 415 cm^3
8. The area of the three adjacent faces of a cuboidal box are 120 cm^2 , 72 cm^2 and 60 cm^2 respectively. The length, breadth and height of the box is
 (a) 8 cm (b) 10 cm
 (c) 12 cm (d) 6 cm

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

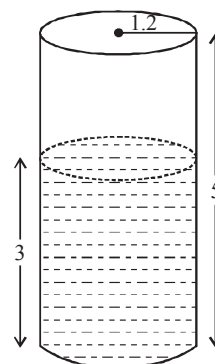
PASSAGE - I

A wire bent in the form of a square encloses an area of 5929 cm^2 .

1. The length of each side of the square is
 (a) 77 cm (b) 308 cm
 (c) 803 cm (d) 830 cm
2. The length of the wire is
 (a) 830 cm (b) 308 cm
 (c) 803 cm (d) 83 cm
3. The area enclosed, if the same wire is bent to form a circle, is
 (a) 5746 cm^2 (b) 7564 cm^2
 (c) 5764 cm^2 (d) 7546 cm^2

PASSAGE - II

A well is a vertical open cylinder of radius 1.2 m and height 5 m. The well contains water to a depth of 3 m. (use : $\pi = 3.142$)



4. The total internal area of the curved surface of the well and the bottom of the well is
 (a) 42.2 to 42.25 m^2 (b) 40.2 to 40.25 m^2
 (c) 24.2 to 24.25 m^2 (d) 42.25 to 42.00 m^2
5. The volume of water in the well, is ($1 \text{ m}^3 = 1000$ litres)
 (a) 13564 (b) 31560
 (c) 53160 (d) 51360

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** The length and breadth of a hall is 20 m by 10 m. 800 tiles of the size $50\text{ cm} \times 50\text{ cm}$ are required to pave the floor of the hall.

$$\text{Reason : Number of tiles} = \frac{\text{Area of hall}}{\text{Area of 1 tile}}$$

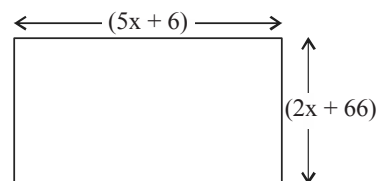
2. **Assertion :** If the side of a rhombus is 10 cm and one diagonal is 16 cm, then area of the rhombus is 96 cm^2 .
Reason : Area of rhombus = $2 \times$ area of triangle.
 3. **Assertion :** If the radius of a cylinder is doubled and curved surface area is not changed then height must be doubled.
Reason : If the radius of a cylinder is doubled and curved surface area is not changed then height must be halved.
 4. **Assertion :** The axis is always perpendicular to the base of a right circular cylinder.

Reason : The line segment joining the two ends of circular bases is called the axis of the cylinder.

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- The area of a rhombus whose diagonals are 10 cm and 12 cm is $x\text{ cm}^2$. The value of $x/15$ is
- The area of a triangular garden is 9520 m^2 . If its base is 340 m, the altitude is a m. The value of $\frac{a}{8}$ is
- Unit digit of the surface area of a chalk box, whose length, breadth and height are 16 cm, 8 cm and 6 cm respectively is
- The area of a parallelogram with base 14 cm and altitude 8 cm is $y\text{ cm}^2$. The value of $(y - 110)$ is
- For what value of x is the perimeter of shape 186 cm?



4

ADVANCED EXERCISE

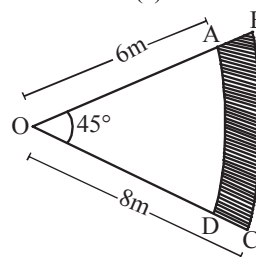
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1-5) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which **ONLY ONE** is correct.

- A chord of a circle of radius 14 cm makes a right angle at the centre. The areas of the minor and the major segment of the circle are respectively :
 (a) 56 cm^2 and 560 cm^2 (b) 560 cm^2 and 56 cm^2
 (c) 56 cm^2 and 616 cm^2 (d) None of these
- A horse is tethered to a corner of a field which is in the shape of an equilateral triangle. If the length of the rope through which it is tied be 7 m, then the area of the field over which it can graze is :
 (a) 49 sq. cm (b) $22\frac{2}{3} \text{ sq. cm}$
 (c) $25\frac{2}{3} \text{ sq. cm}$ (d) None of these
- Four equal sized maximum circular plates are cut off from a square paper sheet of area 784 cm^2 . The circumference of

each plate is

- 20 cm
 - 32 cm
 - 44 cm
 - 64 cm
- If the radius of a circle is increased by 1 cm, its area increases by 22 cm^2 , then original radius of the circle is
 (a) 4 cm (b) 3 cm
 (c) 3.5 cm (d) 5 cm
 - Area of the shaded region of the below given figure is
 (a) 10 m^2 (b) 11 m^2
 (c) 15 m^2 (d) 19 m^2



SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

1. Quadrilateral
2. 3, length, breadth, height
3. faces
4. cuboid
5. n -triangles
6. cuboid, cube
7. length
8. perimeter
9. rectangular
10. Let the original volume of cube = a^3 , where a is the side.
Now when side is doubled, the volume of cube
= $(2a)^3 = 8a^3$. So, the volume becomes 8 times.

True / False :

1. True
2. True
3. True
4. False
5. True
6. True
7. True
8. False, $2\pi r$ = circumference
9. **True :** $l = 6$ cm, $b = 5$ cm, $h = 4$ cm

$$\begin{aligned}\text{Surface area of chalk box} &= 2(lb + bh + hl) \\ &= 2(6 \times 5 + 5 \times 4 + 4 \times 6) \\ &= 2(30 + 20 + 24) = 2(74) = 148 \text{ cm}^2\end{aligned}$$

10. **False :** $\frac{d}{2} = \frac{14}{2} = 7$, $h = 12$ cm

$$\text{Volume of the cylinder} = \pi r^2 h$$

$$\begin{aligned}&= \frac{22}{7} \times 7 \times 7 \times 12 \\ &= 1848 \text{ cm}^3.\end{aligned}$$

Match the Following :

1. (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q)
(A) Base = 24 cm, height = 1.5 dm = 15 cm

$$\begin{aligned}\text{Area of triangle} &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times 24 \times 15 = 180 \text{ cm}^2\end{aligned}$$

$$(B) \text{ Radius of the wheel} = \frac{d}{2} = \frac{1.26}{2} = 0.63 \text{ m}$$

$$\text{Circumference of the wheel} = 2\pi r$$

$$= 2 \times \frac{22}{7} \times 0.63 = 3.96 \text{ m}$$

$$\text{Distance travelled in 500 revolution}$$

$$= 3.96 \times 500 = 1980 \text{ m}$$

$$(C) \text{ Volume of the cuboid} = 275 \text{ cm}^3$$

$$\text{Base area of cuboid} = 25 \text{ cm}^2$$

$$\text{i.e., } l \times b = 25 \text{ cm}^2$$

$$\text{Let } h \text{ be the height of cuboid}$$

$$\Rightarrow l \times b \times h = 275$$

$$\Rightarrow 25 \times h = 275$$

$$\Rightarrow h = \frac{275}{25} = 11 \text{ cm}$$

$$(D) \text{ Circumference of the base, } 2\pi r = 132 \text{ cm}$$

$$\Rightarrow r = \frac{132}{2\pi} = \frac{132 \times 7}{2 \times 22} \Rightarrow r = 21 \text{ cm}$$

$$\text{Height of the cylinder, } h = 25 \text{ cm}$$

$$\text{Volume of cylinder} = \pi r^2 h$$

$$= \frac{22}{7} \times 21 \times 21 \times 25 = 34650 \text{ cm}^3$$

2. (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p)

$$(A) \text{ Lateral surface area of cube} = 4l^2$$

$$(B) \text{ Area of circular base of cylinder} = \pi r^2$$

$$(C) \text{ Volume of cuboid} = l \times b \times h$$

$$(D) \text{ Total surface area of cylinder} = 2\pi r(r + h)$$

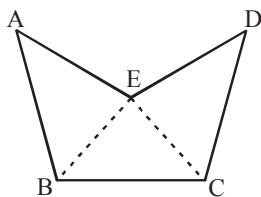
Very Short Answer Questions :

1. $\text{Volume} = \pi r^2 h = \frac{22}{7} \times 8 \times 8 \times 21 = 4224 \text{ cm}^3$
2. Word Capacity is also used instead of word Volume.
We will use :

- (i) Volume when we will refer to the amount of space occupied by an object.
 (ii) Capacity when we will refer to the quantity that a container holds.
3. Length of side of the cube = 1.2 m
 $= (1.2 \times 10) \text{ dm} = 12 \text{ dm}$ [$\because 1 \text{ m} = 10 \text{ dm}$]
 $\therefore$ Volume of the cube = $(\text{Side})^3 = (12)^3 \text{ dm}^3$
 $= 1728 \text{ dm}^3$.
4. Here, $r = 4.2 \text{ m}$, $h = 3.5 \text{ m}$
 $\therefore$ Volume of the drum = $(\pi r^2 h) \text{ m}^3$
 $= (3.14 \times (4.2)^2 \times 3.5) \text{ m}^3 = 193.86 \text{ m}^3$.
5. Number of paving stones required
 $= \frac{\text{length} \times \text{breadth of courtyard}}{\text{length} \times \text{breadth of each tile}}$
 $= \frac{30 \times 16.5}{2.5 \times 2} = 99$
6. Ratio of the circumference to the diameter of the circle
 $= \frac{2\pi r}{2r} = \pi$ Independent of the radius. So, change in radius does not affect the ratio.
7. (i) 1000 millilitres (ii) 5 dm^3
 (iii) 1 ml (iv) $10,00,000 \text{ cm}^3$, 1000 l
 (v) 1000 cm^3

Short Answer Questions :

1. Given ABCDE is a Polygon.



This may be divided into triangles by joining the points B, E and E, C. Then the areas of the triangles can be found. Sum of the areas of these triangles is the area of the polygon.

2. A quadrilateral having one pair of opposite sides parallel and the other pair of opposite sides non-parallel is called a trapezium.
3. Let sides be a, b, c
 $\therefore a + b + c = 324$
 $\Rightarrow a = c = 85, b = 154$

It is an isosceles triangle.

$$\therefore \text{Area} = \frac{b}{4} \sqrt{4a^2 - b^2}$$

4. Length of || sides = $3x$ and $5x$.

$$\text{Area} = \frac{1}{2}(3x + 5x) \times 8$$

$$\Rightarrow 128 = 32x \Rightarrow x = 4$$

Length of parallel sides is 12 cm & 20 cm

5. Volume of log = 625000 cm^3
 Volume of each cube to be cut = 15625 cm^3
 $\therefore$ No. of cubes to be cut = $\frac{625000}{15625} = 40$

6. Number of cartons = $\frac{\text{Vol. of godown}}{\text{Vol. of carton}} \Rightarrow 1000$

7. Width of canal = 6 m., Depth of canal = 1.5 m.
 Length of water column per hour = 10 km.
 $\therefore$ Length of water column in 30 minutes

$$= \frac{10}{2} \text{ km} = 5 \text{ km} = 5000 \text{ m.}$$

$\therefore$ Volume of water that flows in 30 minutes

$$= 6 \text{ m} \times 1.5 \text{ m} \times 5000 \text{ m.}$$

Since, $8 \text{ cm} = \frac{8}{100} \text{ m} = .08 \text{ m}$ standing water is desired, therefore required area of the field that can be irrigated

$$= \frac{\text{Volume of cuboid of water}}{\text{Height of water}} = \frac{6 \times 1.5 \times 5000 \text{ m}^3}{.08 \text{ m}}$$

$$= 562500 \text{ m}^2$$

8. We have,
 (i) a rectangle and with length, $l = 25 \text{ cm}$ and breadth, $b = 7 \text{ cm}$
 (ii) two triangles with base, $a = 5 \text{ cm}$ and $h = 7 \text{ cm}$

Then, Area of shaded region

$$= \text{Area of rectangle ABCD} - 2 \times \text{Area of } \triangle ADE$$

$$\Rightarrow A = l \times b - 2 \times \frac{1}{2} \times b \times h$$

$$= 25 \times 7 - 2 \times \frac{1}{2} \times 5 \times 7$$

$$= (175 - 35) \text{ cm}^2 = 140 \text{ cm}^2$$

Thus, the area of shaded region is 140 cm^2 .

9. Let the edge of the solid cube be a units. Since the cube is cut into two cuboids of equal volumes. Therefore, the dimensions of each of the cuboid are :

Length = a units, breadth = a units

and height = $\frac{a}{2}$ units

Now, S = Total surface area of cube

$$= 6a^2 \text{ sq. units}$$

S_1 = Total surface area of one cuboid

$$= 2 \left(a \times a + a \times \frac{a}{2} + \frac{a}{2} \times a \right) = 4a^2 \text{ sq. units}$$

$$\therefore S : S_1 = 6a^2 : 4a^2 = 3 : 2$$

10. Let the length and breadth of the base of each wooden block be x cm each.

Since eight identical blocks are stacked one on top of the other and the height of each block is 1 cm.

So, height of the solid formed = 8 cm

$$\therefore \text{Volume of the solid formed} = (x \times x \times 8) \text{ cm}^3 = 8x^2 \text{ cm}^3$$

But, the volume of solid formed is 128 cm^3 (given).

$$\therefore 8x^2 = 128 \Rightarrow x^2 = \frac{128}{8} \Rightarrow x^2 = 16 \Rightarrow x = 4 \text{ cm}$$

Hence, each wooden block is of dimension $4 \text{ cm} \times 4 \text{ cm} \times 1 \text{ cm}$.

11. Let length = 100 units and breadth = 100 units

Then Area = $100 \times 100 = 10000$ square units

New length = 110 units

New breadth = 88 units

New Area = $110 \times 88 = 9680$ sq. units

$$\therefore \% \text{ change in area} = \frac{320}{10000} \times 100 = 3.2\%$$

Long Answer Questions :

- 3600 m^2
- (i) 188 cm^2 (ii) 200 cm^2
- Since $AB = 9 \text{ m}$ and $BC = 40 \text{ m}$, $\angle B = 90^\circ$, we have
 $AC = \sqrt{9^2 + 40^2} \text{ m} = \sqrt{81 + 1600} \text{ m} = \sqrt{1681} \text{ m} = 41 \text{ m}$
 Therefore, the first group has to clean the area of triangle ABC, which is right angled.
 Area of $\triangle ABC = \frac{1}{2} \times \text{base} \times \text{height}$

$$= \frac{1}{2} \times 40 \times 9 \text{ m}^2 = 180 \text{ m}^2$$

The second group has to clean the area of triangle ACD, which is scalene having sides 41 m, 15 m and 28 m.

$$\text{Here, } s = \frac{41 + 15 + 28}{2} \text{ m} = 42 \text{ m}$$

Therefore, area of $\triangle ACD$

$$\begin{aligned} &= \sqrt{s(s-a)(s-b)(s-c)} \\ &= \sqrt{42(42-41)(42-15)(42-28)} \text{ m}^2 \\ &= \sqrt{42 \times 1 \times 27 \times 14} \text{ m}^2 = 126 \text{ m}^2 \end{aligned}$$

So first group cleaned 180 m^2 which is $(180 - 126) \text{ m}^2$, i.e., 54 m^2 more than the area cleaned by the second group.

Total area cleaned by all the students

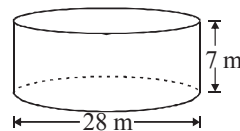
$$= (180 + 126) \text{ m}^2 = 306 \text{ m}^2.$$

4. Volume of earth dug out to sink a well = $\pi r^2 h$

$$= \frac{22}{7} \times \left(\frac{28}{2} \right)^2 \times 7 \text{ m}^3 \left[\because r = \frac{28}{2} \text{ m} \right]$$

$$= \frac{22}{7} \times 14 \times 14 \times 7 \text{ m}^3$$

$$= 22 \times 14 \times 14 \text{ m}^3 = 4312 \text{ m}^3$$



Now, the area of the curved surface of the well = $2\pi rh$

$$= 2 \times \frac{22}{7} \times 14 \times 7 \text{ m}^2 = 2 \times 22 \times 14 \text{ m}^2$$

$$= 616 \text{ m}^2$$

$\therefore$ Cost of plastering the curved surface at the rate of ₹ 3.50 per sq. m.

$$= ₹ 3.50 \times 616 = ₹ 2156.$$

- 673.2 m^2
- Water consumed by the inhabitants in one day
 = Number of inhabitants
 $\times$ Water consumed by an inhabitant in one day
 $= (25000 \times 125) \text{ litres} = 3125000 \text{ litres}.$
 $\therefore$ Water consumed by the inhabitants in two days
 $= (3125000 \times 2) \text{ litres}$
 $= 6250000 \text{ litres} = \frac{6250000}{1000} \text{ m}^3 \left[\because 1000 \text{ litres} = 1 \text{ m}^3 \right]$
 $= 6250 \text{ m}^3$

∴ Volume of the tank = 6250 litres

$$\begin{aligned}\therefore \text{Depth of the tank} &= \frac{\text{Volume}}{\text{Length} \times \text{Breadth}} \\ &= \frac{6250}{1250} \text{ m} = 5 \text{ m}\end{aligned}$$

7. (i) Required depth = $\frac{48000}{150 \times 20}$

(ii) Required rate (Drop) = $\frac{20000}{\pi \times 24^2}$ or $\frac{20000}{20 \times 150}$

Now, Divide drop by 2.5 -ie- $\frac{20000}{2.5}$

⇒ Required rate in cm/hr = 4.42 cm/hr

8. The sails are both triangles.

$$\begin{aligned}A_1 &= \frac{1}{2} \times b \times h & A_2 &= \frac{1}{2} \times b \times h \\ &= \frac{1}{2} \times 10.8 \times 25 = 135 & &= \frac{1}{2} \times 8.5 \times 24 = 102\end{aligned}$$

Total Area = 135 + 102 = 237

The total area of material is 237 m².

2 EXERCISE

Text-Book Exercise :

- Area = 129.5 m² ; Perimeter = 48 m
- 45000 tiles
- 7 cm
- ₹ 810
- Area using Jyoti's way

$$= 2 \times \frac{1}{2} \times \frac{15}{2} \times (30 + 15) \text{ m}^2 = 377.5 \text{ m}^2,$$

Area using Kavita's way

$$= \frac{1}{2} \times 15 \times 15 + 15 \times 15 = 337.5 \text{ m}^2$$

- Let the length of the room = $l = 12$ m
Width of the room = $b = 8$ m
Height of the room = $h = 4$ m
Area of the four walls of the room

= Perimeter of base $\times$ Height of the room

$$= 2(l + b) \times h = 2(12 + 8) \times 4$$

$$= 2 \times 20 \times 4 = 160 \text{ m}^2.$$

Cost of white washing per m² = ₹ 5

Hence the total cost of white washing four walls of the room = ₹ (160 $\times$ 5) = ₹ 800

Area of ceiling is $12 \times 8 = 96 \text{ m}^2$

Cost of white washing the ceiling = ₹ (96 $\times$ 5) = ₹ 480

So the total cost of white washing = ₹ (800 + 480)
= ₹ 1280

7. Radius of cylinder pillar, $r = 28 \text{ cm} = 0.28 \text{ m}$

height, $h = 4 \text{ m}$

curved surface area of a cylinder = $2\pi rh$

$$\text{curved surface area of a pillar} = 2 \times \frac{22}{7} \times 0.28 \times 4 = 7.04 \text{ m}^2$$

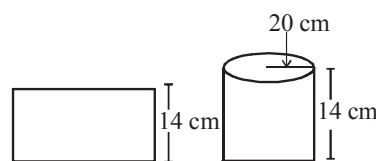
curved surface area of 24 such pillar

$$= 7.04 \times 24 = 168.96 \text{ m}^2$$

cost of painting an area of 1 m² = ₹ 8

Therefore, cost of painting 1689.6 m²
= $168.96 \times 8 = ₹ 1351.68$

- 5 cans
- 322 cm
- A cylinder is formed by rolling a rectangle about its width. Hence the width of the paper becomes height and radius of the cylinder is 20 cm.



Height of the cylinder = $h = 14 \text{ cm}$

Radius = $r = 20 \text{ cm}$

Volume of the cylinder = $V = \pi r^2 h$

$$= \frac{22}{7} \times 20 \times 20 \times 14 = 17600 \text{ cm}^3$$

Hence, the volume of the cylinder is 17600 cm³.

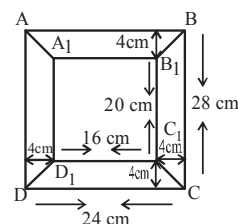
- Volume of cylinder B is greater, Surface area of cylinder B is greater.
- 49500 L
- 140 m

Exemplar Questions :

- ₹ 528 2. 10m 3. 84m
- $V = 11440 \text{ cm}^3$, Weight = 91520 g
- $h = 20 \text{ cm}$
- 6752 cm^3
- 0.78m
- 1 day
- 25 dm, 20 dm, 15 dm
- $r = 8 \text{ cm}$, $A = 603.428 \text{ cm}^2$
- $h = 8 \text{ m}$, $b = 10 \text{ m}$

HOTS Questions :

- Diameter = 98 cm
Radius = 49 cm
Circumference = 308 cm
 $\therefore$ Distance covered by the wheel in one revolution = 308 cm
Total distance to be covered = 770000
Number of revolutions = 2500
 $\therefore$ Distance covered in 500 revolution = 1.54 km
- The area grazed by the calf = πr^2 , where r = length of the rope.
Increase in area = $\pi (23^2 - 12^2)$
 $= \pi \times 35 \times 11$
 $= \frac{22}{7} \times 35 \times 11 = 1210 \text{ m}^2$
- Area of the rectangle = $L \times B$
Let the new width be B_1 .
Then, $L \times B = \frac{4}{3} L \times B_1$
 $\therefore B_1 = \frac{3}{4} B = 0.75 B$
Thus, there should be a reduction of 25% in the width.
- We have,
Area of section $AB B_1 A_1$ = Area of section $CD D_1 C_1$
 $= \frac{1}{2} (24 + 16) \times 4 \text{ cm}^2$
 $= 20 \times 4 \text{ cm}^2 = 80 \text{ cm}^2$



Area of section $ADD_1 A_1$ = Area of section $BCC_1 B_1$

$$= \frac{1}{2} (28 + 20) \times 4 \text{ cm}^2 = 24 \times 4 \text{ cm}^2 = 96 \text{ cm}^2$$

Area of section $A_1 B_1 C_1 D_1$ = $A_1 B_1 \times B_1 C_1 = 16 \times 20 \text{ cm}^2$
 $= 320 \text{ cm}^2$.

- Diameter of the graphite cylinder = 1 mm = $\frac{1}{10} \text{ cm}$

$$\therefore \text{Radius} = \frac{1}{20} \text{ cm}$$

Length of the graphite cylinder = 10 cm

$$\text{Volume of the graphite cylinder} = \left(\frac{22}{7} \times \frac{1}{20} \times \frac{1}{20} \times 10 \right) \text{ cm}^3$$

Weight of graphite = Volume $\times$ Specific gravity

$$= \left(\frac{22}{7} \times \frac{1}{20} \times \frac{1}{20} \times 10 \times 2.1 \right) \text{ gm}$$

$$= \left(\frac{22}{7} \times \frac{1}{20} \times \frac{1}{20} \times 10 \times \frac{21}{10} \right) \text{ gm} = 0.165 \text{ gm}$$

$$\text{Diameter of pencil} = 7 \text{ mm} = \frac{7}{10} \text{ cm}$$

$$\therefore \text{Radius of pencil} = \frac{7}{20} \text{ cm and,}$$

Length of pencil = 10 cm

$$\therefore \text{Volume of pencil} = \left(\frac{22}{7} \times \frac{7}{20} \times \frac{7}{20} \times 10 \right) \text{ cm}^3$$

Volume of wood

$$= \left(\frac{22}{7} \times \frac{7}{20} \times \frac{7}{20} \times 10 - \frac{22}{7} \times \frac{1}{20} \times \frac{1}{20} \times 10 \right) \text{ cm}^3$$

$$= \frac{22}{7} \times \frac{1}{20} \times \frac{1}{20} \times 10 (7 \times 7 - 1) \text{ cm}^3$$

$$= \frac{11}{7} \times \frac{1}{20} \times 48 \text{ cm}^3$$

$$\therefore \text{Weight of wood} = \left(\frac{11}{7} \times \frac{1}{20} \times 48 \times 0.7 \right) \text{ gm}$$

$$= \left(\frac{11}{7} \times \frac{1}{20} \times 48 \times \frac{7}{10} \right) \text{ gm} = 2.64 \text{ gm}$$

$$\text{Total weight} = (2.64 + 0.165) \text{ gm} = 2.805 \text{ gm}$$

6. We have, $v = abc$ and $s = 2(ab + bc + ca)$

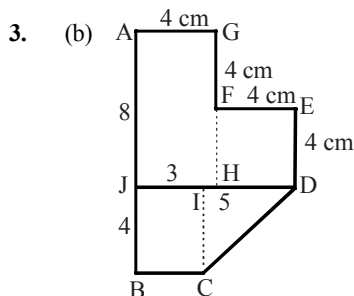
$$\text{R.H.S.} = \frac{2}{s} \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) = \frac{2}{s} \left(\frac{bc + ab + ca}{abc} \right)$$

$$= \frac{2}{s} \cdot \frac{s}{v} = \frac{1}{v} = \text{L.H.S}$$

3 EXERCISE

Single Option Correct :

1. (c)
2. (b) 12 cm



Area ABCDEFGH

$$= \text{Area AJHG} + \text{Area HFED} + \text{Area JICB} + \text{Area ICD}$$

$$= (8 \times 4) + (4 \times 4) + (4 \times 3) + \frac{1}{2} (5 \times 4) = 32 + 16 + 12 + 10$$

$$= 70 \text{ cm}^2$$

4. (a)
5. (b)

Area ABCDEFGH

$$= \text{Area (ABEF)} + \text{Area (FGH)} + \text{Area (CDE)} \dots (1)$$

$$FH = \sqrt{FG^2 - GH^2} = \sqrt{25 - 16} = 3 \text{ cm}$$

$$CE = \sqrt{DE^2 - CD^2} = \sqrt{25 - 16} = 3 \text{ cm}$$

From (1)

$$\begin{aligned} \text{Area ABCDEFGH} &= (6 \times 7) + \frac{1}{2} \times 4 \times 3 + \frac{1}{2} \times 4 \times 3 \\ &= 42 + 6 + 6 = 54 \text{ cm}^2 \end{aligned}$$

6. (c) Length of room = 6.75 m = 675 cm
Breadth of room = 5.75 m = 575 cm
Square tiles are to be used to pave the room.
The side of the square (tile) must be a factor of both length & breadth of the room
 $\therefore$ HCF of 675 and 575 = 25 cm

$$\therefore \text{No of tiles} = \frac{\text{Area of room}}{\text{Area of one tile}}$$

$$\text{No of tiles} = \frac{675 \times 575}{25 \times 25} = 621$$

7. (c) 8. (c) 9. (c) 10. (d)
11. (d) 12. (a) 13. (c) 14. (d)
15. (c)

More Than One Option Correct :

1. (a, b, c) 2. (a, b, c) 3. (a, b, c, d) 4. (a, b, c)
5. (a, b)
6. (a, b, c): Let length = x , breadth = $2x$

and height = $3x$

Surface area of cuboid = 88 cm²

$$\text{So, } 2(lb + bh + hl) = 88 \Rightarrow lb + bh + hl = 44$$

$$\Rightarrow x \times 2x + 2x \times 3x + 3x \times x = 44$$

$$\Rightarrow 2x^2 + 6x^2 + 3x^2 = 44$$

$$\Rightarrow 11x^2 = 44 \Rightarrow x^2 = 4 \Rightarrow x = 2.$$

Length of the cuboid = $x = 2$ cm

Breadth of the cuboid = $2x = 2 \times 2 = 4$ cm

Height of the cuboid = $3x = 3 \times 2 = 6$ cm

7. (a, b, c): Volume = $\pi r^2 h = \left(\frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times 40 \right) \text{ cm}^3$
 $= 1540 \text{ cm}^3$

Curved surface area = $2\pi rh$

$$= \left(2 \times \frac{22}{7} \times \frac{7}{2} \times 40 \right) \text{ cm}^2 = 880 \text{ cm}^2$$

Total surface area = $2\pi rh + 2\pi r^2 = 2\pi r(h + r)$

$$= \left(2 \times \frac{22}{7} \times \frac{7}{2} \times (40 + 3.5) \right) \text{ cm}^2 = 957 \text{ cm}^2$$

8. (b, c, d) : $lb = 120 \text{ cm}^2$, $bh = 72 \text{ cm}^2$
and $hl = 60 \text{ cm}^2$

$$\text{Volume of the box} = lb \times bh \times hl \\ = 120 \times 72 \times 60$$

$$\Rightarrow (lbh)^2 = 518400$$

$$\Rightarrow lbh = \sqrt{518400} = 720 \text{ cm}^3$$

$$\text{length} = \frac{720}{72} = 10, \text{ breadth} = \frac{720}{60} = 12 \text{ cm}$$

$$\text{height} = \frac{720}{120} = 6 \text{ cm}$$

Passage Based Questions :

1. (a) 2. (b) 3. (d)
4. (a) Required area $= 2\pi \times 1.2 \times 5 + \pi \times (1.2)^2$
 $= 42.2$ to 42.25 m^2
5. (a) Required volume $= \pi \times (1.2)^2 \times 3$
 $= 13.564 \text{ m}^3$
 $= 1 \text{ m}^3 = 1000 \text{ l} \Rightarrow 13564 \text{ l}$

Assertion & Reason :

1. (a) Assertion and Reason both are correct and Reason is the correct explanation of Assertion.
Length of hall = 20 m
Breadth = 10 m
Area = 20×10
Length of tile = $\frac{50}{100} \text{ m}$
Breadth = $\frac{50}{100} \text{ m}$
 $\therefore$ Area of each tile $= \left(\frac{50}{100}\right)^2$
2. (a) Both Assertion & Reason are correct.
3. (d) Assertion is incorrect and Reason is correct.
4. (c) The line segment joining the centres of two circular bases is called axis of the cylinder.

Integer Type Questions :

1. (4) Area of rhombus

$$= \frac{1}{2} \times d_1 \times d_2 = \frac{1}{2} \times 10 \times 12 = 60 \text{ cm}^2$$

$$\text{So, } x = 60 \text{ cm}^2. \text{ The value of } \frac{x}{15} = \frac{60}{15} = 4$$

2. (7) Area of triangular garden

$$= \frac{1}{2} \times \text{base} \times \text{altitude}$$

$$\Rightarrow \frac{9520 \times 2}{340} = \text{Altitude} \Rightarrow \text{Altitude} = 56 \text{ m}$$

$$\text{The value of } \frac{a}{8} = \frac{56}{8} = 7$$

3. (4) Surface area of a cuboid $= 2(lb + bh + hl)$

$$= 2(16 \times 8 + 8 \times 6 + 6 \times 16) \text{ cm}^2$$

$$= 2(128 + 48 + 96) \text{ cm}^2 = 544 \text{ cm}^2.$$

4. (2) Area of parallelogram = base $\times$ altitude

$$= 14 \times 8 = 112 \text{ cm}^2$$

$$\text{So, } y = 112 \text{ cm}^2$$

$$\text{The value of } (y - 110) = (112 - 110) = 2 \text{ cm}^2.$$

5. (3) Given shape is a rectangle.

$$\text{Perimeter of rectangle} = 2[\text{Length} + \text{Breadth}]$$

$$186 = 2[5x + 6 + 2x + 66]$$

$$186 = 2[7x + 72]$$

$$7x + 72 = \frac{186}{2}$$

$$7x + 72 = 93$$

$$7x = 93 - 72$$

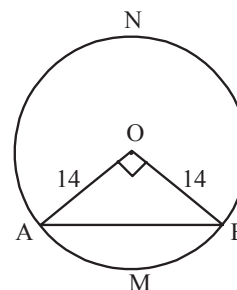
$$7x = 21$$

$$x = \frac{21}{7}$$

$$x = 3 \text{ cm}.$$

4 ADVANCED EXERCISE BASED ON CONNECTING TOPICS

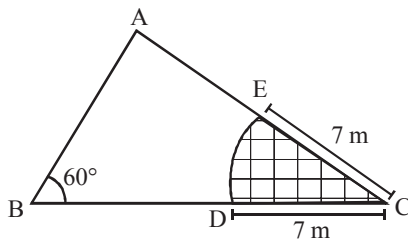
1. (a)



$$\text{Area of } \triangle AOB = \frac{1}{2} \times 14 \times 14 = 98 \text{ cm}^2$$

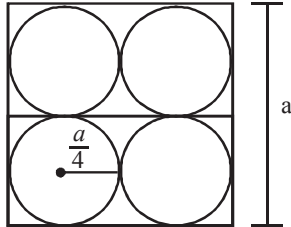
$$\begin{aligned}\text{Area of segment AMBO} &= \frac{90^\circ}{360^\circ} \times \pi(14)^2 = 154 \text{ cm}^2 \\ \text{Area of circular segment AMBA} &= 154 - 98 = 56 \text{ cm}^2 \\ \text{Area of circular segment ABNA} \\ &= \pi(14)^2 - 56 = 560 \text{ cm}^2.\end{aligned}$$

2. (c) Each angle of equilateral triangle is $= 60^\circ$, area of field over which the horse can graze = area of sector CDE with central angle 60° .



$$= \frac{60^\circ}{360^\circ} \times \pi r^2 = \frac{60}{360} \times \frac{22}{7} \times 7^2 = 25\frac{2}{3} \text{ sq.cm}$$

3. (c)



Side of square paper sheet

$$= \sqrt{784} = 28 \text{ cm}$$

$$\text{Radius of each circular plate} = \frac{28}{4} = 7 \text{ cm}$$

$\therefore$ Circumference of each plate $= 2\pi \times \text{Radius}$

$$= 2 \times \frac{22}{7} \times 7 = 44 \text{ cm}$$

4. (b) Let the original radius $= r$

$$\therefore \text{Area } A = \pi r^2$$

$$\text{increased area } A' = \pi(r+1)^2$$

$$\text{Now, } A' = A + 22$$

$$\pi(r+1)^2 = \pi r^2 + 22$$

$$\Rightarrow \pi[(r+1)^2 - r^2] = 22$$

$$\Rightarrow \pi[(r+1+r)(r+1-r)] = 22$$

$$\Rightarrow \pi(2r+1) = 22$$

$$2r+1 = \frac{22 \times 7}{22} \quad \left[\because \pi = \frac{22}{7} \right]$$

$$\Rightarrow 2r+1 = 7$$

$$\Rightarrow 2r = 6$$

$$\Rightarrow r = 3 \text{ cm}$$

5. (b) Area of a sector $= \pi r^2 \times \frac{\theta}{360}$,
 θ = included angle

$$\therefore \text{Area of sector OCBO} = \pi \times 8^2 \times \frac{45}{360}$$

$$= 8\pi \text{ m}^2$$

Area of sector OADO

$$= \pi \times 6^2 \times \frac{45}{360} = \frac{9\pi}{2} \text{ m}^2$$

$$\therefore \text{Area of shaded region} = \left(8\pi - \frac{9\pi}{2} \right) \text{ m}^2$$

$$= \frac{7\pi}{2} \text{ m}^2 = \frac{7 \times 22}{2 \times 7} \text{ m}^2$$

$$\text{Area of shaded region} = 11 \text{ m}^2$$

Chapter 12

Exponents and Power

INTRODUCTION

Some of the numbers like 300,000,000, 1,526,000 are incredibly large and some numbers like 0.000009, 0.00000926 are unbelievable small. It is difficult to comprehend such numbers. We express such numbers in exponential form, so that it becomes very convenient to read, understand and compare.

In the exponential form, a number is multiplied by itself repeatedly. Example: $81 = 3 \times 3 \times 3 \times 3 = 3^4$.

Here, 81 is expressed as 3^4 , which is a shorter form.

It is read as 3 raised to the power 4. Here the base is 3 and the exponent is 4.

EXPONENTS AND POWERS [INDICES]

For any non-zero rational number ' a ' and natural number ' n ' the product $a \times a \times a \times \dots \times a$ (n times) [continued product of ' a ' with itself n times] is written as a^n . It is known as n^{th} power of ' a ' and is read as ' a ' raised to the power ' n '. The rational number ' a ' is called the **base** and ' n ' is called **exponent** or **index**. The notation of writing the product of a rational number by itself several times is called **exponential notation** or **power notation**.

Example : $2 \times 2 \times 2 \times 2 \times 2 = 2^5$, here 2 is base and 5 is exponent or index.

Initially we will consider index as positive integer and then apply this knowledge to understand the meaning when indices are 0 or negative.

Laws of Indices :

(i) $a^m \times a^n = a^{m+n}$ [Multiplication with same base]

' a ' is any rational number, ' m ' and ' n ' are integers.

For example:

(a) $(3)^5 \times (3)^4 = (3)^{5+4} = (3)^9$

(b) $(-6)^{-2} \times (-6)^3 = (-6)^{-2+3} = (-6)$

(ii) $a^m \div a^n = a^{m-n}$, $a \neq 0$ [Division with same base]

For example:

(a) $(4)^5 \div (4)^3 = (4)^{5-3} = (4)^2$

(b) $\left(\frac{2}{3}\right)^3 \div \left(\frac{2}{3}\right)^1 = \left(\frac{2}{3}\right)^{3-1} = \left(\frac{2}{3}\right)^2$

IMPORTANT : Let $m = n$, then by law (ii) we can write

$$a^m \div a^m = a^{m-m}$$

i.e. $1 = a^0$ here $a \neq 0$

If the number itself is zero, then $0^{m-m} = \frac{0^m}{0^m} = \frac{0}{0}$ which is undefined

$\Rightarrow 0^0$ is not defined.

(iii) $(a^m)^n = (a^n)^m = a^{mn}$ [power of a power]

For example:

(a) $(4^2)^3 = 4^{2 \times 3} = 4^6$

(b) $\left[\left(\frac{2}{3}\right)^3\right]^4 = \left(\frac{2}{3}\right)^{3 \times 4} = \left(\frac{2}{3}\right)^{12}$

(iv) $a^m \times b^m = (ab)^m$ [Multiplication with same exponents]

For example:

(a) $3^4 \times 6^4 = (18)^4$

(b) $\left(\frac{1}{2}\right)^3 \times \left(\frac{3}{5}\right)^3 = \left(\frac{3}{10}\right)^3$

(v) $a^m \div b^m = \left(\frac{a}{b}\right)^m$ [Division with same exponent]

For example:

(a) $3^2 \div 5^2 = \left(\frac{3}{5}\right)^2$ (b) $\frac{(5/2)^3}{(4/25)^3} = \left(\frac{125}{8}\right)^3$

We know that $a^m \times a^n = a^{m+n}$ [m & n are integers]

If $m = -n$, then

$a^{-n} \times a^n = a^{-n+n} = a^0 = 1$ [Any number (except 0) raised to the power '0' is 1. For e.g. $(-6)^0 = 1$]

i.e. $a^n \times a^{-n} = 1$

or $a^{-n} = \frac{1}{a^n}$ i.e. a^{-n} is reciprocal of a^n

(vi) $\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n$

For example:

(a) $\left(\frac{2}{5}\right)^{-4} = \left(\frac{5}{2}\right)^4$ (b) $\left(\frac{1}{4}\right)^{-1} = \left(\frac{4}{1}\right)^1 = 4$

ILLUSTRATION : 1

Find the value of $(32)^{\frac{2}{5}}$.

SOLUTION :

$$\begin{aligned} \Rightarrow & \left[(32)^2\right]^{\frac{1}{5}} \quad \text{or} \quad \left[(32)^{1/5}\right]^2 \\ = & (1024)^{\frac{1}{5}} \quad \text{or} \quad \left[(2^5)^{\frac{1}{5}}\right]^2 \\ = & \left[(2)^{10}\right]^{\frac{1}{5}} \quad \text{or} \quad (2)^2 \\ = & (2^2) \quad \text{or} \quad (2)^2 \\ = & 4 \quad \text{or} \quad 4 \end{aligned}$$

Negative Rational Exponents

We know that $a^{-n} = \frac{1}{a^n}$. If exponent is in the form of $\frac{p}{q}$ then $a^{\frac{-p}{q}} = \frac{1}{a^{p/q}} = \left(\frac{1}{a}\right)^{\frac{p}{q}}$ i.e. $a^{\frac{-p}{q}}$ is reciprocal of $a^{p/q}$.

Now if 'a' is also a rational number in the form $\frac{x}{y}$ where both x and y are greater than 0 then we can write $\left(\frac{x}{y}\right)^{\frac{-p}{q}} = \left(\frac{y}{x}\right)^{\frac{p}{q}}$

NOTE: $a^{-1} + b^{-1}$ does not equal $(a + b)^{-1}$.

ILLUSTRATION : 2(a) Find the value of $\left(\frac{8}{27}\right)^{-\frac{2}{3}}$.(b) Evaluate : $\left(\frac{8}{125}\right)^{-\frac{1}{3}} \times \left(\frac{8}{125}\right)^{-\frac{2}{3}}$ **SOLUTION :**

$$(a) \quad \left(\frac{8}{27}\right)^{-\frac{2}{3}} = \left(\frac{27}{8}\right)^{\frac{2}{3}} = \left[\left(\frac{27}{8}\right)^{\frac{1}{3}}\right]^2 = \left[\left(\frac{3^3}{2^3}\right)^{\frac{1}{3}}\right]^2 = \left[\left(\frac{3}{2}\right)^{\frac{1}{3}}\right]^2 = \left(\frac{3}{2}\right)^2 = \frac{9}{4}$$

$$(b) \quad \left(\frac{8}{125}\right)^{-\frac{1}{3}} \times \left(\frac{8}{125}\right)^{-\frac{2}{3}} = \left(\frac{8}{125}\right)^{-\frac{1}{3} + \left(-\frac{2}{3}\right)} = \left(\frac{8}{125}\right)^{-\frac{3}{3}} = \left(\frac{8}{125}\right)^{-1} = \frac{125}{8}$$

USE OF EXPONENTS TO EXPRESS SMALL NUMBERS IN STANDARD FORM

There are three steps:

Step I: Obtain the number and see whether the number is between 1 and 10 or it is less than 1.**Step II:** If the number is between 1 and 10, then write it as the product of the number itself and 10^0 .**Step III :** If the number is less than one, then move the decimal point to the right so that there is just one digit on the left side of the decimal point. Write the given number as the product of the number so obtained and 10^{-n} , where n is the number of places the decimal point has been moved to the right. The number so obtained is the standard form of the given number.**ILLUSTRATION : 3****Write the following numbers in standard form:**

(i) 0.4579

(ii) 216000000

(iii) 0.0000529 $\times 10^4$ (iv) 9573 $\times 10^{-4}$ **SOLUTION :**

- (i) To express 0.4579 in standard form the decimal point is moved through one place only to the right so that there is just one digit on the left of the decimal point. So standard form is 4.579×10^{-1}
- (ii) $216000000 = 2.16 \times 10^8$ [$\therefore$ The decimal point is moved 8 places to the left]
- (iii) $0.0000529 \times 10^4 = 5.29 \times 10^{-5} \times 10^4 = 5.29 \times 10^{-5+4} = 5.29 \times 10^{-1}$
- (iv) $9573 \times 10^{-4} = 9.573 \times 10^3 \times 10^{-4} = 9.573 \times 10^{3+(-4)} = 9.573 \times 10^{-1}$

NOTE: The exponent properties deal only with products and quotients, not with sums.

MISCELLANEOUS SOLVED EXAMPLES

1. Simplify : $(6^{-1} - 8^{-1})^{-1} + (2^{-1} - 3^{-1})^{-1}$

Sol. $(6^{-1} - 8^{-1})^{-1} + (2^{-1} - 3^{-1})^{-1}$

$$= \left(\frac{1}{6} - \frac{1}{8}\right)^{-1} + \left(\frac{1}{2} - \frac{1}{3}\right)^{-1}$$

$$= \left(\frac{4-3}{24}\right)^{-1} + \left(\frac{3-2}{6}\right)^{-1}$$

$$= \left(\frac{1}{24}\right)^{-1} + \left(\frac{1}{6}\right)^{-1} \quad \left[\text{using } n^{-1} = \frac{1}{n}\right]$$

$$= 24 + 6 = 30$$

2. By what number should $(-6)^{-1}$ be multiplied so that the product is equal to $(9)^{-1}$.

Sol. Let the required number be 'x'

Now, $(-6)^{-1} \times x = (9)^{-1}$

$$\therefore x = (9)^{-1} \div (-6)^{-1}$$

$$= \frac{1}{9} \div \frac{1}{-6}$$

$$= \frac{1}{9} \times \frac{-6}{1} = \frac{-6}{9} \quad \left[\text{using } n^{-1} = \frac{1}{n}\right]$$

$$= \frac{-2}{3}$$

$\therefore$ The required number is $\frac{-2}{3}$.

3. Simplify : $\left[\left\{\left(\frac{-1}{5}\right)^{-2}\right\}^2\right]^{-1}$

Sol. Given expression can be written as

$$\left[\left(\frac{-1}{5}\right)^{-2}\right]^{2 \times (-1)} \quad [\text{using } (x^m)^n = x^{m \times n}]$$

$$\Rightarrow \left[\left(\frac{-1}{5}\right)^{-2}\right]^{-2} = \left(\frac{-1}{5}\right)^{(-2) \times (-2)} = \left(\frac{-1}{5}\right)^4$$

$$= \frac{(-1)^4}{(5)^4} \quad \left[\text{using } \left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}\right]$$

$$= \frac{1}{625}$$

4. Simplify: $\frac{3^{-5} \times 10^{-5} \times 125}{5^{-7} \times 6^{-5}}$

Sol. Given expression can be written as

$$\begin{aligned} & \frac{3^{-5} \times (2 \times 5)^{-5} \times 5^3}{5^{-7} \times 6^{-5}} \\ &= \frac{3^{-5} \times 2^{-5} \times 5^{-5} \times 5^3}{5^{-7} \times 2^{-5} \times 3^{-5}} \\ &= 3^{-5 - (-5)} \times 2^{-5 - (-5)} \times 5^{-5 + 3 - (-7)} \\ &= 3^0 \times 2^0 \times 5^5 = 1 \times 1 \times 5^5 \\ &= 5^5 = 3125 \end{aligned}$$

$$[\because (a \times b)^n = a^n \times b^n]$$

$$\left[\because \frac{a^n}{a^m} = a^{n-m}, a^m \times a^n = a^{m+n} \right]$$

$$[\because a^0 = 1]$$

5. Find 'x' so that $\left(\frac{7}{4}\right)^{-4} \times \left(\frac{7}{4}\right)^{-12} = \left(\frac{7}{4}\right)^{4x}$

Sol. Given $\left(\frac{7}{4}\right)^{-4} \times \left(\frac{7}{4}\right)^{-12} = \left(\frac{7}{4}\right)^{4x}$

$$\Rightarrow \left(\frac{7}{4}\right)^{-4+(-12)} = \left(\frac{7}{4}\right)^{4x} \quad [\because a^m \times a^n = a^{m+n}]$$

$$\Rightarrow \left(\frac{7}{4}\right)^{-16} = \left(\frac{7}{4}\right)^{4x}$$

$$\Rightarrow 4x = -16 \Rightarrow x = -4$$

6. Evaluate: $(0.000343)^{\frac{-3}{2}} \times (0.49)^{\frac{-3}{2}}$

Sol. $(0.000343)^{\frac{-3}{2}} \times (0.49)^{\frac{-3}{2}}$

$$= \left(\frac{343}{1000000}\right)^{\frac{-3}{2}} \times \left(\frac{49}{100}\right)^{\frac{-3}{2}}$$

$$= \left(\frac{1000000}{343}\right)^{\frac{3}{2}} \times \left(\frac{100}{49}\right)^{\frac{3}{2}}$$

$$= \left(\frac{10^6}{7^3}\right)^{\frac{3}{2}} \times \left(\frac{10^2}{7^2}\right)^{\frac{3}{2}}$$

$$= \left(\frac{10^6 \times 10^2}{7^3 \times 7^2}\right)^{\frac{3}{2}} = \left(\frac{10^8}{7^5}\right)^{\frac{3}{2}} = \frac{10^{12}}{7^{\frac{15}{2}}}$$

7. Evaluate : $\left(1^3 + 2^3 + 3^3 + 4^3\right)^{\frac{-3}{2}}$

Sol. $\left(1^3 + 2^3 + 3^3 + 4^3\right)^{\frac{-3}{2}} = (1 + 8 + 27 + 64)^{\frac{-3}{2}} = (100)^{\frac{-3}{2}} = \left(10^2\right)^{\frac{-3}{2}} = 10^{-3} = \frac{1}{1000}.$

8. Express each of the following as a rational number of the form $\frac{p}{q}$.

(a) $\left(\frac{3}{8}\right)^{-2} \times \left(\frac{4}{5}\right)^{-3}$ (b) $\left(-\frac{2}{7}\right)^{-4} \times \left(\frac{-7}{5}\right)^2$

Sol. (a) $\left(\frac{3}{8}\right)^{-2} \times \left(\frac{4}{5}\right)^{-3}$

$$= \frac{3^{-2}}{(2^3)^{-2}} \times \frac{(2^2)^{-3}}{5^{-3}} = \frac{2^{3 \times 2}}{3^2} \times \frac{5^3}{2^{3 \times 2}} = \frac{2^6 \times 5^3}{3^2 \times 2^6} = \frac{5^3}{3^2} = \frac{125}{9}.$$

(b) $\frac{(-2)^{-4}}{7^{-4}} \times \frac{(-7)^2}{5^2}$

$$\frac{1}{(-2)^4} \times \frac{7^4 \times (-1)^2 \times (7)^2}{5^2}$$

$$= \frac{1}{(-1)^4 \times (2)^4} \times \frac{7^{4+2}}{25} = \frac{7^6}{400} = \frac{117649}{400}$$

9. Write the following numbers in usual form :

(a) 1.785×10^7 (b) 5.1×10^{-7}

Sol. (a) $1.785 \times 10^7 = 1.785 \times 10000000 = 17850000$

(b) $5.1 \times 10^{-7} = \frac{5.1}{10^7} = \frac{5.1}{10000000} = 0.00000051$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- The scientific notation for 0.000065 is
- $\left(\frac{25}{21}\right)^{3/2}$ can be written as
- The usual form of 6.96×10^5 is
- $(2^5 \div 2^8) \times 2^4$ is equal to
- For any two rational numbers a and b, $a^5 \times b^5$ is equal to
- The value of $(512)^{-2/9}$ is
- $4^{-3/2} + 8^{2/3} = \dots\dots\dots$
- The value of expression $(8^0 - 3^0) \times (8^0 + 3^0)$ is equal to ...
- $(2^{-1} - 4^{-1})^2 = \dots\dots\dots$
- $\frac{25 \times a^{-4}}{5^{-3} \times 10 \times a^{-8}} = \dots$

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- In 6^3 , the number 6 is called the base and 3 is called the exponent.
- A reciprocal is also known as multiplicative inverse.
- $2^2 \times 2^3 = 2^6$
- $6^8 \div 3^4 = 2^4$
- $(2x^2)^3 = 8x^5$
- If $ab = 1$, then the value of $(1 + a - b^{-1})^{-1} + (1 + b - a^{-1})^{-1}$ is 1.
- If $ab = 1$, then the value of $\frac{1}{1+a^{-1}} + \frac{1}{1+b^{-1}}$ is equal to 2.
- If $\frac{10}{3} \times 3^x - 3^{x-1} = 81$, then $x = 4$.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.

- | | |
|--|---------------------|
| 1. Column-I | Column-II |
| (A) $3^{-9} \times 3^x = 3$
then the value of x is | (p) 2 |
| (B) If $5^x \times 5^{-2} \times 5^4 = 5^{-3}$
then the value of x is | (q) -1 |
| (C) If $\left(\frac{-3}{5}\right)^{-4} \times \left(\frac{-3}{5}\right)^6 = \left(\frac{5}{3}\right)^{2x}$ then the
value of x is | (r) 10 |
| (D) If $5^{2x+1} \div 25 = 125$
then the value of x is | (s) -5 |
| 2. Column-I | Column-II |
| (A) By what number should 5^{-1}
be multiplied so that the
product may be equal
to $(-7)^{-1}$ | (p) $\frac{1}{3}$ |
| (B) By what number should $(-15)^{-1}$ be divided so that
the quotient may be equal
to $(-5)^{-1}$? | (q) $-\frac{2}{3}$ |
| (C) By what number should $(-6)^{-1}$
be multiplied so that product
becomes 9^{-1} ? | (r) $-\frac{2}{27}$ |
| (D) By what number should $\left(\frac{-2}{3}\right)^{-3}$ be divided so that the
quotient may $\left(\frac{4}{27}\right)^{-2}$? | (s) $-\frac{5}{7}$ |

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- If $\left(\frac{x^m \times x^n}{x^p}\right)^q = x^K$ then what is the value of K?
- Simplify $\left(\frac{1}{6^2}\right)^{\frac{1}{3}}$
- Write each of the following in radical form.
 - cube root of 972
 - fifth root of 1980
 - seventh root of 1760
 - square root of 2001
- Write the following polynomials in standard form
 - $y - 2 + 8y^3 - 3y^2$
 - $x^2 - 11 + 3x^3 + 8x^4 - 2x$
 - $-5x^2 + 2 - 6x^3 + x^4$
- Write the following monomials according to their degrees in descending order.
 $-4x^2, \frac{1}{7}x^5, -6x^{11}, 7x^6, 2x^3$
- The value of 'x', when $3^x = \frac{\sqrt[3]{3}}{9}$ is:
- The value of $\left\{8^{-\frac{2}{3}}\right\}^{\frac{(27)^{\frac{1}{3}}}{2}}$ is
- Each of the numbers 1, 2, 3 and 4 is substituted, in some order for p, q, r and s. Find the greatest possible value of $p^q + r^s$.
- The length of a room is 9.0085×10^{-2} cm. Which number is equivalent to this length?

Short Answer Questions :

DIRECTIONS : Give answer in two to three sentences.

- If $\frac{x}{y} = \left(\frac{-1}{3}\right)^{-2} \div \left(\frac{2}{3}\right)^{-3}$, find the value of $\left(\frac{x}{y}\right)^{-3}$.
- Evaluate :
 - $\frac{2^0 + 5^0 \times 8^0}{2^0 + 5^0 + 8^0}$
 - $\frac{2}{5} \times \left(\frac{2}{5}\right)^3 \div \left(\frac{3}{5}\right)^4$
 - $(5^{-1} - 7^{-1})^{-1} + (3^{-1} - 5^{-1})^{-1}$
- By what number should $\left(\frac{-3}{2}\right)^{-3}$ be divided, so that the quotient may be $\left(\frac{4}{27}\right)^{-2}$?

- Find x so that

$$\left(\frac{2}{3}\right)^{-9} \times \left(\frac{2}{3}\right)^7 \times \left(\frac{2}{3}\right)^{2x+2} = 1$$

- If $\frac{p}{q} = \left(\frac{2}{3}\right)^2 \div \left(\frac{6}{7}\right)^0$ then find the value of $(q/p)^3$.
- If $3^x = 500$ then find the value of 3^{x-2} .
- Evaluate $(1^3 + 2^3 + 3^3 + 4^3)^{-3/2}$.
- If the mean of three numbers a, b and c is 3, then find the value of $\sqrt[3]{(7^{a+b-c})(7^{b+c-a})(7^{c+a-b})}$.
- If $\sqrt{0.04 \times 0.4 \times a} = 0.004 \times 0.4 \times \sqrt{b}$, then find $\frac{a}{b}$.
- Given that $9^n + 9^n + 9^n = 3^{2013}$, what is the value of n?

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

- Find the value of $\frac{\left(\frac{-1}{2}\right)^5}{\left(\frac{-1}{2}\right)^4} \div \frac{\left(\frac{-1}{8}\right)}{\left(\frac{-1}{4}\right)}$
- If $4^x + 4^x + 4^x + 4^x + 4^x + 4^x + 4^x + 4^x = \frac{1}{512}$, then what is the value of $\frac{-3}{x}$?
- If $x = \left(\frac{-2}{5}\right)^{30} \div \left(\frac{-2}{5}\right)^{28}$ and $y = \left(\frac{-2}{5}\right)^2 \times \left(\frac{-2}{5}\right)^3$ then find the value of $(x^3 \div y)^2$.
- Simplify:

$$\left[\frac{\frac{a}{x^{a-b}}}{\frac{a}{x^{a+b}}} \div \frac{\frac{b}{x^{b-a}}}{\frac{b}{x^{b+a}}} \right]^{a+b}$$
- Find the value of $\frac{(0.6)^0 - (0.1)^{-1}}{\left(\frac{3}{2}\right)^{-1} \left(\frac{3}{2}\right)^3 + \left(\frac{-1}{3}\right)^{-1}}$
- If $x^{x^{\frac{3}{2}}} = (x^{\frac{3}{2}})^x$, then find the value of x.
- Simplify: $\sqrt{\frac{8^{10} + 4^{10}}{64^2 + 4^9 \times 16}}$

8. Find the value of $\frac{1}{(216)^{-2/3}} + \frac{1}{(256)^{-3/4}} + \frac{1}{(32)^{-1/5}}$
9. If $\frac{3^x}{1+3^x} = \frac{1}{9}$, find the value of $\frac{9^x}{1+9^x}$ is

10. Find the value of the expression

$$\frac{(0.3)^{5/6} \cdot \left(\frac{1}{27}\right)^{1/6} \cdot (9)^{1/6} \cdot (0.81)^{2/3}}{(0.9)^{2/3} \cdot (3)^{-1/2} \cdot \left(\frac{1}{3}\right)^{-2} \cdot (243)^{1/4}}$$

2

EXERCISE

Text Book Exercise :

- Express 4^{-3} as a power with the base 2.
- Simplify and write the answer in the exponential form.
 - $(2^5 \div 2^8)^5 \times 2^{-5}$
 - $(-4)^{-3} \times (5)^{-3} \times (-5)^{-3}$
- Find m so that $(-3)^{m+1} \times (-3)^5 = (-3)^7$
- Simplify
 - $\left\{ \left(\frac{1}{3}\right)^{-2} - \left(\frac{1}{2}\right)^{-3} \right\} \div \left(\frac{1}{4}\right)^{-2}$
 - $\left(\frac{5}{8}\right)^{-7} \times \left(\frac{8}{5}\right)^{-5}$
- Find the value of.
 - $(3^0 + 4^{-1}) \times 2^2$
 - $\left(\frac{1}{2}\right)^{-2} + \left(\frac{1}{3}\right)^{-2} + \left(\frac{1}{4}\right)^{-2}$
 - $(3^{-1} + 4^{-1} + 5^{-1})^0$
- Evaluate
 - $\frac{8^{-1} \times 5^3}{2^{-4}}$
 - $(5^{-1} \times 2^{-1}) \times 6^{-1}$
- Simplify.
 - $\frac{25 \times t^{-4}}{5^{-3} \times 10 \times t^{-8}} (t \neq 0)$
 - $\frac{3^{-5} \times 10^{-5} \times 125}{5^{-7} \times 6^{-5}}$
- Express the following numbers in standard form.
 - 0.000035
 - 4050000
- Express the following numbers in usual form.
 - 3.52×10^5
 - 7.54×10^{-4}
 - 3×10^{-5}
- In a stack there are 5 books each of thickness 20mm and 5 paper sheets each of thickness 0.016 mm. What is the total thickness of the stack.

Exemplar Questions :

- Find the product of the cube of (-2) and the square of $(+4)$.
In question 2 and 3, find the value of n .
- If $\frac{6^n}{6^{-2}} = 6^3$, then find the value of n .
- If $\frac{2^n \times 2^6}{2^{-3}} = 2^{18}$, then find the value of n .
- If $\frac{5^m \times 5^3 \times 5^{-2}}{5^{-5}} = 5^{12}$, find m .
- The volume of the Earth is approximately 7.67×10^{-7} times the volume of the Sun. Express this figure in usual form.
- By what number should $(-15)^{-1}$ be divided so that the quotient may be equal to $(-5)^{-1}$?
- Find x .
 - $2^x + 2^x + 2^x = 192$
 - $\left(\frac{-6}{7}\right)^{x-7} = 1$
 - $2^{3x} = 8^{2x+1}$
- If $a = -1$, $b = 2$, then find the value of the following:
 - $a^b + b^a$
 - $a^b - b^a$
 - $a^b \times b^a$
 - $a^b \div b^a$
- Simplify:
 - $\left[\left(\frac{4}{3}\right)^{-2} - \left(\frac{3}{4}\right)^2 \right]^{-2}$
 - $\left(\frac{1}{5}\right)^{45} \times \left(\frac{1}{5}\right)^{-60} - \left(\frac{1}{5}\right)^{28} \times \left(\frac{1}{5}\right)^{-43}$
 - $\frac{(3^{-2})^2 \times (5^2)^{-3} \times (t^{-3})^2}{(3^{-2})^5 \times (5^3)^{-2} \times (t^{-4})^3}$
- Simplify $\frac{(-2)^3 \times (-2)^7}{3 \times 4^6}$
- Find x so that $(-5)^{x+1} \times (-5)^5 = (-5)^7$

HOTS Questions :

- $\frac{1}{1+x^{b-a}+x^{c-a}} + \frac{1}{1+x^{a-b}+x^{c-b}} + \frac{1}{1+x^{b-c}+x^{a-c}} = ?$
- Find the value of $\left(\frac{x^a}{x^b}\right)^{a+b} \times \left(\frac{x^b}{x^c}\right)^{b+c} \times \left(\frac{x^c}{x^a}\right)^{c+a}$
- If $2^x - 2^{x-1} = 4$, then find the value of x^x
- If $a = x^{\frac{1}{3}} + x^{-\frac{1}{3}}$ then find the value of $a^3 - 3a$

- Express the given expression as a power of a rational number with

- a positive exponent
- a negative exponent

$$\left\{ \left(\frac{1}{3}\right)^{-4} \times \left(\frac{3}{7}\right)^5 \div 21^{-3} \right\} \div 7^{10}$$

- Simplify : $\sqrt{\frac{1}{16}} + (0.09)^{-\frac{1}{2}} - (64)^{\frac{5}{6}} \times 7^0$

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- If $4^x - 4^{x-1} = 24$, find the value of $(2x)^x$
 - $5^{\frac{5}{2}}$
 - $\sqrt{5^5}$
 - $5 \times 5^{\frac{3}{2}}$
 - All of these
- If $9^{x+2} = 240 + 9^x$, then the value of x is
 - $\frac{1}{2}$
 - 1
 - 0
 - 2
- If m and n are whole numbers such that $m^n = 121$, then the value of $(m-1)^{n+1}$ is
 - 1
 - 10
 - 1000
 - 121
- If $xy = 1$, then $\frac{1}{1+x^{-1}} + \frac{1}{1+y^{-1}} =$
 - 1
 - xy
 - $\frac{1}{xy}$
 - None of these
- If $a^b = b^a$, then $\left(\frac{a}{b}\right)^{\frac{a}{b}} =$
 - 1
 - $\frac{a}{b}$
 - 2
 - $a^{\frac{a}{b}-1}$
- Simplify: $(5an^{-2})^{-1}$.
 - $\frac{n^2}{5a}$
 - $\frac{n}{5a}$
 - $\frac{n^3}{5a}$
 - $\frac{n^4}{5a}$
- The scientific notation of 16500000000000 is given by:
 - 16.5×10^{13}
 - 16.5×10^{12}
 - 1650×10^{11}
 - 1.65×10^{14}
- If $\frac{p}{q} = \left(\frac{2}{3}\right)^3 \div \left(\frac{3}{2}\right)^{-3}$ then the value of $\left(\frac{p}{q}\right)^{-10}$ is
 - 1
 - 5
 - 2^{-2}
 - 3
- The solution of $3^{3x-5} = \frac{1}{9^x}$ is.
 - 1
 - 3
 - 9^x
 - 3^2
- Value of the expression $\frac{x+y+z}{x^{-1}y^{-1}+y^{-1}z^{-1}+z^{-1}x^{-1}}$ is
 - $\frac{1}{x+y+z}$
 - $\frac{xyz}{x+y+z}$
 - $\frac{1}{xyz}$
 - xyz
- Simplify : $(6^{-1} - 8^{-1})^{-1} + (2^{-1} - 3^{-1})^{-1}$
 - $\frac{5}{2}$
 - $\frac{1}{6}$
 - 30
 - $\frac{1}{24}$
- Solve for y if $\frac{\left(\frac{1}{9}\right)^{2y-1} \cdot (.0081)^{1/3}}{\sqrt{243}} = \left(\frac{1}{3}\right)^{2y-5} \sqrt[3]{\frac{27^{y-1}}{10000}}$
 - 19×8
 - $\frac{-19}{8}$
 - $\frac{18}{9}$
 - $\frac{-19}{18}$

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

- If $800 = 8 \times 10^8 \times x^{-3/2}$, then x equals
 (a) 100^2 (b) 10^3
 (c) 10^4 (d) 100^3
- $\left(-\frac{5}{6}\right)^{3/4}$ when divided by $\left(-\frac{5}{6}\right)^{7/6}$ becomes $\left(-\frac{5}{6}\right)^{7-x}$, the value of x is
 (a) $\frac{70}{12}$ (b) $\frac{89}{12}$
 (c) $\left(\sqrt{\frac{89}{12}}\right)^2$ (d) $\left(\sqrt{\frac{7}{12}}\right)^2$
- $\left(\frac{64}{125}\right)^{-2/3} \div \frac{1}{\left(\frac{256}{625}\right)^{1/4}} + \left(\frac{\sqrt{25}}{\sqrt[3]{64}}\right)^0 =$
 (a) $\frac{9}{2}$ (b) $\frac{9}{4}$
 (c) $\left(\frac{3}{2}\right)^2$ (d) $\left(\frac{8}{4}\right)^2$
- $\frac{4^{n+1} \cdot 2^n - 8^n}{2^{2n}} = \frac{3}{8}$, then n equals
 (a) 2 (b) $(-27)^{1/3}$
 (c) $(3)^3$ (d) -3
- Which of the following is/are correct?
 (a) $\left\{\left(\frac{-7}{9}\right)^2\right\}^3 = \left(\frac{7}{9}\right)^6$
 (b) The reciprocal of $\left\{\left(\frac{-2}{19}\right)^5\right\}^4$ is $\left(\frac{19}{2}\right)^{20}$
 (c) $\left(\frac{-3}{2}\right)^{33} = \frac{3^{33}}{2^{33}}$
 (d) $33^9 - 33^8 = 33$
- Which of the following is/are incorrect?
 (a) $4^5 + 4^3 = 4^8$
 (b) $(2+5)^4 = 2^4 + 5^4$
 (c) $\left(\frac{10}{11}\right)^8 \times \left(\frac{10}{11}\right)^5 = \left(\frac{10}{11}\right)^3 \times \left(\frac{10}{11}\right)^m \Rightarrow m = 10$
 (d) $\left(\frac{-3}{2}\right)^{33} = -\frac{3^{33}}{2^{33}}$

- Which of the following statement is/are incorrect?
 (a) $x^y > y^x$, if $x = 2$ and $y = 4$
 (b) $x = 1$ for all values of x
 (c) $x^{-y} > y^{-x}$ if $x = -103$ and $y = -104$
 (d) $x^y > y^x$, if $x = 4$ and $y = 2$.

- Which of the following values are equal?
 (a) 1^4 (b) 4^0
 (c) 0^4 (d) 4^1

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE - I

Exponents are very useful to express very large number such as 5,000,000,000 which is $5 \times 1000,000,000$ as 5×10^9 and small

numbers such as 0.000,000,005 as $\frac{5}{1,000,000,000} = 5 \times 10^{-9}$.

When a number is expressed with one whole number and the remaining part as exponents of 10, then the number is in standard form. Convert the numbers into standard form.

- The milky way is 100,000 light years away.
 (a) 10×10^5 (b) 2×10^5
 (c) 100 (d) 1×10^4
- The age of our universe is 8000,000,000 years nearly.
 (a) 8×10^{-5} (b) 8.0×10^9
 (c) 9×10^9 (d) 8×10^8
- The coefficient of linear expansion of copper is 0.0000187.
 (a) 1.87×10^4 (b) 2×10^3
 (c) 1.87×10^{-5} (d) 1.87

PASSAGE - II

$$a^m \times a^n \times a^{-p} = \frac{a^m \times a^n}{a^p}$$

- The value of $6^2 \times 6^{-4} \times 6^8$ is
 (a) 6^6 (b) 6^{-6}
 (c) 6^{12} (d) 6^{-10}
- Evaluate: $\left(\frac{1}{2}\right)^{-4} \times \left(\frac{1}{2}\right)^{-8} \times \left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^{+2} \times \left(\frac{1}{4}\right)^{-6} \times \left(\frac{1}{4}\right)^2$
 (a) $\left(\frac{1}{2}\right)^{-10} + \left(\frac{1}{4}\right)^{-2}$ (b) $\left(\frac{1}{2}\right)^{+12} + \left(\frac{1}{4}\right)^{-1}$
 (c) $\left(\frac{1}{2}\right)^{-14} + \left(\frac{1}{4}\right)^0$ (d) $\left(\frac{1}{2}\right)^{+12}$
- Find the value of x if $4^{2x-3} = 4^2 \times 2^3 \times 4$
 (a) 0 (b) 4
 (c) $\frac{15}{4}$ (d) $\frac{-9}{8}$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statement.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** We should multiply $(-7)^{-1}$ to $\frac{-7}{4}$ to get the product as 4^{-1} .

Reason : If $\frac{x}{y} = \left(\frac{5}{2}\right)^{-1} \times \left(\frac{8}{9}\right)^0$ then the value of $\left(\frac{x}{y}\right)^{-2}$ is $\left(\frac{2}{5}\right)^2$.

2. **Assertion :** $(2y^3)^2 = 2y^6$

Reason : $(2y^3)^2 = 4y^6$

3. **Assertion :** The value of $\left(\frac{5}{8}\right)^{-2} \times \left(\frac{5}{8}\right)^{-3} \times \left(\frac{5}{8}\right)^5$ is 0.

Reason : $x^m \times x^n \times x^p = x^{m+n+p}$

4. **Assertion:** $\left\{\left(\frac{1}{2}\right)^6\right\}^6$ is the reciprocal of 2^{36} .

Reason : $\left\{(a^m)^n\right\} = a^{m \times n}$, for any value of a .

5. **Assertion:** The value of $\left[\left(\frac{2}{3}\right)^2\right]^3 \times \left(\frac{1}{3}\right)^{-2} \times 3^{-1} \times \frac{1}{6}$ is $\frac{32}{729}$

Reason : Property I = $(a^{-m})^n = a^{m \times n}$ and property

II = $a^{-1} = \left(\frac{1}{a}\right)^{-1}$ holds.

Integer Type Question :

- Find the value of $\frac{(243)^{\frac{n}{5}} \times 3^{2n+1}}{9^n \times 3^{n-1}}$
- Find the least value of $5^x + 5^{-x}$.
- Find the value of $\frac{1}{1+a^{(n-m)}} + \frac{1}{1+a^{(m-n)}} = ?$
- Find the value of $\left(\frac{x^b}{x^c}\right)^{b+c-a} \times \left(\frac{x^c}{x^a}\right)^{c+a-b} \times \left(\frac{x^a}{x^b}\right)^{a+b-c}$
- Find the value of $\left(\frac{x^a}{x^b}\right)^{\frac{1}{ab}} \times \left(\frac{x^b}{x^c}\right)^{\frac{1}{bc}} \times \left(\frac{x^c}{x^a}\right)^{\frac{1}{ac}}$
- If $27^k = \frac{9}{3^k}$, then find the value of $\frac{1}{k^2}$
- If $4^{\sqrt{x}} \sqrt{x} = 256$ then find the value of x .
- If $2^{x-1} + 2^{x+1} = 320$, then find the value of x .
- If $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{b}{a}\right)^{x-3}$, then find the value of x is equal to
- Given that $4^{n+1} = 256$, find the value of n .

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- $6.5 \times 10^{-5}; 0.000065 = \frac{65}{100000} = 6.5 \times 10^{-5}.$
- $\sqrt{\left(\frac{25}{21}\right)^3}; \left(\frac{25}{21}\right)^{3/2} = \sqrt{\left(\frac{25}{21}\right)^3}$
- $696000; 6.96 \times 10^5 = 6.96 \times 100000 = 696000.$
- $\left(\frac{2^5}{2^8}\right) \times 2^4 = 2^{-3} \times 2^4 = 2$
- $a^5 \times b^5 = (ab)^5.$
- $(512)^{-2/9} = [(2)^9]^{-2/9} = 2^{-2} = \frac{1}{4}$
- $4^{-3/2} + 8^{2/3} = 2^{-3} + 2^2 = \frac{1}{8} + 4 = \frac{33}{8}$
- $(8^0 - 3^0) \times (8^0 + 3^0) = (1 - 1)(1 + 1) = 0$
- $(2^{-1} - 4^{-1})^2 = \left(\frac{1}{2} - \frac{1}{4}\right)^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}$
- $\frac{25 \times a^{-4}}{5^{-3} \times 10 \times a^{-8}} = \frac{25 \times 5^3 \times a^{(8-4)}}{10} = \frac{625}{2} a^4$

True / False :

- True
- True
- False, the exponents is multiplied instead of added.
- False, the exponents were subtracted, but the bases were not the same.
- False, the exponents were added instead of multiplied.
- False: $(1 + a - b^{-1})^{-1} + (1 + b - a^{-1})^{-1}$

$$= \frac{1}{1 + a - \frac{1}{b}} + \frac{1}{1 + b - \frac{1}{a}} = \frac{b}{b + ab - 1} + \frac{a}{a + ab - 1}$$

$$= \frac{b}{b} + \frac{a}{a} = 1 + 1 = 2$$

Its value is 2.

- False: $\frac{1}{1 + \frac{1}{a}} + \frac{1}{1 + \frac{1}{b}} = \frac{a}{1 + a} + \frac{b}{1 + b}$

$$= \frac{a + ab + b + ab}{1 + a + b + ab}$$

$$= \frac{2 + a + b}{2 + a + b} = 1 \quad (\because ab = 1).$$
- False, $\frac{10}{3} \times 3^x - 3^{x-1} = 81$

$$\Rightarrow 3^{x-1} \times 9 = 81$$

$$\Rightarrow 3^{x-1} = 9$$

$$\Rightarrow x = 3$$

Match the Column :

- (A) $\rightarrow$ (r); (B) $\rightarrow$ (s); (C) $\rightarrow$ (q); (D) $\rightarrow$ (p)
 A. $-9 + x = 1$
 B. $x - 2 + 4 = -3$
 C. $\left(\frac{-3}{5}\right)^2 = \left(\frac{5}{3}\right)^{2x}, \frac{9}{25} = \left(\frac{5}{3}\right)^{2x}$

$$\Rightarrow \frac{25}{9} = \left(\frac{5}{3}\right)^{-2x}$$

$$\Rightarrow \left(\frac{5}{3}\right)^2 = \left(\frac{5}{3}\right)^{-2x} \Rightarrow x = -1$$
- (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)

Very Short Answer Questions :

- $K = (m + n - p)^q$
- $\sqrt[9]{6}$
- (i) $\sqrt[3]{972} = \sqrt[3]{2^2 \times 3^5} = 3\sqrt[3]{36}$
 (ii) $\sqrt[5]{1980}$
 (iii) $\sqrt[7]{1760}$
 (iv) $\sqrt{2001}$
- (i) $8y^3 - 3y^2 + 4 - 2$
 (ii) $8x^4 + 3x^3 + x^2 - 2x - 11$
 (iii) $x^4 - 6x^3 - 5x^2 + 2$
- $-6x^4, 7x^6, \frac{1}{7}x^5, 2x^3, -4x^2$

$$6. \quad 3^x = \frac{\sqrt[3]{3}}{9}$$

$$9 \cdot 3^x = \sqrt[3]{3}$$

$$3^2 \cdot 3^x = (3)^{1/3}$$

$$3^{2+x} = (3)^{1/3}$$

$$x + 2 = \frac{1}{3}$$

$$x = \frac{1}{3} - 2 = \left(\frac{-5}{3}\right)$$

$$7. \quad \left\{ (8)^{-\frac{2}{3}} \right\}^{\left(\frac{27}{2}\right)^{\frac{1}{3}}}$$

$$= \left\{ (2^3)^{-2/3} \right\}^{\left[\frac{3^3}{2} \right]^{\frac{1}{3}}}$$

$$= \left\{ (2^3)^{-2/3} \right\}^{3/2}$$

$$= 2^{3 \times \frac{-2}{3} \times \frac{3}{2}}$$

$$= 2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

$$8. \quad p^q + r^s = (1)^2 + (3)^4 = 1 + 81 = 82$$

$$9. \quad 9.0085 \times 10^{-2} = 0.090085$$

Short Answer Questions :

$$1. \quad \frac{x}{y} = \left(\frac{-1}{3}\right)^{-2} \div \left(\frac{2}{3}\right)^{-3}$$

$$= \frac{\left(\frac{-1}{3}\right)^{-2}}{\left(\frac{2}{3}\right)^{-3}} = \left(\frac{-1}{3}\right)^{-2} \times \left(\frac{3}{2}\right)^{-3}$$

$$\text{Hence, } \left(\frac{x}{y}\right)^{-3} = \frac{27}{512}$$

$$2. \quad \text{(i) } \frac{1+1 \times 1}{1+1+1} = \frac{2}{3} \quad \text{(ii) } \left(\frac{2}{5} \times \frac{5}{3}\right)^4 = \left(\frac{2}{3}\right)^4$$

$$\text{(iii) } \frac{50}{2} = 25$$

$$3. \quad \frac{-2^7}{3^9}$$

$$4. \quad \left(\frac{2}{3}\right)^{2x} = 1$$

$$\Rightarrow 2x = 0$$

$$\Rightarrow x = 0$$

$$5. \quad \frac{p}{q} = \left(\frac{2}{3}\right)^2 \div \left(\frac{6}{7}\right)^\circ$$

$$= \left(\frac{2}{3}\right)^2 \div 1 [a^\circ = 1]$$

$$\frac{p}{q} = \frac{4}{9}$$

$$\frac{q}{p} = \frac{9}{4} \Rightarrow \left(\frac{q}{p}\right)^3 = \left(\frac{9}{4}\right)^3 = \frac{729}{64}$$

$$6. \quad 3^x = 500$$

$$3^{x-2} = \frac{3^x}{3^2} = \frac{500}{9}$$

$$7. \quad (1^3 + 2^3 + 3^3 + 4^3)^{-3/2}$$

$$= (1 + 8 + 27 + 64)^{-3/2}$$

$$= (100)^{-3/2}$$

$$= \left[(100)^{1/2}\right]^{-3} = (10)^{-3}$$

$$8. \quad \sqrt[3]{(7^{a+b-c})(7^{b+c-a})(7^{c+a-b})}$$

$$= \sqrt[3]{(7^{a+b-c+b+c-a+c+a-b})}$$

$$= \sqrt[3]{(7^{a+b+c})}$$

$$= \left(7^{\frac{a+b+c}{3}}\right) = 7^3 \quad \left[\because \frac{a+b+c}{3} = \text{mean} = 3\right]$$

$$9. \quad \sqrt{0.04 \times 0.4 \times a} = (0.004) \times (0.4) \times \sqrt{b}$$

$$0.04 \times 0.4 \times a = (0.004)^2 \times (0.4)^2 \times b \quad [\text{Squaring both sides}]$$

$$\Rightarrow \frac{a}{b} = \frac{(0.004)^2 \times (0.4)^2}{0.04 \times 0.4} = 4 \times 4 \times 10^{-5}$$

$$= 16 \times 10^{-5}$$

$$10. \quad 9^n + 9^n + 9^n = 3^{2013}$$

$$3 \cdot 9^n = 3^{2013}$$

$$3 \cdot 3^{2n} = 3^{2013}$$

$$3^{(1+2n)} = 3^{2013}$$

$$1 + 2n = 2013$$

$$2n = 2013 - 1$$

$$n = \frac{2012}{2} = 1006$$

Long Answer Questions :

$$1. \frac{\left(\frac{-1}{2}\right)^5}{\left(\frac{-1}{2}\right)^4} \div \frac{\left(\frac{-1}{8}\right)}{\left(\frac{-1}{4}\right)}$$

$$= \left(\frac{-1}{2}\right)^{5-4} \div \left\{\left(\frac{-1}{2^3}\right) \times \left(\frac{-1}{2^{-2}}\right)\right\}$$

$$= \left(\frac{-1}{2}\right)^1 \div \left(\frac{1}{2}\right)$$

$$= \left(\frac{-1}{2}\right) \times \left(\frac{2}{1}\right) = -1$$

$$2. 4^x + 4^x + 4^x + 4^x + 4^x + 4^x + 4^x + 4^x = \frac{1}{512}$$

$$8(4^x) = \frac{1}{512}$$

$$4^x = \frac{1}{512 \times 8}$$

$$4^x = (512 \times 8)^{-1}$$

$$4^x = 4^{-6}$$

$$\therefore x = -6$$

$$\Rightarrow \frac{-3}{x} = \frac{-3}{-6} = \frac{1}{2} = 0.50$$

$$3. x = \left(\frac{-2}{5}\right)^2, y = \left(\frac{-2}{5}\right)^5$$

$$x^3 = \left(\frac{-2}{5}\right)^6$$

$$x^3 \div y = \frac{-2}{5}$$

$$\therefore (x^3 \div y)^2 = \frac{4}{25}$$

$$4. \left[\frac{x^{\frac{a}{a-b}} \times x^{\frac{b}{b-a}}}{x^{\frac{a}{a+b}} \times x^{\frac{b}{b+a}}} \right]^{a+b}$$

$$= \left[x^{\frac{a}{a-b} - \frac{a}{a+b}} \times x^{\frac{b}{b-a} - \frac{b}{b+a}} \right]^{a+b}$$

$$= \left[x^{\frac{a^2+ab-a^2+ab}{a^2-b^2}} \times x^{\frac{b^2+ab-b^2+ab}{b^2-a^2}} \right]^{a+b}$$

$$= \left[x^{\frac{2ab}{a^2-b^2}} \times x^{\frac{2ab}{b^2-a^2}} \right]^{a+b}$$

$$= \left[x^{\frac{2ab}{a-b}} \times x^{\frac{-2ab}{a-b}} \right]^{\frac{1}{a+b} \times a+b}$$

$$= \left[x^{\frac{2ab-2ab}{a-b}} \right] = x^0 = 1$$

$$5. \frac{(0.6)^0 - (0.1)^{-1}}{\left(\frac{3}{2^3}\right)^{-1} \left(\frac{3}{2}\right)^3 + \left(\frac{-1}{3}\right)^{-1}} = \frac{1 - \frac{1}{0.1}}{\frac{2^3}{3} \times \frac{3^3}{2^3} + (-3)}$$

$$= \frac{1-10}{9-3} = \frac{-9}{6} = \frac{-3}{2}$$

$$6. x^{x^{3/2}} = (x^{3/2})^x$$

$$\Rightarrow x^{x^{3/2}} = x^{3x/2}$$

$$\Rightarrow x^{3/2} = \frac{3x}{2}$$

$$\Rightarrow x \times x^{1/2} = \frac{3x}{2}$$

$$\Rightarrow x^{1/2} = \frac{3}{2}$$

$$\Rightarrow x = \frac{9}{4}$$

$$7. \sqrt{\frac{8^{10} + 4^{10}}{64^2 + 4^9 \times 16}} = \sqrt{\frac{2^{30} + 2^{20}}{2^{12} + 2^{18} \times 2^4}}$$

$$= \sqrt{\frac{2^{30} + 2^{20}}{2^{12} + 2^{22}}} = \sqrt{\frac{2^{18} + 2^8}{1 + 2^{10}}} = (2^8)^{1/2} = 2^4 = 16$$

$$8. \frac{1}{(216)^{-2/3}} + \frac{1}{(256)^{-3/4}} + \frac{1}{(32)^{-1/5}}$$

$$= \frac{1}{(6^3)^{-2/3}} + \frac{1}{(4^4)^{-3/4}} + \frac{1}{(2^5)^{-1/5}}$$

$$= \frac{1}{6^{-2}} + \frac{1}{4^{-3}} + \frac{1}{2^{-1}}$$

$$= 6^2 + 4^3 + 2^1 = 36 + 64 + 2 = 102$$

$$9. \quad \frac{3^x}{1+3^x} = \frac{1}{9}$$

$$\Rightarrow 3^x \cdot 9 = 1 + 3^x \Rightarrow 9 \cdot 3^x - 3^x = 1$$

$$\Rightarrow 3^x(9-1) = 1$$

$$\Rightarrow 3^x = \frac{1}{8} \Rightarrow 9^x = \frac{1}{64}$$

$$\therefore \frac{9^x}{1+9^x} = \frac{\frac{1}{64}}{1+\frac{1}{64}}$$

$$= \frac{1/64}{65/64} = \frac{1}{65}$$

$$10. \quad \frac{(0.3)^{1/3} \left(\frac{1}{27}\right)^{1/4} (9)^{1/6} (0.81)^{1/5}}{(0.9)^{2/3} (3)^{-1/2} \left(\frac{1}{3}\right)^{-2} (243)^{-1/4}}$$

$$= \frac{\left(\frac{3}{10}\right)^{1/3} \left(\frac{1}{3}\right)^{3/4} (3^2)^{1/6} \left(\frac{81}{100}\right)^{2/3}}{\left(\frac{9}{10}\right)^{2/3} (3)^{-1/2} (3^{-1})^{-2} (3^5)^{-1/4}}$$

$$= \frac{10^{-1/3} 3^{1/3} (3)^{-3/4} (3)^{1/3} (3)^{4 \times 2/3} (10^{-2})^{2/3}}{3^{4/3} \times 10^{-2/3} \times 3^{-1/2} \times 3^2 \times 3^{-5/4}}$$

$$= \frac{3^{\frac{1}{3} - \frac{3}{4} + \frac{1}{3} + \frac{8}{3} - \frac{4}{3} - \frac{1}{3}} 10^{-\frac{2}{3}}}{3^{\frac{4}{3} - \frac{1}{2} + 2 - \frac{5}{4}} 10^{-\frac{2}{3}}} = \frac{(3)^{\frac{31}{12}} (10)^{-\frac{5}{3}}}{(3)^{\frac{19}{12}} (10)^{-\frac{2}{3}}}$$

$$= 3^{\frac{31}{12} - \frac{19}{12}} 10^{-\frac{5}{3} + \frac{2}{3}} = 3^1 \times 10^{-1} = \frac{3}{10} = 0.3$$

2 EXERCISE

Text Book Exercise :

$$1. \quad \text{We have, } 4 = 2 \times 2 = 2^2$$

$$\text{Therefore, } (4)^{-3} = (2 \times 2)^{-3} = (2^2)^{-3} = 2^{2 \times (-3)} = 2^{-6}$$

$$[\because (a^m)^n = a^{mn}]$$

$$2. \quad (i) \quad (2^5 \div 2^8)^5 \times 2^{-5} = (2^{5-8})^5 \times 2^{-5} = (2^{-3})^5 \times 2^{-5}$$

$$= 2^{-15-5} = 2^{-20} = \frac{1}{2^{20}}$$

$$(ii) \quad (-4)^{-3} \times (5)^{-3} \times (-5)^{-3} = [(-4) \times 5 \times (-5)]^{-3} = [100]^{-3}$$

$$= \frac{1}{100^3} \quad [\text{using the law } a^m \times b^m = (ab)^m, a^{-m} = \frac{1}{a^m}]$$

$$3. \quad (-3)^{m+1} \times (-3)^5 = (-3)^7$$

$$(-3)^{m+1+5} = (-3)^7$$

$$(-3)^{m+6} = (-3)^7$$

$$\text{Therefore, } m+6=7$$

$$\text{or } m=7-6=1$$

$$4. \quad (i) \quad \left\{ \left(\frac{1}{3}\right)^{-2} - \left(\frac{1}{2}\right)^{-3} \right\} \div \left(\frac{1}{4}\right)^{-2} = \left\{ \frac{1^{-2}}{3^{-2}} - \frac{1^{-3}}{2^{-3}} \right\} \div \frac{1^{-2}}{4^{-2}}$$

$$= \left\{ \frac{3^2}{1^2} - \frac{2^3}{1^3} \right\} \div \frac{4^2}{1^2} = \{9-8\} \div 16 = \frac{1}{16}$$

$$(ii) \quad \left(\frac{5}{8}\right)^{-7} \times \left(\frac{8}{5}\right)^{-5} = \frac{5^{-7}}{8^{-7}} \times \frac{8^{-5}}{5^{-5}} = \frac{5^{-7}}{5^{-5}} \times \frac{8^{-5}}{8^{-7}} = 5^{(-7)-(-5)} \times 8^{(-5)-(-7)}$$

$$5^{-2} \times 8^2 = \frac{8^2}{5^2} = \frac{64}{25}$$

$$5. \quad (i) \quad \left(1 + \frac{1}{4}\right) \times 4 = 5$$

$$(ii) \quad 2^2 + 3^2 + 4^2 = 4 + 9 + 16 = 29$$

$$(iii) \quad (3^{-1} + 4^{-1} + 5^{-1})^0 = 1$$

$$6. \quad (i) \quad \frac{8^{-1} \times 5^3}{2^{-4}} = 2^{-3} \times 2^4 \times 5^3 = 2 \times 5^3 = 250$$

$$(ii) \quad (5^{-1} \times 2^{-1}) \times 6^{-1} = \left(\frac{1}{5 \times 2}\right) \times \frac{1}{6} = \frac{1}{60}$$

$$7. \quad (i) \quad \frac{5^5}{10} t^4 \quad (ii) \quad 5^5$$

$$8. \quad (i) \quad 0.000035 = 3.5 \times 10^{-5} \quad (ii) \quad 4050000 = 4.05 \times 10^6$$

$$9. \quad (i) \quad 3.52 \times 10^5 = 3.52 \times 100000 = 352000$$

$$(ii) \quad 7.54 \times 10^{-4} = \frac{7.54}{10^4} = \frac{7.54}{10000} = 0.000754$$

$$(iii) \quad 3 \times 10^{-5} = \frac{3}{10^5} = \frac{3}{100000} = 0.00003$$

$$10. \quad 5 \times 20 + 5 \times 0.016 = 100.08 \text{ mm}$$

Exemplar Questions :

$$1. \quad (-2)^3 \times 4^2 = -128$$

$$2. \quad n = 1$$

$$3. \quad n = 9$$

$$4. \quad m = 6$$

5. 0.000000767

6. $\frac{1}{3}$

7. (i) $x=6$ (ii) $x=7$
(iii) $(5)^x = -1$

8. (i) $\frac{3}{2}$ (ii) $\frac{1}{2}$

(iii) $\frac{1}{2}$ (iv) 2

9. (i) 0
(ii) 0
(iii) $(3t)^6$

10.
$$\frac{(-2)^3 \times (-2)^7}{3 \times 4^6} = \frac{(-2)^{3+7}}{3 \times (2^2)^6} \{a^m \times a^n = a^{m+n}\}$$
$$= \frac{(-2)^{10}}{3 \times 2^{12}} \{(a^m)^n = a^{m \times n}\}$$
$$= \frac{(-2)^{10}}{3 \times 2^{12}} = \frac{2^{10-12}}{3} \{a^m \div a^n = a^{m-n}, (-2)^{10} = 2^{10}\}$$
$$= \frac{2^{-2}}{3} = \frac{1}{3 \times 2^2} = \frac{1}{12}$$

11. $(-5)^{x+1} \times (-5)^5 = (-5)^7$
 $(-5)^{x+1+5} = (-5)^7 \{a^m \times a^n = a^{m+n}\}$
 $(-5)^{x+6} = (-5)^7$

On both sides, powers have the same base, so their exponents must be equal.

Therefore, $x+6=7$

$$x = 7 - 6 = 1$$

$$x = 1$$

HOTS Questions :

1. 1

2. 1

3. 27

4. $x + x^{-1}$

5. Given expression

$$= 3^{4+5} \times 7^{-5} \times (3 \times 7)^3 \times 7^{-10}$$

$$= 3^{9+3} \times 7^{-5+3-10}$$

$$= \left(\frac{3}{7}\right)^{12} = \left(\frac{7}{3}\right)^{-12}$$

$$\therefore \text{Given expression} = \left(\frac{3}{7}\right)^{12} \text{ or } \left(\frac{7}{3}\right)^{-12}$$

6. $\frac{-341}{12}$

3 EXERCISE

Single Option Correct :

1. (d) Given $4^x - 4^{x-1} = 24$

$$\Rightarrow 4^x - 4^x \times 4^{-1} = 24$$

$$\Rightarrow 4^x \left(1 - \frac{1}{4}\right) = 24 \Rightarrow 4^x \times \frac{3}{4} = 24$$

$$\Rightarrow 4^x = 32 \Rightarrow 2^{2x} = 2^5 \Rightarrow 2x = 5 \Rightarrow x = \frac{5}{2}$$

$$\text{now, } (2x)^x = \left(2 \times \frac{5}{2}\right)^{\frac{5}{2}}$$

$$= (5)^{5/2} = \sqrt{5^5} = 5^{1+\frac{3}{2}} = 5 \times 5^{\frac{3}{2}}$$

2. (a) 3. (c) 4. (a) 5. (d)

6. (a) $(5a^{-2})^{-1} = \frac{n^2}{5a}$

7. (d) 1.65×10^{14}

8. (a) $\frac{p}{q} = \left(\frac{2}{3}\right)^3 \div \left(\frac{3}{2}\right)^{-3} = \left(\frac{2}{3}\right)^3 \div \left(\frac{2}{3}\right)^3 = 1$

$$\therefore \left(\frac{p}{q}\right)^{-10} = (1)^{-10} = 1$$

9. (a) $3^{3x-5} = \frac{1}{9^x}$

$$3^{3x-5} = 3^{-2x}$$

$$3x - 5 = -2x$$

$$x = 1.$$

10. (d) $\frac{x+y+z}{x^{-1}y^{-1}+y^{-1}z^{-1}+z^{-1}x^{-1}}$

$$= \frac{x+y+z}{\frac{1}{xy} + \frac{1}{yz} + \frac{1}{zx}}$$

$$= \frac{x+y+z}{\frac{z+xy}{xyz}}$$

$$= \frac{(x+y+z) \times (xyz)}{(x+y+z)}$$

$$= (xyz)$$

11. (c) $(6^{-1} - 8^{-1})^{-1} + (2^{-1} - 3^{-1})^{-1}$

$$= \left(\frac{1}{6} - \frac{1}{8}\right)^{-1} + \left(\frac{1}{2} - \frac{1}{3}\right)^{-1}$$

$$= \left(\frac{4-3}{24}\right)^{-1} + \left(\frac{3-2}{6}\right)^{-1}$$

$$= \left(\frac{1}{24}\right)^{-1} + \left(\frac{1}{6}\right)^{-1} \quad \left[\text{using } n^{-1} = \frac{1}{n}\right]$$

$$= 24 + 6 = 30$$

12. (d) The given equation is

$$\frac{(3^{-2})^{2y-1} \cdot (3^4 \cdot 10^{-4})^{1/3}}{3^{5/2}}$$

$$= \frac{3^{-(2y-5)} \cdot 3^{3\left(\frac{y-1}{3}\right)}}{10^{4/3}}$$

$$\Rightarrow \frac{3^{-4y+2+4/3-5/2}}{10^{4/3}} = \frac{3^{-2y+5+y-1}}{10^{4/3}}$$

$$\Rightarrow 3^{-4y+5/6} = 3^{-y+4}$$

$$\Rightarrow -4y+5/6 = -y+4$$

$$\Rightarrow y = \frac{-19}{18}$$

More Than One Option Correct :

1. (a,c) $800 = 8 \times 10^8 \times x^{-3/2}$

$$\Rightarrow \frac{800}{8} = 10^8 \times x^{-3/2}$$

$$\Rightarrow \frac{800}{800} = 10^6 \times x^{-3/2}$$

$$\Rightarrow 10^{-6} = x^{-3/2}$$

$$\Rightarrow x = 10^{-6 \times 2/3}$$

$$\Rightarrow x = 10^4$$

2. (b,c) $\left(-\frac{5}{6}\right)^{3/4} \div \left(-\frac{5}{6}\right)^{7/6} = \left(-\frac{5}{6}\right)^{7-x}$

$$\left[-\frac{5}{6}\right]^{\frac{3}{4}-\frac{7}{6}} = \left[-\frac{5}{6}\right]^{7-x}$$

$$\left[-\frac{5}{6}\right]^{\frac{9-14}{12}} = \left[-\frac{5}{6}\right]^{7-x}$$

$$\Rightarrow \frac{-5}{12} = 7-x$$

$$\Rightarrow -5 = 84 - 12x$$

$$\Rightarrow -5 - 84 = -12x \quad \Rightarrow \frac{89}{12} = x$$

3. (b,c) $\left(\frac{64}{125}\right)^{-2/3} \div \frac{1}{\left(\frac{256}{625}\right)^{1/4}} + \left(\frac{\sqrt{25}}{\sqrt[3]{64}}\right)^0$

$$= \left(\frac{4}{5}\right)^{3 \times \left(-\frac{2}{3}\right)} \div \left(\frac{5}{4}\right)^{\frac{4}{4}} + 1$$

$$= \left(\frac{4}{5}\right)^{-2} \div \left(\frac{5}{4}\right) + 1$$

$$= \frac{\frac{5}{4} \times \frac{5}{4}}{\frac{5}{4}} + 1 = \frac{5}{4} + 1 = \frac{9}{4}$$

4. (b,d) $\frac{4^{n+1} \cdot 2^n - 8^n}{2^{2n}} = \frac{3}{8}$

$$\frac{2^{2(n+1)} \cdot 2^n - 2^{3n}}{2^{2n}} = \frac{2^{2n+2+n} - 2^{3n}}{2^{2n}}$$

$$= \frac{2^{2n+n+2-2n} - 2^{3n-2n}}{2^{2n}}$$

$$= \frac{2^{n+2} - 2^n}{2^{2n}} = \frac{2^n(4-1)}{2^{2n}} = 3 \cdot 2^{-n}$$

$$3 \cdot 2^n = \frac{3}{8} \Rightarrow 2^n = \frac{1}{8}$$

$$\Rightarrow 2^n = 2^{-3} \Rightarrow n = -3$$

5. (a,b)

6. (a, b)

7. (a, b, d)

8. (a, b) 1^4 and 4^0 both are equal to 1

Passage Based Questions :

1. (a) 1.0×10^5 light years

2. (b) 8.0×10^9 years

3. (c) 1.87×10^{-5}

4. (a) $6^2 \times 6^{-4} \times 6^8 = \frac{6^2 \times 6^8}{6^4} = \frac{6^{10}}{6^4} = 6^{10-4} = 6^6$

5. (a) $\left(\frac{1}{2}\right)^{-4} \times \left(\frac{1}{2}\right)^{-8} \times \left(\frac{1}{2}\right)^2 + \left(\frac{1}{4}\right)^{+2} \times \left(\frac{1}{4}\right)^{-6} \times \left(\frac{1}{4}\right)^2$

$$= \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{1}{2}\right)^{12}} + \frac{\left(\frac{1}{4}\right)^4}{\left(\frac{1}{4}\right)^6} = \left(\frac{1}{2}\right)^{-10} + \left(\frac{1}{4}\right)^{-2}$$

6. (c) $4^{2x-3} = 4^2 \times 2^3 \times 4$

$$\Rightarrow 4^{2x-3} = 4^3 \times 2^3$$

$$\Rightarrow (2^2)^{2x-3} = (2^2)^3 \times 2^3$$

$$\Rightarrow 2^{4x-6} = 2^9$$

$$\Rightarrow 4x - 6 = 9 \Rightarrow 4x = 15 \Rightarrow x = \frac{15}{4}$$

Assertion & Reason :

1. (c) Assertion is true. Reason is false.

$$\frac{x}{y} = \left(\frac{5}{2}\right)^{-1} \times \left(\frac{8}{9}\right)^0 = \left(\frac{5}{2}\right)^{-1} = \frac{2}{5}$$

$$\therefore \left(\frac{x}{y}\right)^{-2} = \left(\frac{2}{5}\right)^{-2}$$

2. (d) Assertion is false. The coefficient should have been squared.

$\therefore$ Reason is true.

3. (d) $\left(\frac{5}{8}\right)^{-2} \times \left(\frac{5}{8}\right)^{-3} \times \left(\frac{5}{8}\right)^5$

$$= \left(\frac{5}{8}\right)^{-2-3+5} = \left(\frac{5}{8}\right)^{-5+5} = \left(\frac{5}{8}\right)^0 = 1$$

$$x^m \times x^n \times x^p = x^{m+n+p}$$

$\therefore$ Assertion is false and Reason is true.

4. (a) Reciprocal of $2^{36} = 2^{-36}$ or $\left(\frac{1}{2}\right)^{36}$

$$= \left(\left(\frac{1}{2}\right)^6\right)^6 \text{ as } (a^m)^n = a^{m \times n} \text{ for any value of } a$$

$\therefore$ Assertion is true and Reason is also true and correctly explains Assertion.

5. (c) $\left[\left(\frac{2}{3}\right)^2\right]^3 \times \left[\frac{1}{3}\right]^{-2} \times 3^{-1} \times \frac{1}{6}$

$$= \left(\frac{2}{3}\right)^6 \times (3)^2 \times \frac{1}{3} \times \frac{1}{6} = \frac{2^6}{3^6} \times \frac{3^2}{3} \times \frac{1}{6}$$

$$= \frac{2^5}{3^6} = \frac{32}{729}$$

$\therefore$ Assertion is true

$$\text{Now } (a^{-m})^n = a^{-mn}$$

$$\therefore \text{Property I is false and } a^{-1} = \frac{1}{a}$$

Property II is also false.

Assertion is true, Reason is false.

Integer Type Question :

1. (9) $\frac{(243)^{\frac{n}{5}} \times 3^{2n+1}}{9^n \times 3^{n-1}} = \frac{[(3)^5]^{\frac{n}{5}} \times 3^{2n+1}}{(3^2)^n \times 3^{n-1}}$

$$= \frac{3^n \times 3^{2n+1}}{3^{2n} \times 3^{n-1}} [a^m \times a^n = a^{m+n}]$$

$$= \frac{3^{3n+1}}{3^{3n-1}} = 3^{3n+1-3n+1} \left[\frac{a^n}{a^m} = a^{n-m} \right]$$

$$= 3^2 = 9$$

2. (2)

3. (1) $\frac{1}{1+a^{n-m}} + \frac{1}{1+a^{m-n}} = \frac{1}{1+\frac{a^n}{a^m}} + \frac{1}{1+\frac{a^m}{a^n}}$

$$= \frac{a^m}{a^m + a^n} + \frac{a^n}{a^n + a^m}$$

$$= \frac{a^m + a^n}{a^m + a^n} = 1 \quad \left[\text{using } \frac{x^p}{x^q} = x^{p-q} \right]$$

4. (1)

5. (1)

6. (4) If $27^k = \frac{9}{3^k}$

$$\Rightarrow 3^{3k} = \frac{9}{3^k} \Rightarrow 3^{4k} = 9 \quad [a^m \times a^n = a^{m+n}]$$

$$\Rightarrow 9^{2k} = 9 \Rightarrow k = \frac{1}{2} \quad [a^m = a^n \text{ then } m = n]$$

$$\Rightarrow \frac{1}{k^2} = 4$$

7. (4) $4^{\sqrt{x}\sqrt{x}} = 256 = 4^4$

$$\Rightarrow \sqrt{x}\sqrt{x} = 4 = 2^2 \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4$$

8. (6) $2^{x-1} + 2^{x+1} = 320$

$$\Rightarrow 2^x [2^{-1} + 2] = 320$$

$$\Rightarrow 2^x \left[\frac{1}{2} + 2 \right] = 320 \Rightarrow 2^x \left[\frac{1+4}{2} \right] = 320$$

$$\Rightarrow 2^x = \frac{320 \times 2}{5} = 128$$

$$\Rightarrow 2^x = 2^6 \Rightarrow x = 6$$

9. (2) $\left(\frac{a}{b}\right)^{x-1} = \left(\frac{a}{b}\right)^{-(x-3)}$

$$\Rightarrow x-1 = -x+3$$

$$\Rightarrow 2x = 3+1 \Rightarrow 2x = 4 \Rightarrow x = 2$$

10. (3) $4^{n+1} = 256$

$$\Rightarrow 4^{n+1} = 4^4 \Rightarrow n+1 = 4$$

$$\Rightarrow n = 4-1 = 3$$

Chapter

13

Direct & Inverse Proportion

INTRODUCTION

In our daily life, we usually encounter the following variety of questions:

A man walks seven kilometers in two hours. What is his average speed? or how far the man walks in one hour? Or how much time he will take to walk 15 kilometers?

A train takes 15 hours to cover a distance of 1500 kilometers from Delhi to Mumbai. What is its average speed? or how far the train runs in one hour? Or how much time it will take to reach Pune which is 150 kilometers from Mumbai?

The unitary method is a technique in elementary algebra for solving a class of problems in variation. It consists of altering one of the variables to a single unit, i.e. 1, and then performing the operation necessary to alter it to the desired value.

To calculate the percentage of a certain quantity, unitary method is also used. For instance, if you are asked to calculate 9% of 1000 kilograms, all you need to do is to find out the amount equivalent to 1% of 1000 kg. Once it is calculated then you can easily find out 9% of 1000 kg by multiplying the result of 1% by 9. In most of the cases, the quantities are directly proportional to each other but in case of man and work they are inversely proportional to each other. For instance, if a piece of work is completed in 5 days by 2 men, then it will be obviously completed earlier if the work is done by more number of men.

VARIATIONS

We come across many situations in our day to day life where we see change in one quantity bringing change in the other quantity. For example

- If the number of items purchased increases, the cost also increases.
 - If the number of workers working to complete a job increases, then days required to complete the job will decrease.
- Here we observe that change in one quantity leads to change in other quantity. This is called variation.

Direct Variation

A direct variation is indicated when two quantities are so related that

- An increase in one causes a corresponding increase in the other or when a decrease in one causes corresponding decrease in the other and
- They increase or decrease in the same ratio.

If x and y are two variables, then y varies directly as x if the ratio $\frac{y}{x}$ is a constant. 'y varies directly as x' can be represented as $y \propto x$

Here symbol ' $\propto$ ' means 'varies as'.

This representation can be converted to an equation $y = kx$
 k is a constant called constant of proportionality.

$$k = \frac{y}{x}$$

If y_1, y_2 are two values of y corresponding to two values x_1 and x_2 of x , then $\frac{y_1}{x_1} = \frac{y_2}{x_2}$

ILLUSTRATION : 1

A machine takes 5 hours to cut 120 tools. How many tools will it cut in 20 hours?

SOLUTION :

Here more time, more number of tools. i.e. time and number of tools vary directly.

Let number of tools cut in 20 hours be 'x'.

$$\therefore \frac{5}{120} = \frac{20}{x} \Rightarrow x = \frac{20 \times 120}{5}$$

$$x = 480 \text{ tools.}$$

NOTE: If x_1 and x_2 are two values of x and y_1, y_2 are two values of y such that $\frac{x_1}{y_1} \neq \frac{x_2}{y_2}$ then x and y are not in direct variation

Inverse Variation

An inverse variation is indicated when two quantities are so related that

- an increase in one causes a corresponding decrease in the other or vice versa and
- the ratio of any two values of one quantity is the inverse of the ratio of the corresponding values of the other.

If x and y are variables, then y varies inversely as x if $y \propto \frac{1}{x}$ or $y = \frac{k}{x}$, k is constant of proportionality, Also $xy = k$

ILLUSTRATION : 2

If 900 persons can finish the construction of a building in 40 days, how many persons are needed to complete the construction of building in 25 days.

SOLUTION :

Let the required number of persons be 'x'. As the number of days required to complete the job is less, so more number of persons will be required. It is a case of inverse variation.

$$\text{So } 900 \times 40 = x \times 25$$

$$\Rightarrow x = \frac{900 \times 40}{25} = 1440 \text{ persons.}$$

NOTE: If y_1, y_2 are two values of y corresponding to the values x_1, x_2 of x respectively such that $x_1 y_1 = x_2 y_2$, then x and y are in inverse variation.

CONNECTING TOPIC

Time and Work

If A does a work in 'a' days, then in 1 day A does $\frac{1}{a}$ of the work.

If B does a work in 'b' days, then in 1 day B does $\frac{1}{b}$ of the work.

Then in 1 day, if A and B work together, their combined work is $\frac{1}{a} + \frac{1}{b}$ or $\frac{a+b}{ab}$.

The work will be completed when 1 unit of work is completed.

Now using Unitary Method

Time required to complete $\frac{a+b}{ab}$ work = 1 day

$$\therefore \text{Time required to complete 1 work} = \frac{1}{\frac{a+b}{ab}} = \frac{ab}{a+b}$$

Here we should recollect our knowledge of variation.

NOTE: We take the total work to be done as "1 unit of work".

Time and Distance

Motion / Movement occurs when a body of any shape and size changes its position with respect to any external stationary point.

The mathematical equation that describe the motion has three variables Speed, Time and Distance, which are connected by the following formula

$$\text{Distance} = \text{Speed} \times \text{Time}$$

From the above equation, we can have the following conclusions:

(a) If speed is constant, then distance and time are directly proportional to each other. i.e.

$$\text{Distance} \propto \text{Time.}$$

(b) If time is constant, then distance and speed are directly proportional to each other i.e

$$\text{Distance} \propto \text{Speed.}$$

(c) When distance is constant, then speed and time are inversely proportional to each other i.e.

$$\text{Speed} \propto \frac{1}{\text{Time}}$$

Normally speed is measured in km/hr or m/s.

$$1 \text{ km/hr} = \frac{1000 \text{ m}}{3600 \text{ s}} = \frac{5}{18} \text{ m/s}$$

or $1 \text{ m/s} = \frac{18}{5} \text{ km/hr.}$

NOTE: Speed of an object is distance covered by it in unit time.

ILLUSTRATION : 3

Three persons A, B and C can do a job alone in 10 days, 12 days and 15 days respectively. In how many days they can finish the job working together?

SOLUTION :

One day work of A, B and C are $\frac{1}{10}$, $\frac{1}{12}$ and $\frac{1}{15}$ respectively.

One day work if they work together = $\left(\frac{1}{10} + \frac{1}{12} + \frac{1}{15}\right) = \frac{15}{60} = \frac{1}{4}$

$\therefore$ Time required to complete the job = $\frac{1}{1/4} = 4$ days

ILLUSTRATION : 4

An increase in the speed of car by 10 km per hour saves 1 hour in a journey of 200 km, find the initial speed of the car.

SOLUTION :

Let the initial speed of car be x km per hour.

Time taken to cover distance 200 km, ' t_1 ' = $\frac{200}{x}$

Time taken to cover 200 km with increased speed $t_2 = \frac{200}{x+10}$

Now, $t_1 - t_2 = 1$

$$\therefore \frac{200}{x} - \frac{200}{x+10} = 1 \Rightarrow x(x+10) = 2000 \Rightarrow x(x+10) = 40 \times 50$$

From here $x = 40$ km/hr.

$\therefore$ Initial speed of the car is 40 km/hr.

MISCELLANEOUS SOLVED EXAMPLES

1. If 15 books cost ₹ 35, what do 21 books cost ?

Sol. Let cost of 21 books is ₹ x .

No. of books	15	21
Cost of books (in ₹)	35	x

As we know, more number of books, more is the cost, so it is an example of direct variation

$$\therefore 15 : 35 = 21 : x$$

$$\Rightarrow x = \frac{35 \times 21}{15} = ₹ 49$$

$\therefore$ Cost of 21 books will be ₹ 49.

2. If 15 men can reap a field in 28 days. In how many days will 5 men reap it.

Sol. Let the required number of days be ' x '

No. of men	15	5
No. of days	28	x

Here less is the number of men, more is the number of days. So it is a case of indirect variation.

$$\therefore 15 \times 28 = 5 \times x$$

$$\Rightarrow x = \frac{15 \times 28}{5} = 3 \times 28 = 84 \text{ days}$$

$\therefore$ 5 men will reap the field in 84 days.

3. A fort had provisions for 150 men for 50 days. After 15 days 25 men left the fort. How long will food last at the same rate for remaining men ?

Sol. Let required number of days be ' x '. Here the number of men are reduced so the food will last longer. It is a case of indirect variation.

No. of men	150	125
No. of days	35	x

Now, after 15 days the remaining food will last for 150 men for $(50 - 15) = 35$ days.

Also 25 men left the fort so remaining men are $(150 - 25) = 125$

$$\therefore 150 \times 35 = 125 \times x$$

$$\Rightarrow x = \frac{150 \times 35}{125} = 42 \text{ days.}$$

4. Due to bad road, the speed of a tourist bus is reduced by 12 km/hr and it now takes $2\frac{1}{2}$ hours more to cover the same distance of 600 km. Find the time it now takes to cover the distance.

Sol. Let the initial speed of bus = x km/hr

$$\text{Time taken by bus to cover 600 km originally} = \frac{600}{x} \text{ hr} = t_1 \text{ say.}$$

$$\text{Time taken by bus after reduction in speed} = \frac{600}{x-12} \text{ hr} = t_2 \text{ say.}$$

$$\text{Now } t_2 - t_1 = \frac{5}{2}$$

$$\therefore \frac{600}{x-12} - \frac{600}{x} = \frac{5}{2}$$

$$\Rightarrow 2 \times 600(x - x + 12) = 5x(x - 12)$$

$$\Rightarrow x(x - 12) = 2880 = 60 \times 48$$

$$\therefore x = 60.$$

$$\therefore \text{Original time taken } t_1 = \frac{600}{60} = 10 \text{ hr.}$$

$$\text{With reduced speed time taken} = 10 + 2\frac{1}{2} = 12\frac{1}{2} \text{ hours}$$

5. A car travels 270 km in $4\frac{1}{2}$ hrs.

(i) How much time is required to cover 90 km.

(ii) Find the distance covered in 6 hours.

Sol. (i)

Distance (in km)	270	90
Time taken (in hr)	$4\frac{1}{2}$	x

It is a case of direct variation.

$$\Rightarrow \frac{270}{4\frac{1}{2}} = \frac{90}{x}$$

$$\Rightarrow x = \frac{90 \times 4\frac{1}{2}}{270}$$

$$= \frac{90 \times 9}{2 \times 270} = \frac{3}{2} = 1\frac{1}{2} \text{ hrs.}$$

(ii)

Distance (in km)	270	x
Time taken (in hr)	$4\frac{1}{2}$	6

It is a case of direct variation.

$$\therefore \frac{270}{4\frac{1}{2}} = \frac{x}{6}$$

$$\Rightarrow \frac{270 \times 6 \times 2}{9} = x \Rightarrow x = 360 \text{ km.}$$

6. Reema types 540 words during half an hour. How many words would she type in 6 minutes?

Sol. Suppose she types x words in 6 minutes. Then, the given information can be represented in the following tabular form:

Number of words	540	x
Time (in minutes)	30	6

Since in more time more words can be typed, it is case of direct variation.

$$\therefore \frac{540}{30} = \frac{x}{6}$$

$$\Rightarrow x = \frac{6 \times 540}{30}$$

$$\Rightarrow x = 108.$$

Hence, she types 108 words in 6 minutes.

SOLVED EXAMPLES BASED ON CONNECTING TOPICS

7. Two persons A and B can do a piece of work alone in 10 days and 15 days respectively. If A works for 2 days and B works for 5 days, find the total amount of work done.

Sol. 1 day work of A and B are $\frac{1}{10}$ and $\frac{1}{15}$ respectively.

$$2 \text{ days work of A} = \frac{2}{10} = \frac{1}{5}$$

$$5 \text{ days work of B} = \frac{5}{15} = \frac{1}{3}$$

$\therefore$ they complete $\left(\frac{1}{5} + \frac{1}{3}\right) = \frac{8}{15}$ work if they work for 2 and 5 days respectively.

8. If 5 men or 9 women can do a piece of work in 19 days. In how many days 3 men and 6 women will do the same work ?

Sol. 5 men = 9 women

$$\therefore 1 \text{ men} = \frac{9}{5} \text{ women}$$

$$\therefore 3 \text{ men} + 6 \text{ women} = 3 \times \frac{9}{5} + 6 = \frac{57}{5} \text{ women}$$

$\therefore$ 9 women can do the work in 19 days

$$\therefore \frac{57}{5} \text{ women can do the work in } \frac{19 \times 9}{57} \times 5 = 15 \text{ days.}$$

9. A car travels a distance of 170 km in 2 hours partly at a speed of 100 km/hr and partly at 50 km/hr. Find the distance travelled at speed of 100 km/hr.

Sol. Let distance travelled at 100 km/hr be 'x' km.

$\therefore$ Distance travelled at 50 km/hr is $(170 - x)$ km.

Total time taken to cover 170 km is 2 hrs.

$$\therefore \frac{x}{100} + \frac{170 - x}{50} = 2$$

$$\Rightarrow x + 340 - 2x = 200$$

$$\Rightarrow x = 140 \text{ km}$$

$\therefore$ Distance travelled at 100 km/hr is 140 km.

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

1. If a man covers 10.2 km in 3 hours, the distance covered by him in 5 hours is _____.
2. A truck runs 420 km in 20 litres of diesel. It will go _____ kms in 17 litres of diesel.
3. The weight of 48 similar books is 30 kg. The weight of 8 similar books is _____ kg.
4. A tap can fill a water tank in 8 hours. Then the time taken by the tap to fill $\frac{1}{4}$ th part of the tank is _____.
5. Hashim completes $\frac{2}{3}$ rd of a work in 18 days. He can complete $\frac{1}{9}$ th of the work in _____ days.
6. Two quantities are said to vary _____ with each other if they increases (decrease) together in such a way that the ratio of the corresponding values remains same.
7. x and y are said to vary directly with each other if for some positive number k , _____ = k .
8. If $u = 3v$, then u and v vary with each other.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

1. In direct variation, $\frac{x}{y} = k$.
2. If two quantities x and y vary such that their product xy remains constant, then we say that x and y vary inversely.
3. If the cost of 15 pens of the same value is ₹600, then the cost of 20 pens is ₹700.
4. Journey (x km) undertaken by a car and the petrol (y litres) consumed by it, is a case of inverse proportion.
5. If a car travels 432 km on 48 litres of petrol, then it would travel 180 km on 20 litres of petrol.
6. Cost of 12 balls is ₹ 48, the cost of 15 ball is ₹ 60.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | 1. | Column-I | Column-II |
|-----|---|-----------|
| (A) | If the cost of 93 m of a certain kind of plastic sheet is ₹ 1395, then what would it cost to buy 105 m of such plastic sheet? | (p) 42 |
| (B) | 55 cows can graze a field in 16 days. How many cows will graze the same field in 10 days? | (q) 18 |
| (C) | 18 men can reap a field in 35 days. For reaping the same field in 15 days; how many men are required? | (r) 88 |
| (D) | Suneeta types 1080 words in one hour. What is her GWAM (gross words per minute)? | (s) 1575 |

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

1. Rahul drives his car 140 km in $3\frac{1}{2}$ hours. Find the number of kilometres, he drive in one hour.
2. An aeroplane flies 2100 km in 3 hours. At the same speed, how long will it take to fly 9100 km ?
3. If 7 kg of barley costs ₹ 126. Find the cost of 28 kg of barley.
4. If m varies directly as n and $m = 19.6$ when $n = 2$. Then find the value of m when $n = 3$.
5. Raveena types 54 words in 30 minutes. How many words will she type in 105 minutes.
6. 15 table costs ₹ 12500. Find the cost of such 3 tables.
7. A family's earning for 21 days is ₹ 770. Find its earning for 6 weeks and 3 days.
8. At a party 84 bottles of Coca-cola are served to 28 persons. Find the number of bottles of Coca-cola are required for 15 persons.
9. 16 men can do a piece of work in 14 days. How long will take 7 men to do the same work.
10. A school has enough food for 200 children for 6 days. How long will the food last for 40 more children.

Short Answer Questions :**DIRECTIONS :** Give answer in two to three sentences.

- $\frac{1}{4}$ kg of a sweet costs ₹ 32. Find the cost of $2\frac{1}{2}$ kg of the same sweet.
- The cost of bus tickets for 5 people for a journey was ₹ 95. What is the number of tickets that can be bought for ₹ 342 for the same journey?
- If 16 men can make 320 art pieces in one day, how many men will be needed to make 960 such art pieces in one day?
- Rashmi has a road map with a scale of 1 cm representing 18 km. She drives on a road for 72 km. What would be her distance covered in the map?
- Spinning 3 hours daily, Kanta can spin 2 kg cotton-balls in 12 days. Spinning 4 hours daily, how many days will when take to spin 10 kg cotton balls?
- The amount of extension in an elastic spring varies directly as the weight hung on it. If a weight of 150 gm produces an extension of 2.9 cm, then what weight would produce an extension of 17.4 cm?
- In 10 days, the earth picks up 2.6×10^8 pounds of dust from the atmosphere. How much dust will it pick up in 45 days?

- Raghu has enough money to buy 75 machines worth ₹ 200 each. How many machines can he buy if he gets a discount of ₹ 50 on each machine?

Long Answer Questions :**DIRECTIONS :** Give answer in four to five sentences.

- A loaded truck travels 14 km in 25 minutes. If the speed remains the same, how far can it travel in 5 hours?
- It takes three gardeners 90 minutes to weed a garden. If the job is to be done in only 15 minutes, how many more gardeners need to be put on the job?
- If x and y vary directly, find the missing entries in the following table:

x	3	x_1	15	x_2
y	1.2	3.6	y_1	10.8

- If a and b vary inversely, then find the value of the missing numbers in the following table:

a	16	32	8	a_1
b	4	b_1	b_2	0.5

- Shalu cycles to her school at an average speed of 12 km/hr. It takes her 20 minutes to reach the school. If she wants to reach her school in 15 minutes, what should be her average speed?

2

EXERCISE

Text-Book Exercise :

- A train is moving at a uniform speed of 75 km/hour.
 - How far will it travel in 20 minutes?
 - Find the time required to cover a distance of 250 km.
- The scale of a map is given as 1:30000000. Two cities are 4 cm apart on the map. Find the actual distance between them.
- If 15 workers can build a wall in 48 hours, how many workers will be required to do the same work in 30 hours?
- A photograph of a bacteria enlarged 50,000 times attains a length of 5 cm as shown in the diagram. What is the actual length of the bacteria? If the photograph is enlarged 20,000 times only, what would be its enlarged length?
- Suppose 2 kg of sugar contains 9×10^6 crystals. How many sugar crystals are there in (i) 5 kg of sugar? (ii) 1.2 kg of sugar?
- A 5 m 60 cm high vertical pole casts a shadow 3 m 20 cm long. Find at the same time
 - the length of the shadow cast by another pole 10 m 50 cm high
 - the height of a pole which casts a shadow 5 m long.
- A contractor estimates that 3 persons could rewire Jasminder's house in 4 days. If, he uses 4 persons instead of three, how long should they take to complete the job?
- A factory requires 42 machines to produce a given number of articles in 63 days. How many machines would be required to produce the same number of articles in 54 days?
- Two persons could fit new windows in a house in 3 days.
 - One of the persons fell ill before the work started. How long would the job take now?
 - How many persons would be needed to fit the windows in one day?

Exemplar Questions :

- Which of the following is in inverse proportion?
 - One side of a cuboid and its volume.
 - Speed of a vehicle and the time taken to travel a fixed distance.
 - Number of pipes and the quantity of water.
- In a scout camp, there is food provision for 300 cadets for 42 days. If 50 more persons join the camp, for how many days will the provision last?
- If two cardboard boxes occupy 500 cubic centimetres space, then how much space is required to keep 200 such boxes?
- Under the condition that the temperature remains constant, the volume of gas is inversely proportional to its pressure. If the volume of gas is 630 cubic centimetres at a pressure of 360 mm of mercury, then what will be the pressure of the gas if its volume is 720 cubic centimetres at the same temperature?
- A contractor undertook a contract to complete a part of a stadium in 9 months with a team of 560 persons. Later on, it was required to complete the job in 5 months. How many extra persons should he employ to complete the work?
- The mass of an aluminium rod varies directly with its length. If a 16 cm long rod has a mass of 192 g, find the length of the rod whose mass is 105 g.
- A swimming pool can be filled in 4 hours by 8 pumps of the same type. How many such pumps are required if the pool is to be filled in $2\frac{2}{3}$ hours?
- Shabnam takes 20 minutes to reach her school if she goes at a speed of 6 km/h. If she wants to reach school in 24 minutes, what should be her speed?

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- Three pumps working 8 hours a day can empty a tank in 2 day. How many hours a day must 4 pumps work to empty the tank in 1 day.
 - 10 hours
 - 12 hours
 - 8 hours
 - None of these
- In a camp there is provision for 1600 participants for 60 days. Actually 1200 participated, how many days will the provision last for?
 - 70 days
 - 80 days
 - 83 days
 - 95 days.
- If a person walks at 14 km/hr instead of 10 km/hr, he would have walked 20 km more. The actual distance travelled by him is
 - 56 km
 - 80 km
 - 70 km
 - 50 km
- The weight of a 13 m long iron rod is 23.4 kg. The weight of 6 m long of such rod will be _____.
 - 7.2 kg
 - 12.4 kg
 - 10.8 kg
 - 18 kg
- Asha can stitch 'x' shirts in $\frac{3}{4}$ minutes. At this rate, how many shirts can she stitch in $\frac{3}{4}$ th of an hour?
- 50x
 - $\frac{9}{16}x$
 - 60x
 - $\frac{16}{9}x$
- A university has its own hostel for its students. It provides fooding and lodging to the students. Due to festive season some of the students are left for their home and 100 students stays in the hostel. There is a food provision for 20 days for these students. How long the food will last if 25 more students decide to stay back in the hostel?
 - 12 days
 - 13 days
 - 16 days
 - 14 days
- An agent receives a commission of ₹73.00 on sales of ₹1000.00. The commission will be get on sales of ₹100.00 is _____.
 - ₹7.30
 - ₹7.00
 - ₹6.00
 - ₹6.30
- A shopkeeper has just enough money to buy 52 cycles worth ₹525 each. If each cycle were to cost ₹21 more, then number of cycles, he will be able to buy with that amount of money, is _____.
 - 40
 - 30
 - 50
 - 20
- A gardener uses pipes to water his garden from the tank. For his entire garden he uses 6 pipes to water and it takes him 1 hour 20 minutes to completely empty the tank. Now if he uses only 5 pipes to water his garden, how long it will take him to empty the tank?
 - 1 hour 30 min
 - 1 hour 36 min
 - 1 hour 45 min
 - 1 hour 55 min

Direct & Inverse Proportion

10. A company makes 5 blue cars for every 3 white cars it makes. If the company makes 15 white cars in one day, how many blue cars will it make?
- (a) 9 (b) 13
(c) 17 (d) 25
11. A person travels at a speed of 24 km/hr and reaches a town in 32 hours. How much time does he take to cover the same distance if his speed is 32 km/hr?
- (a) 08 (b) 12
(c) 24 (d) 28
12. If 100 students take 20 days to complete the project how much time will be taken by 125 students?
- (a) 63 (b) 16
(c) 25 (d) 7
13. A girl 1.2 metre tall casts a shadow 1.1 metre at the time when a building casts a shadow 6.6 metre long. The height of the building is
- (a) 2.7 metre (b) 7.2 metre
(c) 6.0 metre (d) 5.5 metre
14. A contractor undertook a contract to complete a part of a stadium in 9 months with a work force of 560 men. Later on it was required to complete the job in 7 months. Extra men employed by him is
- (a) 160 (b) 150
(c) 100 (d) 120

More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. In which of the following tables, a and b vary directly. Also, find the constant of variation if a and b are in direct variation.
- (a)

a	4	7	21	28
b	12	21	63	84
- (b)

a	2.5	5	7.5	10
b	10	20	30	40
- (c)

a	1	2	3	4
b	2	1	6	3
- (d)

a	1	4	5	6
b	8	3	2	1
2. Which of the following quantities vary directly with each other?
- (a) number of articles (x) and their price (y).
(b) Weight of articles (x) and their cost (y).
(c) Wages (y) and number of hours (x) of work.
(d) Speed (x) and time (y) (distance covered remaining the same).
3. In which of the following tables a and b vary inversely.
- (a)

a	8	16	32	256
b	32	16	8	1

(b)

a	9	18	2	12
b	8	4	30	6

(c)

a	2	16	8	4
b	40	5	10	20

(d)

a	4	8	12	1
b	6	3	2	26

4. If a and b vary inversely, fill in blanks:

a	8	2	...	5	1
b	10	...	20	...	80

- (a) 40 (b) 16
(c) 24 (d) 32

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE - I

In our daily life, we come across some phrases such as :

- (i) The faster the speed of a car, the lesser is the time taken to cover a given distance.
(ii) The efficiency of a machine decreases with time
(iii) The more men working on a project, the shorter is the time to complete it.

Read the above information and answer the following questions.

1. Which of the above phrases are the examples of direct variation?
- (a) (i) (b) (ii)
(c) (iii) (d) None of these
2. If x varies inversely as y , then complete the table.

x	---	6
y	2	3

- (a) 4 (b) 8
(c) 9 (d) 6

PASSAGE - II

A train is moving with a uniform speed of 105 km/hr.

3. How far will it travel in 20 minutes?
- (a) 15km (b) 25km
(c) 35km (d) 250km
4. Find the time required to cover a distance of 210 km.
- (a) 2 hours (b) 200min
(c) 7 hours (d) 4900 sec
5. How far will it travel in 18 hours?
- (a) 1890km (b) 189000m
(c) 189km (d) 18900m

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** The weight of 85 books of same type is 51 kg. The weight of such 35 books is 21 kg.

Reason : Two quantities are said to be in direct proportion to each other, if the increase in first implies increase in second quantity.

2. **Assertion :** If 24 men can dig a trench in 7 days, it takes 12 days by 10 men to dig a similar trench.

Reason : Since it is a case of inverse variation *i.e.*, when one decreases other increases or vice versa.

Multiple Matching Questions :

DIRECTIONS : Following question has three statements (A, B and C) given in Column-I and five statements (p, q, r, s and t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1.	Column – I	Column – II
	(A) Quantities vary directly with each other are	(p) The number of men hired to construct a wall and the time taken to finish the job.
	(B) Quantities vary indirectly with each other are	(q) Speed of a moving object to the time taken to cover a fixed distance.
	(C) If the thickness of a pile of 12 cardboards is 35 mm, find the thickness of a pile of 294 cardboards.	(r) Number of articles and their price. (s) More men do more work and less men do less work. (t) 85.75 cm

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- A factory produces fast foods and many others eatables. It got an order from a government agency to manufacture biscuits for the school children. It has 6 machines for the work and has to complete the task in 9 days. The number of machines require to complete the work if they are asked to complete the same work in 18 days.
- x varies inversely as square of y . Given that $y = 2$ for $x = 1$. The value of x for $y = 6$ will be equal to $= \frac{1}{m}$, where $m = ?$
- If a quarter kg of potato costs 60 paise, what is the unit digit of paise will 200 gm cost?
- An industrial loom weaves 0.128 metres of cloth every second. Approximately, $190 + n$ seconds will it take for the loom to weave 25 metres cloth. Find n .
- A flagstaff 17.5 m high casts a shadow of length 40.25 m. The height of the building, which casts a shadow of length 28.75 m under similar conditions will be $(12 + 0.n)$. Find n .

4

ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

DIRECTIONS (Qs. 1-8) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. A car travels at a speed of 72 km/h. Express this speed in metres/second.
(a) 22 m/s (b) 20 m/s
(c) 25 m/s (d) 30 m/s
2. With a uniform speed a car covers a distance in 8 hours. Had the speed been increased by 4 km/hr, the same distance could have been covered in $7\frac{1}{2}$ hours. The distance covered is
(a) 400km (b) 450km
(c) 480km (d) 380km
3. A train does a journey without stopping in 8 hours. If it had travelled 5 km an hour faster, it would have done the journey in 6 hours 40 min, its slower speed is
(a) 32 km/hr (b) 25 km/hr
(c) 28 km/hr (d) 40 km/hr
4. A train moving at the speed of 40 km/h takes 2 hours more to cover a certain distance than a train moving at 60 km/h. The distance is
(a) 200km (b) 240km
(c) 250km (d) 300km
5. A alone can do $\frac{1}{5}$ th of a piece of work in 3 days and B alone can do $\frac{1}{5}$ th of the same work in 6 days. In how many days working together will they finish the work ?
(a) 2 days (b) 5 days
(c) 9 days (d) 10 days
6. A truck travels a distance of 240 km in 6 hours, partly at a speed of 60 km/hr and partly at 30 km/hr. Find the time for which it travels at 60 km/hr.
(a) 130km (b) 120km
(c) 115km (d) 125km
7. X can do a piece of work in 24 days, while Y alone can do it in 16 days. With the help of Z they finish the work in 8 days. Find in how many days Z can do the work alone?
(a) 48 days (b) 36 days
(c) 24 days (d) 12 days
8. A and B can do a piece of work in 18 days, B and C in 24 days, A and C in 36 days. In what time can they do it all working together?
(a) 18 (b) 17
(c) 16 (d) 15

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- 17 km.
In 3 hours, man covers the distance = 10.2 km
In 1 hours, man covers the distance = $\frac{10.2}{3}$ km
In 5 hours, man covers the distance = $\frac{5 \times 10.2}{3}$ km
 $= 5 \times 3.4 \text{ km} = 17 \text{ km}$
- 357 kms
In 20 litres, the distance travelled by truck = 420 km
In 1 litres, the distance travelled by truck = $\frac{420}{20}$ km.
In 17 litres, the distance travelled by truck = $\frac{420}{20} \times 17 \text{ km}$.
 $= 357 \text{ km}$.
- 5 kg
The weight of 48 books = 30 kg.
The weight of 1 books = $\frac{30}{48}$ kg.
The weight of 8 books = $\frac{8 \times 30}{48}$ kg = 5 kg.
- 2 hours; $\frac{8}{4} = 2 \text{ hr}$
- 3 days
 $\frac{2}{3}$ rd part of work is completed in 18 days.
1 part of work is completed in $\frac{18 \times 3}{2}$ day.
 $\frac{1}{9}$ part of work is completed in $\frac{1}{9} \times \frac{18 \times 3}{2}$ days = 3 days.
- Directly
- $\frac{x}{y}$
- directly

True/False :

- True
- True
- False
- False
- True
- True

Match The Columns :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (r); (C) $\rightarrow$ (p); (D) $\rightarrow$ (q)

- (A) Let the cost be ₹ x

Cost of sheet	1395	x
Length of sheet	93	105

It is the case of direct variation.

$$\frac{1395}{93} = \frac{x}{105} \Rightarrow x = \frac{1395 \times 105}{93} = 1575$$

So, cost of plastic sheet is ₹ 1575.

- (B) Let the number of cows be x

No. of cows	55	x
No. of days	16	10

It is the case of inverse variation.

$$\Rightarrow 55 \times 16 = x \times 10$$

$$\Rightarrow \frac{55 \times 16}{10} = 88 \text{ cows.}$$

- (C) Let the number of men = x

No. of men	18	x
No. of days	35	15

It is the case of inverse variation.

$$18 \times 35 = x \times 15$$

$$\Rightarrow x = \frac{18 \times 35}{15} = 42$$

- (D) Let the number of words = x

No. of words	1080	x
Time	60	1

It is the case of direct variation.

$$\Rightarrow x = \frac{1080}{60} = 18$$

Very Short Answer Questions :

- 40 km/hr
- 13 hours
- Cost = ₹ $\frac{126}{7} \times 28 = ₹ 504$
- 29.4
- 189
- Earning = ₹ 1650
- ₹ 1650
- 45
- 32 days
- 5 days

Short Answer Questions :

- ₹ 320
- 18 tickets
- 48 men
- Let the distance covered in the map = x
Map with a scale and distance covered in the map vary directly.

Scale (in cm)	1	x
Distance (in km)	18	72

Now, $\frac{1}{18} = \frac{x}{72}$

$$\Rightarrow 18 \times x = 72 \times 1$$

$$\Rightarrow x = \frac{72}{18} = 4 \text{ cm}$$

Hence required distance covered in the map = 4 cm.

- Spinning 3 hours daily, Kanta can spin 2 kg cotton-ball in 12 days.
 $\therefore$ Spinning 3 hours daily, Kanta can spin 1 kg cotton-ball in $\left(\frac{12}{2}\right) = 6$ days
 $\Rightarrow$ Spinning 1 hour daily, Kanta can spin 1 kg cotton-ball in $(6 \times 3) = 18$ days.
 $\Rightarrow$ Spinning 1 hour daily, Kanta can spin 10 kg cotton-ball in $(18 \times 10) = 180$ days.
 $\Rightarrow$ Spinning 4 hour daily, Kanta can spin 10 kg cotton-ball in $\frac{180}{4}$ days = 45 days
- Let the required weight be x grams.

Weight (in grams)	150	x
Extension (in cm)	2.9	17.4

It is given that the amount of extension in the spring varies directly as the weight hung on it.

So, it is a case of direct variation.

$$\Rightarrow 150 : x = 2.9 : 17.4$$

$$\Rightarrow \frac{150}{x} = \frac{2.9}{17.4} \Rightarrow x = 900$$

Hence, a weight of 900 grams would produce an extension of 17.4 cm.

- 11.7×10^8 pound 23.60 days.
- 100 machines

Long Answer Questions :

- 60 minutes = 1 hours

$$\therefore 25 \text{ minutes} = \frac{1}{60} \times 25 = \frac{5}{12} \text{ hour}$$

Let distance of travelling truck = x

Distance of travelling truck and time vary directly.

Distance (in km)	14	x
Time (in hour)	$\frac{5}{12}$	5

Now, $\frac{14}{5/12} = \frac{x}{5}$

$$\Rightarrow \frac{5}{12} \times x = 14 \times 5$$

$$\Rightarrow x = \frac{14 \times 5}{5/12} = \frac{14 \times 5 \times 12}{5}$$

$$\Rightarrow x = 168 \text{ km}$$

Hence distance of travelling track = 168 km.

- Let x gardeners be needed to weed the garden in 15 minutes.

No. of gardeners	3	x
Time (in minutes)	90	15

More gardeners on the job means less time required to complete the job.

Hence, it is a case of inverse variation.

$$\text{So, } 3 \times 90 = x \times 15$$

$$x = \frac{3 \times 90}{15} = 18 \text{ gardeners}$$

Since, three gardeners are already working.

Thus, $18 - 3 = 15$ more gardeners need to be put on the job.

- Here x and y vary directly.

$$\text{So, } \frac{x}{y} = k \text{ (constant)}$$

$$k = \frac{3}{1.2} = \frac{30}{12} = \frac{5}{2}$$

$$\therefore k = \frac{5}{2}$$

$$\text{Now, } \frac{x_1}{3.6} = \frac{5}{2} \Rightarrow x_1 = \frac{5 \times 3.6}{2} = 9$$

$$\frac{15}{y_1} = \frac{5}{2} \Rightarrow y_1 = \frac{15 \times 2}{5} = 6$$

$$\frac{x_2}{10.8} = \frac{5}{2} \Rightarrow x_2 = \frac{10.8 \times 5}{2} = 27$$

Hence, $x_1 = 9$, $y_1 = 6$, and $x_2 = 27$.

4. $ab = k$

$$k = 16 \times 4 = 64$$

$$\text{Now, } 32 \times b_1 = 64 \Rightarrow b_1 = \frac{64}{32} = 2$$

$$8 \times b_2 = 64 \Rightarrow b_2 = \frac{64}{8} = 8$$

$$a_1 \times 0.5 = 64 \Rightarrow a_1 = \frac{64}{0.5} = 128$$

Hence, $b_1 = 2$, $b_2 = 8$, and $a_1 = 128$.

5. Let the required speed be x km/hr. Then, the given information may be presented in the following tabular form.

Speed in km/hr.	12	x
Time (in minutes)	20	15

We note that more the speed, less will be the time taken to cover the given distance. So, it is a case of inverse variation.

$$\therefore \text{Ratio of speeds} = \text{Inverse ratio of time taken}$$

$$\Rightarrow 12 : x = 15 : 20$$

$$\Rightarrow \frac{12}{x} = \frac{15}{20}$$

$$\Rightarrow 15 \times x = 12 \times 20$$

$$\Rightarrow x = \frac{12 \times 20}{15} = 16$$

Hence, Shalu's average speed should be 16 km/hr.

2 EXERCISE

Text-Book Exercise :

1. Let the distance travelled (in km) in 20 minutes be x and time taken (in minutes) to cover 250 km be y .

Distance travelled (in km)	75	x	250
Time taken (in minutes)	60	20	y

Since the speed is uniform, therefore, the distance covered would be directly proportional to time.

(i) We have $\frac{75}{60} = \frac{x}{20}$

$$\text{or } \frac{75}{60} \times 20 = x$$

$$\text{or } x = 25$$

So, the train will cover a distance of 25 km in 20 minutes.

(ii) Also, $\frac{75}{60} = \frac{250}{y}$

$$\text{or } y = \frac{250 \times 60}{75} = 200 \text{ minutes or 3 hours 20 minutes.}$$

2. Let the map distance be x cm and actual distance be y cm, then $1:30000000 = x:y$

$$\text{or } \frac{1}{3 \times 10^7} = \frac{x}{y}$$

$$\text{Since } x = 4 \text{ so, } \frac{1}{3 \times 10^7} = \frac{4}{y}$$

$$\text{or } y = 4 \times 3 \times 10^7 = 12 \times 10^7 \text{ cm} = 1200 \text{ km.}$$

Thus, two cities, which are 4 cm apart on the map, are actually 1200 km away from each other.

3. Let the number of workers employed to build the wall in 30 hours be y .

We have the following table.

Number of hours	48	30
Number of workers	15	y

Obviously more the number of workers, faster will they build the wall.

So, the number of hours and number of workers vary in inverse proportion.

$$\text{So, } 48 \times 15 = 30 \times y$$

$$\text{Therefore, } \frac{48 \times 15}{30} = y \text{ or } y = 24$$

i.e., to finish the work in 30 hours, 24 workers are required.

4. 10^{-4} cm; 2 cm
 5. (i) 2.25×10^7 crystals (ii) 5.4×10^6 crystals
 6. (i) 6 m (ii) 8 m 75 cm
 7. 3 days
 8. 49 machines
 9. (i) 6 days (ii) 6 persons

Exemplar Questions :

1. (2) The correct answer is (2).
 2. More the persons, the sooner would be the provision exhausted. So, this is a case of inverse proportion.

Let the required number of days be x .

$$\text{Hence, } 300 \times 42 = (300 + 50) \times x$$

$$300 \times 42 = 350 \times x$$

$$\frac{300 \times 42}{350} = x$$

$$x = 36$$

Direct & Inverse Proportion

3. As the number of boxes increases, the space required to keep them also increases.

So, this is a case of direct proportion.

Number of boxes	2	200
Space occupied (in cubic centimetres)	500	x

$$\text{So } \therefore \frac{2}{500} = \frac{200}{x} \Rightarrow 2x = 500 \times 200$$

$$\Rightarrow x = \frac{500 \times 200}{2} \Rightarrow x = 50,000$$

Thus, the required space is 50,000 cubic centimetres.

4. Given that, at constant temperature pressure and volume of a gas are inversely proportional. Let the required pressure be x .

Volume of gas (in cubic centimetres)	630	720
Pressure of gas (in mm)	360	x

$$\text{Then, } 630 \times 360 = 720 \times x \Rightarrow \frac{630 \times 360}{720} = x$$

$$\Rightarrow x = 315$$

Therefore, the required pressure is 315 mm of mercury.

5. 448 persons 6. 8.75 cm
7. 12 pumps 8. 5 km

3 EXERCISE

Single Option Correct :

1. (b) Let the required number of working hours/day = x
More Pumps, less working hrs per day (Indirect)
Less days, more working hrs per day (Indirect)

$$\left. \begin{array}{l} \text{Pumps } 4 : 3 \\ \text{Days } 1 : 2 \end{array} \right\} \therefore 8 : x$$

$$\therefore 4 \times 1 \times x = 3 \times 2 \times 8$$

$$\Rightarrow x = \frac{3 \times 2 \times 8}{4} = 12$$

2. (b) 3. (d)
4. (c) Let weight 6 m long rod = x kg

Length of the rod (m)	13	6
Weight (kg)	23.4	x

As length of the rod increases, weight also increase.

$\therefore$ It is direct variation.

$$\Rightarrow \frac{13}{23.4} = \frac{6}{x} \Rightarrow x = \frac{23.4 \times 6}{13} = 10.8 \text{ kg}$$

5. (c)

No. of shirts	Time
x	$\frac{3}{4}$ min
s	$\frac{3}{4} \times 60 = 45$ min

$\therefore$ It is direct variation

$$x : \frac{3}{4} :: s : 45$$

$$s = \frac{45x}{3/4} = 60x$$

6. (c) Initially the number of students = 100.
Provision for food = 20 days.
Finally number of students = 125.
Number of days food will last

$$= \frac{100 \times 20}{125} = 16 \text{ days.}$$

7. (a) Let commission on ₹ 100 be x .
Then

Sale (₹)	1000	100
Commission (₹)	73	x

As the amount of sales increases commission also increases.

$\therefore$ It is direct variation.

$$\Rightarrow \frac{1000}{73} = \frac{100}{x} \Rightarrow x = \frac{73 \times 100}{1000} = 7.3$$

$\therefore$ Commission will be get on ₹ 100 is ₹ 7.30.

8. (c) Total cost of 52 cycles = 27300.
Let number of cycles be x .

Then

Cost of 1 cycle (₹)	525	525 + 21
No. of cycles	52	x

As cost of one cycle increases, number of cycles decreases.

$\therefore$ It is inverse variation.

$$\Rightarrow 525 \times 52 = 546 \times x \Rightarrow x = \frac{525 \times 52}{546} = 50$$

$\therefore$ Number cycles can be purchased on increased price is 50.

9. (b) Let the time required be x .

No. of pipes	6	5
Time taken (min)	80	x

$\therefore$ It is inverse variation.

$$\Rightarrow 6 \times 80 = 5 \times x$$

$$\Rightarrow x = \frac{6 \times 80}{5} = 96 \text{ min} \Rightarrow x = 1 \text{ hour } 36 \text{ min.}$$

10. (d) Let company makes x blue cars for 15 white cars

Blue Cars	5	x
White Cars	3	15

$$\Rightarrow \frac{5}{3} = \frac{x}{15}$$

$$\Rightarrow \frac{5 \times 15}{3} = x$$

$$x = 25$$

11. (c)

Speed (in km/hr)	24	32
Time (in hrs)	32	x

It is a case of inverse variation
 $24 \times 32 = 32 \times x \Rightarrow x = 24$ hrs

12. (b)

No. of Students	100	125
Days	20	x

$$x = \frac{20 \times 100}{125} = 16 \text{ days}$$

13. (b) 1.2 m tall girl's shadow = 1.1
 x m tall building's shadow = 6.6

$$\frac{1.2}{x} = \frac{1.1}{6.6}$$

$$\frac{1.2}{x} = \frac{1}{6} \Rightarrow x = 7.2 \text{ m}$$

14. (a)

Time taken	9	7
No. of Men	560	x

It is the case of inverse variations.

$$\Rightarrow x = \frac{9 \times 560}{7} = 720$$

Extra men to complete the work = $720 - 560 = 160$.

More than One Option Correct :

1. (a) We have,

$$\frac{4}{12} = \frac{7}{21} = \frac{21}{63} = \frac{28}{84} = \frac{1}{3}$$

Thus, the ratio of the corresponding values of a and b is constant and is equal to $\frac{1}{3}$.

Hence, a and b are in direct variation with the constant of variation equal to $\frac{1}{3}$.

- (b) We have,

$$\frac{2.5}{10} = \frac{5}{20} = \frac{7.5}{30} = \frac{10}{40} = \frac{15}{60} = \frac{1}{4}$$

This shows that the ratio of the corresponding values of a and b is constant and is equal to $\frac{1}{4}$. Thus, a and b vary directly. The constant of variation is $\frac{1}{4}$.

(c, d) It is evident from the table that the ratio of the corresponding values of a and b is not constant. So, a and b are not in direct variation i.e. they do not vary directly.

2. (a, b and c)

3. (a, c) We know that if a and b vary inversely, then the product ab remains same for all values of a and b .

Here,

$$8 \times 32 = 256, 16 \times 16 = 256, 32 \times 8 = 256 \text{ and } 256 \times 1 = 256.$$

$$2 \times 40 = 80, 16 \times 5 = 80, 8 \times 10 = 80 \text{ and } 4 \times 20 = 80.$$

4. (a, b) (i) Since a and b vary inversely, so the product ab remains constant and is equal to $8 \times 10 = 80$.

$$\therefore \text{First blank space is to be filled by } \frac{80}{2} = 40$$

$$\text{Second blank space is to be filled by } \frac{80}{20} = 40$$

$$\text{Third blank space is to be filled by } \frac{80}{5} = 16.$$

Passage Based Questions :

Passage - I

1. (d) None of these
 2. (c) 9

Passage - II

Let the distance travelled (in km) in 20 minutes be x and time taken (in minutes) to cover 210 km be y . Since speed is uniform, therefore, the distance covered would be directly proportional to time.

3. (c) We have $\frac{105}{60} = \frac{x}{20} \Rightarrow x = \frac{105 \times 20}{60} = 35$ km.

So, the train will cover a distance of 35 km in 20 minutes.

4. (a) Also, $\frac{105}{60} = \frac{210}{y} \Rightarrow y = \frac{210 \times 60}{105} = 120$

5. (a) The distance in 1 hr. = 150 km
 So, the distance covered in 18 hrs
 $= 105 \times 18 = 1890$ km

Assertion & Reason :

1. (a)

Books	Weight (kg)
85	51
35	x

More book : More weight, direct variation

$$\text{So, } x = \frac{51 \times 35}{85} = 21 \text{ kg}$$

2. (d)

No. of men	24	x
No. of days	7	12

It is the case of inverse variation.
 $24 \times 7 = x \times 12 \Rightarrow x = 14$

Multiple Matching :

1. (A) $\rightarrow$ (r, s); (B) $\rightarrow$ (p, q); (C) $\rightarrow$ (t)

Integer Type Questions :

1. (3) Let number of machines required be x , then

No. of machine	6	x
No. of days	9	18

$\therefore$ It is inverse variation.

$$\Rightarrow 6 \times 9 = 18 \times x$$

$$\Rightarrow x = \frac{6 \times 9}{18} = 3$$

$\therefore$ No. of days = 3

2. (9) Given $x = \frac{k}{y^2}$, where k is a constant.

Now, $y = 2$ and $x = 1$ gives $k = 4$.

$$\therefore x = \frac{4}{y^2} \Rightarrow x = \frac{4}{6^2}, \text{ when } y = 6$$

$$\Rightarrow x = \frac{4}{36} = \frac{1}{9}. m = 9$$

3. (8) Let the required price be ₹ x .
Less weight, Less cost (Direct Proportion)
 $\therefore 250 : 200 :: 60 : x \Leftrightarrow 250 \times x = (200 \times 60)$

$$\Leftrightarrow x = \frac{(200 \times 60)}{250} \Leftrightarrow x = 48.$$

Unit digit is 8.

4. (5) Let the required time be x seconds. Then, More metres, more time (Direct Proportion)
 $\therefore 0.128 : 25 :: 1 : x$

$$\Leftrightarrow 0.128 \times x = 25 \times 1 \Leftrightarrow x = \frac{25}{0.128} = \frac{25 \times 1000}{128}$$

$$\Leftrightarrow x = 195.31.$$

$\therefore$ Required time = 195 sec (approximately). $n = 5$

5. (5) Let the height of the building be x metres.
Less lengthy shadow, Less is the height (Direct Proportion)

$$\therefore 40.25 : 28.75 :: 17.5 : x$$

$$\Leftrightarrow 40.25 \times x = 28.75 \times 17.5$$

$$\Leftrightarrow x = \frac{(28.75 \times 17.5)}{40.25} \Leftrightarrow x = 12.5. n = 5$$

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

1. (b) $\frac{72 \times 1000}{3600} = 72 \times \frac{5}{18} = 20 \text{ m/s}$

2. (c) Let the distance be x km.

According to question

$$\frac{x}{7\frac{1}{2}} - \frac{x}{8} = 4$$

$$\Rightarrow \frac{2x}{15} - \frac{x}{8} = 4 \Rightarrow \frac{16x - 15x}{120} = 4$$

$$\Rightarrow x = 480 \text{ km}$$

3. (b) Let slower speed = u km/hr

As the distance is fixed

$$u \times 8 = (u + 5) \times \frac{20}{3} \quad [\because 6 \text{ hr } 40 \text{ min} = 6 \text{ hr} + \frac{40}{60} \text{ hr}]$$

$$= 6\frac{2}{3} = \frac{20}{3} \text{ hrs}]$$

$$\Rightarrow 24u = 20u + 100 \Rightarrow 4u = 100$$

$$\Rightarrow u = 25 \text{ km/hr}$$

4. (b) Distance covered = $\frac{d}{40} - \frac{d}{60} = 2$.
 $d = 240 \text{ km}$

5. (d) Together they can do $\frac{1}{5}$ of the work in $\frac{3 \times 6}{3+6}$ days
 $= 2$ days.

$\therefore$ They can do the work in $2 \times 5 = 10$ days.

6. (b) 120 km

7. (a) $X + Y + Z = \frac{1}{8}$

$$Z = \frac{1}{8} - X - Y$$

$$= \frac{1}{8} - \frac{1}{24} - \frac{1}{16} = \frac{1}{48}$$

$\therefore$ Z can do the work in 48 days.

8. (c) One day work of (A and B) = $\frac{1}{18}$

$$\text{One day work of (B and C)} = \frac{1}{24}$$

$$\text{One day work of (C and A)} = \frac{1}{36}$$

2(A + B + C)'s one day work

$$= \frac{1}{18} + \frac{1}{24} + \frac{1}{36} = \frac{9}{72} = \frac{1}{8}$$

$$\text{So one day work of (A + B + C)} = \frac{1}{16}$$

So total work = 16 days.

Chapter

14

Factorisation

INTRODUCTION

FACTORISATION OF POLYNOMIALS

Let us take any natural number say 18 and write it as product of other natural numbers

$$18 = 1 \times 18 = 2 \times 9 = 3 \times 6$$

Here 1, 2, 3, 6, 9, 18 are factors of 18 and 2, 3 are prime factors of 18. We can express a number as a product of prime factors. For example, $18 = 2 \times 3 \times 3$

Similarly, we can express algebraic expressions as product of their factors.

In algebraic expressions we use the term 'irreducible' in place of prime; e.g. 2, x , y are irreducible factors of $2xy$. Similarly for expression $5x(x+3)$ are irreducible factors of $5x(x+3)$.

Thus we can say that factorisation of an algebraic expression is the process of expressing the expression as product of irreducible factors.

FACTORISATION

We know, factorisation of a number is expressing the number as the product of other numbers. for example, $6 = 2 \times 3 = 1 \times 6$.

Thus, 1, 2, 3, 6 are factors of 6.

Similarly, we can express algebraic expressions as products of their factors. These factors may be numbers, algebraic variables or algebraic expressions. For example, $5y^2 = 5 \times y \times y$

The factors of $5y^2$ are 1, 5, y and $5y$.

The method of expressing a given algebraic expression as the product of some numbers and algebraic expressions is called **factorisation**.

In an algebraic expression, each term is formed as the product of its factors. For example, in the algebraic expression $2xy + 5x$, the term $2xy$ has been formed by the factors 2, x and y and $5x$ is formed by 5 and x .

Method of Common Factors

Case I. When each term of the given expression contains a common monomial factor. In this case, we take out the common multiplier and use the distributive property.

ILLUSTRATION : 1

Factorise : $4x^3 + 12x^2 + 20x$.

SOLUTION :

The greatest common factor of the numerical coefficients 4, 12 and 20 is 4.

The greatest common factor of variables is x .

So, $4x$ is the greatest common factor of the given expression.

Dividing $4x^3 + 12x^2 + 20x$ by $4x$, we get the quotient as $(x^2 + 3x + 5)$.

Hence, $4x^3 + 12x^2 + 20x = 4x(x^2 + 3x + 5)$

Case II. When a polynomial is a common multiplier of each term of the given expression. In this case, we take out the common multiplier and use the distributive property. Sometimes, we have to make some arrangement of terms so as to have a common polynomial.

ILLUSTRATION : 2

Factorise : $3(3x - 4y) - 4(3x - 4y)^2$.

SOLUTION :

$(3x - 4y)$ is common in both terms.

$3(3x - 4y) - 4(3x - 4y)^2 = (3x - 4y) [3 - 4(3x - 4y)]$

Hence, $3(3x - 4y) - 4(3x - 4y)^2 = (3x - 4y)(3 - 12x + 16y)$

ILLUSTRATION : 3

Factorisation $x^2 - xy + y^3 - xy^2$

SOLUTION :

$$x^2 - xy - xy^2 + y^3 = x(x - y) - y^2(x - y)$$

$$= (x - y)(x - y^2) \quad [\text{Taking } (x - y) \text{ common}]$$

Factorisation by Regrouping Terms

In certain expressions, it is not possible to take out a single common factor from all the terms of the expression. In such cases, we rearrange and group the terms so that each group has some common factor.

ILLUSTRATION : 4

Factorise : $a^2b + b - a - ab^2$.

SOLUTION :

The above expression can be rearranged as:

$$a^2b - ab^2 + b - a = ab(a - b) - 1(a - b)$$

$$= (a - b)(ab - 1)$$

$$\text{Thus, } a^2b + b - a - ab^2 = (a - b)(ab - 1)$$

ILLUSTRATION : 5**Factorise :**

(i) $a^2 + 2a + ab + 2b$ (ii) $x^2 - xz + xy - yz$

SOLUTION :

We have,

$$\begin{aligned}
 \text{(i)} \quad a^2 + 2a + ab + 2b &= (a^2 + 2a) + (ab + 2b) && \text{[Grouping the terms]} \\
 &= a(a + 2) + (a + 2)b && \text{[Taking (a + 2) common]} \\
 &= (a + 2)(a + b) && \text{[Grouping the terms]} \\
 \text{(ii)} \quad x^2 - xz + xy - yz &= (x^2 - xz) + (xy - yz) && \text{[Grouping the terms]} \\
 &= x(x - z) + y(x - z) \\
 &= (x + y)(x - z) && \text{[Taking (x - z) common]}
 \end{aligned}$$

ILLUSTRATION : 6**Factorise each of the following expressions.**

(i) $a^2x^2 + (ax^2 + 1)x + a$ (ii) $3ax - 6ay - 8by + 4bx$

SOLUTION :

$$\begin{aligned}
 \text{(i)} \quad a^2x^2 + (ax^2 + 1)x + a &= a^2x^2 + ax^3 + x + a \\
 &= ax^2(x + a) + (x + a) \\
 &= (x + a)(ax^2 + 1) \\
 \text{(ii)} \quad 3ax - 6ay - 8by + 4bx &= 3ax + 4bx - 6ay - 8by \\
 &= (3ax + 4bx) - (6ay + 8by) \\
 &= x(3a + 4b) - 2y(3a + 4b) \\
 &= (3a + 4b)(x - 2y)
 \end{aligned}$$

Factorisation by Using Identities

Sometimes given algebraic expression can be factorised using standard identities:

$$\begin{aligned}
 \text{(i)} \quad (a + b)^2 &= a^2 + 2ab + b^2 \\
 \text{(ii)} \quad (a - b)^2 &= a^2 - 2ab + b^2 \\
 \text{(iii)} \quad a^2 - b^2 &= (a + b)(a - b)
 \end{aligned}$$

When the given expression is in the form of perfect trinomial square, we use identities (i) & (ii) and when given expression is in the form of difference of two squares, we use identity (iii).

Factorising Perfect Trinomial Squares:

- (i) First determine whether the trinomial is a perfect square trinomial or not by following method:
 - (a) Two terms must be perfect squares and must be preceded by plus sign.
 - (b) The remaining term must be twice the product of the square roots of perfect square terms preceded by either a plus or a minus sign.
- (ii) If the trinomial is a perfect square, then factorise it as follows:
 - (a) Find the square root of the perfect square terms.
 - (b) Connect them by the sign of remaining term.
 - (c) Now express the trinomial as product of binomial obtained in step (b) twice

ILLUSTRATION : 7

Factorise : $16b^2 - 40by + 25y^2$

SOLUTION :

$$16b^2 - 40by + 25y^2 = (4b)^2 - 2(4b)(5y) + (5y)^2 = (4b - 5y)^2$$

$$16b^2 - 40by + 25y^2 = (4b - 5y)(4b - 5y)$$

ILLUSTRATION : 8

Factorise : $x^2 + 6x + 9$

SOLUTION :

By observation we find the x^2 and 9 are perfect square terms preceded by plus sign and $6x$ can be expressed as $2(3)(x)$ which is twice the product of square root of perfect square terms

$$\text{thus } x^2 + 6x + 9 = x^2 + 2x(3) + 3^2$$

$$x^2 + 6x + 9 = (x + 3)^2 = (x + 3)(x + 3) \text{ [using identity } (a + b)^2 = a^2 + 2ab + b^2 \text{]}$$

Factorising Difference of Two Squares

Algebraic expression in the form of difference of two squares is factorised as follows:

- Find square roots of two square terms.
- Write sum of the two square roots as one of the factors and difference of two square roots as the other factor.

ILLUSTRATION : 9

Factorise $x^2 - 4$.

SOLUTION :

Given expression can be written in difference of two squares as $x^2 - 4 = x^2 - 2^2$

Now square roots of x^2 and 2^2 are x & 2 respectively.

$$\therefore x^2 - 4 = x^2 - 2^2 = (x + 2)(x - 2)$$

ILLUSTRATION : 10

Factorise : $\frac{36}{49}p^2 - 81$

SOLUTION :

$$\frac{36}{49}p^2 - 81 = \left(\frac{6}{7}p\right)^2 - (9)^2$$

$$\frac{36}{49}p^2 - 81 = \left(\frac{6}{7}p + 9\right)\left(\frac{6}{7}p - 9\right)$$

ILLUSTRATION : 11

Factorise: $3 - 12x^2$

SOLUTION :

$$3 - 12x^2 = 3(1 - 4x^2)$$

$$3 - 12x^2 = 3[1^2 - (2x)^2]$$

$$3 - 12x^2 = 3(1 - 2x)(1 + 2x)$$

Factors of the Form $(x + a)(x + b)$

We know that $(x + a)(x + b) = x^2 + (a + b)x + ab$

So, we can also factorise the algebraic expressions using the above identity. It can also be generalised as

$(x + a)(x + b) = x^2 + (\text{sum of the constants})x + \text{Product of the constants}$

Consider : $(x + 3)(x + 5)$

$$= x^2 + (3 + 5)x + 15 = x^2 + 8x + 15$$

ILLUSTRATION : 12**Factorise : $x^2 + 6x + 8$** **SOLUTION :**

In order to factorise $x^2 + 6x + 8$, we find two numbers p and q such that $p + q = 6$ and $pq = 8$.

Clearly, $2 + 4 = 6$ and $2 \times 4 = 8$

We now split the middle term $6x$ in the given quadratic as $2x + 4x$.

$$\therefore x^2 + 6x + 8 = x^2 + 2x + 4x + 8$$

$$= (x^2 + 2x) + (4x + 8)$$

$$= x(x + 2) + 4(x + 2)$$

$$= (x + 2)(x + 4)$$

DIVISION OF ALGEBRAIC EXPRESSIONS

We have learnt how to add and subtract algebraic expressions. We also know how to multiply two expressions. In this section we shall learn the division of the algebraic expressions. We know division is the inverse process of multiplication.

Division of a Monomial by Another Monomial

To divide a monomial by another monomial, follow these steps:

Step 1 : Find the quotient of the numerical coefficients.

Step 2 : Find the quotient of the variables.

Step 3 : Find the product of the results obtained in step 1 and 2.

ILLUSTRATION : 13

Divide : (i) $12x^3y^3$ by $3x^2y$ (ii) $-15a^2bc^3$ by $3ab$

SOLUTION :

(i) We have,

$$\frac{12x^3y^3}{3x^2y} = \frac{12 \times x \times x \times x \times y \times y \times y}{3 \times x \times x \times y} = 4 \times x \times y = 4xy$$

Aliter $\frac{12x^3y^3}{3x^2y} = \frac{12}{3}x^{3-2}y^{3-1} = 4xy$

(ii) We have,

$$\frac{-15a^2bc^3}{3ab} = \frac{-15 \times a \times a \times b \times c \times c \times c}{3 \times a \times b} = -5ac^3$$

Aliter $\frac{-15a^2bc^3}{3ab} = -\frac{15}{3}a^{2-1}b^{1-1}c^3$
 $= -5ab^0c^3 = -5ac^3$

Division of a Polynomial by a Monomial

To divide a polynomial by a monomial, divide each term of the polynomial by the given monomial separately as explained below.

Consider $(35x^2y^3 - 15xy^2 + 20y) \div 5y$

$$= \frac{35x^2y^3}{5y} - \frac{15xy^2}{5y} + \frac{20y}{5y}$$

$$= 7x^2y^2 - 3xy + 4$$

ILLUSTRATION : 14**Divide:** $6x^4yz - 3xy^3z + 8x^2yz^4$ by $2xyz$ **SOLUTION :**

$$\text{We have, } \frac{6x^4yz - 3xy^3z + 8x^2yz^4}{2xyz} = \frac{6x^4yz}{2xyz} - \frac{3xy^3z}{2xyz} + \frac{8x^2yz^4}{2xyz} = 3x^3 - \frac{3}{2}y^2 + 4xz^3$$

Division of a Polynomial by a PolynomialConsider : $(7x^2 + 21x) \div (x + 3)$ We shall factorise $(7x^2 + 21x)$ first to check and match factors with the denominator:

$$\begin{aligned} 7x^2 + 21x &= (7 \times x \times x) + (3 \times 7 \times x) \\ &= 7 \times x \times (x + 3) = 7x(x + 3) \end{aligned}$$

$$(7x^2 + 21x) \div (x + 3) = \frac{7x^2 + 21x}{x + 3}$$

$$= \frac{7x(x + 3)}{x + 3} = 7x \quad (\text{cancelling the factor } (x + 3))$$

ILLUSTRATION : 15**Divide:** $35a^2 + 32a - 99$ by $7a - 9$ **SOLUTION :**

We have,

$$35a^2 + 32a - 99$$

$$= 35a^2 + 77a - 45a - 99$$

$$= 7a(5a + 11) - 9(5a + 11) = (5a + 11)(7a - 9) \quad \dots(i)$$

$$\therefore (35a^2 + 32a - 99) \div (7a - 9)$$

$$= \frac{35a^2 + 32a - 99}{7a - 9}$$

$$= \frac{(5a + 11)(7a - 9)}{(7a - 9)} \quad [\text{using (i)}]$$

$$= (5a + 11)$$

ILLUSTRATION : 16**Divide:** $a^{12} + a^6b^6 + b^{12}$ by $a^6 - a^3b^3 + b^6$ **SOLUTION :**

We have,

$$a^{12} + a^6b^6 + b^{12}$$

$$= a^{12} + 2a^6b^6 + b^{12} - a^6b^6 \quad [\text{adding and subtracting } a^6b^6]$$

$$= (a^6 + b^6)^2 - (a^3b^3)^2$$

$$= (a^6 + b^6 - a^3b^3)(a^6 + b^6 + a^3b^3) = (a^6 - a^3b^3 + b^6)(a^6 + a^3b^3 + b^6) \quad \dots(ii)$$

$$\therefore \frac{a^{12} + a^6b^6 + b^{12}}{a^6 - a^3b^3 + b^6} = (a^6 + a^3b^3 + b^6) \frac{(a^6 - a^3b^3 + b^6)}{(a^6 - a^3b^3 + b^6)}$$

$$= a^6 + a^3b^3 + b^6 \quad [\text{Cancelling } a^6 - a^3b^3 + b^6 \text{ from } N^r \text{ and } D^r]$$

CONNECTING TOPIC

Division of Polynomial: When a polynomial $P(x)$ is divided by another polynomial $g(x)$ ($g(x) \neq 0$), we get quotient $q(x)$ and remainder $r(x)$ such that $P(x) = g(x) \times q(x) + r(x)$

Here degree of remainder $r(x) <$ degree of divisor $g(x)$.

Let us learn the division of polynomials by long division method.

Steps of Division: Let dividend = $p(x)$, divisor = $g(x)$

Step-I: Arrange the dividend and divisor in the standard form i.e. arrange the terms in descending order of their degrees.

This step is required if the dividend and divisor are not in standard form

Step-II: Divide first term of dividend by first term of divisor to get first term of quotient.

Step-III: Multiply the divisor by first term of quotient and subtract this product from dividend to get the remainder.

Step-IV: Now treat the remainder obtained in step-III as new dividend. Divisor remains the same. Repeat step-II to get next term of quotient.

Step-V: Repeat step-III to get a new remainder. The process continues till the remainder is zero (0) or the degree of the new dividend (remainder) is less than the degree of the divisor. At this point the new dividend is the remainder of division and sum of quotient gives us the whole quotient.

ILLUSTRATION : 1

Divide $f(x) = -2x + 2x^3 - x^2 - 7$ by $g(x) = -2 + x$

SOLUTION :

As $f(x)$ and $g(x)$ are not in standard form we arrange them in descending order of degree terms.

Now $f(x) = 2x^3 - x^2 - 2x - 7$, $g(x) = x - 2$

$2x^2 + 3x + 4$	
$(x - 2) \overline{) 2x^3 - x^2 - 2x - 7}$	$\frac{2x^3}{x} = 2x^2$
$\underline{2x^3 - 4x^2}$	$2x^2(x - 2) = 2x^3 - 4x^2$
$ + 3x^2 - 2x - 7$	
$ \underline{3x^2 - 6x}$	$\frac{3x^2}{x} = 3x$
$ + 4x - 7$	$3x(x - 2) = 3x^2 - 6x$
$ \underline{4x - 8}$	$\frac{4x}{x} = 4$
$ + 1$	$4(x - 2) = 4x - 8$
$ \underline{1}$	

Here, quotient = $(2x^2 + 3x + 4)$ and remainder = 1

MISCELLANEOUS SOLVED EXAMPLES

1. Factorise : $m^2n^2 - 2m^3n^3 + m^4n^4$

Sol. $m^2n^2 [1 - 2mn + m^2n^2]$

2. Factorise : $a^2 - 10ab + 25b^2 - 36$

Sol. The first three terms of the given expression are of the form $(a - b)^2$ and the last term is also a perfect square.

$$\begin{aligned} & [(a)^2 - 2(a)(5b) + (5b)^2] - (6)^2 \\ &= (a - 5b)^2 - (6)^2 \\ &= (a - 5b + 6)(a - 5b - 6) \quad [\text{using } a^2 - b^2 = (a + b)(a - b)] \\ &= (a - 5b + 6)(a - 5b + 6) \end{aligned}$$

3. Divide $a(10a^2 - 160)$ by $10a(a + 4)$.

Sol. $a(10a^2 - 160)$
 $= a[(10 \times a^2) - (10 \times 16)]$
 $= a \times 10 \times (a^2 - 16) \quad [\text{Taking 10 as common factor}]$
 $= 10a \times (a + 4)(a - 4) \quad [\text{using } a^2 - b^2 = (a + b)(a - b)]$

$$\text{Thus, } a(10a^2 - 160) \div 10a(a + 4) = \frac{10a \times (a + 4)(a - 4)}{10a(a + 4)}$$

$$= (a - 4)$$

4. Factorise $2x^2 + 13x + 20$ and divide it by $x + 4$.

Sol. $2x^2 + 13x + 20 = 2x^2 + 8x + 5x + 20$
 $= 2x(x + 4) + 5(x + 4)$
 $= (2x + 5)(x + 4)$

$$\begin{aligned} \text{Now, } \frac{2x^2 + 13x + 20}{x + 4} &= \frac{(2x + 5)(x + 4)}{(x + 4)} \\ &= (2x + 5) \end{aligned}$$

5. Factorise : $x^2 + 7y + xy + 7x$

Sol. $x^2 + 7y + xy + 7x$
 $= x^2 + xy + 7y + 7x$
 $= x(x + y) + 7(y + x)$
 $= (x + y)(x + 7)$

6. Factorise : (i) $(x + y)(2x + 3y) - (x + y)(x + 1)$ (ii) $(x + y)(2a + 2b) - (3x - 2y)(2a + b)$

Sol. We have,

(i) $(x + y)(2x + 3y) - (x + y)(x + 1)$
 $= (x + y)\{(2x + 3y) - (x + 1)\}$ [Taking $(x + y)$ common]
 $= (x + y)(2x + 3y - x - 1)$
 $= (x + y)(x + 3y - 1)$

$$(ii) (x+y)(2a+2b) - (3x-2y)(2a+b)$$

$$= \{(x+y) - (3x-2y)\} (2a+b)$$

[Taking $(2a+b)$ common]

$$= (x+y-3x+2y)(2a+b)$$

$$= (-2x+3y)(2a+b)$$

$$7. \text{ Factorise : (i) } 80a^2 - 45b^2$$

$$(ii) (3a-b)^2 - 9c^2$$

$$\text{Sol. (i) } 80a^2 - 45b^2 = 5(16a^2 - 9b^2)$$

$$= 5\{(4a)^2 - (3b)^2\}$$

$$= 5(4a+3b)(4a-3b)$$

[Using: $a^2 - b^2 = (a+b)(a-b)$]

$$(ii) (3a-b)^2 - 9c^2 = (3a-b)^2 - (3c)^2$$

$$= \{(3a-b) + 3c\} \{(3a-b) - 3c\}$$

$$= (3a-b+3c)(3a-b-3c)$$

$$8. \text{ Factorise : } x(x+z) - y(y+z)$$

$$\text{Sol. } x(x+z) - y(y+z) = x^2 + xz - y^2 - yz$$

$$= (x^2 - y^2) + (xz - yz)$$

$$= (x-y)(x+y) + z(x-y)$$

$$= (x-y)\{(x+y) + z\}$$

$$= (x-y)(x+y+z)$$

$$9. \text{ Divide } 8y^3 - 6y^2 + 4y - 1 \text{ by } 4y + 2. \text{ Also, write the quotient and the remainder.}$$

Sol. We have,

$$8y^3 - 6y^2 + 4y - 1$$

$$= 2y^2(4y+2) - \frac{5}{2}y(4y+2) + \frac{9}{4}(4y+2) - \frac{11}{2}$$

$$= \left\{ 2y^2(4y+2) - \frac{5}{2}y(4y+2) + \frac{9}{4}(4y+2) \right\} - \frac{11}{2}$$

$$= (4y+2) \left(2y^2 - \frac{5}{2}y + \frac{9}{4} \right) - \frac{11}{2}$$

$$\text{Hence, Quotient} = 2y^2 - \frac{5}{2}y + \frac{9}{4} \text{ and, Remainder} = -\frac{11}{2}.$$

$$10. \text{ Divide } (x^2 + 7x + 6) \text{ by } (x+1)$$

$$\text{Sol. } \frac{(x^2 + 7x + 6)}{(x+1)} = \frac{x^2 + 6x + x + 6}{x+1}$$

$$= \frac{x(x+6) + 1(x+6)}{x+1} = \frac{(x+6)(x+1)}{(x+1)}$$

$$= (x+6)$$

SOLVED EXAMPLES BASED ON CONNECTING TOPICS

11. Divide $6 + x - 4x^2 + x^3$ by $x - 3$.

Sol.

$$\begin{array}{r}
 x^2 - x - 2 \\
 x - 3 \overline{) x^3 - 4x^2 + x + 6} \\
 \underline{x^3 - 3x^2} \\
 -x^2 + x + 6 \\
 \underline{-x^2 + 3x} \\
 -2x + 6 \\
 \underline{-2x + 6} \\
 0
 \end{array}$$

$$\therefore \frac{6 + x - 4x^2 + x^3}{x - 3} = x^2 - x - 2$$

12. Divide $x^3 - 6x^2 + 11x - 6$ by $x^2 - 4x + 3$.

Sol. On dividing, we get

$$\begin{array}{r}
 x - 2 \\
 x^2 - 4x + 3 \overline{) x^3 - 6x^2 + 11x - 6} \\
 \underline{x^3 - 4x^2 + 3x} \\
 -2x^2 + 8x - 6 \\
 \underline{-2x^2 + 8x - 6} \\
 0
 \end{array}$$

$$\therefore x^3 - 6x^2 + 11x - 6 = (x - 2)(x^2 - 4x + 3)$$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- $\frac{9x^2 - 16}{6x + 8}$ is written in its lowest terms as _____.
- $x^2 + x - 56 = (x + 8)(\quad)$.
- Every _____ polynomial has one and only one zero.
- The number of algebraic expressions that may be multiplied to obtain given algebraic expressions are called _____ of the given algebraic expression.
- $100 - 81a^2 = \quad$.
- $(a^2 - b^2) \div (a - b) = \quad$.
- Degree of any constant term is _____.
- $(5x + 9y)^2 + 12(5x + 9y) = \quad$.
- The value of $52xyz(xy + yz + xz + y^2)(z + x) \div 104xy(xy + xz + yz + y^2)$ is _____.
- $(49x^2yz + 35p) \div 7 = \quad$.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- $2x - 1$ is a factor of $6x^2 + x - 1$.
- The highest common factor of $4x^2yz$ and $63x^3y^2z^2$ is x^2yz .
- Factors of $x^2 - 14xy + 49y^2$ will be $(x - 7y)(x - 7y)$.
- $(a + b)(a - b) = a^2 - b^2$
- The factors of $a^2 - 2ab + b^2$ are $(a + b)$ and $(a + b)$.
- An equation is true for all values of its variables.
- The value of p for $51^2 - 49^2 = 100p$ is 2.
- The value of $(a + 1)(a - 1)(a^2 + 1)$ is $a^4 - 1$.
- $abc + bca + cab$ is a monomial.

Match the Following :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- | Column-I | Column-II |
|--------------------|------------------------------------|
| (A) $3a^5 - 48a^3$ | (p) $2a(a-2)(a+2)(a^2+4)$ |
| (B) $2a^5 - 32a$ | (q) $(2xy + 3x - 1)(2xy - 3x + 5)$ |

- (C) $4(xy + 1)^2 - 9(x - 1)^2$ (r) $-xy(x^2 + y^2)(x + y)$
 $(x - y)(x^4 + y^4)$
 (D) $xy^9 - yx^9$ (s) $3a^3(a - 4)(a + 4)$

Very Short Answer Questions:

DIRECTIONS : Give answer in one word or one sentence.

- Factorise : $6ab - b^2 + 12ac - 2bc$
- Divide : $25x^3y^2$ by $-15x^2y$
- Factorise : $x^4 - 4$
- Factorise : $ax^2y + bxy^2 + cxyz$
- The value of $(3x^3 + 9x^2 + 27x) \div 3x$ is:
- Simplified form of $\frac{(-6a^3b^5)(2a^2b^3)}{-18a^4b^8c^3}$.
- Find the value of $x^2 + 8y^3$ divided by $x + 2y$.
- Simplify the given expression $39y^2(50y^2 - 98) \div 26y^2(5y + 7)$.
- The value of a in $14a = 45^2 - 25^2$ is
- $x^2 - y^2$ for $x = 200$ and $y = 199$ is

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- Factorise :
 (i) $x^2 - 14x + 49$ (ii) $d^2 + 4d + 4$
 (iii) $x^2 + 8x - 20$ (iv) $m^2n^2 - 28mn - 60$
 (v) $p^2 - p - 90$
- Simplify $[4x^2 - 25y^2 + 30y - 9] \div [2x + 5y - 3]$.
- Factorise each of the following expression:
 $a^3x + a^2(x - y) - a(y + z) - z$
- Factorise : $25a^2 - 4b^2 + 28bc - 49c^2$.
- Solve by factorising : $[a^2 + b^2 + 2(ab + bc + ca)] \div (a + b + 2c)$.
- Find the factors of $25p^2 - 80pq + 64q^2$
- Factorise:
 (i) $5a(2x + 3y) - 2b(2x + 3y)$
 (ii) $8(5x + 9y)^2 + 12(5x + 9y)$

Factorisation

8. Factorise each of the following algebraic expressions:

(i) $(2x-3y)(a+b) + (3x-2y)(a+b)$

(ii) $4(x-y)(3a-b) + 6(x-y)(2b-3a)$

9. Factorise:

(i) $16x^4-81$

(ii) $2x-32x^5$

(iii) $3a^4-48b^4$

(iv) $81x^4-121x^2$

10. Factorise:

(i) $(x-1)^2-(x-2)^2$

(ii) $16a^2-\frac{25}{4a^2}$

(iii) $16a^2b-\frac{b}{16a^2}$

Long Answer Questions :

DIRECTIONS : Give answer in four to five sentences.

1. Divide $-4p^2-4q^2-12r^2+10pq+16qr-14pr$ by $2p-q+3r$.

2. Simplify:

(i) $\frac{3x^2+6x}{x^2-4x} \times \frac{x-4}{x^2+9x+14}$

(ii) $\frac{2x}{x^3-1} \times (x^2+x+1)$

(iii) $\frac{x-3}{xy^2} \times \frac{4x^2y^3}{x^2+x-12}$

(iv) $\frac{-x}{x^3-8} \times (x^2+2x+4)$

3. Simplify:

(i) $\frac{x^2+4x+3}{4x^2+12x} \div \frac{x+1}{3}$ (ii) $\frac{20x^2+x-1}{7x^3+x} \div \frac{4x+1}{x}$

(iii) $\frac{x^3-4x}{2} \div (x^4-16)$

4. Factorise:

(i) $9-a^6+2a^3b^3-b^6$

(ii) $x^{16}-y^{16}+x^8+y^8$

(iii) $(p+q)^2-(a-b)2+p+q-a+b$

5. Simplify:

(i) $\frac{6x^2+15x+9}{2x+3}$

(ii) $\frac{p^2-p-6}{p-3}$

2**EXERCISE****Text Book Exercise**

1. Factorise $6xy-4y+6-9x$.

2. Find the common factors of the given terms.

(i) $16x^3, -4x^2, 32x$

(ii) $3x^2y^3, 10x^3y^2, 6x^2y^2z$

3. Factorise the following expressions.

(i) $5x^3y-15xy^2$

(ii) $x^2yz+xy^2z+xyz^2$

(iii) $ax^2y+bx^2y+cxyz$

4. Factorise:

$15pq+15+9q+25p$

5. Factorise: m^4-256

6. Find the factors of $y^2-7y+12$.

7. Obtain the factors of $z^2-4z-12$.

8. Factorise the following expression.

$a^4+2a^2b^2+b^4$

9. Factorise:

(i) $(1+m)^2-(1-m)^2$

(ii) $25a^2-4b^2+28bc-49c^2$

10. Factorise the expressions.

(i) $am^2+bm^2+bn^2+an^2$ (ii) $5y^2-20y-8z+2yz$

11. Factorise the following expressions.

(i) p^2+6p+8

(ii) $q^2-10q+21$

(iii) $p^2+6p-16$

12. $8(x^3y^2z^2+x^2y^3z^2+x^2y^2z^3) \div 4x^2y^2z^2$

13. Factorise the expressions and divide them as directed.

(i) $12xy(9x^2-16y^2) \div 4xy(3x+4y)$

(ii) $39y^3(50y^2-98) \div 26y^2(5y+7)$

14. Divide $44(x^4-5x^3-24x^2)$ by $11x(x-8)$

15. Do the following divisions.

(i) $-20x^4 \div 10x^2$

(ii) $7x^2y^2z^2 \div 14xyz$

16. Solve: $x(3x+2)$

17. Solve: $(3x+2)^2$

18. Solve: $(a+4)(a+2)$

Exemplar Question

1. Factorised form of $r^2-10r+21$ is

2. Divide $p(4p^2-16)$ by $4p(p-2)$

3. Factorise the following, using the identity

$a^2+2ab+b^2=(a+b)^2$

(i) $4x^4+12x^3+9x^2$

(ii) $16x^2+40x+25$

4. Factorise the following.
 (i) $18 + 11x + x^2$ (ii) $p^2 - 13p - 30$
5. Factorise the following using the identity
 $a^2 - b^2 = (a + b)(a - b)$.
 (i) $\frac{4x^2}{9} - \frac{9y^2}{16}$ (ii) $\frac{x^3y}{9} - \frac{xy^3}{16}$
6. Perform the following division
 $(x^3y^3 + x^2y^3 - xy^4 + xy) \div xy$
7. Factorise the expressions and divide them as directed:
 (i) $(3x^2 - 48) \div (x - 4)$ (ii) $(3x^4 - 1875) \div (3x^2 - 75)$
8. The area of a square is given by $4x^2 + 12xy + 9y^2$. Find the side of the square.
9. (i) $63p^2q^2r^2s - 9pq^2r^2s^2 + 15p^2qr^2s^2 - 60p^2q^2rs^2$
 (ii) $24x^2yz^3 - 6xy^3z^2 + 15x^2y^2z - 5xyz$
 (iii) $2ax^2 + 4axy + 3bx^2 + 2ay^2 + 6bxy + 3by^2$
10. The area of a rectangle is $x^2 + 7x + 12$. If its breadth is $(x + 3)$, then find its length.

11. The following expressions are the areas of rectangles. Find the possible lengths and breadths of these rectangles.
 $x^2 - 7x + 10$
12. Carry out the following divisions:
 $-121p^3q^3r^3 \div (-11xy^2z^3)$

HOTS Questions :

1. Solve :
 (i) $x \div \frac{1}{x} \times x + x$ of $(x \div x)$
 (ii) $a^3 \times b^2 \div a - ab + a^2 - a(a - b)$
 (iii) $\frac{5}{2}p - \left\{ 2p - p^3 \div p \times \frac{1}{p} + p^2 \div \left(p^2 \times \frac{2}{p} \right) \right\}$
2. If $(x - 5)$ notebooks cost ₹ $(x^2 - 13x + 40)$, what is the cost of one book?
3. One factor of $x^3 - 7x + 6$ is $x - 1$. The other factors are:
4. Factorise : $a^{12} - b^{12}$.

3

EXERCISE

Single Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

1. One of the factors of $x^2 + \frac{1}{x^2} + 2 - 2x - \frac{2}{x}$ is
 (a) $x - \frac{1}{x}$ (b) $x + \frac{1}{x} - 1$
 (c) $x + \frac{1}{x}$ (d) $x^2 + \frac{1}{x^2}$
2. The factors of $x^4 + y^4 + x^2y^2$ are
 (a) $(x^2 + y^2)(x^2 + y^2 - xy)$
 (b) $(x^2 + y^2)(x^2 - y^2)$
 (c) $(x^2 + y^2 + xy)(x^2 + y^2 - xy)$
 (d) Factorization is not possible
3. The factors of $x^2 + xy - 2xz - 2yz$ are
 (a) $(x - y)(x + 2z)$ (b) $(x + y)(x - 2z)$
 (c) $(x - y)(x - 2z)$ (d) $(x + y)(x + 2z)$
4. Factorisation of $15x^2 - 26x + 8$ is
 (a) $(3x - 4)(5x - 2)$ (b) $(3x + 4)(5x + 2)$
 (c) $(3x - 4)^2$ (d) $(5x - 2)^2$
5. Evaluate $x^4 \div x^4$
 (a) 1 (b) 0
 (c) 2 (d) -1

6. Divide $-35x^5y^2$ by $-7x^3y$
 (a) $5xy^2$ (b) $25x^2y$
 (c) $5xy$ (d) $5x^2y$
7. The common factor of $72x^3y^4z^4$, $120z^2d^4x^4$ and $96y^3z^4d^4$ is
 (a) $96z^3$ (b) $120z^3$
 (c) $72z^3$ (d) $24z^2$
8. Divide $(32x^4y^3 - 16x^3y^4)$ by $(-8x^2y)$
 (a) $4x^3y^2 + 2xy^3$ (b) $4x^3y - 2xy$
 (c) $-4x^2y^2 + 2xy^3$ (d) $-4xy^2 + 2xy^3$
9. Value of $(99.8)^2 - (0.2)^2$ is:
 (a) 9980 (b) 9960
 (c) 9860 (d) 9680
10. The real factors of $x^2 + 4$ are:
 (a) $(x^2 + 2)(x^2 - 2)$ (b) $(x + 2)(x - 2)$
 (c) does not exist (d) None of these

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

1. If $(x^2 + 3x + 5)(x^2 - 3x + 5) = m^2 - n^2$, then m and n is
 (a) $x^2 - 3x$ (b) $3x$
 (c) $x^2 + 5$ (d) $x^2 + 3x$

Factorisation

- Factors of $3a^2 - 9ab$ is
(a) $3a$ (b) a
(c) $(a-3b)$ (d) $(a-9b)$
- Factors of $2a^2 + 11a - 21$ is
(a) $(a-7)$ (b) $(2a-3)$
(c) $(a+7)$ (d) $(2a+3)$
- The factors of $3m^2 + 9m + 6$ are not equal to
(a) $3(m+1)(m+2)$ (b) $(3m+1)(3m-1)$
(c) $3(m+2)(m-1)$ (d) $(3m-1)(3m-2)$
- On factorisation $32a^2 - 50b^2$ is not equal to.
(a) $2(4a-5)(4a-5)$ (b) $2(4a-5b)(4a+5b)$
(c) $(8a-5b)(4a+5b)$ (d) $16ab(a+b)$
- Which of the following is not correct?
(a) $4(x^2-1) = 4x^2-1$
(b) $2(x-3) = 2x-3$
(c) $15(x^3-x^2) = 15x^3-x^2$
(d) $2(x+4) = 2x+8$

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE

Factors are of the form $(x+a)(x+b)$ where $(x+a)(x+b) = x^2 + (a+b)x + ab$ or, $x^2 + (\text{sum of constants})x + (\text{product of constants})$

- Factorise $8a^2 - 22ab + 15b^2$.
(a) $(4a-5b)(2a-3b)$ (b) $(4a+5b)(2a+3b)$
(c) $(4a-5b)(2a+3b)$ (d) $(4a+5b)(2a-3b)$
- Factorise $10a^2 - 83a - 17$.
(a) $(2a+17)(5a-1)$ (b) $(2a+17)(5a+1)$
(c) $(2a-17)(5a-1)$ (d) $(2a-17)(5a+1)$
- Factorise $2x^2 - 35x - 18$.
(a) $(x+18)(2x-1)$ (b) $(z+18)(2x+1)$
(c) $(x+18)(2x-1)$ (d) $(x-18)(2x-1)$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
- If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
- If **Assertion** is **correct** but **Reason** is **incorrect**.
- If **Assertion** is **incorrect** but **Reason** is **correct**.

- Assertion :** The possible factors of $5xy$ is 1, $5xy$, 5, xy , $5x$, y , $5y$, x

Reason : If an algebraic expression is written as the product of numbers of algebraic expression, then each of these numbers and expression are called the factors of the given algebraic expression.

- Assertion :** $16x^4 - 81 = (4x^2 + 9)(4x^2 - 9)$

Reason : $a^2 - b^2 = (a+b)(a-b)$

- Assertion :** The greatest common factor of $7x$, $21x^2$ and $14xy^2$ is 7.

Reason : The greatest common factor of given monomials is the common factor having greatest coefficient and highest power of the variable.

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- When we divide $(38a^3b^3c^2 - 19a^4b^2c)$ by $19a^2bc$, the result is $ka^2bc - a^2b$. Find k .
- $\frac{x^6}{x^2} = x^a$, where a is
- $5x^2 - 13x - 6 = (mx+2)(x-n)$, where $m+n$ is

4**ADVANCED EXERCISE
BASED ON CONNECTING TOPICS**

DIRECTIONS (Qs. 1-4) : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which **ONLY ONE** is correct.

- Divide $10x^4 + 17x^3 - 62x^2 + 30x - 3$ by $2x^2 + 7x - 1$
(a) $5x^2 + 9x + 3$ (b) $5x^2 - 9x + 3$
(c) $5x^2 - 3x + 9$ (d) $5x^2 + 3x - 9$
- Using long division, find the quotient :
 $(6y^5 + 15y^4 + 16y^3 + 4y^2 + 10y - 35) \div (3y^2 + 5)$

- $2y^2 + 5y^3 - 7y + 2$ (b) $5y^3 + 2y^2 - 5y + 7$
- $2y^3 + 5y^2 + 2y - 7$ (d) $2y^3 - 5y^2 - 2y + 7$

- Divide : $-21 + 71x - 31x^2 - 24x^3$ by $3 - 8x$.

- $3x^2 + 5x - 7$ (b) $3x^2 - 5x - 7$
- $3x^2 - 7x + 5$ (d) $3x^2 - 5 + 7x$

- Divide $3y^4 - 3y^3 - 4y^2 - 4y$ by $y^2 - 2y$.

- $4y^2 + 4y + 1$ (b) $3y^2 + 3y + 2$
- $2y^2 + 3y + 3$ (d) $3y^2 + y + 3$

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- $\frac{3x-4}{2}$
- $x-7$
- linear
- factors
- $(10-9a)(10+9a)$;
 $100-81a^2=(10)^2-(9a)^2=(10-9a)(10+9a)$
- $a+b$; $\frac{(a^2-b^2)}{a-b} = \frac{(a-b)(a+b)}{(a-b)} = a+b$
- 0; Degree of any constant term is zero.
- $(5x+9y)(5x+9y+12)$;
 $(5x+9y)^2+12(5x+9y)$
 $= (5x+9y)(5x+9y+12)$
- $\frac{z(z+x)}{2}$
- $7x^2yz+5p$

True / False :

- False: $6x^2+x-1=6x^2-2x+3x-1$
 $=2x(3x-1)+1(3x-1)=(3x-1)(2x+1)$
 $\therefore 2x-1$ is not a factor.
- True: $4x^2yz$ and $63x^3y^2z^2$ has common factor
 $4x^2yz=4 \times x \times x \times y \times z$
 $63x^3y^2z^2=63 \times x \times x \times x \times y \times y \times z \times z$
Common factor $=x^2yz$
- True: $x^2-14xy+49y^2$
 $=x^2-2 \times 7x \times 7y+(7y)^2=(x-7y)^2$
- True
- False
- False
- True
- True
- True

Match the Following :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)
(A) $3a^5-48a^3=3a^3(a^2-16)=3a^3(a-4)(a+4)$
(B) $2a^5-32a=2a(a^4-16)$
 $=2a(a^2-4)(a^2+4)$
 $=2a(a-2)(a+2)(a^2+4)$

$$\begin{aligned} \text{(C)} \quad & 4(xy+1)^2-9(x-1)^2 \\ & = [2(xy+1)]^2 - [3(x-1)]^2 \\ & = (2xy+2-3x+3)(2xy+2+3x-3) \\ & = (2xy-3x+5)(2xy+3x-1) \\ \text{(D)} \quad & xy^9-4x^9=xy(y^8-x^8) \\ & = xy[(y^4)^2-(x^4)^2] \\ & = xy(y^4+x^4)(y^4-x^4) \\ & = xy(y^4+x^4)(y^2+x^2)(y+x)(y-x) \end{aligned}$$

Very Short Answer Questions :

- We have, $6ab-b^2+12ac-2bc$
 $=b(6a-b)+2c(6a-b)=(6a-b)(b+2c)$
- We have,
 $\frac{25x^3y^2}{-15x^2y} = \frac{25 \times x \times x \times x \times y \times y}{-15 \times x \times x \times y} = -\frac{5}{3}xy$
- $x^4-4=(x^2)^2-(2)^2$
 $= (x^2-2)(x^2+2) = (x+\sqrt{2})(x-\sqrt{2})(x^2+2)$
- We have, $ax^2y+bx^2y+cx^2y$
 $=xy(ax+by+cz)$
- $(3x^3+9x^2+27x) \div 3x$
Factorising, $3x^3+9x^2+27x$
 $=3x(x^2+3x+9)$
Then, $\frac{\cancel{3x}(x^2+3x+9)}{\cancel{3x}}$
 $=x^2+3x+9.$
- $\frac{(-6a^3b^5)(2a^2b^3)}{-18a^4b^8c^3}$
 $= \frac{12a^{3+2}b^{5+3}}{18a^4b^8c^3}$
 $= \frac{2a^5b^8}{3a^4b^8c^3} = \frac{2a}{3c^3}$
- $\frac{x^3+8y^3}{x+2y} = \frac{(x)^3+(2y)^3}{(x+2y)}$

Applying the identity

$$a^3 + b^3 = (a + b)(a^2 - ab + b^2) \text{ to } (x)^3 + (2y)^3$$

Where, $a = x$ and $b = 2y$, we get,

$$\text{R.H.S} = \frac{(x+2y)[x^2 - x \times 2y + (2y)^2]}{(x+2y)}$$

$$\therefore \frac{x^3 + 8y^3}{x + 2y} = x^2 - 2xy + 4y^2$$

$$\begin{aligned} 8. \quad 39y^2(50y^2 - 98) \div 26y^2(5y + 7) &= \frac{39y^2(50y^2 - 98)}{26y^2(5y + 7)} = \frac{3(50y^2 - 98)}{2(5y + 7)} \\ &= \frac{3 \times 2(25y^2 - 49)}{2(5y + 7)} = \frac{3[(5y)^2 - (7)^2]}{(5y + 7)} \\ &= \frac{3[(5y - 7)(5y + 7)]}{5y + 7} = 3(5y - 7) \\ &= 15y - 21 \\ \therefore \text{Simplified value of the given expression is } 15y - 21 \end{aligned}$$

$$9. \quad 100$$

$$10. \quad 399$$

Short Answer Questions :

1. (i) $(x - 7)(x - 7)$
(ii) $(d + 2)(d + 2)$
(iii) $(x - 2)(x + 10)$
(iv) $(mn - 30)(mn + 2)$
(v) $(1 - 10)(1 + 9)$

$$\begin{aligned} 2. \quad \frac{4x^2 - 25y^2 + 30y - 9}{2x + 5y - 3} &= \frac{(2x)^2 - [25y^2 - 30y + 9]}{2x + 5y - 3} \\ &= \frac{(2x)^2 - [(5y)^2 - 2 \times 5y \times 3 + (3)^2]}{2x + 5y - 3} = \frac{(2x)^2 - (5y - 3)^2}{2x + 5y - 3} \\ &= \frac{(2x - 5y + 3)(2x + 5y - 3)}{2x + 5y - 3} = 2x - 5y + 3 \end{aligned}$$

3. We have,

$$\begin{aligned} a^3x + a^2(x - y) - a(y + z) - z \\ &= a^3x + a^2x - a^2y - ay - az - z \\ &= (a^3x + a^2x - a^2y + ay) - (az + z) \\ &= a^2x(a + 1) - ay(a + 1) - z(a + 1) \\ &= (a + 1)(a^2x - ay - z) \end{aligned}$$

$$\begin{aligned} 4. \quad 25a^2 - 4b^2 + 28bc - 49c^2 \\ &= 25a^2 - (4b^2 - 28bc + 49c^2) \\ &= 25a^2 - [(2b)^2 - 2 \times 2b \times 7c + (7c)^2] \\ &= 25a^2 - (2b - 7c)^2 = (5a)^2 - (2b - 7c)^2 \\ &= (5a + 2b - 7c)(5a - 2b + 7c) \end{aligned}$$

$$\begin{aligned} 5. \quad \frac{a^2 + b^2 + 2(ab + bc + ca)}{a + b + 2c} \\ &= \frac{a^2 + b^2 + 2ab + 2bc + 2ca}{a + b + 2c} \\ &= \frac{(a^2 + b^2 + 2ab) + 2c(b + a)}{a + b + 2c} \\ &= \frac{(a + b)[(a + b) + 2c]}{a + b + 2c} = a + b \end{aligned}$$

$$6. \quad (5p - 8q)(5p - 8q)$$

7. We have,

$$(i) \quad 5a(2x + 3y) - 2b(2x + 3y) = (2x + 3y)(5a - 2b).$$

(Taking $(2x + 3y)$ common)

$$(ii) \quad 8(5x + 9y)^2 + 12(5x + 9y) = 4(5x + 9y) \{2(5x + 9y) + 3\} \\ = 4(5x + 9y)(10x + 18y + 3)$$

$$8. \quad (i) \quad 5(a + b)(x - y)$$

$$(ii) \quad 2(x - y)(4b - 3a)$$

$$\begin{aligned} 9. \quad (i) \quad 16x^4 - 81 &= (4x^2)^2 - (9)^2 \\ &= (4x^2 - 9)(4x^2 + 9) \\ &\quad \text{[Using } a^2 - b^2 = (a - b)(a + b)] \\ &= \{(2x)^2 - (3)^2\}(4x^2 + 9) \\ &= (2x - 3)(2x + 3)(4x^2 + 9) \\ &\quad \text{[Using } a^2 - b^2 = (a - b)(a + b)] \end{aligned}$$

$$\begin{aligned} (ii) \quad 2x - 32x^5 &= 2x(1 - 16x^4) \\ &= 2x\{1^2 - (4x^2)^2\} \\ &= 2x(1 + 4x^2)\{1 - 4x^2\} \\ &= 2x(1 + 4x^2)\{1 - (2x)^2\} \\ &= 2x(1 + 4x^2)(1 - 2x)(1 + 2x) \end{aligned}$$

$$\begin{aligned} (iii) \quad 3a^4 - 48b^4 &= 3(a^4 - 16b^4) \\ &= 3\{(a^2)^2 - (4b^2)^2\} \\ &= 3(a^2 - 4b^2)(a^2 + 4b^2) \\ &= 3\{a^2 - (2b)^2\}(a^2 + 4b^2) \\ &= 3(a - 2b)(a + 2b)(a^2 + 4b^2) \end{aligned}$$

$$\begin{aligned} (iv) \quad 81x^4 - 121x^2 &= x^2(81x^2 - 121) \\ &= x^2\{(9x)^2 - (11)^2\} \\ &= x^2(9x - 11)(9x + 11) \end{aligned}$$

$$\begin{aligned} 10. \quad (i) \quad (x - 1)^2 - (x - 2)^2 &= \{(x - 1) + (x - 2)\} \{(x - 1) - (x - 2)\} \\ &= (2x - 3)(x - 1 - x + 2) \\ &= (2x - 3) \times 1 \\ &= 2x - 3 \end{aligned}$$

$$(ii) \quad 16a^2 - \frac{25}{4a^2} = (4a)^2 - \left(\frac{5}{2a}\right)^2 = \left(4a + \frac{5}{2a}\right)\left(4a - \frac{5}{2a}\right)$$

$$\begin{aligned} (iii) \quad 16a^2b - \frac{b}{16a^2} &= b\left(16a^2 - \frac{1}{16a^2}\right) \\ &= b\left\{\left(4a\right)^2 - \left(\frac{1}{4a}\right)^2\right\} = b\left(4a + \frac{1}{4a}\right)\left(4a - \frac{1}{4a}\right) \end{aligned}$$

Long Answer Questions :

1. $-2p + 4q - 4r.$

2. (i) $\frac{3}{x+7}$ (ii) $\frac{2x}{x-1}$

(iii) $\frac{4xy}{x+y}$ (iv) $\frac{x}{2-x}$

3. (i) $\frac{3}{4}x$ (ii) $\frac{5x-1}{7x^2+1}$

(iii) $\frac{x}{2(x^2+4)}$

4. (i) $9 - a^6 + 2a^3b^3 - b^6$
 $= 9 - (a^6 - 2a^3b^3 + b^6)$
 $= (3)^2 - \{(a^3)^2 - 2 \times a^3 \times b^3 + (b^3)^2\}$
 $= 3^2 - (a^3 - b^3)^2$
 $= (3 + (a^3 - b^3)) \{3 - (a^3 - b^3)\}$

(ii) $x^{16} - y^{16} + x^8 + y^8$
 $= \{(x^8)^2 - (y^8)^2\} + (x^8 + y^8)$
 $= (x^8 + y^8)(x^8 - y^8) + (x^8 + y^8)$
 $= (x^8 + y^8)(x^8 - y^8 + 1)$

(iii) $(p+q)^2 - (a-b)^2 + p+q - a+b$
 $= \{(p+q)^2 - (a-b)^2\} + (p+q) - (a-b)$
 $= \{(p+q) + (a-b)\} \{(p+q) - (a-b)\} + \{(p+q) - (a-b)\}$
 $= (p+q+a-b)(p+q-a+b) + (p+q-a+b)$
 $= (p+q-a+b)(p+q-a+b+1)$

5. (i) $\frac{6x^2+15x+9}{2x+3}$
 $= \frac{3(2x^2+5x+3)}{2x+3} = \frac{3(2x^2+3x+2x+3)}{2x+3}$
 $= \frac{3[x(2x+3)+1(2x+3)]}{2x+3}$
 $= \frac{3(x+1)(2x+3)}{2x+3} = 3(x+1)$

(ii) $\frac{p^2-p-6}{p-3} = \frac{p^2-3p+2p-6}{p-3}$
 $= \frac{p(p-3)+2(p-3)}{p-3} = \frac{(p+2)(p-3)}{p-3} = p+2$

2 EXERCISE

Text-Book Exercise :

1. $6xy - 4y + 6 - 9x = 6xy - 4y - 9x + 6$
 $= 2y(3x-2) - 3(3x-2)$
 $= (3x-2)(2y-3)$
 The factors of $(6xy - 4y + 6 - 9x)$ are $(3x-2)$ and $(2y-3)$.

2. (i) $4x$ (ii) x^2y^2

3. (i) $5xy(x^2-y)$

(ii) $xyz(x+y+z)$

(iii) $xy(ax+by+cz)$

4. $(5p+3)(3q+5)$

5. $\Rightarrow m^4 - 256 = (m^2)^2 - (16)^2$
 $= (m^2 - 16)(m^2 + 16)$
 $\Rightarrow m^2 - 16 = m^2 - 4^2$
 $= (m-4)(m+4)$

Therefore, $m^4 - 256 = (m-4)(m+4)(m^2 + 16)$

6. We note $12 = 3 \times 4$ and $3 + 4 = 7$. Therefore,
 $y^2 - 7y + 12 = y^2 - 3y - 4y + 12$
 $= y(y-3) - 4(y-3) = (y-3)(y-4)$

7. $z^2 - 4z - 12 = z^2 - 6z + 2z - 12$
 $= z(z-6) + 2(z-6)$
 $= (z-6)(z+2)$

8. $(a^2 + b^2)^2$

9. (i) $4lm$ (ii) $(5a-2b+7c)(5a+2b-7c)$

10. (i) $(m^2+n^2)(a+b)$ (ii) $(5y+2z)(y-4)$

11. (i) $(p+2)(p+4)$ (ii) $(q-3)(q-7)$
 (iii) $(p+8)(p-2)$

12. $2(x+y+z)$

13. (i) $3(3x-4y)$ (ii) $3y(5y-7)$

14. Factorising $44(x^4 - 5x^3 - 24x^2)$, we get
 $44(x^4 - 5x^3 - 24x^2) = 2 \times 2 \times 11 \times x^2(x^2 - 5x - 24)$
 (taking the common factor x^2 out of the bracket)
 $= 2 \times 2 \times 11 \times x^2(x^2 - 8x + 3x - 24)$
 $= 2 \times 2 \times 11 \times x^2[x(x-8) + 3(x-8)]$
 $= 2 \times 2 \times 11 \times x^2(x+3)(x-8)$
 Therefore, $44(x^4 - 5x^3 - 24x^2) \div 11x(x-8)$

$$= \frac{2 \times 2 \times 11 \times x \times x \times (x+3) \times (x-8)}{11 \times x \times (x-8)}$$

$$= 2 \times 2 \times x(x+3) = 4x(x+3)$$

15. (i) $-20x^4 = -2 \times 2 \times 5 \times x \times x \times x \times x$
 $10x^2 = 2 \times 5 \times x \times x$
 Therefore, $(-20x^4) \div 10x^2 =$
 $= \frac{-2 \times 2 \times 5 \times x \times x \times x \times x}{2 \times 5 \times x \times x} = -2 \times x \times x = -2x^2$

(ii) $7x^2y^2z^2 \div 14xyz = \frac{7 \times x \times x \times y \times y \times z \times z}{2 \times 7 \times x \times y \times z}$
 $= \frac{x \times y \times z}{2} = \frac{1}{2}xyz$

16. $x(3x+2) = 3x^2 + 2x$

17. $(3x+2)^2 = 9x^2 + 12x + 4$

18. $(a+4)(a+2) = a^2 + 6a + 8$

Exemplar Questions :

- $(r-7)(r-3)$
- $(p+2)$
- (i) $x^2(2x+3)(x+6)$ (ii) $(4x+5)(4x+5)$
- (i) $(9+x)(2+x)$ (ii) $(p-15)(p+2)$
- (i) $\left(\frac{2x}{3} + \frac{3y}{4}\right)\left(\frac{2x}{3} - \frac{3y}{4}\right)$
(ii) $xy\left(\frac{x}{3} + \frac{y}{4}\right)\left(\frac{x}{3} - \frac{y}{4}\right)$
- $x^2y^2 + xy^2 - y^3 + 1$
- (i) $3(x+4)$ (ii) x^2+25
- $2x+3y$
- (i) $3pqrs(21pqr - 3qrs + 5prs - 20pqs)$
(ii) $xyz(24xz^2 - 6y^2z + 15xy - 5)$
(iii) $(2a+3b)(x+y)^2$
- $x+8$
- $x-2$ and $x-5$
- $\frac{11p^3q^3r^3}{xy^2z^3}$

HOTS Questions :

- (i) x^3+x (ii) a^2b^2
(iii) $\frac{5}{2}p - \left\{2p - p^3 \div p \times \frac{1}{p} + p^2 \div \left(p^2 \times \frac{2}{p}\right)\right\}$
Apply principle of BODMAS
$$= \frac{5}{2}p - \left\{2p - p^3 \div p \times \frac{1}{p} + p^2 \div 2p\right\}$$
$$= \frac{5}{2}p - \left\{2p - p^3 \div p \times \frac{1}{p} + \frac{p}{2}\right\}$$
$$= \frac{5}{2}p - \left\{2p - p^2 \times \frac{1}{p} + \frac{p}{2}\right\} = \frac{5}{2}p - \left\{2p - p + \frac{p}{2}\right\}$$
$$= \frac{5p}{2} - \frac{3p}{2} = \frac{2p}{2} = p$$
- $x-8$
Hint: $(x^2 - 13x + 40) \div (x-5)$
- $(x+3)(x-2)$
- $a^{12} - b^{12} = a^{12} - b^{12} = (a^6)^2 - (b^6)^2 = (a^6 - b^6)(a^6 + b^6)$
 $[\therefore a^2 - b^2 = (a-b)(a+b)]$
$$= (a^6 - b^6)((a^3)^2 - (b^3)^2)$$
$$= (a^6 + b^6)(a^3 + b^3)(a^3 - b^3)$$
$$= (a^6 + b^6)(a+b)(a^2 - ab + b^2)(a-b)(a^2 + ab + b^2).$$

3 EXERCISE

Single Option Correct :

- (c) $x^2 + \frac{1}{x^2} + 2 - 2x - \frac{2}{x}$
$$= \left(x^2 + \frac{1}{x^2} + 2 \times x \times \frac{1}{x}\right) - 2\left(x + \frac{1}{x}\right)$$
$$= \left(x + \frac{1}{x}\right)^2 - 2\left(x + \frac{1}{x}\right) = \left(x + \frac{1}{x}\right)\left(x + \frac{1}{x} - 2\right)$$
- (c) $x^4 + y^4 + 2x^2y^2$
$$= (x^2)^2 + (y^2)^2 + x^2y^2 = (x^2 + y^2)^2 - 2x^2y^2 + x^2y^2$$
$$= (x^2 + y^2)^2 - x^2y^2$$
$$= (x^2 + y^2 - xy)(x^2 + y^2 + xy).$$
- (b) $x^2 + xy - 2xz - 2yz$
$$\Rightarrow x^2 + xy - 2z(x+y)$$
$$= x(x+y) - 2z(x+y)$$
$$= (x+y)(x-2z).$$
- (a) $15x^2 - 26x + 8 = 15x^2 - 20x - 6x + 8$
$$= 5x(3x-4) - 2(3x-4) = (5x-2)(3x-4)$$
- (a) $\frac{x^4}{x^4} = x^{4-4} = x^0 = 1$
- (d) $\frac{-35x^4y^2}{-7x^3y} = 5x^{5-3}y^{2-1} = 5x^2y$
- (d) $72x^3y^4z^4 = 2 \times 2 \times 2 \times 3 \times 3 \times x \times x \times x \times y \times y \times y \times y \times z \times z \times z \times z$
 $120z^2d^4x^4 = 2 \times 2 \times 2 \times 3 \times 5 \times z \times z \times d \times d \times d \times d$
 $96y^3z^4d^4 = 2 \times 2 \times 2 \times 2 \times 2 \times 3 \times y \times y \times y \times z \times z \times z \times z \times d \times d \times d \times d$
 $\therefore$ Common factor is $24z^2$
- (c) $\frac{32x^4y^3 - 16x^3y^4}{-8x^2y}$
$$= \frac{32x^4y^3}{-8x^2y} + \frac{16x^3y^4}{8x^2y}$$
$$= -4x^2y^2 + 2xy^3$$
- (b) $(99.8)^2 - (0.2)^2$
$$= (99.8 + 0.2)(99.8 - 0.2)$$
$$= 100 \times 99.6 = 9960$$
- (c) Real factors do not exist for $x^2 + 4$.

More Than One Option Correct :

- (b, c) $(x^2 + 3x + 5)(x^2 - 3x + 5) = m^2 - n^2$
 $\Rightarrow ((x^2 + 5) + 3x)((x^2 + 5) - 3x) = m^2 - n^2$
 $\Rightarrow (x^2 + 5)^2 - (3x)^2 = m^2 - n^2$
 $\therefore m = (x^2 + 5), n = 3x$
- (a, b, c) $3a^2 - 9ab = 3a(a - 3b)$
- (b, c) $2a^2 + 11a - 21 = 2a^2 + 14a - 3a - 21$
 $= 2a(a + 7) - 3(a + 7) = (2a - 3)(a + 7)$
- (b, c, d) 5. (a, c, d) 6. (a, b, c)

Passage Based Questions :

- (a) $8a^2 - 22ab + 15b^2$
 $= 8a^2 - 10ab - 12ab + 15b^2$
 $= 2a(4a - 5b) - 3b(4a - 5b)$
 $= (4a - 5b)(2a - 3b)$
- (d) Product $= 10(-17) = -170$
Sum $= -83$
 $10a^2 - 83a - 17$
 $= 10a^2 - 85a + 2a - 17 = 5a(2a - 17) + 1(2a - 17)$
 $= (2a - 17)(5a + 1)$
- (b) The product is (-36) .
The sum is -35 . So the numbers are -36 and 1 .
 $2x^2 - 36x + x - 18 = 2x(x - 18) + 1(x - 18)$
 $= (x - 18)(2x + 1)$

Assertion & Reason :

- (a) Factors of $5xy = 1, 5xy, 5, x, y, 5x, 5y$
- (a) $16x^2 - 81 = (2x)^4 - (3)^4 = ((2a^2)^2 - (3)^2)^2$
 $= (4x^2 + 9)(4x^2 - 9)$
- (d) Common factor of $7x, 21x^2, 14xy^2$ is $7x$

Integer Type Questions :

- (2) $\frac{38a^3b^3c^2 - 19a^4b^2c}{19a^2bc} = \frac{38a^3b^3c^2}{19a^2bc} - \frac{19a^4b^2c}{19a^2bc}$
 $= 2ab^2c - a^2b$
- (4) $\frac{x^6}{x^2} = x^{6-2} = x^4 \Rightarrow a = 4$

- (8) $5x^2 - 13x - 6 = 5x^2 - 15x + 2x - 6$
 $= 5x(x - 3) + 2(x - 3) = (5x + 2)(x - 3)$
 $\Rightarrow m = 5, n = 3,$
 $m + n = 8$

4 ADVANCED EXERCISE
BASED ON CONNECTING TOPICS

- (b) On dividing, we get

$$\begin{array}{r}
 5x^2 - 9x + 3 \\
 2x^2 + 7x - 1 \overline{) 10x^4 + 17x^3 - 62x^2 + 30x - 3} \\
 \underline{10x^4 + 35x^3 - 5x^2} \\
 -18x^3 - 57x^2 + 30x - 3 \\
 \underline{-18x^3 - 63x^2 + 9x} \\
 6x^2 + 21x - 3 \\
 \underline{6x^2 + 21x - 3} \\
 0
 \end{array}$$

$$\therefore 10x^4 + 17x^3 - 62x^2 + 30x - 3 = (2x^2 + 7x - 1)(5x^2 - 9x + 3)$$

- (c)

$$\begin{array}{r}
 2y^3 + 5y^2 + 2y - 7 \\
 3y^2 + 5 \overline{) 6y^5 + 15y^4 + 16y^3 + 4y^2 + 10y - 35} \\
 \underline{6y^5} \\
 15y^4 + 6y^3 + 4y^2 + 10y - 35 \\
 \underline{15y^4} \\
 6y^3 - 21y^2 + 10y - 35 \\
 \underline{6y^3} \\
 -21y^2 - 35 \\
 \underline{-21y^2 - 35} \\
 0
 \end{array}$$

- (a) $3x^2 + 5x - 7$
- (b) $3y^2 + 3y + 2$

Chapter

15

Introduction to Graphs

INTRODUCTION

Any complicated information when represented in the form of a graph becomes simple to understand. Graph is a pictorial way of representing relationships between various quantities.

Earlier, we have learnt about some types of graphs namely a bar graph, a pictograph and a histogram. In this chapter, we will learn about line graphs and linear graphs.

LINE GRAPH

A line graph displays data that changes continuously over periods of time. It uses points connected by lines to show how something changes in value as the time passes.

For example:

The following table shows the number of runs scored per over by a cricket team.

Over number	1	2	3	4	5
Runs scored per over	8	13	10	5	15

Here, horizontal axis (i.e. x-axis) shows the over number and vertical axis (i.e. y-axis) shows runs scored per over. The information given can be shown in a line graph which is shown below :

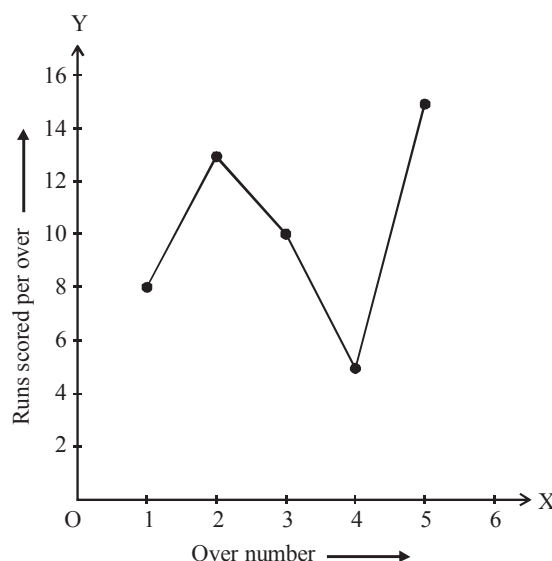


ILLUSTRATION : 1

The given line graph shows the distance travelled by a motorcyclist from a city A to B at different times. These two cities are 500 km apart. Study the graph and answer the following :

- What information is given on the two axes?
- From where and when did the motorcyclist begin its journey?
- How far did the motorcyclist go in the first hour?
- How far did the motorcyclist go during 3rd hour?
- When did the motorcyclist reach city B.

SOLUTION :

- The horizontal (x – axis) shows the time. The vertical (y – axis) shows the distance of the motorcyclist from city A.
- The motorcyclist started from city A at 9 a.m.
- The motorcyclist travelled 50 km during the first hour.
- During 3rd hour (i.e., from 11 a.m. to 12 noon) motorcyclist covered 150 km ($300 - 150 = 150$).
- The motorcyclist reach city B at 4 p.m.

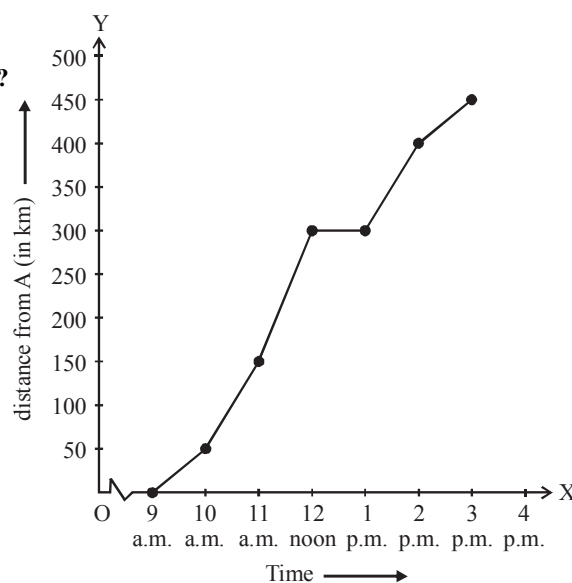


ILLUSTRATION : 2

Following table gives the temperature at 12 : 00 noon on seven successive days in a city:

Day (November)	1	2	3	4	5	6	7
Temperature (in °C)	14	18	14	16	20	15	18

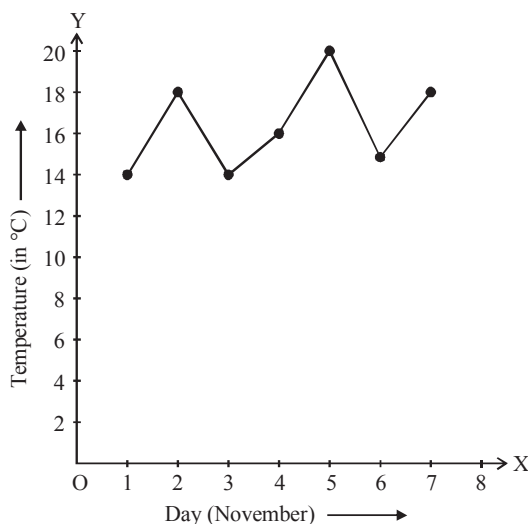
Plot a graph to illustrate this information.

SOLUTION :

Here number of days is an independent variable and temperature is dependent variable. So, we take number of days on x-axis and temperature on y-axis. Let us choose the following scale :

On y-axis : 1 unit = 2°C

Now, we plot each ordered pair (1, 14), (2, 18), (3, 14), (4, 16), (5, 20), (6, 15) and (7, 18). These points are joined to get the graph representing the given information as shown in the below figure.



LINEAR GRAPHS

A linear graph is a graph that is a whole unbroken line. In order to draw such graphs we have to locate some points on the graph sheet.

We know, what is a number line. On the number line, distances from a fixed point are marked in equal units positively in one direction and negatively in other direction. The point from which the distances are marked is called the origin O .

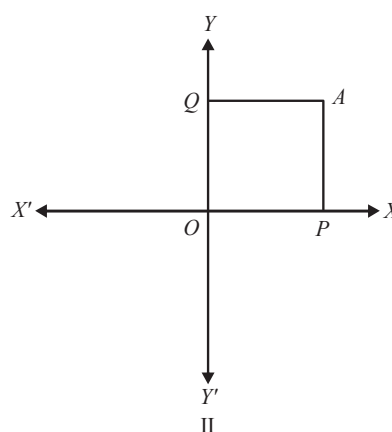
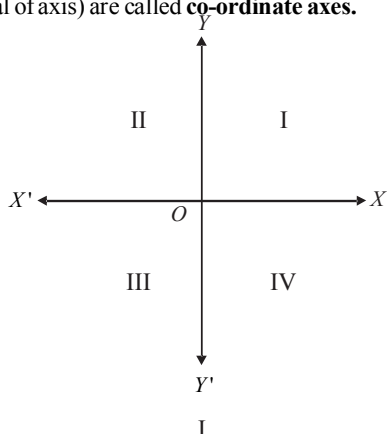
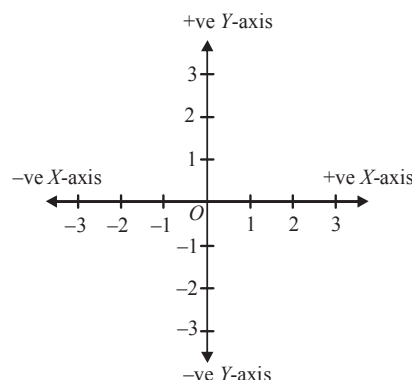
To locate a point on a plane we require two such number lines, both perpendicular to each other and meeting each other at origin or zero. One number line is kept horizontal marked XX' and other perpendicular marked YY' . Numbers are written on both of them just as written on number line.

XX' is called **X-axis** and YY' is called **Y-axis**.

From the **fig.** it is clear that positive numbers lie on OX and OY , OX and OY are called positive direction of X -axis and Y -axis respectively. Similarly OX' and OY' are called negative direction of X -axis and Y -axis respectively.

Here we see that the two axes divide the plane into four parts, these parts are called **quadrants**.

The two axes (plural of axis) are called **co-ordinate axes**.

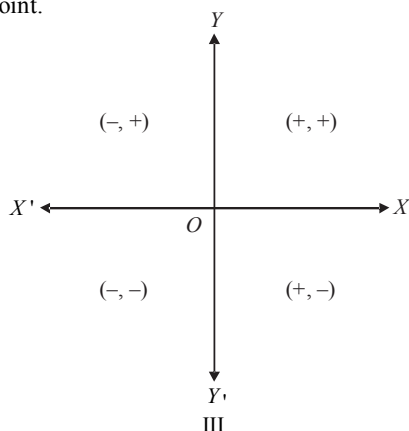


Let us take a point A in XY -plane as shown in **fig. II** and draw two perpendicular AP and AQ on X and Y -axis respectively.

Distance $AQ = OP$ is called **x-co-ordinate** or **abscissa** of point A and distance $AP = OQ$ is called **y-co-ordinate** or **ordinate** of point P .

The abscissa and ordinate of a point taken together is known as **co-ordinate of a point**. Thus if $OP = x$ and $OQ = y$ then (x, y) are co-ordinates of point A . Note that $(x, y) \neq (y, x)$.

Now, we know about the co-ordinates of a point.



The **fig. III** shows the sign of the co-ordinates of a point in different quadrants.

We can clearly see from figure that

- (a) All the points in first quadrant have both abscissa and ordinate positive.
- (b) In second quadrant abscissa is negative and ordinate is positive.
- (c) In third quadrant both abscissa and ordinate are negative
- (d) In fourth quadrant abscissa is positive and ordinate is negative.

NOTE : The order of x and y are important. So (x, y) is an ordered pair.

CO-ORDINATES OF ORIGIN

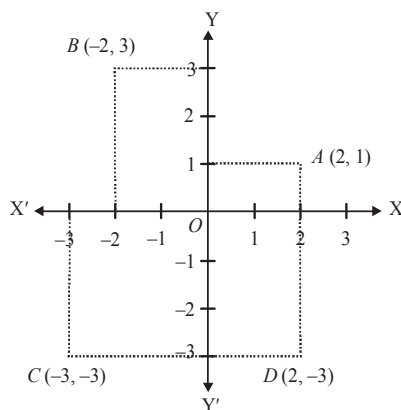
Origin is the point of inter-section of X -axis and Y -axis. Now the distance of origin from both the axes is zero, so origin has zero abscissa and zero ordinate. Therefore the co-ordinates of origin is $(0, 0)$.

ILLUSTRATION : 3

Locate $(2, 1)$, $(-2, 3)$, $(-3, -3)$, $(2, -3)$ on graph paper.

SOLUTION :

Let A, B, C, D respectively denote the points $(2, 1)$, $(-2, 3)$, $(-3, -3)$ and $(2, -3)$.



- (i) As abscissa and ordinate of point A are both positive, So A lies in the first-quadrant.
- (ii) For point B abscissa is negative and ordinate is positive. So point B lies in second quadrant.
- (iii) For point C both abscissa and ordinate are negative, so it lies in third quadrant.
- (iv) For point D, abscissa is positive and ordinate is negative, so it lies in fourth quadrant.

ILLUSTRATION : 4

Find out the quadrants in which the following points lie : (a) $A = (3, -4)$ (b) $B = (-3, 4)$ (c) $C = (-3, -4)$

SOLUTION :

- (a) Point A lies in 4th quadrant, since its abscissa is positive and ordinate is negative.
 (b) Point B lies in second quadrant, since its abscissa is negative and ordinate is positive.
 (c) Point C lies in third quadrant since its abscissa as well as ordinate both are negative.

NOTE : The process of marking a point with given co-ordinates in the cartesian plane is known as plotting a point.

ILLUSTRATION : 5

Plot the following set of points on graph paper.

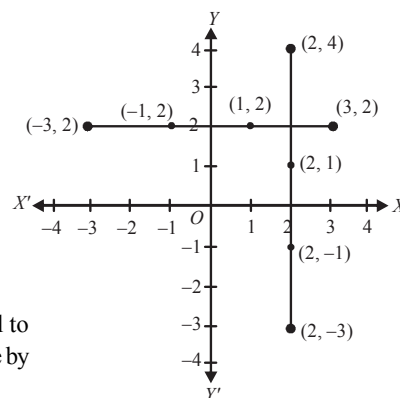
(a) $(-3, 2)$, $(-1, 2)$, $(1, 2)$, $(3, 2)$

(b) $(2, 4)$, $(2, 1)$, $(2, -1)$, $(2, -3)$

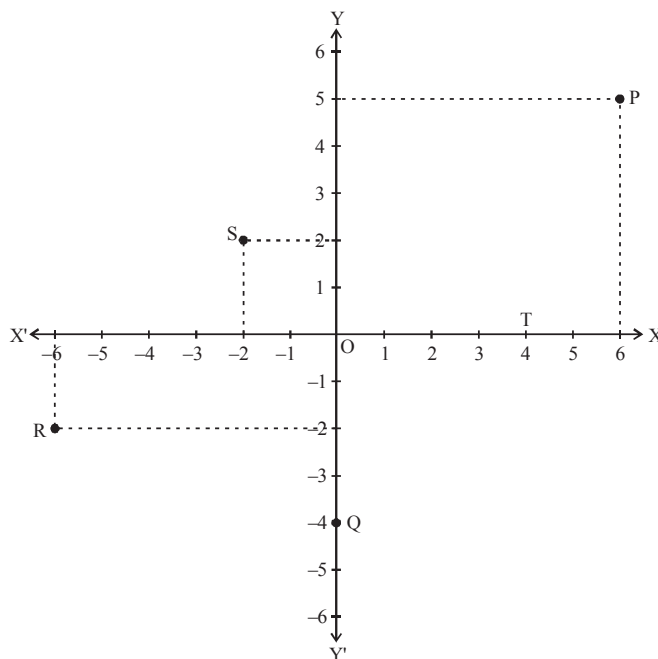
What do you observe?

SOLUTION :

- (a) After plotting the points we observe the following
 (i) Ordinate of all points is '2'
 (ii) After plotting the points, if we join them, we get a straight line, parallel to X -axis which is at a distance of 2 units from X -axis we can represent this line by $y = 2$
- (b) After plotting the set of points we observe that
 (i) Abscissa of all points is 2.
 (ii) By joining all the points, we get a straight line which is parallel to Y -axis and at a distance of 2 units from Y -axis. We can represent such line by $x = 2$.

**ILLUSTRATION : 6**

Find out coordinates of the points P, Q, R, S, T from the following graph.

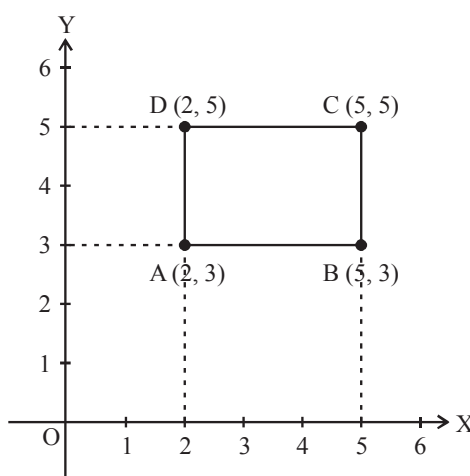


SOLUTION :

- (i) To locate P, we have to move 6 units rightward on x-axis and 5 units upward in vertical direction, Therefore coordinates of P are (6, 5).
- (ii) Point Q lies downward 4 units from origin, therefore coordinates of Q are (0, -4).
- (iii) Point R corresponds to -6 units on x-axis and -2 units on y-axis, therefore coordinates of R are (-6, -2).
- (iv) Point S corresponds to -2 units on x-axis and 2 units on y-axis, i.e., coordinates of S are (-2, 2).
- (v) Since point T lies on x-axis. Therefore, coordinates of point T are (4, 0).

ILLUSTRATION : 7

Plot each point. Connect the points in order i.e., A to B, B to C and so on. A (2, 3), B (5, 3), C (5, 5), D (2, 5).

SOLUTION :**CONNECTING TOPICS****GRAPH OF A LINEAR EQUATION IN TWO VARIABLES**

Consider a linear equation $x + 2y = 5$ in two variables x and y . We can say that (1, 2) is a solution of this equation because by putting $x = 1$ and $y = 2$ in L.H.S. we get the result equal to R.H.S.

$$\text{L.H.S.} = 1 + 2 \times 2 = 5 = \text{R.H.S.}$$

Similarly, (5, 0) is also a solution of equation $x + 2y = 5$. In fact we can get many more solution of given equation by choosing some value of x and getting corresponding value of y from the equation. For example, let $x = 2$ now, by putting $x = 2$ in $x + 2y = 5$ we get

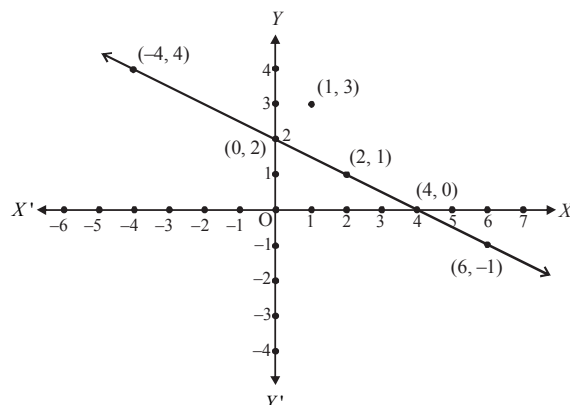
$$2 + 2y = 5 \Rightarrow 2y = 3 \Rightarrow y = \frac{3}{2}$$

$\therefore \left(2, \frac{3}{2}\right)$ is also a solution of given equation. This way we can go on finding many solutions of a linear equation in two variables.

We can say that a linear equation in two variables has infinitely many solutions. The following table shows some of possible solutions of equation $x + 2y = 4$

x	0	2	4	-4	
y	2	1	0	4	

Now, plot the above points $(0, 2)$, $(2, 1)$, $(4, 0)$, $(-4, 4)$... on XY -plane.



If we join any two of the plotted points, we get a straight line. We also observe that the other two points also lie on the line. Let us take another point on the line $(6, -1)$, which lies on the line. We observe that it is also a solution of the given equation.

Now take a point $(1, 3)$, which does not lie on the line say. We see that $(1, 3)$ is not a solution of given equation $x + 2y = 4$ so, we observe the following :

- (a) Every point whose co-ordinates satisfy the given equation lie on the line.
- (b) Every point on the line given a solution of given equation.
- (c) Any point which does not lie on the line, is not a solution of the given equation.

SOLUTION OF A LINEAR EQUATION OF ONE VARIABLE

The value of variable which satisfy a given equation is called solution of the equation. The solution is also known as Root of the equation. e.g. for linear equation $2x - 8 = 0$, $x = 4$ is a solution or root, because by putting $x = 4$ in the equation, both sides of the equation become equal. In other words,

$x = 2$ satisfies equation $2x - 8 = 0$

ILLUSTRATION : 8

Verify that $x = 3$ is a solution of equation $3x + 5 = 14$

SOLUTION :

If $x = 3$ is a solution of the given equation then by putting $x = 3$ in the given equation will make both sides of equation equal

∴ Put $x = 3$ in the equation, we get

$$\text{L. H. S.} = 3 \times 3 + 5 = 9 + 5$$

$$= 14 = \text{RHS}$$

Hence $x = 3$ is a solution of $3x + 5 = 14$

NOTE : Solution of an equation is also known as root of the equation.

ILLUSTRATION : 9

Verify that $x = 2$ is a solution of $2x + 3 = 5x - 6$

SOLUTION :

By putting $x = 2$, we get

$$\text{L. H. S.} = 2x + 3 = 2 \times 2 + 3 = 7$$

$$\text{R. H. S.} = 5x - 6 = 5 \times 2 - 6 = 4$$

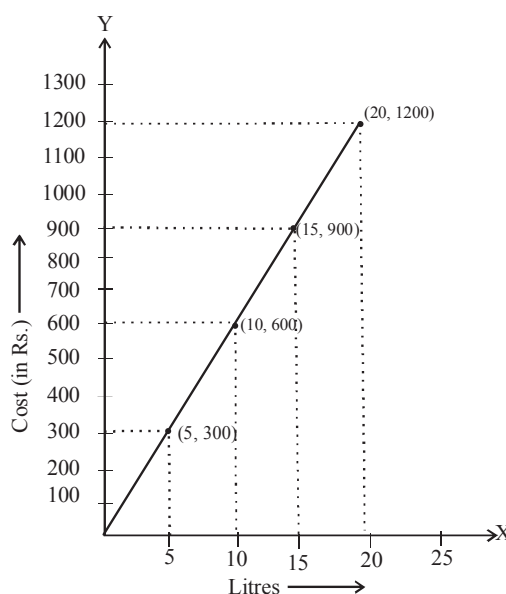
So $x = 2$ is not a solution of $2x + 3 = 5x - 6$

MISCELLANEOUS SOLVED EXAMPLES

1. The given table shows the quantity of petrol and its cost.

No. of litres of petrol	5	10	15	20
Cost of petrol in RS.	300	600	900	1200

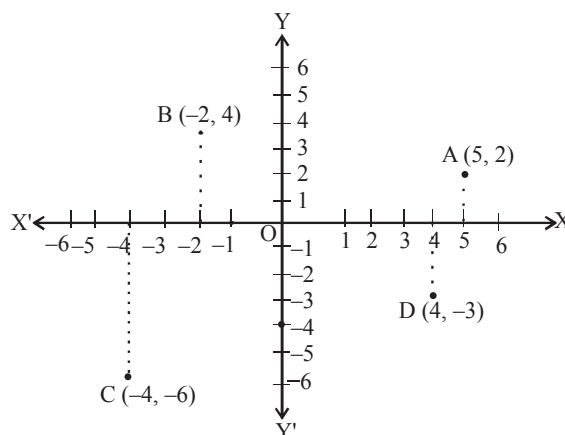
Sol. Let us take a suitable scale on both the axes.



2. Plot each of the following points on a graph paper :

(i) A (5, 2) (ii) B (-2, 4) (iii) C (-4, -6) (iv) D (4, -3)

Sol. Let XOX' and YOY' be the coordinates axes.



- (i) On x-axis take 5 units to the right of the y-axis and then on the y-axis, take 2 units above the x-axis. Thus, we get the point A (5, 2).
- (ii) On the x-axis, take 2 units to the left of the y-axis and then on the y-axis, take 4 units above the x-axis. Thus, we get the point B (-2, 4).

(iii) On the x-axis, take 4 units to the left of the y-axis and then on the y-axis, take 6 units below the x-axis. Thus, we get the point C $(-4, -6)$.

(iv) On the x-axis take 4 units to the right of the y-axis and then on the y-axis, take 3 units below the x-axis. Thus we get the point D $(4, -3)$.

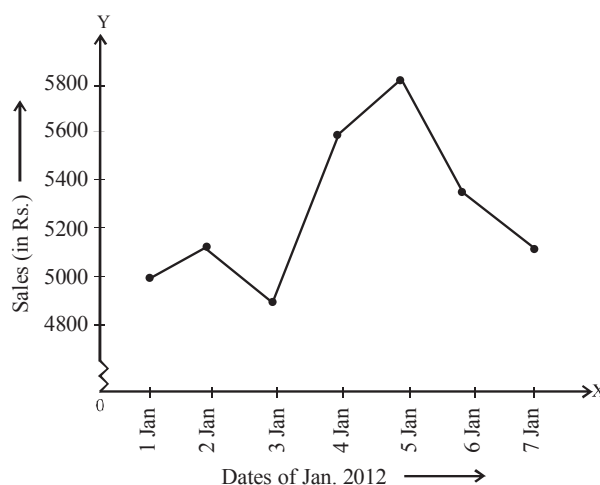
3. The sales of a shopkeeper in the week of January 2012, are given below :

Date	1	2	3	4	5	6	7
Sales (in ₹)	5000	5100	4900	5600	5800	5300	5100

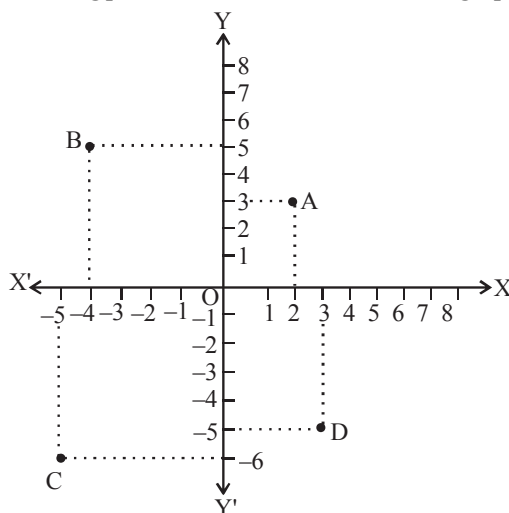
Draw a graph representing the above data.

Sol. In order to represent the above data graphically, we represent dates on x-axis and sales (in ₹) on y-axis. 1 unit on Y-axis represents a scale of ₹ 200.

The ordered pairs $(1, 5000)$, $(2, 5100)$ etc are plotted as points and are joined by the line segments as shown in below figure which gives the required graph.



4. Write down the co-ordinates of the following points A, B, C and D marked on the graph paper shown in below figure.



Sol. Clearly, the distance of A from y – axis is 2 units and that from x – axis is 3 units. Since A lies in the first quadrant, so, its co-ordinates are $(2, 3)$.

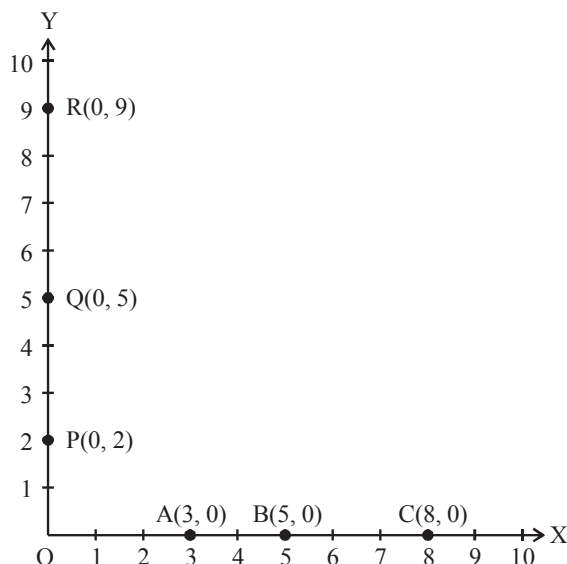
Point B lies in the second quadrant and its distance from y and x–axis are -4 and 5 units respectively. So, its co-ordinates are $(-4, 5)$.

Clearly, point C lies in the third quadrant and its distance from y and x – axis are -5 and -6 units respectively. So, its co-ordinates are $(-5, -6)$.

Clearly, point D lies in the fourth quadrant. The distance of D from y –axis is 3 units and from x – axis is -5 units. So, the co-ordinates of D are $(3, -5)$.

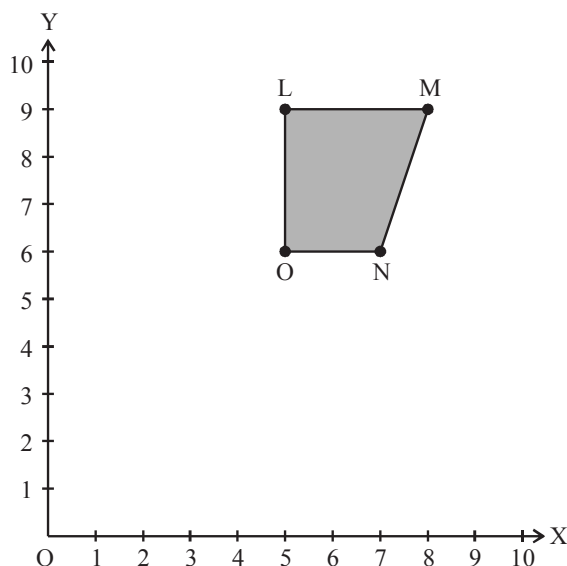
5. Plot the points A (3, 0), B (5, 0) and C (8, 0). What do you observe? Where do they lie? Also, plot the points P (0, 2), Q (0, 5) and R (0, 9). Do they lie on x – axis?

Sol. The points A, B, C, P, Q and R are plotted on the graph paper as shown in figure below. Clearly, points A, B and C lie on x-axis and point P, Q and R lie on y – axis.



6. Draw a quadrilateral LMNO where L (5, 9), M (8, 9), N (7, 6) and O (5, 6).

Sol. Let us mark the coordinates of L (5, 9), M (8, 9), N (7, 6) and O (5, 6) on a graph sheet. Join LM, MN, NO and OL to obtain the desired quadrilateral.



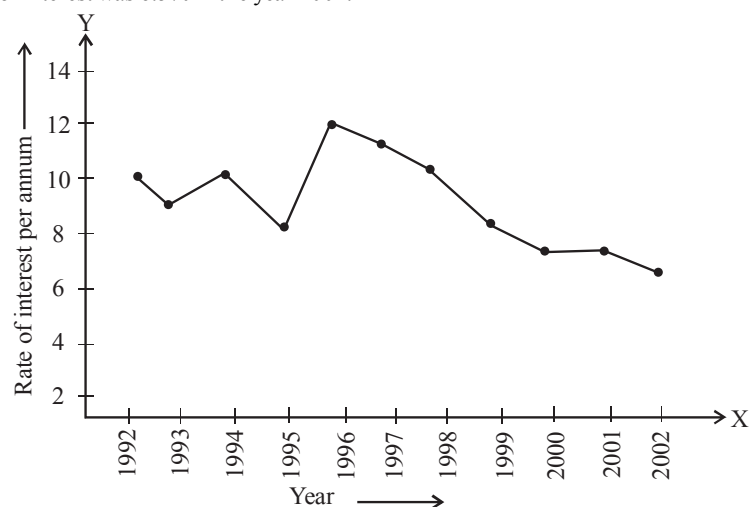
7. The graph shown in fig exhibits the rate of interest on fixed deposits upto one year announced by the Reserve Bank of India in different years. Read the graph and find.

- In which period was the rate of interest maximum?
- In which period was the rate of interest minimum?

Sol. In the graph, we find that years are represented on x – axis and the rate of interest per annum along y – axis. From the graph, we find that

- The rate of interest was maximum (12%) in 1996.

- (ii) The minimum rate of interest was 6.5% in the year 2002.



8. The perimeter P and sides of a square are connected by the relation $P = 4s$.

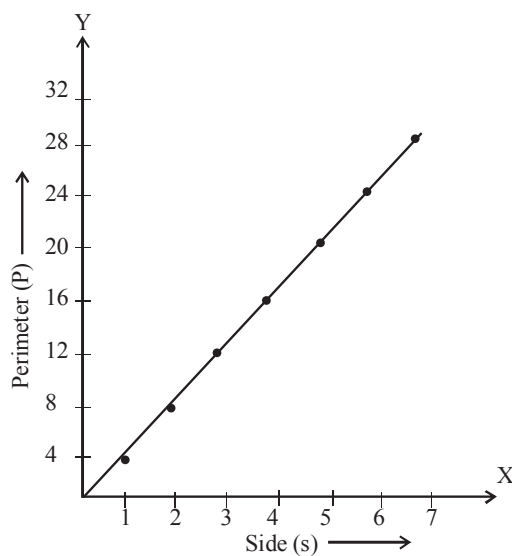
Draw the graph of this relation on the graph paper.

Sol. The values of P for different values of s are given in the following table :

Sides (s) :	1	2	3	4	5	6	7
Perimeter :	$4 \times 1 = 4$	$4 \times 2 = 8$	$4 \times 3 = 12$	$4 \times 4 = 16$	$4 \times 5 = 20$	$4 \times 6 = 24$	$4 \times 7 = 28$
$P = 4s$							

Let us take s on x -axis and P on y -axis. 1 unit on y -axis represents 4 units of perimeter.

Plot the points (ordered pairs (1, 4), (2, 8), (3, 12), (4, 16), (5, 20), (6, 24) and (7, 28)) on the graph paper and join them to get the graph representing the given relation as shown in fig.



1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- The relation between dependent and independent variables is shown through a _____.
- The process of fixing a point with the help of the coordinates is known as _____ of the point.
- We need _____ coordinates for representing a point on the graph sheet.
- The point where the two axes intersect is called the _____.
- For the point (5, 2), the distance from the x - axis is _____ units.
- To draw the graph of a line, the least number of points required is _____.
- (0, -3) lies on _____.
- _____ are used to represent quantitative data.
- In the coordinate of a point, first entry is always _____ coordinate and second entry is always _____ co-ordinate.
- Sum of Abscissa and ordinate of (7, 3) is _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- A line graph can also be a whole unbroken line.
- The points (3, 5) and (5, 3) represent the same point.
- In the point (2, 3), 3 denotes the y-coordinate.
- The y-coordinate of any point lying on the x-axis will be zero.
- The distance of the point (3, 5) from the y-axis is 5.
- Points whose x and y coordinates are equal, lie on a line passing through the origin.
- Data represented in a circular form is called a line graph.
- The graph of a linear equation is always a straight line.
- The process of fixing a point with the help of the coordinates is known as plotting a point in the plane.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column I have to be matched with statements (p, q, r, s) in column II.

- Identify where these points lie.

Column-I

Points

- (2, 3)
- (2, 0)
- (0, 3)
- (0, 0)

Column-II

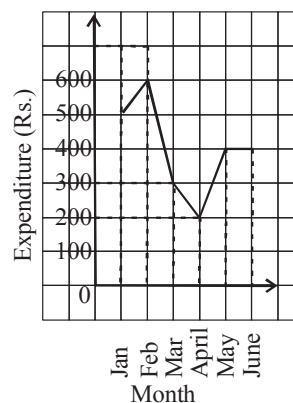
Axis

- y-axis
- 1st quadrant
- origin
- x-axis

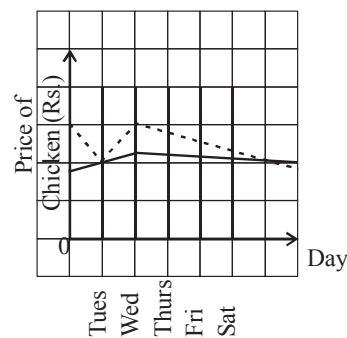
Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- The line graph shows the monthly expenditure of the the Vasu family. Find the difference between their highest and lowest monthly expenditure.



- The line graph shows that price of chickens in towns M and N. On which days are the price of chickens the same in both the towns?

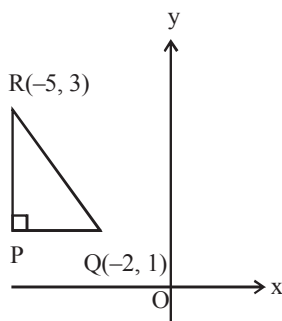


Key: ... Town M
— Town N

Introduction to Graphs

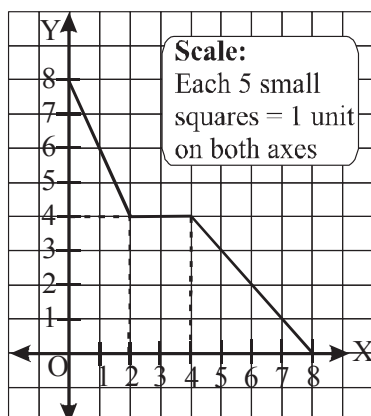
339

3. On the Cartesian plane, PQR is a right-angled triangle.



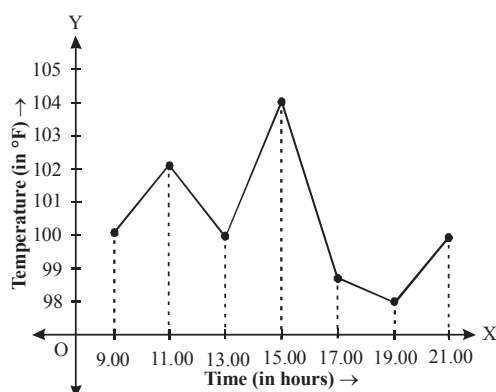
Find the coordinates of P.

4. Observe the adjoining graph and answer the following questions :



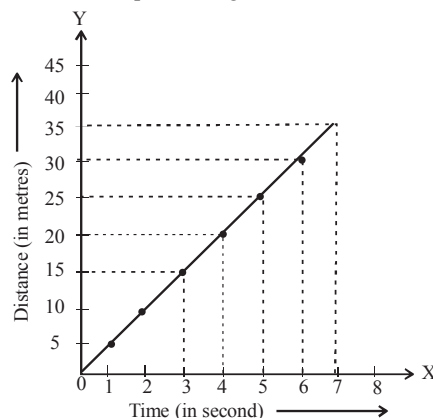
Changes in y when x changes from 0 to 2.

5. What is the information given by the graph shown below?



6. Given below is the distance vs time graph $d = 5t$.

Read the graph carefully where d is distance and t is time and answer the questions given below :



- Find the distance covered when time is 3 sec.
 - Find the distance covered in 5 seconds.
 - Find the time in which the body covers 30m .
7. Plot the following points on a graph paper :
- (3, 2)
 - (2, 2)

Short Answer Questions :

DIRECTIONS : Give answer in two to three sentences.

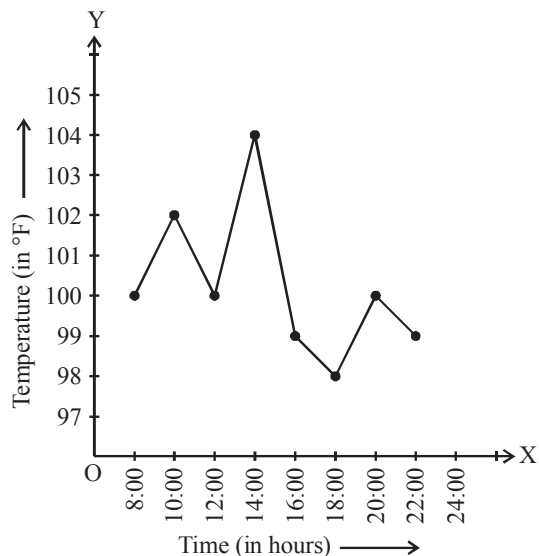
- Write the x and y coordinates for the following points.
 - A (-3, 2)
 - B (2, -1)
 - C (0, -7)
- Plot the following points : (5, 0) (5, 1) (5, 8). Do they lie in a line? What are your observations?
- The maximum temperature on 10 days of June, 2002, in Delhi is given below :

Days	1	2	3	4	5	6	7	8	9	10
Max. Temp. (in °C)	40.5	41	39	40	42	43	44	40	38	39

Draw a graph representing the above data.

- Simple interest on a certain sum is Rs. 40 per year. Then, $S = 40 \times x$, where x is the number of years.
 - Draw a graph of the above function.
 - From the graph find the value of S, when
 - $x = 5$
 - $x = 6$

5. Given below is the temperature chart of a patient.



- Find the temperature of the patient at 12 : 00 hours and 18 : 00 hours.
- At what time is the temperature (i) highest? (ii) lowest?

Long Answer Questions :

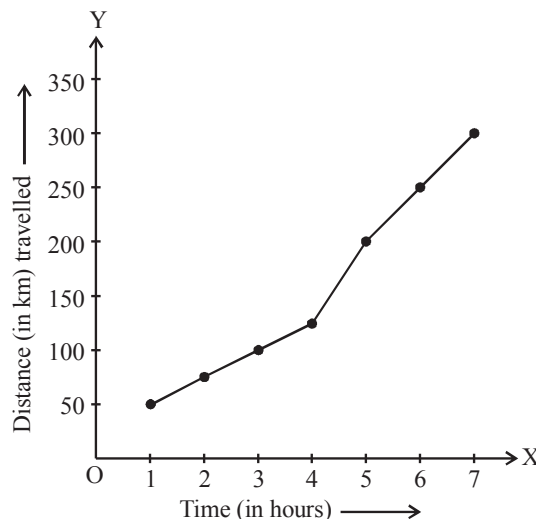
DIRECTIONS : Give answer in four to five sentences.

1. Following is the cost of rice per kg :

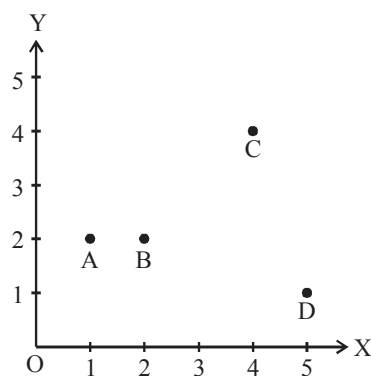
Weight (kg)	10	15	20	25
Cost	500	750	1000	1250

- Plot this information on a graph paper.
 - What will be the cost of 12 kg rice? Find the solution using graph.
 - How much rice can be purchased for Rs. 1500 ? Find the solution using the graph.
2. Plot the following points in a rectangular coordinate plane: (0, 0), (1, 1), (3, 3), (6, 6), (8, 8), (-1, -1), (-4, -4).
- What is the position of all the points? Do they lie in a straight line?
 - What is the relationship between x-coordinates and y-coordinates?

3. Observe the given graph which shows the distance travelled by a car after certain hours and answer the following given question :



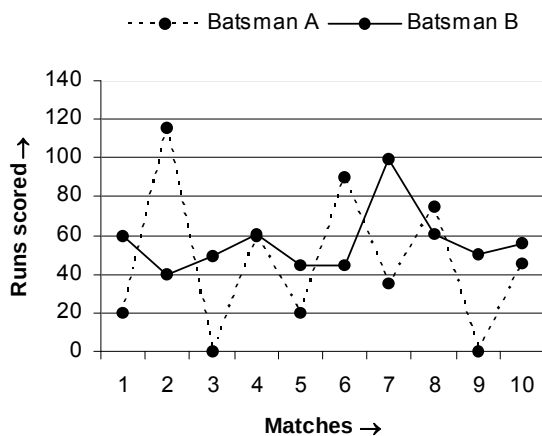
- What is the scale on the vertical and horizontal lines?
 - How much distance is covered after 3 hours?
 - Find the distance covered between 1 hour and 4 hours.
 - Can you tell from the graph, the distance covered by the car between 3 hours and 5 hours?
4. Find the co-ordinates of points A, B, C and D in figure.



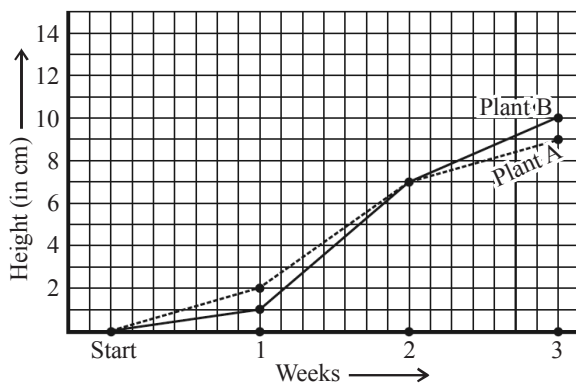
2 EXERCISE

Text-Book Exercise :

1. The given graph (Fig) represents the total runs scored by two batsmen A and B, during each of the ten different matches in the year 2007. Study the graph and answer the following question.



- What information is given on the two axes?
 - Which line shows the runs scored by batsman A?
 - Were the run scored by them same in any match in 2007? If so, in which, match?
 - Among the two batsmen, who is steadier? How do you judge it?
2. For an experiment in Botany, two different plants, plant A and B were grown under similar laboratory conditions. their heights were measured at the end of each week for 3 weeks. the results are shown by the following graph.



- How high was Plant A after (i) 2 weeks (ii) 3 weeks?
- How high was Plant B after (i) 2 weeks (ii) 3 weeks?

- How much did Plant A grow during the 3rd week?
- How much did Plant B grow from the end of the 2nd week to the end of the 3rd week?
- During which week did Plant A grow most?
- During which week did Plant B grow least?
- Were the two plants of the same height during any week shown here? Specify.

3. Use the tables below to draw linear graph.

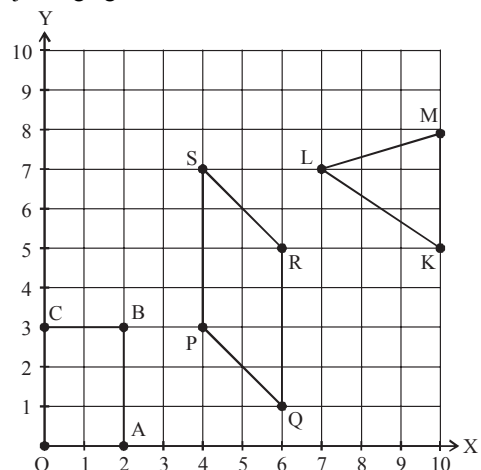
- The number of days a hill side city received snow different years.

Year	2003	2004	2005	2006
Days	8	10	5	12

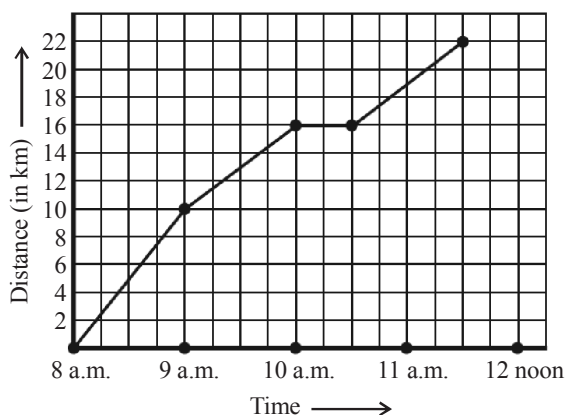
- Population (in thousand) of men and women in a village in different years.

Year	2003	2004	2005	2006	2007
Number of Men	12	12.5	13	13.2	13.5
Number of Women	11.3	11.9	13	13.6	12.8

4. Plot the following points and verify if they lie on a line. If they lie on a line, name it.
- (0, 2), (0, 5), (0, 6), (0, 3.5)
 - A (1, 1), B (1, 2), C (1, 3), D (1, 4)
 - K (1, 3), L (2, 3), M (3, 3), N (4, 3)
 - W (2, 6), X (3, 5), Y (5, 3), Z (6, 2)
5. Draw the line passing through (2, 3) and (3, 2). Find the coordinates of the points at which this line meets the x-axis and y-axis.
6. Write the coordinates of the vertices of each of these adjoining figures.



7. A bank gives 10 % Simple Interest (S.I) on deposits by senior citizens. Draw a graph to illustrate the relation between the sum deposited and simple interest earned. Find from your graph
- The annual interest obtainable for an investment of ₹ 250
 - The investment one has to make to get an annual simple interest of ₹ 70
8. Ajit can ride a scooter constantly at a speed of 30 km/hour. Draw a time – distance graph for this situation. Use it to find
- the time taken by Ajit to ride 75 km.
 - the distance covered by Ajit in $3\frac{1}{2}$ hours.
9. A courier – person cycles from a town to a neighbouring suburban area to deliver a parcel to a merchant. His distance from the town at different times is shown by the following graph.
- What is the scale taken for the time axis?
 - How much time did the person take for the travel?
 - How far is the place of the merchant from the town?
 - Did the person stop on his way? Explain.
 - During which period did he ride fastest?

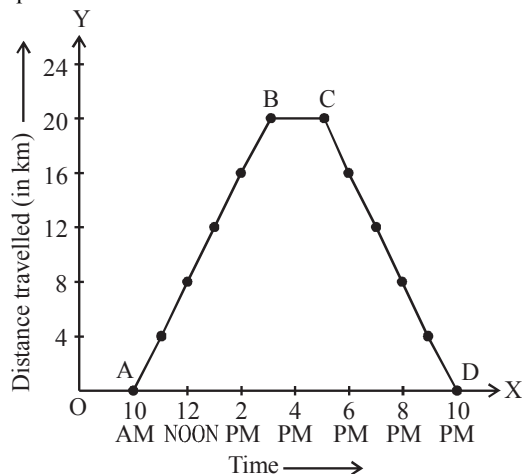


Exemplar Questions :

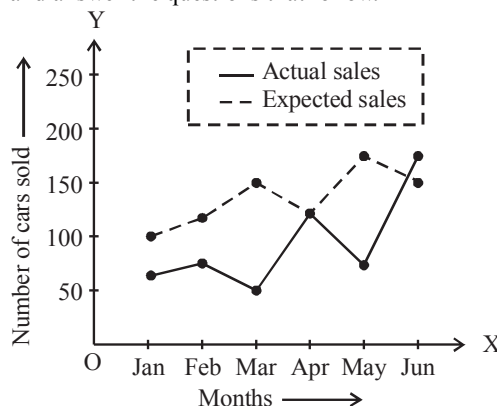
- Plot the given points on a graph sheet and check if the points lie on a straight line. If not, name the shape they form when joined in the given order.
 - (1, 2), (2, 4), (3, 6), (4, 8).
 - (1, 1), (1, 2), (2, 2), (2, 1).
 - (4, 2), (2, 4), (5, 4).
- The cost of a note book is Rs 10. Draw a graph after making a table showing cost of 2, 3, 4, note books. Use it to find
 - the cost of 7 notebooks.
 - The number of note books that can be purchased with Rs. 50.
- Plot a line graph for the variables p and q where p is two times q i.e., the equation is $p = 2q$. Then find

- the value of p when $q = 3$
- the value of q when $p = 8$

4. Study the graph given below of a person who started from his home and returned at the end of the day. Answer the questions that follow.



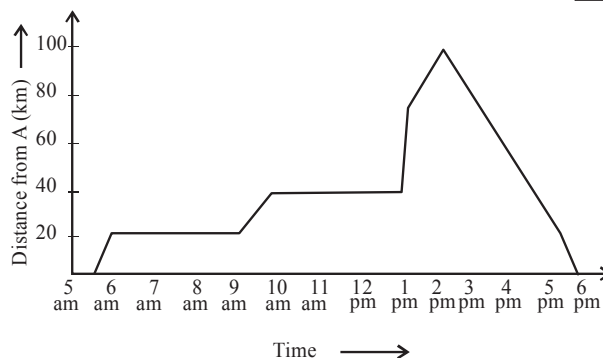
- At what time did the person start from his home?
 - How much distance did he travel in the first four hours of his journey?
 - What was he doing from 3 pm to 5 pm?
 - What was the total distance travelled by him throughout the day?
 - Calculate the distance covered by him in the first 8 hours of his journey.
 - At what time did he cover 16 km of his journey?
 - Calculate the average speed of the man from
 - A to B
 - B to C
 - At what time did he return home?
5. Locate the points A(1, 2), B(3, 4) and C(5, 2) on a graph sheet taking suitable axes. Write the coordinates of the fourth point D to complete the rhombus ABCD. Measure the diagonals of this rhombus and find whether they are equal or not.
6. The graph given below gives the actual and expected sales of cars of a company for 6 months. Study the graph and answer the questions that follow.



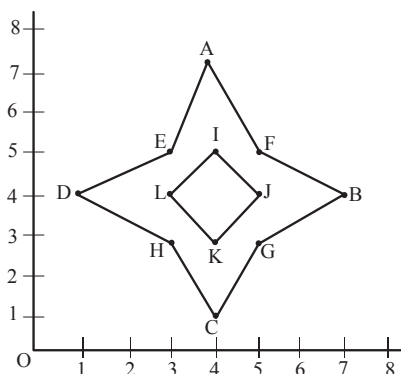
- In which month was the actual sales same as the expected sales?
- For which month(s) was (were) the difference in actual and expected sales the maximum?
- For which month(s) was (were) the difference in actual and expected sales the least?
- What was the total sales of cars in the months – Jan, Feb and March?
- What is the average sales of cars in the last three months?
- Find the ratio of sales in the first three months to the last three months.

7. A man started his journey on his car from location A and came back. The given graph shows his position at different times during back. The given graph shows his position at different times during the whole journey.

- At what time did he start and end his journey?
- What was the total duration of journey?
- Which journey, forward or return, was of longer duration?
- For how many hours did he not move?



8. Write the coordinates of all the points in the given graph.



3

EXERCISE

Single Option Correct :

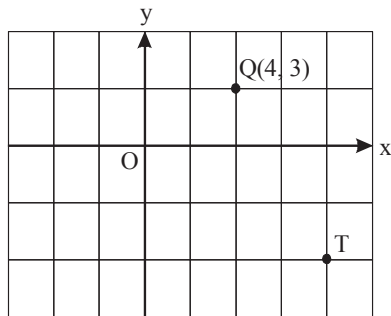
DIRECTIONS : This section contains 30 multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The number of coordinate axes is
 - 1
 - 2
 - 4
 - 3
- The point where both axes meet is called
 - Origin
 - Abcissa
 - Ordinate
 - Quadrant
- The y-axis is a
 - Horizontal line
 - Vertical line
 - Oblique line
 - Vertical segment
- The coordinate of origin is
 - (0, 1)
 - (1, 0)
 - (0, 0)
 - (0, -1)
- The ordinate of the point $(-4, 3)$ is
 - 4
 - 3
 - 0
 - 1
- The abscissa of the point $(-3, -2)$ is
 - 3
 - 2
 - 5
 - 1
- The point $(3, 0)$ lies on
 - x-axis
 - y-axis
 - both axis
 - None of these
- To draw the graph of a line, the least number of points required is
 - One
 - Two
 - Three
 - Four
- Do the points $(2, 3)$ and $(3, 2)$ have the same location on the graph?
 - sometimes
 - No
 - Yes
 - None of these
- If a point is taken on y-axis, x-coordinate of this point is _____.
 - 0
 - 4
 - 3
 - 5
- Which of the points below is a point on the x-axis?
 - $(-5, 0)$
 - $(0, 5)$
 - $(-5, 3)$
 - $(3, -5)$

12. Which of the points below is a point on the y-axis?

- (a) (7,0) (b) (6,-5)
(c) (3,12) (d) (0,11)

13. In the diagram, the coordinates of Q are (4, 3).



What are the coordinates of T?

- (a) (6,-3) (b) (8,-6)
(c) (10,-6) (d) (12,-9)

14. A point both of whose coordinates are negative will lie in _____ quadrants.

- (a) I (b) III
(c) IV (d) II

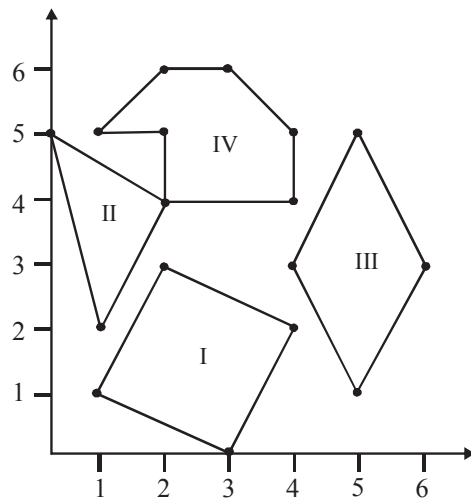
15. Point (0, -7) lies

- (a) on the x-axis (b) in the second quadrant
(c) on the y-axis (d) in the fourth quadrant

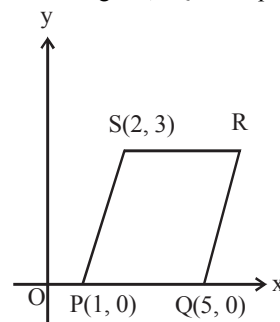
More than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE OR MORE may be correct.

- The x-axis is also known as
 - Abscissa
 - Horizontal axis
 - Ordinate
 - Vertical axis
- The y-axis is also known as
 - Abscissa
 - Horizontal axis
 - Ordinate
 - Vertical axis
- Which of the following statements is true?
 - The x-coordinates of all points to the right of the y-axis are positive
 - The y-coordinates of all points above the x-axis are positive
 - The y-coordinates of all points below the x-axis are positive
 - The x-coordinates of all points to the left of the y-axis are negative
- Which of the following figures is/are not formed by joining the points (1,1), (3,0), (4,2) and (2,3)

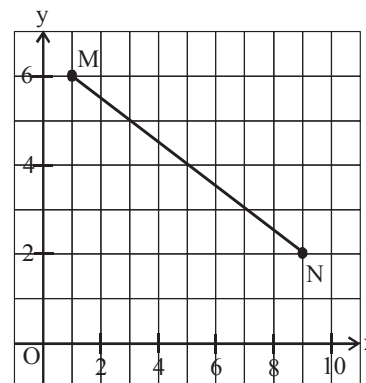


- (a) Figure III (b) Figure I
(c) Figure IV (d) Figure II
5. In the diagram, PQRS is a parallelogram.



The coordinates of R are not equal to

- (a) (5,2) (b) (5,3)
(c) (6,2) (d) (6,3)
6. The distance of the point P(4,3) from the origin is not equal to
- (a) 5 (b) 4
(c) 7 (d) 3
7. The diagram shows a straight line MN.



Which of the following coordinates lie on the line MN?

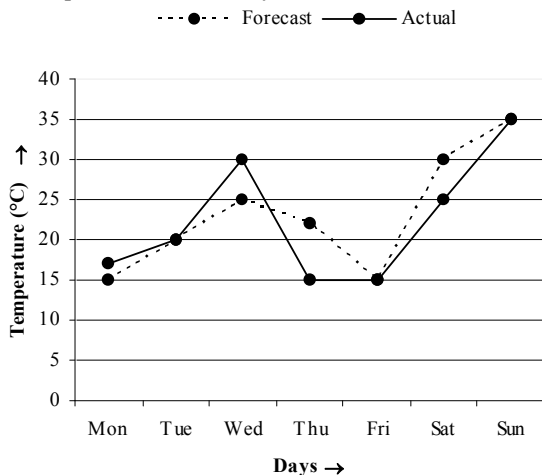
- (a) (1,6) (b) (3,5)
(c) (4,5) (d) (7,3)

Passage Based Question :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE - I

The following graph shows the temperature forecast and the actual temperature for each day of a week.



- On which days was the forecast temperature same as the actual temperature?
 - Tuesday, Friday, Saturday
 - Monday, Friday, Sunday
 - Tuesday, Wednesday, Thursday
 - Tuesday, Friday, Sunday
- What was the maximum forecast temperature during the week?
 - 25°C
 - 30°C
 - 35°C
 - 32.5°C
- On which day did the actual temperature differ the most from the forecast temperature?
 - Wednesday
 - Thursday
 - Saturday
 - Monday

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statement.

- If both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **assertion**.
- If both **Assertion** and **Reason** are true but **Reason** is not the correct explanation of **Assertion**.
- If **Assertion** is true but **Reason** is false.
- If **Assertion** is false but **Reason** is true.

- Assertion :** Coordinate of a point on the y-axis is (0, y)
Reason : Abscissa (x-axis) is 0 on y-axis.
- Assertion :** In a Distance-Time graph, time represents the independent variable.
Reason : Time is represented on y-axis and is dependent on x-axis.
- Assertion :** P(5, 4) is at 5 units from y-axis.
Reason : Abscissa is the distance of a point from y-axis.

Multiple Matching Questions :

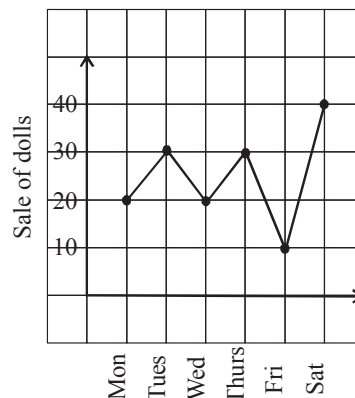
DIRECTIONS : Following question has four statements (A, B, C and D) given in Column-I and four statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Column-I	Column-II
(A) y-axis	(p) (4, 3)
(B) both the axes	(q) (2, 0)
(C) origin	(r) (0, 4)
(D) x-axis	(s) (0, 0)
	(t) (-5, 0)

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- If the co-ordinates of the two points are P(-2, 3) and Q(-3, 5), then. Find (abscissa of P) - (abscissa of Q)
- The line graph shows the sale of dolls by Suhas from Monday to Saturday on a particular week. Given that cost of one doll is ₹ 35, Amount Suhas receive from the sale of dolls on Saturday is 700 k. ₹. Find the value of k.



3. Coordinate of a point A is (1, 2). Coordinate of a point B is (3, 4). Find the product of abscissae of A and ordinate of point B.
4. Coordinate of a point A is (–5, 2). Coordinate of a point B is (8, 0). Find the sum of Abscissae of A and Abscissae of B.
5. Find distance of the point (3, 4) from y-axis

ADVANCED EXERCISE

DIRECTIONS : Give answer in four to five sentences.

- | | |
|---|--|
| Long Answer Questions : | 3. Give the geometric representation of $y = 3$. |
| DIRECTIONS : <i>Give answer in four to five sentences.</i> | 4. Solve each of the following system by graphing :
<div style="display: flex; justify-content: space-around; margin-top: 10px;"> <div style="text-align: center;"> (a) $x - 2y = 2$
 $x + y = 5$ </div> <div style="text-align: center;"> (b) $x + 2y = -4$
 $2x + 4y = 8$ </div> </div> |
| 1. Write an equation of line parallel to y - axis
2. Find the value of 'p', if $x = 2$ and $y = 1$, is a solution of the equation $2x + 3y = p$ | |

SOLUTIONS

Brief Explanations
of
Selected Questions

1 EXERCISE

Fill in the Blanks :

- | | |
|------------|-------------|
| 1. graph | 2. plotting |
| 3. pair of | 4. origin |
| 5. 2 | 6. two |
| 7. y-axis. | 8. graphs |
| 9. $x ; y$ | 10. 10 |

True/False :

- | | |
|----------|----------|
| 1. True | 2. False |
| 3. True | 4. True |
| 5. False | 6. True |
| 7. False | 8. True |
| 9. True | |

Match The Columns :

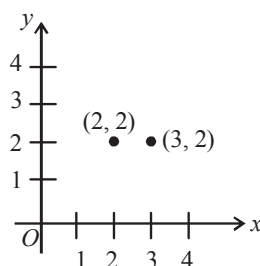
1. (A) $\rightarrow$ (q); (B) $\rightarrow$ (s); (C) $\rightarrow$ (p); (D) $\rightarrow$ (r)
 (A) (2, 3) lies in the 1st quadrant
 (B) (2, 0) lies on x-axis.
 (C) (0, 3) lies on y-axis.
 (D) (0, 0) lies on origin

Very Short Answer Questions :

- Month with highest expenditure = February
 Month with lowest expenditure = April
 Difference in expenditure = $600 - 200 = ₹ 400$
- Price of chickens in towns M and N are same on Tuesday.
- 4 Perpendicular distance of P from x-axis is 1
 Perpendicular distance of P from y-axis is 5
 But P lies in the II quadrant
 $\therefore$ Co-ordinates of P are $(-5, 1)$
- 4, Graph decreases steadily from 8 to 4
- x-axis shows time (in hours) and
 y-axis shows Temperature (in $^{\circ}\text{F}$)
- From the graph, it is clear that :
 (i) When $t = 3$ sec, then $d = 15$ m

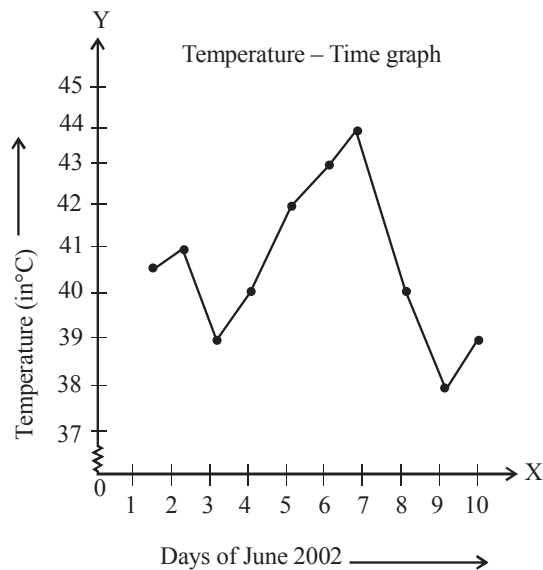
- When $t = 5$ sec, then $d = 25$ m
- When $d = 30$ m, then $t = 6$ sec.

7.



Short Answer Questions :

- (i) A. $x = -3, y = 2$ (ii) B. $x = 2, y = -1$
 (iii) C. $x = 0, y = -7$.
- Yes, they lie in a line parallel to y-axis.
- In order to draw the required temperature-time graph, we represent dates of the month on x-axis and maximum temperature on y-axis. We assume the following scale:
 On X-axis, 1 unit = 1 day and on Y-axis, 1 unit = 1°C .
 We plot the ordered pairs (1, 40.5) (2, 41) ..., (10, 39) as points and join them by line segment to obtain the required graph as shown in the below figure.



4. The given function is $S = 40 \times x$
Putting $x = 1, 2, 3, 4$ successively and getting the corresponding value of S , we get the table given below.

x	1	2	3	4
$S = 40 \times x$	40	80	120	160

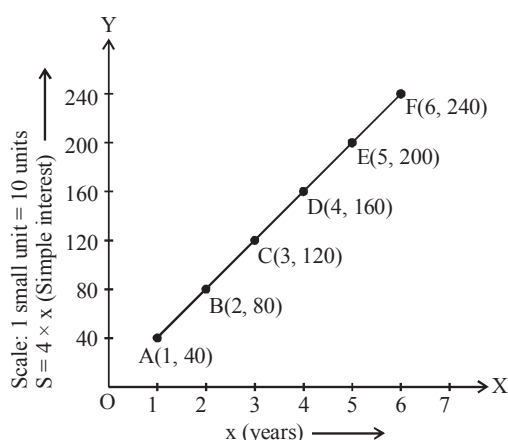
Along the x-axis : Take 1 unit = 1 year.

Along the y-axis : Take 1 unit = ₹ 10.

Now, on a graph paper, plot the points

A(1, 40), B(2, 80), C(3, 120) and D(4, 160)

Join them successively to get the required line graph, shown below.



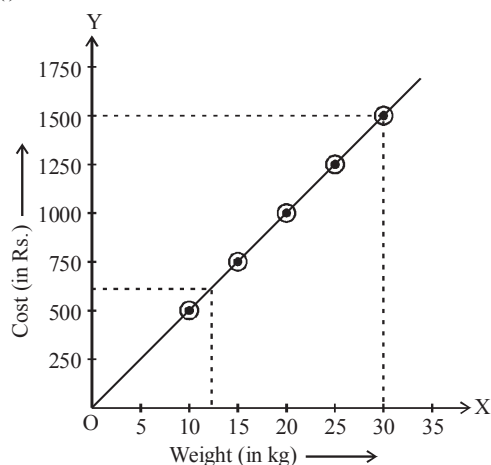
From the graph, we see (i) $x = 5$ then $S = 200$

(ii) $x = 6$ then $S = 240$

5. (i) The temperature of the patient at 12:00 hours is 100°F and at 18:00 hours the temperature is 98°F .
(ii) The maximum temperature of the patient is 104°F at 14:00 hours and the minimum temperature of the patient is 98°F at 18:00 hours.

Long Answer Questions :

1. (i)



(ii) ₹ 600

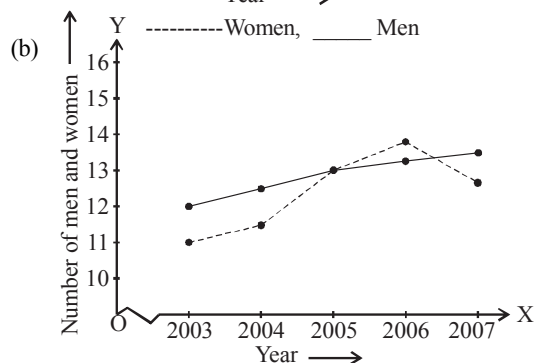
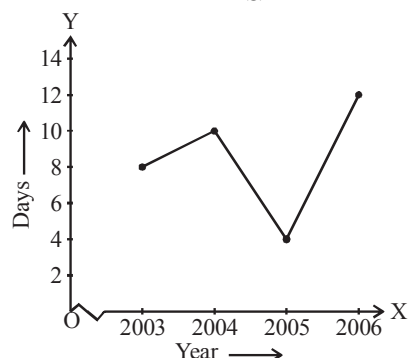
(iii) 30 kg.

2. (i) The points are at same position from both the ordinate axes. Yes, they are in straight line.
(ii) Relationship between x and y :
 $x = y$, i.e., $x - y = 0$.
3. (i) On vertical scale : 1 cm = 50 km,
on horizontal scale : 1 cm = 1 hr.
(ii) 100 km
(iii) $(125 - 50) = 75$ km
(iv) 100 km.
4. A(1, 2), B(2, 2), C(4, 4), D(5, 1).

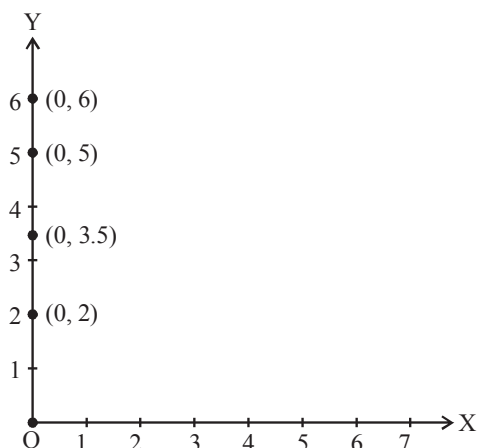
2 EXERCISE

Text-Book Exercise :

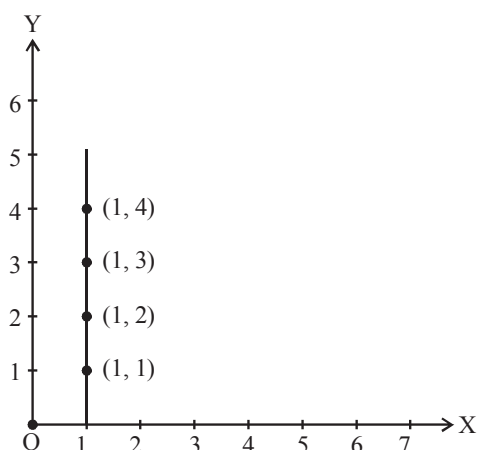
1. (i) The horizontal axis (or the x -axis) indicates the matches during the year 2007. The vertical axis (or the y -axis) shows the total runs scored in each match.
(ii) The dotted line shows the runs scored by Batsman A.
(iii) During 4th match they have scored the same 60 runs. (This is indicated by the point at which both graph meet).
(iv) Batsman A has great "peak" but many deep "valleys". He does not appear to be consistent, on the other hand B has never scored below a total of 40 runs, even though his highest score is only 100 in comparison to 115 of A. Also A has scored a zero in two matches and in a total of 5 matches he has scored less than 40 runs. Since A has a lot ups and down, B is a more constant and reliable batsman.
2. (a) (i) 7 cm (ii) 9 cm
(b) (i) 7 cm (ii) 10 cm
(c) 2 cm (d) 3 cm
(e) Second week (g) At the end of the 2nd week
3. (a)



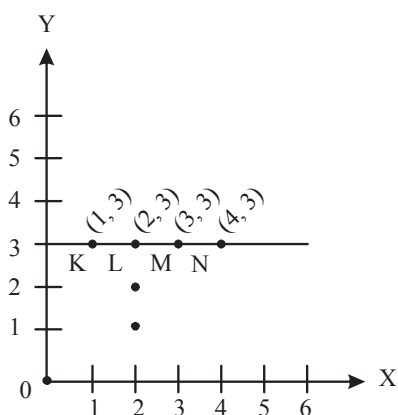
4. (i) These lie on a line. The line is y-axis.



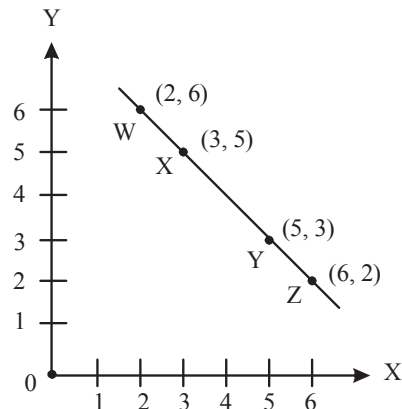
- (ii) These lie on a line. The line is AD. (You may also use other ways of naming it). It is parallel to the y-axis.



- (iii) These lie on a line. We can name it as KL or KM or MN etc. It is parallel to x-axis.

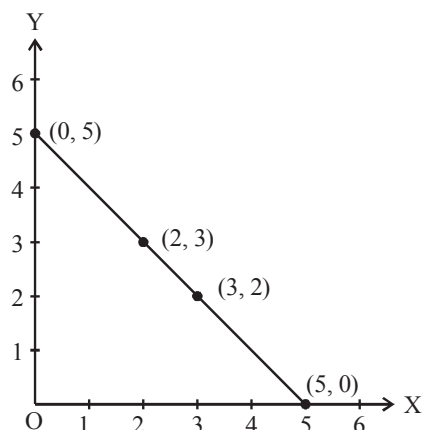


- (iv) These lie on a line. We can name it as XY or WY or YZ etc.



Note that in each of the above cases, graph obtained by joining the plotted points by a line. Such graphs are called linear graphs.

5. The line will cut x-axis at (5, 0) and y-axis at (0, 5)



6. O(0, 0), A(2, 0), B(2, 3), C(0, 3), P(4, 3), Q(6, 1), R(6, 5), S(4, 7), K(10, 5), L(7, 7), M(10, 8)

7.

Sum deposited	Simple interest for a year
₹ 100	$\text{₹ } \frac{100 \times 1 \times 10}{100} = \text{₹ } 10$
₹ 200	$\text{₹ } \frac{200 \times 1 \times 10}{100} = \text{₹ } 20$
₹ 300	$\text{₹ } \frac{300 \times 1 \times 10}{100} = \text{₹ } 30$
₹ 500	$\text{₹ } \frac{500 \times 1 \times 10}{100} = \text{₹ } 50$
₹ 1000	₹ 100

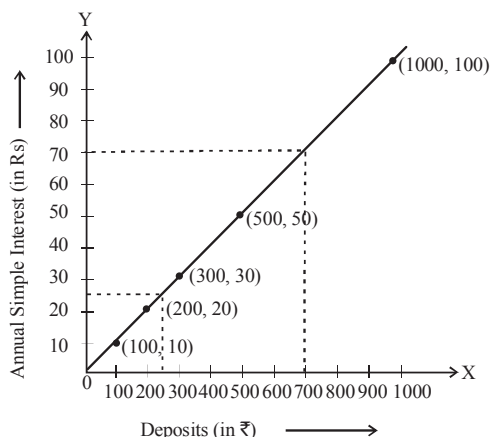
We get a table of values.

Deposit (in ₹)	100	200	300	500	1000
Annual S.I. (in ₹)	10	20	30	50	100

Scale : 1 unit = ₹ 100 on horizontal axis; 1 unit = ₹ 10 on vertical axis.

Plot the points : (100, 10), (200, 20), (300, 30), (500, 50) etc.

- (a) Corresponding to ₹ 250 on horizontal axis, we get the interest to be ₹ 25 on vertical axis.
 (b) Corresponding to ₹ 70 on the vertical axis, we get the sum to be ₹ 700 on the horizontal axis.



8.

Hours of ride	Distance covered
1 hour	30 km
2 hours	$2 \times 30 \text{ km} = 60 \text{ km}$
3 hours	$3 \times 30 \text{ km} = 90 \text{ km}$
4 hours	$4 \times 30 \text{ km} = 120 \text{ km}$ and so on.

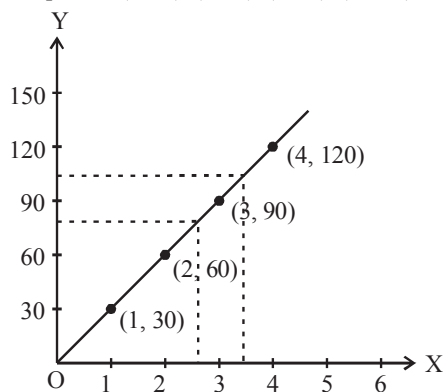
We get a table of values.

Time (in hours)	1	2	3	4
Distance covered (in km)	30	60	90	120

Scale : Horizontal : 1 unit = 1 hour

Vertical : 1 unit = 10 km

Plot the points : (1, 30), (2, 60), (3, 90), (4, 120).



Join the points. We get a linear graph.

- (a) Corresponding to 75 km on the vertical axis, we get the time to be 2.5 hours on the horizontal axis. Thus 2.5 hours are needed to cover 75 km.
 (b) Corresponding to $3\frac{1}{2}$ hours on the horizontal axis, the distance covered is 105 km on the vertical axis.
 9. (a) 4 units = 1 hour
 (b) $3\frac{1}{2}$ hours
 (c) 22 km
 (d) Yes; This is indicated by the horizontal part of the graph (10 a.m. - 10.30 a.m.)
 (e) Between 8 a.m. and 9 a.m.

Exemplar Questions :

- (a) Yes, they lie on a straight line
 (b) No, square
 (c) No, triangle
- (a) Rs. 70 (b) 5
- (a) $p = 6$ (b) $q = 4$
- (a) 10 a.m. (b) 16 km
 (c) not travelling (d) 40 km
 (e) 24 km (f) 2 p.m.
 (g) 4 km/h, 0 km/h (h) 10 p.m.
- D (3, 0) Yes
- (a) April (b) March
 (c) April (d) $75 + 100 + 75 = 250$
 (a) $\frac{125 + 100 + 150}{3} = 125$ (b) $250 : 375 = 2 : 3$
- (a) 5.30 a.m. and ends at 6 p.m.
 (b) 12 hours 30 min.
 (c) forward journey was of longer duration.
 (d) 6 hours
- (A) (4, 7) (B) (7, 4)
 (C) (4, 1) (D) (1, 4)
 (E) (3, 5) (F) (5, 5)
 (G) (5, 3) (H) (3, 3)
 (I) (4, 5) (J) (5, 4)
 (K) (4, 3) (L) (3, 4)

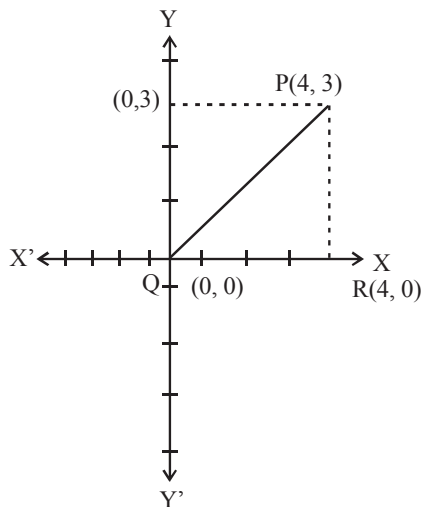
3 EXERCISE

Single Option Correct :

- (b)
- (a)
- (b)
- (c)
- (b)
- (a)
- (a)
- (b) A line is formed by joining minimum two points.
- (b) Location of both points are different.
- (a) The x-coordinate of a point lies on y-axis is 0 for example (0, 3) lies on y-axis.
- (a) y co-ordinate of a point on x-axis is zero.
- (a) x-co-ordinate of a point on y-axis is zero.
- (b) Co-ordinates of T are (8, -6)
- (b) Abscissa and ordinate of a point are negative in III quadrants.
- (c) Point (0, -7) lies on the y-axis.

More than One Option Correct :

- (a, b)
- (c, d)
- (a, b, d) y-coordinates of all the points below the x-axis are negative.
- (a, c, d) Figure II, III, IV are not formed by the given co-ordinates.
- (a, b, c) Opposite sides of parallelogram are equal
 Length of PQ = (abscissa of Q) - (Abcissa of P)
 $= 5 - 1 = 4$
 $\therefore$ Length of SR = 4
 $\Rightarrow$ abscissa of R is 6
 Ordinate of R from X-axis = 3
 $\therefore$ Coordinates of R are (6, 3)
- (b, c, d)



$$QP = \sqrt{QR^2 + PR^2} = \sqrt{(4)^2 + (3)^2}$$

$$= \sqrt{16+9} = \sqrt{25} = 5$$

- (a, b, d) (3, 5), (1, 6), (7, 3)

Passage Based Questions :

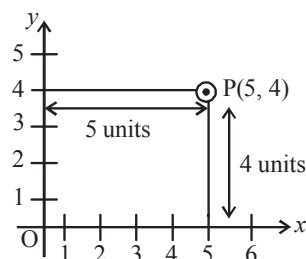
PASSAGE

Days	Forecast Temp.	Actual Temp.	Difference
Mon.	15	17.5	2.5
Tues.	20	20	0
Wed.	25	30	5
Thurs.	15	22.5	7.5
Fri.	15	15	0
Sat.	25	30	5
Sun.	35	35	0

- (d) On Tuesday, Friday and Sunday, the forecast and actual temperature are same.
- (c) Maximum forecast temperature is 35°C on Sunday.
- (b) On Thursday, the difference between the actual and forecast temperature is maximum.

Assertion & Reason :

- (a) On y-axis the abscissa of every point is 0.
 $\therefore$ Coordinate of all points on y-axis are of the form (0, y)
 $\therefore$ Assertion and reason both are true and reason correctly explains assertion
- (c) In distance time graph, time is independent variable and distance depends on time.
 $\therefore$ Time is plotted on x-axis, distance on y-axis.
 Assertion is true, reason is false.
- (a)



Abscissa is the distance of a point from y-axis

Multiple Matching :

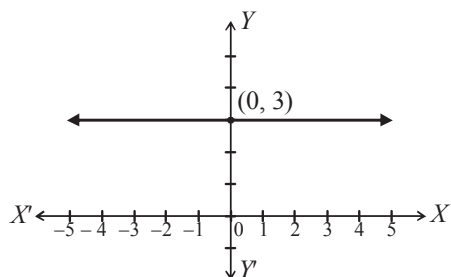
- (A) $\rightarrow$ (r), (s); (B) $\rightarrow$ (s); (C) $\rightarrow$ (s); (D) $\rightarrow$ (q), (s), (t)

Integer Type Question :

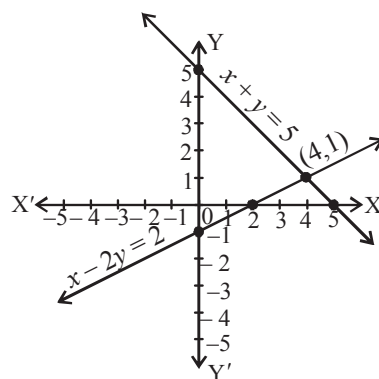
- (abscissa of P) – (abscissa of Q)
 $= (-2) - (-3) = 1$
- Amount received = ₹ 1400 = $700 \times 2 \Rightarrow k = 2$
- Abcissae of A is 1
 Ordinate of B is 4.
 So, $1 \times 4 = 4$.
- Abcissae of A + Abcissae of B = $-5 + 8 = 3$.
- The distance of point (3, 4) from y-axis = 3.

4 **ADVANCED EXERCISE**
 BASED ON CONNECTING TOPICS

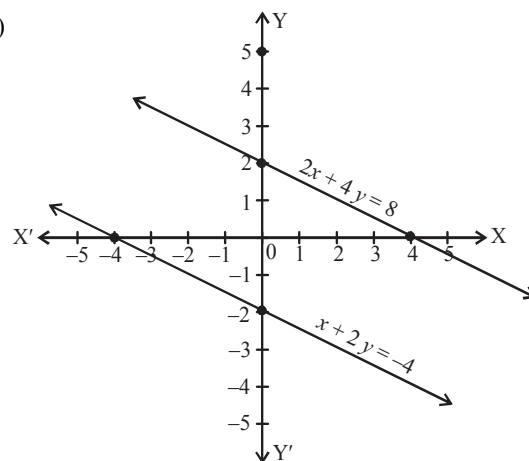
- $x = 3$ is the line parallel to y-axis.
- $p = 7$
-



4. (a)



(b)



Chapter 16

Playing with Numbers

INTRODUCTION

In earlier classes, you have studied various types of numbers such as natural numbers, whole numbers, integers and rational numbers and their properties. In this chapter, we will explore numbers in more detail and learn about making and solving the mathematical puzzles. We shall also deduce the divisibility test rules of 2, 3, 4, 5, 6, 8, 9, 10 and 11 for two-digit and three-digit numbers expressed in general form.

NUMBERS IN GENERAL FORM

General form of any two digit number (ab) is $10 \times a + b$, Where a is tens place digit and b is units place digit.

Consider $58 = 50 + 8 = 10 \times 5 + 8$

General form of any three digits (abc) number is $100 \times a + 10 \times b + c$,

Where a is hundreds place digit, b is tens place digit and c is units place digit.

Let us expand a 3-digit number say, 541

$$541 = 100 \times 5 + 10 \times 4 + 1$$

NUMBER GAMES

Using the general form of two digit and three digit numbers, we can formulate certain mathematical tricks. Look at the few number games given below:

Trick 1

- (i) Think of any two digit number.
- (ii) Add the reverse of the two digit number to the original number.
- (iii) The result is divisible by 11.

ILLUSTRATION : 1

- (i) Let the number be 34
- (ii) Reversed number + original number = $34 + 43 = 77$
- (iii) 77 is divisible by 11

SOLUTION:

- (i) Let the number be $10a + b$
- (ii) Reversed number + original number = $(10b + a) + (10a + b)$
 $= 11a + 11b = 11(a + b)$
- (iii) Clearly, $11(a + b)$ is divisible by 11.

Trick 2

- (i) Think of any three digit number.
- (ii) Subtract the reversed number from the original number.
- (iii) The result is divisible by 99.

ILLUSTRATION : 2

- (i) Let the number be 362
- (ii) Original number – reversed number = $362 - 263 = 99$
- (iii) 99 is divisible by 99

SOLUTION:

- (i) Let the number be $100a + 10b + c$
- (ii) Original number – reversed number
 $= (100a + 10b + c) - (100c + 10b + a)$
 $= 99a - 99c$
 $= 99(a - c)$
- (iii) Clearly, $99(a - c)$ is divisible by 99

CRYPTARITHMS

In this section we will see mathematical puzzles in which the digits are replaced by letters in an arithmetic calculation. The objective of the puzzle is to break the code used.

Cryptarithm Rules

- Each letter represents a unique digit.
- The first digit of a number cannot be zero. For example, twenty five can be only written as 25 not 025.

NOTE: Cryptarithms may have several solutions.

ILLUSTRATION : 3

Find the value of A and B if

$$\begin{array}{r} 41A \\ + B4 \\ \hline 512 \end{array}$$

SOLUTION:

Form ones column $A + 4$ gives a number whose ones digit is 2. So, $A = 8$. The value of B can be obtained by solving $2 + B$ is a number whose ones digit is 1. So, $B = 9$

$$\begin{array}{r} 418 \\ + 94 \\ \hline 512 \end{array}$$

ILLUSTRATION : 4

Solve the Cryptarithm :

$$\begin{array}{r} B A \\ \times B 3 \\ \hline 57A \end{array}$$

SOLUTION:

Here, we have to find the values of A and B.

Since ones digit of $3 \times A$ is A. Therefore, $A = 0$ or $A = 5$.

Now, $\overline{BA} \times \overline{B3} = \overline{57A} \quad \dots(i)$

$\Rightarrow \overline{BA} \times \overline{B3}$ is a three digit number between 500 and 600.

If $B = 1$, then $\overline{BA} \times \overline{B3}$ can have maximum value $19 \times 13 = 247$. Therefore, $B \neq 1$. If $B = 3$, then $\overline{BA} \times \overline{B3}$ can have minimum value $30 \times 33 = 990$. Therefore, $B \neq 3$. Thus, we have $B = 2$.

Putting $B = 2$ in (i), we get

$$\begin{aligned} \overline{2A} \times \overline{23} &= \overline{57A} \\ \Rightarrow (20 + A) \times 23 &= 500 + 70 + A \\ \Rightarrow 460 + 23A &= 570 + A \\ \Rightarrow 22A &= 110 \\ \Rightarrow A &= 5 \end{aligned}$$

Hence, $A = 5$ and $B = 2$ and, $\begin{array}{r} 25 \\ \times 23 \\ \hline 575 \end{array}$

TESTS OF DIVISIBILITY

In this section we will learn how to check the divisibility by numbers 2, 3, 4, 5, 6, 8, 9, 10 and 11. Let us learn how these work.

DIVISIBILITY BY 2

A given number is divisible by 2, if its unit digit is any of 0, 2, 4, 6, 8 e.g. 4268134 is divisible by 2, while 31267429 is not.

Explanation of the rule :

Consider a three digit number, $cba = 100c + 10b + a$

Also, $100c + 10b + a = 2(50c + 5b) + a$

Clearly, $2(50c + 5b)$ is divisible by 2

Thus, for a number to be divisible by 2, a has to be either 0, 2, 4, 6 or 8.

DIVISIBILITY BY 3

A number is divisible by 3, if sum of its digits is divisible by 3 *e.g.* 252771 is divisible by 3 as sum of its digits $(2 + 5 + 2 + 7 + 7 + 1 = 24)$ is divisible by 3.

Explanation of the rule :

Consider a three digit number, $cba = 100c + 10b + a$

$$\begin{aligned}\text{Also, } 100c + 10b + a &= (99 + 1)c + (9 + 1)b + a \\ &= 99c + c + 9b + b + a \\ &= 3(33c + 3b) + a + b + c\end{aligned}$$

Clearly, $3(33c + 3b)$ is divisible by 3

Thus, for a number to be divisible by 3, $(a + b + c)$ should be a factor of 3.

DIVISIBILITY BY 4

A number is divisible by 4 if the last two digits of the number is divisible by 4 or the number ends with '00'. *e.g.* 213428 is divisible by 4 as last two digits is 28 which is divisible by 4. Again, 1246800 is also divisible by 4 as the number ends with 00.

DIVISIBILITY BY 5

A number is divisible by 5 if its unit place digit is either 0 or 5. *e.g.* 123465 and 1243950 are both divisible by 5 while 214672 is not divisible by 5.

Explanation of the rule :

Consider a three digit number, $cba = 100c + 10b + a$

$$\text{Also, } 100c + 10b + a = 5(20c + 2b) + a$$

Clearly, $5(20c + 2b)$ is divisible by 5

Thus, for a number to be divisible by 5, a has to be either 0 or 5.

DIVISIBILITY BY 6 AND 8

A number is divisible by 6 if it is divisible by 2 and 3 both. *e.g.*, 254784 is divisible by 6 because it is an even number and hence divisible by 2, also the sum of digits i.e., $2 + 5 + 4 + 7 + 8 + 4 = 30$ is divisible by 3. Hence the number is divisible by 6. 214351 is not divisible by 6 because it is not divisible by 2.

A given number is divisible by 8 if the number formed by last three digits of the number is divisible by 8 or the number ends with '000' *e.g.*, 342840 is divisible by 8 because 840 is divisible by 8. 29342000 is also divisible by 8.

DIVISIBILITY BY 9

A number is divisible by 9 if the sum of digits of the number is divisible by 9. *e.g.* 284796 is divisible by 9 because sum of digits $2 + 8 + 4 + 7 + 9 + 6 = 36$ is completely divisible by 9.

Explanation of the rule :

Consider a three digit number, $cba = 100c + 10b + a$

$$\begin{aligned}\text{Also, } 100c + 10b + a &= (99 + 1)c + (9 + 1)b + a \\ &= 99c + c + 9b + b + a \\ &= 9(11c + b) + a + b + c\end{aligned}$$

Clearly, $9(11c + b)$ is divisible by 9

Thus, for a number to be divisible by 9, $(a + b + c)$ should be a factor of 9.

DIVISIBILITY BY 10

A number is divisible by 10 if the unit place digit of given number is '0' *e.g.*, 21380, 3142900 are divisible by 10, whereas 214385, 329212, 46843 are not divisible by 10.

Explanation of the rule :

Consider a three digit number, $cba = 100c + 10b + a$

where a is the ones digit, b is the tens digit, c is the hundreds digit.

Since $10b$, $100c$, are multiples of 10, they are divisible by 10,

Hence, a number is divisible by 10, if its ones digit is 0.

DIVISIBILITY BY 11

A number is divisible by 11 if the difference between the sum of the digits in the odd places and the sum of the digits in the even places is either 0 or completely divisible by 11. *e.g.*, 6584919 is divisible by 11 because. (Sum of digits at odd places) – (Sum of digits at even places)

$$\Rightarrow (6 + 8 + 9 + 9) - (5 + 4 + 1)$$

$$\Rightarrow 32 - 10 = 22, \text{ which is divisible by 11.}$$

ILLUSTRATION : 5

Which of the following numbers are divisible by 3?

(a) 21436925

(b) 5875203

SOLUTION:

(a) Sum of digits of 21436925 = $(2 + 1 + 4 + 3 + 6 + 9 + 2 + 5) = 32$, which is not divisible by 3, hence 21436925 is not divisible by 3.

(b) Sum of digits of 5875203 = $(5 + 8 + 7 + 5 + 2 + 0 + 3) = 30$

30 is divisible by 3, hence 5875203 is divisible by 3.

ILLUSTRATION : 6

Is 60643824 divisible by 24?

SOLUTION:

A number is divisible by 24 if it is divisible by both 3 and 8.

Digits sum of 60643824 = $(6 + 0 + 6 + 4 + 3 + 8 + 2 + 4) = 33$

33 is divisible by 3, hence 60643824 is divisible by 3.

Number formed by last 3 digits of 60643824 i.e., 824 is divisible by 8 and hence 60643824 is divisible by 8.

$\therefore$ 60643824 is divisible by both 3 and 8. Hence it is divisible by 24.

MISCELLANEOUS SOLVED EXAMPLES

1. If $24x$ is a multiple of 3, where x is a digit, what is the value of x ?

Sol: It is given that $24x$ is a multiple of 3.

$\therefore 2 + 4 + x$ is a multiple of 3

$\Rightarrow 6 + x$ is a multiple of 3.

$\Rightarrow 6 + x = 0, 3, 6, 9, 12, 15, 18, 21, \dots \dots (i)$

But, x is a digit of the number $24x$.

$\therefore x$ can take values 0, 1, 2, 3, ..., 9

$\Rightarrow 6 + x$ can take values 6, 7, 8, 9, 10, ..., 15 $\dots (ii)$

From (i) and (ii), we get

$6 + x = 6$ or, 9 or, 12 or, 15

$\Rightarrow 6 + x = 6$ or, $6 + x = 9$ or, $6 + x = 12$ or, $6 + x = 15$

$\Rightarrow x = 0$ or, $x = 3$ or, $x = 6$ or, $x = 9$

Hence, $x = 0, 3, 6, 9$.

1 A B

2. If $\begin{array}{r} +CCA \\ 697 \end{array}$ and there is no carry on addition, then the value of B is ____.

Sol. The given conditions are as follows:

$$B + A = 7 \quad \dots (i)$$

$$A + C = 9 \quad \dots (ii)$$

$$1 + C = 6 \Rightarrow C = 5 \quad \dots (iii)$$

Putting the value of C from (iii) to (ii), we get

$$A + 5 = 9 \Rightarrow A = 4$$

Putting $A = 4$ in (i), we get

$$B + 4 = 7 \Rightarrow B = 3$$

3. Solve the following cryptarithm

$$\begin{array}{r} A \quad B \\ \times \quad 6 \\ \hline B \quad B \quad B \end{array}$$

Sol. We have,

$$\begin{array}{r} A \quad B \\ \times \quad 6 \\ \hline B \quad B \quad B \end{array}$$

This means that $6 \times B$ is a number having its ones digit as B. Such values of B are 2, 4, 6 and 8, because $6 \times 2 = 12$, $6 \times 4 = 24$, $6 \times 6 = 36$ and $6 \times 8 = 48$. So, we have following cases:

Case I When $B = 2$

In the case, we have

$$\overline{AB} \times 6 = \overline{BBB}$$

$$\begin{aligned}\Rightarrow \overline{A2} \times 6 &= \overline{222} & \Rightarrow (10A + 2) \times 6 &= 222 \\ \Rightarrow 60A + 12 &= 222 & \Rightarrow 60A &= 210 \\ \Rightarrow 2A &= 7 & \Rightarrow A &= \frac{7}{2}, \text{ which is not possible.}\end{aligned}$$

Case II When $B = 4$

In this case, we have

$$\begin{aligned}\overline{AB} \times 6 &= \overline{BBB} \\ \Rightarrow \overline{A4} \times 6 &= \overline{444} & \Rightarrow (10A + 4) \times 6 &= 444 \\ \Rightarrow 10A + 4 &= \frac{444}{6} & \Rightarrow 10A + 4 &= 74 \\ \Rightarrow 10A &= 70 & \Rightarrow A &= 7\end{aligned}$$

$A = 7$ and $B = 4$ is the required solution.

Case III When $B = 6$

In this case, we have

$$\begin{aligned}\overline{AB} \times 6 &= \overline{BBB} \\ \Rightarrow \overline{A8} \times 6 &= \overline{888} & \Rightarrow \overline{A8} &= 148\end{aligned}$$

This is not possible as LHS is a two digit number and RHS is a three digit number.

4. Write the following numbers in expanded form and then do the same for the number you get by reversing the digits

- (i) 71 (ii) 68 (iii) 432 (iv) 667

Sol. (i) $\overline{ab} = 71 = 7 \times 10 + 1 \times 1$

$$\overline{ba} = 17 = 1 \times 10 + 7 \times 1$$

(ii) $\overline{ab} = 68 = 6 \times 10 + 8 \times 1$

$$\overline{ba} = 86 = 8 \times 10 + 6 \times 1$$

(iii) $\overline{abc} = 432 = 4 \times 100 + 3 \times 10 + 2 \times 1$

$$\overline{cba} = 234 = 2 \times 100 + 3 \times 10 + 4 \times 1$$

(iv) $\overline{abc} = 667 = 6 \times 100 + 6 \times 10 + 7 \times 1$

$$\overline{cba} = 766 = 7 \times 100 + 6 \times 10 + 6 \times 1$$

5. Without performing actual division, find the remainder left when 192837465 is divided by

- (i) 9 (ii) 11

Sol. (i) Any natural number n , can be written as $n = \text{a multiple of } 9 + \text{Sum of the digits of } n$

$$\therefore 192837465 = \text{a multiple of } 9 + (1 + 9 + 2 + 8 + 3 + 7 + 4 + 6 + 5)$$

$$\Rightarrow 192837465 = \text{a multiple of } 9 + 45$$

$$\Rightarrow 192837465 = \text{a multiple of } 9 + 9 \times 5$$

$$\Rightarrow 192837465 = \text{a multiple of } 9$$

So, The remainder left when 192837465 is divided by 9 is zero.

- (ii) Any odd digit natural number n can be written as
 $n = \text{a multiple of } 11 + \text{Sum of its digits in odd place} - \text{Sum of its digits in even places}$
 $\Rightarrow 192837465 = \text{a multiple of } 11 + (1 + 2 + 3 + 4 + 5) - (9 + 8 + 7 + 6)$
 $\Rightarrow 192837465 = \text{a multiple of } 11 + 15 - 30$
 $\Rightarrow 192837465 = \text{a multiple of } 11 - 15$
 $\Rightarrow 192837465 = \text{a multiple of } 11 - 2 \times 11 + 7$
 $\Rightarrow 192837465 = (\text{a multiple of } 11) + 7$

So, the remainder left when 192837465 is divided by 11 is 7.

6. Put the value of star so that the number is

- (i) divisible by 3

9 2 5 * 6

- (ii) divisible by 3

8 2 4 * 6 7

Sol. (i) $* = 5$ (ii) $* = 3$

(Answer may vary)

7. If $13p4$ is a multiple of 6 where p is a digit, what might be the value of p ?

Sol. $p = 4$

1 EXERCISE

Fill in the Blanks :

DIRECTIONS : Complete the following statements with an appropriate word / term to be filled in the blank space(s).

- 3845 is divisible by _____.
 - 78024 is divisible by 24 since it is divisible by 3 and _____.
 - The digit which can replace * in the number $29 * 2$ is _____, so that it is divisible by 9.
 - 3134673 is divisible by 3 and _____.
- $$\begin{array}{r} 2 \quad B \\ + A \quad B \\ \hline 8 \quad A \end{array}$$
- If $\begin{array}{r} 2 \quad B \\ + A \quad B \\ \hline 8 \quad A \end{array}$ then $A = \underline{\hspace{2cm}}$ and $B = \underline{\hspace{2cm}}$.
 - 1×35 is divisible by 9 if $x = \underline{\hspace{2cm}}$.
 - A number is divisible by 11 if the differences between the sum of digits at its odd places and that of digits at the even places is either 0 or divisible by _____.
 - A 2-digit number is divisible by 11, if its two digits are _____.
 - The example of a number which is divisible by 2 but not by 4 is _____.

True / False :

DIRECTIONS : Read the following statements and write your answer as true or false.

- If a number is divisible by 8, it must be divisible by 4.
- The number formed by writing a non-zero digit six times (e.g., 888888) is always divisible by 11.
- If a number is divisible by 3 and 6, it must be divisible by 18.
- If a number is divisible by 15, it must be divisible by 5 and 3.
- A two-digit number ab is always divisible by 2 if b is an even number.
- If $213x27$ is divisible by 9, then the value of x is 0.
- If $N \div 5$ leaves remainder 3 and $N \div 2$ leaves remainder 0, then $N \div 10$ leaves remainder 4.
- If a number is prime, it is not even.

Match the Columns :

DIRECTIONS : Each question contains statements given in two columns which have to be matched. Statements (A, B, C, D) in column-I have to be matched with statements (p, q, r, s) in column-II.

- Match the following cryptarithms:

Column-I

$$\begin{array}{r} (A) \quad 3 \quad 1 \quad A \\ + \quad 1 \quad A \quad 3 \\ \hline \quad 5 \quad 0 \quad 1 \end{array}$$

$$\begin{array}{r} (B) \quad B \quad 9 \\ + \quad 4 \quad A \\ \hline \quad A \quad 5 \end{array}$$

$$\begin{array}{r} (C) \quad 9 \quad A \quad 5 \\ + \quad 9 \quad 4 \quad A \\ \hline 1A \quad 4 \quad 4 \end{array}$$

$$\begin{array}{r} (D) \quad A \\ + A \\ + A \\ \hline BA \end{array}$$

Column-II

(p) $A = 6, B = 1$

(q) $A = 9$

(r) $A = 5, B = 1$

(s) $A = 8$

Very Short Answer Questions :

DIRECTIONS : Give answer in one word or one sentence.

- In a two-digit number, the digit in the units place is four times the digit in the tens place and sum of the digits is equal to 10. What is the number?
- Check whether 923 is divisible by 11.
- Number 90 is divisible by which numbers?
- If $\overline{21y5}$ is a multiple of 9, where y is a digit, what is the value of y ?
- If the division $N \div 5$ leaves a remainder of 2, what might be the ones digit of N ?

Short Answer Questions :

DIRECTIONS : Give answer in 2-3 sentences.

- Test the divisibility of 2346501 by 11.
- If $24a$ is a multiple of 5, where a is a digit, what is the value of a ?
- Find the value of y in each of the following:

$$\begin{array}{r} (i) \quad y \quad 8 \quad 5 \quad y \\ + \quad 1 \quad 9 \quad 1 \quad 5 \\ \hline \quad 9 \quad y \quad y \quad 2 \end{array}$$

$$\begin{array}{r} (ii) \quad y \quad 9 \quad y \quad 9 \\ + \quad 5 \quad y \quad 7 \quad y \\ \hline \quad 8 \quad y \quad 0 \quad 1 \end{array}$$

4. Write the number 927643 in the expanded form
5. Put the value in the star so that the numbers are divisible by 11.
6 2 6 * 7.
6. Without performing actual division, find the remainder when 9249826 is divided by 11.
7. In the given multiplication, the value of A is

$$\begin{array}{r} A \ B \\ \times \ 5 \\ \hline CAB \end{array}$$
8. A three-digit number $42x$ is divisible by 9. Find the value of x .

2 EXERCISE

Text-Book Exercise :

1. Write the following in the usual form.
 - (i) $10 \times 5 + 6$ (ii) $100 \times 7 + 10 \times 1 + 8$
 - (iii) $100 \times a + 10 \times c + b$
2. Find the value of the letters in each of the following and give reasons for the steps involved.
 - (i)

$$\begin{array}{r} 3 \ A \\ + 2 \ 5 \\ \hline B \ 2 \end{array}$$
 - (ii)

$$\begin{array}{r} 1 \ 2 \ A \\ + 6 \ A \ B \\ \hline A \ 0 \ 9 \end{array}$$
3. Check the divisibility of 152875 by 9.
4. Check the divisibility of 2146587 by 3.
5. If $31z5$ is a multiple of 3, where z is a digit, what might be the values of z ?
4. Find A , B and C :

$$\begin{array}{r} C \ B \ A \\ + \ C \ B \ A \\ \hline 1 \ A \ 3 \ 0 \end{array}$$
5. Find the value of k where $31k2$ is divisible by 6.
6. A three-digit number $2a3$ is added to the number 326 to give a three-digit number $5b9$ which is divisible by 9. Find the value of $b - a$.
7. Find A , B , C and D :

$$\begin{array}{r} 8 \ A \ B \ C \\ - A \ B \ C \ 5 \\ \hline D \ 4 \ 8 \ 8 \end{array}$$
8. Let $E = 3$, $B = 7$ and $A = 4$. Find the other digits in the sum.

$$\begin{array}{r} B \ A \ S \ E \\ + B \ A \ L \ L \\ \hline G \ AM \ E \ S \end{array}$$
9. Find the value of the letters in each of the following:

$$\begin{array}{r} P \ Q \\ \times \ 6 \\ \hline Q \ Q \ Q \end{array}$$

Exemplar Questions :

1. Suppose that the division $x \div 5$ leaves a remainder 4 and the division $x \div 2$ leaves a remainder 1. Find the ones digit of x .
2. Find the least value that must be given to number a so that the number 91876a2 is divisible by 8.

$$\begin{array}{r} 1 \ P \\ \times \ P \\ \hline Q \ 6 \end{array}$$
3. If $\frac{1 \ P}{Q \ 6}$ where $Q - P = 3$, then find the values of P and Q .
10. If 123123A4 is divisible by 11, find the value of A .

HOTS Questions :

1. If $31z5 + 51z3$ is divisible by 3, where z is a digit less than 5, then the values of z are ____.
2. Find what @ and # stand for in each of the following.
 - (i) $1 @ 0 \# 3 = 4$ (ii) $2 @ 2 \# 4 = 5$
 - (iii) $3 @ 3 \# 3 = 30$ (iv) $5 @ 5 \# 3 = 28$
 - (v) $2 @ 5 \# 3 = 30$ (vi) $2 @ 3 \# 2 = 64$
3. What is the value of the digit a if the number $a37382$ is divisible by 11?

3 EXERCISE

Single Option Correct:

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONLY ONE is correct.

- The generalised form of the number 234 is
 (a) $2 \times 100 + 3 \times 10 + 4$ (b) $3 \times 100 + 4 \times 10 + 2$
 (c) $4 \times 100 + 2 \times 10 + 3$ (d) None of these
- The number $5 \times 100 + 7 \times 10 + 9$ in usual form is
 (a) 795 (b) 759
 (c) 579 (d) 597
- Find the values of A, B in the following:

$$\begin{array}{r} A \quad B \\ \times \quad 2 \\ \hline B \quad 02 \end{array}$$
 (a) 5, 1 (b) 5, 5
 (c) 1, 1 (d) 1, 5
- Which of the following numbers is not divisible by 22?
 (a) 200 (b) 40
 (c) 66 (d) 83
- If the number $1x8$ is divisible by 3, then x is equal to
 (a) 0 or 3 or 6 or 9 (b) 4
 (c) 5 (d) 7
- If the three-digit number $6y8$ is divisible by 9, then the value of y is
 (a) 1 (b) 2
 (c) 3 (d) 4
- Find the value of A in the following:

$$\begin{array}{r} 1 \quad A \\ \times \quad A \\ \hline A \quad 9 \end{array}$$
 (a) 1 (b) 2
 (c) 3 (d) 4
- Which of the following numbers is not divisible by 10?
 (a) 120 (b) 340
 (c) 760 (d) 1246
- If $P \div 2$ leaves a remainder of 1, what might be the ones digit of P?
 (a) 2 (b) 7
 (c) 4 (d) 8
- If a number divided by 5 leaves remainder 1, what might be the ones digit of number?
 (a) 6 (b) 2
 (c) 5 (d) 0

More Than One Option Correct :

DIRECTIONS : This section contains multiple choice questions. Each question has 4 choices (a), (b), (c) and (d) out of which ONE or MORE may be correct.

- Which of the following numbers are divisible by 2?
 (a) 63 (b) 47
 (c) 120 (d) 332
- If $\overline{273x6}$ is a multiple of 3, where x is a digit, what might be the value of x ?
 (a) 0 (b) 3
 (c) 6 (d) 9
- Which of the following numbers are divisible by 5?
 (a) 27450 (b) 28594
 (c) 18755 (d) 18743
- The number 211539 is not divisible by
 (a) 2 (b) 5
 (c) 3 (d) 9
- Which of the following numbers is divisible by 5?
 (a) 2035 (b) 490
 (c) 32100 (d) 123456
- Which of the following numbers is not divisible by 11?
 (a) 303457 (b) 25679
 (c) 890612 (d) 23456
- If a number is divisible by 12, then it is divisible by
 (a) 3 (b) 2
 (c) 8 (d) 4
- The digit, which can not replace * in $26 * 84$ so that it is divisible by 3, is
 (a) 0 (b) 1
 (c) 2 (d) 4
- Which of the following numbers is not divisible by 3 and 2?
 (a) 1023 (b) 1032
 (c) 4029 (d) 2512

Passage Based Questions :

DIRECTIONS : Study the given paragraph(s) and answer the following questions.

PASSAGE

Solve the game of addition.

- Find the value of R :

$$\begin{array}{r} 2 \quad 1 \quad R \\ + 1 \quad R \quad 3 \\ \hline 3 \quad 6 \quad 8 \end{array}$$
 (a) $R = 2$ (b) $R = 4$
 (c) $R = 6$ (d) $R = 5$

2. Solve the cryptarithm

$$\overline{AB} + \overline{BA} = \overline{DAD}$$

- (a) $A=2, B=9, D=1$ (b) $A=1, B=9, D=2$
 (c) $A=2, B=8, D=1$ (d) $A=9, B=8, D=1$

3. Solve : $\overline{ON} + \overline{ON} = \overline{GO}$

- (a) $O=4, N=2, G=7$ (b) $O=3, G=9, N=4$
 (c) $O=2, N=6, G=5$ (d) $O=2, N=1, G=5$

Assertion & Reason :

DIRECTIONS : Each of these questions contains an Assertion followed by Reason. Read them carefully and answer the question on the basis of following options. You have to select the one that best describes the two statements.

- (a) If both **Assertion** and **Reason** are **correct** and Reason is the **correct explanation** of Assertion.
 (b) If both **Assertion** and **Reason** are correct, but Reason is **not the correct explanation** of Assertion.
 (c) If **Assertion** is **correct** but **Reason** is **incorrect**.
 (d) If **Assertion** is **incorrect** but **Reason** is **correct**.

1. **Assertion :** 678 is not divisible by 12.

Reason : A number is said to be divisible by 12 if it is divisible by 3.

2. **Assertion :** The generalised form of a two digit number is $10a + b$.

Reason : A number is said to be in a generalized form if it is expressed as the sum of the product of its digit with respect to its place values.

Multiple Matching Question :

DIRECTIONS : Following question has four statements (A, B, C and D) given in Column I and five statements (p, q, r, s, t) in Column-II. Any given statement in Column-I can have correct matching with one or more statement(s) given in Column-II.

1. Match the following

Column-I

- (A) 60643824
 (B) 1246800
 (C) 6584919
 (D) 4268134

Column-II

- (p) Divisible by 2
 (q) Divisible by 3
 (r) Divisible by 4
 (s) Divisible by 5
 (t) Divisible by 11

Integer Type Questions :

DIRECTIONS : Answer the following questions. The answer to each of the question is a single digit integer, ranging from 0 to 9.

- Without subtracting, $98 - 89$ is divisible by
- In the expansion of $394 = 3 \times 100 + a \times 10 + 4$, value of a is
- $\overline{31x}$ is divisible by 2 if $x =$
- $A \times A = \overline{3A}$ when A
- $1x3y6$ is a five digit number where x, y are digits and y exceeds x by 6. If this number is divisible by 18, then the value of $\frac{y}{x}$ is

SOLUTIONS

Brief Explanations of Selected Questions

1 EXERCISE

Fill in the Blanks :

- 5 2. 8 3. 5 4. 9
- A = 6, B = 3 6. $x = 0$ 7. 11
- Same : Two digit number is divisible by 11 be it is 11, 22, 33, 44, ..., 99 i.e. its digits are same.
- 6 : 6 is a number divisible by 2 but not by 4.

True / False :

- True 2. True 3. False 4. True
- True 6. False 7. False
- False : 2 is a prime number which is even also.

Match the Columns :

- (A) $\rightarrow$ (s); (B) $\rightarrow$ (p); (C) $\rightarrow$ (q); (D) $\rightarrow$ (r)

(A) We have,

$$\begin{array}{r} 3 \quad 1 \quad A \\ + 1 \quad A \quad 3 \\ \hline 5 \quad 0 \quad 1 \end{array}$$

Hence, we have to find the value of A which can take values from 0 to 9.

As A take values from 0 to 9. Therefore, A + 3 can take value from 3 to 12. Since, digit at the units place of the sum of the two digit A and 3 is 1. Therefore either A + 3 is equal of 1 or A + 3 is a number between 3 and 12 whose units digit is 1. Clearly such a number between 3 and 12 is 11.

$$\therefore A + 3 = 11 \Rightarrow A = 8$$

This value of A satisfies the addition in tens and hundred column. Hence, A = 8.

(B) We have,

$$\begin{array}{r} B \quad 9 \\ + 4 \quad A \\ \hline A \quad 5 \end{array}$$

Clearly, 9 + A is a number taking values from 9 to 18. Also, either 9 + A is 5 or it is a two digit number whose units digit is 5. But, 9 + A is greater than or equal to 9.

$$\therefore 9 + A = 15 \Rightarrow A = 6$$

Now, considering tens column, we have

$$B + 4 + 1 = 6 \Rightarrow B + 5 = 6 \Rightarrow B = 1$$

Hence, A = 6 and B = 1.

(C) We have,

$$\begin{array}{r} 9 \quad A \quad 5 \\ + 9 \quad 4 \quad A \\ \hline 1A \quad 4 \quad 4 \end{array}$$

In the ones column the sum of 5 and A is 4. This means that the sum of 5 and A is a two digit number between 10 and 19 whose units digit is 4. Clearly, such a number is 14.

$$\therefore 5 + A = 14 \Rightarrow A = 9$$

Clearly, this value of A satisfies the addition in tens and hundreds columns.

(D) We have,

$$\begin{array}{r} A \\ + A \\ \hline A \\ \hline BA \end{array}$$

Here, values of two letters A and B are to be found.

In ones column the sum of three A's is a number whose ones digit is A, which is possible only when the units digit of the sum of two A's is 0. This happens only for A = 0 and A = 5.

If A = 0, then the sum of three A's is 0.

Since $A + A + A = \overline{BA}$. Therefore, B = 0.

This is not possible as $\overline{BA}$ is a two digit number.

So, A = 5.

$$\therefore A + A + A = \overline{BA}$$

$$\therefore 3A = \overline{BA} \Rightarrow 3 \times 5 = 10B + A$$

$$\Rightarrow 15 = 10B + 5 \Rightarrow 10B = 10 \Rightarrow B = 1$$

Hence, A = 5 and B = 1.

Very Short Answer Questions :

- Let the original number be $10a + b$.
Then, $b = 4a$ and $a + b = 10$.
We put $b = 4a$ in $a + b = 10$
so that $a + 4a = 10$, i.e., $5a = 10$, i.e., $a = 2$.
 $\therefore a = 2$ and $b = 4a = 8$.
Hence, the number is $10a + b = 20 + 8 = 28$.
- 923 fails the test, because $(9 + 3) - 2 = 10$, which is not 0 or 11. And indeed it is not divisible by 11.
- If a number is divisible by 'a' and 'b', where 'a' & 'b' are co-primes, then that number is divisible by 'ab'.
Here 2, 45; 5, 18; 9, 10 are co-primes.
In each case, the product is also equal to 90.

4. Since $\overline{21y5}$ is a multiple of 9. Therefore, the sum of its digits is a multiple of 9.
 i.e. $2 + 1 + y + 5$ is a multiple of 9
 $\Rightarrow y + 8$ is a multiple of 9
 $\Rightarrow y + 8 = 0$, or 9 or 18 or 27, 36, ...
 But, y is a digit. So, y can take values 0, 1, 2, ..., 9
 $\therefore y + 8$ can take values 8, 9, 10, 11, ..., 17.
 From (i) and (ii), we get
 $y + 8 = 9 \Rightarrow y = 9 - 8 = 1$
 Hence $y = 1$
5. The ones digit when divided by 5, must leave a remainder of 2. So, ones digit must be 7.

Short Answer Questions :

- Not completely divisible.
- $a = 0, 5$
- (i) $y = 7$ (ii) $y = 2$
- $927643 = 9 \times 100000 + 2 \times 10000 + 7 \times 1000 + 6 \times 100 + 4 \times 10 + 3$
- $* = 6$
- Remainder = 3
- $$\begin{array}{r} 25 \\ \times 5 \\ \hline 125 \end{array}$$
 True
 So, $A = 2, B = 5, C = 1$
- Since $42x$ is divisible by 9, the sum of its digits, i.e. $4 + 2 + x$ must be divisible by 9.
 i.e. $6 + x$ is divisible by 9
 i.e. $6 + x = 9$ or 18,
 Since x is a digit, therefore, $6 + x = 9$
 or $x = 3$.

2 EXERCISE

Text-Book Exercise :

- (i) 56 (ii) 718 (iii) acb
- (i) $A = 7, B = 6$ (ii) $A = 8, B = 1$
- The sum of the digits of 152875 is $1 + 5 + 2 + 8 + 7 + 5 = 28$. This number is not divisible by 9. We conclude that 152875 is not divisible by 9.
- The sum of the digits of 2146587 is $2 + 1 + 4 + 6 + 5 + 8 + 7 = 33$. This number is divisible by 3 (for $33 \div 3 = 11$). We conclude that 2146587 is divisible by 3.
- 0, 3, 6 or 9

Exemplar Questions :

- Since $x \div 5$ leaves a remainder 4, so ones digit of x can be 4 or 9. Also, since $x \div 2$ leaves a remainder 1, so ones digit must be 9 only.
- $a = 3$

- $P = 6$ and $Q = 9$
- $A = 5, B = 6, C = 7$
- k is either 0 or 3, 6, 9
- 2
- $A = 7, B = 2, C = 3, D = 1$
- $S = 8, L = 5, M = 9, G = 1$
- $P = 7, Q = 4$
- $A = 4$

HOTS Questions :

- $31z5 + 51z3$ is divisible by 3
 $\Rightarrow 82(2z)8$ is divisible by 3 where $2z$ is the digit at the tens place.
 A number is divisible by 3 if the sum of its digits is divisible by 3.
 $\Rightarrow 8 + 2 + 2z + 8$ is divisible by 3
 $\Rightarrow 18 + 2z$ is divisible by 3.
 Given that $z < 5$
 So, maximum value of $18 + 2z = 26$, when $z = 4$
 And minimum value of $18 + 2z = 18$, when $z = 0$
 From 18 to 26, the numbers divisible by 3 are 18, 21 & 24.
 $18 + 2z$ can have the values 18 & 24 which are divisible by 3 for $z = 0$ & $z = 3$ respectively.
- (i) $@ = -, \# = +$ (ii) $@ = \div, \# = +$
 (iii) $@ =$ raised to the power, $\# = +$
 (iv) $@ = \times, \# = +$ (v) $@ = \times, \# = \times$
 (vi) $@ =$ raised to the power, $\# =$ raised to the power
- $a = 4$

3 EXERCISE

Single Option Correct :

- (a)
- (c)
- (a)

$$\begin{array}{r} 51 \\ \times 2 \\ \hline 102 \end{array}$$
- (d) In 83, one's digit is not divisible by 2.
- (a) $1 + 0 + 8 = 9$
 $1 + 3 + 8 = 12$; 9, 12, 15, 18 each is divisible by 3.
- (d) $6 + 4 + 8 = 18$ is divisible by 9.
- (c)

$$\begin{array}{r} 13 \\ \times 3 \\ \hline 39 \end{array}$$
- (d) In 1246, one's digit is not 0.
- (b)
- (a)

More Than One Option Correct :

- (c, d) We know that a number is divisible by 2 if its units digit is 0 or 2 or 4 or 6 or 8.
- (a, c, d) A number is a multiple of 3, if sum of values of its digits are also multiple of 3.
Here $2 + 7 + 3 + x + 6 = 18 + x$ will be multiple of 3 if $x = 0, 3, 6, 9$.
- (a, c) A number is divisible by 5 if its units digit is either 0 or 5.
- (a, b, d)
- (a, b, c)
- (b, c, d)
- (a, b, d)
- (a, c, d)
- (a, c, d)

Passage Based Questions :

- (d) Here there is only one letter R, which we need to find.
Now, $R + 3 = 8$
Which is only possible when $R = 5$
Also $1 + R = 6$, This is only possible when $R = 5$.
Hence $R = 5$.
- (a) Clearly, $\overline{AB}$ and $\overline{BA}$ are two digit numbers
So, maximum value of their sum is $99 + 99 = 198$. This means that the number $\overline{DAD}$ is at most equal to 198. So, D must be equal to 1. Note that D can not be zero as $\overline{DAD}$ is a three digit number.
Now, $\overline{AB} + \overline{BA} = \overline{DAD}$
 $\Rightarrow (10A + B) + (10B + A) = \overline{1A1}$
 $\Rightarrow 11A + 11B = \overline{1A1}$
 $\Rightarrow 11(A + B) = \overline{1A1} \quad \dots(i)$
Clearly, LHS of this equation is a multiple of 11.
So, RHS must be a multiple of 11 having digits at units and hundreds place are unity. RHS can take ten values viz. 101, 111, 121, 131,, 191.
Out of these values only 121 is a multiple of 11.
Therefore, $A = 2$.
Substituting $A = 2$ in (i), we get
 $11(2 + B) = 121 \Rightarrow 2 + B = 11 \Rightarrow B = 9$
Hence, $A = 2$, $B = 9$ and $D = 1$.
- (c) Here, $\overline{ON} + \overline{ON} = \overline{GO}$
 $\Rightarrow 2 \times \overline{ON} = \overline{GO}$
So, $2 \times \overline{ON}$ is two-digit number, it follows that $\overline{ON}$ is at most 49 (because $2 \times 50 = 100$) therefore, O is at most equal to 4. We have to consider all four possibilities i.e., O is equal to 4, 3, 2, 1.

Case I : If O is 4 then $N = 2$ or 7.

This makes $ON = 42$.

or $ON = 47$ (but this is not possible)

$$2(ON) = 2(42) = 84$$

Case II : If O is 3 and N cannot take any value as we cannot get 3 in the units place.

Case III : If O is 2 then $N = 1$ or 6.

This makes $ON = 21$ or $ON = 26$

$$2(ON) = 2(21) = 42 \text{ or } 2(ON) = 2(26) = 52$$

Case IV : If O is 1 then N cannot take any value as we cannot get 1 in the units place. So, possible options are

$$2 \times 42 = 84 \text{ with } O = 4, N = 2, G = 8;$$

$$2 \times 21 = 42 \text{ with } O = 2, N = 1, G = 4;$$

$$2 \times 26 = 52 \text{ with } O = 2, N = 6, G = 5.$$

Assertion & Reason :

- (c) Assertion : Correct; Reason : Incorrect.
We have, 678 is not divisible by 4 as 78 is not divisible by 4.
But 678 is divisible by 3.
 $\therefore$ 678 is not divisible by 12.
So, a number is said to be divisible 12 if it is divisible by 3 and 4.
- (a) Assertion : Correct; Reason : Correct.
Reason correctly explains assertion.

Multiple Matching Questions :

- (A) $\rightarrow (p, q, r)$; (B) $\rightarrow (p, q, r, s)$; (C) $\rightarrow (q, t)$; (D) $\rightarrow (p)$
(A) 60643824 is divisible by 2 since it has unit's digit 4.
Sum of it's digit = $6 + 0 + 6 + 4 + 3 + 8 + 2 + 4 = 33$ which is divisible by 3.
It is divisible by 4 as its last 2 digits 24 is divisible by 4.
(sum of digits at odd places) – (sum of digits at even places)
 $= (2 + 3 + 6 + 6) - (4 + 8 + 4 + 0)$
 $= (17 - 16) = 1$ which is not divisible by 11.
(B) 1246800 is divisible by 2 as it has 0 at unit's place
1246800 is also divisible by 4 as it has 00 as last 2 digits.
Sum of it's digit = $1 + 2 + 4 + 6 + 8 + 0 + 0 = 21$ which is divisible by 3.
It is divisible by 5 also as it has 0 at units place.
(sum of digits at odd places) – (sum of digits at even places)
 $= (1 + 4 + 8 + 0) - (2 + 6 + 0)$
 $= (13 - 8)$
 $= 5$
(C) 6584919 is not divisible by 264 and 5

Sum of digits = $6 + 5 + 8 + 4 + 9 + 1 + 9 = 42$ which is divisible by 3.

(sum of digits at odd places) – (sum of digits at even places)
 $= (6 + 8 + 9 + 9) - (5 + 4 + 1)$
 $= 32 - 10$
 $= 22$ which is divisible by 11

(D) 4268134 is divisible by 2 as units digit is even. It is not divisible by 4.

Sum of its digits = $4 + 2 + 6 + 8 + 1 + 3 + 4 = 28$ which is not divisible by 3.

(sum of digits at odd places) – (sum of digits at even places)
 $= (4 + 1 + 6 + 4) - (2 + 8 + 3)$
 $= (15) - (13) = 2$

Integer Type Questions :

1. (9) We know that if $\overline{ab}$ and $\overline{ba}$ are any two digit numbers. Then

$$\overline{ab} = 10a + b \text{ and } \overline{ba} = 10b + a$$

$$\begin{aligned} \therefore \overline{ab} - \overline{ba} &= 10a + b - 10b - a \\ &= 9a - 9b = 9(a - b) \end{aligned}$$

This shows that it is divisible by 9.

$\therefore 98 - 89 = 9$, which are divisible by 9.

2. (9) Since $394 = 3 \times 100 + 9 \times 10 + 4$

Compare this with $3a4 = 3 \times 100 + a \times 10 + 4$, we get $a = 9$.

3. (0) $\overline{31x}$ is divisible by 2 if the units digit of number is 0, 2, 4, 6 or 8.
 $\therefore 310$ is divisible by 2.

4. (6) $6 \times 6 = 36 \therefore A = 6$.

5. (7) $1x3y6$ is divisible by 18 $\Rightarrow$ divisible by 9

$$1 + x + 3 + y + 6 = 10 + x + y \quad (x + y = 8)$$

$$y - x = 6 \quad \dots(1)$$

[Given in the problem]

$$y + x = 8 \quad \dots(2)$$

Subtracting (1) from (2)

$$y + x = 8$$

$$y - x = 6$$

$$2y = 14$$

$$y = 7 \Rightarrow x = 1$$

$$\frac{y}{x} = 7$$

